Topology and Geometry

Parts I and II

Lecture notes

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About these notes

These notes are written to be read alone, one lecture at a time. Each lecture opens with its objectives and the outcomes it promises. It proves its results at the level shown by a tag beside each statement, works at least one example in full, and ends with self-checks. The answer to each self-check follows it. Longer lectures are divided into units, such as Unit 8A and Unit 8B. A unit ends where a new task begins. A section marked *(optional)* is not used later in the course.

A statement with no tag, or with the tag *proved*, is proved in full. The tag *sketched* means that the central idea is given correctly and completely, and routine details are omitted. The tag *cited* means that the proof belongs elsewhere. The source and a precise statement are then named. The tag *imported* marks a result of the prerequisite course *Linear Algebra for Physics*. A result proved with a cited tool carries the same qualification. Physical laws are not theorems of these notes. A lecture states the laws it uses in a paragraph headed *Physical input*.

The course has two parts. We first compare spaces by continuous maps, and by loops that cannot be shrunk to a point. We then add coordinates, tangent vectors and differential forms, so that calculus makes sense on a curved space. Integrating closed forms joins this calculus to the earlier comparisons: a closed form with a nonzero integral around a loop has no global primitive. In Part II a metric adds lengths and angles, and a connection compares vectors at different points. The failure of this comparison around a small loop is curvature. The same construction with a phase in place of a vector is a gauge field. Integers then appear as normalised integrals of curvature over closed manifolds. These integers constrain quantum phases, defects in ordered media and zero modes of the Dirac operator.

Part I

Topology and Manifolds

Part I asks when two spaces are the same, and how to detect that they are not. Lectures 1 and 2 fix what a space and a continuous map are. Lectures 3 to 6 build detectors: loops, covering spaces and homology. Lectures 7 to 12 add smoothness: charts, tangent vectors, flows and forms, ending with Stokes' theorem. Lectures 13 and 14 turn integrals of closed forms back into detectors, as de Rham cohomology and the degree.

Part I assumes *Linear Algebra for Physics*: duals, bilinear forms, signature, the exterior algebra with the determinant as a top wedge, and metric spaces. It also assumes multivariable calculus, including the inverse and implicit function theorems and the existence and uniqueness of solutions of ordinary differential equations, which are cited. The matrix groups $O(n)$, $U(n)$ and $SU(n)$ are assumed only as sets of matrices. Where Lectures 4 to 7 use quantum mechanics, they state it as physical input.

Lecture 1

Spaces and continuity

Objectives. *We compare spaces using bijections that are continuous in both directions. An open interval and the whole line are the same in this sense, despite their different lengths. A line and a plane are different, as Lecture 2 proves. We define continuity using open sets, without choosing a distance. We then construct such bijections for an interval and a line, a punctured circle and a line, and an open disc and a plane.*

By the end of this lecture you will be able to:

- Construct a topology from a metric or a basis.
- Test continuity by inverse images and verify a homeomorphism.
- Use the continuity tests for subspaces and finite products.
- Separate points in a Hausdorff space and prove uniqueness of sequence limits.

1.1 Open sets from a metric

We recall the distance function from *Linear Algebra for Physics*, Lecture 13.

Definition 1.1 (Metric)**.** A *metric* on a set X is a function $d\colon X \times X \to [0,\infty)$ such that $d(x,y) = 0$ exactly when $x = y$, $d(x,y) = d(y,x)$, and $d(x,z) \le d(x,y) + d(y,z)$ for all $x, y, z \in X$.

The last condition is the triangle inequality.

Definition 1.2 (Open ball)**.** Let d be a metric on a set X. The *open ball* with centre $x \in X$ and radius $r > 0$ is

$$B_r(x) = \{y \in X : d(x,y) < r\}.$$

Definition 1.3 (Open set in a metric space)**.** In a metric space (X,d), a subset $U \subseteq X$ is *open* if every $x \in U$ has a ball $B_r(x) \subseteq U$ with $r > 0$.

The radius may depend on the point.

For example, in $\mathbb{R}$ with $d(x,y) = |x-y|$, an interval (a,b) is open. At $x \in (a,b)$ take $r = \min(x-a, b-x) > 0$. Neither $[a,b]$ nor $[a,b)$ is open when $a < b$, because no ball about a stays in the interval. The empty set is open because it has no points to test.

Every point of an open ball has a smaller ball about it inside the original ball. The triangle inequality explains the inclusion in the figure.

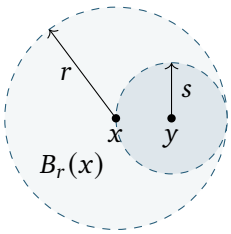


Figure 1.1: If $y \in B_r(x)$, the ball about y of radius $s = r - d(x,y)$ lies inside $B_r(x)$. The smaller boundary is also excluded.

Indeed, if $z \in B_s(y)$, then

$$d(x,z) \le d(x,y) + d(y,z) < d(x,y) + s = r.$$

Thus every open ball is open.

Proposition 1.4 (Open sets of a metric space). *The open subsets of a metric space include $\varnothing$ and X. Arbitrary unions and finite intersections of open sets are open.*

Proof. The whole set contains every ball. If x belongs to a union of open sets, it belongs to one member of the union. That member contains a ball about x, so the union does too. If $x \in U \cap V$ with U and V open, choose $B_r(x) \subseteq U$ and $B_s(x) \subseteq V$. Then $B_{\min(r,s)}(x) \subseteq U \cap V$. Repeating this argument gives any nonempty finite intersection. The intersection of no subsets is X. ∎

The restriction to finite intersections matters. In $\mathbb{R}$ the intersection

$$\bigcap_{n \ge 1} (-1/n, 1/n) = \{0\}$$

is not open.

1.2 Topological spaces and bases

Definition 1.5 (Topology). A *topology* on a set X is a collection $\mathcal{T}$ of subsets of X that contains $\varnothing$ and X and is closed under arbitrary unions and finite intersections. The pair $(X, \mathcal{T})$ is a *topological space*. Its members are called *open sets*.

Definition 1.6 (Closed set). A subset of a topological space X is *closed* if its complement in X is open.

The preceding proposition constructs the *metric topology* of d. The standard topology on $\mathbb{R}^n$ is the topology of the Euclidean distance. We use this topology unless another one is specified. The same set may carry other topologies.

Definition 1.7 (Discrete topology). The *discrete topology* on a set X contains every subset of X.

Definition 1.8 (Indiscrete topology). The *indiscrete topology* on a set X contains only $\varnothing$ and X.

Both satisfy the axioms directly.

A set can be both open and closed. The sets X and $\varnothing$ always are. A set can be neither open nor closed. The set $[0, 1)$ in the standard real line is one example. Its complement is not open at 1.

Definition 1.9 (Neighbourhood)**.** A *neighbourhood* of $x \in X$ is a set containing an open set that contains x.

A neighbourhood itself need not be open.

Definition 1.10 (Interior)**.** For a subset $A \subseteq X$, its *interior* $\operatorname{int} A$ is the union of all open subsets of A.

Definition 1.11 (Closure)**.** For a subset $A \subseteq X$, its *closure* $\overline{A}$ is the intersection of all closed sets containing A.

The interior is open by the union axiom. The closure is closed because its complement is a union of open sets. Every open subset of A is one of the sets being united, so lies in $\operatorname{int} A$. Every closed superset of A contains the defining intersection, so contains $\overline{A}$. Thus the interior is the largest open subset of A, and the closure is the smallest closed superset of A.

The closure describes approximation using open sets.

Proposition 1.12 (Closure criterion)**.** *For $A \subseteq X$, a point $x \in X$ lies in $\overline{A}$ exactly when every open set containing x meets A.*

Proof. If an open set U contains x and misses A, its complement is a closed superset of A omitting x. Hence $x \notin \overline{A}$. Conversely, if $x \notin \overline{A}$, some closed superset C of A omits x. Its complement is an open set containing x and missing A. ∎

A topology therefore determines which points can be approximated by each subset, without assigning distances. In the discrete topology, a point outside a set has the singleton containing that point as an open neighbourhood missing the set. Thus every subset of a discrete space is its own closure. In the indiscrete topology, every open neighbourhood of any point meets every nonempty set. Thus every nonempty subset of an indiscrete space has the whole space as its closure.

In the standard real line, $\operatorname{int}[0, 1) = (0, 1)$ and $\overline{[0, 1)} = [0, 1]$. To check the second equality, note that every open interval about 1 meets $[0, 1)$, so no closed superset can omit 1. The closed set $[0, 1]$ itself contains $[0, 1)$. The figure distinguishes the included and excluded endpoints.

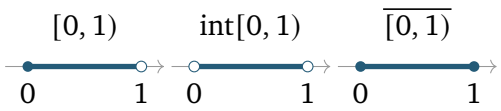


Figure 1.2: The half-open interval, its interior and its closure in $\mathbb{R}$. Filled endpoints belong to the set, and hollow endpoints do not.

Listing all open sets is usually unnecessary. Open balls already determine the metric topology because every open set is a union of balls.

Definition 1.13 (Basis). A collection $\mathcal{B}$ of subsets of X is a *basis* if it covers X and has the following refinement property: whenever $x \in B_1 \cap B_2$ with $B_1, B_2 \in \mathcal{B}$, there is $B_3 \in \mathcal{B}$ such that

$$x \in B_3 \subseteq B_1 \cap B_2.$$

Proposition 1.14 (Topology from a basis). *The unions of members of a basis form a topology. A set U belongs to this topology exactly when every $x \in U$ lies in some $B \in \mathcal{B}$ with $B \subseteq U$.*

Proof. The empty union is $\varnothing$, and the covering property gives X. A union of unions is a union. For two unions U and V, a point $x \in U \cap V$ lies in basis members $B_1 \subseteq U$ and $B_2 \subseteq V$. Refine their intersection to B_3 as in the definition. The union of all these refinements is $U \cap V$. Finite intersections follow by repetition. The final assertion is precisely the condition that U be a union of basis members. ∎

The collection $\mathcal{B}$ is a *basis for a topology* $\mathcal{T}$ when $\mathcal{T}$ is exactly the topology it generates. Open balls form a basis for the metric topology. They cover the space, and the open intersection of two balls contains a ball about each of its points.

For example, open intervals form a basis for $\mathbb{R}$. Intervals with rational endpoints suffice: for $x \in (a, b)$ choose rational p and q with $a < p < x < q < b$.

The rational intervals are countable.

Definition 1.15 (Second countability). A space is *second countable* if its topology has a countable basis.

This is a size restriction: every open set is a union of members of one countable family.

In $\mathbb{R}^n$, boxes with rational endpoints give a countable basis as well. Every open ball contains such a box about each of its points, and every open box contains a ball about each of its points. Thus boxes and balls give the same topology. Second countability will be a condition in the definition of a manifold in Lecture 7.

1.3 Continuity and homeomorphisms

Definition 1.16 (Continuity). A map $f : X \to Y$ of topological spaces is *continuous* if $f^{-1}(V)$ is open in X for every open set $V \subseteq Y$. Here the *inverse image* is the set

$$f^{-1}(V) = \{x \in X : f(x) \in V\}.$$

An inverse image exists for every map. The notation does not assume that the map is invertible. It is enough to test basis members of Y, since inverse images preserve unions.

The inverse image can have several pieces even when the target set is an interval. For the map $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$, we obtain

$$f^{-1}((1, 4)) = (-2, -1) \cup (1, 2).$$

The figure shows the two pieces. Both are open, as required by continuity.

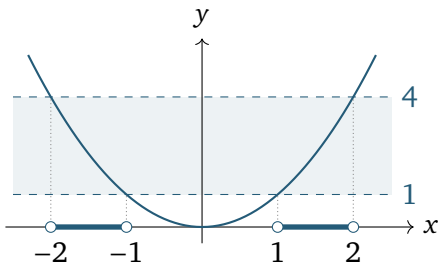


Figure 1.3: The horizontal band $1 < y < 4$ meets the graph $y = x^2$ above two open intervals on the x-axis. Endpoints are excluded.

The same squaring map sends the open interval $(-1, 1)$ onto $[0, 1)$, which is not open in $\mathbb{R}$. Thus continuity does not require images of open sets to be open. The inverse-image condition instead fixes an open set in the target and tests the inputs whose values lie in it.

Proposition 1.17 (Metric and topological continuity). *For metric spaces X and Y, the map $f : X \to Y$ is continuous exactly when, for every $x \in X$ and every $\varepsilon > 0$, there is $\delta > 0$ such that*

$$d_X(x, x') < \delta \quad \Longrightarrow \quad d_Y(f(x), f(x')) < \varepsilon \qquad (x' \in X).$$

The figure shows the ball inclusion in this criterion for the squaring map at $x = 1$, with $\varepsilon = 1$ and $\delta = 1/3$.

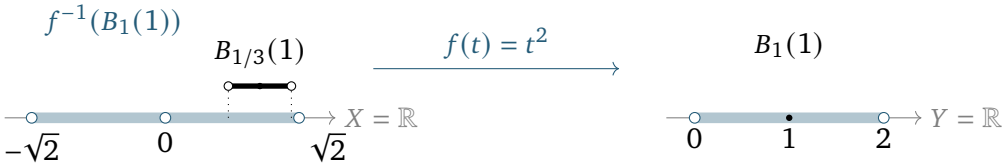


Figure 1.4: For $f(t) = t^2$, the inverse image of $B_1(1) = (0, 2)$ is $(-\sqrt{2}, 0) \cup (0, \sqrt{2})$. It contains $B_{1/3}(1)$.

Proof. Suppose inverse images of open sets are open. The inverse image of $B_\varepsilon(f(x))$ is an open set containing x. It contains a ball $B_\delta(x)$, which gives the implication.

Conversely, suppose the displayed condition holds. For an open set $V \subseteq Y$ and $x \in f^{-1}(V)$ choose $B_\varepsilon(f(x)) \subseteq V$. The condition supplies a ball $B_\delta(x) \subseteq f^{-1}(V)$. Thus $f^{-1}(V)$ is open. ∎

Continuity therefore depends on a metric only through its open sets.

Definition 1.18 (Equivalent norms). Two norms $\|\cdot\|_a$ and $\|\cdot\|_b$ on $\mathbb{R}^n$ are *equivalent* if there are constants $c, C > 0$ such that

$$c\,\|v\|_a \le \|v\|_b \le C\,\|v\|_a \qquad (v \in \mathbb{R}^n).$$

Equivalent norms induce the same topology. Indeed, their balls satisfy

$$B^a_{r/C}(x) \subseteq B^b_r(x) \subseteq B^a_{r/c}(x).$$

Applying these inclusions at each point gives the same open sets. Equivalent norms give the same continuous maps, even though they may assign different lengths.

For a vector $v = (v^1, v^2) \in \mathbb{R}^2$, the superscripts label coordinates. The maximum norm and the Euclidean norm are given by

$$\|v\|_\infty = \max(|v^1|, |v^2|), \qquad \|v\|_2 = \sqrt{(v^1)^2 + (v^2)^2}.$$

The sum of the squared coordinates is at least the larger squared coordinate. It is at most twice that squared coordinate. Taking square roots gives

$$\|v\|_\infty \le \|v\|_2 \le \sqrt{2}\,\|v\|_\infty .$$

We write $B_r^\infty(x)$ and $B_r^2(x)$ for the balls of these two norms. The figure shows the resulting ball inclusions at the origin.

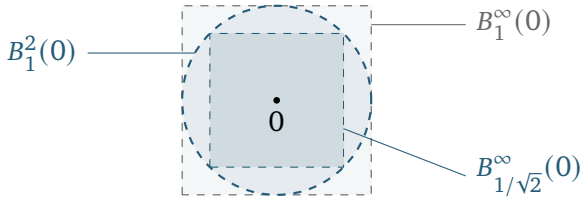


Figure 1.5: The open maximum-norm ball of radius $1/\sqrt{2}$ lies inside the open Euclidean unit ball, which lies inside the open maximum-norm unit ball. All boundaries are excluded.

Identity maps are continuous, and composites of continuous maps are continuous. The latter follows from $(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W))$ for maps $f\colon X \to Y$ and $g\colon Y \to Z$ and an open set $W \subseteq Z$.

Definition 1.19 (Homeomorphism)**.** A *homeomorphism* is a continuous bijection $f\colon X \to Y$ whose inverse $f^{-1}\colon Y \to X$ is continuous. Spaces joined by such a map are *homeomorphic*, written $X \cong Y$.

For example, $f\colon (0,1) \to \mathbb{R}$ given by $f(t) = \tan(\pi(t - \frac{1}{2}))$ is a homeomorphism. Its inverse is given by $f^{-1}(s) = \frac{1}{2} + \pi^{-1}\arctan s$. Both maps are continuous by ordinary one-variable calculus and the metric-continuity proposition. This homeomorphism identifies a bounded interval with an unbounded line. Boundedness and length therefore depend on more than the topology. Continuity of the inverse makes images of open sets open. Thus a homeomorphism preserves open sets in both directions. Every property defined solely from open sets is therefore shared by the two spaces. The graph shows how the values tend to infinity near the omitted endpoints.

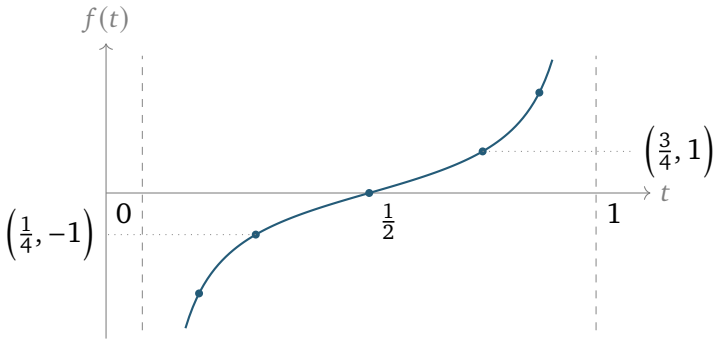


Figure 1.6: The map $f(t) = \tan(\pi(t - \frac{1}{2}))$ on $(0,1)$. The dashed vertical lines mark the omitted endpoints.

1.4 Subspaces and products

The circle $S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ is a subset of the plane. Its open sets must describe motion along the circle, although no plane ball lies inside the circle.

Definition 1.20 (Subspace topology). The *subspace topology* on $A \subseteq X$ consists of the sets $A \cap U$ with U open in X.

Definition 1.21 (Product topology). For topological spaces X and Y, the *product topology* on $X \times Y$ has basis the rectangles $U \times V$ with $U \subseteq X$ and $V \subseteq Y$ open. The topology on a finite product has basis the products of open sets in its factors.

Intersections with A preserve unions and finite intersections, so the subspace sets form a topology. Rectangles cover the product, and the intersection of two rectangles is a rectangle. Hence the basis proposition constructs the product topology.

The restricted metric has balls $A \cap B_r(x)$ for $x \in A$. They are subspace-open. Conversely, if $x \in A \cap U$ with U open in X, a ball $B_r(x) \subseteq U$ gives a restricted ball inside $A \cap U$. Thus the two topologies agree.

For $\mathbb{R}^n$, every product of open sets is a union of open boxes, by choosing an interval about each coordinate. The box-and-ball comparison therefore identifies the finite product and Euclidean topologies.

The figure compares the two constructions. An open arc on the circle is open relative to the circle. An open rectangle in the plane is open in the product of two real lines.

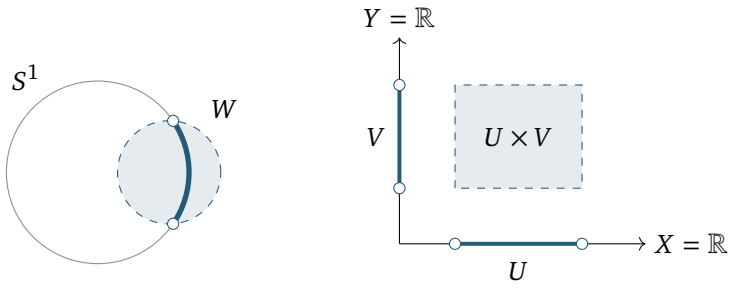


Figure 1.7: An open disc W cuts out an open arc on the circle. Open intervals U and V determine an open rectangle $U \times V$ in the plane.

Proposition 1.22 (Continuity tests for subspaces and products). *For a map $f\colon Z \to A \subseteq X$, continuity into the subspace A is equivalent to continuity of $i \circ f\colon Z \to X$, where $i\colon A \to X$ is inclusion. A map $h\colon Z \to X \times Y$ is continuous exactly when its components $\pi_X \circ h$ and $\pi_Y \circ h$ are continuous. Here $\pi_X(x, y) = x$ and $\pi_Y(x, y) = y$.*

Proof. For an open set $U \subseteq X$, the inverse image under inclusion is $i^{-1}(U) = A \cap U$, so i is continuous. Continuity of f then gives continuity of the composite. Conversely, $f^{-1}(A \cap U) = (i \circ f)^{-1}(U)$, proving continuity into A.

The projections are continuous because $\pi_X^{-1}(U) = U \times Y$ and $\pi_Y^{-1}(V) = X \times V$ are open. Thus continuous h has continuous components. Conversely, for a basis rectangle the inverse image is

$$h^{-1}(U \times V) = (\pi_X \circ h)^{-1}(U) \cap (\pi_Y \circ h)^{-1}(V).$$

Continuous components make this intersection open. Testing basis members proves continuity of h. ∎

In particular, restricting a continuous map $f\colon X \to Y$ to $A \subseteq X$ preserves continuity, since the restriction is $f \circ i$.

For example, the map $t \mapsto (\cos t, \sin t)$ is continuous into $\mathbb{R}^2$ because both components are continuous. Its values lie in S^1, so the subspace test makes it continuous into the circle too. The same proof, with one projection and one inverse image per factor, gives the test for every finite product.

A continuous bijection need not have continuous inverse. Restrict the preceding map to $f\colon [0, 2\pi) \to S^1$. Each point of the circle has exactly one angle in this interval, so f is bijective. Openness depends on the ambient space. The identity

$$[0, \pi) = (-1, \pi) \cap [0, 2\pi)$$

shows that $[0, \pi)$ is open in $[0, 2\pi)$. It is not open in $\mathbb{R}$, since every real open ball about 0 contains negative numbers. Write $g\colon S^1 \to [0, 2\pi)$ for the inverse of f. Its inverse image of this relatively open set is

$$g^{-1}([0, \pi)) = f([0, \pi)).$$

This arc contains $(1, 0)$ but omits the nearby circle points below the horizontal axis. Every open neighbourhood of $(1, 0)$ in the circle contains some of those points. Thus the arc is not open in S^1, so g is not continuous. The figure shows the angles approaching 2π and their images approaching $(1, 0)$ from below. The dashed circle is the boundary of an open disc about $(1, 0)$. The intersection of this open disc with S^1 is an open neighbourhood of $(1, 0)$.

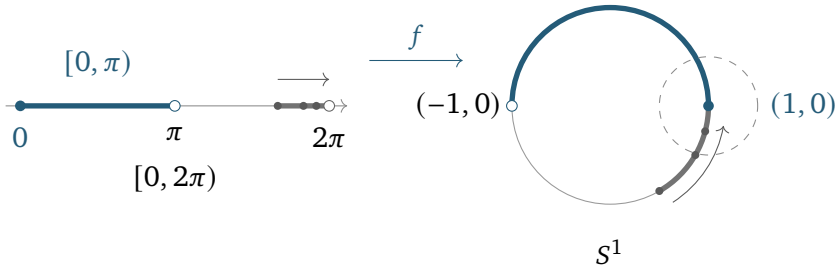


Figure 1.8: The highlighted interval $[0, \pi)$ maps to the upper arc, including $(1, 0)$ and excluding $(-1, 0)$. The grey points approach the excluded endpoint 2π in the interval and the included point $(1, 0)$ on the circle.

Lecture 2 supplies a useful sufficient condition for a continuous bijection to be a homeomorphism.

Self-check. Let $A = (-2, -1] \cup \{0\} \cup [1, 3)$ have the subspace topology from $\mathbb{R}$. Define $h\colon A \to [0, 9)$ by $h(t) = t^2$, with the subspace topology on the target. Find $h^{-1}([0, 4))$. Prove that $[0, 4)$ is open in the target and that its inverse image is open in A.

Answer. The target set is relatively open because $[0, 4) = (-1, 4) \cap [0, 9)$. Solving $0 \le t^2 < 4$ and intersecting with A gives

$$h^{-1}([0, 4)) = (-2, -1] \cup \{0\} \cup [1, 2) = A \cap (-2, 2).$$

The final expression is the intersection of A with an open set of $\mathbb{R}$, so the inverse image is open in A.

Self-check. State in one sentence what a topology records and why we introduced it. Use the subspace definition to decide whether $[0, 1)$ is open in each of $\mathbb{R}$, $[0, 1]$ and $[0, 2]$.

Answer. A topology records the open subsets of a space so that we can define continuity without choosing a distance. The set $[0, 1)$ is not open in $\mathbb{R}$, because no real ball about 0 stays in the set. It is open in both subspaces: intersect either $[0, 1]$ or $[0, 2]$ with $(-1, 1)$.

1.5 Hausdorff spaces and invariants

Limits in a metric space are unique. We identify the separation property of open sets that ensures this uniqueness without a metric.

Definition 1.23 (Hausdorff space)**.** A space X is *Hausdorff* if any two distinct points have disjoint open neighbourhoods.

Definition 1.24 (Convergence)**.** A sequence $(x_n)_{n\geq 1}$ *converges* to $x \in X$ if every open neighbourhood of x contains all x_n for sufficiently large n.

Proposition 1.25 (Separation and limits)**.** *Metric spaces are Hausdorff. Subspaces and finite products of Hausdorff spaces are Hausdorff. A sequence in a Hausdorff space has at most one limit.*

Proof. For distinct points x and y of a metric space, put $r = d(x, y)/3$. The balls $B_r(x)$ and $B_r(y)$ are disjoint. A point in both would give $d(x, y) < 2r < d(x, y)$ by the triangle inequality. Intersecting disjoint open neighbourhoods with a subspace separates its points. Two distinct points in a product differ in some coordinate. Separate those coordinate values and take inverse images under the corresponding projection. The resulting product neighbourhoods are disjoint.

If a sequence had distinct limits x and y, disjoint open neighbourhoods U and V of those points would both contain every term after some common index. This would put a term in $U \cap V = \varnothing$. ∎

The Hausdorff condition can fail.

Definition 1.26 (Cofinite topology)**.** The *cofinite topology* on $\mathbb{R}$ consists of $\varnothing$ and the subsets with finite complement.

It is a topology. A nonempty union has complement contained in the finite complement of any one nonempty member. A finite intersection of nonempty open sets has complement a finite union of finite sets. Two nonempty open sets cannot be disjoint, since their intersection has finite complement in the infinite set $\mathbb{R}$. This space is not Hausdorff. The sequence given by $x_n = n$ converges to every $x \in \mathbb{R}$ in this topology. The terms of this sequence are distinct. Every open neighbourhood of x omits only finitely many real numbers, so it omits only finitely many terms of the sequence. Thus limits need not be unique without the Hausdorff property.

Definition 1.27 (Topological invariant)**.** A *topological invariant* is a property shared by homeomorphic spaces.

Hausdorffness is one example.

Proposition 1.28 (Transport of open sets). *A homeomorphism $f\colon X \to Y$ gives a bijection $U \mapsto f(U)$ between their topologies, preserving unions, intersections and inclusion. Hausdorffness and second countability are topological invariants.*

Proof. If $f\colon X \to Y$ is a homeomorphism and $U \subseteq X$ is open, then $f(U) = (f^{-1})^{-1}(U)$ is open by continuity of the inverse. Continuity of f gives the converse correspondence. A bijection preserves the stated set operations. Images of disjoint open neighbourhoods separate the image points, proving Hausdorff invariance. If $\mathcal{B}$ is a countable basis for X, its images form a countable family of open sets. Bijectivity carries the covering and refinement properties across. Every open set of Y is the image of a union of members of $\mathcal{B}$, hence a union of these image sets. They form a basis for Y. ∎

The standard real line and the cofinite real line are therefore not homeomorphic. They have the same underlying set but different separation properties.

1.6 Explicit homeomorphisms

We compare the open interval, the line, a circle with a point removed, the open disc and the plane. All subsets have their subspace topologies. The interval and the line were identified above. To treat the punctured circle, remove $N = (0, 1)$ and define

$$p\colon S^1 \setminus \{N\} \longrightarrow \mathbb{R}, \qquad p(x, y) = \frac{x}{1 - y}.$$

The line through N and (x, y) meets the horizontal axis at $(p(x, y), 0)$. This construction is called *stereographic projection*. The figure shows this construction.

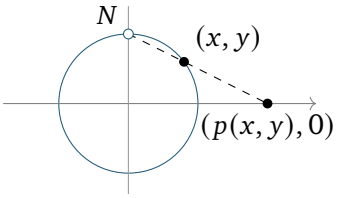


Figure 1.9: Stereographic projection from N onto the horizontal axis. The projection point N is omitted from the circle.

To find the inverse, we parameterise the line through N and $(t, 0)$ by $(\lambda t, 1 - \lambda)$, with $\lambda \in \mathbb{R}$. Substitution in the circle equation gives

$$\lambda^2 t^2 + (1 - \lambda)^2 = 1 \quad \Longleftrightarrow \quad \lambda\big(\lambda(1 + t^2) - 2\big) = 0.$$

The root $\lambda = 0$ gives the omitted point N. The other root is $\lambda = 2/(1 + t^2)$, which gives the inverse

$$p^{-1}\colon \mathbb{R} \longrightarrow S^1 \setminus \{N\}, \qquad p^{-1}(t) = \left(\frac{2t}{1 + t^2}, \frac{t^2 - 1}{1 + t^2}\right).$$

The squared coordinates sum to 1, and the second coordinate is less than 1. Substitution gives $p(p^{-1}(t)) = t$. Conversely, using $x^2 = 1 - y^2$ and $y \neq 1$ gives $p^{-1}(p(x, y)) = (x, y)$. Both maps have continuous coordinate functions with nonzero denominators on their domains. The subspace and product tests therefore make p a homeomorphism. By composition the punctured circle is homeomorphic to $(0, 1)$ as well.

For the open disc $D = \{u \in \mathbb{R}^2 : \|u\| < 1\}$, move each point outward along its radius. Define

$$f_D : D \to \mathbb{R}^2, \qquad f_D(u) = \frac{u}{1 - \|u\|}, \qquad f_D^{-1} : \mathbb{R}^2 \to D, \qquad f_D^{-1}(v) = \frac{v}{1 + \|v\|}.$$

The norm here is Euclidean and continuous. The denominators are positive, so both formulas are continuous, including at 0. For $r = \|u\| < 1$, the norm of $f_D(u)$ is $r/(1 - r)$, giving $f_D^{-1}(f_D(u)) = u$. For $s = \|v\|$, the inverse formula has norm $s/(1 + s) < 1$, so lands in D. It gives $f_D(f_D^{-1}(v)) = v$. The subspace test proves continuity into D. Thus f_D is a homeomorphism. The figure follows concentric circles under this radial map.

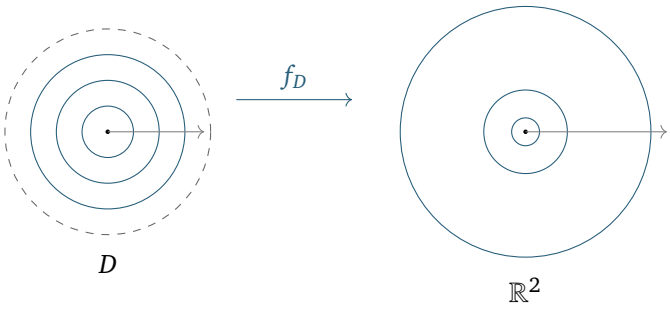


Figure 1.10: The circles of radii 1/4, 1/2 and 3/4 in the open disc map to circles of radii 1/3, 1 and 3 in the plane, with the panels drawn at different scales. The dashed unit circle is excluded from the domain.

The results proved so far form two families:

Spaces identified	Maps used
$(0, 1) \cong \mathbb{R} \cong S^1 \setminus \{N\}$	tangent and stereographic projection
$D \cong \mathbb{R}^2$	radial map f_D

Nothing proved here distinguishes the two families. Lecture 2 proves that a line and a plane are not homeomorphic by examining what happens when a point is removed. It also distinguishes the circle from the line by compactness, and the circle from the closed interval by removing a point. Those distinctions are not assumed here.

Self-check. Here x^i and u^i label coordinates. A squared coordinate is written $(x^i)^2$. For the unit sphere $S^2 = \{(x^1, x^2, x^3) \in \mathbb{R}^3 : (x^1)^2 + (x^2)^2 + (x^3)^2 = 1\}$ and $N = (0, 0, 1)$, prove that the map

$$p_N : S^2 \setminus \{N\} \to \mathbb{R}^2, \qquad p_N(x) = \frac{(x^1, x^2)}{1 - x^3}$$

is a homeomorphism by finding its inverse.

Answer. For $u = (u^1, u^2) \in \mathbb{R}^2$, put $r^2 = (u^1)^2 + (u^2)^2$. The line from N to $(u^1, u^2, 0)$ has points $(\lambda u^1, \lambda u^2, 1 - \lambda)$. The sphere equation becomes

$$\lambda^2 r^2 + (1 - \lambda)^2 = 1 \quad \Longleftrightarrow \quad \lambda\big(\lambda(1 + r^2) - 2\big) = 0.$$

Discarding $\lambda = 0$, which gives N, leaves $\lambda = 2/(1 + r^2)$. The inverse is

$$p_N^{-1}(u) = \frac{(2u^1, 2u^2, r^2 - 1)}{1 + r^2}.$$

Its squared norm is $[4r^2 + (r^2 - 1)^2]/(1 + r^2)^2 = 1$ and its last coordinate is less than 1. Substitution gives $p_N(p_N^{-1}(u)) = u$. For $x \in S^2 \setminus \{N\}$, the sphere equation gives $\|p_N(x)\|^2 = (1+x^3)/(1-x^3)$. Hence $1+\|p_N(x)\|^2 = 2/(1-x^3)$. Substitution into the first two coordinates of the inverse recovers (x^1, x^2), and the last coordinate recovers x^3. Thus $p_N^{-1}(p_N(x)) = x$. The coordinate formulas and the subspace and product tests prove continuity of both maps.

Summary. A topology records which subsets are open. Continuity means that inverse images of open sets are open, and agrees with metric continuity when a metric is present. A homeomorphism requires continuity in both directions. Subspace and product topologies let us test continuity using inclusion and coordinate projections. Hausdorffness is preserved by homeomorphisms and ensures uniqueness of sequence limits. Lecture 2 introduces compactness and connectedness to distinguish more spaces.

Reading. Munkres, *Topology*, second edition, §§12–13 and 15–18: topologies, bases, products, subspaces, separation and continuity.

Lecture 2

Compactness, connectedness and gluing

Objectives. *Open sets let us define properties that distinguish spaces. Compactness expresses a finiteness condition on open covers. Connectedness rules out a separation into two open pieces. We prove how continuous maps preserve these properties, then use them to identify spaces formed by gluing. The main criterion turns a continuous bijection into a homeomorphism when its domain is compact and its target is Hausdorff.*

By the end of this lecture you will be able to:

- Prove the compact-to-Hausdorff criterion for a homeomorphism.
- Distinguish spaces by compactness, and by connectedness after removing a point.
- Test continuity out of a quotient and identify its points correctly.
- Describe the torus, Möbius band, projective plane and rotation group as quotients.

Unit 2A Compactness and connectedness

2.1 Compactness

Definition 2.1 (Compactness). An *open cover* of a space X is a family of open sets $(U_\alpha)_{\alpha\in I}$ whose union is X. A *subcover* is a subfamily that still covers X. The space is *compact* if every open cover has a finite subcover.

Compactness lets us replace data given on open sets near individual points by data on finitely many of those sets covering the whole space.

Definition 2.2 (Sequential compactness). A metric space is *sequentially compact* if every sequence in it has a subsequence converging to a point of that space.

This is the notion used in *Linear Algebra for Physics*, Lecture 13. It is equivalent to open-cover compactness for metric spaces (Munkres, Theorem 28.2). We use the open-cover definition on spaces without a specified metric.

A subset $K \subseteq X$ is called compact when its subspace topology is compact. Equivalently, every cover of K by open sets of X has a finite subcover. This follows because the open sets of K are intersections $K \cap U$ with ambient open sets U (§1.4).

The bounded interval $(0, 1)$ is not compact. The open sets

$$U_n = (1/n, 1), \qquad n \geq 2,$$

cover it, since every $x \in (0, 1)$ satisfies $1/n < x$ for some $n \geq 2$. A nonempty finite selection has a largest index N and misses $(0, 1/N]$. The empty selection covers no points. The figure compares this cover with a particular cover of $[0, 1]$.

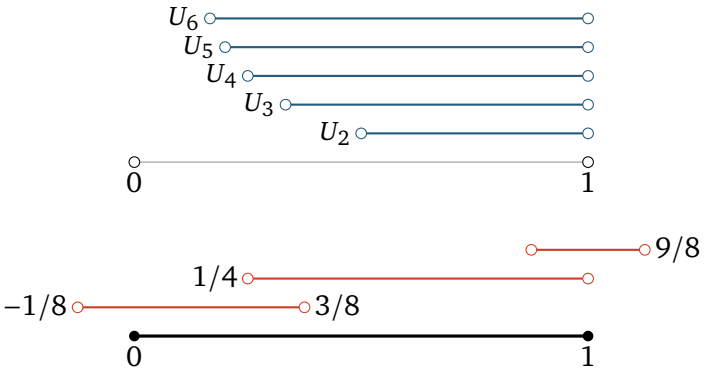


Figure 2.1: The sets $(1/n, 1)$ cover $(0, 1)$ with no finite subcover. The lower panel shows one finite cover of $[0, 1]$; compactness requires a finite subcover for every open cover.

Every finite space is compact. Choose one cover member containing each point. The resulting selection is finite. The empty space needs no members. In contrast, the real line is not compact. The open cover $\{(-n, n) : n \geq 1\}$ has no finite subcover, since a finite selection is contained in its member with the largest index.

Theorem 2.3 (Continuous images and closed subsets). *A continuous image of a compact space is compact. A closed subset of a compact space is compact.*

Proof. Let $f : X \to Y$ be continuous with X compact. Take an ambient open cover (V_α) of $f(X)$. Its inverse images cover X and are open. Choose finitely many of them covering X. The corresponding sets V_α cover $f(X)$.

Now let $C \subseteq X$ be closed, with X compact. Take an ambient open cover of C and add the open set $X \setminus C$. This covers X and therefore has a finite subcover. Discarding $X \setminus C$, if selected, leaves finitely many members covering C. ∎

We use two results about metric spaces with their proofs cited. *Heine–Borel* says that a subset of $\mathbb{R}^n$ is compact exactly when it is closed and bounded (Munkres, Theorem 27.3). Bounded means contained in some ball of finite radius. Thus closed intervals such as $[0, 1]$, closed balls and unit spheres are compact. The hypotheses refer to Euclidean space. Closed and bounded need not imply compact in a general metric space.

Compactness also gives extrema. For a continuous function $f : X \to \mathbb{R}$ on a nonempty compact space X, the image $f(X)$ is nonempty and compact. By Heine–Borel it is closed and bounded. Its supremum and infimum belong to its closure, hence to $f(X)$. Thus f attains a maximum and a minimum.

The circle is compact by Heine–Borel, whereas the real line is not. They cannot be homeomorphic. A homeomorphism would take the compact circle onto the line. This settles one distinction promised in Lecture 1.

2.2 Compact spaces and Hausdorff targets

Lecture 1 gave a continuous bijection with a discontinuous inverse. We seek hypotheses that force the inverse to be continuous. We follow closed sets: a closed subset of a compact domain is compact, and its continuous image is compact. We need that image to be closed. Hausdorffness supplies this last step. Recall that distinct points in a Hausdorff space have disjoint open neighbourhoods (§1.5).

In the figure, finitely many neighbourhoods cover the compact set K. Their corresponding neighbourhoods of y have an open intersection missing K.

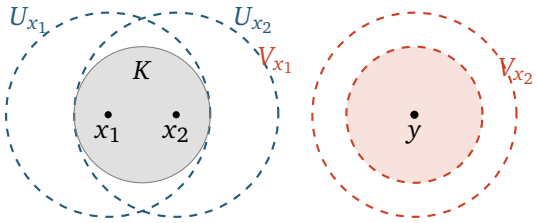


Figure 2.2: The blue neighbourhoods cover K. The red neighbourhoods of y each avoid the corresponding blue set, so their intersection avoids K.

Proposition 2.4 (Compact subsets are closed). *A compact subset of a Hausdorff space is closed.*

Proof. Let $K \subseteq Y$ be compact and $y \in Y \setminus K$. For each $x \in K$, choose disjoint open neighbourhoods U_x of x and V_x of y. Finitely many sets $U_{x_1}, \dots, U_{x_m}$ cover K. Finiteness matters here because finite intersections of open sets are open. The open neighbourhood $V_{x_1} \cap \cdots \cap V_{x_m}$ of y misses their union, so misses K. Thus $Y \setminus K$ is open. If K is empty, it is closed directly. ∎

Continuity can also be tested on closed sets. Inverse images commute with complements, so the open-set test is equivalent to the closed-set test.

Corollary 2.5 (Compact-to-Hausdorff criterion). *A continuous bijection from a compact space to a Hausdorff space is a homeomorphism.*

Proof. Let $f : X \to Y$ satisfy the hypotheses. For a closed subset $C \subseteq X$, compactness of X makes C compact. Its image $f(C)$ is compact, hence closed in Y. This image is the inverse image of C under $f^{-1} : Y \to X$. The closed-set test proves continuity of the inverse. ∎

Lecture 1's continuous bijection $[0, 2\pi) \to S^1$ does not meet the criterion. Its domain is not compact, since it is not closed in $\mathbb{R}$. We will instead use $[0, 1]$ and identify its two endpoints.

2.3 Connectedness

Definition 2.6 (Connectedness). A *separation* of a space X is a pair of disjoint, nonempty open sets whose union is X. The space is *connected* if it has no separation.

Definition 2.7 (Path). A *path* from x to y in X is a continuous map $\gamma : [0, 1] \to X$ with $\gamma(0) = x$ and $\gamma(1) = y$.

Definition 2.8 (Path-connectedness)**.** The space is *path-connected* if such a path exists for every pair $x, y \in X$.

A subset is connected or path-connected when its subspace topology has the corresponding property.

Theorem 2.9 (Intervals, images and paths)**.** *The interval* $[0,1]$ *is connected. Continuous images preserve connectedness and path-connectedness. Every path-connected space is connected.*

Proof. We argue by contradiction. A separation would put a supremum of points on one side into one of two open sets, and either choice contradicts that supremum. Suppose $[0,1] = U \cup V$ is a separation. Choose $a \in U$ and $b \in V$, with $a < b$ after exchanging the names if necessary. Put $c = \sup(U \cap [a,b])$. If $c \in U$, then $c < b$, and openness of U gives a point of $U \cap [a,b]$ larger than c, contradicting that c is an upper bound. If $c \in V$, then $c > a$, and openness of V forbids points of U immediately below c. This contradicts that points of $U \cap [a,b]$ approach their supremum from below. Both cases are impossible, so no separation exists.

For a continuous map $f\colon X \to Y$, a separation of $f(X)$ pulls back to a separation of X. Thus a connected domain has connected image. If X is path-connected, composing its paths with f gives paths between any two image points.

Finally, a path-connected space cannot have a separation $U \cup V$. A path joining a point of U to a point of V would have connected image, since $[0,1]$ is connected. Its intersections with U and V would separate that image. ∎

For example, a square and a ball are path-connected. The segment $\gamma(t) = (1-t)x + ty$ stays in the set and joins any two of its points.

Definition 2.10 (Connected component)**.** A *connected component* is a connected subset maximal under inclusion.

The component of $x \in X$ is the union of all connected subsets containing x. This union is connected. In any proposed separation, each of its connected subsets must lie wholly on the side containing x. It is maximal by its definition. Two components that meet therefore coincide, so components partition X.

Removing a point distinguishes a line from a plane, as the figure shows.

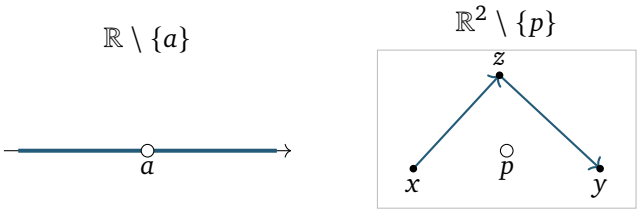


Figure 2.3: Removing a point separates a line into two open rays. In a plane, a broken line can join two points while avoiding the removed point.

The space $\mathbb{R} \setminus \{a\}$ has a separation into its two open rays. In $\mathbb{R}^n \setminus \{p\}$, with $n \geq 2$, take points x and y and choose z outside the two lines through p, x and through p, y. The segments from x to z and from z to y avoid p. They give the path

$$\gamma(t) = \begin{cases} (1-2t)x + 2tz, & 0 \leq t \leq \frac{1}{2}, \\ (2-2t)z + (2t-1)y, & \frac{1}{2} \leq t \leq 1. \end{cases}$$

The two formulas agree at $t = 1/2$. Each affine formula is continuous, so the preimage of a closed set is closed on its half-interval. It is also closed in $[0, 1]$, since both half-intervals are closed there. Their finite union is closed, proving continuity of γ. Thus the punctured space is path-connected.

We can now distinguish the line from the plane, as proposed in Lecture 1. Suppose $f : \mathbb{R} \to \mathbb{R}^2$ were a homeomorphism. Its restriction would give a homeomorphism

$$\mathbb{R} \setminus \{0\} \longrightarrow \mathbb{R}^2 \setminus \{f(0)\}.$$

The domain is disconnected and the target is path-connected, hence connected. A homeomorphism preserves connectedness, so $\mathbb{R} \not\simeq \mathbb{R}^2$. Removing a point cannot distinguish $\mathbb{R}^2$ from $\mathbb{R}^3$, since both remain path-connected. Lecture 3 introduces loops and their continuous deformations to make finer distinctions.

The radial map $x \mapsto x/\|x\|$ also shows that S^n is path-connected for $n \geq 1$, as an image of $\mathbb{R}^{n+1} \setminus \{0\}$.

Remark 2.11 (Optional: topologist's sine curve)**.** The closure of $\{(x, \sin(1/x)) : 0 < x \leq 1\}$ in $\mathbb{R}^2$ is connected but not path-connected (cited: Munkres, §24, Example 7). Later arguments do not depend on this example.

The figure shows part of the graph and the segment added by taking its closure.

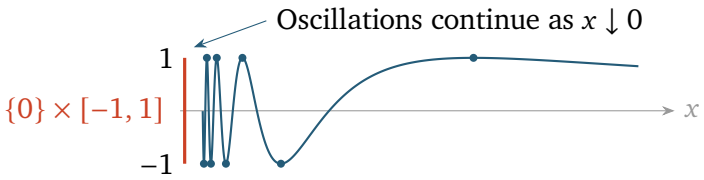


Figure 2.4: The plotted graph has $1/(8\pi) \leq x \leq 1$; its oscillations continue towards the added vertical segment. The marked extrema have heights 1 and -1.

Self-check. Prove that S^1 and $[0, 1]$ are not homeomorphic by removing one point. Explain why deleting an endpoint of the interval is not a useful test.

Answer. Delete the midpoint of $[0, 1]$. The two resulting half-intervals give a separation. Deleting any point from the circle leaves a space homeomorphic to $\mathbb{R}$. Rotate that point to the north pole, then use Lecture 1's stereographic projection. The line is path-connected by straight segments. A homeomorphism would restrict to a homeomorphism between these two punctured spaces, which is impossible. Deleting an endpoint leaves a path-connected half-open interval, so connectedness would not distinguish them.

Unit 2B Gluing

2.4 Quotients and orbit spaces

A periodic coordinate identifies the two ends of an interval. For $[0,1]$ with $0 \sim 1$, a neighbourhood of the identified point must include points near both ends. For $0 < \varepsilon < 1/2$, the set

$$[0,\varepsilon) \cup (1-\varepsilon, 1]$$

is open in $[0,1]$. We want it to be the inverse image of an open neighbourhood after gluing. We define openness in the quotient by this inverse-image rule.

Definition 2.12 (Quotient topology). For an equivalence relation $\sim$ on X, the *quotient set* $X/\!\sim$ consists of the equivalence classes. The projection $q\colon X \to X/\!\sim$ sends each point to its class. The *quotient topology* is defined by

$$V \subseteq X/\!\sim \text{ is open} \quad\Longleftrightarrow\quad q^{-1}(V) \text{ is open in } X.$$

Inverse images preserve unions and finite intersections, so this is a topology. It is the largest topology on the quotient set that makes the projection continuous. The projection is also surjective. Hence a quotient of a compact or path-connected space retains that property, as a continuous image (Theorems 2.3 and 2.9).

Proposition 2.13 (Continuity out of a quotient). *A map $\bar{f}\colon X/\!\sim \to Y$ is continuous exactly when $\bar{f}\circ q$ is continuous. A map $f\colon X \to Y$ that is constant on equivalence classes induces a unique map $\bar{f}$ with $f = \bar{f}\circ q$.*

Proof. Composition with continuous q preserves continuity. Conversely, if $\bar{f}\circ q$ is continuous, then for open $W \subseteq Y$ the set $q^{-1}(\bar{f}^{-1}(W))$ is open. The quotient definition says that $\bar{f}^{-1}(W)$ is open. The induced map is given by $\bar{f}([x]) = f(x)$, which is well-defined precisely when f is constant on each class. Surjectivity of q makes it unique. ∎

Definition 2.14 (Fibre). A *fibre* is the inverse image of one point.

Method for identifying a quotient. To identify $X/\!\sim$ with Y, we proceed as follows.

1. Construct a continuous surjection $f\colon X \to Y$.
2. Prove that its fibres are exactly the equivalence classes.
3. Prove that X is compact and Y is Hausdorff.

The quotient is compact as an image of X. The induced map is a continuous bijection by Proposition 2.13, so Corollary 2.5 makes it a homeomorphism.

Define the circle map $c\colon [0,1] \to S^1$ by

$$c(t) = (\cos(2\pi t), \sin(2\pi t)).$$

The figure depicts its endpoint identification and the inverse image of an open arc through the identified point, with $\varepsilon = 1/8$.

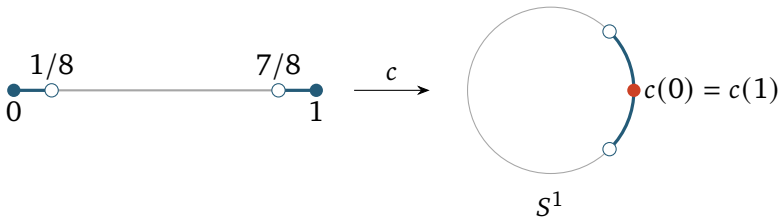


Figure 2.5: The coloured open arc has inverse image $[0, 1/8) \cup (7/8, 1]$, open in $[0, 1]$. The filled interval endpoints have the same image under c.

This map is continuous and surjective onto S^1. Its only repeated value on $[0, 1]$ is the common endpoint value. The induced map $[0, 1]/(0 \sim 1) \to S^1$ is therefore a continuous bijection. Its domain is compact and its target Hausdorff, so it is a homeomorphism by Corollary 2.5.

Group actions describe quotients by symmetries. We first define a group.

Definition 2.15 (Group). A *group* is a set G with an associative product $(g, h) \mapsto gh$, an *identity* $e \in G$ with $eg = ge = g$ for every $g \in G$, and, for each $g \in G$, an *inverse* $g^{-1} \in G$ with $gg^{-1} = g^{-1}g = e$.

Definition 2.16 (Group action). An *action by homeomorphisms* of a group G on X assigns a homeomorphism $x \mapsto gx$ to each $g \in G$, with $ex = x$ and $(gh)x = g(hx)$.

Definition 2.17 (Orbit). The *orbit* of x is $\{gx : g \in G\}$.

Definition 2.18 (Orbit space). The *orbit space* X/G has the quotient topology for these orbits.

For example, the additive group $\mathbb{Z}$ acts on $\mathbb{R}$ by integer translations. The orbit of $x \in \mathbb{R}$ is $\{x + n : n \in \mathbb{Z}\}$. Every orbit meets $[0, 1]$. The only distinct points of $[0, 1]$ in the same orbit are its two endpoints. The figure marks one orbit and its representative in $[0, 1]$.

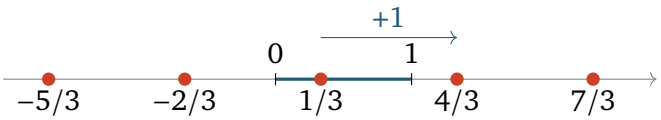


Figure 2.6: The orbit of $1/3$ consists of its integer translates. Exactly one of these points lies in $[0, 1]$.

We prove Hausdorffness for the projective and Möbius quotients by constructing a metric on their orbits. We first verify the metric axioms. We then show that its open sets are exactly the quotient-open sets.

Definition 2.19 (Isometry). An *isometry* preserves a metric.

Suppose $a : X \to X$ is an isometry with $a^2 = \mathrm{id}$. Its orbit classes are $[x] = \{x, ax\}$, with duplicates omitted. The orbit distance is

$$d_{\mathrm{orb}}([x], [y]) = \min\{d(x, y), d(x, ay)\}.$$

The isometry and $a^2 = \mathrm{id}$ give $d(ax, y) = d(a^2x, ay) = d(x, ay)$. Thus changing either

representative interchanges the two distances. The minimum vanishes exactly when the orbits agree. Symmetry follows from the isometry.

For the triangle inequality, choose maps $g, h \in \{\mathrm{id}, a\}$ such that

$$d_{\mathrm{orb}}([x],[y]) = d(x, gy), \qquad d_{\mathrm{orb}}([y],[z]) = d(y, hz).$$

The composition gh is again either id or a. By the triangle inequality and the fact that g is an isometry,

$$d(x, ghz) \le d(x, gy) + d(gy, ghz) = d(x, gy) + d(y, hz).$$

This bounds the orbit distance from $[x]$ to $[z]$ by the sum of the other two. Thus the formula defines a metric.

For $r > 0$, its ball centred at $[x]$ is $q(B_r(x))$: a class lies in that ball exactly when one of its representatives lies in $B_r(x)$. The inverse image of this ball is $B_r(x) \cup B_r(ax)$. Thus orbit-metric balls are quotient-open. Conversely, the preimage of a quotient-open set containing $[x]$ contains some $B_r(x)$. Its image is an orbit-metric ball inside that set, so the set is metric-open. The topologies agree. The orbit space is therefore Hausdorff by Proposition 1.25.

Definition 2.20 (Real projective space)**.** For $n \ge 1$, the *real projective space* $\mathbb{R}\mathrm{P}^n$ is $S^n/(x \sim -x)$. Its points are the lines through the origin in $\mathbb{R}^{n+1}$.

The antipodal map is an isometry, so $\mathbb{R}\mathrm{P}^n$ is Hausdorff. It is compact and path-connected as an image of S^n. Write $D^n = \{v \in \mathbb{R}^n : \|v\| \le 1\}$ for the closed unit disc. Each line meets S^n in an antipodal pair. Keeping the upper hemisphere retains exactly one point of each pair off the equator. On the equator both points remain. Projection onto the first n coordinates identifies the upper hemisphere with a closed disc. Thus only boundary points of that disc still have partners. The figure shows this construction for $n = 2$. The blue point $p = (3/5, 0, 4/5)$ projects to $v = (3/5, 0)$ in the disc. The red points $b = (1, 0, 0)$ and $-b$ remain paired.

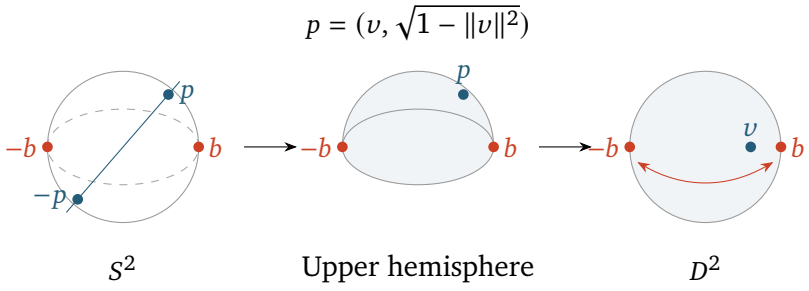


Figure 2.7: The upper hemisphere retains one representative of a non-equatorial pair and both representatives of an equatorial pair. Projection to D^2 leaves only boundary antipodes paired; the same construction gives D^3 for $\mathbb{R}\mathrm{P}^3$.

The map

$$D^n \longrightarrow \mathbb{R}\mathrm{P}^n, \qquad v \longmapsto [(v, \sqrt{1 - \|v\|^2})]$$

is continuous and surjective. The upper hemisphere consists of unit vectors whose last coordinate is nonnegative. Every line has a representative there. That representative is unique unless its last coordinate vanishes, when there are two antipodal representatives. Thus this realizes $\mathbb{R}\mathrm{P}^n$ as D^n with antipodal points of its boundary sphere identified, again by the compact-to-Hausdorff criterion.

The Hausdorff conclusion fails for general orbit spaces. Let the additive group $\mathbb{Q}$ act on $\mathbb{R}$ by translation. A nonempty open union U of orbits contains an interval. For any $x \in \mathbb{R}$, density of the rationals supplies $r \in \mathbb{Q}$ with $x - r$ in that interval. Invariance under translation by r gives $x \in U$. Hence the only open sets of $\mathbb{R}/\mathbb{Q}$ are the empty set and the whole space. There are distinct classes, for example those of 0 and $\sqrt{2}$, so this quotient is not Hausdorff. The Hausdorff condition will matter for the spaces constructed in Lectures 7 and 8.

2.5 Rotations as a glued ball

Definition 2.21 (Rotation group)**.** The rotation group is the matrix set

$$\mathrm{SO}(3) = \{R \in \mathbb{R}^{3\times 3} : R^T R = \mathbb{1},\ \det R = 1\},$$

with its Euclidean subspace topology.

A vector v with $\|v\| \le \pi$ will specify an axis and an angle. The figure shows the identification that occurs at angle π, a half-turn. The vector n marks a unit axis.

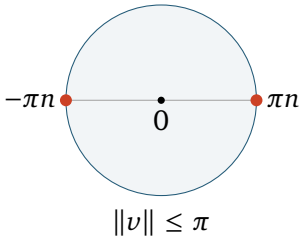


Figure 2.8: A cross-section of the closed rotation ball. Interior vectors are distinct rotations. The matching red boundary points πn and $-\pi n$ represent the same half-turn.

Proposition 2.22 (Axis–angle ball)**.** *The closed ball* $\{v \in \mathbb{R}^3 : \|v\| \le \pi\}$ *with antipodal boundary points identified is homeomorphic to* $\mathrm{SO}(3)$*. Consequently,* $\mathrm{SO}(3) \cong \mathbb{R}\mathrm{P}^3$.

Proof. We construct a continuous rotation map from the ball, prove it is onto, and determine its fibres. The compact-to-Hausdorff criterion then identifies the quotient.

For $v = rn$ with $r > 0$ and $\|n\| = 1$, split a vector $w \in \mathbb{R}^3$ into its axis and perpendicular parts:

$$w = (n \cdot w)n + w_\perp.$$

A rotation around n fixes the axis part. On the perpendicular plane it sends $w_\perp$ to $(\cos r)w_\perp + (\sin r)n \times w_\perp$. The figure shows these parts for $n = (0, 0, 1)$, $w = (1, 0, 1)$ and $r = \pi/3$.

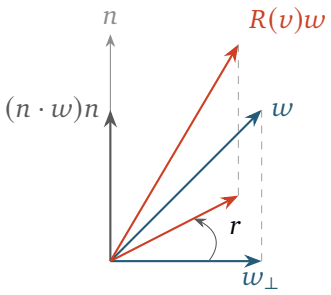


Figure 2.9: The axis part remains fixed. The perpendicular part rotates through $\pi/3$, sending $w = (1, 0, 1)$ to $(1/2, \sqrt{3}/2, 1)$.

The resulting axis–angle formula is

$$R(v)w = (\cos r)w + (1 - \cos r)n(n \cdot w) + (\sin r)n \times w.$$

We need a formula continuous at $v = 0$, where n is undefined. Define $A_v(w) = v \times w$ for $w \in \mathbb{R}^3$. Put $r = \|v\|$ and define

$$R(v) = \mathbb{1} + \frac{\sin r}{r} A_v + \frac{1 - \cos r}{r^2} A_v^2.$$

At $r = 0$, the coefficients have limiting values 1 and $1/2$. The map is therefore continuous there and $R(0) = \mathbb{1}$. For $v = rn$ with $\|n\| = 1$, the vector identity $n \times (n \times w) = n(n \cdot w) - w$ recovers the axis–angle formula. This fixes n and rotates its perpendicular plane through angle r. Thus it is orthogonal with determinant 1. At $r = \pi$ it equals $2nn^T - \mathbb{1}$, unchanged by $n \mapsto -n$.

Conversely, every $R \in \mathrm{SO}(3)$ has a unit fixed vector n. Indeed, orthogonality forces all eigenvalues to have modulus 1. Its nonreal eigenvalues occur in conjugate pairs of product 1. Its real eigenvalues are ± 1. After removing the conjugate pairs, an odd number of real eigenvalues remains. Their product is 1, so an even number are -1. At least one is therefore $+1$.

The perpendicular plane is invariant. Since $Rn = n$ and R is orthogonal, for $w \perp n$ we have $Rw \cdot n = Rw \cdot Rn = w \cdot n = 0$. The restriction has determinant 1, so is a plane rotation. Choosing its angle in $[-\pi, \pi]$ represents R as $R(\theta n)$.

There are no further identifications. The trace is $1 + 2\cos r$, so it determines $r \in [0, \pi]$. For $0 < r < \pi$, the difference $R(v) - R(v)^T = 2 \sin r\, A_n$ determines n. At $r = \pi$, the matrix $2nn^T - \mathbb{1}$ determines the line spanned by n, hence just the pair $\{n, -n\}$. At $r = 0$ there is only the vector 0.

We have a continuous bijection from the compact quotient ball to the Hausdorff matrix space $\mathrm{SO}(3)$. It is a homeomorphism. Rescaling the ball to D^3 and using the hemisphere construction above identifies it with $\mathbb{R}\mathrm{P}^3$. ∎

Lecture 4 will use this model to distinguish loops representing full rotations. No property of loops is needed for the identification here.

2.6 Three quotients of a square

We determine compactness, path-connectedness and Hausdorffness for the spaces obtained by the edge pairings in the figure. All start from $Q_0 = [0, 1]^2$. Matching arrows indicate which points are identified. Apply also all identifications forced transitively by these pairings.

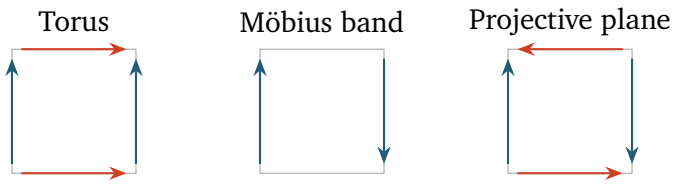


Figure 2.10: The torus pairs opposite edges without reversal. The Möbius band pairs only the vertical edges, with reversal. The projective plane pairs antipodal points of the square boundary.

Each quotient is compact and path-connected, since the square has both properties and the projection is continuous. Hausdorffness needs a separate argument. It need not survive a quotient, as $\mathbb{R}/\mathbb{Q}$ showed.

The following figure shows the surfaces described by the edge pairings. The torus closes both pairs of edges. The Möbius band closes one pair with a half-twist and retains one boundary component. In the cylinder panel, z is a unit complex number and $u \in [-1, 1]$ is the height.

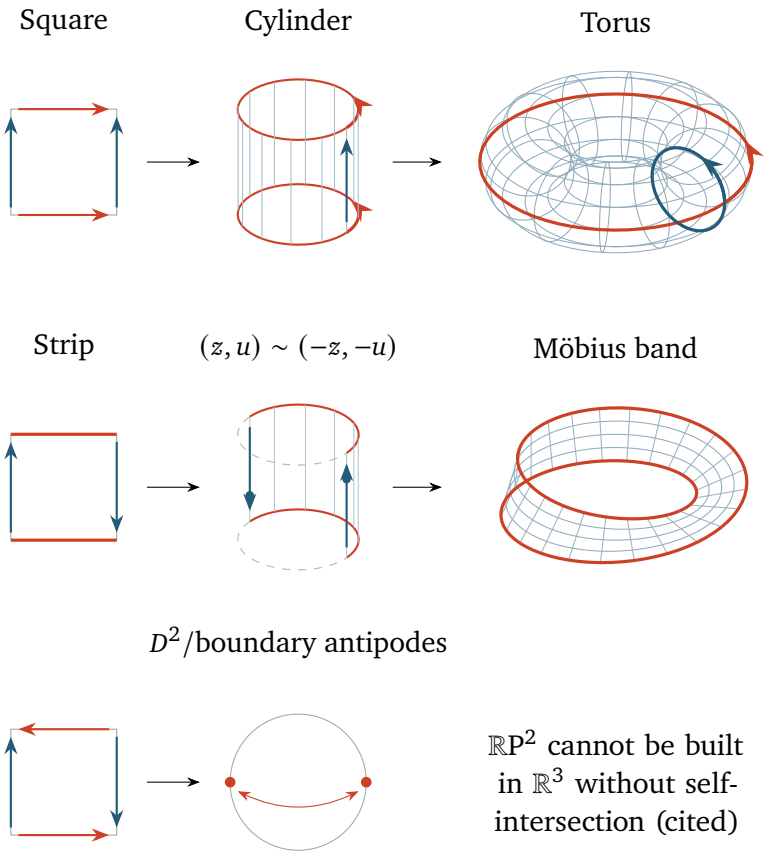


Figure 2.11: The square gives a cylinder and then a torus; the strip with reversed ends gives a Möbius band with one boundary component. The projective-plane row shows identifications; its non-embedding statement is cited from Hatcher, *Algebraic Topology*, Corollary 3.46, p. 256.

For the *torus*, the pairings are

$$(0, t) \sim (1, t) \quad \text{and} \quad (s, 0) \sim (s, 1).$$

Use the circle map c from §2.4.

Definition 2.23 (Torus). The product $T^2 = S^1 \times S^1 \subseteq \mathbb{R}^4$ is the torus.

The map $Q_0 \to T^2$ given by $(s,t) \mapsto (c(s), c(t))$ is continuous and onto. The familiar torus of revolution is the image of Q_0 under the map to $\mathbb{R}^3$ given by

$$\big((2+\cos(2\pi t))\cos(2\pi s),\ (2+\cos(2\pi t))\sin(2\pi s),\ \sin(2\pi t)\big).$$

Its distance from the vertical axis is $2+\cos(2\pi t) > 0$. This distance and the height recover $c(t)$, and the horizontal direction recovers $c(s)$. Hence its fibres are the same as those of $(c(s), c(t))$. The compact-to-Hausdorff criterion identifies the product torus with this surface.

Equality in the product is coordinatewise. In each coordinate, c identifies only the two endpoints. Thus the map's fibres are exactly the stated equivalence classes, including the single class of all four corners. The induced continuous bijection is a homeomorphism, because the quotient is compact and T^2 is Hausdorff.

Definition 2.24 (Möbius band). For the *Möbius band*, the pairing is $(0,t) \sim (1, 1-t)$.

We construct the half-twist from a cylinder by keeping its upper semicircle. This gives a rectangle of representatives. Identifying the two ends of that semicircle must reverse the height coordinate to give the stated pairing. Identify S^1 with unit complex numbers. On the cylinder $S^1 \times [-1,1] \subseteq \mathbb{R}^3$, the map $(z,u) \mapsto (-z,-u)$ is an isometry whose square is the identity. Its orbit space is therefore Hausdorff by the orbit metric above. The map

$$Q_0 \longrightarrow (S^1 \times [-1,1])/\{\pm 1\}, \qquad (s,t) \longmapsto [(e^{i\pi s}, 2t-1)]$$

is continuous and onto. Every circle point or its negative lies on the upper semicircle.

For $0 < s < 1$ that representative is unique. At the two ends, changing 1 to -1 reverses the other coordinate, giving precisely the stated edge pairing. The induced bijection is a homeomorphism by the same compact-to-Hausdorff criterion.

For the *projective plane*, the pairings are

$$(0,t) \sim (1, 1-t) \quad \text{and} \quad (s,0) \sim (1-s, 1).$$

The radial map F in the next figure sends concentric squares to concentric circles. Each point stays on its ray, so antipodes remain antipodes.

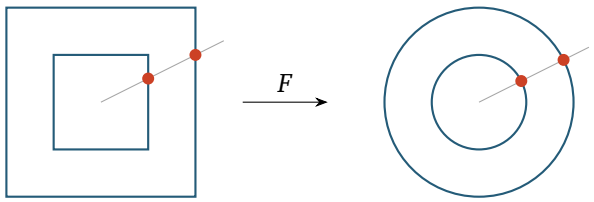


Figure 2.12: The contours $\|w\|_\infty = 1/2$ and $\|w\|_\infty = 1$ map to circles of radii $1/2$ and 1. The marked points lie on the ray through $(1, 1/2)$.

Put $w = (2s-1, 2t-1)$. The centred square maps homeomorphically to D^2 by radial rescaling:

$$w \longmapsto \frac{\|w\|_\infty}{\|w\|} w \quad (w \neq 0), \qquad 0 \longmapsto 0,$$

where $\|w\|_\infty$ is the maximum absolute coordinate and $\|w\|$ is Euclidean. The inverse sends nonzero u to $\|u\|\ u/\|u\|_\infty$. Their norms tend to zero at the origin, so both maps are continuous there as well. The square boundary maps to the unit circle and antipodes map to antipodes. The hemisphere construction therefore identifies this square quotient with $\mathbb{R}\mathrm{P}^2$, whose Hausdorffness was proved using its orbit metric.

The results, including the rotation ball, are as follows.

Quotient	Compact and path-connected	Hausdorff witness
Torus	yes	subspace of $\mathbb{R}^4$
Möbius band	yes	isometric orbit space
Projective plane	yes	antipodal orbit metric
Rotation ball	yes	$\mathrm{SO}(3) \subseteq \mathbb{R}^9$

Self-check. Let Q_0 have the torus edge pairings. Determine whether the continuous surjection $f\colon Q_0 \to S^1 \times S^1$ given by

$$f(s,t) = (c(s+2t), c(3t))$$

induces a homeomorphism from that quotient. Here the same formula for c is defined on all of $\mathbb{R}$. Compare the classes of $(0,0)$ and $(1/3,1/3)$, and identify which condition of the quotient method decides the answer.

Answer. Changing s or t by 1 leaves f unchanged, so it descends to the quotient. It is onto: choose $t \in [0,1/3]$ for the second circle coordinate, then choose $s \in [0,1]$ for the first. But

$$f(0,0) = (c(0), c(0)) = f(1/3, 1/3),$$

since both arguments of c at $(1/3,1/3)$ equal 1. The interior point $(1/3,1/3)$ is not equivalent to a corner. Thus a fibre contains two different equivalence classes. The induced map is not injective, so it is not a homeomorphism. The map $(s,t) \mapsto (c(s), c(t))$ has exactly the required fibres instead.

Self-check. State in one sentence what the quotient topology is and why we introduce it. Explain what its rule requires at the glued point of $[0,1]/(0 \sim 1)$.

Answer. The quotient topology makes a set of equivalence classes open exactly when its inverse image under the projection is open, so we can construct spaces by gluing and test continuous maps out of them before gluing. At the glued point an open inverse image contains both endpoints, hence contains $[0,\varepsilon) \cup (1-\varepsilon, 1]$ for some $0 < \varepsilon < 1/2$. A set whose inverse image contains only a neighbourhood of 0 cannot be a neighbourhood of the glued point, since that inverse image must also contain 1.

Summary. Continuous images preserve compactness and connectedness. A continuous bijection from compact to Hausdorff is a homeomorphism. A quotient topology tests continuity after composition with its projection. Quotients preserve compactness and path-connectedness, but Hausdorffness needs its own proof. The rotation group is the closed three-ball with antipodal boundary points identified, hence $\mathbb{R}\mathrm{P}^3$. Lecture 3 begins the study of loops.

Reading. Munkres, *Topology*, second edition, §§22–28; Heine–Borel is Theorem 27.3, metric sequential compactness is Theorem 28.2, and the sine curve is §24, Example 7.

Lecture 3

Homotopy and the fundamental group

Objectives. *We study which loops can be continuously deformed to a constant loop. A loop around a missing point in the plane cannot be contracted without meeting that point. We prove this obstruction by following a continuous phase around the circle. We compare paths by deformations that keep their endpoints fixed. A path returning to its starting point is a loop. The classes of loops at one fixed point form the fundamental group. For the circle this group is the integers: a continuously chosen angle records how many turns a loop makes. We prove this by lifting paths and homotopies, then use the integer to prove the fundamental theorem of algebra.*

By the end of this lecture you will be able to:

- Construct the fundamental group, induced homomorphisms and base-point isomorphisms.
- Prove that lifting computes the fundamental group of the circle as $\mathbb{Z}$.
- Compute the group of the punctured plane and prove the disc has no retraction onto its circle.
- Prove that a nonconstant complex polynomial has a root.

Unit 3A Homotopies and continuous phases

3.1 Homotopy and fixed endpoints

Write $I = [0, 1]$. Recall that a path in X is a continuous map $\gamma : I \to X$ (Lecture 2, connectedness and paths).

Definition 3.1 (Loop and base point)**.** A *loop* is a path with equal endpoints. Their common value is its *base point*.

Definition 3.2 (Homotopy)**.** A *homotopy* from $f : X \to Y$ to $g : X \to Y$ is a continuous map $h : X \times I \to Y$ with $h(x, 0) = f(x)$ and $h(x, 1) = g(x)$ for every $x \in X$. We write $f \simeq g$. For paths from x_0 to x_1, a homotopy *relative to the endpoints* also satisfies

$$h(0, s) = x_0, \qquad h(1, s) = x_1 \qquad (s \in I).$$

Write $h_s(x) = h(x, s)$ for a fixed-parameter map in the homotopy. Continuity is required jointly in the path parameter t and deformation parameter s. For example, the paths $\gamma(t) = (t, 0)$ and $\delta(t) = (t, t(1 - t))$ in $\mathbb{R}^2$ are homotopic relative to their endpoints by

$$h(t, s) = (t, s\,t(1 - t)).$$

This map is continuous because its coordinates are polynomials. At $t = 0$ it is $(0, 0)$ and at $t = 1$ it is $(1, 0)$, independently of s. The figure shows the intermediate paths.

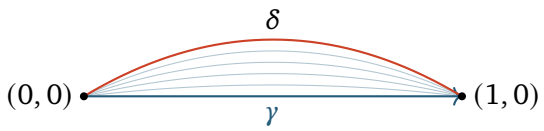


Figure 3.1: The homotopy keeps both endpoints fixed. Increasing s moves the straight path γ to the parabola δ.

We will glue formulas on finitely many closed pieces. If continuous maps on a finite closed cover agree on overlaps, their union is continuous. Indeed, the preimage of a closed set is a finite union of sets closed in those pieces, hence closed in the whole domain. This is the pasting argument of §2.3, now also applied to squares.

Homotopy relative to endpoints is an equivalence relation. The constant family proves reflexivity, and $h(t, 1 - s)$ reverses a homotopy. To join two homotopies, run the first on $0 \le s \le 1/2$ and the second on $1/2 \le s \le 1$, rescaling each parameter interval to I. They agree at the common edge, so the pasting argument proves transitivity.

The following loops illustrate how removing a point can obstruct a contraction. In the punctured plane, the circle once around the origin cannot contract. We prove this in §3.5. The circle centred at 2 with radius $2/5$ contracts by

$$h(t, s) = 2 + (1 - s)\frac{2}{5}e^{2\pi i t}, \qquad (t, s) \in I^2.$$

Its distance from 0 is at least $8/5$. A circle in three-space can instead contract through latitudes:

$$h(t, s) = \big(\cos(\pi s/2)\cos(2\pi t), \cos(\pi s/2)\sin(2\pi t), \sin(\pi s/2)\big).$$

Its norm is 1, and at $s = 1$ it is the constant north pole. This contracts this particular circle in $\mathbb{R}^3 \setminus \{0\}$. The figure shows these examples.

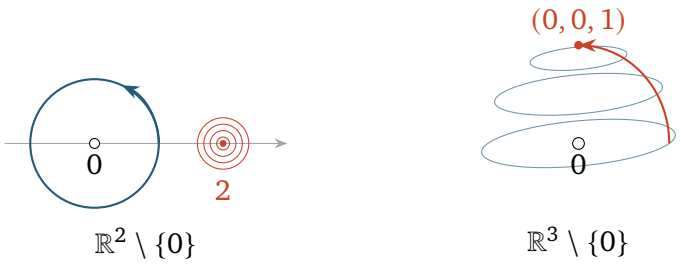


Figure 3.2: In the punctured plane, the circle around 0 and the circles centred at 2 behave differently under contraction. In three-space, the equatorial circle contracts to a pole without meeting 0.

Definition 3.3 (Free homotopy of loops). In a *free homotopy of loops* $h\colon I^2 \to X$, each h_s is a loop, but its base point may move. The defining endpoint condition is

$$h(0, s) = h(1, s) \qquad (s \in I).$$

3.2 Lifting phases on the circle

A circle point determines an angle only up to whole turns. We follow its angle continuously along a path, without resetting after a full turn. We measure this continuous phase in turns. Identify S^1 with the unit complex numbers, and set

$$p\colon \mathbb{R} \to S^1, \qquad p(u) = e^{2\pi i u}.$$

On I, this is the circle map c of Lecture 2, written with complex numbers. Here p denotes the exponential projection, rather than the stereographic projection of Lecture 1. We have $p(u) = p(v)$ exactly when $u - v \in \mathbb{Z}$.

Definition 3.4 (Lift and phase). For a continuous map $f\colon Y \to S^1$, a *lift* through p is a continuous map $\tilde{f}\colon Y \to \mathbb{R}$ with $p \circ \tilde{f} = f$. For this projection, a lift is also called a *phase*.

For example, the loop $\gamma(t) = p(2t)$ has lift $\tilde{\gamma}(t) = 2t$ starting at 0. It ends at 2, although the loop ends at 1 on the circle. Counter-clockwise travel $p(t)$ increases the phase by one turn. A principal phase in $(-1/2, 1/2]$ has its cut at -1 on the circle. For $p(2t)$ it is discontinuous at $t = 1/4$ and $t = 3/4$. It takes the value $+1/2$ at each cut time. The continuous lift has no such jumps.

The following figure compares these phases. The helix is the image of the map

$$H\colon \mathbb{R} \to \mathbb{R}^3, \qquad H(u) = (\cos(2\pi u), \sin(2\pi u), u),$$

whose projection onto the first two coordinates is $p(u)$. The loop $p(t + \sin(2\pi t))$ has lift $t + \sin(2\pi t)$. This lift takes the values $5/4$ at $t = 1/4$ and $-1/4$ at $t = 3/4$, but ends at 1.

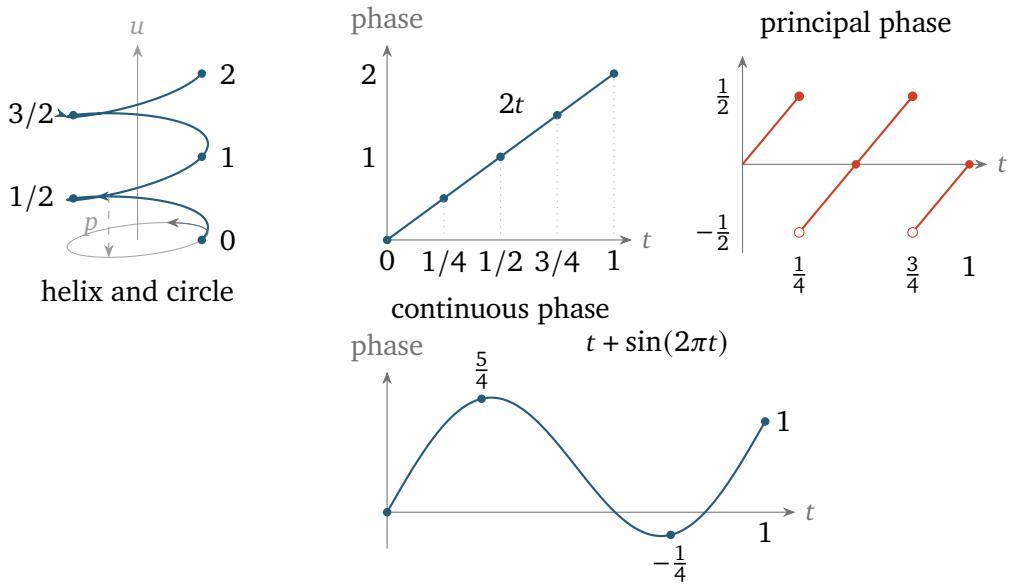


Figure 3.3: The loop $p(2t)$ makes two counter-clockwise turns: its continuous phase rises from 0 to 2, while its principal phase resets at the cut. The phase $t + \sin(2\pi t)$ reverses direction during the path and has total change 1.

For $r \in \mathbb{R}$ put

$$J_r = (r - 1/4, r + 1/4), \qquad U_r = \{z \in S^1 : \mathrm{Re}(p(-r)z) > 0\}.$$

The interval J_r has length $1/2$ and centre r. The set U_r is the open semicircle centred at $p(r)$. Multiplication by $p(-r)$ rotates it to the right half-circle U_0. These open semicircles cover the circle. The restriction $p\colon J_r \to U_r$ has the continuous inverse branch

$$b_0(z) = \frac{1}{2\pi} \arctan \frac{\operatorname{Im} z}{\operatorname{Re} z}, \qquad b_r(z) = r + b_0(p(-r)z).$$

For $z = e^{i\theta} \in U_0$ with $|\theta| < \pi/2$, the quotient $\operatorname{Im} z/\operatorname{Re} z$ is $\tan\theta$. The principal arctangent returns θ, so $b_0(z) = \theta/(2\pi) \in J_0$ and $p(b_0(z)) = z$. Conversely, for $u \in J_0$ the angle $2\pi u$ is in this range, so $b_0(p(u)) = u$. Rotating gives the inverse on U_r:

$$p(b_r(z)) = p(r)\,p(-r)z = z \qquad (z \in U_r).$$

The figure shows repeated real intervals for one semicircle.

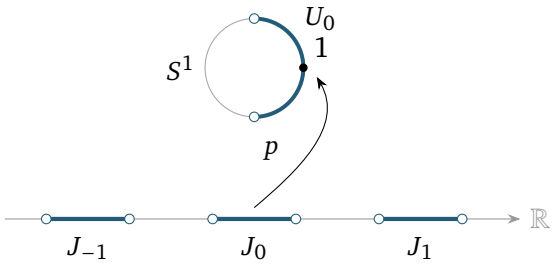


Figure 3.4: The intervals J_{-1}, J_0 and J_1 map to the same open semicircle U_0. Its boundary points are excluded.

We construct a lift by subdividing the path into pieces whose images lie in single semicircles. On each piece we use an inverse branch, then add an integer to match the preceding endpoint. For a homotopy we use the same construction on a square grid. Compactness of I and I^2 supplies a subdivision fine enough for every piece. The following lemma gives the uniform bound on their sizes. The *diameter* of a bounded nonempty set is the supremum of its pairwise distances.

Lemma 3.5 (Lebesgue number). *An open cover of a nonempty compact metric space K has a number $\delta > 0$ such that each nonempty subset of diameter less than δ lies in one cover member.*

Proof. For each $x \in K$ choose $\varepsilon_x > 0$ with $B_{2\varepsilon_x}(x)$ inside a cover member. Finitely many balls $B_{\varepsilon_x}(x)$ cover K. Let δ be the minimum of their radii. For a set A of diameter less than δ, choose $a \in A$ and one selected ball $B_{\varepsilon_x}(x)$ containing a. For every $z \in A$,

$$d(z,x) \le d(z,a) + d(a,x) < \delta + \varepsilon_x \le 2\varepsilon_x.$$

Thus A lies in the associated cover member. ∎

Integer principle. A continuous integer-valued function on a connected space is constant. Indeed, if two different integers occurred, a noninteger threshold between them would separate the domain into two nonempty open inverse images. We apply this principle to differences of lifts and then to their endpoint differences.

Lemma 3.6 (Path and homotopy lifting for the circle). *A path $\gamma\colon I \to S^1$ has a unique lift after specifying $\tilde\gamma(0) \in p^{-1}(\{\gamma(0)\})$. A homotopy $h\colon I^2 \to S^1$ has a unique lift after a lift of its bottom edge $h(t,0)$ has been specified.*

Proof. The interval and square are compact by Heine–Borel. Apply Lemma 3.5 to the cover of I by $\gamma^{-1}(U_r)$. For its Lebesgue number δ, choose N with $1/N < \delta$ and put $t_i = i/N$ for $i = 0, \dots, N$. Each subinterval maps into some U_r. On $[t_i, t_{i+1}]$, set $\tilde{\gamma} = b_r \circ \gamma + k$, with

$$k = \tilde{\gamma}(t_i) - b_r(\gamma(t_i)) \in \mathbb{Z}.$$

This matches the value already assigned at t_i. The pieces agree at their endpoints and paste continuously.

For a homotopy, use the cover of I^2 by $h^{-1}(U_r)$. Choose a square grid whose cells have diameter less than its Lebesgue number. Extend the given bottom lift across cells, row by row, from left to right. On a cell choose $b_r \circ h + k$ to match the assigned value at its lower-left corner. The only previously assigned points of a new cell are on its bottom and left edges. Along each already assigned bottom or left edge, the two lifts differ by a continuous integer-valued function. That function is constant by connectedness, and zero at the corner. Thus the new piece agrees on shared edges and pastes continuously.

Finally, two lifts on I or I^2 differ by a continuous integer-valued function. The integer principle makes that difference constant. Agreement at one point therefore proves uniqueness. ∎

The figure shows one extension step of the grid construction.

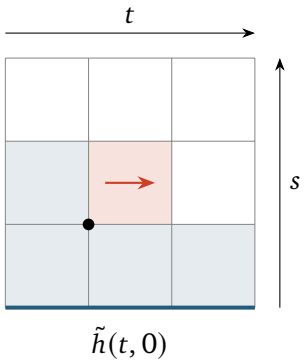


Figure 3.5: Previously assigned cells are shaded. A branch on the next cell is shifted by an integer to agree at the marked corner and hence on the shared edges.

Definition 3.7 (Winding number)**.** For any loop γ in the circle, define its *winding number* by

$$\operatorname{wind} \gamma = \tilde{\gamma}(1) - \tilde{\gamma}(0) \in \mathbb{Z}.$$

Different initial lifts differ by an integer, so this number is independent of that choice. In a free homotopy of loops, lift the whole square. The endpoint difference is a continuous integer-valued function of s, so it is constant. Thus winding is invariant even when the base point moves.

For $p(2t)$ the endpoint difference is 2. For $p(t + \sin(2\pi t))$ it is 1, despite the changes of direction. A constant loop has a constant lift and winding 0. Hence neither of these two loops is freely homotopic to a constant loop.

Self-check. Find the winding number of the circle loop

$$\gamma(t) = \frac{3/5 + p(2t)}{|3/5 + p(2t)|}\, p(-5t), \qquad t \in I.$$

Construct a free homotopy to a loop with an explicit lift. Justify that its normalization is defined throughout, and decide whether γ contracts in the circle.

Answer. We use the free homotopy

$$h(t,s) = \frac{3s/5 + p(2t)}{|3s/5 + p(2t)|}\, p(-5t).$$

The numerator in the fraction has modulus at least $1 - 3s/5 \geq 2/5$. Thus the formula is continuous, and its endpoints in t agree for every $s \in I$. At $s = 0$ it is $p(2t)p(-5t) = p(-3t)$, with lift $-3t$. At $s = 1$ it is γ. Free-homotopy invariance gives $\operatorname{wind}\gamma = -3$. Since a constant loop has winding 0, the loop cannot contract in the circle.

Unit 3B Loop classes and the fundamental group

3.3 The fundamental group

For paths γ from x to y and δ from y to z, define

$$(\gamma \cdot \delta)(t) = \begin{cases} \gamma(2t), & 0 \leq t \leq \frac{1}{2}, \\ \delta(2t-1), & \frac{1}{2} \leq t \leq 1. \end{cases}$$

The convention is γ **first, then** δ, as in Hatcher, §1.1. Define the reversed path by $\bar{\gamma}(t) = \gamma(1-t)$, and the constant path by $e_x(t) = x$. All these paths are continuous, by pasting at the matching endpoints. Write $[\gamma]$ for the endpoint-fixed homotopy class of γ.

Theorem 3.8 (Fundamental group). *The endpoint-fixed homotopy classes of loops at $x_0 \in X$ form a group, written $\pi_1(X, x_0)$, with*

$$[\gamma][\delta] = [\gamma \cdot \delta], \qquad 1 = [e_{x_0}], \qquad [\gamma]^{-1} = [\bar{\gamma}].$$

Proof. We verify the group laws using endpoint-fixed homotopies. Concatenation of paths is not literally associative: the bracketings allot different times to the three paths. We make their classes associative by changing those times continuously.

Products respect classes. Concatenate two endpoint-fixed homotopies in the t direction. Their formulas agree along $t = 1/2$. Reversing the t parameter similarly respects classes.

A change of timing does not change a path class. For any continuous $\rho\colon I \to I$ with $\rho(0) = 0$ and $\rho(1) = 1$, the formula $h(t,s) = \gamma((1-s)\rho(t) + st)$ joins $\gamma \circ \rho$ to γ while fixing the endpoints. For associativity, the following timing changes $((\gamma \cdot \delta) \cdot \eta) \circ \rho$ into $\gamma \cdot (\delta \cdot \eta)$:

$$\rho(t) = \begin{cases} t/2, & 0 \leq t \leq \frac{1}{2}, \\ t - \frac{1}{4}, & \frac{1}{2} \leq t \leq \frac{3}{4}, \\ 2t - 1, & \frac{3}{4} \leq t \leq 1. \end{cases}$$

The formulas agree at the break times and fix 0 and 1. On the first interval the parameter in γ becomes $4\rho(t) = 2t$. On the middle interval it becomes $4\rho(t) - 1 = 4t - 2$ in δ. On the

last it becomes $2\rho(t) - 1 = 4t - 3$ in η. These are precisely the timings of $\gamma \cdot (\delta \cdot \eta)$. The preceding homotopy therefore proves associativity of classes.

The paths $e_x \cdot \gamma$ and $\gamma \cdot e_y$ are γ with timings $\max(2t-1, 0)$ and $\min(2t, 1)$. Thus their classes equal $[\gamma]$. Finally, the homotopy

$$h(t,s) = \gamma\big((1-s)\min(2t, 2-2t)\big)$$

contracts $\gamma \cdot \bar{\gamma}$ to e_x, fixing both endpoints. At deformation time s, it follows γ only as far as parameter $1-s$ and then returns along the same part of γ. Applying it to $\bar{\gamma}$ gives the other inverse identity. These homotopies also apply to joined paths with different endpoints. ∎

The figure records the two timings and the shortening of the return path.

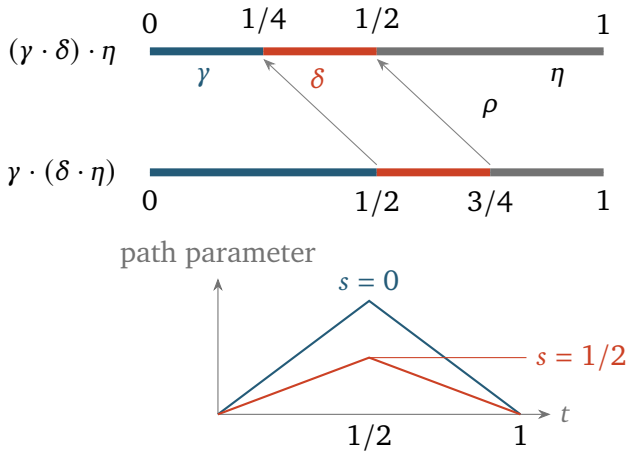


Figure 3.6: The map ρ aligns the break times of the two bracketings. In the return-path homotopy, the path parameter rises only to $1-s$ and then falls to 0.

Definition 3.9 (Homomorphism and isomorphism)**.** A *homomorphism* preserves group products. A bijective homomorphism is an *isomorphism*.

A continuous map $f\colon X \to Y$ induces

$$f_*\colon \pi_1(X, x_0) \longrightarrow \pi_1(Y, f(x_0)), \qquad [\gamma] \longmapsto [f \circ \gamma].$$

Proposition 3.10 (Induced homomorphisms)**.** *The induced map f_* is a homomorphism. Induced maps preserve composition and identity maps.*

Proof. Composing a path homotopy with f proves well-definedness. Composition also respects the two path pieces, so f_* preserves products. For continuous $g\colon Y \to Z$, this gives $(g \circ f)_* = g_* \circ f_*$. Identity maps give $(\mathrm{id}_X)_* = \mathrm{id}$. ∎

Theorem 3.11 (Fundamental group of the circle)**.** *Winding gives an isomorphism* $\mathrm{wind}\colon \pi_1(S^1, 1) \to \mathbb{Z}$. *The loop $t \mapsto p(mt)$ has winding m for every $m \in \mathbb{Z}$.*

Proof. Winding is invariant under free homotopy, hence under endpoint-fixed homotopy. It therefore defines a map on classes. Choose a lift starting at 0. If it ends at m, the homotopy

$$\tilde{h}(t,s) = (1-s)\tilde{\gamma}(t) + smt$$

fixes the endpoints 0 and m. Its projection gives an endpoint-fixed homotopy from γ to $p(mt)$. Thus winding determines the class, and every integer occurs. For a product of loops

with windings m and n, join the first lift to the second lift shifted by m. The joined lift ends at $m + n$, so winding preserves multiplication as addition. ∎

The translated second lift in the proof is shown for windings 1 and 2. Its final height is 3, the sum of the two endpoint changes.

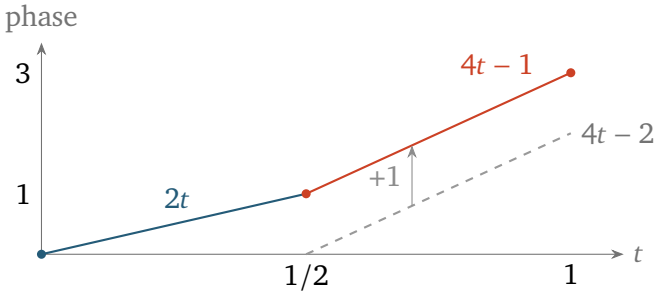


Figure 3.7: For $p(t) \cdot p(2t)$, the second lift starts at height 1. The joined lift is $2t$ on the first half and $4t - 1$ on the second half.

3.4 Base points and free homotopy

The base point requires care. For a path a from x_0 to x_1, the base-point change map is

$$c_a \colon \pi_1(X, x_0) \to \pi_1(X, x_1), \qquad c_a([\gamma]) = [\bar{a} \cdot \gamma \cdot a].$$

The figure gives the traversal order.

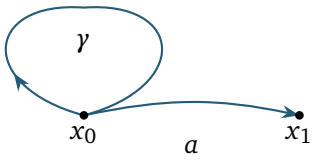


Figure 3.8: A loop at x_1 follows $\bar{a}$ to x_0, traverses γ, then follows a back.

Proposition 3.12 (Change of base point)**.** *For every path a from x_0 to x_1, the map c_a is an isomorphism.*

Proof. Concatenating with fixed paths respects homotopy, so c_a is well-defined. Cancellation of $a \cdot \bar{a}$ proves that c_a preserves products. The inverse is $c_{\bar{a}}$, by the same cancellation. Thus it is an isomorphism. ∎

Definition 3.13 (Conjugate elements)**.** Elements u and v of a group are *conjugate* if $v = k^{-1}uk$ for some element k of that group.

When $x_0 = x_1$, the map c_a is conjugation by $[a]$. If $[a][\gamma] \neq [\gamma][a]$, then $c_a([\gamma]) \neq [\gamma]$, whereas the constant choice of a gives the identity map. Thus the isomorphism can depend on the chosen path.

For example, $h(t, s) = p(t + s/4)$ is a free homotopy from $p(t)$ to $p(t + 1/4)$. Its base point follows $a(s) = p(s/4)$ from 1 to i. Its continuous phase $t + s/4$ has endpoint difference 1 for every $s \in I$.

We now convert a free homotopy into a homotopy with a fixed base point. Let h be a free homotopy of loops. Let $\gamma_0 = h_0$ be based at x_0, and put $\gamma_1 = h_1$. The base point follows the path $a(s) = h(0, s)$. Put $a_s(t) = a(st)$. The loops $a_s \cdot h_s \cdot \bar{a}_s$ keep base point x_0. Each first follows a_s from x_0, then traverses h_s, then returns along $\bar{a}_s$. At $s = 0$, the inserted constant paths do not change the class, by the unit laws. At $s = 1$ this gives $[\gamma_0] = [a \cdot \gamma_1 \cdot \bar{a}]$. If γ_1 is also based at x_0, then a is a loop at x_0. The equality therefore makes $[\gamma_0]$ and $[\gamma_1]$ conjugate.

Conversely, let a be a loop at x_0. The family $\bar{a}_s \cdot \gamma \cdot a_s$ consists of loops with base point $a(s)$. Its end loops represent $[\gamma]$ and $[\bar{a} \cdot \gamma \cdot a]$. For each family, the three formulas are jointly continuous in t and s. They agree on the closed boundaries between their parameter pieces, so pasting proves joint continuity. Free homotopy is an equivalence relation by the same pasting argument as before. Based homotopies are free homotopies, so conjugate classes are freely homotopic. Thus two loops based at x_0 are freely homotopic exactly when their classes are conjugate. In a path-connected space, move any base point to x_0 by a path. Free homotopy classes then correspond to conjugacy classes in $\pi_1(X, x_0)$, rather than to individual group elements.

The parameter cylinder in the figure records the moving base point and the inserted paths. Its image under h need not be an embedded annulus in X.

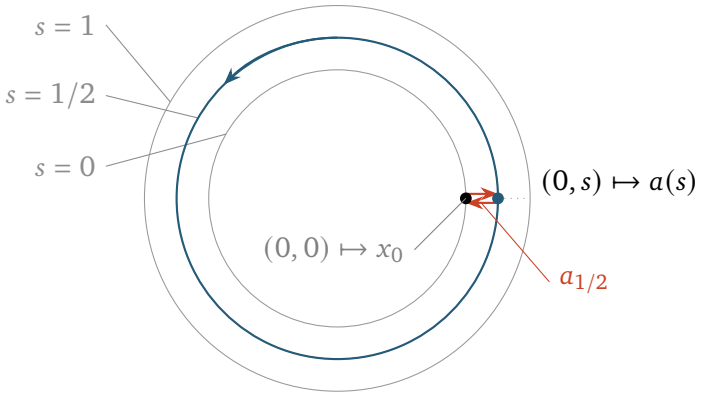


Figure 3.9: The annulus represents the parameter cylinder by $(t, s) \mapsto (1 + s/2)p(t)$. The marked radial segment represents $a_{1/2}$, followed outward and then back in the based construction.

For the circle group, conjugation changes no class, since addition in $\mathbb{Z}$ commutes. Thus two loops based at 1 are freely homotopic exactly when they have the same winding. In a noncommutative group, $k^{-1}uk$ can differ from u. Lecture 5 computes such fundamental groups using reduced words.

3.5 Retractions and homotopy invariance

Definition 3.14 (Retraction and deformation retraction)**.** A *retraction* onto $A \subseteq X$ is a continuous map $r\colon X \to A$ with $r|_A = \mathrm{id}_A$. A *deformation retraction* also has a homotopy from id_X to $i \circ r$, where $i\colon A \to X$ is inclusion, that keeps every point of A fixed throughout.

Proposition 3.15 (Deformation retractions preserve the group). *For a deformation retraction and $a_0 \in A$, the induced maps r_* and i_* are inverse isomorphisms of fundamental groups.*

Proof. We have $r \circ i = \mathrm{id}_A$. The other composite is homotopic to id_X while fixing a_0. Composing this homotopy with any loop at a_0 is an endpoint-fixed path homotopy. Hence $i_* r_* = \mathrm{id}$ as well as $r_* i_* = \mathrm{id}$. ∎

Identify $\mathbb{C}$ with $\mathbb{R}^2$. For example, on $\mathbb{C} \setminus \{0\}$ the formula

$$h(z, s) = \left(1 - s + \frac{s}{|z|}\right) z$$

deforms the identity to $r(z) = z/|z|$. Its coefficient is positive, so the image never reaches 0. The homotopy fixes every point of S^1. Consequently $\pi_1(\mathbb{R}^2 \setminus \{0\}, 1) \cong \mathbb{Z}$. For a loop in the punctured plane, its winding means the winding of its normalization $\gamma/|\gamma|$. The loop $p(t)$ has winding +1, whereas a constant loop has winding 0. Therefore it cannot contract in the punctured plane. This proves the obstruction pictured in Unit 3A.

Definition 3.16 (Simply connected space). A nonempty space is *simply connected* if it is path-connected and all based loops are endpoint-fixed homotopic to the constant loop.

Definition 3.17 (Contractible space). A space is *contractible* if its identity is homotopic to a constant map with value x_* in the space.

A nonempty convex subset of $\mathbb{R}^n$ is contractible: for a chosen point x_*, the straight-line homotopy $h(x, s) = (1 - s)x + s x_*$ stays in the subset.

Proposition 3.18 (Contractible spaces). *Every contractible space is simply connected.*

Proof. The contraction gives paths from all points to x_*, so the space is path-connected. For a loop γ at x_0, it also gives a free homotopy to e_{x_*}. The moving-basepoint construction above converts this to a based homotopy from γ to $a \cdot e_{x_*} \cdot \bar{a}$, which cancels to e_{x_0}. ∎

The punctured plane is not contractible, since its fundamental group is $\mathbb{Z}$.

Definition 3.19 (Homotopy equivalence). Maps $f : X \to Y$ and $g : Y \to X$ give a *homotopy equivalence* if $g \circ f \simeq \mathrm{id}_X$ and $f \circ g \simeq \mathrm{id}_Y$.

The punctured plane and the circle are homotopy equivalent by the radial retraction and inclusion. Their two composites are the identity on the circle and a map homotopic to the identity on the punctured plane, as the radial homotopy proves. The general theorem that such f induces an isomorphism on fundamental groups at every base point is cited from Hatcher, Proposition 1.18. The deformation-retraction case used here was proved directly above.

3.6 No retraction of the disc onto its circle

A retraction would assign a boundary point continuously to every point of the disc while fixing each boundary point. The obstruction is the loop that contracts through the disc but cannot contract along its boundary. Let $D^2 = \{z \in \mathbb{C} : |z| \le 1\}$ be the closed disc.

Corollary 3.20. *There is no retraction $r : D^2 \to S^1$.*

Proof. The homotopy $(z, s) \mapsto (1 - s)z$ contracts the disc. Thus $\pi_1(D^2, 1)$ is the trivial group, consisting only of its identity. If r were a retraction and $i\colon S^1 \to D^2$ the inclusion, then $r \circ i = \mathrm{id}_{S^1}$. Therefore $r_* i_*$ is the identity of $\pi_1(S^1, 1) \cong \mathbb{Z}$. But i_* takes values in the trivial group, so the composite sends every class to the identity element. It cannot be the identity map of this nontrivial group. ∎

The base point here is 1 on the circle. Contractibility gives the trivial group there even though the displayed contraction ends at the centre.

The corollary gives the fixed-point theorem in the plane.

Corollary 3.21 (Brouwer's theorem in the plane)**.** *Every continuous map $f\colon D^2 \to D^2$ has a fixed point.*

Proof. We argue by contradiction. Suppose that $f(z) \neq z$ for every $z \in D^2$, and put $u(z) = z - f(z)$, so that $u(z) \neq 0$. We follow the ray from $f(z)$ through z until it meets the circle. The point $z + tu(z)$ with $t \geq 0$ lies on the circle exactly when

$$|u(z)|^2 t^2 + 2\,\mathrm{Re}\big(z\overline{u(z)}\big)t + |z|^2 - 1 = 0.$$

The product of the two roots is $(|z|^2 - 1)/|u(z)|^2 \leq 0$, so the larger root is nonnegative. We take $t(z)$ to be this root:

$$t(z) = \frac{-\,\mathrm{Re}\big(z\overline{u(z)}\big) + \sqrt{\mathrm{Re}\big(z\overline{u(z)}\big)^2 + |u(z)|^2(1 - |z|^2)}}{|u(z)|^2}.$$

This formula is continuous in z, because $u(z) \neq 0$ and the quantity under the root is nonnegative. For $|z| = 1$ we have $\mathrm{Re}\big(z\overline{u(z)}\big) = 1 - \mathrm{Re}\big(z\overline{f(z)}\big) \geq 1 - |f(z)| \geq 0$, so $t(z) = 0$. Hence $r(z) = z + t(z)u(z)$ is a retraction of D^2 onto S^1, which contradicts Corollary 3.20. ∎

Lecture 14 proves the same theorem for smooth maps of balls in every dimension.

Self-check. State in one sentence what the fundamental group of (X, x_0) is and why we introduced it. Explain why paths themselves are replaced by classes.

Answer. The fundamental group is the group of endpoint-fixed homotopy classes of loops at x_0, introduced to distinguish loops that cannot deform into one another while retaining concatenation. The two bracketings of three paths have different timings, but a change of timing is an endpoint-fixed homotopy. Thus associativity holds for classes.

Self-check. Compute $\pi_1(S^1 \times S^1, (1, 1))$ using the coordinate projections. Prove the group identification, not just its surjectivity.

Answer. Send $[(\gamma, \delta)]$ to $(\mathrm{wind}\,\gamma, \mathrm{wind}\,\delta) \in \mathbb{Z}^2$. Projecting a product homotopy gives endpoint-fixed homotopies of the coordinates. Winding is invariant under these, so the map is well-defined. Projection respects concatenation, so the map preserves multiplication coordinatewise. Conversely, the component test of Lecture 1 makes a pair of loops a loop in the product, and a pair of endpoint-fixed homotopies a homotopy in the product. For every $(m, n) \in \mathbb{Z}^2$, the pair is represented by $t \mapsto (p(mt), p(nt))$. If both windings vanish, each coordinate loop has an endpoint-fixed contraction. Their pair contracts the product loop. If two classes have the same image, their product with the inverse of the second has both windings zero, so is trivial. The two classes therefore agree. The group is $\mathbb{Z}^2$.

Unit 3C Polynomial roots and winding

3.7 The fundamental theorem of algebra

Corollary 3.22 (Fundamental theorem of algebra)**.** *Every complex polynomial of degree at least one has a complex root.*

Proof. We compare the same large-circle loop in two ways. On a sufficiently large circle the leading term dominates, so removing the lower terms without meeting zero gives winding equal to the degree. If the polynomial had no root anywhere, shrinking the input circle to its centre would also avoid zero and give winding zero. We prove these two claims.

Write $f(z) = a_n z^n + \sum_{j=0}^{n-1} a_j z^j$ with $n \geq 1$ and $a_n \neq 0$. Choose

$$R > \max\left\{1, \frac{\sum_{j=0}^{n-1} |a_j|}{|a_n|}\right\}.$$

The diagram separates the two deformations. The second row assumes that f has no zero anywhere in $\mathbb{C}$.

$$\begin{array}{lccc} \text{large circle} & a_n R^n p(nt) & \xrightarrow{\text{add lower terms}} & f(Rp(t)) \\ & \text{winding } n & & \text{winding } n \\ \\ \text{no-root assumption} & f(Rp(t)) & \xrightarrow{\text{shrink input circle}} & f(0) \\ & \text{winding } n & & \text{winding } 0 \end{array}$$

Figure 3.10: The large-circle comparison keeps winding n while adding the lower terms. Under the no-root assumption, shrinking the input circle would instead give winding 0 for the same loop.

The large circle. For $|z| = R$ the lower terms satisfy

$$\sum_{j=0}^{n-1} |a_j| R^j \leq R^{n-1} \sum_{j=0}^{n-1} |a_j| < |a_n| R^n.$$

Thus the loops

$$a_n R^n p(nt) + s \sum_{j=0}^{n-1} a_j R^j p(jt)$$

avoid zero for every $s \in I$. Normalizing gives a free homotopy of circle loops from the leading term to $f(Rp(t))$. Choose $b \in \mathbb{R}$ with $p(b) = a_n/|a_n|$. The leading loop has phase $b + nt$, so its winding is n.

The figure illustrates this comparison for a quadratic.

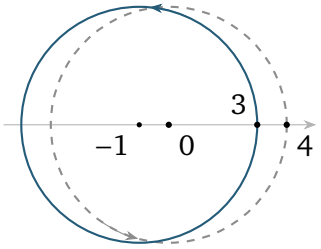


Figure 3.11: For $f(z) = z^2 - 1$ and $R = 2$, the comparison loops are $4p(2t) - s$, each traversed twice. The dashed loop ($s = 0$) and solid loop ($s = 1$) start at 4 and 3 and have centres 0 and −1, respectively.

The shrinking circle. Suppose f had no zero anywhere in $\mathbb{C}$. The map $h_f \colon I^2 \to \mathbb{C} \setminus \{0\}$ given by

$$h_f(t, s) = f((1 - s)Rp(t))$$

would be a homotopy of nonzero loops from $f(Rp(t))$ to the constant $f(0)$. These families may move the base point. After normalization, free-homotopy invariance would preserve winding and give its final value zero. This contradicts its value $n \geq 1$. Hence f has a root. ∎

The quadratic $f(z) = z^2 - 1$ shows where a shrinking family can meet zero. For an input circle of radius ρ, its image is $\rho^2 p(2t) - 1$. At $\rho = 1$ it meets 0 at $t = 0$ and $t = 1/2$, corresponding to the roots 1 and −1. The figure compares these input and image circles.

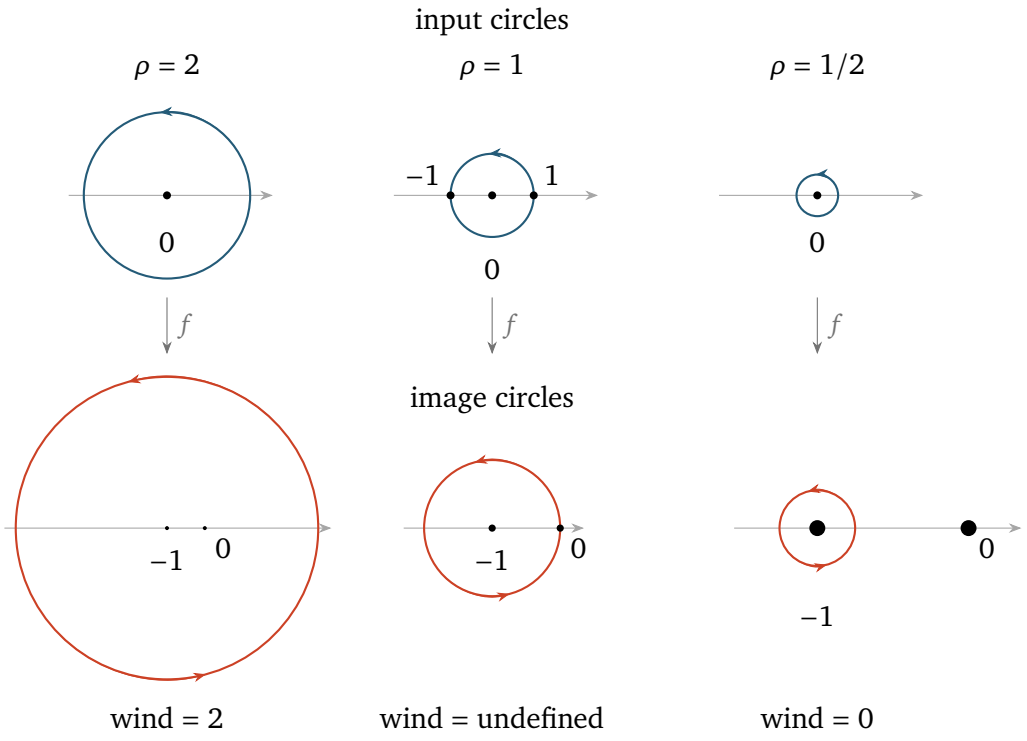


Figure 3.12: For input radii 2, 1 and 1/2, the image circles, drawn at different scales, have radii 4, 1 and 1/4 about −1 and are each traversed twice. The middle image meets 0, so its winding about 0 is undefined.

Summary. Classes of loops at one fixed base point form the fundamental group. Changing the base point along a path gives an isomorphism that can depend on that path. In a

path-connected space, free homotopy keeps only conjugacy classes. Lifting computes $\pi_1(S^1, 1) \cong \mathbb{Z}$. Deformation retraction gives the same group for the punctured plane. This integer rules out a disc-to-circle retraction and proves the fundamental theorem of algebra. Lecture 4 extends the lifting construction to other maps.

Reading. Hatcher, *Algebraic Topology*, §1.1, pp. 25–38 (Theorem 1.8 and Proposition 1.18); Munkres, *Topology*, §§51–56, §58 and Lemma 27.5.

Lecture 4

Covering spaces and rotations

Objectives. *We generalise the circle projection of Lecture 3 to covering maps. Their local inverse branches lift paths and homotopies. For a simply connected covering space, lifted endpoints determine loop classes. We apply this to rotations: one full turn is nontrivial, and two full turns contract.*

By the end of this lecture you will be able to:

- Prove path and path-homotopy lifting for a covering map.
- Recover loop classes from a simply connected covering space.
- Compute $\pi_1(\mathbb{R}\mathrm{P}^n)$ for $n \geq 2$ and $\pi_1(\mathrm{SO}(3))$.
- Distinguish the lifted paths of a 2π and a 4π rotation.

Unit 4A Coverings, lifted endpoints and rotations

4.1 Coverings and local inverse branches

Definition 4.1 (Covering map)**.** A *covering map* is a continuous surjection $p\colon \widetilde{X} \to X$ such that each $x \in X$ has an open neighbourhood U with

$$p^{-1}(U) = \coprod_{\alpha} V_\alpha, \qquad p|_{V_\alpha}\colon V_\alpha \longrightarrow U \text{ a homeomorphism.}$$

The symbol $\coprod$ denotes a disjoint union. The sets V_α are open in $\widetilde{X}$. We call U *evenly covered* and each V_α a *sheet over* U.

The fibre $p^{-1}(\{x\})$ is the inverse image of the point, as in Lecture 2. It contains one point in each sheet over U. In particular, the fibre has the discrete subspace topology: every one of its points is open in the fibre. Its cardinality is constant as x varies in an evenly covered set. The figure shows the local structure.

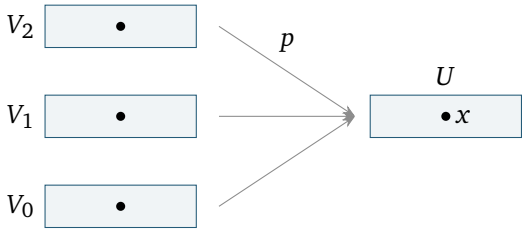


Figure 4.1: Each sheet maps homeomorphically onto the same open set U. The marked points belong to one fibre.

For example, the projection $u \mapsto e^{2\pi i u}$ from $\mathbb{R}$ to S^1 is a covering. The semicircles U_r of §3.2 have inverse images $\coprod_{k\in\mathbb{Z}}(J_r + k)$. Each restriction has inverse $b_r + k$. The map $z \mapsto z^m$ on S^1, for an integer $m \geq 1$, is an m-sheeted covering. An open arc of angle less than 2π has m disjoint inverse arcs, obtained by dividing its angles by m and adding $2\pi k/m$ for $k = 0, \ldots, m-1$. Lecture 3 chose a continuous phase along a circle path, rather than a single inverse branch on the whole circle. A loop of winding one raises that phase by one turn. In the helix model, the real coordinate is the height, and projection forgets the height. For the double cover, following a loop of winding one exchanges the two fibre points. A second traversal returns to the initial point upstairs. The figure compares these global behaviours.

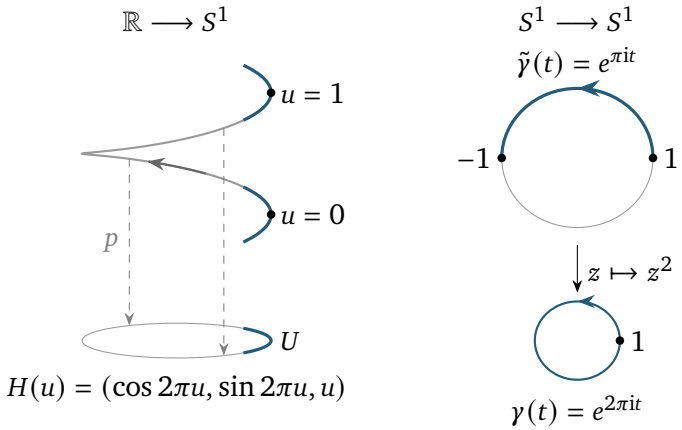


Figure 4.2: The helix projects vertically to the circle and rises one unit per positive turn. For $z \mapsto z^2$, a positive semicircle upstairs projects to a full circle downstairs and ends at the other point of the fibre.

Definition 4.2 (Local homeomorphism)**.** A *local homeomorphism* restricts near each domain point to a homeomorphism onto an open subset of the target.

Every covering has this property. The converse fails, even for a surjection. Consider

$$q\colon (0, 3\pi) \longrightarrow S^1, \qquad q(\theta) = e^{i\theta}.$$

Small intervals give local inverses. The fibre over 1 is $\{2\pi\}$, but the fibre over $e^{i\varepsilon}$ has two points whenever $0 < \varepsilon < \pi$. Every neighbourhood of 1 contains such points. Thus 1 has no evenly covered neighbourhood. The figure marks the missing endpoint of the extra inverse branch.



Figure 4.3: The truncated angle interval contains one preimage of 1, but two of $e^{i\varepsilon}$. Open endpoints prevent a second branch at 1.

The missing endpoint also prevents a path lift from existing on the whole interval. For $\gamma(t) = e^{-2\pi i t}$, the only possible initial value is 2π. Any lift would differ from $2\pi(1-t)$ by a continuous integer multiple of 2π. This integer is zero at the start, so it is zero everywhere. The forced endpoint is 0, outside the domain of q. Thus local inverse branches alone do not guarantee path lifting.

4.2 Lifting paths and their homotopies

Definition 4.3 (Lift). A *lift* of $f\colon Y \to X$ is a continuous map $\tilde{f}\colon Y \to \widetilde{X}$ with $p \circ \tilde{f} = f$.

Write $I = [0,1]$.

Theorem 4.4 (Path and path-homotopy lifting). *For a covering p, a path $\gamma\colon I \to X$ has a unique lift after specifying $\tilde{\gamma}(0)$ above $\gamma(0)$. A homotopy $h\colon I^2 \to X$ has a unique lift after a lift of its bottom edge has been specified. If h fixes its path endpoints, so does its lift.*

Proof. First, uniqueness of a path lift follows locally. On a subinterval mapping into an evenly covered U, any lift stays in one sheet: the interval is connected and the sheets are disjoint and open. The sheet is determined by one lifted value. Its inverse branch determines the whole subinterval.

For existence, pull back evenly covered sets to an open cover of I. The Lebesgue-number lemma of Lemma 3.5 gives a finite subdivision such that each closed subinterval maps into one such set. On the first subinterval use the sheet containing the prescribed initial point. On each successive subinterval use the sheet containing the preceding endpoint. The formulas agree at the joins and paste continuously. Local uniqueness on successive subintervals gives uniqueness on I.

For a homotopy, pull the same open cover back to the compact square. Choose a grid with each cell mapping into an evenly covered set. Extend the bottom lift row by row, from left to right. Choose on each cell the inverse branch whose sheet contains the already assigned lower-left value. On its bottom and left edges, path-lift uniqueness forces agreement with every previously assigned value. There are no other previously assigned points on the cell. Finite pasting gives a continuous lift on the square. The same cell order proves uniqueness. This is the grid construction of Lemma 3.6, with sheets in place of integer translates.

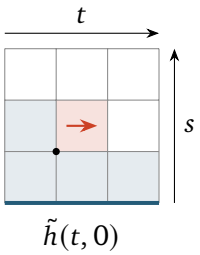


Figure 4.4: The new inverse branch is chosen at the marked corner. Uniqueness on shared edges makes it agree with the shaded region.

Finally, if an edge of h is constant, its lift is a path in a discrete fibre. A continuous image of the connected interval is connected, so this path is constant. Apply this to the two endpoint edges. ∎

In particular, endpoint-fixed homotopic paths have lifts with the same endpoint when their initial lifts agree. The induced homomorphism p_* is injective: lift a based contraction of a projected loop. The lift is a based contraction of the original loop. Also,

$$[\gamma] \in p_*\pi_1(\widetilde{X}, \tilde{x}_0) \quad \Longleftrightarrow \quad \text{the lift of } \gamma \text{ from } \tilde{x}_0 \text{ closes.}$$

For the reverse implication project the closed lift. For the forward implication compare γ with the projection of a representative loop upstairs and lift their based homotopy.

4.3 Fibres, loop classes and deck transformations

For the circle projection, translations $u \mapsto u + k$ with $k \in \mathbb{Z}$ preserve the projected point. Composing translations adds their integers, just as concatenating loops adds winding numbers. We use transformations preserving a covering projection to recover the multiplication of loop classes from lifted endpoints.

Definition 4.5 (Deck transformation). A *deck transformation* of p is a homeomorphism $D\colon \widetilde{X} \to \widetilde{X}$ with $p \circ D = p$. Their group under composition is written $\mathrm{Deck}(p)$.

On a path-connected covering space, a deck transformation is determined by its value at one point. Indeed, for a path a from that point to any other point, $D \circ a$ is the unique lift of $p \circ a$ from the specified image point. In particular, a deck transformation fixing one point is the identity. The deck group acts *transitively* on a fibre if for any two points of that fibre some deck transformation sends the first to the second.

Corollary 4.6 (Counting a simply connected covering). *Let $p\colon \widetilde{X} \to X$ be a covering with $\widetilde{X}$ simply connected. Fix $\tilde{x}_0$ above x_0. Sending a loop class to its lifted endpoint gives a bijection*

$$\pi_1(X, x_0) \longrightarrow p^{-1}(\{x_0\}).$$

If the deck group acts transitively on this fibre, it is isomorphic to $\pi_1(X, x_0)$.

Proof. The endpoint depends only on the based class by homotopy lifting. Every fibre point can be joined to $\tilde{x}_0$, since simple connectedness includes path-connectedness. Projecting this path proves surjectivity. If two lifted paths have the same endpoint, join the first to the reverse of the second. This loop contracts in $\widetilde{X}$. Its projection shows that the two original loop classes agree. This proves injectivity.

Under transitivity, let D_γ be the unique deck transformation taking $\tilde{x}_0$ to the endpoint of the lift $\tilde{\gamma}$. The preceding bijection and deck uniqueness give a bijection of groups. The path $D_\gamma \circ \tilde{\delta}$ projects to δ, because $p \circ D_\gamma = p$. It starts at the endpoint of $\tilde{\gamma}$. Path-lift uniqueness therefore makes it the continuation of the lift of $\gamma \cdot \delta$ after $\tilde{\gamma}$. The concatenated lift ends at

$$D_\gamma(D_\delta(\tilde{x}_0)), \qquad D_{\gamma\cdot\delta} = D_\gamma \circ D_\delta.$$

Thus the bijection preserves multiplication with the course's first-path-first convention. The isomorphism uses the chosen lifted base point. ∎

The figure shows this continuation for circle loops of windings 2 and -1.

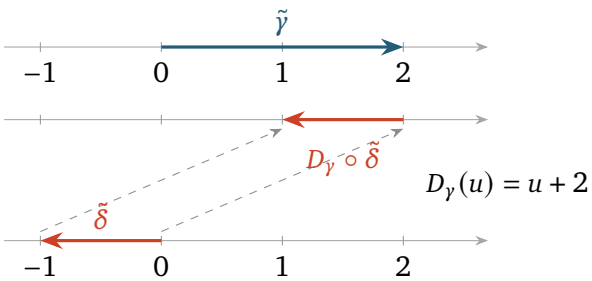


Figure 4.5: The lift of the second loop starts at 0 and ends at -1. Translation by 2 moves it to a path from 2 to 1, which continues the first lift in the order γ then δ.

For example, the deck transformations of $\mathbb{R} \to S^1$ are exactly $u \mapsto u + k$ for $k \in \mathbb{Z}$. These are deck transformations, act transitively on the fibre, and exhaust the possibilities by uniqueness from the image of 0. They recover $\pi_1(S^1, 1) \cong \mathbb{Z}$. The deck group $\mathbb{Z}$ is the first example of two notions used from Lecture 5 onward.

Definition 4.7 (Subgroup)**.** A *subgroup* of a group G is a subset $H \subseteq G$ containing the identity and closed under multiplication and inverses.

Definition 4.8 (Abelian group)**.** A group G is *abelian* if $gh = hg$ for every $g, h \in G$.

For example, $m\mathbb{Z}$ is a subgroup of the additive group $\mathbb{Z}$ for every integer $m \geq 0$. The group $\mathbb{Z}$ is abelian because addition commutes.

4.4 The antipodal cover and rotations

Proposition 4.9 (Spheres of dimension at least two · *sketched*)**.** *The sphere S^n is simply connected for $n \geq 2$.*

Proof sketch. Short spherical arcs join nonantipodal points. A path via a third point joins antipodes, so the sphere is path-connected.

A loop missing a point contracts after stereographic projection to $\mathbb{R}^n$. An arbitrary continuous loop need not miss a point. We first replace the loop by a homotopic spherical polygon, whose image lies in finitely many proper linear subspaces.

First, cover the loop by open hemispheres. Compactness and Lemma 3.5 give a finite subdivision whose pieces each lie in one hemisphere. Replace each piece by the shortest spherical arc joining its endpoints in that hemisphere. Normalizing their straight interpolation gives the homotopy: its scalar product with the hemisphere centre stays positive.

Next, the resulting polygon lies in finitely many two-dimensional linear subspaces. For $n \geq 2$ these are proper subspaces of $\mathbb{R}^{n+1}$ and their union misses a point of the sphere. Finally, stereographic projection from that point identifies its complement with $\mathbb{R}^n$, where the polygon contracts with its base point fixed. The formulas of Lecture 1 apply in n coordinates. This sketches the sphere result of Hatcher, Proposition 1.14, using polygonal approximation.

This also distinguishes the plane from higher-dimensional Euclidean spaces, given Proposition 4.9 at its sketched level. For every $n \geq 2$, the radial homotopy is given by

$$h\colon (\mathbb{R}^n \setminus \{0\}) \times I \longrightarrow \mathbb{R}^n \setminus \{0\}, \qquad h(x,t) = \left(1 - t + \frac{t}{|x|}\right) x.$$

Its positive scalar factor keeps the image away from the origin. It fixes the unit sphere. The final map's value at x is $x/|x|$. Thus $\mathbb{R}^n \setminus \{0\}$ has the fundamental group of S^{n-1}, by Lecture 3's homotopy-invariance argument. For $n \geq 3$ this group is trivial, whereas the punctured plane has group $\mathbb{Z}$ by Lecture 3. A homeomorphism from $\mathbb{R}^2$ to $\mathbb{R}^n$ would restrict to a homeomorphism after deleting 0 and its image. Translation identifies the latter puncture with 0, contradicting these different groups. Hence $\mathbb{R}^2$ is not homeomorphic to $\mathbb{R}^n$ for $n \geq 3$.

Recall from Lecture 2 that $\mathbb{R}\mathrm{P}^n$ identifies antipodal points of S^n. The quotient map is a covering. To see this, for $x \in S^n$ take the cap $V = \{u \in S^n : u \cdot x > 1/2\}$. The inverse image of its quotient image is $V \coprod (-V)$. The two caps are disjoint. Write p for this quotient map. For every open $W \subseteq S^n$,

$$p^{-1}(p(W)) = W \cup (-W).$$

This set is open, so $p(W)$ is open by the quotient topology. A map taking open sets to open sets is called an *open map*. Each cap therefore maps homeomorphically onto the same open quotient set. These sets cover $\mathbb{R}\mathrm{P}^n$. The figure shows the two caps over one neighbourhood.

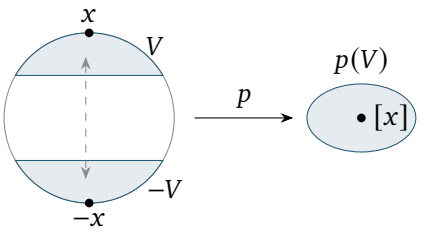


Figure 4.6: The two disjoint caps V and $-V$ lie above the same quotient neighbourhood. The antipodal map exchanges the two sheets.

Write $\mathbb{Z}_2$ for the integers modulo two.

Theorem 4.10 (Projective space and rotations · *proved given S^n simply connected*)**.** *For $n \geq 2$, the antipodal cover gives $\pi_1(\mathbb{R}\mathrm{P}^n) \cong \mathbb{Z}_2$. Consequently $\pi_1(\mathrm{SO}(3)) \cong \mathbb{Z}_2$.*

Proof. The sphere is simply connected by Proposition 4.9, at the sketched level just given. The identity and antipodal map act transitively on the two-point fibre. Deck uniqueness excludes any other transformation. The antipodal map has square the identity, so Corollary 4.6 gives $\mathbb{Z}_2$. The homeomorphism $\mathbb{R}\mathrm{P}^3 \cong \mathrm{SO}(3)$ was proved in Proposition 2.22 by the axis–angle construction. ∎

The nontrivial class has a representative with a direct geometric meaning. Take $x = e_1$ and the path

$$\tilde{\gamma}(t) = \cos(\pi t)e_1 + \sin(\pi t)e_2 \qquad (t \in I)$$

on S^n. Its endpoint is $-x$, so its projection to $\mathbb{R}\mathrm{P}^n$ is a loop whose lift does not close. A point of projective space is a line through the origin. This projected loop turns the line through a half-turn. Traversing it twice lifts to a full great circle, which closes and contracts on S^n for $n \geq 2$. The figure compares the sphere path with its lines.

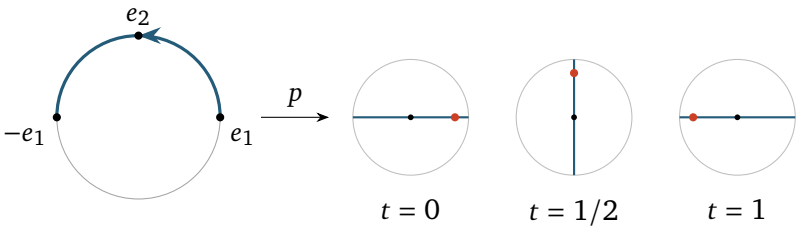


Figure 4.7: The upper semicircle joins e_1 to $-e_1$ on S^n. The lines at $t = 0$, $t = 1/2$ and $t = 1$ show the projected loop returning to its initial line after a half-turn.

The circle case is different: $\mathrm{SO}(2) \cong S^1$ has fundamental group $\mathbb{Z}$. The general result $\pi_1(\mathrm{SO}(m)) \cong \mathbb{Z}_2$ for $m \geq 3$ is cited from Nakahara, Table 4.1; only $m = 3$ is proved in this lecture.

4.5 The half-angle covering and Pauli matrices

The group computed in Theorem 4.10 has two elements. To determine the class of a particular rotation loop, we need an explicit covering formula and its lifted path. The half-angle coordinates associate a rotation through angle φ about a unit vector $n \in \mathbb{R}^3$ with

$$(a, v) = \left(\cos\frac{\varphi}{2}, \sin\frac{\varphi}{2}\, n\right) \in S^3.$$

Substituting $\varphi = 2\theta$ in Lecture 2's axis–angle formula gives the following rotation. Write a point of S^3 as (a, v), where $a \in \mathbb{R}$, $v \in \mathbb{R}^3$ and $a^2 + |v|^2 = 1$. The corresponding rotation is

$$w \longmapsto (a^2 - |v|^2)w + 2(v \cdot w)v + 2a(v \times w).$$

When $v = 0$, this is the identity. Otherwise, substituting $a = \cos\theta$ and $v = \sin\theta\, n$ gives Lecture 2's rotation $R(2\theta n)$ about the unit axis n. The formula is unchanged by $(a, v) \mapsto (-a, -v)$, so it induces a continuous map from $\mathbb{R}\mathrm{P}^3$ to $\mathrm{SO}(3)$.

To identify that map, choose representatives with $a \geq 0$ and $0 \leq \theta \leq \pi/2$. The corresponding axis–angle vector is

$$v \longmapsto \frac{2\arcsin|v|}{|v|} v \quad (v \neq 0), \qquad 0 \longmapsto 0.$$

Its radius increases strictly and continuously from 0 to π, so it is a bijection from the hemisphere to the closed angle ball. Continuity at $v = 0$ follows because the image norm is $2\arcsin|v| \to 0$. By Proposition 2.22, composing it with the angle-ball rotation identifies only antipodal boundary points. Thus the induced map $\mathbb{R}\mathrm{P}^3 \to \mathrm{SO}(3)$ is a continuous bijection and is a homeomorphism by Corollary 2.5. The displayed rotation formula is therefore the antipodal covering followed by this homeomorphism, and hence is itself a covering $S^3 \to \mathrm{SO}(3)$. Its radial coordinates differ from Lecture 2's linear hemisphere rescaling.

Definition 4.11 (Special unitary group)**.** The *special unitary group* $\mathrm{SU}(2)$ consists of complex unitary 2×2 matrices of determinant one.

Its matrices have the form below, identifying it with S^3:

$$U(a, v) = \begin{pmatrix} a - \mathrm{i}v_3 & -v_2 - \mathrm{i}v_1 \\ v_2 - \mathrm{i}v_1 & a + \mathrm{i}v_3 \end{pmatrix}, \qquad a^2 + |v|^2 = 1.$$

Indeed, orthonormal columns and determinant one give the form $\left(\begin{smallmatrix} z & w \\ -\bar{w} & \bar{z} \end{smallmatrix}\right)$ with $|z|^2 + |w|^2 = 1$. Taking real and imaginary parts gives the displayed coordinates and a continuous inverse.

We express the same matrix using the Pauli matrices, with the conventions of *Linear Algebra*:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

For $w = (w_1, w_2, w_3) \in \mathbb{R}^3$, write $w \cdot \sigma = w_1\sigma_x + w_2\sigma_y + w_3\sigma_z$. Matrix multiplication gives

$$(v \cdot \sigma)(w \cdot \sigma) = (v \cdot w)\mathbb{1} + \mathrm{i}(v \times w) \cdot \sigma.$$

In particular, $(n \cdot \sigma)^2 = \mathbb{1}$ for $|n| = 1$. In the half-angle coordinates, the even and odd powers in the exponential series therefore give

$$U(a, v) = a\mathbb{1} - \mathrm{i}v \cdot \sigma = \cos\frac{\varphi}{2}\mathbb{1} - \mathrm{i}\sin\frac{\varphi}{2}\, n \cdot \sigma = \exp\left(-\frac{\mathrm{i}\varphi}{2}\, n \cdot \sigma\right).$$

The same multiplication rule relates the matrix to its spatial rotation. Its triple-product consequence is

$$(v \cdot \sigma)(w \cdot \sigma)(v \cdot \sigma) = 2(v \cdot w)v \cdot \sigma - |v|^2 w \cdot \sigma.$$

Expanding the conjugation with this identity gives

$$\begin{aligned} U(w \cdot \sigma)U^\dagger &= \big((a^2 - |v|^2)w + 2(v \cdot w)v + 2a(v \times w)\big) \cdot \sigma \\ &= (Rw) \cdot \sigma. \end{aligned}$$

Here R is exactly the rotation displayed at the start of the section. Replacing U by $-U$ leaves the conjugation unchanged.

4.6 A full turn and a double turn

We compute the class of rotation through angle $2\pi t$ about the third axis, for $t \in I$.

The lifted path from the identity for the chosen rotation is

$$(a(t), v(t)) = (\cos \pi t,\ \sin \pi t\, e_3), \qquad U(t) = \begin{pmatrix} e^{-\pi i t} & 0 \\ 0 & e^{\pi i t} \end{pmatrix}.$$

At $t = 1$ it ends at $(-1, 0)$, or $-\mathbb{1}$ in matrix form. Its projection is closed, but its lift is not. By Corollary 4.6, this is the nontrivial loop class in $\mathrm{SO}(3)$.

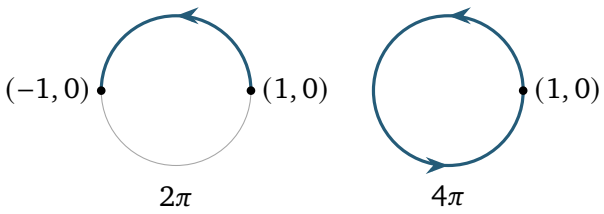


Figure 4.8: A full turn lifts from $(1, 0)$ to $(-1, 0)$ along a semicircle in S^3. A double turn completes the great circle and closes upstairs.

For a 4π rotation, replace t by $2t$ in the lifted formula. The lift ends at $(1, 0)$ and is a loop in the simply connected sphere. It contracts there, so its projection contracts in $\mathrm{SO}(3)$. The rotation ball of Lecture 2 represents the same full-turn loop by the piecewise representatives

$$b(t) = \begin{cases} 2\pi t\, e_3, & 0 \le t \le \frac{1}{2}, \\ (2\pi t - 2\pi)e_3, & \frac{1}{2} < t \le 1. \end{cases}$$

At the join, the two one-sided values are antipodal boundary representatives of the same rotation. Thus the path runs from the centre to πe_3, continues from $-\pi e_3$, and returns to the centre. A double turn traverses this loop twice. The spherical contraction can be written explicitly. In coordinates (a, v_1, v_2, v_3), put

$$q_s(t) = \left(\cos \frac{\pi s}{2} \cos(2\pi t),\ 0,\ \sin \frac{\pi s}{2},\ \cos \frac{\pi s}{2} \sin(2\pi t)\right) \quad (s, t \in I).$$

Every q_s is a loop on S^3. The family starts at the double-turn lift and ends at the constant point $(0, 0, 1, 0)$. Its base point moves. Lecture 3's change-of-base-point argument turns this free contraction into a based contraction, so its projection is the identity loop class. The covering proof above also supplies a based contraction directly by simple connectedness.

The figure shows the identified boundary points and the corresponding closed lift of the double turn.

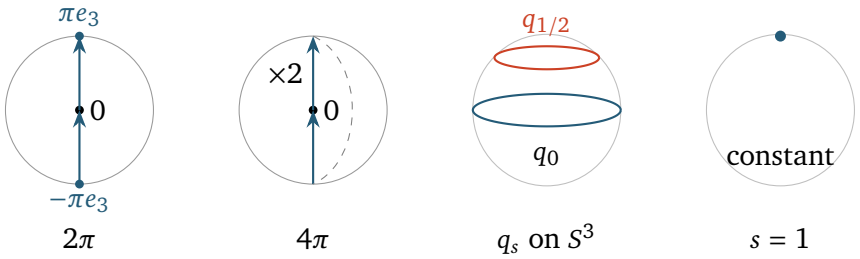


Figure 4.9: The ball section uses its antipodal boundary identification. A full turn passes through that identification once, while a double turn lifts to a full great circle that contracts on S^3.

A belt can be modelled by an oriented frame along its centre line. For a straight belt with k full twists, an explicit strip is

$$B_k(t,r) = \big(t, r\cos(2\pi kt), r\sin(2\pi kt)\big), \qquad t \in I, \quad -\tfrac{1}{8} \le r \le \tfrac{1}{8}.$$

At $r = 0$, its unit tangent is e_1. The width direction rotates through $2\pi kt$ about e_1. The tangent, width and their cross product form a path of oriented frames, hence a path in $\mathrm{SO}(3)$. For integer k the end frames agree, so it is a loop. A continuous motion of such frames fixing their end orientations gives an endpoint-fixed homotopy of that loop. The figure compares one and two twists with their frame paths.

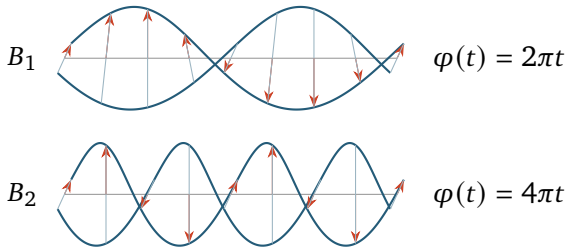


Figure 4.10: The strips are the exact models B_1 and B_2, with width directions marked along their centre lines. Their frame loops have respectively nonclosed and closed lifts to S^3.

For $k = 1$ the nontrivial loop class obstructs a deformation to constant frames. For $k = 2$ the frame loop contracts. This frame statement alone does not construct a motion of an embedded belt with fixed end positions.

Physical input. For a spin-1/2 degree of freedom, a pure state is represented by a nonzero vector $\psi \in \mathbb{C}^2$. Vectors differing by an overall phase represent the same isolated state. A rotation through φ about a unit axis n acts by

$$\psi \longmapsto \exp\left(-\frac{\mathrm{i}\varphi}{2}\, n\cdot\sigma\right)\psi.$$

The spin operators are $(\hbar/2)\sigma_x$, $(\hbar/2)\sigma_y$ and $(\hbar/2)\sigma_z$. Coherent alternatives combine by addition of their state amplitudes.

We can now derive the transformation of the spin expectation vector. Define its dimensionless components by

$$r_j = \frac{\psi^\dagger \sigma_j \psi}{\psi^\dagger \psi} \qquad (j = x, y, z).$$

For every $w \in \mathbb{R}^3$, the conjugation identity applied to $U^\dagger$ gives

$$w \cdot r' = \frac{\psi^\dagger U^\dagger (w \cdot \sigma) U \psi}{\psi^\dagger \psi} = (R^{-1} w) \cdot r = w \cdot (Rr).$$

Thus the expectation vector rotates by R. The computed matrices at the end of one and two full turns are $-\mathbb{1}$ and $\mathbb{1}$. A full turn therefore multiplies ψ by -1, although the classical rotation has returned to the identity. This overall sign leaves the isolated state unchanged. It changes a relative phase if one coherent alternative undergoes the rotation while another does not: the combined amplitude changes from $\psi_1 + \psi_2$ to $-\psi_1 + \psi_2$.

Self-check. Consider rotation through $6\pi t^2$ about $n = (2, -1, 2)/3$ for $t \in I$. Find its lift from $(1, 0)$, compute its matrix at $t = 1/2$, and determine whether the rotation loop contracts.

Answer. The axis has norm one. The half-angle formula gives

$$(a(t), v(t)) = \big(\cos(3\pi t^2), \sin(3\pi t^2) n\big), \qquad U(1/2) = \frac{\sqrt{2}}{6} \begin{pmatrix} -3 - 2\mathrm{i} & 1 - 2\mathrm{i} \\ -1 - 2\mathrm{i} & -3 + 2\mathrm{i} \end{pmatrix}.$$

At $t = 1$ the lifted point is $(-1, 0)$, since the half-angle is 3π. The projected rotation is the identity, but its lift does not close. The closed-lift test therefore excludes a contraction.

Self-check. State in one sentence what a covering map is and why we introduced it to study loop classes. Explain why specifying one starting point upstairs is essential.

Answer. A covering map is a continuous surjection with disjoint local inverse sheets, introduced to lift paths and homotopies and to recover loop classes from their endpoints when the covering space is simply connected. The initial point selects a sheet on the first subinterval. Each successive endpoint selects the next sheet, which is the decisive step in path-lift existence and uniqueness.

Unit 4B Optional extensions

4.7 General map lifting and classification *(optional)*

The later computations of fundamental groups by open pieces do not depend on the general map-lifting criterion or the classification theorem in this section.

Definition 4.12 (Local path-connectedness)**.** A space is *locally path-connected* if every neighbourhood of each point contains a path-connected open neighbourhood of that point.

Let $p(z) = z^3$ and $f(z) = z^6$ on S^1, with base point 1. The induced maps multiply winding by 3 and 6 respectively. A lift is $\tilde{f}(z) = z^2$, since $(z^2)^3 = z^6$ and $\tilde{f}(1) = 1$. In contrast, $z \mapsto z^5$ has no such lift. Dividing the angle by three along the generating loop ends at $e^{2\pi \mathrm{i} 5/3} \neq 1$ and therefore does not define a loop upstairs. The closed-lift test rules out a global lift, although every small arc has inverse branches.

The following *lifting criterion* is stated with its proof sketched. For path-connected, locally path-connected Y, a continuous map $f\colon (Y, y_0) \to (X, x_0)$ lifts with value $\tilde{x}_0$ at y_0 exactly when

$$f_*\pi_1(Y, y_0) \subseteq p_*\pi_1(\widetilde{X}, \tilde{x}_0).$$

Necessity follows from $f_* = p_*\tilde{f}_*$. For sufficiency, choose a path from y_0 to y and lift its image under f. Define $\tilde{f}(y)$ as the lifted endpoint. The inclusion and the closed-lift test make this independent of the chosen path. On a small path-connected open set mapping into an evenly covered set, this rule is one inverse branch composed with f, so is continuous. This is Hatcher, Proposition 1.33.

For these power maps, the criterion reads $6\mathbb{Z} \subseteq 3\mathbb{Z}$ in the first case. In the second case, $5\mathbb{Z}$ is not contained in $3\mathbb{Z}$.

Definition 4.13 (Universal cover)**.** A simply connected covering space is called a *universal cover*.

The circle projection $\mathbb{R} \to S^1$, the antipodal covers $S^n \to \mathbb{R}\mathrm{P}^n$ for $n \geq 2$, and the rotation cover $S^3 \to \mathrm{SO}(3)$ are examples.

Definition 4.14 (Semilocal simple connectedness)**.** A space is *semilocally simply connected* if each $x \in X$ has an open neighbourhood U such that every loop in U based at x contracts in X.

The neighbourhood itself need not be simply connected.

Definition 4.15 (Covering isomorphism)**.** A *covering isomorphism* is a homeomorphism between the total spaces commuting with their projections to X. In the based version it also preserves the chosen lifted point.

Definition 4.16 (Conjugate subgroups)**.** Two subgroups are *conjugate* if they have the form H and $k^{-1}Hk$ for some group element k.

Theorem 4.17 (Existence and classification · *cited*)**.** *Let X be nonempty, connected, locally path-connected and semilocally simply connected. It has a universal cover. Isomorphism classes of based connected coverings correspond to subgroups of $\pi_1(X, x_0)$ by $p_*\pi_1(\widetilde{X}, \tilde{x}_0)$. Without chosen base points, they correspond to conjugacy classes of subgroups.*

This is Hatcher, Theorem 1.38. For the circle, the covers $z \mapsto z^m$ with $m \geq 1$ give the subgroups $m\mathbb{Z}$, whereas $\mathbb{R} \to S^1$ gives the zero subgroup. Local path-connectedness makes path components open, so connectedness implies path-connectedness here, both downstairs and in a connected cover.

Self-check. Classify the connected coverings of the circle up to covering isomorphism.

Answer. The circle satisfies the classification theorem's hypotheses: small open arcs are intervals. Every subgroup of $\mathbb{Z}$ is $m\mathbb{Z}$ for exactly one integer $m \geq 0$. For a nonzero subgroup H, let m be its least positive element. For every $h \in H$, integer division writes $h = qm + r$ with $0 \leq r < m$. Closure gives $r = h - qm \in H$, so minimality forces $r = 0$. Thus $H = m\mathbb{Z}$. The zero subgroup is $0\mathbb{Z}$. For $m \geq 1$, the map $z \mapsto z^m$ on S^1 induces multiplication by m on winding, so its image subgroup is $m\mathbb{Z}$. The cover $\mathbb{R} \to S^1$ has subgroup zero because $\mathbb{R}$ is contractible. The cited classification proves these exhaust the connected coverings. Conjugation in additive notation gives $-k + H + k = H$ for every subgroup $H \subseteq \mathbb{Z}$ and every $k \in \mathbb{Z}$, because addition commutes. Thus distinct subgroups give distinct unbased covering

isomorphism classes.

Summary. Local inverse sheets lift paths and their homotopies uniquely. For a simply connected covering space, based loop classes correspond to fibre points. A transitive deck group gives the same group structure after choosing a lifted base point. The antipodal cover computes $\pi_1(\mathbb{R}\mathrm{P}^n) \cong \mathbb{Z}_2$ for $n \geq 2$, using the sketched sphere result. In $\mathrm{SO}(3)$, a full turn is nontrivial and a double turn is trivial. Lecture 5 computes fundamental groups by joining open pieces.

Reading. Hatcher, *Algebraic Topology*, Proposition 1.14, §1.3, especially Propositions 1.30–1.34 and Theorem 1.38; Munkres, *Topology*, §§53–54. Nakahara, *Geometry, Topology and Physics*, §4.6.5 and Table 4.1.

Lecture 5

Van Kampen, surfaces and braids

Objectives. *We compute loop classes in a space assembled from open pieces. Van Kampen's theorem records how loops in the overlap become relations. For a surface presented by a polygon, its boundary gives the relation. For identical particles, the space consists of unordered configurations of distinct points. Loops in this space describe exchanges. Planar exchanges retain more information than their endpoint permutations. Under a scalar statistical-transport assumption, these exchanges allow arbitrary unit phases.*

By the end of this lecture you will be able to:

- Apply van Kampen with path-connected open sets and intersection.
- Read the fundamental-group presentation of a polygonal surface.
- Relate planar exchanges to braids and three-dimensional exchanges to permutations.
- Derive the allowed one-dimensional exchange phases in each case.

Unit 5A Surfaces

5.1 Free groups and relations

Definition 5.1 (Wedge of circles). The *wedge* $S^1 \vee S^1$ identifies one point of each of two circles to a common point x_0.

We call the loops round the two circles a and b. A loop can traverse these circles in an ordered sequence. Traversing a circle and immediately retracing it gives a removable inverse pair. The following figure records the order of traversals, including the four traversals of the word $aba^{-1}b^{-1}$, called the *commutator*.

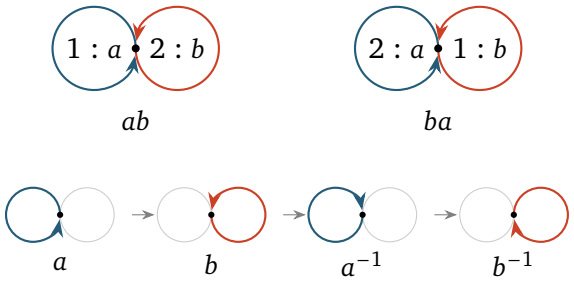


Figure 5.1: The numerals in ab and ba give the order of traversal. The commutator traverses each circle once in each direction from the common point.

Definition 5.2 (Word and reduced word)**.** A *word* in letters $a_1, \dots, a_k$ and their formal inverses is a finite string of these symbols. A *reduced word* has no adjacent inverse pairs.

Delete adjacent inverse pairs until none remain. The resulting reduced word is unique. One algorithm reads from left to right, keeping a stack: cancel the last stored letter when the incoming letter is its inverse, and append it otherwise. Inserting an adjacent inverse pair leaves the final stack unchanged, even when the first inserted letter cancels a stored letter. The second then restores it. Thus reduction is independent of the order of deletions. A reduced word passes through the stack unchanged. Hence every complete sequence of deletions ends at the same stack output.

Definition 5.3 (Free group)**.** The *free group* F_k consists of reduced words. Multiply by concatenating and reducing. The empty word is the identity. Reverse a word and invert each letter to get its inverse.

Reduction is unchanged by inserting or deleting inverse pairs in any context, so reducing after either stage of a triple concatenation gives the same result. This proves associativity. For example,

$$ab^{-1}ba^{-1}b = b, \qquad aba^{-1}b^{-1} \neq 1 \quad \text{in } F_2.$$

The second word is already reduced. A choice of images for the generators in any group extends uniquely to a homomorphism, by substituting in words. Inverse cancellations do not change the substituted group element.

Recall from Lecture 4 that a subgroup contains the identity and is closed under products and inverses. The subgroup generated by a set consists of finite products of its elements and their inverses.

Definition 5.4 (Kernel)**.** The *kernel* of a homomorphism $f\colon G \to H$ is the subgroup

$$\ker f = \{g \in G : f(g) = 1_H\}.$$

The homomorphism laws make this set closed under products and inverses. They also give $f(gng^{-1}) = 1_H$ for $g \in G$ and $n \in \ker f$.

Definition 5.5 (Normal subgroup)**.** A subgroup N of G is *normal* if $gNg^{-1} = N$ for every $g \in G$.

Thus every kernel is normal. Applying the preceding calculation with g^{-1} gives equality of the conjugate subgroup with the kernel.

Definition 5.6 (Coset and quotient group)**.** For a normal subgroup N of G, the *coset* of $g \in G$ is

$$gN = \{gn : n \in N\}.$$

The *quotient group* G/N is the set of cosets, with product

$$(gN)(hN) = ghN.$$

Two cosets agree exactly when $g^{-1}h \in N$. If $h = gn$ with $n \in N$, then $hN = gnN = gN$. Conversely, equality of the cosets gives $h \in hN = gN$, so $h = gn$ for some $n \in N$. To check independence of representatives, write $g' = gn$ and $h' = hm$ with $n, m \in N$. Then

$$g'h' = gnhm = gh(h^{-1}nh)m \in ghN.$$

Normality puts $h^{-1}nh$ in N. Hence the product is well defined. Associativity follows from

associativity in G. The identity coset is N, and the inverse of gN is $g^{-1}N$.

A surjective homomorphism $f\colon G \to H$ with kernel N identifies G/N with H by $gN \mapsto f(g)$. Indeed, $f(g) = f(h)$ holds exactly when $g^{-1}h \in N$. This proves that the induced map is both well defined and injective. It is surjective because f is surjective, and it preserves products. For any homomorphism the same argument identifies $G/\ker f$ with its image.

Definition 5.7 (Normal closure)**.** The *normal closure* of specified words is the smallest normal subgroup containing them, equivalently the subgroup generated by all their conjugates and inverses.

Imposing a relation $r = 1$ also imposes $uru^{-1} = 1$ for every word u. This explains why the whole normal closure is imposed.

Definition 5.8 (Presentation)**.** A *presentation* is written

$$\langle a_1, \dots, a_k \mid r_1, \dots, r_s \rangle = F_k / \langle\langle r_1, \dots, r_s \rangle\rangle.$$

It imposes $r_j = 1$ in every word context, not merely deletion of one printed copy of a relator. For example, $\langle a \mid a^m \rangle \cong \mathbb{Z}_m$ for $m \geq 1$, where $\mathbb{Z}_m$ is the group of integers modulo m. For a two-generator example, impose $aba^{-1}b^{-1} = 1$. Multiplying on the right by ba gives $ab = ba$. Thus every word becomes $a^m b^n$ for $m, n \in \mathbb{Z}$. The exponent-sum map to $(m, n) \in \mathbb{Z}^2$ kills the relator and is inverse to $(m, n) \mapsto a^m b^n$. Hence

$$\langle a, b \mid aba^{-1}b^{-1} \rangle \cong \mathbb{Z}^2.$$

In the free group the commutator remains a nonempty reduced word.

Definition 5.9 (Free product)**.** The *free product* $G * H$ joins two groups without additional relations: words alternate nonidentity elements of the two groups, combining adjacent elements from the same group.

In particular, $\mathbb{Z} * \mathbb{Z} = F_2$.

5.2 Van Kampen's theorem

Van Kampen uses loops from both open pieces as generators and identifies the two images of every overlap loop. Let $X = U \cup V$, with $x_0 \in U \cap V$. The inclusions induce maps from the fundamental groups of these sets. Write i_{U*} and i_{V*} for the maps from the intersection to the two pieces.

Theorem 5.10 (Van Kampen · *cited; surjectivity sketched*)**.** *Suppose U and V are open and path-connected, and $U \cap V$ is path-connected. The natural homomorphism*

$$\pi_1(U, x_0) * \pi_1(V, x_0) \longrightarrow \pi_1(X, x_0)$$

is surjective. Its kernel is the normal closure of $i_{U*}(w) i_{V*}(w)^{-1}$ for $w \in \pi_1(U \cap V, x_0)$.

This is Hatcher, Theorem 1.20. It identifies the two images of each overlap loop. The quotient by the stated kernel is therefore the group of X, by the quotient construction in §5.1. We sketch the surjectivity construction. Subdivide a loop into pieces γ_j in U or V, using the Lebesgue-number lemma. Merge successive pieces in the same set. At a change of set the joining point is in $U \cap V$. Choose a path a_j in the intersection from x_0 to each joining

point, with constant paths at the start and end. The based loops $a_{j-1} \cdot \gamma_j \cdot \bar{a}_j$ each lie in one piece. Their product represents the original loop, since the inserted reverse paths cancel. The kernel statement is cited, not proved by this construction.

The connected-intersection hypothesis matters. On S^1, take the north point $N = (0, 1)$ and south point $S = (0, -1)$. Cover the circle by their complements. Both sets are open intervals and have trivial fundamental group. Their intersection has two components. The circle group is $\mathbb{Z}$, so the conclusion of the theorem cannot hold without the hypothesis.

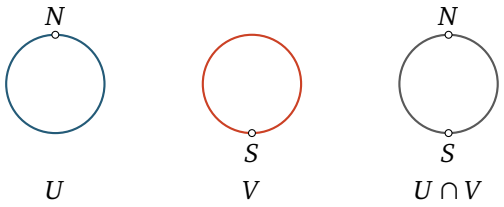


Figure 5.2: Removing N or S gives an open arc. Removing both leaves two components, so this circle cover does not satisfy van Kampen's hypotheses.

For the wedge of two circles, remove a non-base point y_b of the second circle to get U, and a non-base point y_a of the first to get V. These are open in the wedge. The first retracts onto the first circle, the second onto the second circle. Collapse the remaining open arcs linearly to x_0; these homotopies fix their retained circles. The formulas agree at x_0, so finite closed pasting proves continuity. The intersection is four open arms joined at x_0 and contracts there. Hence the cited theorem gives

$$\pi_1(S^1 \vee S^1, x_0) \cong \mathbb{Z} * \mathbb{Z} = F_2.$$

The loops a and b run round the two circles. Inductively the same open-neighbourhood construction gives F_r for a finite wedge of r circles. The figure marks the points removed to construct the two open sets.

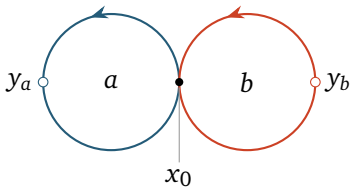


Figure 5.3: Removing the marked point from the opposite circle leaves an open neighbourhood that retracts onto the chosen circle. Both circles share x_0.

For a sphere S^n with $n \geq 2$, remove its north pole to obtain U and its south pole to obtain V. Both pieces are homeomorphic to $\mathbb{R}^n$, so their groups are trivial. Their overlap is path-connected: it is homeomorphic to $S^{n-1} \times \mathbb{R}$. Van Kampen gives a trivial fundamental group for S^n. This recovers Lecture 4's sphere result from the cited theorem.

For $\mathbb{R}\mathrm{P}^2$, use Lecture 2's disc with antipodal boundary points identified. Let U be its quotient minus the centre and let V be a small open centre disc. The set U retracts radially onto the boundary quotient, a circle with generator a. The overlap is a punctured disc. Its counter-clockwise generator runs round the full disc boundary, so runs twice round the boundary quotient. Its image in U is a^2, and its image in V is trivial. Thus van Kampen gives

$$\pi_1(\mathbb{R}\mathrm{P}^2) \cong \langle a \mid a^2 \rangle \cong \mathbb{Z}_2.$$

This agrees with the antipodal-cover calculation in Lecture 4.

5.3 The torus square

We compute the group of the square with opposite sides identified in the same direction. The quotient is the torus T^2 of Lecture 2. All four corners give one vertex x_0, and the two edge loops are a and b. Choose the centre o in the face. Let $U = T^2 \setminus \{o\}$ and let V be a small open disc about o inside the face. These are path-connected open sets covering the torus. The set U retracts radially onto the edge wedge, and V is contractible. Their overlap is the punctured disc $V \setminus \{o\}$, whose group is $\mathbb{Z}$.

Choose a base point y in this overlap and a connector from the vertex x_0 to y in U. We orient the overlap generator counter-clockwise in the displayed square, rather than as an inner boundary of a punctured surface. Following the connector, the overlap loop and the reverse connector gives a loop based at x_0 in U. The figure shows its radial retraction to the square boundary.

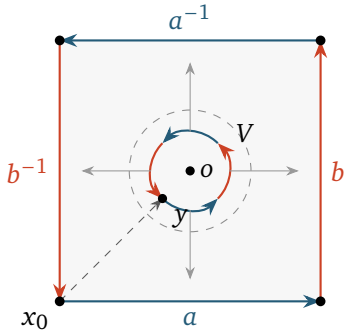


Figure 5.4: Edge arrows show the counter-clockwise boundary traversal $aba^{-1}b^{-1}$ of the retracted overlap loop, with opposite edges identified in the same direction. The dashed connector joins the vertex to its base point y.

The overlap generator maps to $aba^{-1}b^{-1}$ in the edge wedge and to the identity in the disc. A change of base point along the connector identifies the wedge group with $\pi_1(U, y)$. Van Kampen imposes exactly the normal closure of this relator. Therefore

$$\pi_1(T^2) \cong \langle a, b \mid aba^{-1}b^{-1} \rangle \cong \mathbb{Z}^2.$$

The last isomorphism is the two-generator quotient computed in §5.1. This agrees with the coordinate-winding calculation of Lecture 3.

5.4 Polygonal surfaces

Definition 5.11 (Surface). A *surface* is a Hausdorff space with a countable basis in which every point has a neighbourhood homeomorphic to an open disc in $\mathbb{R}^2$. A compact surface with these local models is called *closed*.

Definition 5.12 (Genus surface). For $g \geq 1$, let Σ_g be the standard closed surface obtained from a $4g$-gon by pairing equally labelled edges in the boundary word

$$a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1}.$$

The integer g is its *genus*.

The torus has one handle. Joining g tori along boundary circles of removed discs gives the surface with g handles. Its identification with the polygon model is cited from Hatcher, §1.2. The following figure shows the genus-two model and its polygon. The numbers $0, \dots, 7$ name the octagon corners. In the inset we rescale the corner sectors to show their cyclic order around the identified vertex. On the surface, the dashed paths join a common point x_0 to each displayed curve. Following a connector, its curve and the reverse connector gives the based generator with the corresponding polygon label.

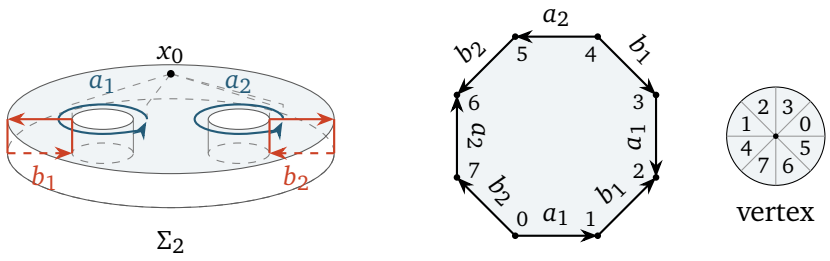


Figure 5.5: A surface with two handles and its octagon model. The paired edges give four loops, and the corner sectors join to one disc.

The paired arrows specify the same direction along each identified edge. Edge interiors join two half discs, and the cyclic corner identifications join to a disc. All vertices become one vertex, and the edges become a wedge of $2g$ circles. This is the polygonal model used in Hatcher, §1.2.

Proposition 5.13 (Surface groups · *proved given van Kampen*). *For $g \geq 1$, the fundamental group is*

$$\pi_1(\Sigma_g) \cong \left\langle a_1, b_1, \dots, a_g, b_g \;\middle|\; \prod_{j=1}^{g} [a_j, b_j] \right\rangle, \qquad [a, b] = aba^{-1}b^{-1}.$$

Proof. Choose a point o in the face interior. Let $U = \Sigma_g \setminus \{o\}$ and let V be a small open disc about o inside that face. The punctured polygon retracts radially to its boundary, fixing it. This deformation respects the edge identifications and retracts U onto the wedge of $2g$ circles. The set V is contractible. The overlap is a punctured disc, with fundamental group $\mathbb{Z}$. Its positive generator follows the face boundary once after radial retraction. Choose a path from its base point to the polygon vertex to read the word in the stated generators. The result is $[a_1, b_1] \cdots [a_g, b_g]$. Van Kampen kills exactly its normal closure, giving the presentation. A different connecting path conjugates the relator and leaves its normal closure unchanged. ∎

For $g = 1$ this is the torus presentation computed in §5.3.

Definition 5.14 (Permutation group). A permutation of $\{1, \dots, n\}$ is a bijection of this set to itself. These form the group S_n, with right-to-left composition. In cycle notation, (12) exchanges 1 and 2, while (123) sends 1 to 2, 2 to 3 and 3 to 1.

Self-check. Prove that the group of $S^1 \vee S^1$ is nonabelian by finding a surjective homomorphism onto S_3.

Answer. Send a to (12) and b to (13). The universal property of F_2 gives a homomorphism. Its image contains $(23) = (12)(13)(12)$ and the two three-cycles, as well as the identity.

Thus it contains all six permutations. Since $(12)(13) \neq (13)(12)$, the image is nonabelian, and F_2 is nonabelian. In particular, its commutator $aba^{-1}b^{-1}$ cannot be removed before a face imposes that relation. Its image is $(132)^2 = (123) \neq 1$, which verifies nontriviality directly.

Self-check. Suppose $X = U \cup V$ satisfies van Kampen's hypotheses, with $\pi_1(U) = \langle a \rangle \cong \mathbb{Z}$, $\pi_1(V) = \langle b \rangle \cong \mathbb{Z}$ and $\pi_1(U \cap V) = \langle w \rangle \cong \mathbb{Z}$. The inclusion maps send w to a^3 and b^2 respectively. Give a presentation of $\pi_1(X)$. Determine whether assigning $a \mapsto e^{2\pi \mathrm{i}/7}$ and $b \mapsto e^{3\pi \mathrm{i}/7}$ defines a homomorphism to the group of unit complex numbers.

Answer. The two images of w must agree, so

$$\pi_1(X) \cong \langle a, b \mid a^3 b^{-2} \rangle.$$

A homomorphism from the free group descends to this quotient exactly when it kills the relator. For the proposed images, its value on the relator is

$$(e^{2\pi \mathrm{i}/7})^3 (e^{3\pi \mathrm{i}/7})^{-2} = e^{(6-6)\pi \mathrm{i}/7} = 1.$$

Thus the assignment does define a homomorphism.

Unit 5B Particles and braids

5.5 Configuration spaces and endpoint permutations

We model n identical particles by their positions in $\mathbb{R}^d$. We restrict to histories with no coincident positions. An ordered list keeps labels, whereas its unordered set forgets them. Fix integers $n \geq 2$ and $d \geq 2$.

Definition 5.15 (Configuration spaces)**.** The *ordered configuration space* and *unordered configuration space* are

$$\mathrm{Conf}_n^{\mathrm{ord}}(\mathbb{R}^d) = \{(x_1, \ldots, x_n) \in (\mathbb{R}^d)^n : x_i \neq x_j \text{ for } i \neq j\},$$
$$\mathrm{Conf}_n(\mathbb{R}^d) = \mathrm{Conf}_n^{\mathrm{ord}}(\mathbb{R}^d)/S_n.$$

We keep the dimension in the notation to distinguish the plane from space. In the planar case we also write $C_n = \mathrm{Conf}_n(\mathbb{R}^2)$. The permutation group S_n acts by relabelling the coordinates. Its action is $(s \cdot x)_i = x_{s^{-1}(i)}$, which forgets labels in the quotient. The projection is a covering: take disjoint small balls around the n points. Their product and its $n!$ permuted copies are disjoint sheets over the same quotient neighbourhood. The quotient map is open because the inverse image of the image of an open set is the finite union of its permuted copies.

For two particles in either dimension, write their positions as $c + u$ and $c - u$, where $c \in \mathbb{R}^d$ and $u \in \mathbb{R}^d \setminus \{0\}$. The inverse formulas are $c = (x_1 + x_2)/2$ and $u = (x_1 - x_2)/2$. The centre is unrestricted. Collision occurs exactly when $u = 0$. Forgetting labels identifies u with $-u$. The centre, radius and line through u therefore give a homeomorphism

$$\mathrm{Conf}_2(\mathbb{R}^d) \cong \mathbb{R}^d \times (0, \infty) \times \mathbb{RP}^{d-1}.$$

The inverse sends $(c, r, [v])$, with $|v| = 1$, to $\{c + rv, c - rv\}$. This is independent of the representative v. Local sections of the antipodal cover make the inverse continuous.

In space, the last factor is $\mathbb{R}\mathrm{P}^2$. Its fundamental group is $\mathbb{Z}_2$ by Lecture 4, given the sphere's simple connectedness. One exchange lifts to a half great circle from v to $-v$ on S^2. Two exchanges lift to the full great circle. The following figure compares this loop with the planar one.

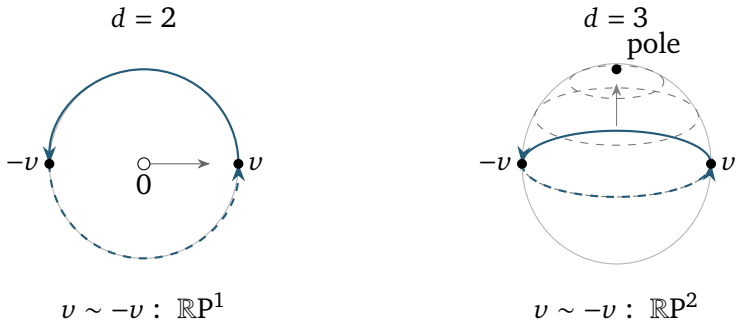


Figure 5.6: One exchange joins opposite directions, identified in projective space. The full circle contracts on S^2 through latitudes, whereas the planar double exchange has winding two in the squared direction and does not contract.

For an explicit free contraction of the lifted spatial double exchange, set

$$H(t,s) = \big(\cos(\pi s/2)\cos(2\pi t),\ \cos(\pi s/2)\sin(2\pi t),\ \sin(\pi s/2)\big), \qquad t,s \in [0,1].$$

Each latitude lies on the unit sphere. At $s = 0$ it is the great circle, and at $s = 1$ it is the north pole. The base point moves during this contraction. Attaching a path along the moving base point and its reverse, as in Lecture 3, converts it into a based contraction. Composing with the antipodal projection and keeping the centre and radius fixed contracts the double exchange in $\mathrm{Conf}_2(\mathbb{R}^3)$.

For two planar particles, write the positions as $c+u$ and $c-u$ in $\mathbb{C}$, where $u \neq 0$. Forgetting labels identifies u with $-u$. The centre c, radius $|u|$ and squared direction $(u/|u|)^2$ give

$$\mathrm{Conf}_2(\mathbb{R}^2) \cong \mathbb{R}^2 \times (0,\infty) \times S^1.$$

The inverse chooses either square root of the direction; the two choices give the same unordered pair. Locally a square-root branch is continuous, so the inverse is continuous. Contracting the first two factors gives $\pi_1(\mathrm{Conf}_2(\mathbb{R}^2)) \cong \mathbb{Z}$. The counter-clockwise exchange $u(t) = e^{\pi i t}$ makes the squared direction wind once. Two exchanges have winding two, even though their endpoint permutation is the identity.

A loop downstairs lifts to n labelled particle trajectories. Label their initial positions $1, \dots, n$.

Definition 5.16 (Outgoing permutation)**.** The *outgoing permutation* $p_\gamma(i)$ is the final position of the particle starting at i.

With the first-path-first convention,

$$p_{\gamma\cdot\delta} = p_\delta \circ p_\gamma, \qquad \rho([\gamma]) = p_\gamma^{-1}, \qquad \rho([\gamma][\delta]) = \rho([\gamma])\rho([\delta]).$$

Paths run first to last, while permutations compose right to left. Taking the inverse reverses the order again, so ρ is a homomorphism. Homotopy lifting makes it well-defined on loop classes. For the noncommuting test $p_\gamma = (12)$ and $p_\delta = (13)$, it gives

$$p_{\gamma\cdot\delta} = (123), \qquad \rho([\gamma][\delta]) = (132) = (12)(13).$$

In the plane, fix an unordered base configuration x_0.

Definition 5.17 (Geometric braid and braid group)**.** A *geometric braid* is n collision-free trajectories $(z_j(t), t)$ with that fixed set of positions at both $t = 0$ and $t = 1$. Time increases along every strand. Deformations keep the endpoint positions fixed and preserve the collision-free condition. Write B_n for their classes, with multiplication by running the first braid and then the second.

Proposition 5.18 (Exchange classes · *sketched; proved in* $\mathbb{R}^3$)**.** *In three dimensions,* $\pi_1(\mathrm{Conf}_n(\mathbb{R}^3)) \cong S_n$. *In the plane, the fundamental group is the braid group* B_n.

For $\mathbb{R}^3$, we prove that the ordered space is simply connected. We approximate a loop by a polygon, then contract it to a suitable apex. The dimension bound allows a single apex avoiding every pairwise collision. Its collision sets are linear subspaces $L_{ik} = \{x_i = x_k\}$, for $1 \le i < k \le n$, of $\mathbb{R}^{3n}$, each of dimension $3n - 3$. A finite union of proper linear subspaces misses a point. Indeed, for each subspace choose a nonzero linear functional vanishing on it. On the curve $(1, t, \dots, t^{3n-1})$ each functional becomes a nonzero polynomial, with finitely many roots. Choose t outside those finite root sets.

First, a loop γ has positive distance ε from the closed union of collision sets, by compactness. Uniform continuity gives a subdivision $0 = t_0 < \cdots < t_m = 1$ with each image piece of diameter less than $\varepsilon/4$. Join successive vertices $v_j = \gamma(t_j)$ by straight segments to form η. At matching parameters, the interpolation between η and γ stays within $\varepsilon/2$ of $\gamma(t)$. This is a based homotopy avoiding collisions.

Next, choose an apex z_* outside all subspaces $L_{ik} + \mathrm{span}\{v_j, v_{j+1}\}$. Each has dimension at most $3n - 1$, so the finite-union argument applies. With two spatial coordinates, that sum can fill the ambient space, so the apex argument does not apply. The triangle with vertices v_j, v_{j+1}, z_* misses L_{ik}. Otherwise a collision point with nonzero apex coefficient would express z_* as an element of that forbidden subspace. A collision with zero apex coefficient would lie on the already collision-free polygon edge. Thus the map $h\colon [0,1]^2 \to \mathrm{Conf}_n^{\mathrm{ord}}(\mathbb{R}^3)$ given by

$$h(t, s) = (1 - s)\eta(t) + sz_*$$

contracts the loop freely. Lecture 3's moving-base-point construction turns this into a based contraction by attaching the path of the moving base point and its reverse. Apply the same apex choice to two configurations instead of polygon vertices to join them by two straight segments. Hence the ordered space is path-connected and simply connected. The relative-coordinate triangle below displays the collision-free cone for one edge and contrasts it with a planar triangle that hits the origin.

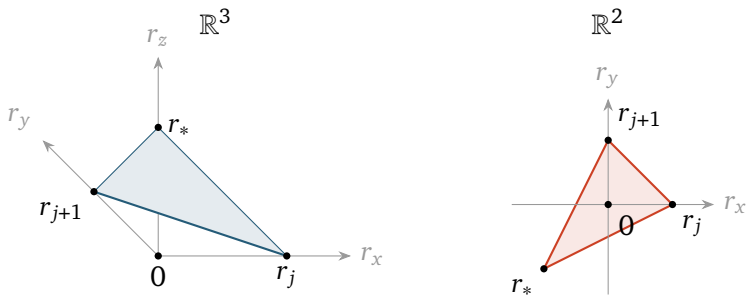


Figure 5.7: For two particles, project to relative coordinates $r = x_1 - x_2$. The spatial triangle misses 0, whereas the planar comparison contains 0.

Coordinate permutations are homeomorphisms commuting with the projection, so they are deck transformations. They act transitively on a fibre and exhaust the deck group by uniqueness from one point. By Corollary 4.6, a simply connected cover whose deck group acts transitively on a fibre has that deck group as the fundamental group of the base. Hence the unordered space has group S_n. The permutation reaching the lifted endpoint is $\rho([\gamma])$, by our chosen action, so this is the isomorphism stated above. Simon, *Topological Quantum*, §3.3.2, describes the exchange-space conclusion.

For the plane we sketch the identification. Lifting a configuration loop labels its strands. Conversely, their unordered positions give a configuration loop. The same correspondence applies to homotopies. Thus the two groups are identified. This is Simon, §3.3.1, in the course's path convention. In particular, the two-particle calculation gives $B_2 \cong \mathbb{Z}$.

5.6 Braid relations and exchange phases

Arrange the base positions on a real line. Let σ_i exchange positions i and $i+1$ counter-clockwise, fixing the other positions. Its inverse exchanges them clockwise. We cite Artin's presentation:

$$B_n = \left\langle \sigma_1, \dots, \sigma_{n-1} \;\middle|\; \begin{array}{ll} \sigma_i\sigma_j = \sigma_j\sigma_i, & |i-j| \geq 2, \\ \sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}, & 1 \leq i \leq n-2 \end{array} \right\rangle .$$

The first relation exchanges separated pairs in either order. The second changes the order of three adjacent exchanges. The completeness of these relations is cited from Simon, §3.3.1 and Exercise 3.1. Our words are read leftmost first and positive exchanges are counter-clockwise.[1] The elementary crossings below fix the projection convention.

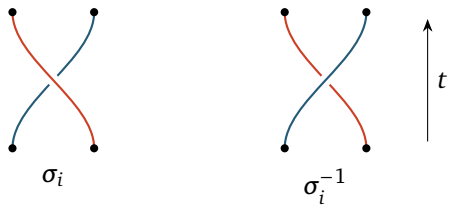


Figure 5.8: Time runs upward and the real axis to the right. At a crossing, the strand with larger imaginary coordinate is drawn on top.

The next figure shows both braid relations with the same labelled strands. At a crossing in the drawing, the particles have different imaginary coordinates, so they do not collide.

[1] Simon reads words rightmost first and takes clockwise exchanges as positive. His positive generator is our σ_i^{-1}. A homomorphism to the group of unit complex numbers with value $e^{i\theta}$ on our positive exchange has value $e^{-i\theta}$ on his positive exchange.

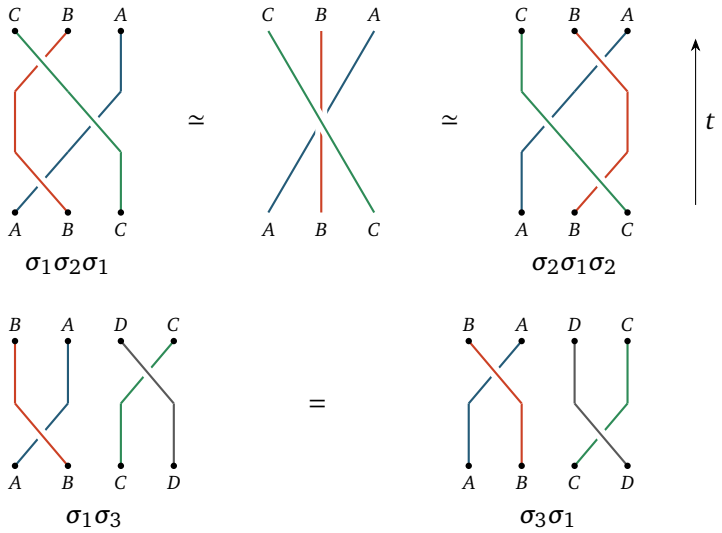


Figure 5.9: Time increases upward, so each word starts at the bottom. The leftward-moving strand is on top at each crossing.

For the three-strand relation, each pair crosses once: C over B, C over A and B over A. Moving the middle crossing past the third strand deforms one diagram into the other with endpoints fixed. The map $\rho\colon B_n \to S_n$ sends σ_i to $(i\ i+1)$. It is surjective because adjacent transpositions generate S_n. It forgets the motion and retains the endpoint permutation. Two positive exchanges already give a nontrivial element of B_2 with identity permutation, by the winding calculation above.

Physical input. We consider a scalar statistical-transport sector for identical particles. The statistical factor multiplies a quantum amplitude, a complex number whose squared modulus gives a probability in the quantum description. Fix a base configuration $x_0 \in \mathrm{Conf}_n(\mathbb{R}^d)$. We assume that each collision-free loop based at x_0 has a unit statistical factor. We assume that this factor depends only on endpoint-fixed homotopy, that successive loops multiply their factors, and that the constant loop has factor 1. This specifies the statistical part of transport after any phase depending on the detailed dynamics has been separated.

These assumptions define a homomorphism from the configuration-space fundamental group to the unit complex numbers. They do not follow from the return of the particle positions. A particular physical system must supply such a sector before the classification applies. For scalar factors, reversing composition order leaves the product unchanged.

Definition 5.19 (Unitary character and exchange phase)**.** A one-dimensional unitary representation is a homomorphism $\chi\colon G \to U(1)$, where $U(1) = \{z \in \mathbb{C} : |z| = 1\}$. Such a representation is also called a *unitary character*. In the scalar statistical sector, its value on an exchange is the *exchange phase*.

Corollary 5.20 (Scalar exchange phases · *proved given the presentations*)**.** *For $n \geq 2$, the one-dimensional unitary representations of S_n give exchange phases $+1$ or -1. For B_n, any phase $e^{i\theta}$, with $\theta \in \mathbb{R}$, is possible, with the same phase on every σ_i.*

Proof. Adjacent transpositions generate S_n and are conjugate to one another. Scalar images of conjugate elements agree. Each transposition squares to the identity, so its common image q satisfies $q^2 = 1$. Both possibilities occur: the constant character and the sign character, the latter given by the determinant of the permutation matrix.

For B_n, write $q_i = \chi(\sigma_i)$. All q_i are nonzero scalars. The adjacent braid relation gives

$$q_i^2 q_{i+1} = q_i q_{i+1}^2, \qquad q_i = q_{i+1}.$$

For $n \geq 3$ this makes all images equal. For $n = 2$ there is only one generator. Conversely, assigning the same $q \in U(1)$ to every generator satisfies both families of defining relations. Thus every $q = e^{i\theta}$ defines a character. ∎

The choices $q = 1$ and $q = -1$ describe scalar bosonic and fermionic statistics. The other planar choices describe *abelian anyons*. Higher-dimensional unitary representations of S_n give matrix-valued exchange operators and the possibility of parastatistics. The scalar conclusion does not classify those representations.

Self-check. State in one sentence what scalar statistical transport is and why it was introduced. For a planar sector with $\chi(\sigma_i) = e^{\pi i/3}$, compute the factor for $\beta = \sigma_1^2\sigma_2^{-1}\sigma_1\sigma_2^{-3}$ in B_3. Determine whether this sector is possible in three dimensions.

Answer. Scalar statistical transport assigns deformation-invariant multiplicative unit factors to collision-free loops, so that topology can classify the statistical factors multiplying quantum amplitudes. The exponent sum of β is $2 - 1 + 1 - 3 = -1$, so

$$\chi(\beta) = e^{-\pi i/3}.$$

In space a double exchange is trivial. The proposed factor fails this relation because $(e^{\pi i/3})^2 = e^{2\pi i/3} \neq 1$. Thus this scalar sector does not descend to three-dimensional exchanges.

Summary. Van Kampen combines path-connected open pieces with path-connected intersection by imposing the overlap relations. A polygonal surface has one relation, the product of its handle commutators. Planar configuration loops are braids. In three dimensions the ordered-space contraction reduces loop classes to permutations. The inverse outgoing permutation is the homomorphism in our path convention. Under the scalar statistical-transport assumptions, exchange phases are arbitrary for braids and restricted to ± 1 for permutations. Lecture 6 counts holes of every dimension by linear algebra.

Reading. Hatcher, *Algebraic Topology*, §1.2; Nakahara, *Geometry, Topology and Physics*, §4.4; Simon, *Topological Quantum*, §§3.3–3.5 and Exercise 3.1.

Lecture 6

Homology and the Euler characteristic

Objectives. *Homology distinguishes the sphere from a point using the invariance theorem of §6.5. Both spaces have trivial fundamental group (Lecture 4). Homology compares closed chains with boundaries to count holes in each dimension. We compute it from boundary matrices for the torus and projective plane. Its alternating dimensions give the Euler characteristic. Binary cycle classes also describe closed strings in the toric code.*

By the end of this lecture you will be able to:

- Construct boundary matrices from ordered simplices.
- Compute cycles, boundaries and homology with specified coefficients.
- Prove the Euler characteristic formula for a finite complex.
- Identify the two independent binary string classes on a torus.

6.1 Simplices and their identifications

Definition 6.1 (Simplex). A *k-simplex* is the convex hull of $k+1$ affinely independent points, with a chosen ordering of these vertices. The standard simplex is

$$\Delta^k = \{(t_0, \dots, t_k) \in \mathbb{R}^{k+1} : t_i \geq 0,\ t_0 + \cdots + t_k = 1\}.$$

Deleting one vertex gives a *face*, with its inherited order.

A point, an interval and a triangle are simplices of dimensions zero, one and two. The order of the vertices orients an edge from the first vertex to the second. It also specifies the signs of higher-dimensional faces.

We identify the endpoints of one interval, as in Lecture 2, to obtain a circle. The domain endpoints v_0 and v_1 are distinct, but both map to the same vertex v. The interval interior maps injectively to the open edge. This circle has one vertex and one edge. The quotient map $q\colon \Delta^1 \to S^1$ sends the interval to the circle in the drawing, which also shows a three-edge decomposition.

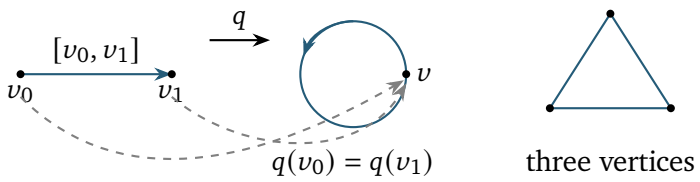


Figure 6.1: The interval's distinct domain endpoints both map to v. The triangle outline is a simplicial decomposition of the same circle.

Definition 6.2 (Δ-complex)**.** A Δ-*complex structure* on X is a collection of continuous maps $\sigma\colon \Delta^k \to X$, called its simplices, with the following properties. Each map is injective on the interior of its domain. These interiors have disjoint images and together cover X. The restriction to each face is a designated lower-dimensional simplex, using the affine identification preserving the vertex order. The topology is specified by these maps: a subset $A \subseteq X$ is closed exactly when $\sigma^{-1}(A)$ is closed for every simplex σ.

The closed-set condition gives the quotient topology of Lecture 2. Thus the space is obtained by attaching the faces in the specified way.

Different vertices of one domain simplex may have the same image in X. An edge can therefore become a loop. By contrast, in a *simplicial complex* each simplex has distinct vertices in the space and any two simplices intersect in a common face or not at all. The torus model below has one vertex and two triangles. It is a Δ-complex, not a simplicial complex. Vertex names such as $[v_0, v_1, v_2]$ refer to ordered domain vertices even when some images coincide.

6.2 Chains and the boundary map

Fix a *coefficient ring* R. Our default is $R = \mathbb{Z}$. The alternatives used here are $\mathbb{Z}_2$, the field with arithmetic modulo two, and the real field $\mathbb{R}$. Integers form a ring, not a field.

Definition 6.3 (Chains)**.** A k-*chain* is a finite formal sum of the k-simplices with coefficients in R. These sums form the *chain group* $C_k(X; R)$.

Chains are added and multiplied by coefficients componentwise. Every chain has a unique expression in the simplex generators. When R is a field, they form a vector-space basis. Write $C_{-1} = 0$.

Definition 6.4 (Boundary map)**.** The *boundary map* $\partial_k\colon C_k(X;R) \to C_{k-1}(X;R)$ is the linear map defined on a simplex by alternating deletion:

$$\partial_k[v_0, \dots, v_k] = \sum_{i=0}^{k} (-1)^i [v_0, \dots, \widehat{v_i}, \dots, v_k] \quad (k \geq 1), \qquad \partial_0 = 0.$$

The hat means that the vertex is omitted. Each face is read as the designated simplex of the complex. This ∂ acts on chains. It is not a coordinate derivative.

For an ordered triangle, the formula is

$$\partial[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1].$$

The chain records the circuit from v_0 to v_1, then to v_2, and back to v_0. The negative sign reverses the contribution of the middle edge. An integer chain such as $2a - b$ records two copies of edge a and one oppositely oriented copy of edge b. A chain need not form a single continuous path. Over $\mathbb{Z}_2$ the negative sign equals a positive sign. Repeated occurrences of an edge still have to be added in the chosen ring. The drawing separates the ordered edges from their boundary circuit.

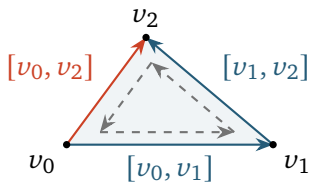


Figure 6.2: The ordered edges point from lower to higher vertex index. The dashed boundary circuit traverses the middle edge with a negative coefficient.

Theorem 6.5 (The boundary of a boundary)**.** *For every $k \geq 1$, the composition $\partial_{k-1}\partial_k$ is zero.*

Proof. For $k = 1$ the target boundary is $\partial_0 = 0$. For $k \geq 2$, fix indices $i < j$ in an ordered k-simplex. Deleting v_i first moves v_j to position $j - 1$. The resulting face has coefficient $(-1)^i(-1)^{j-1}$. Deleting v_j first leaves v_i in position i and gives coefficient $(-1)^j(-1)^i$. These are opposite signs. Every face with two deleted vertices occurs in exactly this pair. The pairs cancel before the face identifications, so they also cancel after those identifications. Linearity proves the assertion for every chain. ∎

For the triangle, applying the boundary once more gives

$$(v_2 - v_1) - (v_2 - v_0) + (v_1 - v_0) = 0.$$

Here a vertex name denotes its generator in C_0. For a one-cycle, zero boundary means that the signed endpoint contributions cancel. The boundary of a triangle has no remaining endpoints.

Definition 6.6 (Chain complex)**.** A sequence of groups linked by homomorphisms whose consecutive compositions vanish is called a *chain complex*.

6.3 Cycles, boundaries and coefficients

Definition 6.7 (Cycles)**.** A *cycle* is a chain with zero boundary. Its group is

$$Z_k(X; R) = \ker \partial_k.$$

Definition 6.8 (Boundaries)**.** A *boundary* is a chain that is the boundary of a chain one dimension higher. Its group is

$$B_k(X; R) = \operatorname{im} \partial_{k+1}.$$

The boundary-of-boundary theorem gives $B_k \subseteq Z_k$.

Definition 6.9 (Homology)**.** The *homology group* in degree k is

$$H_k(X; R) = Z_k(X; R)/B_k(X; R).$$

Two cycles are *homologous* when their difference is a boundary.

Homology records which cycles remain after this identification. The symbols Z_k and B_k denote groups, distinct from the coefficient ring $\mathbb{Z}$ and the braid group B_n of Lecture 5.

For example, take the boundary of a triangle, with no filled face. The sum z of its cyclically directed edges is a cycle. A general cycle has equal coefficients on those three directed edges, since its boundary vanishes at each vertex. There are no two-dimensional chains, so this cycle generates $H_1 \cong R$. Filling the triangle makes that same sum a boundary and gives $H_1 = 0$.

Since $\partial_0 = 0$, every zero-chain is a cycle. The boundary of an edge from v_i to v_j is $v_j - v_i$, so the endpoint classes become equal in H_0. Edge paths therefore identify all vertices in one component.

In a finite Δ-complex, these edge-path components are exactly the path components: each simplex joins its vertices by edges, and its points by paths to those vertices. Distinct edge-path components are disjoint finite unions of closed simplices, so they are also open.

Summing vertex coefficients separately in each component shows that there is no relation between different components. Thus H_0 has one independent generator for each path component. Over $\mathbb{Z}$ this is a free abelian group. Over a field it is a vector space.

The one-edge circle has zero edge boundary and no two-simplices, so its zeroth and first homology groups are both R. The triangle outline has the same answer. Filling changes its first homology but not its component count, as the following panels show.

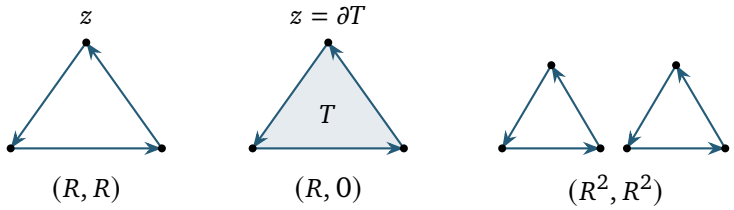


Figure 6.3: With $R \in \{\mathbb{Z}, \mathbb{Z}_2, \mathbb{R}\}$, the panels show $(H_0, H_1) = (R, R)$, $(R, 0)$ and (R^2, R^2). In the filled panel the cycle is the boundary of the face T.

One dimension higher, the boundary of a tetrahedron is a model of S^2. Its ordered vertices are $0, 1, 2, 3$. The oriented surface chain is

$$z_2 = [123] - [023] + [013] - [012].$$

The boundary-of-boundary theorem gives $\partial z_2 = 0$. Cancellation on shared edges forces every two-cycle to be a multiple of z_2. There are no three-simplices, so $H_2(S^2; R) \cong R$. Every one-cycle is a boundary: subtract boundaries of faces $[0ij]$ to remove all edges not incident to 0. The remaining cycle uses only edges from 0, whose other endpoints force their coefficients to vanish. We obtain

$$H_0(S^2; R) \cong R, \qquad H_1(S^2; R) = 0, \qquad H_2(S^2; R) \cong R.$$

For a point, only H_0 is nonzero. By the invariance theorem of §6.5, this distinguishes the sphere from a point even though their fundamental groups agree. Filling the tetrahedron makes z_2 a boundary. The drawing shows the surface before and after filling.

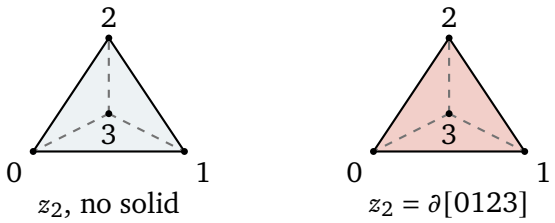


Figure 6.4: The same tetrahedral surface cycle z_2 is not a boundary in the surface complex, but is the boundary of $[0123]$ in the solid complex.

Homologous cycles need not coincide as chains. For example, triangulate a square annulus with every triangle counter-clockwise. Its outer and inner boundary circuits c_1 and c_0 can both be directed counter-clockwise. The sum B of its oriented triangles then satisfies

$$\partial B = c_1 - c_0.$$

The inner induced direction is clockwise, accounting for the minus sign. Thus c_1 and c_0 represent the same homology class. The shaded annulus displays this boundary relation.

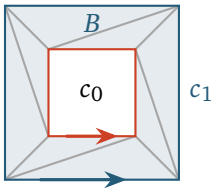


Figure 6.5: Both marked circuits run counter-clockwise. The oriented annulus has boundary $c_1 - c_0$, so their difference is a boundary.

For a finite complex, order the simplices of each dimension. The boundary matrix D_k has a column for each source k-simplex and a row for each target $(k-1)$-simplex. Its entries are the signed face coefficients. Write c_k for the number of k-simplices and r_k for the rank of D_k over a specified field. Rank–nullity and the dimension formula for a quotient, recalled from *Linear Algebra for Physics*, give

$$\dim H_k(X;R) = (c_k - r_k) - r_{k+1} \qquad \text{when } R \text{ is a field.}$$

Over $\mathbb{Z}$, integer relations determine an abelian-group quotient instead. Real row reduction alone does not determine that group.

Definition 6.10 (Torsion)**.** An element is a *torsion element* if it is nonzero and some positive integer multiple is zero.

For an algebraic example, take the chain complex with $C_2 = C_1 = \mathbb{Z}$, boundary ∂_2 given by multiplication by two, and $\partial_1 = 0$. Its first homology is $\mathbb{Z}/2\mathbb{Z} = \mathbb{Z}_2$. With real coefficients, multiplication by two is onto, so the first homology is zero. With $\mathbb{Z}_2$ coefficients it is the zero map, so the first homology is $\mathbb{Z}_2$. This illustrates why coefficients are part of the answer.

Definition 6.11 (Betti number)**.** For a finite complex, the *kth Betti number* means $b_k = \dim_{\mathbb{R}} H_k(X;\mathbb{R})$.

For $k \geq 1$ this counts independent real cycle classes modulo boundaries, often called k-dimensional holes. The number b_0 counts path components. Real coefficients discard torsion information.

6.4 The Euler characteristic

The alternating simplex count can be computed without reducing boundary matrices. We relate it to homology to explain why a different decomposition gives the same count.

Definition 6.12 (Euler characteristic)**.** For a finite Δ-complex of dimension d, its *Euler characteristic* is the alternating simplex count

$$\chi(X) = \sum_{k=0}^{d} (-1)^k c_k.$$

For a triangulated surface this is vertices minus edges plus triangles. The symbol χ denotes a topological count here, not the character of Lecture 5.

Theorem 6.13 (Euler characteristic and homology)**.** *For a finite Δ-complex and any field R, $\chi(X) = \sum_{k=0}^{d} (-1)^k \dim_R H_k(X; R)$. In particular, $\chi(X) = \sum_k (-1)^k b_k$.*

Proof. Set $r_0 = r_{d+1} = 0$. The dimension formula for homology gives $c_k = \dim H_k + r_k + r_{k+1}$. On summing with alternating signs, the rank terms cancel:

$$\sum_{k=0}^{d} (-1)^k c_k = \sum_{k=0}^{d} (-1)^k \dim H_k + \sum_{k=0}^{d} (-1)^k r_k + \sum_{k=0}^{d} (-1)^k r_{k+1} = \sum_{k=0}^{d} (-1)^k \dim H_k.$$

Indeed, the coefficient of each r_j with $1 \le j \le d$ is $(-1)^j + (-1)^{j-1} = 0$. The endpoint ranks vanish. ∎

For a single filled triangle, the counts are 3, 3 and 1, giving $\chi = 1$. Its homology has one degree-zero generator and no higher generators, which gives the same alternating sum.

6.5 Invariance and closed surfaces

The calculation uses a decomposition. Its independence of that choice requires a theorem, rather than a comparison of a few examples.

Theorem 6.14 (Homology invariance · *cited*)**.** *The homology of a Δ-complex agrees with its singular homology. Homotopy equivalences induce isomorphisms of these groups, with the coefficient ring fixed. Consequently the Euler characteristic of a finite Δ-complex is invariant under homotopy equivalence.*

The comparison is Hatcher, Theorem 2.27. Homotopy invariance is Corollary 2.11. Coefficients are treated in §2.2, pp. 153–155. Singular homology uses all continuous maps $\Delta^k \to X$ as generators, without the interior-injectivity requirement. That construction applies without choosing a decomposition. The Euler assertion follows from the theorem above, for example with real coefficients.

Definition 6.15 (Triangulation)**.** A *triangulation* is a presentation of a surface by triangles glued along edges.

Definition 6.16 (Orientability)**.** A triangulated surface is *orientable* if its triangles can be oriented so that their induced directions on every shared edge are opposite.

We use as part of the cited surface classification below that every closed surface admits such a presentation and that this condition is independent of the choice. The torus has this property. The projective plane fails the sign test in §6.6.

The Möbius band of Lecture 2 illustrates the obstruction. After rescaling the transverse coordinate to $t \in [-1, 1]$, its rectangle model identifies $(1, t)$ with $(0, -t)$. Orienting adjacent triangles consistently in the rectangle makes the induced directions opposite on their interior edges. At the seam, the reversal of t makes the induced directions agree instead. Reversing every triangle changes both seam directions and cannot fix this. The drawing compares an interior edge with this seam.

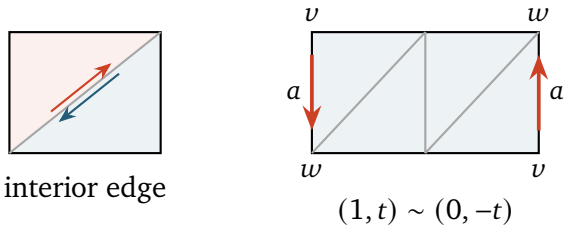


Figure 6.6: Counter-clockwise face orientations cancel on an interior edge. The seam $(1, t) \sim (0, -t)$ identifies the two marked boundary directions, so they agree on the Möbius band.

Theorem 6.17 (Closed surfaces · *cited*)**.** *Closed connected surfaces are classified, up to homeomorphism, by their orientability and Euler characteristic. For the orientable surface Σ_g of genus $g \geq 0$, the Euler characteristic is $2 - 2g$, where $\Sigma_0 = S^2$.*

This is the classification theorem of Kosniowski, Theorem 11.3, together with the Euler counts in Hatcher, §2.2, pp. 146–147. For $g \geq 1$, the polygon model of Lecture 5 has one vertex, $2g$ edges and one face. Using such polygonal cells instead of triangles gives the same Euler characteristic by Hatcher, Theorem 2.44. Thus its count is $1 - 2g + 1$.

There is also a direct connection to Lecture 5. For any path-connected space, $H_1(X; \mathbb{Z})$ is the *abelianization* of $\pi_1(X)$: the quotient by the normal subgroup generated by all commutators $aba^{-1}b^{-1}$. This result is cited from Hatcher, Theorem 2A.1. Homology therefore forgets the order information in noncommuting loop classes. It need not recover the fundamental group.

6.6 Boundary matrices for two surfaces

We compute the torus over $\mathbb{Z}$ and then $\mathbb{Z}_2$. Take a square with corners P, Q, R', S counter-clockwise from the bottom left. The prime distinguishes the corner R' from the coefficient ring R. Pair the horizontal edges to the right and the vertical edges upward. All corners become one vertex v. Call the horizontal edge a, the vertical edge b and the diagonal $P \to R'$ edge c. The ordered triangles are $T_0 = [P, Q, R']$ and $T_1 = [P, S, R']$.

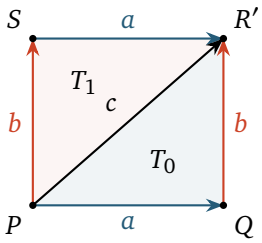


Figure 6.7: The torus has one vertex, three edges and two triangles. The upper triangle's vertex order has the opposite planar orientation to the lower one's.

Example 6.18 (The torus)**.** Both triangle boundaries are $a + b - c$. Each edge has zero boundary, since both endpoints become v. In the bases (a, b, c) and (T_0, T_1), the boundary matrices over $\mathbb{Z}$ are

$$D_1 = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}.$$

$$D_2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ -1 & -1 \end{pmatrix}.$$

Every one-chain is a cycle. The boundaries are integer multiples of $a + b - c$. The list $(a, b, a + b - c)$ is an integer basis of C_1: its change-of-basis determinant is -1, so its inverse has integer entries. Quotienting by its last generator leaves a and b free. The kernel of D_2 consists of pairs $(m, -m)$, and there are no three-chains. The zero-dimensional quotient retains the single vertex. We have therefore obtained

$$H_0(T^2; \mathbb{Z}) = \mathbb{Z}, \qquad H_1(T^2; \mathbb{Z}) = \mathbb{Z}^2, \qquad H_2(T^2; \mathbb{Z}) = \mathbb{Z}.$$

The top generator is $T_0 - T_1$. Its signs give both triangles the same planar orientation, and their edge contributions cancel.

With $\mathbb{Z}_2$ coefficients, every entry -1 becomes 1. The ranks of D_1 and D_2 are zero and one. The homology dimensions are therefore 1, $3 - 0 - 1 = 2$ and $2 - 1 = 1$. The classes of a and b still generate H_1, and the top cycle is $T_0 + T_1$. Both coefficient choices give $\chi(T^2) = 1 - 3 + 2 = 0$.

For the projective plane, pair opposite square edges with opposite arrows. The corners P and R' become v, while Q and S become w. The edge a runs from P to S and from R' to Q. The edge b runs from P to Q and from R' to S. The diagonal c runs from P to R'.

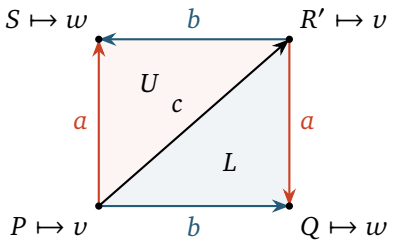


Figure 6.8: The diagonal c is a loop at v in this two-vertex model of the projective plane. Both boundary-edge generators join v to w.

Example 6.19 (The projective plane over $\mathbb{Z}_2$)**.** Use the vertex basis (v, w), edge basis (a, b, c) and ordered triangles $U = [P, R', S]$ and $L = [P, R', Q]$. Work over $\mathbb{Z}_2$ throughout. Both edges a and b have boundary $v+w$, and c has zero boundary. Both triangles have boundary $a+b+c$. Thus

$$D_1 = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}.$$

$$D_2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Both matrices have rank one. The kernel of D_1 is generated by $a + b$ and c. Its boundary subspace is generated by $a + b + c$, so the quotient is one-dimensional, represented by c. The kernel of D_2 is generated by $U + L$. The image of D_1 identifies the two vertex classes in H_0. Consequently $H_k(\mathbb{R}\mathrm{P}^2; \mathbb{Z}_2) \cong \mathbb{Z}_2$ for $k = 0, 1, 2$, and $\chi(\mathbb{R}\mathrm{P}^2) = 2 - 3 + 2 = 1 = 1 - 1 + 1$. These are binary homology groups. They do not assert the same answer with integer or real coefficients.

We now use integer coefficients with the same arrows and ordered triangles. Their boundaries are

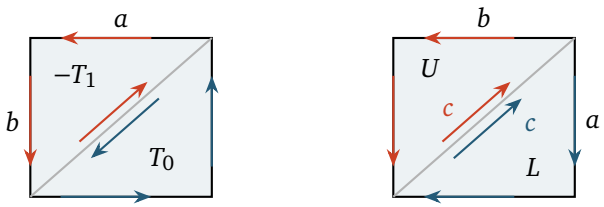
$$\partial a = \partial b = w - v, \qquad \partial c = 0, \qquad \partial U = b - a + c, \qquad \partial L = a - b + c.$$

Put $x = a - b$. The integer one-cycles have basis (x, c). The boundaries are generated by $c - x$ and $c + x$. The first relation identifies x with c. The second then becomes $2c = 0$. The class of c is nonzero: solving $m(c - x) + n(c + x) = c$ forces $m = n$ and $2m = 1$, which has no integer solution. Thus

$$H_0(\mathbb{R}\mathrm{P}^2; \mathbb{Z}) \cong \mathbb{Z}, \qquad H_1(\mathbb{R}\mathrm{P}^2; \mathbb{Z}) \cong \mathbb{Z}_2, \qquad H_2(\mathbb{R}\mathrm{P}^2; \mathbb{Z}) = 0.$$

For the last assertion, a two-cycle $mU + nL$ would require $m = n$ at a and $m = -n$ at c, hence $m = n = 0$. The relation $\partial(U + L) = 2c$ gives a geometric example of torsion. The diagonal joins antipodal boundary points P and R' in the square model. We identify that square with a hemisphere, as in Lecture 2. The corresponding lift to S^2 has distinct antipodal endpoints. By the covering argument of Lecture 4, the diagonal represents the nontrivial fundamental-group class of $\mathbb{R}\mathrm{P}^2$. The fundamental group $\mathbb{Z}_2$ is already abelian, agreeing with the abelianization result in §6.5. Over $\mathbb{R}$, the relations $x = c$ and $2c = 0$ instead give $c = 0$. The real Betti numbers are therefore $(b_0, b_1, b_2) = (1, 0, 0)$.

For an orientation test, the coefficients m and n must each be $+1$ or -1. The conditions $m = n$ and $m = -n$ cannot both hold. Over $\mathbb{Z}_2$, their distinction disappears and $U + L$ is a top cycle. The comparison with the torus shows precisely which edge contributions fail.



$\partial(T_0 - T_1) = 0$ $\quad$ $\partial(U + L) = 2c$, $2c = 0$ over $\mathbb{Z}_2$

Figure 6.9: The torus cycle $T_0 - T_1$ cancels every edge contribution. For $U + L$ on $\mathbb{R}P^2$, the a and b contributions cancel but the two copies of c add to $2c$, which vanishes only after reduction modulo two.

The classification theorem and these computations give the following comparison.

Surface	Orientable	χ
S^2	yes	2
T^2	yes	0
Σ_2	yes	−2
$\mathbb{R}P^2$	no	1
Klein bottle	no	0

The Klein bottle is the square quotient with horizontal edges paired in the same direction and vertical edges paired with reversal. Its reversed seam fails the same orientation test as the Möbius band. Its polygonal cells give $\chi = 1 - 2 + 1 = 0$ by the cited cell-count theorem. Thus the Klein bottle and torus have the same Euler characteristic. Their different orientability explains why the classification needs both data.

Self-check. Compute the Euler characteristic of the sphere from the boundary of a tetrahedron and from the boundary of an octahedron. Confirm the tetrahedral answer from the homology of its boundary.

Answer. The tetrahedron has four vertices. Every pair is an edge, giving $\binom{4}{2} = 6$ edges, and every triple is a face, giving four triangles. The count is $4 - 6 + 4 = 2$. The octahedron has vertices at the positive and negative coordinate-axis unit vectors. Every vertex joins the four vertices other than itself and its opposite, giving $6 \cdot 4/2 = 12$ edges. A face chooses one endpoint from each axis, giving $2^3 = 8$ triangles. Its count is $6 - 12 + 8 = 2$, in agreement with the tetrahedron. The tetrahedral boundary has real Betti numbers $(1, 0, 1)$, so its homological alternating sum is also $1 - 0 + 1 = 2$.

Self-check. Consider the chain complex $C_2 = \mathbb{Z}$, $C_1 = \mathbb{Z}^3$ and $C_0 = \mathbb{Z}^3$, with bases (T), (a, b, c) and (v_0, v_1, v_2). Its only nonzero boundary matrices are

$$D_1 = \begin{pmatrix} -1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \qquad D_2 = \begin{pmatrix} 5 \\ 5 \\ -5 \end{pmatrix}.$$

Compute $\partial(2a - b + 3c)$ and H_1 over $\mathbb{Z}$ and over $\mathbb{R}$.

Answer. The chain has boundary $-5v_0+3v_1+2v_2$, so it is not a cycle. Solving $D_1(u, v, w)^{\mathsf{T}} = 0$ gives $u = v = -w$. Hence the cycles are generated by $z = a + b - c$, and $\partial T = 5z$. The quotient is $H_1 \cong \mathbb{Z}/5\mathbb{Z}$ over $\mathbb{Z}$. Over $\mathbb{R}$, multiplication by five is onto, so $H_1 = 0$. The same real rank does not determine the integer quotient.

6.7 Binary cycles and the toric code

We use homology to distinguish strings that leave stored information unchanged from strings that can change it. The model is Kitaev's toric code (arXiv:quant-ph/9707021v1, §1, equations (1)–(2)).

Physical input. In the ideal finite-qubit model, normalized vectors in a complex tensor product describe pure states. A qubit has orthonormal basis $|0\rangle$ and $|1\rangle$. Measuring a Hermitian operator on an eigenstate returns its eigenvalue. We interpret the joint $+1$ eigenspace defined below as the states used to store quantum information, following Kitaev, §1.

The phase-flip operator Z and bit-flip operator X are given by

$$Z|0\rangle = |0\rangle, \qquad Z|1\rangle = -|1\rangle, \qquad X|0\rangle = |1\rangle, \qquad X|1\rangle = |0\rangle.$$

Thus $X^2 = Z^2 = 1$ and $XZ = -ZX$, as follows by applying them to both basis vectors. The operator Z is distinct from the cycle group Z_k.

Definition 6.20 (Toric code)**.** Take an $N \times N$ square lattice with opposite sides identified, where $N \geq 3$. Put one qubit on each edge. The joint state space is

$$\mathcal{H} = (\mathbb{C}^2)^{\otimes 2N^2}.$$

The notation X_e and Z_e means that the operator acts on edge e and as the identity on every other factor. For each vertex v and each square face, called a *plaquette* p, define the *stabilizer operators*

$$A_v = \prod_{e \ni v} X_e, \qquad B_p = \prod_{e \in \partial p} Z_e.$$

The *toric code subspace* is their joint $+1$ eigenspace

$$C = \{\psi \in \mathcal{H} : A_v\psi = \psi \text{ for every } v,\ B_p\psi = \psi \text{ for every } p\}.$$

All stars A_v commute with one another. All plaquettes B_p also commute. A star and a plaquette share zero or two edges. Every shared edge contributes one minus sign from $XZ = -ZX$, so their even overlap makes $A_vB_p = B_pA_v$.

Definition 6.21 (Z-strings)**.** For a binary one-chain $c = \sum_e c_e e$, the *Z-string* is

$$Z(c) = \prod_e Z_e^{c_e}, \qquad c_e \in \mathbb{Z}_2.$$

Definition 6.22 (Logical operators)**.** A *logical operator* preserves the code subspace and acts on the states stored there.

Adding binary chains toggles selected edges, since $Z_e^2 = 1$. At a vertex v, commuting A_v past the string gives

$$A_vZ(c) = (-1)^{(\partial c)_v} Z(c)A_v.$$

Here $(\partial c)_v$ is represented by 0 or 1. For a code state ψ, the state $Z(c)\psi$ has star eigenvalue -1 exactly at vertices with odd incident string parity. The binary record of violated conditions is the *syndrome* of a Z-type error. It is precisely ∂c. Along a simple open string only the two ends violate star conditions. Every Z-string commutes with every B_p. Consequently strings with $\partial c = 0$ preserve C.

A face-boundary string is $Z(\partial p) = B_p$, which acts as the identity on C. Products of these operators show that any boundary acts as the identity. Homologous closed strings therefore act alike on the code. The drawing separates the local stabilizers from open and closed strings.

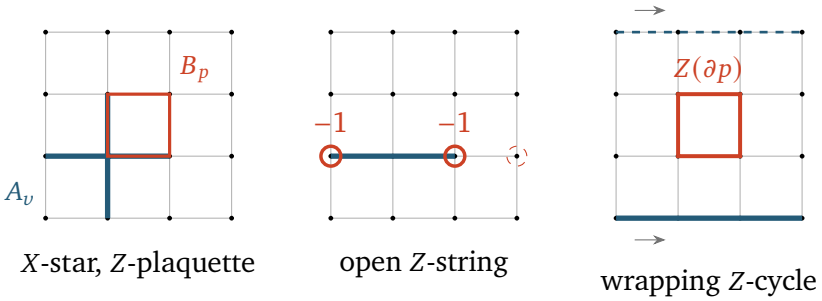


Figure 6.10: The code conditions are $A_v = B_p = +1$, with dashed marks repeating periodic copies. An open Z-string violates stars at its two ends, a plaquette boundary acts identically on the code, and a wrapping string has no endpoint syndrome.

The periodic cycles and a plaquette change can also be drawn on the full fundamental square.

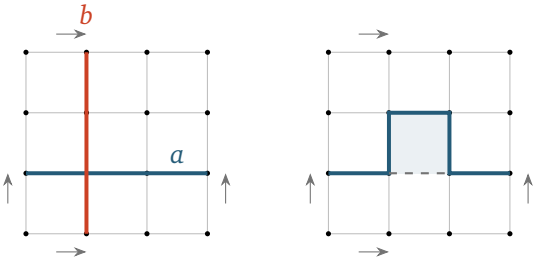


Figure 6.11: On the periodic square with opposite sides identified, the first panel shows the two wrapping cycles. The second panel adds one plaquette boundary, with the dashed edge removed.

The lattice uses square faces, whereas our homology was defined using simplices. We compare the two quotients by adding diagonals. We draw every square's diagonal from lower left to upper right and order its triangles as in the torus example. A triangle boundary replaces each added diagonal in a cycle by its other two edges. Conversely, each diagonal occurs in exactly the two triangles of its square. A binary sum of triangle boundaries with no diagonal term therefore uses the two triangles of each square with equal coefficients. The two triangle boundaries add to the square boundary, so the sum is a sum of square boundaries. This proves that square-lattice cycles modulo square boundaries agree with the homology of this triangulation. By the cited Theorem 6.14, its homology agrees with that of the two-triangle torus in Example 6.18.

Example 6.23 (Toric-code string classes · *cited; two-triangle homology proved*)**.** The binary string classes form $H_1(T^2;\mathbb{Z}_2) \cong \mathbb{Z}_2^2$. Representatives wind once in the horizontal and vertical directions. Every class is a sum of a subset of these two generator classes, including the empty sum. There are therefore four classes. The square-to-triangle comparison was proved above. Its comparison with the two-triangle torus uses the cited homology-invariance

theorem. The identification of these lattice representatives with logical Z operators is Kitaev's construction, with the parity calculation given here.

We can also derive the dimension of C. There are N^2 stars and N^2 plaquettes, with relations

$$\prod_v A_v = 1, \qquad \prod_p B_p = 1.$$

Each edge occurs twice in each product. These are the only relations within each family. A product of selected stars is the identity only if the selection agrees at both ends of every edge. The vertex graph is connected, so either every vertex or no vertex is selected.

The same argument on the connected dual graph, whose vertices are faces and whose edges join neighbouring faces, applies to plaquettes. A product of X operators and a product of Z operators can multiply to a scalar identity only when both supports are empty. Hence there are $r = 2N^2 - 2$ independent conditions, with no relation giving -1.

For independent commuting stabilizers $S_1, \dots, S_r$, the projector onto their joint $+1$ eigenspace is

$$P = 2^{-r} \prod_{j=1}^{r} (1 + S_j).$$

Each factor $(1 + S_j)/2$ is the projector onto its $+1$ eigenspace. Every nonidentity product in the expansion has, on some edge, a factor X, Z or XZ, up to a sign. Each of these has trace zero. The trace of a tensor product is the product of its factor traces. Only the identity contributes to the trace. Thus

$$\dim C = \operatorname{tr} P = 2^{2N^2 - r} = 4.$$

This is the state-space dimension of two logical qubits. The complementary logical operators come from X-strings on the dual lattice, as in Kitaev, §1. A closed dual string applies X on the primal edges it crosses. It meets each plaquette boundary evenly, so it preserves C. For each of the horizontal and vertical wrapping Z-strings, we choose a closed dual string crossing it once. The two operators anticommute, so the wrapping Z-string cannot act as the identity on C.

The distinction matters for errors in this ideal model. A Z-type error supported on fewer than N edges either has a nonzero syndrome or is a boundary and acts identically on the code. For the length bound, decompose a binary cycle into closed edge walks. A walk with nonzero winding lifts to a lattice walk whose endpoint displacement has a coordinate of magnitude at least N. It therefore needs at least N edges. A cycle of smaller total length has no wrapping component and is a boundary. This is the Z-string part of Kitaev's error-detection argument in §1. The interpretation as a quantum code and the complementary operators remain those of the cited construction.

Self-check. State in one sentence what $H_k(X; R)$ is and why it was introduced. State in a second sentence what the toric code subspace is and why it was introduced.

Answer. The group $H_k(X; R)$ consists of k-cycles modulo boundaries, so it distinguishes closed chains that cannot be filled within X. The toric code subspace is the joint $+1$ eigenspace of all A_v and B_p, introduced to store states on which boundary strings act identically while nontrivial wrapping strings can act nontrivially.

Summary. The boundary map has square zero. Homology is the quotient of cycles by boundaries, with the coefficient ring specified. Over a field, boundary-matrix ranks give its dimensions and their alternating sum is the Euler characteristic. Invariance and surface classification are cited. The torus has two independent binary one-cycle classes, which describe the two logical Z-string classes in the toric code. Lecture 7 adds smooth coordinates to spaces so that calculus can be defined on them.

Reading. Hatcher, *Algebraic Topology*, §2.1, pp. 102–107, Theorems 2.27 and 2A.1, and §2.2, pp. 146–147; Nakahara, *Geometry, Topology and Physics*, §3.3; Kosniowski, *A First Course in Algebraic Topology*, Theorem 11.3 and Chapter 29; Kitaev, *Fault-tolerant quantum computation by anyons* (2003); section numbers here refer to arXiv:quant-ph/9707021v1, §1.

Lecture 7

Smooth manifolds

Objectives. *A smooth manifold has local coordinates whose changes are smooth. This allows derivatives to be compared between coordinate systems. We construct coordinates on spheres and projective spaces, then prove that regular constraints define manifolds using the implicit function theorem. The space of pure qubit states gives a concrete example: a sphere described by two complex coordinate charts.*

By the end of this lecture you will be able to:

- Check chart transitions and construct a smooth atlas.
- Test smooth maps in coordinates and apply the chain rule.
- Prove that a nonempty regular level set is an embedded submanifold.
- Relate the two projective qubit coordinates by $w = 1/z$.

Unit 7A Charts and the complex projective line

7.1 Charts and smooth structures

Definition 7.1 (Topological manifold). A *topological n-manifold* is a Hausdorff, second-countable space M locally homeomorphic to open subsets of $\mathbb{R}^n$. Here second-countable means that its topology has a countable basis.

We consider manifolds without boundary.

Definition 7.2 (Chart). A *chart* is a pair (U, φ), with $U \subseteq M$ open and $\varphi : U \to \varphi(U) \subseteq \mathbb{R}^n$ a homeomorphism onto an open subset. The component functions of the chart are its *coordinates*, written $x^1, \dots, x^n$. The superscripts are coordinate labels, not powers.

By Lecture 1, equivalent norms give the same open sets. For $n \geq 1$, the Euclidean and maximum norms on $\mathbb{R}^n$ are equivalent. The sum of the n squared coordinates is at least the largest one and at most n times it. Taking square roots gives

$$\|v\|_\infty \leq \|v\|_2 \leq \sqrt{n}\,\|v\|_\infty \qquad (v \in \mathbb{R}^n).$$

Thus the meaning of an open chart image does not depend on which of these norms is used.

On the right semicircle, the angle θ and the vertical coordinate y describe the same point by

$$(\cos\theta, \sin\theta), \qquad -\pi/2 < \theta < \pi/2, \qquad y = \sin\theta.$$

The transition has the smooth inverse $\theta = \arcsin y$ on $(-1, 1)$. For the scalar function $f(\cos\theta, \sin\theta) = \theta$, the angle expression has derivative 1. Its expression in the vertical

coordinate is $\arcsin y$, whose derivative is $1/\sqrt{1-y^2}$. The chain rule relates these derivatives because $\mathrm{d}y/\mathrm{d}\theta = \cos\theta$. The marked point in the drawing has angle $\pi/6$ and vertical coordinate $1/2$.

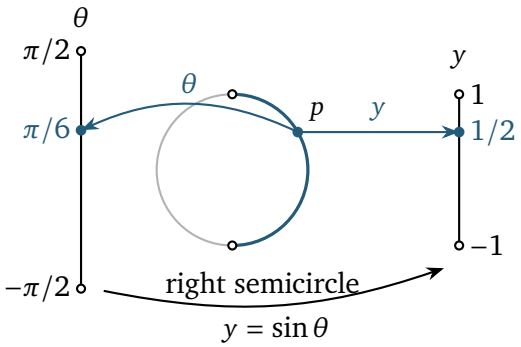


Figure 7.1: The same point has coordinates $\pi/6$ and $1/2$. The transition $y = \sin\theta$ compares coordinates on open intervals.

For a function $f\colon M \to \mathbb{R}$, its expression in a chart is $f \circ \varphi^{-1}$. On a second chart the expression is

$$f \circ \psi^{-1} = (f \circ \varphi^{-1}) \circ (\varphi \circ \psi^{-1}).$$

Smooth transitions make smoothness of the function independent of its coordinate description, by the Euclidean chain rule.

Definition 7.3 (Compatible charts). Two charts are *smoothly compatible* when both coordinate changes on their overlap are smooth. For charts (U, φ) and (V, ψ), one such change is

$$\psi \circ \varphi^{-1}\colon \varphi(U \cap V) \longrightarrow \psi(U \cap V).$$

Smooth means C^∞: all orders of partial derivatives exist and are continuous. An empty overlap imposes no condition.

The charts describe the same points of $U \cap V$. The transition compares their descriptions between open subsets of $\mathbb{R}^n$. The drawing uses $M = \mathbb{R}^2$, the rectangles $U = (-2, 1) \times (-1, 1)$ and $V = (-1, 2) \times (-1, 1)$, and charts $\varphi(x, y) = (x, y)$ and $\psi(x, y) = (x + 2, y)$.

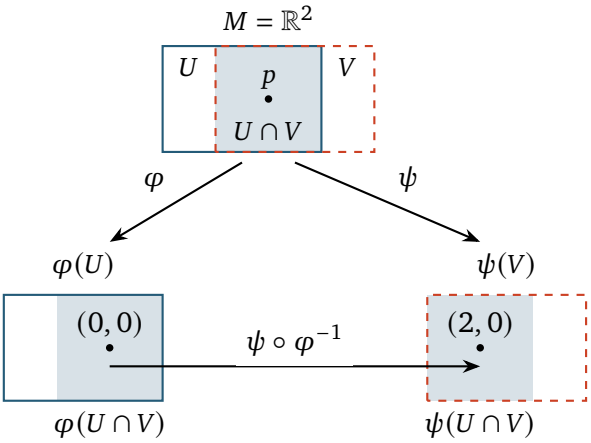


Figure 7.2: Only the shaded overlap images are the source and target of $\psi \circ \varphi^{-1}$. The marked point $p = (0, 0)$ has coordinates $(0, 0)$ and $(2, 0)$.

Definition 7.4 (Smooth atlas)**.** A *smooth atlas* is a collection of pairwise compatible charts covering M.

Example 7.5 (An angle atlas on the circle)**.** Take $U = S^1 \setminus \{(1,0)\}$ with angle $\theta \in (0, 2\pi)$. Take $V = S^1 \setminus \{(-1,0)\}$ with angle $\theta' \in (-\pi, \pi)$. The overlap has an upper arc and a lower arc. On these arcs the transition is respectively

$$\theta' = \theta \quad (0 < \theta < \pi), \qquad \theta' = \theta - 2\pi \quad (\pi < \theta < 2\pi).$$

Both transitions and their inverses are smooth, so these charts form an atlas. A single chart cannot cover the circle. Its image would be a nonempty compact open subset of $\mathbb{R}$. Compactness makes that image closed and bounded. Connectedness of $\mathbb{R}$ would then make the image all of $\mathbb{R}$, contradicting boundedness.

Definition 7.6 (Smooth structure)**.** A *smooth structure* is a maximal smooth atlas, containing every chart compatible with it.

Definition 7.7 (Smooth manifold)**.** A *smooth manifold* is a topological manifold with a smooth structure.

A covering atlas specifies which coordinate descriptions are compatible. Its maximal extension includes every chart compatible with that choice.

Any smooth atlas determines a unique smooth structure. To prove this, adjoin every chart compatible with the given atlas. For two added charts ψ and θ and a point in their overlap, choose an original chart φ containing that point. Locally the transition is

$$\theta \circ \psi^{-1} = (\theta \circ \varphi^{-1}) \circ (\varphi \circ \psi^{-1}).$$

Both factors are smooth. Reversing the roles gives a smooth inverse. Thus the enlarged collection is an atlas. Any chart compatible with the enlarged collection was compatible with the original one and has already been added. This proves maximality. Every chart of any maximal atlas containing the original atlas belongs to this enlarged collection. That maximal atlas must therefore equal the enlarged collection, proving uniqueness.

The smooth structure is extra data. On the topological line, the global charts $x(t) = t$ and $y(t) = t^3$ individually define smooth structures. Their transition $y \circ x^{-1}(s) = s^3$ is smooth, but its inverse is not differentiable at zero. Along $s = 1/k^3$, its difference quotient is $(1/k)/(1/k^3) = k^2$, which is unbounded as $k \to \infty$. Hence the two charts do not belong to one smooth atlas. The function $g(t) = t$ has expression $g \circ x^{-1}(s) = s$ in the first structure and $g \circ y^{-1}(s) = \sqrt[3]{s}$ in the second. Thus the two structures give different sets of smooth real-valued functions. We compare these structures by a smooth map in §7.3.

Local Euclidean coordinates alone do not imply Hausdorffness. Identify two copies of $\mathbb{R}$ at every nonzero coordinate, leaving two origins. Each origin has an interval chart, but any two neighbourhoods of the origins intersect. The sequence with coordinates $1/k$, for positive integers k, converges to both origins. Hausdorffness excludes this failure of uniqueness of limits. The drawing separates the two copies vertically only to distinguish their origins.

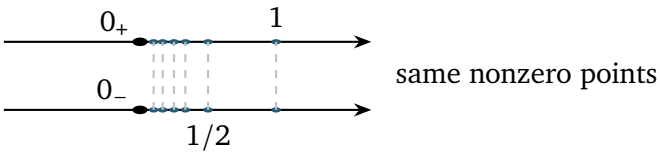


Figure 7.3: Nonzero points in the two rows are identified. Both origins are limits of the same positive sequence, so the quotient is not Hausdorff.

An uncountable disjoint union of lines is locally Euclidean and Hausdorff. It is not second-countable: each component would need a different nonempty basis element contained in it. Second countability supplies the countable chart constructions used for partitions of unity and integration in Lecture 12.

A smooth invertible transition also fixes the dimension. By the chain rule, its derivative has a two-sided inverse, so its source and target have the same dimension.

7.2 Spheres, projective spaces and products

Write a point of the unit sphere $S^n \subset \mathbb{R}^{n+1}$ as (y, t), where $y \in \mathbb{R}^n$ and $t \in \mathbb{R}$. Let $N = (0, 1)$ and $S = (0, -1)$ be its north and south poles. Stereographic projection gives charts

$$\varphi_N(y,t) = \frac{y}{1-t}, \qquad \varphi_S(y,t) = \frac{y}{1+t}$$

on $S^n \setminus \{N\}$ and $S^n \setminus \{S\}$ respectively. Both have image $\mathbb{R}^n$. The projection is shown in a two-dimensional slice. The transition $u \mapsto u/|u|^2$ is inversion in the unit sphere: the two coordinates shown have lengths 2 and $1/2$.

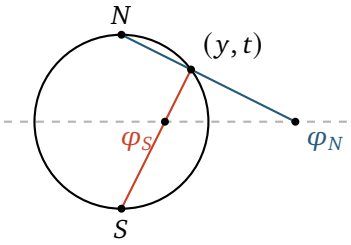


Figure 7.4: The ray from the omitted pole meets the equatorial coordinate plane. The two projections describe the same sphere away from their respective poles.

A line through zero in $\mathbb{R}^3$ with $X^0 \neq 0$ meets the plane $X^0 = 1$ at exactly one point. Its representative there is $(1, u^1, u^2)$, where $(u^1, u^2) = (X^1/X^0, X^2/X^0)$. The drawing shows these ratios as affine coordinates of the intersection.

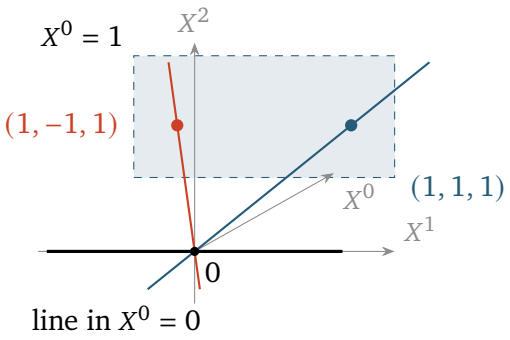


Figure 7.5: Two lines meet the chart plane $X^0 = 1$. The line in $X^0 = 0$ misses that plane and is outside this chart.

On the overlap with $X^1 \neq 0$, we divide the representative by u^1:

$$(1, u^1, u^2) \sim (1/u^1, 1, u^2/u^1), \qquad \varphi_1 \circ \varphi_0^{-1}(u^1, u^2) = (1/u^1, u^2/u^1), \quad u^1 \neq 0.$$

This transition and its inverse are smooth rational functions. For $\mathbb{R}\mathrm{P}^1$ the transition is $u \mapsto 1/u$. This is also the stereographic transition on S^1.

Proposition 7.8 (Standard atlases). *For $n \geq 1$, the sphere S^n and real projective space $\mathbb{R}\mathrm{P}^n$ are smooth n-manifolds. Products of smooth manifolds have smooth product atlases. In particular, $T^2 = S^1 \times S^1$ is a smooth surface.*

Proof. First, the inverse north chart is

$$\varphi_N^{-1}(u) = \left(\frac{2u}{1 + |u|^2}, \frac{|u|^2 - 1}{1 + |u|^2} \right).$$

Its squared norm is one. Substitution in $y/(1 - t)$ gives u. Conversely, for $(y, t) \neq N$, the relation $|y|^2 = 1 - t^2$ gives $|y/(1 - t)|^2 = (1 + t)/(1 - t)$. Substituting this into the displayed formula returns (y, t). Thus it is the continuous inverse. The inverse south chart changes the sign of the last component. On the overlap, substitution gives $\varphi_S \circ \varphi_N^{-1}(u) = u/|u|^2$ for $u \neq 0$. This map is smooth and is its own inverse. The sphere is Hausdorff and second-countable as a subspace of Euclidean space. These charts therefore define its smooth structure.

Next, use Lecture 2's quotient $q \colon S^n \to \mathbb{R}\mathrm{P}^n$ by antipodes. The map q is open: the inverse image of the image of an open set W is $W \cup (-W)$. Distinct two-point orbits have disjoint small antipodally invariant neighbourhoods, proving Hausdorffness of the quotient. Images of a countable sphere basis give a countable quotient basis.

Write a class of unit vectors as $[X^0 : \cdots : X^n]$. A nonzero vector represents the class of its normalisation. Multiplying it by a nonzero scalar leaves the same antipodal class, so the notation also describes its line through zero. The open patch U_i consists of classes with $X^i \neq 0$. Its coordinates and inverse are

$$\varphi_i([X]) = (X^0/X^i, \ldots, \widehat{X^i/X^i}, \ldots, X^n/X^i), \qquad \varphi_i^{-1}(u) = [u^0 : \cdots : 1 : \cdots : u^n],$$

with the 1 in position i. The remaining coordinate labels omit i. The hat in this display means that the ith entry is omitted. The ratios are invariant under antipodes and continuous upstairs. Since q is open, its restriction over U_i is a quotient map, so the ratios are continuous on U_i. For the inverse, normalise the displayed representative vector and apply q. These continuous maps are inverse: normalising does not change ratios, and the ratios recover

the original vector up to a nonzero scalar. On $U_i \cap U_j$, divide every component of the representative with ith component 1 by its nonzero jth component, then omit component j. All new coordinates are smooth rational functions with nonzero denominator. The same description with i and j exchanged gives the inverse transition.

The $n+1$ patches cover, completing the projective-space assertion.

Finally, take products of charts. Their coordinate changes are products of smooth maps, so they are smooth. Finite products preserve Hausdorffness and second countability: products of the respective basis elements give a countable basis. This proves the product assertion and the torus case. ∎

7.3 Smooth maps and the chain rule

Let M and N be smooth manifolds of dimensions n and m.

Definition 7.9 (Smooth map)**.** A continuous map $f\colon M \to N$ is *smooth* if, near every $p \in M$, its coordinate expression is smooth. For charts φ near p and ψ near $f(p)$, this expression is

$$\widehat{f} = \psi \circ f \circ \varphi^{-1}\colon \varphi(U \cap f^{-1}(V)) \longrightarrow \psi(V).$$

The domain is open because f is continuous.

Definition 7.10 (Diffeomorphism)**.** A *diffeomorphism* is a bijective smooth map with a smooth inverse.

We use the following index convention for coordinate calculus. A repeated upper and lower index is summed from 1 to n, unless another range is stated. For a linear functional on $\mathbb{R}^n$ with coefficient row a_i and a vector in $\mathbb{R}^n$ with component column X^i, write $a_i X^i = \sum_{i=1}^{n} a_i X^i$. The index position distinguishes transformation rules. It does not mean that a metric has been used. For example, a linear coordinate change and the corresponding rules are

$$J = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}, \qquad x' = Jx, \qquad X' = JX, \qquad a' = aJ^{-1}, \qquad a'X' = aX.$$

The last equality follows by cancelling $J^{-1}J$. Lower coordinate labels x_i in some prerequisite notes become x^i here. This relabelling does not raise the index of a linear functional.

Proposition 7.11 (Independence of coordinates)**.** *Smoothness is independent of the chosen charts. Compositions of smooth maps are smooth. Their coordinate Jacobians obey the chain rule.*

Proof. With different charts $\widetilde{\varphi}$ and $\widetilde{\psi}$, the new expression, on its common local domain, is

$$\widetilde{\psi} \circ f \circ \widetilde{\varphi}^{-1} = (\widetilde{\psi} \circ \psi^{-1}) \circ \widehat{f} \circ (\varphi \circ \widetilde{\varphi}^{-1}).$$

The outer maps are smooth transitions. This proves independence at each point. For a further smooth map $g\colon N \to P$, use the same middle chart on N. The coordinate expression of $g \circ f$ is $\widehat{g} \circ \widehat{f}$, and the Euclidean chain rule gives

$$D(\widehat{g} \circ \widehat{f})_x = D\widehat{g}_{\widehat{f}(x)}\, D\widehat{f}_x.$$

Both smoothness of the composition and the Jacobian formula follow. ∎

The hat in $\widehat{f}$ denotes a coordinate expression of a map. Here $D\widehat{f}_x$ denotes the $m \times n$ matrix of first partial derivatives. In component notation, the same formula is

$$\frac{\partial (g \circ f)^a}{\partial x^i} = \frac{\partial g^a}{\partial y^b} \frac{\partial f^b}{\partial x^i},$$

with the repeated index b summed. These are coordinate expressions; Lecture 8 defines the differential without choosing coordinates.

Example 7.12 (Testing a height function)**.** Consider $h\colon S^n \to \mathbb{R}$ given by $h(y,t) = t$. Its expressions in the stereographic charts are

$$h \circ \varphi_N^{-1}(u) = \frac{|u|^2 - 1}{1 + |u|^2}, \qquad h \circ \varphi_S^{-1}(u) = \frac{1 - |u|^2}{1 + |u|^2}.$$

Both are smooth on $\mathbb{R}^n$, since their denominators are positive. The charts cover the sphere, so h is smooth everywhere.

For the two smooth structures on the line above, the map $f(t) = \sqrt[3]{t}$ from the x-structure to the y-structure has coordinate expression $y \circ f \circ x^{-1}(s) = s$. Its inverse has the identity coordinate expression too. Thus it is a diffeomorphism, although the identity map between those structures is not a diffeomorphism.

7.4 The qubit's projective state space

Physical input. A two-level system has orthonormal reference states $|0\rangle$ and $|1\rangle$. A normalised column $\psi = (a, b)^T$ represents $a\,|0\rangle + b\,|1\rangle$, with $|a|^2 + |b|^2 = 1$. Measurement in this basis has probabilities $|a|^2$ and $|b|^2$. For a Hermitian matrix A, the expectation in any nonzero column ψ is

$$\langle A \rangle_\psi = \frac{\psi^\dagger A \psi}{\psi^\dagger \psi}.$$

Pure states are identified under a common phase. The Pauli matrices in the reference basis are

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

For a spin-1/2 system the physical spin components are $S_a = (\hbar/2)\sigma_a$, for $a \in \{x, y, z\}$.

A common nonzero scalar cancels between the numerator and denominator of every expectation. Normalisation removes its magnitude, leaving the common phase. Relative phase can affect expectations in other bases.

Definition 7.13 (Complex projective line)**.** The *complex projective line* $\mathbb{C}\mathrm{P}^1$ consists of nonzero columns $(a, b) \in \mathbb{C}^2$ modulo multiplication by a common nonzero complex scalar. Its points are written $[a : b]$. It carries the quotient topology from $\mathbb{C}^2 \setminus \{0\}$.

This is the construction of $\mathbb{R}\mathrm{P}^1$ with $\mathbb{C}$ in place of $\mathbb{R}$. The physical pure-state space is this quotient. The patches with $a \neq 0$ and $b \neq 0$ have coordinates

$$z = b/a, \qquad w = a/b.$$

Their inverses are $z \mapsto [1 : z]$ and $w \mapsto [w : 1]$. For an open set W of nonzero columns, the set of all columns equivalent to some point of W is $\bigcup_{\lambda \in \mathbb{C} \setminus \{0\}} \lambda W$. Each scalar multiplication

is a homeomorphism, so this union is open. The quotient projection is therefore open, as for real projective space. Thus the patches are open and the ratios define continuous charts. On the overlap $z \neq 0$, the transition is $w = 1/z$. For $z = u + \mathrm{i}v$, this is the real smooth map

$$(u, v) \longmapsto \left(\frac{u}{u^2 + v^2}, \frac{-v}{u^2 + v^2} \right).$$

It is its own inverse. The missing point of either chart is the origin of the other chart. Lecture 20 differentiates this transition to describe vectors attached to the sphere.

For $\psi = (a, b)^T \neq 0$, the Pauli expectation numerators are

$$\psi^\dagger \sigma_x \psi = 2\,\mathrm{Re}(\bar{a}b), \qquad \psi^\dagger \sigma_y \psi = 2\,\mathrm{Im}(\bar{a}b), \qquad \psi^\dagger \sigma_z \psi = |a|^2 - |b|^2.$$

Definition 7.14 (Bloch vector)**.** The *Bloch vector* of a pure qubit state is its vector of Pauli expectations:

$$B \colon \mathbb{C}\mathrm{P}^1 \longrightarrow \mathbb{R}^3, \qquad B([a : b]) = \frac{1}{|a|^2 + |b|^2} \left(2\,\mathrm{Re}(\bar{a}b),\ 2\,\mathrm{Im}(\bar{a}b),\ |a|^2 - |b|^2\right).$$

For a spin-1/2 system this vector is $2\langle \mathbf{S} \rangle / \hbar$. For a general two-level system its components are Pauli expectations, without an identification with spatial directions. The north pole has $z = 0$, and the south pole has $w = 0$. Both coordinates are available away from the poles, with $w = 1/z$. The drawing places the two basis states and one further ray on the unit sphere.

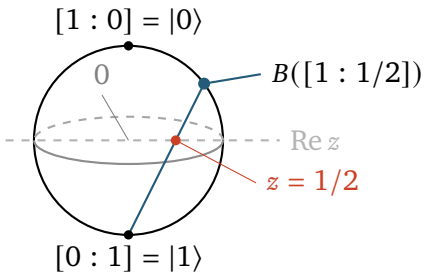


Figure 7.6: The north state $[1 : 0]$ is $|0\rangle$, and the south state $[0 : 1]$ is $|1\rangle$. The ray from the south pole through $B([1 : 1/2]) = (4/5, 0, 3/5)$ meets the coordinate plane at $z = 1/2$.

Example 7.15 (Projective states and the Bloch sphere)**.** We show that the Bloch map takes values in S^2 and is a diffeomorphism. Multiplying a and b by a common nonzero scalar multiplies numerator and denominator by the same positive number, so the map is well-defined. The formula on nonzero columns is continuous with nonvanishing denominator. Lecture 2's quotient criterion therefore makes B continuous. Its squared norm is one because $4|a|^2|b|^2 + (|a|^2 - |b|^2)^2 = (|a|^2 + |b|^2)^2$. In the z chart it becomes

$$B([1 : z]) = \frac{(2\,\mathrm{Re}\, z,\ 2\,\mathrm{Im}\, z,\ 1 - |z|^2)}{1 + |z|^2}.$$

For a sphere point (X, Y, Z) away from the south pole, the inverse coordinate is $z = (X + \mathrm{i}Y)/(1 + Z)$. Away from the north pole it is $w = (X - \mathrm{i}Y)/(1 - Z)$. These formulas are inverse on their patches and agree by $w = 1/z$ on the overlap. Indeed, the displayed z formula for B is exactly the inverse south stereographic chart, with real coordinates $(\mathrm{Re}\, z, \mathrm{Im}\, z)$. On the w patch it is the inverse north chart with coordinates $(\mathrm{Re}\, w, -\mathrm{Im}\, w)$.

The already proved stereographic identities establish both inverse compositions. Hence B is a homeomorphism. In particular, $\mathbb{C}\mathrm{P}^1$ is Hausdorff and second-countable, inherited through this homeomorphism from the sphere. Its compatible charts therefore make it a smooth surface. The same formulas are smooth in both directions, so B is a diffeomorphism.

Take angles $0 \le \theta \le \pi$ and $0 \le \phi < 2\pi$. For the normalised representative $(a, b) = (\cos(\theta/2), e^{i\phi}\sin(\theta/2))$, the z coordinate, when $\theta < \pi$, is $z = \tan(\theta/2)e^{i\phi}$. At $\theta = \pi$ the state is $[0 : 1]$, with $w = 0$. The sphere image for the full angle range is $(\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$. The antipodal sphere point is represented by $(\sin(\theta/2), -e^{i\phi}\cos(\theta/2))$. Its inner product with the displayed representative is zero because

$$\cos(\theta/2)\sin(\theta/2) - \sin(\theta/2)\cos(\theta/2) = 0.$$

Thus antipodal Bloch vectors represent orthogonal pure states. The two-component state and the sphere point describe the same pure state, with no overall phase coordinate.

Self-check. Let $F\colon \mathbb{R}\mathrm{P}^1 \to \mathbb{R}\mathrm{P}^1$ be given by $F([X^0 : X^1]) = [X^0 + X^1 : X^0 - X^1]$. At $[2 : 3]$, compute its expression and derivative in the ratio coordinates $u = X^1/X^0$ on the source and $v = Y^1/Y^0$ on the target. State where this expression is valid and explain why F is smooth everywhere.

Answer. The matrix defining F has determinant -2, so a nonzero source never maps to zero. In the specified charts,

$$v = \frac{1-u}{1+u}, \qquad \frac{\mathrm{d}v}{\mathrm{d}u} = -\frac{2}{(1+u)^2}, \qquad u = \frac{3}{2}, \quad v = -\frac{1}{5}, \quad \left.\frac{\mathrm{d}v}{\mathrm{d}u}\right|_{u=3/2} = -\frac{8}{25}.$$

This expression is valid where $X^0 \neq 0$ and $X^0 + X^1 \neq 0$. In any source patch, choose a nonzero component of the target as denominator. The resulting coordinate expressions are ratios of linear functions with nonzero denominator on that neighbourhood, so they are smooth.

Self-check. State in one sentence what a smooth structure is and why it was introduced. Explain how the two structures on the line show that topology alone is insufficient.

Answer. A smooth structure is a maximal atlas of smoothly compatible charts, introduced to make smoothness of functions independent of coordinate descriptions. For $g(t) = t$, the expressions are s in the x chart and $\sqrt[3]{s}$ in the y chart. The second is not differentiable at zero, although both charts describe the same topological line.

Unit 7B Submanifolds and embeddings

7.5 Regular level sets and embeddings

We construct smooth manifolds as subsets of an ambient manifold. The local coordinates must also describe how the subset sits in that space.

Definition 7.16 (Embedded submanifold). A subset S of a smooth N-manifold M is an *embedded d-submanifold* if, with the subspace topology, it satisfies the following condition. At every $p \in S$ there is an ambient smooth chart (U, φ) containing p such that

$$\varphi(S \cap U) = \varphi(U) \cap (\mathbb{R}^d \times \{0\}).$$

The restricted coordinates give its smooth structure.

Definition 7.17 (Regular point)**.** For a smooth map $f\colon W \to \mathbb{R}^k$, where $W \subseteq \mathbb{R}^N$ is open, a point $p \in W$ is *regular* when Df_p is onto.

Definition 7.18 (Regular value)**.** For a smooth map $f\colon W \to \mathbb{R}^k$, a *regular value* is a point $c \in \mathbb{R}^k$ such that every point of $f^{-1}(c)$ is regular. A value outside the image is regular vacuously.

Surjectivity means that the constraint derivatives are independent. The hypothesis distinguishes two different failures. For $f(x,y) = xy$, the zero set is the union of the coordinate axes. Its derivative is the row (y,x), which vanishes at the crossing. Removing the crossing from a small connected neighbourhood leaves four components. An interval leaves two, and a punctured Euclidean ball of dimension at least two is connected. Thus the crossing has no Euclidean neighbourhood. For $f(x,y) = y^2$, the derivative is $(0, 2y)$. The value zero is again not regular, although its level set is the embedded x-axis. Regularity is sufficient, not necessary.

The graph $t = s^2$ becomes the slice $u = 0$ under $\Phi(s,t) = (s, t - s^2)$, as the drawing shows.

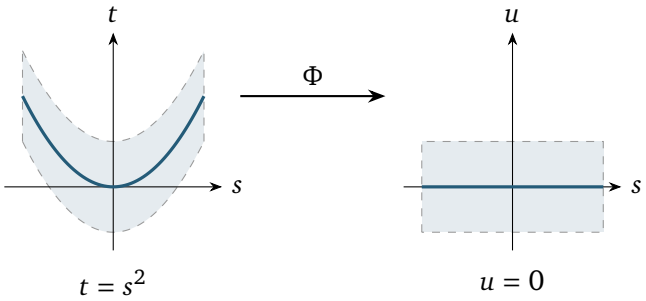


Figure 7.7: The open band $|s| < 1$, $|t - s^2| < 1/2$ maps to $|s| < 1$, $|u| < 1/2$. The graph becomes the horizontal slice.

Theorem 7.19 (Regular level sets · *proved given the implicit function theorem*)**.** *If c is a regular value of f, its nonempty level set $f^{-1}(c)$ is an embedded submanifold of W of dimension $N - k$.*

Proof. We use the implicit function theorem to write the level set as a graph, then subtract that graph in the constrained coordinates. Fix $p \in f^{-1}(c)$. Surjectivity gives a nonsingular $k \times k$ minor of Df_p. Permute the input coordinates and write them as $(s,t) \in \mathbb{R}^{N-k} \times \mathbb{R}^k$, with this minor in the t coordinates. The smooth implicit function theorem, cited in the form of Tu, Theorem B.2, gives open neighbourhoods in which

$$f(s,t) = c \quad \Longleftrightarrow \quad t = h(s)$$

for a smooth function h. In this neighbourhood the ambient coordinate change $\Phi(s,t) = (s, t - h(s))$ has smooth inverse $\Phi^{-1}(s,u) = (s, u + h(s))$. It sends the level set to $u = 0$. These are the required ambient charts. On their overlaps, the restricted transitions are smooth, because they are restrictions of smooth ambient transitions followed by projection onto the first $N - k$ components. The subspace is Hausdorff and second-countable, inherited from W. Thus these charts define the claimed embedded manifold. ∎

This is the Euclidean case of Tu, Theorem 9.9, pp. 105–106. For the sphere, use $f(x) = |x|^2$. At $p \in S^n$, its derivative is $Df_p(v) = 2p \cdot v$, which takes the value 1 on $v = p/2$. Hence 1 is

regular and the sphere has dimension $(n+1)-1=n$. The resulting structure agrees with the stereographic one, since the stereographic maps and their inverses are smooth ambient formulas on their respective domains.

For the orthogonal group, the constraint takes values in the vector space Sym_n of real symmetric matrices:

$$f\colon \mathbb{R}^{n\times n} \to \mathrm{Sym}_n, \qquad f(A) = A^T A.$$

The target has dimension $n(n+1)/2$, since its diagonal and entries above the diagonal determine it. Expanding $f(A+tH)$ to first order in the real parameter t gives

$$Df_A(H) = A^T H + H^T A.$$

At $A \in \mathrm{O}(n)$, any symmetric matrix S is the image of $H = AS/2$, since $A^T A = I$. Thus I is a regular value and $\dim \mathrm{O}(n) = n^2 - n(n+1)/2 = n(n-1)/2$. Taking all n^2 matrix entries as independent target constraints would incorrectly count the repeated off-diagonal equations.

Definition 7.20 (Immersion)**.** A smooth map is an *immersion* if its coordinate Jacobian is injective at every point.

Definition 7.21 (Embedding)**.** An immersion is an *embedding* if it is also a homeomorphism onto its image with the subspace topology.

Chart changes multiply the Jacobian by invertible matrices, so the immersion condition is independent of charts.

The inclusion of an embedded submanifold is an embedding. In slice charts its Jacobian is the inclusion of $\mathbb{R}^d$ into $\mathbb{R}^N$. Its topology is the subspace topology by definition.

Conversely, the image of an embedding is an embedded submanifold. Indeed, a rank-d minor makes d target coordinates local coordinates on the source, by the inverse function theorem. In these coordinates the remaining components form a smooth graph. Subtracting that graph gives a slice chart. The homeomorphism onto the image lets us restrict the ambient neighbourhood so that no other part of the image enters this local graph.

For comparison, the cusp parametrisation $t \mapsto (t^3, t^2)$ has zero velocity at $t=0$, so it is not an immersion. The curve $t \mapsto (\sin t, \sin 2t)$, with t taken modulo 2π, is an immersion: if $\cos t = 0$, then $\cos 2t = -1$. It is not injective, since $t=0$ and $t=\pi$ both map to $(0,0)$. The circle inclusion is an embedding. The drawing compares these maps with the irrational curve considered next.

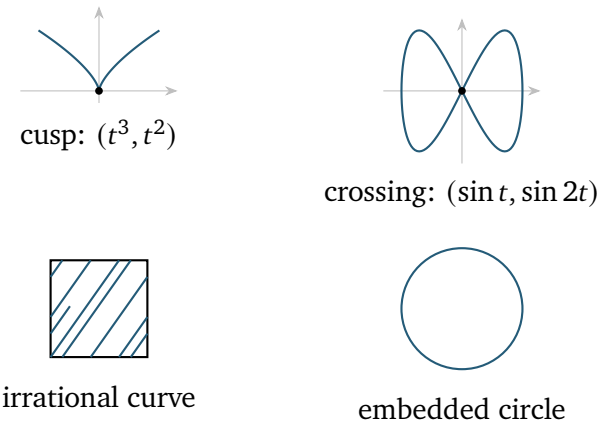


Figure 7.8: The cusp fails the derivative test, and the crossing curve fails injectivity. The irrational curve is injective but fails the subspace-topology test proved below.

An injective immersion need not be an embedding. Fix an irrational number $\alpha > 0$ and consider

$$\gamma : \mathbb{R} \longrightarrow T^2, \qquad \gamma(t) = (e^{it}, e^{i\alpha t}).$$

Equal images require a difference $2\pi m$ with $m \in \mathbb{Z}$ and $\alpha m \in \mathbb{Z}$. Irrationality forces $m = 0$, proving injectivity. The first circle component has nonzero derivative, proving immersion.

We test continuity of the inverse at $\gamma(0)$ by finding large parameters whose images approach $\gamma(0)$. At times $t = 2\pi q$ with a positive integer q, the first component is exactly 1. The second is $e^{2\pi i \alpha q}$, so we seek αq close to an integer. The square model uses coordinates $t/(2\pi)$ and $\alpha t/(2\pi)$ modulo integers. It shows a finite part of these repeated returns.

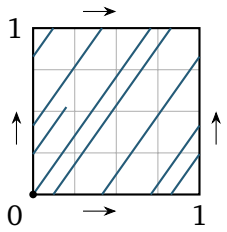


Figure 7.9: A finite segment of the line of slope $\sqrt{2}$ in the square quotient $\mathbb{R}^2/\mathbb{Z}^2$. Each exit continues at the identified opposite edge.

We produce unbounded positive integers q_j whose distances from αq_j to the integers tend to zero. Divide $[0, 1)$ into N equal half-open intervals. Place the $N + 1$ fractional parts of $0, \alpha, \ldots, N\alpha$ into them. Two lie in the same interval. Subtracting their indices gives $1 \le q \le N$ with distance from αq to an integer less than $1/N$. If these choices of q stayed bounded as N increased, one positive integer q would recur for arbitrarily large N. Its fixed distance to the integers would then be zero. This would give $\alpha q \in \mathbb{Z}$, contradicting irrationality. Hence we can extract an unbounded subsequence of choices with $N \to \infty$. Consequently $\gamma(2\pi q_j) \to \gamma(0)$, although $2\pi q_j$ does not tend to zero. The inverse from the image is not continuous. Thus γ is not an embedding. Lecture 9 describes motion along this curve by prescribing its velocity at every point.

For $\alpha = \sqrt{2}$, the returns at $t = 10\pi$ and $t = 58\pi$ have local coordinates $(0, 5\sqrt{2} - 7)$ and $(0, 29\sqrt{2} - 41)$. The drawing compares nearby image points with their separated parameters.

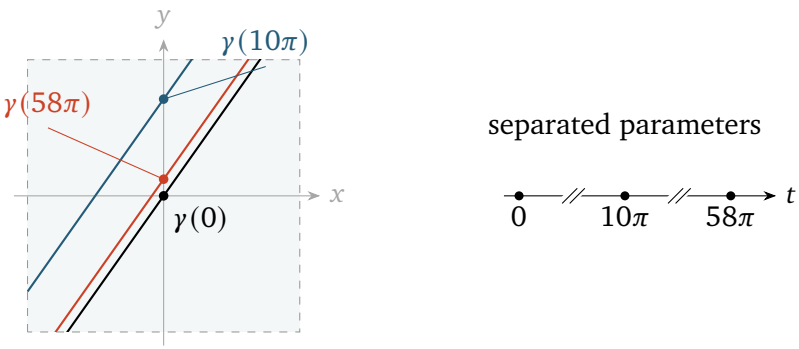


Figure 7.10: In local torus coordinates, return strands have equation $y = \sqrt{2}x + q\sqrt{2} - p$ for $(q, p) = (0, 0), (5, 7), (29, 41)$. The parameter axis uses breaks between the labelled values.

Self-check. Use the regular level set theorem to show that $\mathrm{U}(n)$ is a real manifold of dimension n^2. Specify the real target of the constraint. For the following unitary matrix A and Hermitian matrix S, exhibit H with $Df_A(H) = S$:

$$A = \begin{pmatrix} 3/5 & 4\mathrm{i}/5 \\ 4\mathrm{i}/5 & 3/5 \end{pmatrix}, \qquad S = \begin{pmatrix} 2 & 1+\mathrm{i} \\ 1-\mathrm{i} & -1 \end{pmatrix}.$$

Answer. Use $f(A) = A^\dagger A$ on $\mathbb{C}^{n\times n} \cong \mathbb{R}^{2n^2}$ with target the real vector space of Hermitian matrices. That target has n real diagonal entries and one complex entry for each pair above the diagonal, so its real dimension is $n + 2\binom{n}{2} = n^2$. Differentiating with respect to a real parameter gives $Df_A(H) = A^\dagger H + H^\dagger A$. At a unitary A, every Hermitian S is reached by $H = AS/2$. Thus I is a regular value and its level set has dimension $2n^2 - n^2 = n^2$. Treating this as a complex-linear constraint would be incorrect: conjugation is part of the formula. For the specified data, the same right inverse gives

$$H = \frac{AS}{2} = \begin{pmatrix} 1+2\mathrm{i}/5 & 3/10-\mathrm{i}/10 \\ 3/10+\mathrm{i}/2 & -7/10+2\mathrm{i}/5 \end{pmatrix}.$$

Substitution into $A^\dagger H + H^\dagger A$ returns S.

Summary. A smooth structure is specified by charts with smooth transitions. Smooth maps are tested in these charts, and the Euclidean chain rule makes the test independent of coordinates. The implicit function theorem turns independent smooth constraints into embedded submanifolds. The qubit state space has the two projective charts z and $w = 1/z$, and is diffeomorphic to the Bloch sphere. Lecture 8 uses coordinate changes to define tangent vectors and differentials intrinsically.

Reading. Tu, *An Introduction to Manifolds*, §§5–7, Theorem 9.9 and §11; the irrational-line example is Example 15.9. Nakahara, *Geometry, Topology and Physics*, §5.1.

Lecture 8

Tangent spaces, Lie groups and homogeneous spaces

Objectives. *A tangent vector records the initial velocity of a curve without choosing an ambient Euclidean space. Coordinate changes relate its components by their Jacobian matrices. We use these vectors to differentiate maps and describe the infinitesimal directions of matrix groups. A transitive smooth action of a Lie group presents a manifold as a quotient by the subgroup fixing a point. The sphere and the qubit state space give two concrete quotient descriptions.*

By the end of this lecture you will be able to:

- Define tangent vectors by curves and transform their components.
- Compute differentials and tangent spaces to regular level sets.
- Identify the tangent space of $\mathrm{SO}(3)$ at its identity.
- Determine stabilisers and justify the sphere quotient descriptions.

Unit 8A Tangent spaces

8.1 Tangent vectors as velocities

Let p be a point of a smooth n-manifold M.

Definition 8.1 (Tangent vector)**.** A *smooth curve through* p is a smooth map $\gamma\colon (-\varepsilon,\varepsilon) \to M$ with $\gamma(0) = p$, where $\varepsilon > 0$. In a chart $x = (x^1,\dots,x^n)$ near p, its coordinate velocity is $(x\circ\gamma)'(0) \in \mathbb{R}^n$. Two such curves are equivalent if these velocities agree. A *tangent vector* at p is an equivalence class of curves.

Definition 8.2 (Tangent space)**.** The set of these classes is the *tangent space*, written T_pM.

The drawing shows curves with different second-order behaviour but the same initial velocity in one chart.

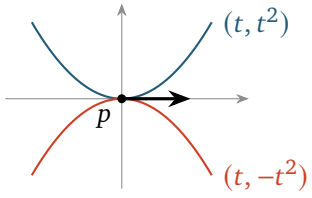


Figure 8.1: The coordinate curves (t, t^2) and $(t, -t^2)$ have the same velocity $(1, 0)$ at the marked point. Their common tangent vector does not record their bending.

Theorem 8.3 (Velocity classes and components)**.** *Curve equivalence is independent of the chart. A chart identifies T_pM with $\mathbb{R}^n$, giving it an n-dimensional vector-space structure independent of that chart. Under a coordinate change $y = y(x)$, components transform by*

$$v_y^a = \frac{\partial y^a}{\partial x^i}(p) v_x^i.$$

Proof. The chain rule gives $(y \circ \gamma)'(0) = D(y \circ x^{-1})_{x(p)}(x \circ \gamma)'(0)$. Thus equal velocities in x give equal velocities in y. The reverse coordinate change proves the converse. This also gives the displayed component formula.

The velocity map $T_pM \to \mathbb{R}^n$ is injective by the definition of equivalence. For any $v \in \mathbb{R}^n$, the curve $\gamma(t) = x^{-1}(x(p) + tv)$ is defined for sufficiently small $|t|$ because the chart image is open. It has velocity v, proving surjectivity. Transport addition and scalar multiplication through this bijection. A change of chart acts by an invertible linear Jacobian, so it preserves these operations. This proves that the vector-space structure is independent of the chart. ∎

The basis vector with coordinate components equal to the ith unit column is written $\partial_i|_p = \partial/\partial x^i|_p$. Thus $v = v^i \partial_i|_p$, with the upper/lower index convention of Lecture 7. The symbols $\partial_i|_p$ denote tangent vectors, not points of M.

A tangent vector differentiates smooth functions near p by

$$v(f) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} f(\gamma(t)) = v^i \frac{\partial (f \circ x^{-1})}{\partial x^i}(x(p)).$$

The right side shows independence of the representative curve.

Definition 8.4 (Point derivation)**.** A *point derivation* at p is a real linear rule on smooth real functions near p satisfying the product rule

$$v(fg) = v(f)g(p) + f(p)v(g).$$

The curve formula is real linear in the function. The ordinary product rule therefore makes every tangent vector a point derivation. Applied to a function, the vector $\partial_i|_p$ gives its partial derivative in the chart. Lecture 9 uses this action to differentiate functions by vector fields.

We obtain the change of coordinate basis by applying the component law to each basis vector. Its x-components form the ith unit column, so

$$\left.\frac{\partial}{\partial x^i}\right|_p = \frac{\partial y^a}{\partial x^i}(p) \left.\frac{\partial}{\partial y^a}\right|_p.$$

The vector stays fixed while its basis and its component column change. The figure shows one vector in Cartesian and polar bases at the same point.

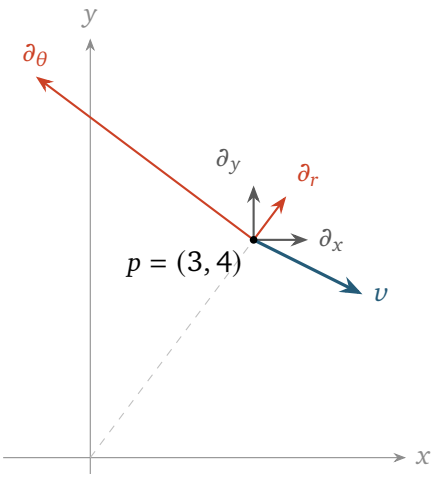


Figure 8.2: The vector with Cartesian components $(2, -1)$ at $p = (3, 4)$ has polar components $(2/5, -11/25)$. The polar angular basis vector has length 5, so it is not a unit vector.

Example 8.5 (One vector in two charts)**.** We express the velocity of a curve in Cartesian and polar coordinates. The point and curve are

$$p = (3, 4), \qquad \gamma(t) = (3 + 2t, 4 - t).$$

We use a local polar chart with $r > 0$ and a smooth angle branch near $\theta = \arctan(4/3)$. The Cartesian velocity column is $(2, -1)^T$. The Jacobian rows of $r = \sqrt{x^2 + y^2}$ and of the local angle give

$$\frac{\partial(r, \theta)}{\partial(x, y)} = \begin{pmatrix} x/r & y/r \\ -y/r^2 & x/r^2 \end{pmatrix}, \qquad \left.\frac{\partial(r, \theta)}{\partial(x, y)}\right|_p = \begin{pmatrix} 3/5 & 4/5 \\ -4/25 & 3/25 \end{pmatrix}.$$

The component law now gives

$$\begin{pmatrix} v^r \\ v^\theta \end{pmatrix} = \begin{pmatrix} 3/5 & 4/5 \\ -4/25 & 3/25 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 2/5 \\ -11/25 \end{pmatrix}.$$

For the basis conversion we use $x = r\cos\theta$ and $y = r\sin\theta$. The columns of this inverse Jacobian give

$$\partial_r|_p = \frac{3}{5}\partial_x|_p + \frac{4}{5}\partial_y|_p, \qquad \partial_\theta|_p = -4\partial_x|_p + 3\partial_y|_p.$$

Substitution recovers the original vector:

$$v = \frac{2}{5}\partial_r|_p - \frac{11}{25}\partial_\theta|_p = 2\partial_x|_p - \partial_y|_p.$$

The two columns describe the same velocity class.

8.2 Germs and the converse derivation characterisation *(optional)*

The differentials, vector fields and group quotients in later lectures do not depend on germs or on the converse characterisation in this section.

Definition 8.6 (Germ)**.** A *germ* at p is an equivalence class of functions defined near p, with two functions equivalent when they agree on some smaller neighbourhood.

Only the function near p matters in the action of a tangent vector. The product rule therefore makes sense on germs. Conversely, all point derivations on smooth germs have the displayed coordinate form for $v(f)$ in §8.1. This comparison is cited from Tu, §8, Proposition 8.9.

8.3 Differentials and regular constraints

Definition 8.7 (Differential). For a smooth map $f\colon M \to N$, its *differential* at p is

$$\mathrm{d}f_p\colon T_pM \longrightarrow T_{f(p)}N, \qquad \mathrm{d}f_p[\gamma] = [f\circ\gamma].$$

In coordinates the image velocity is $D\widehat{f}_{x(p)}v_x$, by the chain rule. This depends only on the velocity class and is linear. It therefore defines the stated linear map. For $g\colon N \to P$, composition on representatives gives the intrinsic chain rule

$$\mathrm{d}(g\circ f)_p = \mathrm{d}g_{f(p)} \circ \mathrm{d}f_p. \tag{8.1}$$

The identity map has identity differential. Applying this formula to a diffeomorphism and its inverse shows that their differentials are inverse linear maps.

For example, if $f\colon \mathbb{R}^2 \to \mathbb{R}^3$ is given by $f(s,t) = (s, t, s^2 + st)$, then

$$\mathrm{d}f_{(s,t)}(u,v) = (u, v, (2s+t)u + sv).$$

Its first two components recover (u,v), so the differential is injective.

Definition 8.8 (Immersion, submersion and embedding). Recall from Lecture 7 that an injective differential defines an *immersion*. An onto differential defines a *submersion*. An immersion that is a homeomorphism onto its image is an *embedding*.

Proposition 8.9 (Tangent space of a level set · *proved given the implicit function theorem*). *Let $S = f^{-1}(c) \subset W \subseteq \mathbb{R}^N$ be a nonempty regular level set of a smooth map $f\colon W \to \mathbb{R}^k$, with W open. The inclusion identifies T_pS with $\ker Df_p$ for every $p \in S$.*

Proof. For a curve in S, differentiating $f(\gamma(t)) = c$ gives $Df_p\gamma'(0) = 0$. In the graph chart of Theorem 7.19, the inclusion has the form $s \mapsto (s, h(s))$. Its differential sends u to $(u, Dh\,u)$, so it is injective. Its image has dimension $N-k$. The kernel of the onto map Df_p has that same dimension by rank–nullity. The inclusion already proved is therefore equality. ∎

For S^2, the constraint $|p|^2 = 1$ gives $T_pS^2 = \{v \in \mathbb{R}^3 : p\cdot v = 0\}$, since the derivative is $v \mapsto 2p\cdot v$. Thus the intrinsic velocity classes agree with the familiar tangent plane in this embedded example. The figure places velocity arrows in the affine planes through their base points.

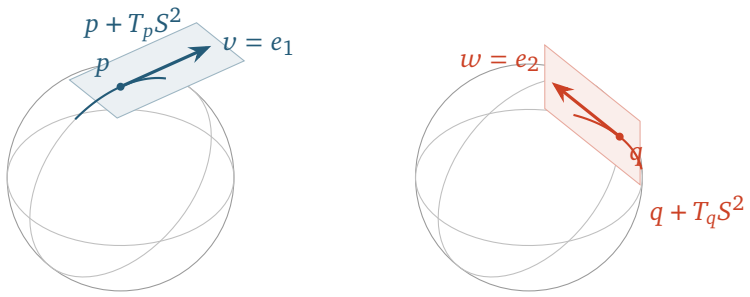


Figure 8.3: The planes $p + T_pS^2$ and $q + T_qS^2$ touch the sphere at $p = e_3$ and $q = e_1$. Each great circle has its velocity arrow in the plane at its own base point.

The vector spaces T_pS^2 and T_qS^2 are different subspaces of $\mathbb{R}^3$. Their affine translates in the figure show where their vectors are based. On a general manifold there need not be a common ambient vector space.

8.4 The tangent bundle

A moving point has both a position and a velocity at that position. We collect these pairs in one space so that the velocity can vary along a curve.

Definition 8.10 (Tangent bundle)**.** The *tangent bundle* is the disjoint union of all tangent spaces:

$$TM = \{(p, v) : p \in M,\ v \in T_pM\}.$$

The projection $\pi_{TM} \colon TM \to M$ is given by $\pi_{TM}(p, v) = p$. Its *fibre* over p is the inverse image $\pi_{TM}^{-1}(p)$, identified with T_pM.

Vectors at different base points belong to different fibres. They cannot be added without a further identification.

Physical input. In a mechanical model with configuration manifold M, the instantaneous position and velocity form a pair (p, v) with $v \in T_pM$.

Thus a time-independent Lagrangian, which assigns a real number to each position and velocity, is a function $L \colon TM \to \mathbb{R}$. For a pendulum the configuration is a point of S^1. Its position–velocity states therefore form TS^1, the cylinder drawn below.

Proposition 8.11 (Manifold structure on the tangent bundle · *sketched*)**.** *If M is a smooth n-manifold, then TM has a natural smooth structure of dimension $2n$. The projection is a smooth submersion.*

We sketch its chart construction. A chart x on $U \subseteq M$ gives coordinates $(x(p), v_x)$ on $\pi_{TM}^{-1}(U)$ with image $x(U) \times \mathbb{R}^n$. If y is another chart, the overlap map is

$$(x, v_x) \longmapsto \big(y(x), Dy_x v_x\big).$$

It is smooth and has the inverse obtained by the inverse coordinate change. The base and velocity coordinates give $2n$ independent real coordinates. They also specify the topology. Distinct base points can be separated downstairs. Distinct vectors over the same point can be separated in a common bundle chart. A countable atlas downstairs gives a countable basis in these product coordinates. This gives Hausdorffness and second countability. In each chart the projection is $(x, v) \mapsto x$, whose derivative is onto. On a triple overlap, the

ordinary chain rule makes these transition maps compose correctly. The inverse transition returns the original velocity because the two coordinate Jacobians are inverse matrices. These checks give compatible bundle charts. Further details are in Tu, §12.

For the circle, write $p = (\cos\theta, \sin\theta)$. Every tangent vector has the unique form $v = r(-\sin\theta, \cos\theta)$ with $r \in \mathbb{R}$. These formulas identify TS^1 with $S^1 \times \mathbb{R}$: the angle is periodic but the velocity coefficient is unrestricted. The figure displays this cylinder and its projection.

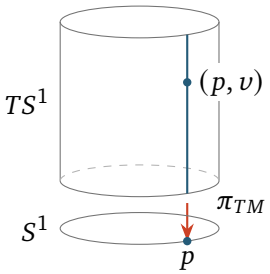


Figure 8.4: A fibre of TS^1 is a line of velocity coefficients. Projection forgets the coefficient and retains the point of the circle.

An identification with $M \times \mathbb{R}^n$ that preserves the projection and is linear on each fibre would choose a basis smoothly in every tangent space. The tangent bundle of S^2 admits no such identification. Indeed, the first basis vector would give a smooth tangent field with no zero. Lecture 14 proves that no such field exists.

Self-check. At $p = (-4, 3)$, let $v = 2\partial_x|_p - 3\partial_y|_p$. Use a local polar chart with angle near the second quadrant to find its polar components and recover v from the polar basis. For the map $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ given by $f(x, y) = (x^2 - y, xy)$, compute $\mathrm{d}f_p(v)$ and state its base point.

Answer. The two Jacobian rows at p are $(-4/5, 3/5)$ and $(-3/25, -4/25)$. Multiplying them by $(2, -3)^T$ gives $(-17/5, 6/25)^T$. The inverse basis vectors are

$$\partial_r|_p = -\frac{4}{5}\partial_x|_p + \frac{3}{5}\partial_y|_p, \qquad \partial_\theta|_p = -3\partial_x|_p - 4\partial_y|_p.$$

Their linear combination with these polar components is v. For the differential we obtain

$$\mathrm{d}f_p(v) = \begin{pmatrix} -8 & -1 \\ 3 & -4 \end{pmatrix} \begin{pmatrix} 2 \\ -3 \end{pmatrix} = \begin{pmatrix} -13 \\ 18 \end{pmatrix} \quad \text{at } f(p) = (13, -12).$$

Self-check. State in one sentence what a tangent vector is and why we introduced it. Then describe a point of TS^1 and what its projection retains.

Answer. A tangent vector is a class of curves through one point with the same coordinate velocity, introduced to define velocity without an ambient space. The chain rule makes equality of those velocities independent of the chart. A point of TS^1 is a circle position together with one tangent velocity there. The projection retains the position.

The tangent space at the identity of a smooth group describes the velocities of curves of symmetries that start at the identity.

Unit 8B Lie groups and homogeneous spaces

8.5 Lie groups and tangent matrices

Definition 8.12 (Lie group)**.** A *Lie group* is a smooth manifold G that is also a group, with smooth multiplication $G \times G \to G$ and smooth inversion $G \to G$. Write e for its identity.

Definition 8.13 (General linear group)**.** For $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, the *general linear group* is

$$\mathrm{GL}(n, \mathbb{K}) = \{A \in \mathbb{K}^{n \times n} : \det A \neq 0\}.$$

Definition 8.14 (Special orthogonal group)**.** The *special orthogonal group* is

$$\mathrm{SO}(n) = \{A \in \mathrm{O}(n) : \det A = 1\}.$$

Definition 8.15 (Matrix subgroup)**.** A *matrix subgroup* is a subgroup of a general linear group.

For complex matrices, the ambient space is regarded as a real manifold.

Proposition 8.16 (Regular matrix groups · *proved given the regular level set theorem*)**.** *A matrix subgroup that is a regular level set has the induced structure of a Lie group. The tangent space $T_I\mathrm{SO}(3)$ is the vector space $\mathfrak{so}(3) = \{A \in \mathbb{R}^{3\times 3} : A^T = -A\}$.*

Proof. The regular level set theorem gives an embedded manifold. Matrix multiplication is polynomial in real entries. Matrix inversion is smooth on invertible matrices, by the adjugate formula divided by the nonzero determinant. These maps preserve the subgroup. In adapted submanifold charts, their restrictions retain the first manifold-coordinate components of their ambient expressions, so the restrictions are smooth. They are the group's multiplication and inversion, as required.

The orthogonal constraint and its surjective derivative were computed in Lecture 7. The determinant-one part is open in $\mathrm{O}(3)$, since its only determinants are $+1$ and -1. It therefore has the same tangent space at I. By Proposition 8.9, that space is the kernel of $A \mapsto A^T + A$. This is precisely the antisymmetric matrices. ∎

Such a matrix has the unique form

$$A(a, b, c) = \begin{pmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{pmatrix}, \qquad (a, b, c) \in \mathbb{R}^3.$$

Thus $\mathfrak{so}(3)$ has basis $A(1, 0, 0)$, $A(0, 1, 0)$ and $A(0, 0, 1)$. These are infinitesimal rotations in the fixed Cartesian basis.

Definition 8.17 (Lie algebra)**.** A *Lie algebra* is a real vector space with an antisymmetric bilinear bracket satisfying the Jacobi identity $[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0$.

For a Lie group embedded as a matrix subgroup, the tangent space $\mathfrak{g} = T_I G$ carries the matrix commutator $[A, B] = AB - BA$. To show closure, choose curves $a(s)$ and $b(t)$ in the group with initial value I and velocities A and B. Differentiating $a(s)b(t)a(s)^{-1}$ first in t gives $a(s)Ba(s)^{-1} \in \mathfrak{g}$. Differentiating in s at zero gives $AB - BA$, since $(a^{-1})'(0) = -A$, obtained by

differentiating $a(s)a(s)^{-1} = I$. This derivative stays in the finite-dimensional vector subspace $\mathfrak{g}$. Bilinearity and antisymmetry follow from the formula. One nested commutator expands as

$$[A, [B, C]] = ABC - ACB - BCA + CBA.$$

In the sum with its two cyclic versions, each product occurs once with each sign. This proves Jacobi. For an abstract Lie group, Lecture 9 constructs the bracket on T_eG using vector fields.

Example 8.18 (The commutator in $\mathfrak{so}(3)$)**.** We compute the bracket of the chosen rotation generators. For a vector $u = (a, b, c)$ and $w \in \mathbb{R}^3$, matrix multiplication gives

$$A(a, b, c)w = u \times w.$$

Thus this matrix represents the cross-product map A_u of Lecture 2. For a unit axis n, differentiating Lecture 2's rotation $R(tn)$ at $t = 0$ gives $A(n)$. This identifies the matrix with an initial rotation velocity. Write $E_1 = A(1, 0, 0)$, $E_2 = A(0, 1, 0)$ and $E_3 = A(0, 0, 1)$. Their products are

$$E_1E_2 = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad E_2E_1 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Subtracting gives $[E_1, E_2] = E_3$. The same multiplication with the cyclic index choices gives $[E_2, E_3] = E_1$ and $[E_3, E_1] = E_2$. Bilinearity now gives the general formula

$$[A(u), A(w)] = A(u \times w), \qquad u, w \in \mathbb{R}^3.$$

The chosen basis therefore identifies this Lie algebra with $\mathbb{R}^3$ equipped with the cross product.

8.6 Actions and homogeneous spaces

Rotations send the north pole $N = e_3$ to every point of S^2. A rotation describing a point is not unique. If $gN = p$ and $hN = N$, then $(gh)N = gN = p$. Thus rotations that differ on the right by a rotation fixing N describe the same point. This gives a way to describe sphere points using rotations and their redundancy. Lecture 2 defined group actions on spaces. We now require the action map to be smooth. Write $R_y(t)$ and $R_z(t)$ for right-handed rotations by t about the respective Cartesian axes. The figure shows the equal images and the corresponding change of frame.

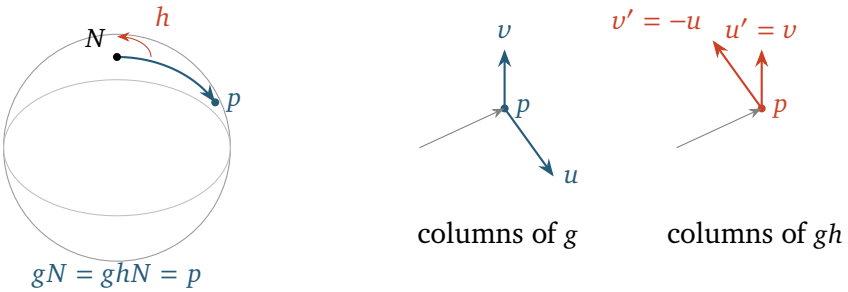


Figure 8.5: For $g = R_y(\pi/3)$ and $h = R_z(\pi/2)$, both g and gh send N to $p = (\sqrt{3}/2, 0, 1/2)$. The first two columns change from (u, v) to $(v, -u)$ while the last column stays fixed.

Definition 8.19 (Smooth left action)**.** A *smooth left action* of a Lie group on M is a smooth map $G \times M \to M$, written $(g, p) \mapsto g \cdot p$, satisfying $e \cdot p = p$ and $(gh) \cdot p = g \cdot (h \cdot p)$.

Definition 8.20 (Orbit and orbit map)**.** The *orbit* of p is $\{g \cdot p : g \in G\}$. The *orbit map* at p is the map $G \to M$ given by $g \mapsto g \cdot p$.

Definition 8.21 (Homogeneous space)**.** The action is *transitive* when the orbit of some point $p \in M$ is all of M. A manifold with a transitive smooth action is a *homogeneous space*.

For the rotation action, every unit vector is the image of N. Thus S^2 is homogeneous under $\mathrm{SO}(3)$.

Definition 8.22 (Stabiliser)**.** The *stabiliser* (or isotropy subgroup) of p is

$$H = \{g \in G : g \cdot p = p\}.$$

It is a subgroup by the action laws. It is closed, being the inverse image of the closed singleton $\{p\}$ under the continuous orbit map.

Definition 8.23 (Cosets and quotient)**.** For a subgroup H, the *left coset* gH is $\{gh : h \in H\}$. The set of left cosets, with the quotient topology, is written G/H.

Two elements have the same coset exactly when $g_1^{-1} g_2 \in H$.

Definition 8.24 (Embedded Lie subgroup)**.** An *embedded Lie subgroup* is a subgroup with an embedded smooth manifold structure and the restricted smooth group operations.

Theorem 8.25 (Closed subgroups and homogeneous spaces · *cited*)**.** *A closed subgroup H of a Lie group is an embedded Lie subgroup. The quotient G/H has a smooth manifold structure of dimension $\dim G - \dim H$ for which the quotient map is a smooth submersion. For a transitive smooth action on M with stabiliser H at p, the map $gH \mapsto g \cdot p$ is a diffeomorphism $G/H \to M$.*

These are Lee, second edition, Theorems 20.12, 21.17 and 21.18. The action identifies cosets with points: if $g_1 \cdot p = g_2 \cdot p$, then $g_1^{-1} g_2 \in H$, and the converse follows from the action law. Transitivity gives surjectivity, completing the bijection. The smooth-manifold and diffeomorphism assertions require the cited theorems. The quotient description supplies a smooth structure and a dimension formula for a space described by symmetries.

Remark 8.26 (A nonclosed subgroup)**.** Closedness cannot be omitted. A quotient by a dense proper subgroup is not Hausdorff, as in Lecture 2's quotient $\mathbb{R}/\mathbb{Q}$. For the irrational line of Lecture 7, the same obstruction occurs in a torus. Write that line as

$$H = \{(e^{it}, e^{i\alpha t}) : t \in \mathbb{R}\} \subset T^2, \qquad \alpha \in \mathbb{R} \setminus \mathbb{Q}.$$

Adding the parameters multiplies the circle components, so H is a subgroup. We prove that it is dense and proper.

The pigeonhole argument of Lecture 7 gives nonzero integer multiples of α arbitrarily close to integers. Irrationality makes each distance positive. By changing the integer's sign when necessary, we obtain a fractional part δ with $0 < \delta < \varepsilon$, for any $\varepsilon > 0$. The numbers $0, \delta, 2\delta, \ldots$, up to the last below 1, approximate every number in $[0, 1)$ to within ε. Each is

the fractional part of an integer multiple of α. Thus the points $e^{2\pi i\alpha m}$, with $m \in \mathbb{Z}$, are dense in the circle.

Now fix a target $(e^{ia}, e^{ib}) \in T^2$. The parameters $t = a + 2\pi m$ give points of H with coordinates

$$(e^{ia},\ e^{i\alpha a}e^{2\pi i\alpha m}), \qquad m \in \mathbb{Z}.$$

Their first coordinate is exactly the target's first coordinate. The density just proved lets their second coordinate approach e^{ib}. This proves density in the torus. Over first coordinate 1, however, H contains only the countably many second coordinates $e^{2\pi i\alpha m}$. The circle is uncountable, so H is proper. It is therefore not closed.

If T^2/H were Hausdorff, its singleton identity coset would be closed. Its inverse image H under the quotient map would then be closed, a contradiction. This quotient is not a manifold of our definition.

8.7 The sphere and the Hopf map

The standard action of $\mathrm{SO}(n+1)$ on S^n is transitive. Given a unit vector, extend it to an oriented orthonormal basis by Gram–Schmidt, recalled from *Linear Algebra for Physics*. Use it as the last column of an orthogonal matrix, completing the other columns with positive orientation. That matrix sends the last coordinate unit vector to the chosen point. The stabiliser consists of matrices with a block in $\mathrm{SO}(n)$ on its orthogonal complement and a last diagonal entry 1. The group $\mathrm{O}(n+1)$ is a regular level set by Lecture 7, so it is a Lie group by Proposition 8.16. Its determinant-one part is open and hence is also a Lie group. The sphere action is the restriction of the polynomial map $(g, p) \mapsto gp$, so it is smooth. Applying the cited theorem gives

$$S^n \cong \mathrm{SO}(n+1)/\mathrm{SO}(n) \qquad (n \geq 1).$$

The identification is a diffeomorphism. In particular, the orbit map $g \mapsto ge_3$ exhibits $S^2 = \mathrm{SO}(3)/\mathrm{SO}(2)$. The subgroup $\mathrm{SO}(2)$ rotates the other two columns while leaving the last one fixed. These quotient descriptions concern smooth manifolds; the stabiliser is not a normal subgroup in general, so the quotient need not be a group.

Lecture 4 identified $\mathrm{SU}(2)$ with S^3 using real coordinates. If we call its real scalar a_4, its first column is $(a_4 - iv_3, v_2 - iv_1)^T$. We now use the complex entries a and b of that column as coordinates. This gives the same identification with S^3.

The Lie group. If $|a|^2 + |b|^2 = 1$, the matrix with that first column is

$$U(a,b) = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}.$$

Its columns are orthonormal and its determinant is one, so $U(a,b) \in \mathrm{SU}(2)$. Conversely, a unitary matrix with first column $(a,b)^T$ has second column $\lambda(-\bar{b}, \bar{a})^T$ for some $|\lambda| = 1$. Its determinant is λ, so determinant one forces $\lambda = 1$. Thus every matrix in $\mathrm{SU}(2)$ has this form. Transporting the smooth structure of S^3 through U makes $\mathrm{SU}(2)$ a smooth manifold. Its product and inverse are smooth: the first columns of the corresponding matrix formulas are polynomial in the real and imaginary coordinates. This makes $\mathrm{SU}(2)$ a Lie group.

The action. We use the left action $g \cdot [a : b] = [g(a,b)^T]$ of $\mathrm{SU}(2)$ on $\mathbb{C}\mathrm{P}^1$. The notation means the line spanned by the image column. It is independent of the nonzero scalar used to represent $[a : b]$. The action is smooth in projective charts: each output ratio is a ratio of complex linear expressions, on the open set where its denominator is nonzero. Every point $[a : b]$ has a unit representative $(a,b)^T$. The matrix $U(a,b)$ sends $[1 : 0]$ to $[a : b]$, so the action is transitive.

The stabiliser. Fixing $[1 : 0]$ requires the first column to be $(e^{it}, 0)^T$. Unitarity and determinant one then force the other diagonal entry. Thus the stabiliser is the subgroup

$$H = \left\{ \begin{pmatrix} e^{it} & 0 \\ 0 & e^{-it} \end{pmatrix} : t \in \mathbb{R} \right\} \cong \mathrm{U}(1).$$

Consequently the cited theorem applies and gives $\mathbb{C}\mathrm{P}^1 \cong \mathrm{SU}(2)/\mathrm{U}(1)$ as smooth manifolds.

The figure places the two orbit maps and their stabilisers side by side.

$$\begin{array}{ccc} \mathrm{SO}(2) & \longrightarrow & \mathrm{SO}(3) \\ & & \downarrow{\scriptstyle g \mapsto ge_3} \\ & & S^2 \end{array} \qquad \begin{array}{ccc} \mathrm{U}(1) & \longrightarrow & \mathrm{SU}(2) \\ & & \downarrow{\scriptstyle g \mapsto [ge_1]} \\ & & \mathbb{C}\mathrm{P}^1 \end{array}$$

Figure 8.6: Each orbit map retains one datum: the last unit column for a rotation, or the line of the first unit column for $\mathrm{SU}(2)$. Multiplication on the right by its stabiliser leaves that datum fixed.

Definition 8.27 (Hopf map). The *Hopf map* sends a unit column to its complex line:

$$S^3 \longrightarrow \mathbb{C}\mathrm{P}^1, \qquad (a,b) \longmapsto [a : b].$$

The two orbit maps have circle fibres. For a fixed image $n = ge_3$, the rotations are $gR_z(t)$ with t taken modulo 2π. Locally, choosing a reference rotation for each nearby direction lets us describe a rotation by that direction and a spin angle about it. For a fixed projective point $[a : b]$, the unit columns are $e^{it}(a,b)^T$ with the same period. The figure draws these parameter circles over two chosen base points.

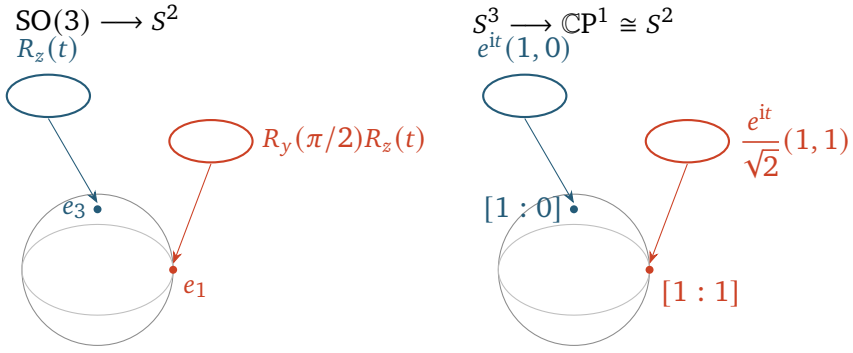


Figure 8.7: The displayed circles are schematic copies of the individual fibres: $g\mathrm{SO}(2)$ for rotations and $\{e^{it}(a,b)^T\}$ for the Hopf map. Their placement above the spheres does not assert a global product.

Example 8.28 (The Hopf map · *proved given the homogeneous-space theorem*)**.** The first-column map identifies SU(2) with $S^3 = \{(a,b) \in \mathbb{C}^2 : |a|^2 + |b|^2 = 1\}$. The inverse is $U(a,b)$, so both maps are smooth. The projection to the first-column line is the Hopf map. Its fibre consists of the unit representatives $(e^{it}a, e^{it}b)$ of the same complex line. Indeed, two unit columns span the same line exactly when the scalar relating them has modulus one. The matrix identity $U(e^{it}a, e^{it}b) = U(a,b)\,\mathrm{diag}(e^{it}, e^{-it})$ shows that these are precisely the left cosets $U(a,b)H$, the fibres of the orbit map $g \mapsto [ge_1]$. Composing with the Bloch diffeomorphism of Example 7.15 gives $S^3 \to S^2$. It forgets the common phase of a normalised qubit column. Lecture 20 develops its structure as a circle bundle and proves that no continuous choice of a normalised representative exists over all projective states.

Self-check. Find the dimension of $\mathrm{SO}(3)/\mathrm{SO}(2)$ using its stabiliser, and compare it with the tangent-plane calculation for the sphere. Then let SO(3) act on $\mathbb{R}\mathrm{P}^2$ by $g \cdot [X] = [gX]$. Find the stabiliser of $[0:0:1]$, its connected components and the quotient dimension.

Answer. The matrices in $\mathfrak{so}(3)$ have three independent parameters. The group SO(2) is a circle, so it has dimension one. The quotient dimension is $3 - 1 = 2$. The plane $p \cdot v = 0$ in $\mathbb{R}^3$ also has dimension two, since $p \neq 0$. The two calculations agree.

The projective action is transitive because the sphere action is transitive. Fixing $[0:0:1]$ means $ge_3 = \pm e_3$. Orthogonality preserves the perpendicular plane. Hence the stabiliser is

$$\left\{ \begin{pmatrix} B & 0 \\ 0 & \det B \end{pmatrix} : B \in \mathrm{O}(2) \right\}.$$

Indeed, the determinant-one condition forces the last entry to be $\det B$. The two components have $\det B = 1$ and $\det B = -1$. The first consists of rotations about e_3. The second consists of half-turns about axes in its perpendicular plane. Both have dimension one, so the quotient again has dimension $3 - 1 = 2$.

Summary. Tangent vectors are velocity classes. A chart identifies them with coordinate columns, and its Jacobian gives their transformation law. Differentials send curve velocities to image velocities and satisfy the chain rule. The tangent space of a regular level set is the kernel of the constraint's derivative. Closed stabilisers give smooth quotient manifolds by cited theorems. The rotation and qubit actions identify the sphere with $\mathrm{SO}(3)/\mathrm{SO}(2)$ and $\mathrm{SU}(2)/\mathrm{U}(1)$ respectively. Lecture 9 studies curves whose velocities are prescribed at every point by a vector field.

Reading. Tu, *An Introduction to Manifolds*, §§8, 12, 15–16; Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 20.12, 21.17 and 21.18.

Lecture 9

Vector fields and flows

Objectives. *A smooth vector field prescribes a velocity at every point of a manifold. Its integral curves describe the resulting motion. Comparing a second field along that motion gives the Lie bracket. We prove the connection between a zero bracket and commuting complete flows, then compute the rotation fields and their signs. Left translation turns a tangent vector at the identity of a Lie group into a vector field. This defines the group's Lie algebra and its exponential map.*

By the end of this lecture you will be able to:

- Find integral curves and identify their maximal time domains.
- Compute a Lie bracket and derive it from backward transport by a flow.
- Prove the commuting-flow criterion for complete fields.
- Reconcile rotation-field and angular-momentum commutator signs.

Unit 9A Fields, flows, brackets and rotations

9.1 Vector fields and related fields

Let M be a smooth n-manifold.

Definition 9.1 (Smooth vector field)**.** A *smooth vector field* on M is a smooth map

$$X\colon M \to TM, \qquad X(p) \in T_pM.$$

Such a map is called a *section* of the tangent bundle: composing it with the bundle projection gives the identity on M.

In local coordinates it has the form $X = X^i\partial_i$, where the coefficient functions X^i are smooth. Indeed, its bundle-chart expression is $x \mapsto (x, X^1(x), \dots, X^n(x))$, so these two smoothness conditions are equivalent.

The field acts on smooth real functions by

$$Xf = X^i\partial_i f.$$

This is a smooth function. The product rule gives $X(fg) = (Xf)g + f(Xg)$, so X acts as a derivation of functions. The value $(Xf)(p)$ is the point derivation $X_p(f)$ of Lecture 8. The same definition applies to smooth functions on any open subset of M.

Definition 9.2 (Push-forward field)**.** For a diffeomorphism $f\colon M \to N$, the *push-forward*

field f_*X on N is given by

$$(f_*X)_q = \mathrm{d}f_{f^{-1}(q)}X_{f^{-1}(q)}.$$

The inverse selects exactly one starting point for each q.

Definition 9.3 (Related fields). For a general smooth map f, fields X on M and Y on N are *f-related* when $\mathrm{d}f_pX_p = Y_{f(p)}$ for every $p \in M$.

We use related fields to compare motion on different manifolds. For the inclusion $i\colon S^1 \hookrightarrow \mathbb{R}^2$, write $i(\theta) = (\cos\theta, \sin\theta)$ in an angular chart. Its differential gives

$$\mathrm{d}i_\theta(\partial_\theta) = (-\sin\theta, \cos\theta).$$

Thus ∂_θ is i-related to $x\partial_y - y\partial_x$.

A field need not descend under a general map. For $f\colon \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$ and $X = \partial_x$, the image vector is $2x\partial_y$ at $y = x^2$. The starting points 1 and -1 both map to 1 but give opposite nonzero vectors there. Thus no f-related field Y exists on the target. Lecture 10 compares this restriction with the pullback of forms.

9.2 Integral curves and maximal flows

Definition 9.4 (Integral curve). An *integral curve* of X is a smooth curve $\gamma\colon I \to M$ on an open time interval satisfying $\gamma'(t) = X_{\gamma(t)}$. Here $\gamma'(t)$ is the tangent class at $\gamma(t)$ of $s \mapsto \gamma(t+s)$, as in Lecture 8. For initial point p, impose $0 \in I$ and $\gamma(0) = p$. In coordinates this is the ordinary differential equation (ODE)

$$\dot{x}^i(t) = X^i(x(t)).$$

Definition 9.5 (Maximal curve and maximal flow). A curve is *maximal* if it has no extension to a larger open time interval as a solution with that initial condition. The resulting solution map is the *maximal flow*, written $\varphi\colon \mathcal{D} \to M$, with $\varphi_t(p) = \gamma_p(t)$. Its time domain at p is the interval $I_p = \{t : (t,p) \in \mathcal{D}\}$.

Definition 9.6 (Complete field). A field is *complete* if $I_p = \mathbb{R}$ for every $p \in M$.

Definition 9.7 (Support). The *support* of a field is $\operatorname{supp} X = \overline{\{p : X_p \neq 0\}}$. The field has *compact support* when this closed set is compact.

For translation and dilation on $\mathbb{R}$, direct differentiation gives

$$X = \partial_x: \quad \varphi_t(x) = x + t, \qquad X = x\partial_x: \quad \varphi_t(x) = e^t x.$$

Both flows are defined for every real time. Their time maps satisfy the addition law by addition of real numbers and multiplication of exponentials.

The planar dilation field $x\partial_x + y\partial_y$ has flow $\varphi_t(x,y) = e^t(x,y)$. The figure shows its velocities and one integral curve.

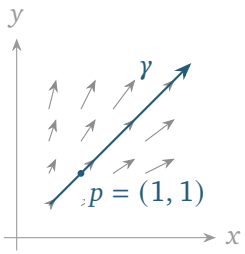


Figure 9.1: The field (x, y) and its integral curve $e^t(1, 1)$. Each arrow gives a velocity at its own point; the marked curve follows those velocities.

On $\mathbb{R}$, the field $X = x^2\partial_x$ has maximal flow

$$\varphi_t(x) = \frac{x}{1 - tx}, \qquad \mathcal{D} = \{(t, x) : 1 - tx > 0\}.$$

Differentiating gives $\partial_t \varphi_t = \varphi_t^2$, and the initial value is x. At $x = 0$ the solution is constant for all time. For $x > 0$ it diverges at $t = 1/x$. For $x < 0$ the divergence is at that negative time. These intervals are maximal because a continuous real-valued extension cannot pass through a divergence. Requiring only a nonzero denominator would include a separate branch beyond the pole. This field has noncompact support. A linear field $X_A(x) = Ax$ on $\mathbb{R}^n$, where $A \in \mathbb{R}^{n\times n}$ is nonzero, also has noncompact support. By contrast, its flow $x \mapsto e^{tA}x$ (§9.3) is defined for all real t. The graph shows the failure at $x = 1$ for $x^2\partial_x$.

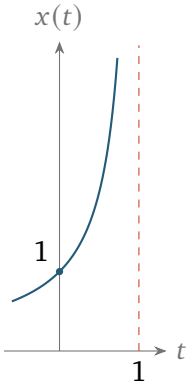


Figure 9.2: The solution starting at $x = 1$ diverges as t approaches 1 from below. The rational expression beyond the pole is not part of this integral curve.

Theorem 9.8 (Flow properties · *cited; laws and completeness proved from the cited flow properties*)**.** *A smooth vector field has a unique smooth maximal flow on an open set $\mathcal{D} \subseteq \mathbb{R} \times M$ containing $\{0\} \times M$. The time domains I_p are intervals. Its time maps satisfy*

$$\varphi_0 = \mathrm{id}, \qquad \varphi_{s+t}(p) = \varphi_s(\varphi_t(p))$$

whenever both sides are defined. Each time map is a diffeomorphism between its open source and image, with inverse φ_{-t}. Every smooth field with compact support is complete.

Existence and uniqueness of maximal curves, openness of $\mathcal{D}$ and joint smoothness in (t, p) are cited from Lee, second edition, Theorem 9.12. Its local input is the ODE theorem in charts. We prove the laws and completeness from these properties.

Proof. First fix $t \in I_p$ and put $q = \varphi_t(p)$. The curve $u \mapsto \varphi_{t+u}(p)$ is an integral curve starting at q. Uniqueness identifies it with $u \mapsto \varphi_u(q)$ on their common interval. Maximality gives $I_q = I_p - t$: either larger interval would, by shifting back and using uniqueness, extend the other solution. This proves the group law and ensures that $\varphi_{-t}(q) = p$. Define $M_t = \{p \in M : t \in I_p\}$, open because $\mathcal{D}$ is open. Applying the inverse identity with $-t$ shows that $\varphi_t \colon M_t \to M_{-t}$ is onto and both inverse compositions are identities. Joint smoothness gives smooth time maps and inverses.

Next we use compactness to obtain a common positive existence time. This permits continuation from every point reached by a solution. Let $K = \operatorname{supp} X$ be compact. If K is empty, the field is zero and every solution is constant. Otherwise, openness of $\mathcal{D}$ gives, near each $p \in K$, an open set U_p and a time $\varepsilon_p > 0$ for which the flow is defined on $(-\varepsilon_p, \varepsilon_p) \times U_p$. Choose a finite subcover of K and let $\delta > 0$ be the minimum of these finitely many times. Solutions starting in K exist for at least time δ in both directions. Outside K, the constant curve is a solution and uniqueness makes it the maximal solution. Thus every starting point has a solution for $|t| < \delta$. If a maximal interval had a finite right endpoint b, restart the solution at a time $t < b$ with $b - t < \delta$. The new solution extends past b, contradicting maximality and uniqueness. The same argument at a finite left endpoint proves completeness. ∎

In particular, every smooth field on a compact manifold is complete, since its support is a closed subset of that manifold.

On the torus $T^2 = S^1 \times S^1$, local angular coordinates (θ^1, θ^2) change by constant multiples of 2π. Their coordinate fields therefore define global fields. Fix $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ and set

$$X = \partial_{\theta^1} + \alpha \partial_{\theta^2}, \qquad \varphi_t(e^{\mathrm{i}\theta^1}, e^{\mathrm{i}\theta^2}) = (e^{\mathrm{i}(\theta^1 + t)}, e^{\mathrm{i}(\theta^2 + \alpha t)}).$$

Differentiating the angular coordinates gives the prescribed velocity. Starting at $(1, 1)$ gives $\gamma(t) = (e^{\mathrm{i}t}, e^{\mathrm{i}\alpha t})$, the irrational curve of Lecture 7. The formula exists for every $t \in \mathbb{R}$. Completeness concerns the time domain, even though this orbit is an injective immersion whose image is not embedded.

9.3 The Lie bracket and matrix signs

We compare two motions by following each and then reversing each in the same order. On $\mathbb{R}^2$, take $X = \partial_x$ and $Y = x\partial_y$. Their flows are $\varphi_u(x, y) = (x + u, y)$ and $\psi_v(x, y) = (x, y + vx)$. Starting at (x, y), the successive endpoints for times $u, v, -u, -v$ are

$$\begin{aligned}(x, y) &\longmapsto (x + u, y) \longmapsto (x + u, y + (x + u)v) \\ &\longmapsto (x, y + (x + u)v) \longmapsto (x, y + uv).\end{aligned}$$

The final gap is exactly $uv\partial_y$. The figure compares this path with commuting coordinate translations. The shear panel uses $x = u = v = 1$ and $y = 0$.



Figure 9.3: Coordinate translations close the rectangle. For $X = \partial_x$ and $Y = x\partial_y$, the return segment is shorter and leaves a positive vertical gap.

Definition 9.9 (Lie bracket). For fields X and Y, define their *Lie bracket* by its action on smooth functions:

$$[X, Y]f = X(Yf) - Y(Xf).$$

The first composition expands as

$$X(Yf) = X^i(\partial_i Y^j)\partial_j f + X^i Y^j \partial_i \partial_j f.$$

In the second composition, swap the indices i and j in its second-derivative term. Equality of mixed partial derivatives makes those terms cancel. The result is

$$[X, Y] = \big(X^i \partial_i Y^j - Y^i \partial_i X^j\big)\partial_j.$$

An individual composition XY can contain second derivatives of f. Their cancellation makes the commutator a first-order derivation. The coefficients are smooth, so this expression is a smooth vector field. Its action on functions is intrinsic, making it independent of coordinates. Bilinearity, antisymmetry and Jacobi follow from the commutator calculation for operators, exactly as in Lecture 8.

For general smooth fields, let $f_{u,v}$ be the map obtained by following X, Y, X, Y for times $u, v, -u, -v$, where these local flows are defined. In a fixed chart near p, the displacement satisfies

$$f_{u,v}(p) - p = uv[X, Y]_p + O\big(|uv|(|u| + |v|)\big) \qquad (u, v \to 0).$$

Indeed, differentiating the four compositions once in each time at zero gives $DY_p X_p - DX_p Y_p$. The displacement vanishes when either time is zero. Smoothness therefore factors it as uv times a smooth function of the times. That function has value $[X, Y]_p$ at zero, which gives the stated remainder. For the shear fields above, the bracket is ∂_y and the remainder vanishes.

The bracket preserves related fields under any smooth map f. Suppose X_j is f-related to Y_j for $j = 1, 2$. For every local smooth function h on the target, the chain rule gives $X_j(h \circ f) = (Y_j h) \circ f$. Applying it twice yields

$$[X_1, X_2](h \circ f) = ([Y_1, Y_2]h) \circ f.$$

Testing coordinate functions proves that the brackets are f-related. For a diffeomorphism this says $[f_* X, f_* Y] = f_*[X, Y]$.

Proposition 9.10 (Linear fields and the matrix commutator). *For $A, B \in \mathbb{R}^{n\times n}$, the fields $X_A(x) = Ax$ on $\mathbb{R}^n$ satisfy*

$$[X_A, X_B] = X_{-[A,B]}, \qquad [A, B] = AB - BA.$$

Proof. For linear fields, $DX_A = A$ and $DX_B = B$. The coordinate formula gives $[X_A, X_B](x) = BAx - ABx$, proving the minus sign. ∎

For example, let $A = E_{12}$ and $B = E_{21}$ in two dimensions, where E_{ij} has entry 1 in position (i, j) and zero elsewhere. Their fields and brackets are

$$X_A = y\partial_x, \qquad X_B = x\partial_y, \qquad [X_A, X_B] = -x\partial_x + y\partial_y.$$

The matrix commutator is $[A, B] = \operatorname{diag}(1, -1)$. Its linear field is the negative of the displayed bracket.

Recall the matrix exponential from *Linear Algebra*. The solution of $\dot{x} = Ax$ with initial value x is $e^{tA}x$. Thus the linear-field flow multiplies the position column on the left. For a smooth function f, the iterated field action is

$$(X_A X_B f)(x) = \left.\frac{\partial^2}{\partial s\, \partial t}\right|_{s=t=0} f(e^{tB}e^{sA}x).$$

The field X_B acts first on f, but the matrix e^{sA} acts first on x. This reversal of composition order explains the minus sign.

9.4 The Lie derivative of a field

Vectors at different points cannot be subtracted intrinsically. For $t \in I_p$, the map $\varphi_{-t}\colon M_{-t} \to M_t$ is a diffeomorphism and $\varphi_t(p) \in M_{-t}$. Thus it pushes $Y_{\varphi_t(p)}$ back to p by $\mathrm{d}\varphi_{-t}|_{\varphi_t(p)}$.

Definition 9.11 (Lie derivative). The *Lie derivative* is the resulting derivative in T_pM:

$$(L_X Y)_p = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_0 ((\varphi_{-t})_* Y)_p.$$

For functions, define $L_X f = Xf$, the derivative of $f(\varphi_t(p))$ at zero.

We compare the two vectors only after both have base point p. The figure shows this comparison for the shear fields at $p = (1, 0)$ and $t = 1$.

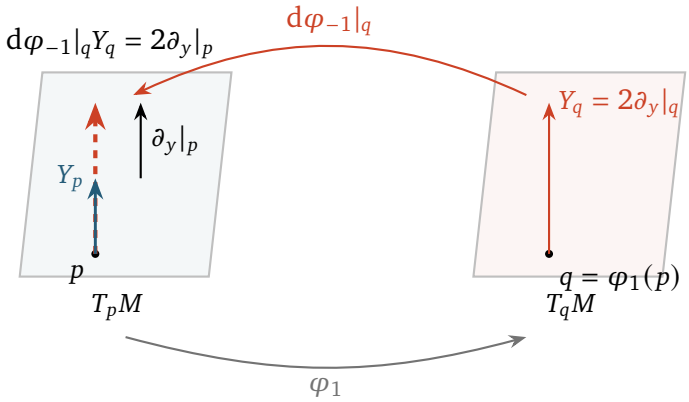


Figure 9.4: The vector Y_q is based at $q = \varphi_t(p)$. Its backward image and Y_p are both based at p; their difference divided by t tends to $L_X Y$.

For $X = \partial_x$ and $Y = x\partial_y$, the flow is $\varphi_t(x, y) = (x + t, y)$. The inverse flow has identity Jacobian, so

$$\mathrm{d}\varphi_{-t}|_{\varphi_t(p)} Y_{\varphi_t(p)} = (x + t)\partial_y|_p, \qquad \frac{(x + t)\partial_y|_p - x\partial_y|_p}{t} = \partial_y|_p \quad (t \neq 0).$$

The derivative is $L_X Y = \partial_y$, in agreement with the bracket.

Theorem 9.12 (Flow derivative and bracket · *proved given local flow existence and smooth dependence*)**.** *The bracket $[X, Y]$ is a smooth vector field. The Lie derivative exists and satisfies $L_X Y = [X, Y]$.*

Proof. The bracket's coordinate formula above proves its smoothness. The change in Y along the trajectory contributes $DY_p X_p$. The changing inverse differential contributes $-DX_p Y_p$. In a coordinate neighbourhood, the ODE gives $\varphi_t(p) = p + tX(p) + O(t^2)$. Differentiating the ODE $\partial_t \varphi_t = X(\varphi_t)$ in p, as joint smoothness permits, gives $\partial_t D\varphi_t = DX(\varphi_t) D\varphi_t$. Hence $D\varphi_t(p) = I + tDX_p + O(t^2)$. Inverting gives the inverse Jacobian $I - tDX_p + O(t^2)$ at $\varphi_t(p)$. Thus

$$\begin{aligned}((\varphi_{-t})_* Y)_p &= \big(I - tDX_p + O(t^2)\big)\big(Y_p + tDY_p X_p + O(t^2)\big) \\ &= Y_p + t\big(DY_p X_p - DX_p Y_p\big) + O(t^2).\end{aligned}$$

The coefficient of t is the bracket's coordinate formula. The definition by intrinsic push-forward makes this computation independent of the chosen neighbourhood. ∎

This is Tu, Theorem 20.4, with the same backward-push convention. Forward push would give the opposite sign.

9.5 Commuting complete flows

Suppose X and Y have flows φ_t and ψ_s.

Definition 9.13 (Commuting fields and flows)**.** The fields *commute* when $[X, Y] = 0$. Their complete flows commute when $\varphi_t \circ \psi_s = \psi_s \circ \varphi_t$ for every $s, t \in \mathbb{R}$.

Theorem 9.14 (Commuting flows · *proved given flow existence, uniqueness and smooth dependence*)**.** *For two complete smooth fields, their bracket vanishes if and only if their flows commute for all real times.*

Proof. The flow group law, the push-forward chain rule $(f \circ g)_* = f_* g_*$ and Theorem 9.12 give

$$\frac{\mathrm{d}}{\mathrm{d}t}(\varphi_{-t})_* Y = (\varphi_{-t})_*[X, Y].$$

Indeed, write the value at $t + h$ as $(\varphi_{-t})_*((\varphi_{-h})_* Y)$ and differentiate in h at zero. If the bracket is zero, this family is constant and equals Y. Hence $(\varphi_t)_* Y = Y$ for every t. For fixed t, both curves $s \mapsto \varphi_t(\psi_s(p))$ and $s \mapsto \psi_s(\varphi_t(p))$ solve the Y equation with the same initial point. Uniqueness makes them equal. Completeness gives the equality for all $s, t \in \mathbb{R}$.

Conversely, differentiate commuting flows in s at zero. The result is $\mathrm{d}\varphi_t|_p Y_p = Y_{\varphi_t(p)}$, so $(\varphi_{-t})_* Y = Y$. Differentiating in t at zero gives $[X, Y] = 0$. ∎

9.6 Distributions and Frobenius's theorem *(optional)*

Later material on rotations, differential forms and connections does not depend on this section. We test whether prescribed tangent planes are tangent to local surfaces. Frobenius gives a criterion for their local existence.

Definition 9.15 (Distribution)**.** A *smooth rank-k distribution* assigns a k-dimensional subspace $D_p \subseteq T_pM$ to every point, locally spanned by k independent smooth fields.

Definition 9.16 (Involutive distribution)**.** A distribution is *involutive* when the bracket of any two local fields taking values in D also takes values in D.

Definition 9.17 (Integral manifold)**.** An *integral manifold* is an immersion $i\colon S \to M$ satisfying

$$\mathrm{d}i_s(T_sS) = D_{i(s)} \qquad (s \in S).$$

Theorem 9.18 (Frobenius · *cited*)**.** *Every smooth involutive distribution of constant rank k has, near each point, coordinates in which it is spanned by $\partial_1, \dots, \partial_k$. Fixing the other coordinates gives local integral manifolds.*

This is Lee, second edition, Theorem 19.12. For example, the planes spanned by ∂_x and ∂_y in $\mathbb{R}^3$ have integral surfaces $z = c$. By contrast, consider the rank-two distribution on $\mathbb{R}^3$ spanned by

$$X = \partial_x + y\partial_z, \qquad Y = \partial_y, \qquad [X, Y] = -\partial_z.$$

The fields are independent at every point, as their x and y components show. Matching those components in $aX + bY = -\partial_z$ would force $a = b = 0$. Thus the bracket lies outside the prescribed plane at every point.

No local immersed surface can have these tangent planes. Restricted to a small enough open set, an immersion is an embedding: by the inverse-function argument of Lecture 7, d of its target coordinates are local coordinates on its d-dimensional source. On such a surface the two fields would restrict to fields related by inclusion to X and Y. Naturality would make their bracket tangent too, a contradiction. Equivalently, projection to (x, y) is locally invertible on a candidate surface. Writing that surface as $z = h(x, y)$ would require $h_x = y$ and $h_y = 0$. The two mixed derivatives would be 1 and 0, contradicting smoothness. The figure illustrates the same obstruction by exact flows in the prescribed planes. The drawing uses $p = (0, 0, 0)$ and $u = v = 1$.

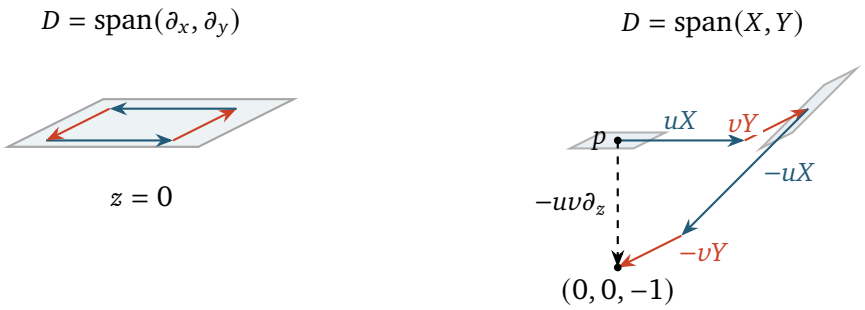


Figure 9.5: Horizontal planes are tangent to $z = c$. For $\mathrm{span}(\partial_x + y\partial_z, \partial_y)$, the four-flow path ends at $(x, y, z - uv)$; the vertical gap illustrates the computed bracket obstruction.

The paths in the second panel have successive endpoints

$$(x,y,z) \longmapsto (x+u,y,z+uy) \longmapsto (x+u,y+v,z+uy)$$
$$\longmapsto (x,y+v,z-uv) \longmapsto (x,y,z-uv).$$

The irrational field on T^2 spans a rank-one distribution. Its local sections have the form fX, and the product rule gives $[fX,gX] = (fXg - gXf)X$, so the distribution is involutive. Its orbits are integral manifolds. The irrational orbit of Lecture 7 is an injective immersion but is not embedded. This example explains why the definition requires only an immersion.

9.7 Rotation fields and angular momentum

In Cartesian coordinates on $\mathbb{R}^3$, the rotation fields are

$$L_x = y\partial_z - z\partial_y, \qquad L_y = z\partial_x - x\partial_z, \qquad L_z = x\partial_y - y\partial_x.$$

For example, $L_x(x^2+y^2+z^2) = 2yz - 2zy = 0$. Cyclic permutation gives the same result for the other fields. Thus each is tangent to every sphere centred at zero. Its integral curves retain that radius. Here L_x names a rotation field. The subscript X in the Lie derivative L_X names the generating vector field. In cylindrical coordinates $x = r\cos\phi$ and $y = r\sin\phi$ with $r > 0$, differentiating at fixed r and z gives $\partial_\phi = -y\partial_x + x\partial_y = L_z$. The angle ϕ is the azimuth. Thus the L_z flow adds t to the azimuth. On S^2, this field vanishes exactly at the two poles. The arrows in the figure show the azimuthal velocities.

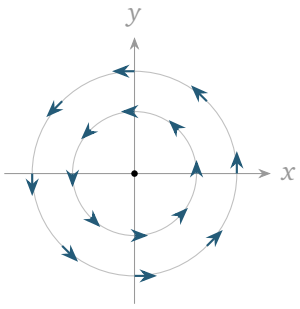


Figure 9.6: The field $L_z = (-y,x,0)$ has circular trajectories in each horizontal plane. The arrows are its tangent velocities; the axis consists of fixed points.

Example 9.19 (Rotation flows and brackets)**.** The three flows are defined for all real t:

$$\varphi_t^x(x,y,z) = (x,\ y\cos t - z\sin t,\ y\sin t + z\cos t),$$
$$\varphi_t^y(x,y,z) = (x\cos t + z\sin t,\ y,\ -x\sin t + z\cos t),$$
$$\varphi_t^z(x,y,z) = (x\cos t - y\sin t,\ x\sin t + y\cos t,\ z).$$

For the first flow, differentiating its last two components gives $(-y\sin t - z\cos t,\ y\cos t - z\sin t)$, which is $(-z(t), y(t))$. At $t = 0$ the components are the initial coordinates. The same differentiation in the other coordinate planes verifies the other flows. Thus all three restrict to complete flows on S^2.

Their linear matrices are the basis matrices $A(1,0,0)$, $A(0,1,0)$ and $A(0,0,1)$ of Lecture 8. For a unit axis $n \in \mathbb{R}^3$, Lecture 8's matrix is $A(n)w = n \times w$. Lecture 2's axis–angle rotation $R(tn)$ has initial value I and derivative $A(n)R(tn)$. Uniqueness of the linear ODE

therefore gives $e^{tA(n)} = R(tn)$. The displayed flows are these matrix exponentials for the three coordinate axes.

One bracket can also be read directly on coordinates: $[L_x, L_y]x = L_x(z) = y$, $[L_x, L_y]y = -L_y(-z) = -x$ and $[L_x, L_y]z = 0$. Therefore

$$[L_x, L_y] = -L_z, \qquad [L_y, L_z] = -L_x, \qquad [L_z, L_x] = -L_y.$$

The matrices have the positive cyclic commutator, so this is precisely the linear-field minus sign of Proposition 9.10.

Physical input. For a scalar particle wavefunction, orbital angular momentum about axis a is represented by the differential operator $\widehat{L_a} = -\mathrm{i}\hbar L_a$ for $a \in \{x, y, z\}$. The constant $\hbar$ is the reduced Planck constant.

On complex smooth functions, these angular-momentum differential operators have the commutator

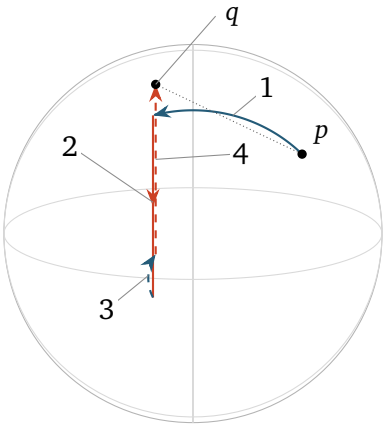
$$[\widehat{L_x}, \widehat{L_y}] = (-\mathrm{i}\hbar)^2(-L_z) = \hbar^2 L_z = \mathrm{i}\hbar\widehat{L_z}.$$

Thus the rotation-field sign agrees with the positive quantum angular-momentum relation. This is the differential-operator identification; no quantum representation theorem is assumed.

The rotation flows about different axes do not commute. For instance, the two orders of quarter-turns give

$$\varphi^x_{\pi/2}\big(\varphi^y_{\pi/2}(1,0,0)\big) = (0,1,0), \qquad \varphi^y_{\pi/2}\big(\varphi^x_{\pi/2}(1,0,0)\big) = (0,0,-1).$$

For small times, following L_x, L_y, L_x, L_y for $u, v, -u, -v$ has displacement $-uvL_z + O(|uv|(|u| + |v|))$ in a local chart. At $(1,0,0)$ its leading term is $(0, -uv, 0)$. The figure shows the corresponding four arcs at finite times on the unit sphere.



$1: L_x, \quad 2: L_y, \quad 3: -L_x, \quad 4: -L_y$

Figure 9.7: Successive rotations about x, y, x, y for times $\pi/3, \pi/3, -\pi/3, -\pi/3$ start at $p = (1,1,1)/\sqrt{3}$ and end at $q \neq p$. Solid arcs show the first two rotations and dashed arcs their reversals. Arcs 2 and 4 lie on nearly the same circle and are told apart by dashing. The dotted chord joins the final point to the initial point; all arcs are exact rotation trajectories.

Self-check. Find the flow of $x\partial_y - y\partial_x$ on $\mathbb{R}^2$ and its value at time $\pi/2$ from $(1,0)$. Include the trajectory from the origin.

Answer. The solution is $(x\cos t - y\sin t, x\sin t + y\cos t)$ for every $t \in \mathbb{R}$. From $(1,0)$ it reaches $(0,1)$ at time $\pi/2$, so positive time gives counter-clockwise rotation. Differentiating the displayed solution gives $(-y(t), x(t))$, which verifies the prescribed field. The origin is a constant solution.

Self-check. For $X = x\partial_x + 2y\partial_y$ and $Y = x^2\partial_x + xy\partial_y$ on $\mathbb{R}^2$, compute $L_X Y$ from backward transport. Verify the result by the coordinate bracket.

Answer. The flow is $\varphi_t(x,y) = (e^t x, e^{2t}y)$. Its inverse differential is $\operatorname{diag}(e^{-t}, e^{-2t})$. Consequently

$$((\varphi_{-t})_* Y)_{(x,y)} = \begin{pmatrix} e^{-t} & 0 \\ 0 & e^{-2t} \end{pmatrix} \begin{pmatrix} e^{2t}x^2 \\ e^{3t}xy \end{pmatrix} = e^t \begin{pmatrix} x^2 \\ xy \end{pmatrix}.$$

Differentiating at zero gives $L_X Y = x^2\partial_x + xy\partial_y$. The bracket has x component $2x^2 - x^2 = x^2$ and y component $3xy - 2xy = xy$, so it agrees. The inverse differential supplies the subtractions in both components.

Unit 9B Left-invariant fields and the exponential

9.8 Left-invariant fields and their bracket

Lecture 8 put the matrix commutator on the tangent space at the identity of a matrix group. On an abstract Lie group, we extend a tangent vector at the identity to a field by left translation. Its field bracket then defines a bracket on that tangent space. For $\mathrm{SO}(3)$ this constructs fields on the group itself, distinct from the rotation fields on $\mathbb{R}^3$.

For a Lie group, write $\ell_g(h) = gh$ for left translation.

Definition 9.20 (Left-invariant field)**.** A field is *left-invariant* when $(\ell_g)_* X = X$ for all $g \in G$.

Here a *matrix Lie group* means an embedded matrix subgroup with the smooth structure of Lecture 8. For tangent matrices, the formula $A^L(g) = gA$ follows by differentiating left multiplication.

Proposition 9.21 (Left-invariant matrix fields)**.** *For tangent matrices A and B at the identity of a matrix Lie group, the left-invariant fields $A^L(g) = gA$ satisfy $[A^L, B^L] = [A,B]^L$.*

Proof. The group fields are related by the inclusion i to the ambient fields $g \mapsto gA$ and $g \mapsto gB$. Their ambient derivatives in direction K are KA and KB. The related-fields result identifies their ambient bracket with the inclusion of the group bracket. Thus

$$\mathrm{d}i_g([A^L, B^L]_g) = (gA)B - (gB)A = g[A,B].$$

The inclusion has injective differential, so this identifies the group bracket. ∎

The map $x \mapsto Ax$ multiplies a column on the left. The map $g \mapsto gA$ multiplies a matrix on the right. The coordinate bracket therefore gives $BA - AB$ for the first map and $g(AB - BA)$ for the second. In particular, $A(1,0,0)$ and $A(0,1,0)$ have positive cyclic matrix commutator, their left-invariant fields have positive bracket, and their rotation fields have negative bracket.

For an abstract Lie group, each $A \in T_eG$ determines a left-invariant field by $A^L(g) = \mathrm{d}(\ell_g)_e A$. Smoothness follows by differentiating the smooth multiplication map in its second argument. Associativity and the chain rule give

$$(\ell_g)_* A^L(h) = \mathrm{d}(\ell_g)_h \mathrm{d}(\ell_h)_e A = \mathrm{d}(\ell_{gh})_e A = A^L(gh).$$

This proves invariance, and evaluation at the identity recovers A. Conversely invariance forces this formula, so the correspondence is a linear bijection. Naturality of the bracket makes $[A^L, B^L]$ left-invariant.

Definition 9.22 (Lie algebra of a Lie group)**.** The Lie algebra of G is T_eG with bracket

$$[A, B] = [A^L, B^L](e).$$

The field bracket's bilinearity, antisymmetry and Jacobi identity give these properties to the bracket at the identity. For matrix groups it agrees with the commutator just computed.

For example, the group of maps $r \mapsto ar+b$, where $a > 0$, has law $(a, b)(a', b') = (aa', b+ab')$. Set $H = \partial_a$ and $N = \partial_b$ at the identity $(1, 0)$. Differentiating left translation gives $H^L = a\partial_a$ and $N^L = a\partial_b$. The coordinate bracket gives $[H^L, N^L] = a\partial_b = N^L$.

9.9 One-parameter subgroups and the exponential

Definition 9.23 (One-parameter subgroup)**.** A *one-parameter subgroup* is a smooth map $c\colon \mathbb{R} \to G$ with $c(s+t) = c(s)c(t)$ and $c(0) = e$.

The integral curve of A^L starting at the identity is a one-parameter subgroup. To prove this, take a local solution on $(-\delta, \delta)$ starting at the identity. Left translation gives a solution on that same interval starting at every group element. The uniform-time continuation argument in the proof of Theorem 9.8 proves completeness. Left invariance and uniqueness identify the two curves $t \mapsto c(s+t)$ and $t \mapsto c(s)c(t)$, proving the group law. Conversely, differentiating a one-parameter subgroup's law makes it an integral curve of A^L, with $A = c'(0)$. For matrices the solution is $c(t) = e^{tA}$, where

$$e^{tA} = \sum_{k=0}^{\infty} \frac{t^k A^k}{k!}.$$

If each $|A_{ij}| \le m$, multiplication of matrices gives $|(A^k)_{ij}| \le n^{k-1} m^k$ for $k \ge 1$, by induction. The resulting scalar exponential bounds the series and its differentiated series uniformly on bounded time intervals. Termwise differentiation gives $c'(t) = c(t)A$ and $c(0) = I$. The group-valued curve, viewed through the inclusion, solves this ambient equation. Ambient uniqueness identifies it with the series. The full field flow is $g \mapsto ge^{tA}$. For this right-multiplication flow, iterated differentiation gives

$$(A^L B^L f)(g) = \left.\frac{\partial^2}{\partial s\, \partial t}\right|_{s=t=0} f(g e^{sA} e^{tB}).$$

The order of the field actions agrees with the order of these right factors. This explains the positive sign of the left-invariant bracket.

Definition 9.24 (Lie group exponential)**.** For $A \in T_eG$, let c_A be the integral curve of A^L with $c_A(0) = e$. The *exponential* is the map

$$\exp\colon T_eG \longrightarrow G, \qquad \exp A = c_A(1).$$

Rescaling time in the integral-curve equation gives $c_A(t) = \exp(tA)$. For a matrix group, the preceding series calculation identifies this map with the matrix exponential.

Self-check. State in one sentence what the Lie algebra of an abstract Lie group is and why left-invariant fields enter its definition.

Answer. The Lie algebra is T_eG with the bracket $[A, B] = [A^L, B^L](e)$, obtained by extending tangent vectors by left translation so that their vector-field bracket is defined. Naturality makes that bracket left-invariant, so evaluation at the identity returns a tangent vector in the same space.

Summary. A vector field prescribes an ordinary differential equation in every chart. Uniqueness gives the flow group law, and compact support gives completeness. The Lie derivative of a field uses backward push-forward and equals its bracket with the generating field. Complete fields have zero bracket exactly when their flows commute. Linear fields Ax have the negative matrix commutator, whereas left-invariant matrix fields gA have the positive one. Lecture 10 introduces forms and their pullback under arbitrary smooth maps.

Reading. Tu, *An Introduction to Manifolds*, §14 and §20.2; Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 9.12 and 19.12.

Lecture 10

Differential forms

Objectives. *A differential form supplies an integrand for curves, surfaces and higher-dimensional pieces of a manifold. At each point it assigns a signed multilinear measurement to tangent vectors. Forms pull back under every smooth map. Their exterior derivative extends the differential of a function and combines the Cartesian gradient, curl and divergence. We construct this derivative in coordinates, prove that it is independent of coordinates, and compute the sphere's area form.*

By the end of this lecture you will be able to:

- Evaluate forms with the determinant wedge convention.
- Compute exterior derivatives and pullbacks under smooth maps.
- Detect nonexactness of a closed 1-form by a nonzero loop integral.
- Derive both angular and stereographic expressions for sphere area.

10.1 Covectors and the cotangent bundle

Let M be a smooth n-manifold.

Definition 10.1 (Cotangent space). The *cotangent space* at $p \in M$ is the dual vector space $T_p^*M = \mathrm{Hom}(T_pM, \mathbb{R})$. Its elements are *covectors*.

Definition 10.2 (Differential). For a smooth function $f : M \to \mathbb{R}$, its *differential* is given by

$$(\mathrm{d}f)_p(v) = v(f) \qquad (v \in T_pM).$$

This is the differential of Lecture 8 with $T_{f(p)}\mathbb{R}$ identified with $\mathbb{R}$ by its standard coordinate. In coordinates $x^1, \dots, x^n$, the covectors $\mathrm{d}x^i$ are dual to ∂_j, since $\mathrm{d}x^i(\partial_j) = \delta^i_j$. Thus a covector has a unique expression $\alpha = \alpha_i \mathrm{d}x^i$, and $\alpha(v) = \alpha_i v^i$. In particular,

$$\mathrm{d}f = \frac{\partial f}{\partial x^i}\mathrm{d}x^i.$$

Definition 10.3 (Cotangent bundle). The *cotangent bundle* is the disjoint union $T^*M = \coprod_{p\in M} T_p^*M$, with projection to p.

Its local coordinates are $(x^1, \dots, x^n, \alpha_1, \dots, \alpha_n)$. If $y = y(x)$ has Jacobian $J = Dy$, invariance of $\alpha(v)$ requires the coefficient row to change by $\alpha' = \alpha J^{-1}$. These are smooth invertible bundle transitions. They give the cotangent manifold by the same construction as sketched for the tangent bundle in Lecture 8. The Hausdorff and countable-basis arguments there apply with these transitions.

Definition 10.4 (Smooth 1-form)**.** A *smooth 1-form* is a smooth section of T^*M. Equivalently, its coefficients α_i are smooth in every chart.

The form $\mathrm{d}f$ is smooth because the partial derivatives of f are smooth.

Physical input. In classical mechanics momentum at a position is represented by a covector that pairs with velocity. Thus a point of T^*M records position and momentum. Lecture 11 uses these pairs as phase-space coordinates.

A covector measures a tangent vector, as the figure illustrates.

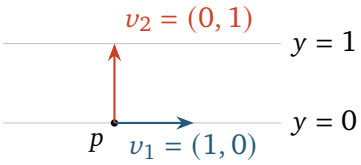


Figure 10.1: The horizontal lines are levels of the coordinate y. The covector $\mathrm{d}y$ gives 0 on the horizontal vector and 1 on the vertical vector.

For example, on $\mathbb{R}^2$ let $\alpha = 2x\mathrm{d}x + \mathrm{d}y$. At $(1, 0)$ it gives $\alpha(3\partial_x - \partial_y) = 6 - 1 = 5$. This form is the differential of $x^2 + y$.

10.2 Alternating tensors and the wedge product

A signed surface integrand measures a parallelogram in a tangent space. Linearity expresses addition and scaling in each edge. Equal edges span no area, so the value is zero. For a bilinear map, expansion on $(v + w, v + w)$ then shows that exchanging the edges reverses the sign. Alternating forms extend this rule to higher dimensions.

Definition 10.5 (Tensor field)**.** A *tensor field of type* (r, s) assigns at each point a multilinear real-valued map with r cotangent arguments and s tangent arguments. Its coordinate coefficients must be smooth. A tensor with $r = 0$ is *covariant*.

Definition 10.6 (Differential form)**.** A *k-form* is a covariant k-tensor field that is alternating: interchanging two arguments reverses its sign. We write $\Omega^k(M)$ for the vector space of smooth k-forms. A 0-form is a smooth function.

The degree k counts tangent arguments, whereas n is the dimension of M. Alternation implies that every k-form is zero if $k > \dim M$.

Recall the exterior powers from *Linear Algebra*, Lecture 12, the wedge-and-determinant example.

Proposition 10.7 (Exterior powers · *imported definition; evaluation proved*)**.** *For a finite-dimensional real vector space V, the space $\Lambda^k V^*$ consists of alternating k-linear forms on V. For covectors $\alpha_1, \dots, \alpha_k$, the determinant convention gives*

$$(\alpha_1 \wedge \cdots \wedge \alpha_k)(v_1, \dots, v_k) = \det\big(\alpha_i(v_j)\big)_{i,j=1}^{k}.$$

Proof of the evaluation formula. LA12 identifies covariant tensors with multilinear maps. Its antisymmetric tensors are therefore alternating maps. Evaluating its unnormalized antisymmetrization gives the sum $\sum_\pi \operatorname{sgn}(\pi) \prod_j \alpha_{\pi(j)}(v_j)$. The Leibniz determinant formula identifies this with the stated determinant. ∎

For two covectors the determinant formula reads

$$(\alpha \wedge \beta)(v, w) = \alpha(v)\beta(w) - \alpha(w)\beta(v).$$

In the plane, the form $\mathrm{d}x \wedge \mathrm{d}y$ measures signed area. The figure also shows the coordinate-plane areas measured by a 2-form in $\mathbb{R}^3$.

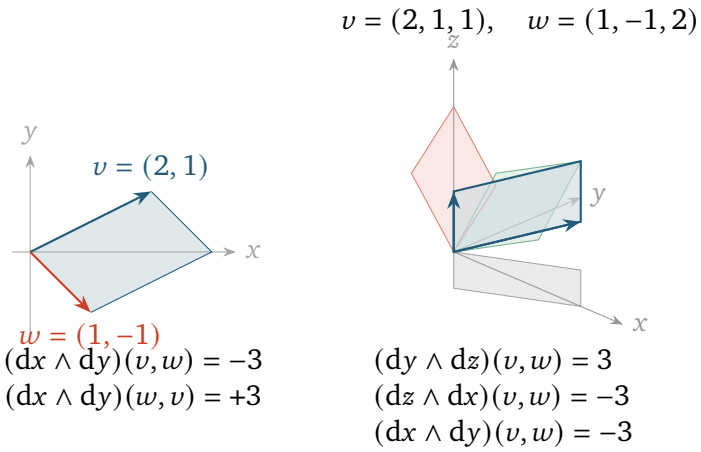


Figure 10.2: The ordered planar edges $(2,1)$ and $(1,-1)$ give signed area -3; exchanging them gives $+3$. The projections of $(2,1,1)$ and $(1,-1,2)$ have signed areas 3, -3 and -3 in the displayed coordinate orders.

Example 10.8 (A numerical 2-form)**.** On $\mathbb{R}^2$ take $\alpha = \mathrm{d}x + 2\mathrm{d}y$ and $\beta = 3\mathrm{d}x - \mathrm{d}y$. Bilinearity and alternation give

$$\alpha \wedge \beta = -\mathrm{d}x \wedge \mathrm{d}y + 6\mathrm{d}y \wedge \mathrm{d}x = -7\mathrm{d}x \wedge \mathrm{d}y.$$

For $v = (2,1)$ and $w = (1,-1)$, the signed area is $(\mathrm{d}x \wedge \mathrm{d}y)(v,w) = 2(-1) - 1 \cdot 1 = -3$. The value of $\alpha \wedge \beta$ is therefore $(-7)(-3) = 21$.

For an increasing index tuple $I = (i_1 < \cdots < i_k)$, write $\mathrm{d}x^I = \mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_k}$. Every k-form has a unique expansion

$$\omega = \sum_I a_I \mathrm{d}x^I, \qquad a_I = \omega(\partial_{i_1}, \dots, \partial_{i_k}).$$

To prove this, the determinant gives $\mathrm{d}x^I(\partial_{j_1},\dots,\partial_{j_k}) = \delta^I_J$ for increasing tuples I and J. This forces the displayed coefficients. Conversely, subtracting the displayed sum from ω gives an alternating form vanishing on every increasing tuple of coordinate vectors. Alternation makes it zero on every ordered tuple, including repeated indices. Multilinearity then makes it zero on arbitrary vectors. This proves existence and uniqueness of the expansion. The sum runs over increasing tuples only, so it has no factorial. For a 2-form with component $\omega_{12} = c$ in two dimensions, alternation gives $\omega_{21} = -c$. The expression with all indices is

$$\frac{1}{2}\big(c\mathrm{d}x^1 \wedge \mathrm{d}x^2 - c\mathrm{d}x^2 \wedge \mathrm{d}x^1\big) = c\mathrm{d}x^1 \wedge \mathrm{d}x^2.$$

If instead every index ranges independently from 1 to n, set $\omega_{i_1\cdots i_k} = \omega(\partial_{i_1},\dots,\partial_{i_k})$. Then

$$\omega = \frac{1}{k!}\sum_{i_1,\dots,i_k} \omega_{i_1\cdots i_k}\mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_k}.$$

Each increasing tuple occurs $k!$ times with compensating signs in the component and basis wedge, which explains the factorial.

Definition 10.9 (Wedge product)**.** The *wedge product* of forms of degrees k and ℓ is the bilinear extension of concatenation of these basis wedges. A repeated index gives zero; otherwise sorting the concatenated indices gives its permutation sign.

For variable coefficients the sorting rule gives

$$(x\mathrm{d}x + y\mathrm{d}y) \wedge (z\mathrm{d}x + x\mathrm{d}y) = (x^2 - yz)\mathrm{d}x \wedge \mathrm{d}y.$$

For a 2-form α and a 1-form β, the intrinsic evaluation is

$$(\alpha \wedge \beta)(u,v,w) = \alpha(u,v)\beta(w) - \alpha(u,w)\beta(v) + \alpha(v,w)\beta(u).$$

In the full permutation sum, each term appears twice because interchanging the two arguments of α reverses both signs. Division by 2! leaves these three terms. The general formula has the same cancellation within each block. This definition agrees with the determinant convention. Intrinsically it is

$$(\alpha \wedge \beta)(v_1,\dots,v_{k+\ell}) = \frac{1}{k!\ell!}\sum_{\pi\in S_{k+\ell}} \operatorname{sgn}(\pi)\alpha(v_{\pi(1)},\dots,v_{\pi(k)})\beta(v_{\pi(k+1)},\dots,v_{\pi(k+\ell)}).$$

Indeed, grouping the permutations by which k arguments enter α leaves $k!\ell!$ identical contributions within each group. By bilinearity take $\alpha = \mathrm{d}x^I$ and $\beta = \mathrm{d}x^J$. Both the permutation expression and the sorted basis wedge are alternating in their arguments, so the preceding basis test reduces their comparison to increasing tuples of coordinate vectors. Only shuffles placing the first index block in its matching slots contribute. They give its sorting sign; a repeated index gives zero. Thus the formula equals the sorting rule. The intrinsic formula proves that wedge is independent of coordinates and has smooth coefficients. Concatenating three index lists proves associativity: both parenthesizations have the same sorting sign, or both vanish because an index repeats. Moving a block of k indices past a block of ℓ indices makes $k\ell$ swaps. Bilinearity therefore gives graded commutativity:

$$\alpha \wedge \beta = (-1)^{k\ell}\beta \wedge \alpha.$$

As a tensor, the determinant wedge is $\alpha\wedge\beta = \alpha\otimes\beta - \beta\otimes\alpha$. The normalized fermionic tensor of *QM-I*, Lecture 14, instead uses $(u\otimes v - v\otimes u)/\sqrt{2}$ for orthonormal one-particle states. For k

orthonormal factors its normalization is $1/\sqrt{k!}$ times the unnormalized antisymmetrization.[1] Our convention gives $(\mathrm{d}x \wedge \mathrm{d}y)(\partial_x, \partial_y) = 1$. This value fixes the normalization used by the Hodge operator in Lecture 15.

Self-check. State in one sentence what a differential k-form is and why it is introduced. Explain why repeated tangent arguments give zero.

Answer. A differential k-form is a smooth alternating multilinear measurement of k tangent vectors at each point, introduced as an integrand for signed k-dimensional integration. Exchanging two equal arguments leaves the value unchanged and negates it, so the value is zero.

10.3 The exterior derivative

Definition 10.10 (Exterior derivative)**.** The *exterior derivative* raises form degree by one. On a coordinate chart define it by

$$\mathrm{d}\omega = \sum_I \sum_j (\partial_j a_I)\mathrm{d}x^j \wedge \mathrm{d}x^I \quad \text{when} \quad \omega = \sum_I a_I \mathrm{d}x^I.$$

On functions this agrees with the differential already defined.

Definition 10.11 (Line integral)**.** For a smooth 1-form α on M and a smooth path $\gamma\colon [a,b] \to M$, define the line integral by

$$\int_\gamma \alpha = \int_a^b \alpha_{\gamma(t)}(\gamma'(t))\,\mathrm{d}t.$$

Here a smooth path on $[a,b]$ means the restriction of a smooth curve on an open interval containing $[a,b]$. For a 1-form $\alpha = a\mathrm{d}x + b\mathrm{d}y$ on an open set in $\mathbb{R}^2$, exterior differentiation gives

$$\mathrm{d}\alpha = (\partial_x b - \partial_y a)\mathrm{d}x \wedge \mathrm{d}y.$$

This coefficient measures circulation per unit area. To see this directly, let C_h be the line integral around a counter-clockwise square of side $h > 0$ based at $p = (x_0, y_0)$, with the square contained in the open set. Pairing the horizontal edges and then the vertical edges gives

$$\begin{aligned} C_h &= \int_0^h \big[a(x_0 + t, y_0) - a(x_0 + t, y_0 + h)\big]\,\mathrm{d}t \\ &\quad + \int_0^h \big[b(x_0 + h, y_0 + t) - b(x_0, y_0 + t)\big]\,\mathrm{d}t \\ &= h^2(\partial_x b - \partial_y a)_p + o(h^2). \end{aligned}$$

The fundamental theorem of calculus expresses each bracket as an integral of a first derivative. Continuity of these derivatives gives the last line as $h \to 0^+$. Its leading term is $\mathrm{d}\alpha_p(h\partial_x, h\partial_y)$. Lecture 12 proves the corresponding equality between boundary and surface integrals in Stokes' theorem.

For $\alpha = -y\mathrm{d}x + x\mathrm{d}y$, the derivative is $2\mathrm{d}x \wedge \mathrm{d}y$. The figure records the four edge integrals on the square $[0,h]^2$.

[1] QM-I, Lecture 14, Many fermions: Slater determinants, uses normalized tensor states. The present wedge acts on covectors and uses determinant normalization.

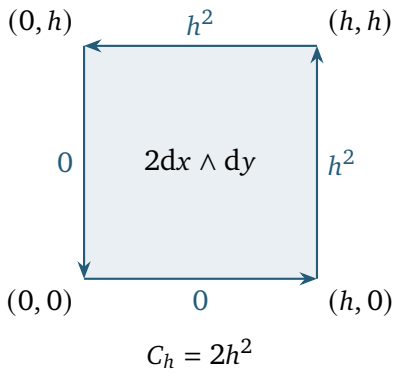


Figure 10.3: For $\alpha = -y\mathrm{d}x + x\mathrm{d}y$, the counter-clockwise edge integrals are 0, h^2, h^2 and 0. Their sum is $2h^2$, in agreement with the exterior derivative.

Theorem 10.12 (Exterior derivative · *sketched; existence and identities proved*)**.** *The coordinate formula defines a coordinate-independent real-linear map* $\mathrm{d}\colon \Omega^k(M) \to \Omega^{k+1}(M)$. *It satisfies* $\mathrm{d}^2 = 0$ *and, for* $\alpha \in \Omega^k(M)$,

$$\mathrm{d}(\alpha \wedge \beta) = \mathrm{d}\alpha \wedge \beta + (-1)^k \alpha \wedge \mathrm{d}\beta.$$

These properties and its value on functions determine it uniquely.

Proof of existence and identities. Work first in one chart. The formula $\mathrm{d}(b\mathrm{d}x^J) = \mathrm{d}b \wedge \mathrm{d}x^J$ also holds for a tuple J that is not increasing: reordering changes both sides by the same sign, and a repeated index makes both sides zero. Applying d twice gives terms $(\partial_r \partial_j a_I)\mathrm{d}x^r \wedge \mathrm{d}x^j \wedge \mathrm{d}x^I$. Terms with $r = j$ vanish. Each remaining term cancels its partner with j and r exchanged, because mixed derivatives agree and wedge is antisymmetric. Thus $\mathrm{d}^2 = 0$.

For the product rule, expand into terms $\alpha = a\mathrm{d}x^I$ and $\beta = b\mathrm{d}x^J$, where $|I| = k$. The ordinary product rule differentiates ab into $b\mathrm{d}a + a\mathrm{d}b$. The first contribution is $\mathrm{d}\alpha \wedge \beta$. In the second, moving $\mathrm{d}b$ past the k factors of $\mathrm{d}x^I$ gives $(-1)^k \alpha \wedge \mathrm{d}\beta$. Summing proves the identity.

Now let $y = y(x)$ be another chart. The chain rule gives $\mathrm{d}y^a = (\partial_i y^a)\mathrm{d}x^i$. The operator defined in x coordinates annihilates $\mathrm{d}y^a$ by the same mixed-derivative cancellation. For a coefficient function b, the chain rule also identifies its x and y differentials. Apply the product rule just proved to a monomial $b\mathrm{d}y^{i_1} \wedge \cdots \wedge \mathrm{d}y^{i_k}$. Only the differentiated coefficient survives, so the result is precisely the y coordinate formula. Such monomials span every form. Thus the definitions agree on chart overlaps and give a global smooth form. ∎

The uniqueness assertion is sketched in the optional §10.8.

For example, $\mathrm{d}(x\mathrm{d}y) = \mathrm{d}x \wedge \mathrm{d}y$, while $\mathrm{d}(x\mathrm{d}x) = \mathrm{d}x \wedge \mathrm{d}x = 0$.

10.4 Pullback under a smooth map

Let $f\colon M \to N$ be any smooth map.

Definition 10.13 (Pullback)**.** For $\omega \in \Omega^k(N)$ its *pullback* is given by

$$(f^*\omega)_p(v_1, \ldots, v_k) = \omega_{f(p)}(\mathrm{d}f_p v_1, \ldots, \mathrm{d}f_p v_k).$$

For functions, set $f^*h = h \circ f$.

The differential sends each argument to the required target tangent space. No inverse of f or of its differential is needed.

Theorem 10.14 (Pullback calculus). *Pullback takes smooth forms to smooth forms and satisfies*

$$f^*(\alpha \wedge \beta) = f^*\alpha \wedge f^*\beta, \qquad \mathrm{d}(f^*\alpha) = f^*(\mathrm{d}\alpha).$$

For smooth $g\colon N \to P$ it also satisfies $(g \circ f)^* = f^*g^*$ and $\mathrm{id}^* = \mathrm{id}$.

Proof. Substituting $\mathrm{d}f_p v_j$ in the permutation formula proves wedge compatibility. The chain rule gives $f^*\mathrm{d}h = \mathrm{d}(h \circ f)$. In local target coordinates y, write $f^a = y^a \circ f$. The pullback of $\alpha = \sum_I a_I \mathrm{d}y^I$ is

$$f^*\alpha = \sum_I (a_I \circ f)\mathrm{d}f^{i_1} \wedge \cdots \wedge \mathrm{d}f^{i_k}.$$

Its coefficients are smooth. Differentiating this expression leaves only $\mathrm{d}(a_I \circ f)$, since $\mathrm{d}^2 f^a = 0$. Pulling back $\mathrm{d}\alpha$ gives the same expression by the chain rule on coefficient functions. This proves commutation with d without using the uniqueness sketch. Finally, substituting $\mathrm{d}(g \circ f) = \mathrm{d}g \circ \mathrm{d}f$ into the definition proves the composition rule, and identity differentials prove the identity rule. ∎

For $f\colon \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$, pullback gives $f^*(\mathrm{d}y) = 2x\mathrm{d}x$, including at $x = 0$. Lecture 9 showed that ∂_x does not descend under this same map. Forms pull back even when a map loses rank or identifies distinct points.

Proposition 10.15 (Determinant rule for top-form pullback). *Let $U, V \subseteq \mathbb{R}^n$ be open and let $f\colon U \to V$ be smooth. For a smooth function $a\colon V \to \mathbb{R}$,*

$$f^*(a\mathrm{d}y^1 \wedge \cdots \wedge \mathrm{d}y^n) = (a \circ f)\det(Df)\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n.$$

Proof. Both sides are top forms on U. Evaluating on $(\partial_1, \ldots, \partial_n)$ gives $(a \circ f)\det(Df)$ on the left by the determinant wedge formula. The right side has this same coefficient, so the forms agree. ∎

We compute a pullback by substituting the component functions of the map and replacing each target differential by the differential of that component.

Example 10.16 (Polar pullback). Let $f(r, \theta) = (r\cos\theta, r\sin\theta)$, where $r > 0$ and θ belongs to an open interval of length less than 2π. The component differentials are

$$f^*\mathrm{d}x = \cos\theta\,\mathrm{d}r - r\sin\theta\,\mathrm{d}\theta, \qquad f^*\mathrm{d}y = \sin\theta\,\mathrm{d}r + r\cos\theta\,\mathrm{d}\theta.$$

Their wedge is

$$f^*(\mathrm{d}x \wedge \mathrm{d}y) = r(\cos^2\theta + \sin^2\theta)\mathrm{d}r \wedge \mathrm{d}\theta = r\mathrm{d}r \wedge \mathrm{d}\theta.$$

The determinant rule identifies the coefficient r as the Jacobian determinant.

Self-check. For $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ given by $f(u, v) = (u^2 - v, uv)$ and the target 1-form $\alpha = x\mathrm{d}y$, compute $f^*\alpha$ and $\mathrm{d}(f^*\alpha)$. Compute $f^*(\mathrm{d}\alpha)$ independently and compare.

Answer. Substitution gives

$$f^*\alpha = (u^2 - v)(v\mathrm{d}u + u\mathrm{d}v).$$

The $\mathrm{d}u \wedge \mathrm{d}v$ coefficient of its derivative is $\partial_u[u(u^2 - v)] - \partial_v[v(u^2 - v)] = 2u^2 + v$. On the target, $\mathrm{d}\alpha = \mathrm{d}x \wedge \mathrm{d}y$. The determinant of the derivative of f is $\det\left(\begin{smallmatrix} 2u & -1 \\ v & u \end{smallmatrix}\right) = 2u^2 + v$. Thus both computations give $(2u^2 + v)\mathrm{d}u \wedge \mathrm{d}v$.

10.5 Gradient, curl and divergence

Fix Cartesian coordinates on Euclidean $\mathbb{R}^3$. Identify a vector (A_1, A_2, A_3) with the 1-form $\alpha = A_1\mathrm{d}x + A_2\mathrm{d}y + A_3\mathrm{d}z$. Identify (B_1, B_2, B_3) with the 2-form $\beta = B_1\mathrm{d}y\wedge\mathrm{d}z + B_2\mathrm{d}z\wedge\mathrm{d}x + B_3\mathrm{d}x\wedge\mathrm{d}y$. Identify a scalar g with the 3-form $g\mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z$. These are specified coordinate identifications. The 1-form evaluates on one tangent vector by the Euclidean dot product. The 2-form evaluates on an ordered pair by the dot product with their cross product. The second identification also uses the orientation (x, y, z).

Physical input. For a force field in Euclidean space, work along a path is the line integral of its dot product with velocity. For a fluid velocity field, volume flux through an oriented surface is the surface integral of its normal component.

The 1-form therefore represents work along curves. The 2-form represents flux through surfaces. The differential $\mathrm{d}f$ exists without a metric. Its associated gradient vector uses the Euclidean dot product through $\mathrm{d}f(v) = \operatorname{grad} f \cdot v$.

Proposition 10.17 (Vector calculus from forms)**.** *Under these identifications, exterior differentiation gives gradient, curl and divergence in successive degrees.*

Proof. The defining formula gives the following coefficients:

$$\begin{aligned}
\mathrm{d}f &= (\partial_x f)\mathrm{d}x + (\partial_y f)\mathrm{d}y + (\partial_z f)\mathrm{d}z,\\
\mathrm{d}\alpha &= (\partial_y A_3 - \partial_z A_2)\mathrm{d}y\wedge\mathrm{d}z + (\partial_z A_1 - \partial_x A_3)\mathrm{d}z\wedge\mathrm{d}x\\
&\quad + (\partial_x A_2 - \partial_y A_1)\mathrm{d}x\wedge\mathrm{d}y,\\
\mathrm{d}\beta &= (\partial_x B_1 + \partial_y B_2 + \partial_z B_3)\mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z.
\end{aligned}$$

For example, the $\mathrm{d}y\wedge\mathrm{d}z$ coefficient receives $\partial_y A_3$ from $\mathrm{d}(A_3\mathrm{d}z)$ and $-\partial_z A_2$ from $\mathrm{d}(A_2\mathrm{d}y)$. Cyclic permutation gives the other two coefficients. For $\mathrm{d}\beta$, repeated factors vanish and each surviving cyclic order has positive sign. The displayed coefficients are the Cartesian definitions of gradient, curl and divergence. ∎

Applying $\mathrm{d}^2 = 0$ in degrees zero and one respectively gives

$$\operatorname{curl}\operatorname{grad} f = 0, \qquad \operatorname{div}\operatorname{curl} A = 0.$$

10.6 Closed and exact forms

Definition 10.18 (Closed form)**.** A form ω is *closed* if $\mathrm{d}\omega = 0$.

Definition 10.19 (Exact form)**.** A k-form with $k \geq 1$ is *exact* if it equals $\mathrm{d}\eta$ for some $(k-1)$-form η.

Every exact form is closed by $\mathrm{d}^2 = 0$. For a smooth 1-form α on M and a smooth path $\gamma\colon [a, b] \to M$, the pullback definition gives $\gamma^*\alpha = \alpha_{\gamma(t)}(\gamma'(t))\mathrm{d}t$, so this is also the integral of the pullback's coefficient. If $\alpha = \mathrm{d}f$, the chain rule and the fundamental theorem of calculus give $\int_\gamma \mathrm{d}f = f(\gamma(b)) - f(\gamma(a))$.

Proposition 10.20 (A closed form that is not exact). *On* $\mathbb{R}^2 \setminus \{0\}$ *the angular form*

$$\omega_{\text{ang}} = \frac{x\mathrm{d}y - y\mathrm{d}x}{x^2 + y^2}$$

is closed but not exact.

Proof. Its exterior derivative has $\mathrm{d}x \wedge \mathrm{d}y$ coefficient

$$\partial_x \frac{x}{x^2+y^2} + \partial_y \frac{y}{x^2+y^2} = \frac{2}{x^2+y^2} - \frac{2(x^2+y^2)}{(x^2+y^2)^2} = 0.$$

For $\gamma(t) = (\cos t, \sin t)$, $0 \le t \le 2\pi$, substitution gives $\gamma^*\omega_{\text{ang}} = \mathrm{d}t$. Its integral is 2π. If it were exact, the endpoint formula would give zero on this closed path. This contradiction proves nonexactness. ∎

For a smooth loop $\gamma(t) = r(t)(\cos\phi(t), \sin\phi(t))$, where $r(t) > 0$, differentiation gives $x\dot{y} - y\dot{x} = r^2\dot{\phi}$. Thus $\gamma^*\omega_{\text{ang}} = \dot{\phi}\mathrm{d}t$. Lecture 3 lifts $\gamma/|\gamma|$ in turns, as drawn in its continuous-phase figure. Multiplying that lift by 2π gives ϕ. The angle ϕ is smooth because it locally differs from a smooth angle branch along γ by a constant multiple of 2π. The integral divided by 2π is therefore the winding number. The same computation with variable r and ϕ gives $\omega_{\text{ang}} = \mathrm{d}\phi$ on an angle chart. There is no global smooth angle function on the punctured plane. On the polar chart of §10.4, substitution gives $f^*\omega_{\text{ang}} = \mathrm{d}\theta$: the numerator pulls back to $r^2\mathrm{d}\theta$ and the denominator to r^2. Thus a counter-clockwise circle of any radius $R > 0$ has integral 2π. The angle branch $\phi_1 \in (-\pi, \pi)$ is defined off the nonpositive x-axis. The branch $\phi_2 \in (0, 2\pi)$ is defined off the nonnegative x-axis. On the upper half-plane the branches satisfy $\phi_2 = \phi_1$. On the lower half-plane they satisfy $\phi_2 = \phi_1 + 2\pi$. Their differentials agree on both components of the overlap. The continuous phase of Lecture 3 records a full turn even when the planar endpoints agree. The two angle charts in the figure show why their local primitives do not give a single global function.

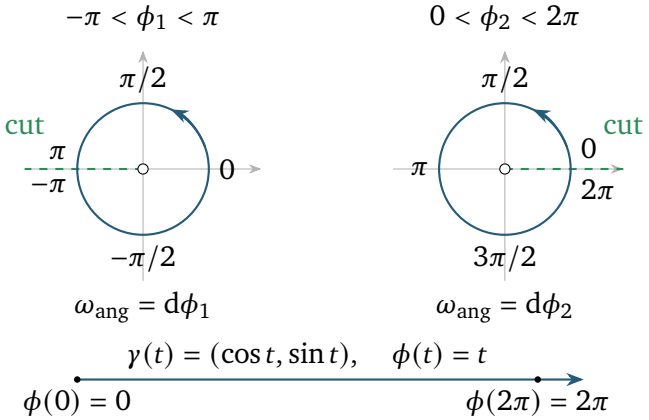


Figure 10.4: The angle branches have different radial cuts and differ by a constant on each component of the overlap. The continuous angle along a positive loop changes by 2π although the planar endpoints agree.

10.7 The area form of the sphere

The vector-calculus dictionary assigns the ambient 2-form β to the radial field (x, y, z). It measures the field's flux through a tangent parallelogram. On the unit sphere this field

is the outward unit normal, so its flux measures signed area. The inclusion $i\colon S^2 \to \mathbb{R}^3$ pulls the ambient form $\beta = x\,dy\wedge dz + y\,dz\wedge dx + z\,dx\wedge dy$ back to $\omega_{S^2} = i^*\beta$. At $p \in S^2$, for $v, w \in T_pS^2$, it evaluates as

$$(\omega_{S^2})_p(v,w) = \det[p\ v\ w] = p\cdot(v\times w).$$

Expanding the determinant along its first column gives

$$x(v_2w_3 - v_3w_2) + y(v_3w_1 - v_1w_3) + z(v_1w_2 - v_2w_1),$$

which is the evaluation of β at $p = (x,y,z)$. This is signed area measured with the outward unit normal p.

Example 10.21 (Angular area)**.** For angular parameters put

$$s(\theta,\phi) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta),$$

with $0 < \theta < \pi$ and ϕ in an open interval of length less than 2π. Its coordinate tangents are

$$s_\theta = (\cos\theta\cos\phi, \cos\theta\sin\phi, -\sin\theta), \qquad s_\phi = (-\sin\theta\sin\phi, \sin\theta\cos\phi, 0).$$

Their cross product is $s_\theta \times s_\phi = \sin\theta\, s$, so

$$s^*\omega_{S^2} = \sin\theta\, d\theta\wedge d\phi.$$

For the inverse stereographic map from the north pole, set $\rho = 1 + u^2 + v^2$ and $\sigma(u,v) = (2u, 2v, u^2+v^2-1)/\rho$. At $(u,v) = (0,0)$ the point is $p = (0,0,-1)$, while the tangent columns are $\sigma_u = 2e_1$ and $\sigma_v = 2e_2$. Their cross product is $4e_3$, opposite to the outward normal p. The figure shows both tangents in the plane $z = -1$.

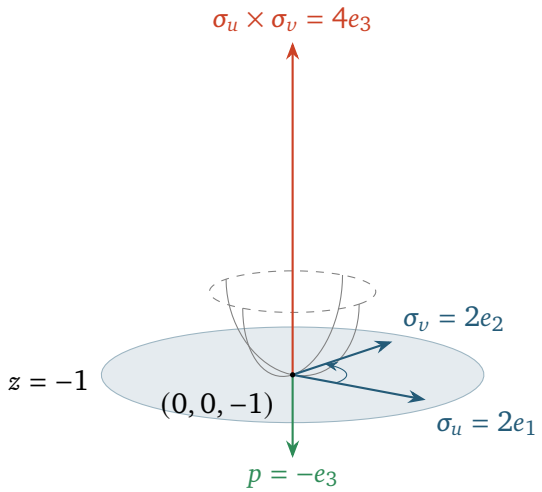


Figure 10.5: The shaded ellipse lies in the tangent plane $z = -1$ at the south pole. The ordered tangents have cross product $+4e_3$, opposite to the outward normal $p = -e_3$.

Example 10.22 (Stereographic area)**.** The derivative matrix of σ is

$$(\sigma_u\ \sigma_v) = \frac{1}{\rho^2}\begin{pmatrix} 2(\rho - 2u^2) & -4uv \\ -4uv & 2(\rho - 2v^2) \\ 4u & 4v \end{pmatrix}.$$

The cross product of these columns is

$$\sigma_u \times \sigma_v = \frac{(-8u\rho, -8v\rho, 4\rho(1 - u^2 - v^2))}{\rho^4} = -\frac{4\sigma}{\rho^2}.$$

Since $\sigma \cdot \sigma = 1$, the pullback is

$$\sigma^*\omega_{S^2} = -\frac{4}{(1 + u^2 + v^2)^2}\mathrm{d}u \wedge \mathrm{d}v.$$

The south-pole sign test agrees with the negative coefficient. Lecture 12 formalizes this comparison as orientation reversal.

Self-check. Compute $\mathrm{d}(x\mathrm{d}y \wedge \mathrm{d}z)$ on $\mathbb{R}^3$ and its pullback to S^2.

Answer. The derivative is $\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z$. Its pullback is zero: an alternating trilinear form vanishes on the two-dimensional tangent space of the sphere.

10.8 Uniqueness of the exterior derivative *(optional)*

The pullback, vector-calculus and area calculations do not depend on this uniqueness argument. Later results on integration and Cartan's formula also do not depend on it.

Sketch of uniqueness. Let d' have the properties in Theorem 10.12. A smooth *cutoff function* equals 1 near a chosen point and 0 outside a chosen chart neighbourhood. Such functions are constructed in a smaller coordinate ball using $e^{-1/t}$ for $t > 0$, extended by zero for $t \le 0$. If α vanishes near p, choose a cutoff χ supported where $\alpha = 0$. Since $\chi\alpha = 0$, the product rule at p gives $(\mathrm{d}'\alpha)_p = 0$. Thus d' depends only on the form near p. Use a cutoff to extend local coefficient and coordinate functions smoothly to M, without changing them near p. The identity $\mathrm{d}'\mathrm{d}f = (\mathrm{d}')^2 f = 0$ for these extensions and the product rule force $\mathrm{d}'(\sum_I a_I \mathrm{d}x^I) = \sum_I \mathrm{d}a_I \wedge \mathrm{d}x^I$ at p. This proves uniqueness, with the cutoff construction left at sketch level. See Tu, §19.4. ∎

Summary. Forms alternate in their tangent arguments and use the determinant wedge. The exterior derivative is independent of coordinates, squares to zero and satisfies the graded product rule. Pullback exists under every smooth map and commutes with this derivative. A closed form need not be exact, as the angular form's nonzero loop integral proves. The sphere's outward area has negative coefficient in the north-pole stereographic coordinates. Lecture 11 differentiates forms along flows.

Reading. Tu, *An Introduction to Manifolds*, §§17–19; *Linear Algebra*, Lecture 12, exterior powers and the determinant wedge.

Lecture 11

Cartan's calculus

Objectives. *Hamiltonian motion moves sets of initial positions and momenta through phase space. We prove that this motion preserves phase-space volume. Pullback compares a form at a moving point with a form at the starting point. Differentiating this comparison gives its Lie derivative. Cartan's formula expresses that derivative using exterior differentiation and insertion of a vector field. We prove the formula, including its time-dependent version, and apply it to volume preservation and Hamiltonian motion.*

By the end of this lecture you will be able to:

- Compute contraction with its first-slot sign convention.
- Prove Cartan's formula and differentiate a time-dependent pullback.
- Determine whether a local flow preserves a chosen volume form.
- Derive Liouville's theorem from closedness and nondegeneracy.

11.1 Phase space and Hamiltonian motion

A particle's position alone does not determine its motion. We also specify its momentum.

Definition 11.1. The *phase space* of a particle moving in $\mathbb{R}^n$ is $T^*\mathbb{R}^n$. A point (q, p) records a position $q \in \mathbb{R}^n$ and a momentum $p \in T_q^*\mathbb{R}^n$.

We write the position coordinates as q^i and the momentum coefficients as p_i. The lower momentum index records its covector transformation law.

Physical input. For a classical Hamiltonian model with smooth energy function $H : T^*\mathbb{R}^n \to \mathbb{R}$, motion obeys Hamilton's equations:

$$\dot{q}^i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q^i}.$$

For a harmonic oscillator of mass $m > 0$ and angular frequency $\nu > 0$, the energy is

$$H(q, p) = \frac{p^2}{2m} + \frac{m\nu^2 q^2}{2}.$$

These equations give $\dot{q} = p/m$ and $\dot{p} = -m\nu^2 q$. Differentiating the first equation gives the oscillator equation $\ddot{q} = -\nu^2 q$. The complete flow is

$$\varphi_t(q, p) = \left(q\cos(\nu t) + \frac{p}{m\nu}\sin(\nu t), -m\nu q\sin(\nu t) + p\cos(\nu t)\right).$$

Differentiation verifies Hamilton's equations and the initial value (q, p). In units with $m = \nu = 1$, the Hamiltonian is $(q^2 + p^2)/2$ and the field is $p\partial_q - q\partial_p$. Its flow becomes

$$\varphi_t(q, p) = (q\cos t + p\sin t, -q\sin t + p\cos t).$$

From $(1, 0)$ it reaches $(0, -1)$ at time $\pi/2$. The figure shows this clockwise motion and the stationary origin.

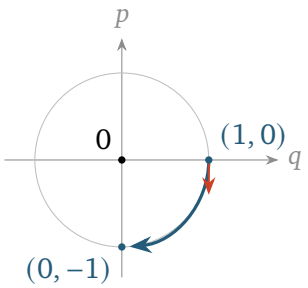


Figure 11.1: Positive-time oscillator motion is clockwise in the (q, p) plane for $m = \nu = 1$. The arc follows the orbit from $(1, 0)$ towards $(0, -1)$, with the straight arrow marking the initial velocity.

We express Hamilton's equations using the forms of Lecture 10.

Definition 11.2. On $T^*\mathbb{R}^n$, the *canonical 1-form* θ evaluates a phase-space tangent vector on its position component using the momentum covector. In these coordinates it is given by

$$\theta = p_i \mathrm{d}q^i.$$

For a position-coordinate change with Jacobian J, write p as a row and $\mathrm{d}q$ as a column. The covector rule of Lecture 10 gives $p' = pJ^{-1}$ and $\mathrm{d}q' = J\mathrm{d}q$. Thus $p'\mathrm{d}q' = p\mathrm{d}q$. The definition is independent of the position coordinates. We denote the following 2-form by ω:

$$\omega = \sum_i \mathrm{d}q^i \wedge \mathrm{d}p_i = -\mathrm{d}\theta.$$

The minus sign follows from $\mathrm{d}\theta = \sum_i \mathrm{d}p_i \wedge \mathrm{d}q^i$. For a tangent field $X = \dot{q}^i \partial_{q^i} + \dot{p}_i \partial_{p_i}$, inserting X into the first argument gives the 1-form

$$\omega(X, \cdot) = \sum_i (\dot{q}^i \mathrm{d}p_i - \dot{p}_i \mathrm{d}q^i).$$

Equating it with $\mathrm{d}H$ recovers Hamilton's equations by comparing coefficients. The form is closed because its coefficients are constant. It is nondegenerate: if this insertion is zero, all coefficients of X vanish. Nondegeneracy lets the energy determine a unique field. We will use closedness to prove preservation of phase-space volume.

11.2 Contraction with a vector field

The insertion used in Hamilton's equations applies to forms of every degree. It also lets us differentiate a pullback without solving for the flow. Let X be a smooth vector field on a smooth manifold M.

Definition 11.3. For $k \geq 1$, the *interior product*, or *contraction*, with X is the map

$$\iota_X \colon \Omega^k(M) \longrightarrow \Omega^{k-1}(M)$$

that fills the first argument. For $p \in M$ and $v_j \in T_pM$ it is given by

$$(\iota_X \alpha)_p(v_1, \ldots, v_{k-1}) = \alpha_p(X_p, v_1, \ldots, v_{k-1}).$$

For a 1-form this is the function $\alpha(X)$. For a function f set $\iota_X f = 0$.

The resulting form is smooth because its coordinate coefficients are sums of products of smooth coefficients of X and α.

Theorem 11.4 (Contraction rules). *Contraction is real-linear, lowers degree by one, and satisfies*

$$\iota_X(\alpha \wedge \beta) = (\iota_X\alpha) \wedge \beta + (-1)^k \alpha \wedge \iota_X\beta, \qquad \iota_X^2 = 0,$$

where α has degree k. The first identity is the graded antiderivation rule. *Contraction is also linear over smooth functions in both X and the form.*

Proof. For 1-forms $\alpha_1, \dots, \alpha_\ell$, expand the determinant defining their wedge along its first column. This gives

$$\iota_X(\alpha_1 \wedge \cdots \wedge \alpha_\ell) = \sum_{j=1}^{\ell} (-1)^{j-1} \alpha_j(X) \alpha_1 \wedge \cdots \wedge \widehat{\alpha_j} \wedge \cdots \wedge \alpha_\ell.$$

The hat means that the factor is omitted. For the wedge of a block of k factors and a second block, split this sum according to the block containing the omitted factor. Omissions in the second block acquire $(-1)^k$. This is the stated graded antiderivation rule. Basis wedges span all forms, so bilinearity extends the identity to arbitrary forms. Applying contraction twice inserts X twice. Alternation makes this zero. On degrees zero and one it is zero by definition. Pointwise multilinearity proves both function-linear assertions and real-linearity. ∎

For example, $\iota_{\partial_x}(\mathrm{d}x \wedge \mathrm{d}y) = \mathrm{d}y$. For the radial field $X = x\partial_x + y\partial_y$,

$$\iota_X(\mathrm{d}x \wedge \mathrm{d}y) = x\mathrm{d}y - y\mathrm{d}x.$$

Filling the last argument instead would reverse this sign. This contraction is the numerator of Lecture 10's angular form.

11.3 Lie derivatives of forms and tensors

Let φ be the maximal local flow of X, supplied by Theorem 9.8. Near each $p \in M$, its maps are defined for all sufficiently small times. Pullback gives a form in the fixed exterior space at p, as in the figure.

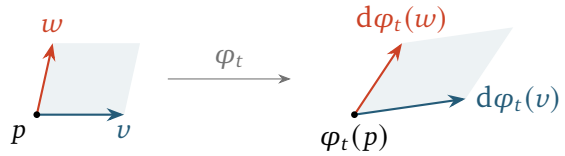


Figure 11.2: Pullback evaluates the form at $\varphi_t(p)$ on the transported vectors $\mathrm{d}\varphi_t(v)$ and $\mathrm{d}\varphi_t(w)$. The pullback at p gives this same evaluation.

Definition 11.5. The *Lie derivative of a form* is the operator

$$L_X \colon \Omega^k(M) \longrightarrow \Omega^k(M)$$

given by

$$(L_X\alpha)_p = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_0 (\varphi_t^*\alpha)_p.$$

In coordinates the pullback has smooth coefficients jointly in time and position. Their time derivatives therefore define a smooth form. For a function, differentiating $f(\varphi_t(p))$ gives $L_X f = Xf$, agreeing with Lecture 9.

Theorem 11.6 (Lie derivative rules · *proved given smooth local flows*)**.** *The Lie derivative is real-linear and preserves degree. It satisfies*

$$L_X(\alpha \wedge \beta) = L_X\alpha \wedge \beta + \alpha \wedge L_X\beta, \qquad L_X \mathrm{d}\alpha = \mathrm{d}L_X\alpha.$$

Proof. Pullback preserves wedge by Theorem 10.14. Differentiating this identity at zero and using the ordinary product rule proves the first identity. No graded sign occurs because the derivative preserves degree. For the second, differentiate $\varphi_t^* \mathrm{d}\alpha = \mathrm{d}(\varphi_t^*\alpha)$. In a fixed chart, exterior differentiation differentiates coefficient functions in space. Time and space derivatives commute by joint smoothness. Thus the time derivative passes through d, proving the assertion. ∎

For example, the field $X = x\partial_x$ has flow $\varphi_t(x) = e^t x$. It gives $\varphi_t^* \mathrm{d}x = e^t \mathrm{d}x$, so $L_X \mathrm{d}x = \mathrm{d}x$.

Remark 11.7 (Tensor pullback)**.** Lie derivatives extend to tensors using the same comparison.

Definition 11.8. For a diffeomorphism $f\colon M \to N$ and a tensor of type (r, s) on N, define f^*T by sending each tangent argument forward by $\mathrm{d}f_p$ and each cotangent argument forward by the dual of $(\mathrm{d}f_p)^{-1}$. These arguments all lie at $f(p)$, where T is evaluated. The *Lie derivative of a tensor* $L_X T$ is the derivative of $\varphi_t^* T$ at zero.

This exists by the same smooth-coordinate argument. For vectors this pullback is $(\varphi_{-t})_* Y$, so it recovers $L_X Y = [X, Y]$ from Lecture 9. General tensor pullback here requires a diffeomorphism. Lecture 10's pullback formula for forms applies verbatim to covariant tensors along any smooth map. For example, on $\mathbb{R}$ the field $X = x\partial_x$ has flow $\varphi_t(x) = e^t x$. The vector field $Y = \partial_x$ pulls back to $e^{-t}\partial_x$. Its derivative at zero is $-\partial_x = [x\partial_x, \partial_x]$. Lecture 16 uses tensor Lie derivatives to characterize flows that preserve a metric.

11.4 Cartan's formula and time dependence

Cartan's formula computes the derivative of a form using only exterior differentiation and contraction. Both are local operations on coefficients.

Theorem 11.9 (Cartan's formula · *proved given smooth local flows*)**.** *On differential forms,*

$$L_X = \mathrm{d}\iota_X + \iota_X \mathrm{d}.$$

Proof. We compare the two operators on functions and their differentials. Write $C_X = \mathrm{d}\iota_X + \iota_X \mathrm{d}$. The two graded product rules imply that C_X is an ordinary derivation. To see the cancellation explicitly, let $\deg \alpha = k$. In $C_X(\alpha \wedge \beta)$ the terms $(\iota_X\alpha) \wedge \mathrm{d}\beta$ have total coefficient $(-1)^{k-1} + (-1)^k = 0$. The terms $\mathrm{d}\alpha \wedge \iota_X\beta$ have coefficient $(-1)^k + (-1)^{k+1} = 0$. The remaining terms are $(C_X\alpha) \wedge \beta + \alpha \wedge C_X\beta$. The calculation also holds when $k = 0$, since contraction of a function is zero.

On functions, $C_X f = \iota_X \mathrm{d}f = Xf = L_X f$. On their differentials, $C_X \mathrm{d}f = \mathrm{d}(Xf) = L_X \mathrm{d}f$, using $\mathrm{d}^2 = 0$ and Theorem 11.6. Both sides depend only on the form near the point in question. For small time the trajectory stays in any chosen chart neighbourhood. Uniqueness makes the restricted field's flow agree with the original flow there. The Lie derivative therefore

commutes with restriction to that chart. Locally every form is a sum of products of functions and coordinate differentials. The derivation rules therefore prove equality on every form in every chart. ∎

This is Tu, Theorem 20.10, with contraction in the first slot. For example, for $X = \partial_x$ and $\alpha = x\mathrm{d}y$, contraction gives zero while $\iota_X \mathrm{d}\alpha = \iota_X(\mathrm{d}x \wedge \mathrm{d}y) = \mathrm{d}y$. Thus $L_X\alpha = \mathrm{d}y$. The flow gives an independent check: $\varphi_t(x, y) = (x + t, y)$, so $\varphi_t^*(x\mathrm{d}y) = (x + t)\mathrm{d}y$ has derivative $\mathrm{d}y$. Cartan's formula acts on forms. It does not replace the vector-field formula $L_X Y = [X, Y]$.

For fixed α, the group law for an autonomous flow also gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\varphi_t^*\alpha = \varphi_t^* L_X \alpha.$$

Indeed, write $\varphi_{t+h}^* = \varphi_t^*\varphi_h^*$ and differentiate in h at zero. This identity holds wherever the indicated local maps exist.

Self-check. For a smooth function $f\colon \mathbb{R} \to \mathbb{R}$ and $X = x\partial_x$, compute $L_X(f\mathrm{d}x)$.

Answer. Cartan gives $\mathrm{d}\iota_X(f\mathrm{d}x) = \mathrm{d}(xf) = (xf' + f)\mathrm{d}x$. The other term is zero because every 2-form on $\mathbb{R}$ vanishes. The flow gives $\varphi_t^*(f\mathrm{d}x) = f(e^t x)e^t\mathrm{d}x$. Differentiating at zero gives the same coefficient $xf' + f$.

11.4.1 Time-dependent transport and fluid volume

A fluid's velocity can change with time. Its trajectories then depend on both their starting time and their current time.

Definition 11.10. A *time-dependent field* V_t has coefficients smooth jointly in $(t, p) \in I \times M$, where $I \subseteq \mathbb{R}$ is an open interval. Its *evolution maps* $\psi_{t,s}$ solve $\partial_t \psi_{t,s}(p) = (V_t)_{\psi_{t,s}(p)}$ with $\psi_{s,s}(p) = p$.

Their local existence and joint smoothness follow by applying Theorem 9.8 to the field $(1, V_t)$ on $I \times M$. Adding time as a coordinate makes this field autonomous. Uniqueness gives the local inverse $\psi_{s,t}$, so each evolution map is a diffeomorphism between its open domain and image.

Definition 11.11. A *smooth family of forms* α_t has coefficients smooth in (t, p). Its derivative $\partial_t \alpha_t$ differentiates those coefficients in fixed, time-independent charts.

The pullback changes because the points move and because the form itself changes.

Proposition 11.12 (Time-dependent pullback · *proved given smooth local evolution maps*). *Fix an initial time $s \in I$ and write $\psi_t = \psi_{t,s}$. On every region where these maps are defined,*

$$\frac{\mathrm{d}}{\mathrm{d}t}\psi_t^*\alpha_t = \psi_t^*\big(L_{V_t}\alpha_t + \partial_t\alpha_t\big) = \psi_t^*\big(\mathrm{d}\iota_{V_t}\alpha_t + \iota_{V_t}\mathrm{d}\alpha_t + \partial_t\alpha_t\big).$$

Proof. Fix t and define $\eta_h = \psi_{t+h} \circ \psi_t^{-1}$ on the current image, shrinking neighbourhoods and the range of h if necessary. Its initial value is the identity and $\partial_h \eta_h|_0 = V_t$. Joint smoothness gives the coordinate expansions

$$\eta_h(q) = q + hV_t(q) + O(h^2), \qquad D\eta_h(q) = I + hDV_t|_q + O(h^2).$$

These agree to first order with the autonomous flow of the fixed field V_t. The coordinate pullback formula uses just these first derivatives in h, so $\partial_h(\eta_h^*\alpha_t)|_0 = L_{V_t}\alpha_t$. Now $\psi_{t+h}^*\alpha_{t+h} =$

$\psi_t^* \eta_h^* \alpha_{t+h}$. The family $\eta_h^* \alpha_{t+k}$ is jointly smooth in h and k. The chain rule along $h = k$ adds its two partial derivatives at zero: $L_{V_t} \alpha_t$ and $\partial_t \alpha_t$. The fixed pullback ψ_t^* carries their sum back to the starting point. Cartan's formula gives the second equality. No autonomous group law for ψ_t was used. ∎

For example, $\psi_t(x, y) = (x + t^2 y, y)$ is generated from time zero by $V_t = 2ty\partial_x$. For $\alpha_t = tx\mathrm{d}y$ its pullback is $t(x + t^2 y)\mathrm{d}y$. The derivative is $(x + 3t^2 y)\mathrm{d}y$. Here $L_{V_t} \alpha_t = 2t^2 y\mathrm{d}y$ and $\partial_t \alpha_t = x\mathrm{d}y$. Pulling their sum back gives precisely that same derivative. Omitting the explicit time derivative would give a different answer.

For a fluid in Euclidean space, let v_t be its smooth velocity field.

Physical input. In the constant-density incompressible-fluid model, the velocity satisfies $\sum_i \partial_i v_t^i = 0$ at every point and time.

With Euclidean volume $\mu = \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^m$, Cartan's formula gives $L_{v_t} \mu = \mathrm{d}\iota_{v_t} \mu$. Its coefficient is $\sum_i \partial_i v_t^i$, because only the derivative in the omitted coordinate survives. Thus this Lie derivative is zero. The time-dependent pullback formula, with $\partial_t \mu = 0$, therefore proves $\psi_t^* \mu = \mu$ wherever fluid trajectories exist. Every bounded parcel keeps its Euclidean volume in physical space throughout any interval on which its trajectories exist. Hamiltonian volume preservation concerns initial conditions in phase space.

Self-check. For the evolution $\psi_t(x, y) = (x + t^2 y, y)$ and the form $\alpha_t = (x^2 + ty)\mathrm{d}y$, compute $\frac{\mathrm{d}}{\mathrm{d}t} \psi_t^* \alpha_t$. Evaluate its $\mathrm{d}y$ coefficient at $t = 1$ and $(x, y) = (2, -1)$.

Answer. The velocity is $V_t = 2ty\partial_x$, so $L_{V_t} \alpha_t = 4txy\mathrm{d}y$ and $\partial_t \alpha_t = y\mathrm{d}y$. Pulling their sum back gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \psi_t^* \alpha_t = \big(4ty(x + t^2 y) + y\big)\mathrm{d}y.$$

The requested coefficient is $4(-1)(2 - 1) - 1 = -5$. The term $y\mathrm{d}y$ comes from the explicit time dependence.

11.5 Volume forms and divergence

Definition 11.13. On an m-manifold, a *volume form* μ is a smooth m-form that is nonzero at every point.

Definition 11.14. For a smooth field X, the *divergence relative to* μ is the function $\mathrm{div}_\mu X$ satisfying

$$L_X \mu = (\mathrm{div}_\mu X)\mu.$$

Since the top exterior space is one-dimensional, this function exists uniquely. It is smooth: divide the local coefficient of $L_X \mu$ by the nowhere-zero coefficient of μ.

Definition 11.15. A flow *preserves* a form if its pullback equals that form on every time map's domain. Such a form is called *invariant* under the flow.

Proposition 11.16 (Divergence and volume preservation · *proved given smooth local flows*)**.** *In local coordinates, for* $\mu = \rho \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^m$*, with* $\rho \neq 0$*,*

$$\operatorname{div}_\mu X = \frac{1}{\rho} \sum_{i=1}^{m} \partial_i(\rho X^i).$$

The local flow of X *preserves* μ *if and only if* $\operatorname{div}_\mu X = 0$ *everywhere.*

Proof. Since there are no forms of degree $m+1$, Cartan gives $L_X\mu = \mathrm{d}\iota_X\mu$. The contraction formula gives

$$\iota_X\mu = \sum_i (-1)^{i-1} \rho X^i \mathrm{d}x^1 \wedge \cdots \wedge \widehat{\mathrm{d}x^i} \wedge \cdots \wedge \mathrm{d}x^m.$$

After differentiation, only the x^i derivative of the ith coefficient survives. Moving $\mathrm{d}x^i$ to its ordered place gives another $(-1)^{i-1}$. Thus the resulting top-form coefficient is $\sum_i \partial_i(\rho X^i)$. Dividing by ρ proves the formula. If divergence vanishes, the finite-time pullback derivative is zero. Along each allowed time interval it is constant and equals μ at time zero. Conversely, differentiating invariance at zero gives $L_X\mu = 0$, hence zero divergence. ∎

At each starting point, Taylor expansion of the pullback gives

$$\varphi_t^*\mu = \big(1 + t \operatorname{div}_\mu X + O(t^2)\big)\mu.$$

Thus divergence is the instantaneous relative rate of volume expansion at that point. Positive divergence means expansion to first order.

For Euclidean volume, $\rho = 1$ gives the usual sum $\sum_i \partial_i X^i$. For example, $X = x\partial_x + y\partial_y$ has divergence 2 relative to $\mathrm{d}x \wedge \mathrm{d}y$. Its flow multiplies that form by e^{2t}. On the punctured plane choose instead $\mu' = (x^2+y^2)^{-1}\mathrm{d}x \wedge \mathrm{d}y$. The same field gives

$$\operatorname{div}_{\mu'} X = (x^2+y^2)\left(\partial_x \frac{x}{x^2+y^2} + \partial_y \frac{y}{x^2+y^2}\right) = 0.$$

The two derivatives sum to $2/r^2 - 2r^2/r^4 = 0$, where $r^2 = x^2+y^2$. Thus its flow preserves μ' while expanding Euclidean area. The choice of volume form matters. In polar coordinates, $\mathrm{d}x \wedge \mathrm{d}y = r\mathrm{d}r \wedge \mathrm{d}\theta$, so $\mu' = (\mathrm{d}r/r) \wedge \mathrm{d}\theta$. Dilation sends r to $e^t r$ and leaves $\mathrm{d}r/r$ unchanged. For an annulus with $r_2 > r_1 > 0$, its weighted area is

$$2\pi \int_{r_1}^{r_2} \frac{\mathrm{d}r}{r} = 2\pi \log(r_2/r_1).$$

Scaling both radii leaves this ratio unchanged, as the figure illustrates.

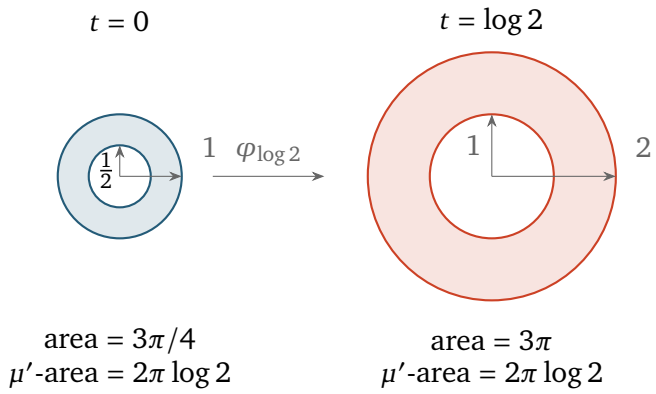


Figure 11.3: Dilation at time $t = \log 2$ sends the annulus with radii $1/2, 1$ to the annulus with radii $1, 2$. The weighted area for μ' stays fixed under this fourfold increase in Euclidean area.

11.6 Symplectic forms and Liouville's theorem

The canonical 2-form in §11.1 is closed and nondegenerate. Closedness will make its Lie derivative vanish. Nondegeneracy makes the Hamiltonian equation determine a unique field.

Definition 11.17. On a $2n$-manifold with $n \geq 1$, a *symplectic form* ω is a closed smooth 2-form that is *nondegenerate*: if $\omega_p(v, w) = 0$ for every $w \in T_pM$, then $v = 0$.

Definition 11.18. For a symplectic form ω on M and a smooth Hamiltonian function $H\colon M \to \mathbb{R}$, the *Hamiltonian field* X_H is determined by

$$\iota_{X_H}\omega = \mathrm{d}H.$$

Nondegeneracy makes $v \mapsto \omega_p(v, \cdot)$ an isomorphism. Its inverse has smooth matrix entries in charts, so the field exists uniquely and is smooth. We use the sign convention $\iota_{X_H}\omega = \mathrm{d}H$, which gives Hamilton's equations in the canonical coordinates above.

Lemma 11.19. *For a symplectic form on a $2n$-manifold, the top form ω^n is nowhere zero.*

Proof. Choose tangent vectors a, b with $\omega(a, b) = 1$. Every tangent vector v splits into a component $\omega(v, b)a - \omega(v, a)b$ in their plane and a component orthogonal to that plane for ω, meaning that its pairing with both a and b is zero. The restriction to the complementary subspace is nondegenerate, since a vector orthogonal to it and to the plane is orthogonal to the whole space. Repeat in the complement. In the resulting dual basis α^j, β^j, $\omega = \sum_j \alpha^j \wedge \beta^j$. Its nth power is $n!\alpha^1 \wedge \beta^1 \wedge \cdots \wedge \alpha^n \wedge \beta^n$, which is nonzero. Hence $\omega^n/n!$ is a volume form. ∎

For the canonical form the same wedge expansion gives

$$\frac{\omega^n}{n!} = \mathrm{d}q^1 \wedge \mathrm{d}p_1 \wedge \cdots \wedge \mathrm{d}q^n \wedge \mathrm{d}p_n.$$

The factor $n!$ counts the orders of the n distinct 2-form factors. Dividing by it gives the usual coordinate volume element.

Corollary 11.20 (Liouville · *proved given smooth local flows*)**.** *For a symplectic form and its Hamiltonian field,*

$$L_{X_H}\omega = 0, \qquad L_{X_H}(\omega^n) = 0.$$

Consequently its local flow preserves both the symplectic form and the volume form $\omega^n/n!$.

Proof. Cartan's formula and the defining Hamiltonian equation give $L_{X_H}\omega = \mathrm{d}^2 H + \iota_{X_H}\mathrm{d}\omega$. Both terms vanish because $\mathrm{d}^2 = 0$ and ω is closed. The derivation rule then gives zero on the n-fold wedge power ω^n. The finite-time pullback identity proves preservation. ∎

A set of initial positions and momenta can change shape under Hamiltonian motion, but its phase-space volume stays constant while the trajectories exist. This conclusion concerns the volume determined by $\omega^n/n!$. For example, the Hamiltonian $H = p^2/2$ on $\mathbb{R}^2$ gives $\dot{q} = p$ and $\dot{p} = 0$. Its flow is the shear

$$\varphi_t(q,p) = (q + tp, p), \qquad D\varphi_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad \det D\varphi_t = 1.$$

The pullback is $(\mathrm{d}q + t\mathrm{d}p) \wedge \mathrm{d}p = \mathrm{d}q \wedge \mathrm{d}p$. The figure compares this changing shape with clockwise oscillator motion.

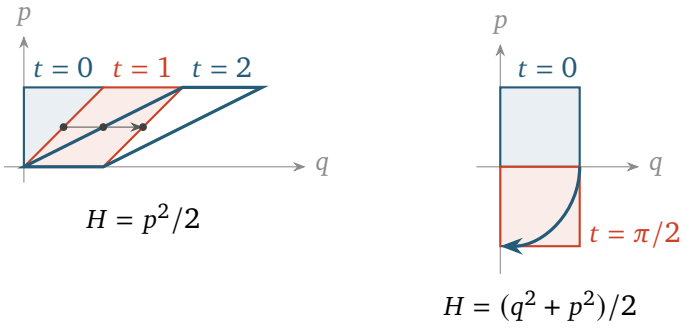


Figure 11.4: The shear sends a unit square to parallelograms of area one at times $0, 1, 2$, with dots tracking $(1/2, 1/2)$. The oscillator rotates the unit square clockwise through a quarter period and also preserves area.

11.6.1 The Poisson bracket and the field bracket

Definition 11.21. The canonical *Poisson bracket* of smooth functions on $T^*\mathbb{R}^n$ is

$$\{f,g\} = \sum_i \left(\frac{\partial f}{\partial q^i}\frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial g}{\partial q^i} \right).$$

The Hamiltonian field acts on a smooth function h by $X_H h = \{h, H\}$, as its coordinate coefficients show.

Example 11.22 (The bracket sign)**.** For one degree of freedom, take $f = q^2/2$ and $g = p^2/2$. Their fields are $X_f = -q\partial_p$ and $X_g = p\partial_q$. The coordinate bracket is $[X_f, X_g] = -q\partial_q + p\partial_p$. The Poisson bracket gives $\{f,g\} = qp$, whose field is $X_{qp} = q\partial_q - p\partial_p$. Thus this pair satisfies $[X_f, X_g] = -X_{\{f,g\}}$, using Lecture 9's bracket $[X,Y]h = X(Yh) - Y(Xh)$.

The minus sign holds for all smooth f and g. To prove it, differentiate the evaluation defining contraction under pullback. The product rule, with the tensor Lie derivative of Y, gives

$$L_X(\iota_Y\omega) = \iota_{[X,Y]}\omega + \iota_Y(L_X\omega).$$

Apply this identity with $X = X_f$ and $Y = X_g$. Liouville makes the last term zero. Since Lie differentiation commutes with d,

$$\iota_{[X_f,X_g]}\omega = L_{X_f}\mathrm{d}g = \mathrm{d}(X_f g) = -\mathrm{d}\{f,g\}.$$

Nondegeneracy therefore gives $[X_f, X_g] = -X_{\{f,g\}}$. This is a reversal of bracket sign, as for the linear fields $X_A(x) = Ax$ in Lecture 9. The Poisson bracket acts on functions, whereas the bracket of Lecture 9 acts on vector fields.

Self-check. State in one sentence what the Lie derivative of a form is and why we introduced it. Explain how Cartan's formula proves Hamiltonian volume preservation.

Answer. The Lie derivative of a form is the infinitesimal change of its flow pullback, introduced to test how a flow changes or preserves that form. For a Hamiltonian field, Cartan gives $L_{X_H}\omega = \mathrm{d}^2H + \iota_{X_H}\mathrm{d}\omega = 0$. The derivation rule gives zero on $\omega^n/n!$, so the flow preserves this volume form.

Summary. Contraction inserts a vector in the first slot and is a graded antiderivation. The Lie derivative of forms differentiates flow pullback. Cartan expresses it as $\mathrm{d}\iota_X + \iota_X\mathrm{d}$. A time-dependent form contributes its explicit time derivative as well. Zero divergence relative to a chosen volume form is equivalent to local flow preservation of that form. A Hamiltonian flow preserves its symplectic form and the associated top wedge volume. Lecture 12 integrates forms and relates their exterior derivatives to boundaries.

Reading. Tu, *An Introduction to Manifolds*, §20.3–§20.5, especially Theorem 20.10, pp. 229–230.

Lecture 12

Orientation, integration and Stokes' theorem

Objectives. *An orientation specifies the sign of a top-degree form. It lets us integrate forms on a manifold. Stokes' theorem identifies the integral of an exterior derivative with the integral on the boundary, using outward-first orientation. We prove it in every dimension and apply it to the flux of the Coulomb form.*

By the end of this lecture you will be able to:

- Determine orientability using charts and top forms.
- Define an integral independent of the oriented charts and partition of unity.
- Prove Stokes' theorem with its support and orientation hypotheses.
- Compute Coulomb flux by removing a ball around its singularity.

Unit 12A Orientation and integration

12.1 Orientations and oriented charts

Definition 12.1 (Orientation of a vector space)**.** For an n-dimensional real vector space with $n \geq 1$, two ordered bases have the same *orientation* if their change-of-basis determinant is positive.

There are two classes.

Definition 12.2 (Smooth frame)**.** A *frame* on an open set is an ordered list of smooth tangent fields that is a basis at every point.

Definition 12.3 (Orientation of a manifold)**.** An *orientation of a manifold* chooses a class in each tangent space, locally represented by a smooth frame. A manifold is *orientable* if it admits such a choice, and *oriented* when a choice has been made.

Definition 12.4 (Oriented atlas)**.** An *oriented atlas* has positive Jacobian determinants on all chart transitions.

Its coordinate frames give compatible orientation classes: the change of frame uses the inverse Jacobian, whose determinant is also positive. Conversely, near any point an orientation is represented by a frame. Its determinant relative to a coordinate frame is continuous and nonzero. Shrink to a connected chart so that the sign is constant. If necessary replace x^1 by $-x^1$. The coordinate frame then represents the chosen orientation. Charts

obtained this way have positive transitions. This proves the equivalence of orientations and oriented atlases.

A nowhere-zero top form μ selects the bases on which its value is positive. These choices are locally represented by coordinate frames, because its coefficient in a connected chart has constant sign.

Definition 12.5 (Orientation-preserving diffeomorphism)**.** For oriented manifolds, a diffeomorphism is *orientation-preserving* when its differential sends positive bases to positive bases.

In positive charts this means its Jacobian determinant is positive.

For example, on $S^n \subset \mathbb{R}^{n+1}$ the form

$$\mu_p(v_1, \dots, v_n) = \det[p\ v_1\ \cdots\ v_n] \qquad (v_j \in T_pS^n)$$

is smooth and nowhere zero: the radial vector p is independent of the n tangent basis vectors. It defines the outward orientation. For $n = 2$ it is the sphere area form of Lecture 10. Lecture 10 computed the north-pole stereographic inverse pullback as $-4\mathrm{d}u\wedge\mathrm{d}v/(1+u^2+v^2)^2$. This chart reverses the outward orientation.

In dimension zero an orientation is a choice of sign $+1$ or -1 at each point. Integration of a function then means its signed sum. This convention includes the endpoints of an oriented interval.

12.2 Partitions of unity and orientability

A form may have support in several overlapping charts. We multiply it by nonnegative smooth functions whose sum is one. Each weighted form then has support inside one chart. A locally finite sum permits this construction on a noncompact manifold. Only finitely many pieces meet a compact support.

Definition 12.6 (Locally finite collection)**.** A collection of sets is *locally finite* if every point has a neighbourhood meeting only finitely many of them.

Definition 12.7 (Partition of unity)**.** A smooth *partition of unity subordinate to an open cover* $\{U_i\}$ consists of nonnegative smooth functions ρ_i such that

$$\sum_i \rho_i = 1, \qquad \operatorname{supp}\rho_i \subset U_i, \qquad \{\operatorname{supp}\rho_i\} \text{ is locally finite.}$$

Several functions may be assigned to the same cover member.

Local finiteness makes each sum finite near any point. It also implies that a compact set meets only finitely many supports: choose finitely many such neighbourhoods covering the compact set.

Theorem 12.8 (Partitions of unity · *sketched; normalization and use proved*)**.** *Every open cover of a smooth manifold admits a subordinate smooth partition of unity.*

Sketch of existence. The second-countability assumption in Lecture 7 is used in this construction. Together with local compactness, it gives a countable exhaustion by compact sets, each lying in the interior of the next. Refining the cover on the successive compact layers gives a locally finite cover by coordinate balls with compact closure, with smaller balls still

covering the manifold. Choose a smooth bump on each ball that is positive on its smaller ball and supported in the larger one. The coordinate construction uses the smooth function $e^{-1/t}$ for $t > 0$, extended by zero for $t \le 0$. The locally finite sum $b = \sum_i b_i$ of these bumps is smooth and strictly positive. Thus $\rho_i = b_i/b$ is smooth and nonnegative, has the required support, and sums to one. The refinement and bump construction are the sketched steps. See Tu, Theorem 13.7 and Appendix C. ∎

Proposition 12.9 (Orientability criteria · *proved given partitions of unity*)**.** *A smooth manifold is orientable if and only if it has an oriented atlas, and if and only if it admits a nowhere-zero smooth top form.*

Proof. The atlas equivalence and the top-form-to-orientation implication were proved above. For the converse, take positively oriented charts and a subordinate partition. Extend $\rho_i \mathrm{d}x_i^1 \wedge \cdots \wedge \mathrm{d}x_i^n$ by zero outside its chart. This is smooth because its support lies inside the chart. Their locally finite sum is smooth. Every summand is nonnegative on a positive tangent basis, and at least one has positive coefficient at every point. The sum therefore never vanishes. In dimension zero the constant function 1 is already a nowhere-zero 0-form, and every discrete manifold has an atlas. ∎

A tangent pair on a Möbius band reverses orientation after one circuit of its central circle. The twisted band in the figure is parametrised by

$$m(t,s) = \big((2 + s\cos(t/2))\cos t, (2 + s\cos(t/2))\sin t, s\sin(t/2)\big), \quad 0 \le t \le 2\pi, \quad |s| \le 1/2.$$

Along the central circle, we follow the pair $(\partial_t m, \partial_s m)$. At the common point $m(0,0) = m(2\pi,0)$, the first vector is unchanged and the second is reversed.

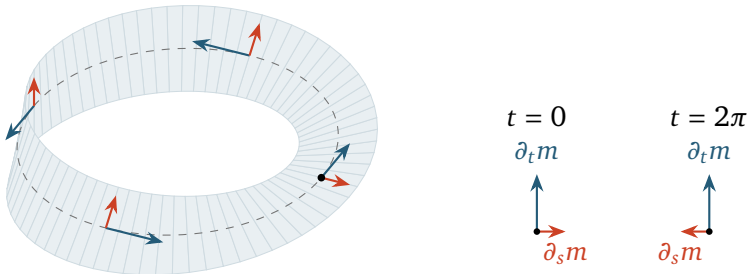


Figure 12.1: The tangent pair follows the central circle of the parametrised band. The two pairs on the right compare its initial and final values at the same point.

The strip model shows the same reversal in quotient coordinates.



Figure 12.2: The red edge arrows mark $(t,s) \mapsto (t+1,-s)$. An upward transverse arrow at $t = 1$ therefore returns at $t = 0$ pointing down.

The open Möbius band is the quotient of $\mathbb{R} \times (-1,1)$ by this identification. On a rectangle of width less than 1, the projection is injective and open: the inverse image of its image is

the open union of all translated copies under the identification. The inverse restrictions give charts on the quotient. Their transitions are restrictions of integer powers of $(t, s) \mapsto (t+1, -s)$, hence smooth. Thus the projection is a local diffeomorphism. If the quotient had a nowhere-zero top form, its pullback would be $h(t, s)\mathrm{d}t \wedge \mathrm{d}s$ with h continuous, nonzero and of constant sign on the connected strip. Invariance under the identification gives $h(t + 1, -s) = -h(t, s)$, contradicting that constant sign. Hence it is not orientable.

For $\mathbb{R}\mathrm{P}^2$, the projection $q\colon S^2 \to \mathbb{R}\mathrm{P}^2$ is a local diffeomorphism by the ratio charts of Lecture 7. On a hemisphere $\sigma X^i > 0$, with $\sigma \in \{+1, -1\}$, its coordinate map is $X \mapsto (X^j/X^i)_{j\neq i}$. Its smooth inverse is $y \mapsto \sigma Y/\|Y\|$, where Y has ith component 1 and remaining components y. This proves the local diffeomorphism assertion. A nonzero top form would pull back to $h\mu$ on S^2, where μ is its outward area form and h has constant nonzero sign. The antipodal map $a(p) = -p$ satisfies

$$(a^*\mu)_p(v, w) = \det[-p\ -v\ -w] = -\mu_p(v, w).$$

Since $q \circ a = q$, invariance forces $h(-p) = -h(p)$, again impossible. Thus $\mathbb{R}\mathrm{P}^2$ is not orientable.

For $n \geq 1$, the ratio charts also make $q\colon S^n \to \mathbb{R}\mathrm{P}^n$ a local diffeomorphism. The same determinant calculation gives

$$a^*\mu = (-1)^{n+1}\mu.$$

For even n, a pulled-back nowhere-zero form would again require a constant-sign function to change sign. For odd n, the outward sphere form is invariant under a. Its local expressions therefore descend through q to a nowhere-zero top form on $\mathbb{R}\mathrm{P}^n$. Hence $\mathbb{R}\mathrm{P}^n$ is orientable exactly for odd $n \geq 1$. In particular, $\mathrm{SO}(3) \cong \mathbb{R}\mathrm{P}^3$ is orientable. This also follows from Lecture 9: left translation carries a basis at the identity to a smooth frame on any Lie group.

12.3 Integrating a top form

Recall the ordinary change-of-variables theorem: under a smooth diffeomorphism $y = f(x)$, integration of a compactly supported function uses the factor $|\det Df|$. In positive coordinates the determinant itself is positive. This is the calculus input to integration of forms.

Definition 12.10 (Integral of a top form)**.** For $n \geq 1$, let ω be a smooth compactly supported n-form on an oriented n-manifold. On a positive chart $x\colon U \to x(U)$ write $(x^{-1})^*\omega = f\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n$. If its support lies in U, define its integral to be the ordinary integral of f on $x(U)$. For a general support, choose a positive atlas and a subordinate partition and set

$$\int_M \omega = \sum_i \int_{U_i} \rho_i\omega.$$

Only finitely many terms meet the compact support. Each has compact support inside its chart, so every term is defined.

We construct a partition on $\mathbb{R}$ with its usual orientation. We take $U = (-\infty, 1)$ with coordinate x and $V = (-1, \infty)$ with coordinate $y = 2x$. We define a smooth step using the bump function above:

$$\eta(u) = \begin{cases} e^{-1/u}, & u > 0, \\ 0, & u \leq 0, \end{cases} \qquad a(x) = \frac{\eta(x + 1/2)}{\eta(x + 1/2) + \eta(1/2 - x)}.$$

The denominator is positive everywhere. The step is zero for $x \le -1/2$ and one for $x \ge 1/2$. The functions $\rho_U = 1 - a$ and $\rho_V = a$ form a subordinate partition. Their supports lie in U and V respectively. The figure shows two complementary weights on the bounded interval $[-1, 1]$. Their overlap has total weight one.

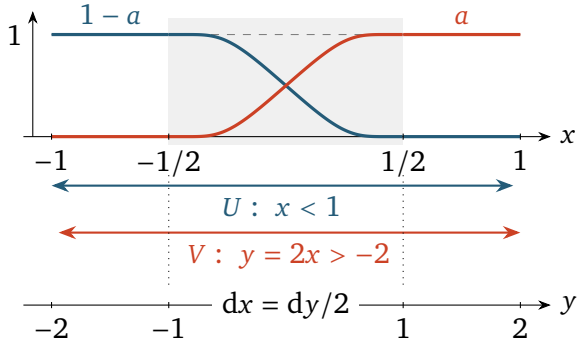


Figure 12.3: The weights $1 - a$ and a sum to one, with $\operatorname{supp} a'$ in the shaded interval. The second coordinate $y = 2x$ doubles lengths, so its form coefficient contains $1/2$.

Example 12.11 (A weighted integral in two charts)**.** We integrate a compactly supported form using these charts and weights. We choose $\omega = a'(x)\mathrm{d}x$, whose support lies in $[-1/2, 1/2]$. In the second chart its expression is $a'(y/2)\mathrm{d}y/2$. The two weighted chart integrals are

$$\int_U \rho_U \omega = \int_{-1/2}^{1/2} (1 - a(x))a'(x)\mathrm{d}x = \left[a - \frac{a^2}{2}\right]_0^1 = \frac{1}{2},$$
$$\int_V \rho_V \omega = \int_{-1}^{1} a(y/2)a'(y/2)\frac{\mathrm{d}y}{2} = \left[\frac{a^2}{2}\right]_0^1 = \frac{1}{2}.$$

The bracket limits are values of the step a. Adding gives $\int_{\mathbb{R}} \omega = 1$, which agrees with $a(+\infty) - a(-\infty) = 1$. The Jacobian factor cancels the coordinate stretching. The weights prevent the overlap from being counted twice.

Lecture 10 used cutoff functions in a sketch. Later lectures extend locally defined fields by zero with them, so we now construct them.

Definition 12.12 (Cutoff function)**.** A smooth *cutoff function* at a point equals 1 near that point and has support inside a chosen neighbourhood.

To construct one, take a chart centred at the point and radii $0 < r_1 < r_2$ such that the closed ball of radius r_2 lies inside the coordinate image of the neighbourhood. Using the smooth step above, set

$$m = \frac{r_1^2 + r_2^2}{2}, \qquad w = r_2^2 - r_1^2, \qquad f(x) = 1 - a\left(\frac{|x|^2 - m}{w}\right).$$

The step's argument is at most $-1/2$ for $|x| \le r_1$ and at least $1/2$ for $|x| \ge r_2$. Hence $f = 1$ on the smaller ball and $f = 0$ outside the larger ball. Extending by zero outside the chart gives the required smooth cutoff on the manifold.

Proposition 12.13 (Independence of choices · *proved given partitions of unity and change of variables*)**.** *The integral is independent of the positive atlas and subordinate partition. Reversing the orientation reverses its sign.*

Proof. On overlapping positive charts, the determinant rule for top-form pullback and ordinary change of variables give the same chart integral. For a smooth coordinate change $y = f(x)$, Lecture 10's wedge convention gives

$$f^*(\mathrm{d}y^1 \wedge \cdots \wedge \mathrm{d}y^n) = \det(Df)\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n.$$

For two partitions ρ_i and χ_j subordinate to positive charts, insert $\sum_j \chi_j = 1$ into each integral for the first partition. The result is the finite double sum of the integrals of $\rho_i\chi_j\omega$. Each term is supported in the chart overlap and has the same value in either chart system. Interchanging the sums and using $\sum_i \rho_i = 1$ gives the second partition's answer. This proves independence. For the opposite orientation, reflect one coordinate. Its derivative has determinant -1, whereas the absolute Jacobian in ordinary integration is 1. The top-form coefficient changes sign, proving the last assertion. ∎

For computations, let $D \subset \mathbb{R}^n$ be a compact parameter region, and let $F\colon D \to M$ be smooth with $F(D) = M$. Here smoothness on D means local smooth extension to open neighbourhoods in $\mathbb{R}^n$. Suppose F restricts to an orientation-preserving diffeomorphism $\operatorname{int} D \to M \setminus E$, where $E = F(\partial D)$. Assume ∂D has Lebesgue measure zero and E has measure zero in every manifold chart. The latter condition is chart-independent by change of variables. Then integration is computed by pullback:

$$\int_M \omega = \int_D F^*\omega.$$

Indeed, insert a subordinate partition ρ_i on M. For each summand, ordinary change of variables on $\operatorname{int} D$ gives the integral of $\rho_i\omega$ on $M \setminus E$. The two null boundaries add zero. Summing the finitely many nonzero chart integrals proves the formula. The angular rectangle for the sphere satisfies these hypotheses. See also Tu, §23.4, Definition 23.10.

Example 12.14 (The angular rectangle). We integrate the outward area form μ on S^2 using

$$D = [0, \pi] \times [0, 2\pi], \qquad F(\theta, \phi) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta).$$

The interior maps diffeomorphically onto the sphere minus the seam $E = F(\partial D)$, a half great circle including the poles. Both null-set hypotheses hold. Lecture 10 gives $F^*\mu = \sin\theta\mathrm{d}\theta\wedge\mathrm{d}\phi$. Since $\sin\theta > 0$ in the interior, the parametrisation preserves orientation. Consequently

$$\int_{S^2} \mu = \int_0^{2\pi}\int_0^{\pi} \sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi = 2\pi[-\cos\theta]_0^{\pi} = 4\pi.$$

Self-check. Compute the area of the unit sphere using its angular area form. Explain why the negative north-pole stereographic pullback does not make this outward-oriented integral negative.

Answer. The angular integral is $2\pi[-\cos\theta]_0^{\pi} = 4\pi$. The north-pole stereographic chart reverses orientation. We therefore reverse the parameter orientation when using its pullback, which has coefficient $-4/(1+u^2+v^2)^2$ in the standard (u, v) order.

Definition 12.15 (Density). A *density* has local expressions $f|\mathrm{d}x^1 \cdots \mathrm{d}x^n|$ that transform with the absolute Jacobian.

Densities can be integrated without orientation because the ordinary change-of-variables factor has the same law. A top form uses a signed determinant, so its signed integral requires

a chosen orientation. The two transformation laws must be distinguished on nonorientable manifolds. For example, $|\mathrm{d}t\,\mathrm{d}s|$ on the Möbius strip is invariant under $(t, s) \mapsto (t + 1, -s)$, since the absolute determinant is one. It defines an area density on the quotient, although no nowhere-zero top form exists there. Lecture 15 constructs such densities from a metric.

Self-check. State in a sentence what the integral of a compactly supported top form on an oriented manifold is and why a partition of unity is introduced.

Answer. The integral is the sum of the ordinary coordinate integrals of $\rho_i\omega$, whose supports lie inside positive charts. The partition localises the form without counting overlaps twice: $\sum_i \rho_i\omega = \omega$.

Unit 12B Boundary and Stokes

12.4 Manifolds with boundary

Three facts are used below. The boundary is a smooth manifold of dimension $n - 1$. The outward transverse direction does not depend on the boundary chart. Tangent spaces at boundary points have dimension n, and forms can be differentiated there.

The closed disc, the closed ball and the annulus have boundary points with half-space neighbourhoods. Their interior points have ordinary open coordinate neighbourhoods. The figure compares these two types of point on the closed unit disc at 0 and $p = (0, 1)$. Near p, the coordinates $u = x$ and $v = \sqrt{1 - x^2} - y$ put the boundary at $v = 0$ and the interior at $v > 0$.

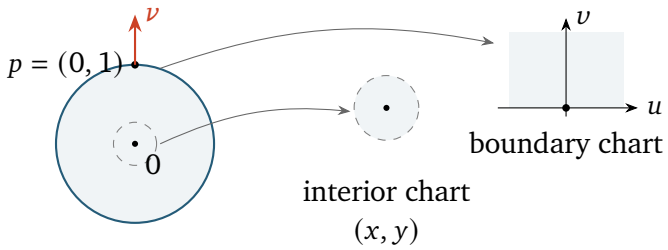


Figure 12.4: An interior point has an open-disc chart. At the top boundary point, the displayed chart sends the disc locally into $v \geq 0$ and the rim into $v = 0$.

The upper and lower closed half-spaces are

$$\mathbb{H}^n = \{x \in \mathbb{R}^n : x^n \geq 0\}, \qquad \mathbb{H}^n_- = \{x \in \mathbb{R}^n : x^n \leq 0\}.$$

Definition 12.16 (Manifold with boundary)**.** A smooth *manifold with boundary* is Hausdorff and second countable, with charts onto relatively open subsets of either half-space. Chart transitions and their inverses extend smoothly to neighbourhoods in $\mathbb{R}^n$ near each point.

Definition 12.17 (Boundary and interior)**.** The *boundary* ∂M consists of chart points with $x^n = 0$; the others form the interior.

This geometric boundary is a set of points. Lecture 6's boundary operator acts on chains. When oriented cells are added, their shared faces cancel, leaving the faces along the geometric boundary.

Allowing either half-space permits the coordinate reflection needed for positive charts also in dimension one. At the right endpoint of the oriented interval $[0, 1]$, the positive chart $u = x - 1$ maps into $(-\infty, 0]$. A positive chart increases with x, so it cannot send this right endpoint to zero with its interior on the positive side. In dimensions $n \geq 2$, reflecting a tangential coordinate can correct the orientation while retaining the upper half-space.

Verification of the boundary facts

The set of boundary points is independent of the chart. Indeed, the chain rule on interior points and continuity make transition derivatives invertible also at boundary points. The inverse function theorem then makes their extensions local diffeomorphisms. Such a map cannot send a half-neighbourhood to a full interior neighbourhood: its inverse would have to take points on both sides of the source hyperplane into the half-space. Restricting charts to $x^n = 0$ gives charts of dimension $n - 1$ on ∂M. The restricted transitions make them smoothly compatible.

A transition's last component vanishes on the boundary, so its tangential derivatives vanish there. Its normal derivative is nonzero by invertibility. It sends inward vectors to inward vectors because it sends interior points to interior points. Thus the outward transverse direction is chart-independent, even when the two charts use opposite half-spaces.

At a boundary point, T_pM is the ambient n-dimensional space modeled by a half-space chart, with changes given by the transition Jacobians. Its vectors need not be velocities of two-sided curves remaining in M. In particular the outward vector $-\partial_n$ is allowed in the upper model. Functions and forms are smooth when their chart coefficients extend smoothly to open sets of $\mathbb{R}^n$. Their derivatives at the boundary are independent of extension, since extensions agree on the interior and all their derivatives agree there and extend continuously to the boundary. The same half-chart construction gives tangent changes and exterior differentiation. Partitions of unity and integration extend using restrictions of bumps to half-balls. The preceding normalization and independence proofs then apply.

Definition 12.18 (Induced boundary orientation)**.** On an oriented manifold the *induced boundary orientation* is *outward first*: a boundary basis $(v_1, \dots, v_{n-1})$ is positive if $(\nu, v_1, \dots, v_{n-1})$ is positive in T_pM, where ν is outward transverse.

Outward-first gives the sign-free formula in Theorem 12.20. Changing ν by a tangent vector or multiplying it by a positive scalar does not change this sign. This proves independence of that choice; no metric is required. For the standard orientation on $\mathbb{H}^n$, take $\nu = -\partial_n$. The boundary orientation is represented by

$$\iota_{-\partial_n}(\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n) = (-1)^n \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^{n-1} \quad \text{on } x^n = 0.$$

On $\mathbb{H}^n_-$ the outward vector is $+\partial_n$, so the boundary orientation is $(-1)^{n-1}\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^{n-1}$. For $n = 1$ the upper model gives the negative initial endpoint, and the lower model gives the positive final endpoint. The figure compares the upper half-plane with an interval.

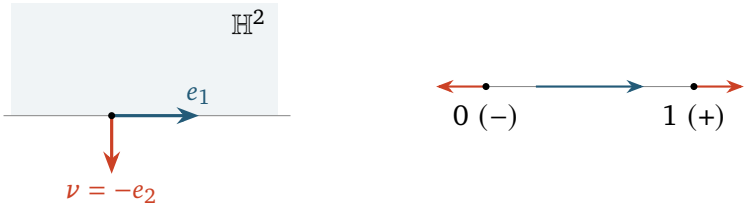


Figure 12.5: On the upper half-plane, the outward vector points down. The boundary tangent points right. On the positively oriented interval, the induced endpoint signs are − and +.

The sphere orientation above is exactly the boundary orientation of the unit ball. The usual Möbius band with its boundary is also nonorientable, since an orientation would restrict to its nonorientable interior.

Definition 12.19 (Closed manifold). A *closed manifold* means a compact manifold without boundary.

12.5 Stokes' theorem in every dimension

For adjacent counter-clockwise squares, a shared edge is traversed in opposite directions. The figure shows this cancellation in a square region with a hole. The remaining outer boundary is counter-clockwise, and the inner boundary is clockwise, as for chains in Lecture 6.

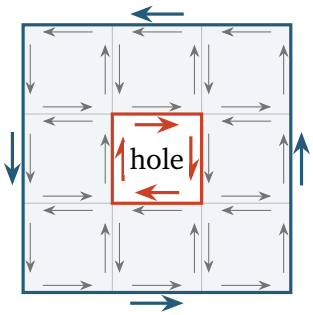


Figure 12.6: Each square has its counter-clockwise boundary orientation. Shared edges cancel in pairs, leaving an outer counter-clockwise boundary and an inner clockwise boundary.

On a coordinate cell the fundamental theorem of calculus gives the integral over its faces. Adding these formulas cancels internal faces. For a general manifold we prove the formula using partitions of unity. The cancellation is expressed by $\sum_i \mathrm{d}\rho_i = 0$.

Theorem 12.20 (Stokes · *proved given partitions of unity*). *Let M be an oriented smooth n-manifold with boundary, where $n \geq 1$. For a smooth compactly supported $(n-1)$-form ω,*

$$\int_M \mathrm{d}\omega = \int_{\partial M} \omega.$$

The right side means restriction by the boundary inclusion, with the outward-first orientation. In particular the statement applies to every smooth $(n-1)$-form when M is compact.

Proof. First work on $\mathbb{H}^n$ with compact support. Write the form as

$$\omega = \sum_{i=1}^{n} (-1)^{i-1} f_i \mathrm{d}x^1 \wedge \cdots \wedge \widehat{\mathrm{d}x^i} \wedge \cdots \wedge \mathrm{d}x^n, \qquad \mathrm{d}\omega = \Big(\sum_i \partial_i f_i\Big) \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n.$$

The derivative formula follows because repeated coordinate factors vanish; the two omission signs cancel for the surviving derivative. Fubini's theorem and the fundamental theorem of calculus integrate $\partial_i f_i$ first in x^i. For $i < n$ the endpoints are outside compact support, so each contribution is zero. The normal contribution is

$$\int_{\mathbb{H}^n} \mathrm{d}\omega = \int_{\mathbb{R}^{n-1}} \int_0^\infty \partial_n f_n(x', x^n)\, \mathrm{d}x^n\, \mathrm{d}x' = -\int_{\mathbb{R}^{n-1}} f_n(x', 0)\, \mathrm{d}x'.$$

On the boundary only the term $i = n$ restricts nontrivially. Its coefficient is $(-1)^{n-1} f_n$. The boundary orientation has the additional sign $(-1)^n$, so its integral is also $-\int f_n(x', 0)\mathrm{d}x'$. For $n = 1$, the same equality uses the negative boundary point and the ordinary endpoint formula. This proves the half-space case in every dimension. On $\mathbb{R}^n$ all coordinate integrations have two endpoints outside support, so the integral of $\mathrm{d}\omega$ is zero, matching its empty boundary. On $\mathbb{H}^n_-$, integrating the normal coordinate from $-\infty$ to 0 gives $+\int_{\mathbb{R}^{n-1}} f_n(x', 0)\mathrm{d}x'$. Its boundary orientation has sign $(-1)^{n-1}$. Multiplying by the same sign in the restricted form gives $+f_n$, so Stokes holds there too. For $n = 1$ this gives the positive final endpoint.

Now choose positively oriented interior and boundary charts on M and a subordinate partition ρ_i. Each $\rho_i \omega$ has compact support inside its chart. In coordinates it extends by zero across the artificial chart edges to a smooth compactly supported form on $\mathbb{R}^n$, $\mathbb{H}^n$ or $\mathbb{H}^n_-$. The local result therefore applies to it. Sum these equalities. Local finiteness and compact support make the sum finite. On M, the product rule gives

$$\sum_i \mathrm{d}(\rho_i \omega) = \Big(\sum_i \mathrm{d}\rho_i\Big) \wedge \omega + \Big(\sum_i \rho_i\Big) \mathrm{d}\omega = \mathrm{d}\omega.$$

Here $\sum_i \rho_i = 1$ and its differential is zero. The restricted partition also sums to one on the boundary. Thus the summed local equalities give the claimed global equality. ∎

The support hypothesis matters on a noncompact manifold. On $\mathbb{R}$, the improper integral of $\mathrm{d}(\tanh x)$ is $1 - (-1) = 2$, while the boundary is empty. The function has no compact support. The nonzero limiting values at infinity remain in the endpoint formula.

12.6 Classical integral theorems and equal fluxes

Definition 12.21 (Compact smooth region)**.** A *compact smooth region* in $\mathbb{R}^n$ means a compact n-manifold with boundary embedded in $\mathbb{R}^n$.

For an oriented manifold S, write $-S$ for the opposite orientation. Stokes includes the integral theorems of vector calculus through Lecture 10's correspondence between vector fields and forms.

- On the interval $[a, b]$ with $a < b$ and increasing orientation, for a smooth function f, the endpoint signs give the fundamental theorem:

$$\int_a^b f'(x)\mathrm{d}x = f(b) - f(a).$$

- On a compact smooth region $D \subset \mathbb{R}^2$ oriented by $\mathrm{d}x \wedge \mathrm{d}y$, for smooth functions P and Q on a neighbourhood of D, Green's theorem is
$$\int_D (\partial_x Q - \partial_y P)\mathrm{d}x \wedge \mathrm{d}y = \int_{\partial D} (P\mathrm{d}x + Q\mathrm{d}y).$$
The outer boundary is counter-clockwise. Boundaries of holes are clockwise.
- On a compact oriented smooth surface $S \subset \mathbb{R}^3$ with smooth boundary, let $\mathbf{A}$ be a smooth vector field on a neighbourhood of S. With the compatible unit normal n, Kelvin–Stokes is
$$\int_S (\operatorname{curl} \mathbf{A}) \cdot n \, \mathrm{d}A = \int_{\partial S} \mathbf{A} \cdot \mathrm{d}\mathbf{x}.$$
Here $\mathrm{d}A$ denotes Euclidean surface area. The positive rim tangent T satisfies that (n, ν, T) is positive in $\mathbb{R}^3$, where ν is the outward tangent normal along the rim. Indeed, (ν, T) is then a positive basis of TS, which is exactly outward-first.

The figure compares these boundary conventions on an interval, a disc and a flat surface in three dimensions.

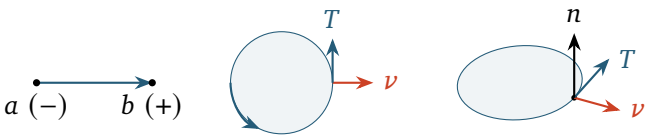


Figure 12.7: Endpoint signs and rim tangents follow the outward-first convention. For the surface panel, the ordered ambient triple is (n, ν, T).

Example 12.22 (Stokes on the unit disc)**.** We take $D = \{x^2 + y^2 \le 1\}$ with orientation $\mathrm{d}x \wedge \mathrm{d}y$ and $\alpha = x\mathrm{d}y$. Its derivative is $\mathrm{d}\alpha = \mathrm{d}x \wedge \mathrm{d}y$, whose disc integral is π by polar coordinates. At $(1, 0)$ the outward vector is e_x, so the positive boundary tangent is e_y. The induced circle runs counter-clockwise. For $c(t) = (\cos t, \sin t)$, $0 \le t \le 2\pi$, we obtain
$$c^*\alpha = \cos^2 t \mathrm{d}t, \qquad \int_{\partial D} \alpha = \int_0^{2\pi} \cos^2 t \mathrm{d}t = \pi.$$
The boundary and area integrals agree.

Corollary 12.23 (Consequences of Stokes · *proved given partitions of unity*)**.** *For a smooth vector field v on a compact smooth region $K \subset \mathbb{R}^3$, with Euclidean volume $\mu = \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z$,*
$$\int_K (\operatorname{div} v)\mu = \int_{\partial K} \iota_v \mu.$$
On a closed oriented manifold the integral of an exact top form is zero. If an oriented compact 3-manifold has boundary $S_1 \sqcup (-S_2)$ and a closed 2-form β is smooth on it, then $\int_{S_1} \beta = \int_{S_2} \beta$.

Proof. The first statement follows by applying Stokes to $\iota_v \mu$: Lecture 11 gives $\mathrm{d}\iota_v \mu = (\operatorname{div} v)\mu$. With an outward unit normal ν and a positively oriented orthonormal boundary basis, $\mu(v, \cdot, \cdot)$ is $(v \cdot \nu)$ times $\iota_\nu \mu$, the induced boundary area form. Thus this is the ordinary normal-flux divergence theorem. The empty boundary gives the second statement. Closedness gives $\int \mathrm{d}\beta = 0$; the two induced boundary orientations give $\int_{S_1} \beta - \int_{S_2} \beta = 0$, proving the third. ∎

12.7 Coulomb flux around a missing point

We distinguish the Coulomb form β_C from Lecture 10's ambient form β.

Definition 12.24 (Coulomb flux form)**.** On $\mathbb{R}^3 \setminus \{0\}$ set $r = \sqrt{x^2+y^2+z^2}$ and define the Coulomb flux form by

$$\beta_C = \frac{x\,dy \wedge dz + y\,dz \wedge dx + z\,dx \wedge dy}{r^3}.$$

The forms β_C and β agree on the unit sphere.

Physical input. In the rationalised electromagnetic units used in Lecture 15, a stationary point charge q at the origin has electric field

$$\mathbf{E}(\mathbf{x}) = \frac{q}{4\pi}\frac{\mathbf{x}}{r^3}, \qquad r = \|\mathbf{x}\| > 0.$$

Under Lecture 10's flux correspondence, β_C is the flux form of $\mathbf{x}/r^3$. Thus the electric flux form is $q\beta_C/(4\pi)$. Closedness of β_C expresses zero divergence away from the charge.

Let K be a compact smooth region with outward-oriented boundary $S = \partial K$ and $0 \notin S$. We compute the outward flux of β_C through S. When the origin lies inside K, a removed small ball has surface S_ε. Its induced inner normal points into the ball, as the section of the region in the figure shows.

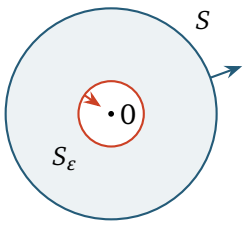


Figure 12.8: The region between S and the small sphere has an inner boundary with normal pointing into the removed ball. Its induced inner orientation is opposite to the ball's outward orientation.

Example 12.25 (Outward Coulomb flux)**.** The derivative of β_C has volume coefficient

$$\partial_x \frac{x}{r^3} + \partial_y \frac{y}{r^3} + \partial_z \frac{z}{r^3} = \frac{3}{r^3} - \frac{3(x^2+y^2+z^2)}{r^5} = 0.$$

Thus β_C is closed on its domain. On the unit sphere it is the area form of Lecture 10, so angular coordinates give

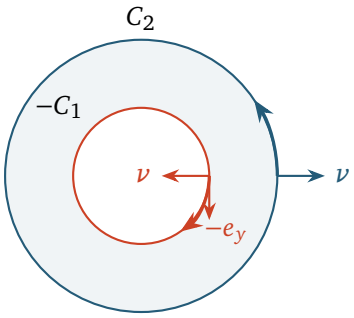
$$\int_{S^2} \beta_C = \int_0^{2\pi}\int_0^{\pi} \sin\theta\, d\theta\, d\phi = 4\pi.$$

Here θ is the polar angle and ϕ the azimuth. The coordinate seam and poles have measure zero and do not change this integral. On the sphere of radius $\varepsilon > 0$, radial scaling contributes ε^3 in the numerator and ε^3 in the denominator, so its outward flux is again 4π.

If $0 \notin K$, Stokes applies directly to the closed form on K and gives zero flux. If 0 is inside K, remove an open ball of radius ε whose closure lies in the interior of K. The remaining compact region has boundary $S \sqcup (-S_\varepsilon)$. The form is smooth and closed on this region. The

equal-flux consequence gives $\int_S \beta_C = \int_{S_\varepsilon} \beta_C = 4\pi$. Thus the outward Coulomb flux detects whether the original region contains the origin. For the electric field above, the outward flux is q when 0 lies inside K, and zero when $0 \notin K$. This is Gauss's law for this point charge.

The annulus A in the figure has radii 1 and 2. We write C_s for a circle of radius $s > 0$ with counter-clockwise orientation. The annulus has outward normals pointing away from the region on both circles. The normal at the inner circle points into the hole.



$$\partial A = C_2 \sqcup (-C_1)$$

Figure 12.9: The annulus has outer counter-clockwise and inner clockwise boundary orientations. At the inner rightmost point, both the outward normal and positive tangent are negative coordinate directions.

Self-check. Let $A = \{(x, y) : 1 \le x^2 + y^2 \le 4\}$, oriented by $\mathrm{d}x \wedge \mathrm{d}y$, and let

$$\alpha = (x^2 + y^2 + \frac{1}{2})(x\mathrm{d}y - y\mathrm{d}x).$$

Determine both boundary directions and verify Stokes for α on A.

Answer. At $(1, 0)$ the outward vector from A is $-e_x$. The tangent $-e_y$ makes $(-e_x, -e_y)$ positive, so the inner circle is clockwise. The outer circle is counter-clockwise. The derivative and its polar integral are

$$\mathrm{d}\alpha = (4x^2 + 4y^2 + 1)\mathrm{d}x \wedge \mathrm{d}y, \qquad \int_A \mathrm{d}\alpha = 2\pi \int_1^2 (4s^2 + 1)s\mathrm{d}s = 33\pi.$$

For the counter-clockwise circle $c_s(t) = (s\cos t, s\sin t)$ with $0 \le t \le 2\pi$ and $s \in \{1, 2\}$,

$$c_s^*\alpha = (s^4 + s^2/2)\mathrm{d}t, \qquad \int_{\partial A} \alpha = 36\pi - 3\pi = 33\pi.$$

The inner contribution has the minus sign required by its induced orientation.

Summary. Orientations fix signs of top forms. Partitions of unity define a chart-independent integral. Stokes applies to compactly supported forms on oriented manifolds with boundary, using outward-first boundary orientation. Its local proof is the fundamental theorem of calculus in the normal coordinate. Closed forms have equal fluxes across two oriented surfaces that together, with the second orientation reversed, bound a compact region. Removing a ball around the Coulomb singularity gives flux 4π. Lecture 13 studies closed forms modulo exact forms.

Reading. Tu, *An Introduction to Manifolds*, §13 and Appendix C; §§21–23, especially Theorem 23.12, pp. 269–271.

Lecture 13

De Rham cohomology

Objectives. *A closed form of positive degree has local primitives, forms whose exterior derivative is the given form, but a global primitive may not exist. De Rham cohomology records this obstruction. We prove the local assertion by integrating along a contraction. We then compare homotopic maps and compute the cohomology of spheres by joining two open sets. The Coulomb form gives a nonzero class with a measurable period.*

By the end of this lecture you will be able to:

- Distinguish closed forms, exact forms and their cohomology classes.
- Construct a primitive of a closed positive-degree form on a star-shaped open set.
- Prove smooth homotopy invariance using the homotopy operator.
- Compute sphere cohomology from Mayer–Vietoris and normalize its area class.

Unit 13A Local primitives and homotopy

13.1 Closed forms modulo exact forms

All manifolds in this lecture are smooth and without boundary. All vector spaces and cohomology groups have real coefficients. Recall from Lecture 10 that a form α is *closed* if $\mathrm{d}\alpha = 0$ and *exact* if $\alpha = \mathrm{d}\eta$ for some form η. The identity $\mathrm{d}^2 = 0$ makes every exact form closed.

Definition 13.1. A *cochain complex* is a sequence of vector spaces with linear maps $\mathrm{d}\colon C^k \to C^{k+1}$ whose successive composites are zero.

The spaces $\Omega^k(M)$ with exterior differentiation form such a complex. We put $\Omega^k(M) = 0$ for $k < 0$.

Definition 13.2. The *de Rham cohomology* in degree k is the quotient

$$H^k(M) = \frac{\ker(\mathrm{d}\colon \Omega^k(M) \to \Omega^{k+1}(M))}{\operatorname{im}(\mathrm{d}\colon \Omega^{k-1}(M) \to \Omega^k(M))}.$$

Definition 13.3. The *cohomology class* $[\alpha]$ of a closed form consists of all closed forms differing from α by an exact form.

In positive degree, $[\alpha] = 0$ means precisely that α has a global primitive. Lecture 6 formed homology as the kernel of a boundary map modulo the image of the preceding boundary map. Here the arrows reverse direction: exterior differentiation raises degree, while the chain boundary lowers it. The identity $\mathrm{d}^2 = 0$ replaces $\partial^2 = 0$. Closed forms play the

role of cycles, and exact forms play the role of boundaries. The superscript distinguishes cohomology from homology.

In degree zero there are no forms of degree -1, so the only exact form is zero. A function is closed exactly when all its partial derivatives vanish. It is then constant on connected coordinate balls, hence locally constant. Conversely a locally constant function has zero derivative. In particular, $H^0(M) = \mathbb{R}$ when M is connected and nonempty. For a disconnected manifold, each component contributes an independently chosen constant. With infinitely many components this is a product of copies of $\mathbb{R}$.

For example, on the punctured plane the angular form is

$$\alpha_{\text{ang}} = \frac{-y\,\mathrm{d}x + x\,\mathrm{d}y}{x^2 + y^2}.$$

Lecture 10 proved that this form is closed and has integral 2π around the counter-clockwise unit circle. An exact form has zero integral around that circle by Stokes. Hence $[\alpha_{\text{ang}}] \neq 0$.

Definition 13.4. A *period* of a closed k-form is its integral over a closed oriented k-manifold mapped smoothly into M, using pullback.

Such a mapped manifold is a geometric cycle. Stokes makes every period of an exact form zero, so periods depend only on the class. These geometric cycles suffice here. Lecture 14 relates forms to the singular cycles of Lecture 6.

Definition 13.5. The *induced pullback* of a smooth map $f\colon M \to N$ is given by

$$f^*\colon H^k(N) \longrightarrow H^k(M), \qquad f^*[\alpha] = [f^*\alpha].$$

Pullback commutes with d, so it preserves closed forms and exact differences. This proves that the induced map is well defined. Pullback reverses composition: $(g \circ f)^* = f^* g^*$, and the identity map induces the identity. The wedge product also descends to classes. For closed β, $(\mathrm{d}\eta) \wedge \beta = \mathrm{d}(\eta \wedge \beta)$. For closed α of degree k, $\alpha \wedge \mathrm{d}\xi = (-1)^k \mathrm{d}(\alpha \wedge \xi)$. Changing either representative therefore changes the product only by an exact form. We write $H^*(M)$ for all the degrees together. For example, on $T^2 = S^1 \times S^1$ let ν_1 and ν_2 be the pullbacks of the period-one angular form from the two factors. In angular coordinates these are $\mathrm{d}\theta/(2\pi)$ and $\mathrm{d}\phi/(2\pi)$. With the product orientation their wedge has period

$$\int_{T^2} \nu_1 \wedge \nu_2 = \frac{1}{(2\pi)^2} \int_0^{2\pi} \int_0^{2\pi} \mathrm{d}\theta\, \mathrm{d}\phi = 1.$$

Thus $[\nu_1] \wedge [\nu_2]$ is a nonzero degree-two class.

13.2 The homotopy operator and the Poincaré lemma

Definition 13.6. An open set $U \subset \mathbb{R}^n$ is *star-shaped about* 0 if $0 \in U$ and $tx \in U$ for every $x \in U$ and $t \in [0, 1]$.

For a closed 1-form α on an open set star-shaped about the origin, integration along the segment from 0 to x gives a potential function:

$$f(x) = \int_0^1 \alpha_{tx}(x)\, \mathrm{d}t.$$

The homotopy operator extends this construction to forms of every degree.

Definition 13.7. A *smooth homotopy* between smooth maps F_0 and F_1 from M to N is a smooth map $F\colon [0,1]\times M \to N$ with $F_t(p) = F(t,p)$ and those endpoint maps. Smoothness at the endpoints means local smooth extension, as in Lecture 12.

No inverse for any F_t is required. For a k-form α on N, the pulled-back form decomposes uniquely as

$$F^*\alpha = \mathrm{d}t \wedge a_t + b_t,$$

where a_t and b_t are forms on M of degrees $k-1$ and k. Here a_t is the restriction of $\iota_{\partial_t} F^*\alpha$ to the slice at time t, and $b_t = F_t^*\alpha$.

Definition 13.8. The *homotopy operator* is given by integrating the time-containing coefficients:

$$K_F\colon \Omega^k(N) \longrightarrow \Omega^{k-1}(M), \qquad K_F\alpha = \int_0^1 a_t\,\mathrm{d}t \quad (k \geq 1),$$
$$K_F f = 0 \quad (k = 0).$$

The integral is smooth. Near each point, compactness of the time interval permits differentiating the smooth coefficients under the integral. The intrinsic definition by contraction makes the chart definitions agree.

Proposition 13.9 (Homotopy operator identity)**.** *For every smooth homotopy F and every differential form α,*

$$F_1^*\alpha - F_0^*\alpha = \mathrm{d}K_F\alpha + K_F\mathrm{d}\alpha.$$

Proof. Write d_M for differentiation in M. The product rule gives

$$\mathrm{d}(F^*\alpha) = \mathrm{d}t \wedge (\partial_t b_t - \mathrm{d}_M a_t) + \mathrm{d}_M b_t.$$

The minus sign comes from moving d_M past $\mathrm{d}t$. Since $\mathrm{d}F^*\alpha = F^*\mathrm{d}\alpha$, the time-containing coefficient on the right is the one integrated in $K_F\mathrm{d}\alpha$. Thus

$$K_F\mathrm{d}\alpha = \int_0^1 \partial_t b_t\,\mathrm{d}t - \mathrm{d}_M \int_0^1 a_t\,\mathrm{d}t = b_1 - b_0 - \mathrm{d}_M K_F\alpha.$$

The fundamental theorem of calculus gives the endpoint difference. This is the claimed identity. For functions, the same computation has $a_t = 0$ and is the ordinary endpoint formula. ∎

For a flow $F_t = \varphi_t$ generated by X, the time coefficient is $a_t = \varphi_t^*(\iota_X\alpha)$. Indeed, $\partial_t F(t,p) = X_{F(t,p)}$ fills the first argument of α. The homotopy identity is then the integral of Lecture 11's Cartan formula $L_X = \mathrm{d}\iota_X + \iota_X\mathrm{d}$. For a smooth oriented path σ in M, Stokes on its swept square gives

$$\int_\sigma (F_1^*\alpha - F_0^*\alpha) = \int_{\partial\sigma} K_F\alpha + \int_{[0,1]\times\sigma} F^*\mathrm{d}\alpha \qquad (\deg\alpha = 1).$$

The square has time-first orientation. Its side edges account for the endpoint values of $K_F\alpha$, as the figure shows.

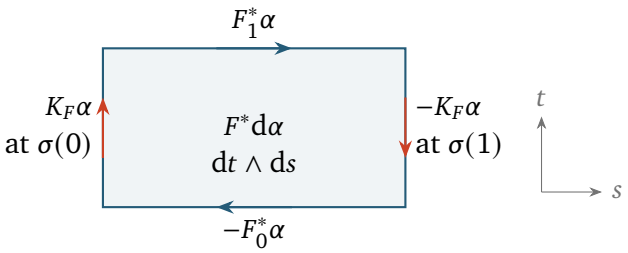


Figure 13.1: The homotopy sweeps a path into a square oriented by $\mathrm{d}t \wedge \mathrm{d}s$. The top and bottom give the endpoint pullbacks, and the sides give the boundary values of K_F.

Definition 13.10. Two maps of complexes whose difference is $\mathrm{d}K + K\mathrm{d}$ are called *cochain homotopic*.

The displayed identity constructs that relation for the two endpoint pullbacks.

For a star-shaped open set U, the radial homotopy is given by $F(t, x) = tx$. The figure shows its segments from 0 to x.

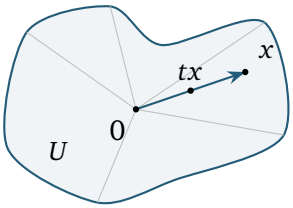


Figure 13.2: In a star-shaped set, each segment from 0 to x stays inside U. The contraction is given by $F(t, x) = tx$.

For this homotopy the operator has the explicit formula, for $k \geq 1$,

$$(K\alpha)_x(v_1, \ldots, v_{k-1}) = \int_0^1 t^{k-1}\alpha_{tx}(x, v_1, \ldots, v_{k-1})\,\mathrm{d}t. \tag{13.1}$$

Indeed, the time derivative of F is x, while each spatial tangent vector is multiplied by t. This gives the power t^{k-1}. Near a fixed $x \in U$, compactness of its segment and openness of U give $\varepsilon > 0$ and a neighbourhood of x such that $tx' \in U$ for every point x' in that neighbourhood and $t \in (-\varepsilon, 1 + \varepsilon)$. This gives the required local smooth extension in time. Differentiating the coefficients under the integral therefore defines a smooth form on U.

Theorem 13.11 (Poincaré lemma). *On a star-shaped open set, every closed k-form with $k \geq 1$ is exact. Such a set has degree-zero cohomology consisting of the constant functions.*

Proof. After translating the star centre, use $F(t, x) = tx$. The map F_1 is the identity. The map F_0 is constant, so its differential is zero and its pullback on positive-degree forms is zero. For closed α the operator identity consequently gives $\alpha = \mathrm{d}K\alpha$. In degree zero it instead gives $f - f(0) = K\mathrm{d}f$. A closed function is therefore constant. ∎

For example, $\alpha = 2x\mathrm{d}x + 2y\mathrm{d}y$ on $\mathbb{R}^2$ has $K\alpha = \int_0^1 2t(x^2+y^2)\,\mathrm{d}t = x^2+y^2$. For a degree-two example on $\mathbb{R}^2$, take $\alpha = \mathrm{d}x \wedge \mathrm{d}y$. For a tangent vector $v = (v_x, v_y)$, the radial formula gives

$$(K\alpha)_{(x,y)}(v) = \int_0^1 t(xv_y - yv_x)\,\mathrm{d}t = \frac{xv_y - yv_x}{2}.$$

The radial vector fills the first argument. The remaining tangent vector contributes the factor t. Hence

$$K(\mathrm{d}x \wedge \mathrm{d}y) = \frac{x\mathrm{d}y - y\mathrm{d}x}{2}, \qquad \mathrm{d}K(\mathrm{d}x \wedge \mathrm{d}y) = \frac{\mathrm{d}x \wedge \mathrm{d}y - \mathrm{d}y \wedge \mathrm{d}x}{2} = \mathrm{d}x \wedge \mathrm{d}y.$$

In $\mathbb{R}^3$, write a closed 2-form as $B_x\mathrm{d}y \wedge \mathrm{d}z + B_y\mathrm{d}z \wedge \mathrm{d}x + B_z\mathrm{d}x \wedge \mathrm{d}y$. Its closedness means $\nabla \cdot \mathbf{B} = 0$, by Lecture 10. On a region star-shaped about 0, its radial primitive is the 1-form $\mathbf{A} \cdot \mathrm{d}\mathbf{x}$, where

$$\mathbf{A}(x) = \int_0^1 t\,\mathbf{B}(tx) \times x\,\mathrm{d}t.$$

This follows from the determinant identity $\mathbf{B} \cdot (x \times v) = (\mathbf{B} \times x) \cdot v$. For a constant field, the integral gives $\mathbf{A}(x) = \mathbf{B} \times x/2$.

The punctured plane is not star-shaped. Its radial paths end at a missing point, so the homotopy does not have its claimed target there. A formal insertion into α_{ang} would even give zero, because $-yx + xy = 0$. It cannot supply a primitive of that nonzero form.

13.3 Homotopy invariance

Theorem 13.12 (Homotopy invariance · *proved given cited smoothing; smooth homotopies proved directly*)**.** *Smoothly homotopic maps induce the same maps on de Rham cohomology. The same holds for smooth maps that are continuously homotopic. Smooth manifolds with the same continuous homotopy type have isomorphic cohomology in every degree.*

Proof. For a smooth homotopy and a closed form α, the operator identity gives $F_1^*\alpha - F_0^*\alpha = \mathrm{d}K_F\alpha$. The two pullbacks therefore represent the same class.

For the continuous case we use the following smooth approximation results. Every continuous map between smooth manifolds is homotopic to a smooth map. Two homotopic smooth maps admit a smooth homotopy. These are cited from Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 6.26 and 6.29, pp. 141–143. Choose a smooth representative of a continuous map's homotopy class and use its pullback. Two choices induce the same map by the second cited result and the smooth proof above. Composition and identities are preserved because compositions of smooth representatives represent the continuous compositions. If continuous maps $f\colon M \to N$ and $g\colon N \to M$ are homotopy inverses, their composites represent identity maps. Contravariance then gives $f^*g^* = \mathrm{id}$ and $g^*f^* = \mathrm{id}$ on the respective cohomology spaces. Thus these maps are inverse isomorphisms. ∎

A contractible manifold consequently has the cohomology of a point: $\mathbb{R}$ in degree zero and zero in positive degrees. For $n \geq 1$, the punctured space $\mathbb{R}^n \setminus \{0\}$ retracts onto S^{n-1} by $r(x) = x/\|x\|$. With i the inclusion, $r \circ i = \mathrm{id}$. A smooth homotopy from the identity to $i \circ r$ is

$$F(t, x) = \left((1-t)^2 + \frac{t^2}{\|x\|}\right) x.$$

We use squares in this coefficient to obtain a homotopy defined for all real t, including smooth extensions past both endpoints. The coefficient is positive for every real t, since its two nonnegative summands never vanish together. Hence the homotopy stays in the punctured space. It has the required endpoints. This proves $H^*(\mathbb{R}^n \setminus \{0\}) \cong H^*(S^{n-1})$.

Self-check. State what a de Rham cohomology class is and why we introduced it. Explain how a nonzero period distinguishes a class from zero.

Answer. A class is a closed form modulo addition of exact forms. In positive degree, the zero class consists of forms with a global primitive. We introduced the quotient to measure the obstruction to such a primitive. Stokes makes every period of an exact form zero, so a nonzero period proves that a closed form represents a nonzero class.

Unit 13B Gluing forms and computing sphere cohomology

13.4 Exact sequences and Mayer–Vietoris

Lecture 5 used van Kampen to compute the fundamental group of a union from two open sets and their overlap. Mayer–Vietoris uses the same cover to compute cohomology. We restrict forms to the pieces, compare them on the overlap, and differentiate local lifts to recover a global class.

Definition 13.13. A sequence of vector spaces and linear maps is *exact* at a term if the image of the incoming map equals the kernel of the outgoing map.

Definition 13.14. A *short exact sequence* has the form $0 \to A \to B \to C \to 0$.

Its first nonzero map is injective and its last is surjective. For a sequence, exactness is a property of adjacent maps. For a form, exactness means that the form is an exterior derivative. The uses are related: a cochain complex is exact at a degree precisely when every closed element in that degree is exact.

Definition 13.15. The *cokernel* of a linear map $T\colon A \to B$ is the quotient

$$\operatorname{coker} T = B/\operatorname{im} T.$$

Let $M = U \cup V$ be an open cover and set $W = U \cap V$. Restriction gives a short exact sequence of form spaces in each degree:

$$0 \longrightarrow \Omega^k(M) \xrightarrow{i} \Omega^k(U) \oplus \Omega^k(V) \xrightarrow{j} \Omega^k(W) \longrightarrow 0, \qquad j(a,b) = b|_W - a|_W.$$

The map i restricts a global form to both sets. It is injective because these sets cover M. A pair is in $\ker j$ exactly when its forms agree on the overlap, which is precisely when they glue to a global form. To prove surjectivity, choose a partition $\rho_U + \rho_V = 1$ subordinate to the cover, supplied by Theorem 12.8. For $\zeta \in \Omega^k(W)$ take

$$a = -\operatorname{ext}_U(\rho_V \zeta), \qquad b = \operatorname{ext}_V(\rho_U \zeta).$$

Here ext means extension by zero from the overlap. This extension is smooth: near a point of $U \setminus V$, the support condition makes ρ_V zero, and similarly on $V \setminus U$. Their difference on W is $(\rho_U + \rho_V)\zeta = \zeta$. All restriction and difference maps commute with differentiation.

For a closed overlap form ζ, choose any lifts with $j(a,b) = \zeta$. Then $\mathrm{d}b - \mathrm{d}a = \mathrm{d}\zeta = 0$ on W. The derivatives glue to a global closed form c of degree $k+1$.

Definition 13.16. The *connecting homomorphism* is given by

$$\delta_{\mathrm{MV}}\colon H^k(W) \longrightarrow H^{k+1}(M), \qquad \delta_{\mathrm{MV}}[\zeta] = [c], \quad c|_U = \mathrm{d}a, \quad c|_V = \mathrm{d}b.$$

There is no additional sign. The subscript identifies this as the Mayer–Vietoris map. Different lifts differ by restrictions of a global form η, so their resulting c differs by $\mathrm{d}\eta$. If ζ changes by $\mathrm{d}\xi$, lift ξ to (a', b') and replace (a, b) by $(a + \mathrm{d}a', b + \mathrm{d}b')$. The resulting c is unchanged. In degree zero there is no such exact change. These observations prove well-definedness. Adding or scaling lifts proves linearity.

Theorem 13.17 (Mayer–Vietoris · *exactness sketched; connecting map and overlap exactness proved given partitions of unity*)**.** *The induced maps and connecting homomorphisms form an exact sequence*

$$\begin{aligned} 0 \longrightarrow H^0(M) \xrightarrow{i^*} H^0(U) \oplus H^0(V) \xrightarrow{j^*} H^0(W) \xrightarrow{\delta_{\mathrm{MV}}} H^1(M) \longrightarrow \cdots, \\ \cdots \longrightarrow H^k(M) \xrightarrow{i^*} H^k(U) \oplus H^k(V) \xrightarrow{j^*} H^k(W) \xrightarrow{\delta_{\mathrm{MV}}} H^{k+1}(M) \longrightarrow \cdots. \end{aligned} \tag{13.2}$$

Proof at the overlap term and sketch elsewhere. If $\zeta = b - a$ for closed lifts, its connecting form is zero. Hence $\operatorname{im} j^* \subset \ker \delta_{\mathrm{MV}}$. Conversely, if $\delta_{\mathrm{MV}}[\zeta] = 0$, its glued form is $c = \mathrm{d}\eta$ for a global η. Subtract the restrictions of η from both lifts. Their derivatives now vanish and their difference remains ζ. Thus $[\zeta] \in \operatorname{im} j^*$, proving exactness at $H^k(W)$.

At the pair term, an exact difference of closed representatives can be removed by lifting its primitive through j. The adjusted representatives agree and glue. At the global term, local primitives of a class restricting to zero have a closed difference on W whose connecting class is the original class. The subsection on the remaining exactness arguments supplies the representative and degree-zero details. This is the cochain construction of Tu, §25–§26, especially (26.1), pp. 288–291. ∎

The remaining exactness arguments *(optional)*

Later sphere and bundle computations do not depend on the proof details in this subsection.

Exactness at the pair and global terms. At the pair term, restrictions of a global closed form have zero difference. Thus $\operatorname{im} i^* \subset \ker j^*$. Conversely, let (a, b) be closed representatives of a class in $\ker j^*$. For $k \geq 1$, their difference is $b - a = \mathrm{d}\xi$ on W. Surjectivity of j supplies forms r and s of degree $k - 1$ with $s - r = \xi$ on W. The adjusted forms satisfy

$$(b - \mathrm{d}s) - (a - \mathrm{d}r) = \mathrm{d}\xi - \mathrm{d}(s - r) = 0.$$

They are closed and represent the original classes on U and V. They therefore glue to a global closed form. This proves the reverse inclusion. For $k = 0$, a zero difference class means $b - a = 0$, since no nonzero 0-form is exact. The functions already agree and glue.

At the global term of degree $k \geq 1$, a connecting form restricts to $\mathrm{d}a$ and $\mathrm{d}b$. Its restrictions are exact, so $\operatorname{im} \delta_{\mathrm{MV}} \subset \ker i^*$. Conversely, let a global closed k-form c have zero restriction classes. Choose local primitives a and b with $c|_U = \mathrm{d}a$ and $c|_V = \mathrm{d}b$. Their difference $\zeta = b - a$ is closed on W, because $\mathrm{d}\zeta = c|_W - c|_W = 0$. These primitives are lifts of ζ, so $\delta_{\mathrm{MV}}[\zeta] = [c]$. This proves the reverse inclusion. In degree zero the incoming map is zero. A closed function with zero restriction classes vanishes on each cover member, hence vanishes on M. Thus i^* is injective also in degree zero. ∎

A circle connecting class

We use the ordered cover $U = S^1 \setminus \{S\}$ and $V = S^1 \setminus \{N\}$, where $S = (0, -1)$ and $N = (0, 1)$. The overlap consists of the left arc W_- and the right arc W_+. The counter-clockwise orientation

traverses W_+ from S to N. Write z for the vertical coordinate in this circle calculation. An explicit smooth partition is $\rho_U = h(z)$ and $\rho_V = 1 - h(z)$, where

$$E(s) = \begin{cases} e^{-1/s}, & s > 0, \\ 0, & s \le 0, \end{cases} \qquad h(z) = \frac{E(z + 1/2)}{E(z + 1/2) + E(1/2 - z)}.$$

For $s > 0$, each derivative of E is a polynomial in $1/s$ times $e^{-1/s}$. Each expression tends to zero as s decreases to zero. Thus these derivatives match the zero derivatives on the negative half-line, so E is smooth. The denominator defining h is positive for every $z \in \mathbb{R}$. The support of ρ_U avoids S, and the support of ρ_V avoids N. Take the overlap function ζ equal to 0 on W_- and 1 on W_+. Write c for its glued connecting form. The figure plots this partition and the coefficient of c on W_+.

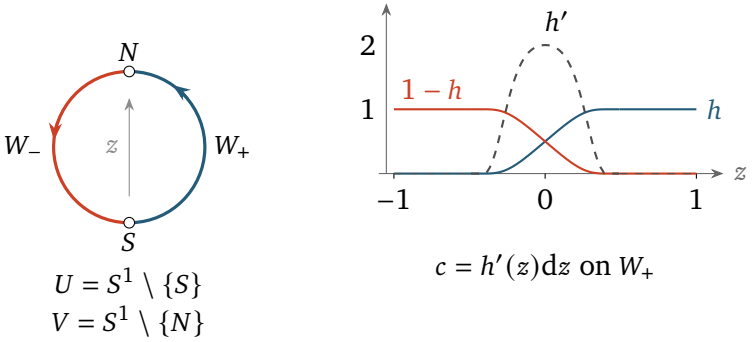


Figure 13.3: On the right overlap arc, increasing height follows the positive circle orientation. The connecting form has coefficient $h'(z)$ there and is zero on the left arc.

Example 13.18 (An overlap function with period-one image)**.** The overlap function ζ is closed, since it is locally constant. On W_+ choose $a = -\rho_V$ and $b = \rho_U$. On W_- choose both lifts to be zero. Extend a across N by zero, and extend b across S by zero. The support conditions make these extensions smooth. Their difference is ζ. On W_+ their derivatives agree:

$$da = -d\rho_V = d\rho_U = db = h'(z)dz.$$

The glued form c vanishes on W_- and near both poles. Its period is

$$\int_{S^1} c = \int_{-1}^{1} h'(z)\,dz = h(1) - h(-1) = 1.$$

Thus $\delta_{\mathrm{MV}}[\zeta] = [c]$ is a nonzero circle class.

13.5 Cohomology of spheres

Remove the south pole to obtain $U \subset S^n$ and the north pole to obtain V, for $n \ge 1$. Each set is diffeomorphic to $\mathbb{R}^n$ by stereographic projection. Their overlap has the product description in the figure.

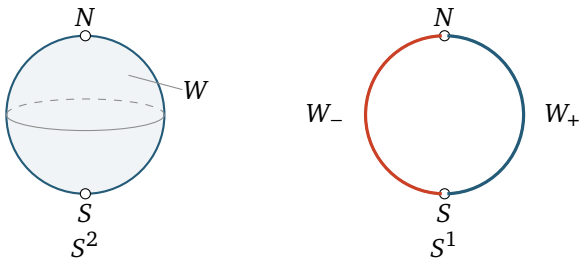


Figure 13.4: Each open set omits one pole. Their overlap is the whole sphere except the two marked poles, an open cylinder for S^2. For S^1 it has two components.

Write a sphere point as $(x', z) \in \mathbb{R}^n \times \mathbb{R}$. On W, the map $(x', z) \mapsto (x'/\sqrt{1-z^2}, z)$ is a diffeomorphism onto $S^{n-1} \times (-1, 1)$, with inverse $(u, z) \mapsto (\sqrt{1-z^2}u, z)$. Contracting the second coordinate to zero retracts W onto S^{n-1}. The two contractible cover members have zero positive-degree cohomology.

Theorem 13.19 (Sphere computation · *proved given Mayer–Vietoris exactness*)**.** *For $n \geq 1$ and $k \geq 0$,*

$$H^k(S^n) \cong \begin{cases} \mathbb{R}, & k = 0 \text{ or } k = n, \\ 0, & \text{otherwise.} \end{cases}$$

For S^0, degree zero is $\mathbb{R}^2$ and all higher degrees vanish. For $n \geq 1$, punctured $\mathbb{R}^n$ has the cohomology of S^{n-1}.

Proof. The space S^0 is two points. A function assigns one value to each point, and all positive-degree forms vanish. This proves the initial case. For $n = 1$, the overlap has two contractible components. The beginning of Mayer–Vietoris is

$$0 \longrightarrow H^0(S^1) \longrightarrow \mathbb{R}^2 \xrightarrow{(a,b)\mapsto(b-a,b-a)} \mathbb{R}^2 \xrightarrow{\delta_{\mathrm{MV}}} H^1(S^1) \longrightarrow 0.$$

The map after δ_{MV} lands in zero, so exactness makes δ_{MV} surjective. Exactness at its source gives

$$\ker \delta_{\mathrm{MV}} = \operatorname{im} j^* = \{(c, c) : c \in \mathbb{R}\}.$$

Hence the induced quotient map is an isomorphism

$$H^1(S^1) \cong \mathbb{R}^2/\{(c, c) : c \in \mathbb{R}\}.$$

This quotient is the cokernel of $j^* : H^0(U) \oplus H^0(V) \to H^0(W)$. In the order (W_-, W_+), the value $v - u$ labels the class of (u, v) modulo the diagonal. Its kernel is precisely that diagonal, so the quotient is one-dimensional. The kernel of j^* is also the diagonal, which gives $H^0(S^1) = \mathbb{R}$. The figure shows the quotient coordinate.

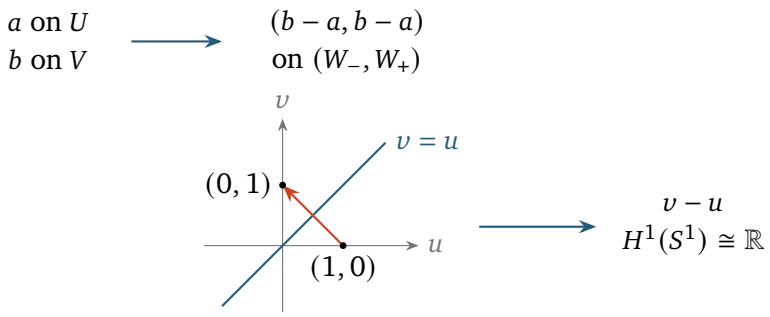


Figure 13.5: Constant functions on U and V give the same difference on both overlap components. The coordinate $v - u$ is unchanged by adding a diagonal vector, so it identifies the quotient with $\mathbb{R}$.

Higher degrees vanish because the circle has dimension one.

For $n \geq 2$, the overlap is connected, so its degree-zero group is $\mathbb{R}$. The difference map $\mathbb{R}^2 \to \mathbb{R}$ is surjective with diagonal kernel. Exactness gives $H^0(S^n) = \mathbb{R}$ and $H^1(S^n) = 0$. For $k \geq 2$, the zero positive-degree groups of U and V give

$$0 \longrightarrow H^{k-1}(W) \xrightarrow{\delta_{\mathrm{MV}}} H^k(S^n) \longrightarrow 0.$$

The left zero forces injectivity, and the right zero forces surjectivity. Thus δ_{MV} is an isomorphism and $H^k(S^n) \cong H^{k-1}(S^{n-1})$. Induction gives precisely one nonzero positive-degree group, in degree n. The explicit retraction in §13.3 proves the punctured-space assertion. In particular, punctured $\mathbb{R}$ has two-dimensional degree-zero cohomology, whereas punctured $\mathbb{R}^n$ for $n \geq 2$ is connected. ∎

13.6 The area class and local primitives on the sphere

On the outward-oriented unit sphere let β be Lecture 10's area form, $\sin\theta\, \mathrm{d}\theta \wedge \mathrm{d}\phi$. It is also the form denoted by μ for $n = 2$ in Lecture 12. Write β_C for Lecture 12's Coulomb flux form on $\mathbb{R}^3 \setminus \{0\}$. Its restriction to the unit sphere is β. We compute its class using the ordered cover above: $U = S^2 \setminus \{\text{south}\}$ and $V = S^2 \setminus \{\text{north}\}$. Write the local 1-forms as a_U and a_V, and their common derivative as c. Their difference and derivatives are related as in the diagram.

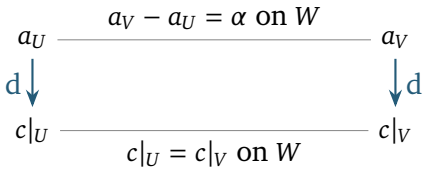


Figure 13.6: Local 1-forms differ by a closed form on W. Their derivatives agree and glue to the 2-form representing the connecting class.

Example 13.20 (The normalized sphere class)**.** The sphere computation gives $H^2(S^2) = \mathbb{R}$ and $H^1(S^2) = 0$. The form $\beta/(4\pi)$ is closed, since every 3-form on a surface vanishes. Its period is

$$\int_{S^2} \frac{\beta}{4\pi} = \frac{1}{4\pi} \int_0^{2\pi} \int_0^{\pi} \sin\theta\, \mathrm{d}\theta\, \mathrm{d}\phi = 1.$$

It is therefore not exact and its class spans $H^2(S^2)$.

The nonzero period prevents a global primitive of $\beta/(4\pi)$. Local primitives still exist. Their overlap difference computes the sign of the connecting map and supplies the cap potentials used in the magnetic monopole model. On W, the globally defined angular 1-form is $\alpha = \mathrm{d}\phi/(2\pi)$, where $\mathrm{d}\phi$ denotes the angular form, not the differential of a globally defined real angle. Choose lifts

$$a_U = -\frac{x\mathrm{d}y - y\mathrm{d}x}{4\pi(1+z)}, \qquad a_V = \frac{x\mathrm{d}y - y\mathrm{d}x}{4\pi(1-z)}.$$

Their denominators do not vanish on their respective domains, so these forms are smooth also at the opposite poles. On the overlap, $x\mathrm{d}y - y\mathrm{d}x = (1-z^2)\mathrm{d}\phi$. Consequently $a_U = -(1-z)\mathrm{d}\phi/(4\pi)$ and $a_V = (1+z)\mathrm{d}\phi/(4\pi)$, giving $a_V - a_U = \alpha$. Their common derivative is

$$\mathrm{d}a_U = \mathrm{d}a_V = \frac{\mathrm{d}z \wedge \mathrm{d}\phi}{4\pi} = -\frac{\sin\theta\,\mathrm{d}\theta \wedge \mathrm{d}\phi}{4\pi} = -\frac{\beta}{4\pi}.$$

Equality extends over each pole by smoothness. Thus $\delta_{\mathrm{MV}}[\mathrm{d}\phi/(2\pi)] = -[\beta/(4\pi)]$, with the order U, V fixed in §13.5. The positive period-one class is $[\beta/(4\pi)]$. A nonzero period obstructs joining its local primitives to a global one. The closed Coulomb form β_C restricts to the area form β. Restriction is an isomorphism on cohomology, inverse to the radial retraction pullback. It sends $[\beta_C/(4\pi)]$ to $[\beta/(4\pi)]$. Hence $[\beta_C/(4\pi)]$ spans $H^2(\mathbb{R}^3 \setminus \{0\})$, with sphere period one.

Cap potentials and the magnetic monopole

Physical input. A magnetic field is represented by a real flux 2-form F_{em}. A local real magnetic potential A_{em} satisfies $F_{\mathrm{em}} = \mathrm{d}A_{\mathrm{em}}$. We model a point magnetic source at the origin by $F_{\mathrm{em}} = (\Phi_{\mathrm{m}}/(4\pi))\beta_C$ on $\mathbb{R}^3 \setminus \{0\}$. The parameter Φ_{m} is its outward sphere flux.

The cap primitives above give real potentials on the unit sphere:

$$A^{(N)}_{\mathrm{em}} = -\Phi_{\mathrm{m}} a_U, \qquad A^{(S)}_{\mathrm{em}} = -\Phi_{\mathrm{m}} a_V, \qquad \mathrm{d}A^{(N)}_{\mathrm{em}} = \mathrm{d}A^{(S)}_{\mathrm{em}} = \frac{\Phi_{\mathrm{m}}}{4\pi}\beta.$$

The north chart is U, and the south chart is V. For nonzero flux, Stokes excludes a global potential on the sphere. The overlap difference $a_V - a_U = \mathrm{d}\phi/(2\pi)$ is closed. It is a differential on each local angle patch, but its period one prevents a single global real angle on the overlap.

For comparison with Lecture 22, let $q \neq 0$ be a particle's signed charge and set $n = q\Phi_{\mathrm{m}}/(2\pi\hbar)$, initially a real parameter. That lecture uses imaginary local potentials, obtained by the rescaling

$$A_N = 2\pi i n\, a_U = -\frac{iq}{\hbar} A^{(N)}_{\mathrm{em}}, \qquad A_S = 2\pi i n\, a_V = -\frac{iq}{\hbar} A^{(S)}_{\mathrm{em}}.$$

Their common derivative is $-in\beta/2$, and their difference is $A_S - A_N = in\mathrm{d}\phi$. Lecture 22 derives the integer condition on n from compatibility of local charged states. Real de Rham cohomology alone allows every real multiple of $[\beta/(4\pi)]$.

Radially extending the northern potential excludes the negative z-axis. This excluded half-axis is called the *Dirac string* in the singular potential description. For $\Phi_{\mathrm{m}} > 0$, a small increasing-azimuth circle near this axis has limiting circulation Φ_{m}. In the thin-solenoid representation this is flux toward the origin along the negative axis. The string represents

the singular choice of potential, not an additional field in the two-chart model. The figure distinguishes the signed circulations on equatorial test circles from this string.

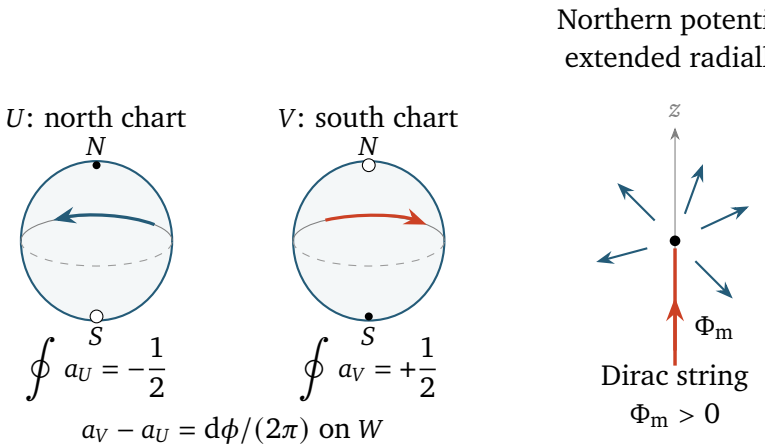


Figure 13.7: The cap arrows show the signs of the real 1-forms on equatorial test circles: a_U has negative azimuthal circulation and a_V positive circulation. For positive flux the singular northern-potential description has a string on the negative z-axis directed toward the origin.

Real coefficients lose torsion information. A torsion element is one annihilated by a positive integer. No real vector space has a nonzero torsion element. In particular, Lecture 4's order-two loop in SO(3) cannot be detected by a closed real 1-form's period. Use smooth loops constant near their endpoints, so their concatenation is smooth. Periods of a closed 1-form are invariant under smooth loop homotopy by Stokes on the cylinder. Continuous homotopies can be smoothed by the cited theorem. Periods add under loop composition, so twice that loop's period is zero and its real period is zero.

Self-check. For the ordered circle cover above, let ζ have values $-5/2$ on W_- and $7/3$ on W_+. Compute the period of $\delta_{\mathrm{MV}}[\zeta]$. Decide whether the overlap values $(-1/2, 13/3)$ give the same class.

Answer. For overlap values (u, v), subtracting (u, u) leaves $(0, v-u)$. The period-one calculation therefore makes the period equal to $7/3 - (-5/2) = 29/6$. The second pair differs from the first by $(2, 2)$, which is in $\operatorname{im} j^*$. The connecting classes are equal.

Self-check. Find $H^1(\mathbb{R}^2 \setminus \{0\})$ and a generator of period one on the counter-clockwise unit circle.

Answer. Radial retraction gives the cohomology of S^1, so this space is one-dimensional. The angular form is closed by Lecture 10. A nonzero period rules out an exact form by Stokes. The class $[\alpha_{\mathrm{ang}}/(2\pi)]$ has period one by Lecture 10. It is nonzero and therefore spans the space.

Summary. De Rham cohomology consists of closed forms modulo exact forms, over $\mathbb{R}$. The homotopy operator proves local exactness in positive degree on star-shaped sets and invariance under smooth homotopy. Cited smoothing extends invariance to continuous homotopy type. Mayer–Vietoris computes sphere cohomology from two contractible sets. On S^2, the area class $[\beta/(4\pi)]$ has period one. Lecture 14 relates periods to degree and singular homology.

Reading. Tu, *An Introduction to Manifolds*, §24–§29, especially §26, pp. 288–291, and §29, pp. 311–317. Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 6.26 and 6.29, pp. 141–143.

Lecture 14

Degree, Poincaré–Hopf and de Rham's theorem

Objectives. *We prove that degree is an integer invariant under homotopy for smooth maps between nonempty closed connected oriented manifolds of the same positive dimension. Degree gives the smooth even-sphere hairy-ball and fixed-point theorems. Two cited theorems then relate vector-field zeros and periods to the homology of Part I.*

By the end of this lecture you will be able to:

- Define degree with its dimension, orientation and compactness hypotheses.
- Compute degree as a signed count over a regular value.
- Prove the smooth hairy-ball theorem on even spheres and the smooth Brouwer theorem.
- Compute the two local indices of the specified field on S^2.

Unit 14A Degree

14.1 Integration and top cohomology

Lecture 3 assigned a winding number to a loop in the circle. Give S^1 its counter-clockwise orientation and normalized angular form $\nu = \alpha_{\mathrm{ang}}/(2\pi)$ from Lecture 10. For each $m \in \mathbb{Z}$, the power map and its pullback are

$$f_m \colon S^1 \longrightarrow S^1, \qquad f_m(e^{\mathrm{i}\phi}) = e^{\mathrm{i}m\phi}, \qquad f_m^* \nu = m\nu.$$

Indeed, the angular lift $m\phi$ has derivative m. The integral of the pullback is therefore m. Degree extends this oriented count to higher dimensions. Independence of the chosen integral-one form requires the following fact: a top form on a nonempty closed connected oriented manifold is exact if and only if its integral is zero.

All manifolds below are smooth. Cohomology has real coefficients. On an oriented n-manifold every n-form is closed, since all $(n+1)$-forms vanish. For a closed manifold, Stokes makes integration zero on exact forms. Integration therefore defines a linear map on $H^n(M)$.

Theorem 14.1 (Top cohomology · *sketched*)**.** *For a nonempty closed connected oriented n-manifold M,*

$$I_M \colon H^n(M) \longrightarrow \mathbb{R}, \qquad I_M[\alpha] = \int_M \alpha$$

is an isomorphism. Thus two top forms have equal integrals exactly when they differ by an exact form.

Sketch. For $n = 0$, connected M is a point and integration is its signed value, an isomorphism. For $n \geq 1$, a positive bump function in one positive chart gives a top form of positive integral. Rescaling it gives integral one, proving surjectivity. For injectivity the local input is this compact-support version of the Poincaré lemma: on $\mathbb{R}^n$, a compactly supported top form of integral zero has a compactly supported primitive.

The local construction uses induction and the fundamental theorem of calculus. For $n = 1$, integrate its coefficient from $-\infty$; zero total integral makes the primitive zero also beyond the support. For $n > 1$, write the coefficient as $f(x', u)$, set $h(x') = \int_{\mathbb{R}} f(x', u)\,\mathrm{d}u$, and choose a compactly supported smooth function $\rho(u)$ of integral one. Split the coefficient as $(f - h\rho) + h\rho$. The first term has zero integral in every u fibre, so integrating in u gives a compactly supported primitive, with the sign from moving $\mathrm{d}u$ past $\mathrm{d}x'$. The second term has a primitive by the induction hypothesis applied to $h\mathrm{d}x'$, whose integral is $\int f = 0$, and then wedged with $\rho\mathrm{d}u$.

On M, choose a finite cover by positive coordinate balls diffeomorphic to $\mathbb{R}^n$. A partition of unity splits a top form into ball-supported pieces. Connectedness permits ordering the balls so that each after the first meets an earlier ball. In each chosen overlap take a top form of integral one. Starting with the last piece, subtract its integral times this overlap form and add the same form to the earlier piece. Repeating backwards makes every piece have integral zero, since the total is zero. The local construction now supplies primitives which extend by zero and sum to a global primitive. The cover and transfer details are the sketched steps. See Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 17.30–17.31, pp. 454–455. ∎

For example, on outward S^2, the form $\beta/(4\pi)$ from Lecture 13 has integral one. A 2-form α of integral $a \in \mathbb{R}$ consequently satisfies $[\alpha] = a[\beta/(4\pi)]$.

14.2 The degree of a map

Definition 14.2 (Degree)**.** Let M and N be nonempty closed connected oriented manifolds of the same dimension $n \geq 1$. Let $f \colon M \to N$ be smooth. Choose a top form ν on N with integral one. The *degree* of f is

$$\deg f = \int_M f^*\nu.$$

Degree compares the orientations and will count inverse images with signs.

Theorem 14.3 (Degree and homotopy · *proved given top cohomology and cited smoothing*)**.** *The degree is independent of ν. For every top form α on N,*

$$\int_M f^*\alpha = (\deg f) \int_N \alpha.$$

Homotopic smooth maps have the same degree. The identity has degree one, and a constant

map has degree zero when $n \geq 1$.

Proof. Two choices of ν differ by $\mathrm{d}\eta$, by the top-cohomology theorem. Their pullbacks differ by $\mathrm{d}f^*\eta$, whose integral is zero by Stokes. For an arbitrary α, write $a = \int_N \alpha$. The form $\alpha - a\nu$ has integral zero, so it is exact. Pulling back and integrating gives the claimed formula.

For smoothly homotopic maps, the homotopy operator of Proposition 13.9 makes the pullbacks of ν differ by an exact form. Their integrals coincide. A continuous homotopy between smooth maps can be smoothed by Lee's Theorem 6.29, recalled in Lecture 13, so the same conclusion holds. The identity pulls ν back to itself. A constant map has zero differential, hence pulls a positive-degree form back to zero. ∎

For example, the reflection $(x, y) \mapsto (x, -y)$ on S^1 reverses the angular form, so its degree is -1. The degree also multiplies under composition. For a smooth map g from N to another such manifold of the same dimension, choose a top form ν of integral one on its target. Apply the integral formula to $g^*\nu$ and then to ν to obtain $\deg(g \circ f) = (\deg g)(\deg f)$.

Closedness matters. The map $x \mapsto x/2$ from $[0, 1]$ to itself pulls $\mathrm{d}x$ back to $\mathrm{d}x/2$. The quotient of the two integrals is $1/2$, but the interval has boundary, so this quotient is not a degree under our definition.

14.3 Integrality and regular values

Definition 14.4 (Regular value)**.** A point $y \in N$ is a *regular value* of f if $\mathrm{d}f_x \colon T_xM \to T_yN$ is surjective for every $x \in f^{-1}(y)$. An empty inverse image satisfies this condition.

In equal dimensions each such differential is invertible.

Definition 14.5 (Local degree)**.** The *local degree* at $x \in f^{-1}(y)$, for a regular value y, is $+1$ if $\mathrm{d}f_x$ preserves orientation and -1 if it reverses orientation. Write this sign as $\operatorname{sgn} \det \mathrm{d}f_x$, using positive bases in both tangent spaces.

An inverse branch is a neighbourhood mapped diffeomorphically onto a small target neighbourhood W. The diagram names three such branches V_1, V_2 and V_3 and shows their signs. The proof below constructs the common neighbourhood W.

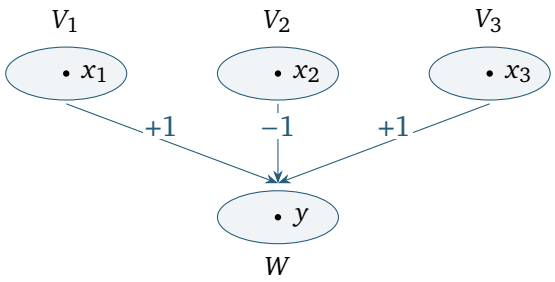


Figure 14.1: Three inverse branches over a small neighbourhood W contribute signs $+1$, -1 and $+1$. Their total contribution is one.

Proposition 14.6 (Signed count · *proved given top cohomology; regular values exist by cited Sard*)**.** *For every regular value $y \in N$, the inverse image is finite and*

$$\deg f = \sum_{x \in f^{-1}(y)} \operatorname{sgn} \det \mathrm{d}f_x.$$

In particular the degree is an integer, independent of the chosen regular value.

Proof. The inverse function theorem isolates each inverse-image point. The fibre is closed in compact M, hence compact. Its isolating neighbourhoods give an open cover with a finite subcover, so the fibre is finite. Write its points as $x_1, \dots, x_m$ when it is nonempty. Choose disjoint neighbourhoods U_i on which f is a diffeomorphism onto a neighbourhood of y. The complement $K = M \setminus \bigcup_i U_i$ is compact and has no point mapping to y. Its image is compact and closed. Choose a small connected coordinate ball W about y inside every $f(U_i)$ and disjoint from $f(K)$. The sets $V_i = U_i \cap f^{-1}(W)$ are then all the inverse branches, each diffeomorphic to W. On a connected branch the orientation sign is constant, since its nonzero Jacobian determinant is continuous.

Choose a smooth top form ν supported in W with integral one. Change of variables on each branch gives

$$\int_M f^*\nu = \sum_{i=1}^{m} \int_{V_i} f^*\nu = \sum_{i=1}^{m} \operatorname{sgn} \det \mathrm{d}f_{x_i}.$$

If the fibre is empty, choose W disjoint from the compact image $f(M)$. The pullback of such a supported ν is zero, giving degree zero and the empty sum. This proves the formula for every regular value.

Sard's theorem says that the critical values of a smooth map have measure zero in the target. We cite Lee, Theorem 6.10, p. 129. Here a critical value is the image of a point with nonsurjective differential. A nonempty coordinate ball has positive measure, so regular values exist. The formula at any of them proves integrality. Since it computes the already defined degree, it also proves independence of the regular value. ∎

For a circle map, the local signs are the signs of the derivative of an angular lift. Consider the lift

$$\psi(\phi) = \phi + \frac{\pi}{2} \sin \phi, \qquad \psi(\phi + 2\pi) = \psi(\phi) + 2\pi.$$

It defines the smooth map $e^{\mathrm{i}\phi} \mapsto e^{\mathrm{i}\psi(\phi)}$. The target point -1 has inverse images at angles $\pi/2$, π and $3\pi/2$. The derivatives there are 1, $1 - \pi/2$ and 1. Their signs sum to one, which is the degree. The graph shows these crossings and a critical level c at a local maximum.

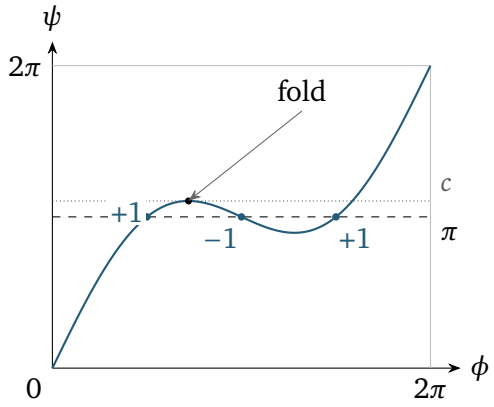


Figure 14.2: The lift crosses level π with signs $+, -, +$. At the critical level c, two crossings meet at a fold; passing below c creates a pair with opposite signs.

The critical angle is $\phi_c = \arccos(-2/\pi)$ and $c = \psi(\phi_c)$. The second derivative there is negative, so the critical point is a nondegenerate local maximum. The signed count stays one on either side.

14.4 Antipodal maps, tangent fields and fixed points

A fixed point of a map f means a point x with $f(x) = x$. Let μ be the outward area form on S^n, characterized in Lecture 12 by $\mu_p(v_1, \ldots, v_n) = \det[p\ v_1\ \cdots\ v_n]$.

Corollary 14.7 (Applications of degree · *proved given top cohomology*)**.** *For $n \geq 1$, the antipodal map $a\colon S^n \to S^n$, given by $a(x) = -x$, has degree $(-1)^{n+1}$. Every smooth tangent field on S^{2m}, with $m \geq 1$, has a zero. Every smooth map from the closed unit ball $\overline{B}^d$ to itself has a fixed point, for $d \geq 1$.*

A nonvanishing smooth tangent field X would give the unit field $u(x) = X_x/\|X_x\|$. Tangency makes x and $u(x)$ orthogonal. The figure shows the semicircle determined by these two unit vectors.

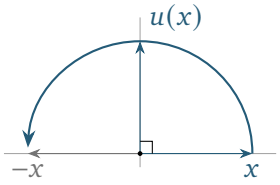


Figure 14.3: In the plane spanned by the orthogonal unit vectors x and $u(x)$, the unit semicircle joins x to $-x$ through $u(x)$.

For the fixed-point argument, the direction from $f(x)$ to x is undefined precisely at a fixed point. The figure displays this direction for $x = (3/4, 1/4)$ and $f(x) = (0, 1/4)$ in the unit disc.

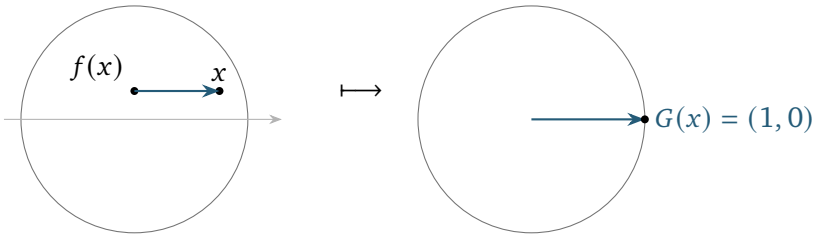


Figure 14.4: The displacement $x - f(x)$ points to the right. Normalizing it gives the sphere-valued map $G(x)$ used under the assumption that no fixed point exists.

Proof. The derivative of the antipodal map sends every tangent vector to its negative. Thus $a^*\mu = (-1)^{n+1}\mu$, because the determinant has $n+1$ columns. The integral definition gives its degree.

Suppose a tangent field X on S^{2m} never vanishes and normalize it to $u(x) = X_x/\|X_x\|$. Tangency gives $x \cdot u(x) = 0$. The map $H\colon [0,1] \times S^{2m} \to S^{2m}$ given by

$$H(t, x) = \cos(\pi t)x + \sin(\pi t)u(x)$$

has squared norm $\cos^2(\pi t) + \sin^2(\pi t) = 1$. It is a smooth homotopy from the identity to the antipodal map. Their degrees would be equal, contradicting $1 \neq -1$. This proves the *hairy-ball theorem* for even spheres.

For the ball of dimension $d \geq 2$, suppose a smooth f has no fixed point. The map $G\colon \overline{B}^d \to S^{d-1}$ given by $G(x) = (x - f(x))/\|x - f(x)\|$ is smooth. Its boundary restriction g

has degree zero: for a normalized top form ν on S^{d-1}, Stokes gives $\int_{S^{d-1}} g^*\nu = \int_{B^d} \mathrm{d}G^*\nu = 0$, since $\mathrm{d}\nu = 0$. On the boundary the formula

$$H(t,x) = \frac{x - tf(x)}{\|x - tf(x)\|}$$

is a smooth homotopy from the identity to g. For $t < 1$ its denominator is nonzero because $\|tf(x)\| \le t < 1 = \|x\|$. At $t = 1$ it is nonzero by the assumption. Homotopy invariance gives $\deg g = 1$, a contradiction. For $d = 1$, the function $f(x) - x$ is nonnegative at -1 and nonpositive at 1, so the intermediate value theorem gives a zero. ∎

Evenness is necessary for the tangent-field conclusion. On $S^{2m-1} \subset \mathbb{R}^{2m}$, with $m \ge 1$, the field

$$u(x) = (-x_2, x_1, -x_4, x_3, \ldots, -x_{2m}, x_{2m-1})$$

has $x \cdot u(x) = 0$ because each coordinate pair cancels. Its squared norm is $\sum_{j=1}^{2m} x_j^2 = 1$, so it never vanishes.

The fixed-point proof extends Lecture 3's no-retraction argument: an extension to the ball forces boundary degree zero, whereas a boundary map homotopic to the identity has degree one. The fixed-point conclusion is the *Brouwer theorem*. It also holds for continuous maps. That extension is cited here from Lee, Theorem 17.39, pp. 459–460, for dimensions $d \ge 2$. Its proof uses Theorem 17.38 and smooth approximation into manifolds, Theorem 6.26. For $d = 1$ the intermediate value argument above applies directly to continuous maps. For $d = 2$, Lecture 3 proved it by the retraction argument.

14.5 The index of a vector-field zero

Definition 14.8 (Index of an isolated zero)**.** An isolated zero of a vector field is a zero with a neighbourhood containing no other zero. In dimension $n \ge 2$, choose coordinates centred at such a zero p and a small outward-oriented coordinate sphere S_ε^{n-1}. Write the coordinate components of the field as $X(x) \in \mathbb{R}^n$. Its *index* is

$$\operatorname{ind}_p X = \deg\left(x \longmapsto \frac{X(x)}{\|X(x)\|} : S_\varepsilon^{n-1} \longrightarrow S^{n-1}\right).$$

In dimension one use half the difference of the component's signs at the right and left endpoints of a small coordinate interval centred at the zero. In dimension zero each point has index one.

Changing the small radius gives homotopic maps through spheres avoiding zeros. Independence of the coordinate chart is cited from Milnor, *Topology from the Differentiable Viewpoint*, §6, Lemma 1, pp. 33–35. In dimension two the index is the winding number of the normalized field on a small counter-clockwise circle. For the source (x, y), sink $(-x, -y)$ and centre $(-y, x)$, the target angles are respectively ϕ, $\phi + \pi$ and $\phi + \pi/2$. Each gives index $+1$. For the saddle $(x, -y)$ the target angle is $-\phi$, so the index is -1. The sampling circles and the target dials show these different turns.

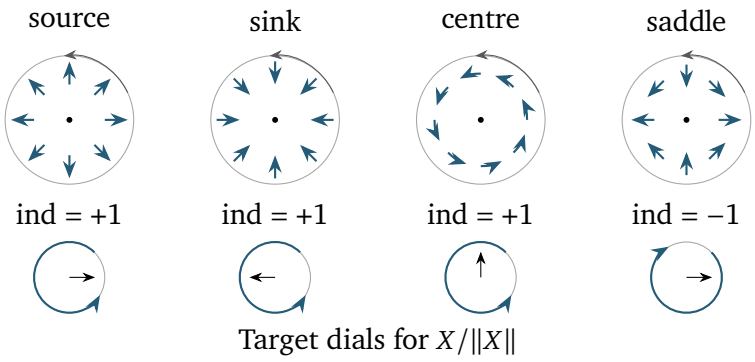


Figure 14.5: Each sampling circle is traversed counter-clockwise. The normalized vectors turn counter-clockwise for the source, sink and centre, and clockwise for the saddle.

In dimension n, the sink $-x$ has index $(-1)^n$ by the antipodal formula on S^{n-1}.

The alternating count of Lecture 6 also has a vector-field interpretation. On a triangulated closed surface, sources at vertices contribute +1, saddles on edges contribute −1, and sinks in faces contribute +1. The triangle illustrates the directions from vertices to edges to faces.

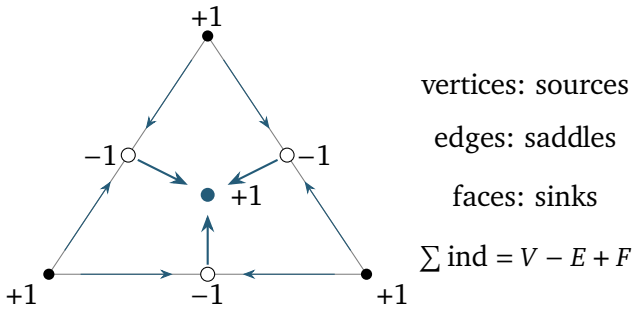


Figure 14.6: One face of a triangulation, with arrows from vertices to edge midpoints and from edge midpoints to the face centre. Counting each vertex, edge and face once gives $V - E + F$.

This is an illustration of the index count, rather than a proof that such local pictures fit together smoothly. The cited theorem gives the same total for every smooth field with isolated zeros. For a closed manifold, the Euler characteristic $\chi(M)$ is the alternating sum of its real Betti numbers, as in Lecture 6. The finiteness and the definition for general closed manifolds follow from the comparison in §14.7.

Theorem 14.9 (Poincaré–Hopf · *cited*)**.** *For a smooth vector field with isolated zeros on a closed manifold M,*

$$\sum_{p : X_p = 0} \operatorname{ind}_p X = \chi(M).$$

Here χ is the Euler characteristic. The theorem is cited from Milnor, §6, p. 35.

On the unit sphere write $p = (x, y, z)$ and take $X_p = e_3 - z\,p$. This field provides the explicit test in the figure.

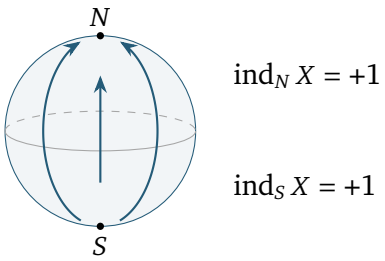


Figure 14.7: The field $X_p = e_3 - zp$ on S^2 points toward the north pole. It vanishes at the north pole N and the south pole S. Each zero has index +1.

The field is tangent because $p \cdot X_p = z - z\,\|p\|^2 = 0$. A zero requires $e_3 = zp$, hence $p = \pm e_3$. Near either pole use the coordinates $p = (x, y, \sigma\sqrt{1 - x^2 - y^2})$, where $\sigma = +1$ at the north pole and $\sigma = -1$ at the south pole. The coordinate components are $(-zx, -zy)$. On the small circle $(x, y) = \varepsilon(\cos\phi, \sin\phi)$, their normalization is $-\sigma(\cos\phi, \sin\phi)$. Both signs give one counter-clockwise turn as ϕ runs from 0 to 2π. Thus both indices are +1, and their sum is $2 = \chi(S^2)$ from Lecture 6. A sink in two dimensions has index +1 here, not −1. For comparison, the planar source (x, y) has index +1, while the saddle $(x, -y)$ normalizes to $(\cos\phi, -\sin\phi)$ and has index −1. Lecture 9's rotation field L_z has a centre at each pole, with the same index sum. Poincaré–Hopf also rules out a field with no zeros on S^2, since its empty index sum would be zero.

14.6 Radial flux and complex power maps

Let $K \subset \mathbb{R}^3$ be a compact smooth region with connected boundary $S = \partial K$, oriented outward, and assume $0 \notin S$.

Definition 14.10 (Radial Gauss map)**.** The *radial Gauss map* $G\colon S \to S^2$ is given by $G(x) = x/\|x\|$.

It records the direction from the origin. This map differs from the map assigning the outward unit normal to a surface point. The figure shows rays through unit spheres centred at the origin and at $(2, 0, 0)$. A ray from an exterior origin first enters with sign −1 and then exits with sign +1.

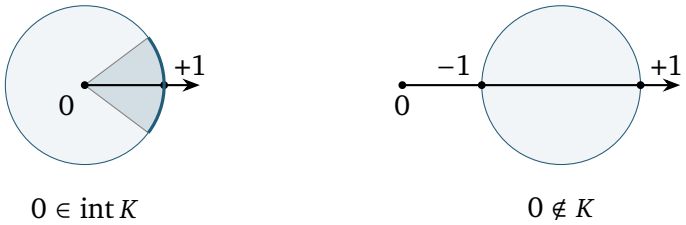


Figure 14.8: A positive ray exits the sphere centred at the origin once. For the sphere centred at $(2, 0, 0)$ it crosses at $(1, 0, 0)$ and $(3, 0, 0)$ with opposite signs; the shaded sector shows the directions of a surface patch.

For a regular target direction, the inverse images of G are the transverse intersections of its positive ray with S. The sign is positive when the ray exits K and negative when it enters. Indeed, for an outward unit normal n_x and a positive tangent basis (v, w), the sign

of $\det[x\ v\ w]$ is the sign of $x \cdot n_x$. Crossings alternate. Starting inside gives one more exit than entry; starting outside gives equal numbers.

Definition 14.11 (Signed solid angle)**.** The *signed solid angle* of an oriented patch in $\mathbb{R}^3 \setminus \{0\}$ is the integral over that patch of the pullback of the unit sphere's area form under the radial projection $x \mapsto x/\|x\|$.

It measures the area of its radial image with orientation and multiplicity.

Physical input. For an electrostatic point charge $q \in \mathbb{R}$ at the origin, the Coulomb field in rationalised units is given, as in Lecture 12, by

$$\mathbf{E}(x) = \frac{q}{4\pi}\frac{x}{\|x\|^3}, \qquad x \in \mathbb{R}^3 \setminus \{0\}.$$

Its outward flux is the integral of $q\beta_C/(4\pi)$, where β_C is the Coulomb flux form of that lecture.

Example 14.12 (Degree of radial flux)**.** Let β_{S^2} be the outward area form on the target sphere and let $r = \|x\|$. For tangent vectors v and w at $x \in S$,

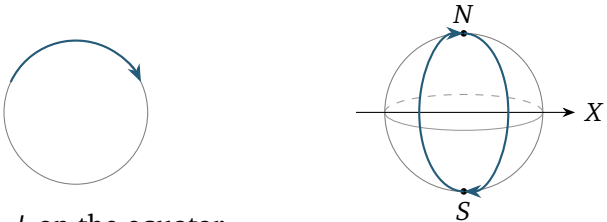
$$\mathrm{d}G_x(v) = \frac{v}{r} - \frac{x(x \cdot v)}{r^3}, \qquad (G^*\beta_{S^2})_x(v, w) = \frac{\det[x\ v\ w]}{r^3}.$$

Indeed, the target form is the determinant with first column x/r. Every additional term parallel to x in the other columns gives zero. This pullback is precisely Lecture 12's Coulomb flux form restricted to S. Since the target area is 4π, its flux calculation gives

$$\deg G = \frac{1}{4\pi}\int_S G^*\beta_{S^2} = \begin{cases} 1, & 0 \in \operatorname{int} K, \\ 0, & 0 \notin K. \end{cases}$$

The flux is therefore an integer multiple of the sphere's area. For the Coulomb field, multiplying this integral by $q/(4\pi)$ gives flux $q \deg G$: it is q when the charge is inside and zero when it is outside.

For complex maps use $\mathbb{C}\mathrm{P}^1$ with its complex orientation: if $z = u + \mathrm{i}v$ is a coordinate, the ordered real coordinates (u, v) are positive. The transition $w = 1/z$ has positive real Jacobian away from zero. This orientation agrees with the outward Bloch sphere of Lecture 7: $z = \tan(\theta/2)e^{\mathrm{i}\phi}$ gives its ordinary angular parametrization. Equivalently, the Bloch map pulls the outward area form back to $4\mathrm{d}u \wedge \mathrm{d}v/(1 + u^2 + v^2)^2$, which is positive. For $z \mapsto 1/z$, the Bloch coordinates change by $(X, Y, Z) \mapsto (X, -Y, -Z)$. This is a rotation by π about the X-axis. The figure compares its equatorial action with its sphere action.



$\phi \mapsto -\phi$ on the equator
degree -1

$(\theta, \phi) \mapsto (\pi - \theta, -\phi)$
degree $+1$

Figure 14.9: The reciprocal map reverses the equatorial direction and interchanges the poles. The two reversals preserve the sphere orientation.

Example 14.13 (Powers on the complex projective line)**.** For $n \in \mathbb{Z} \setminus \{0\}$ consider the map $f_n(z) = z^n$ on $\mathbb{C}\mathrm{P}^1$. For $n > 0$ it fixes zero and infinity. For $n < 0$ it exchanges them. Using $w = 1/z$ near infinity, in the source or target as appropriate, the expression near zero or near infinity is a positive power $w^{|n|}$ or $z^{|n|}$. Thus the extended map is smooth at zero and at infinity.

Fix a finite nonzero target value. Its inverse image has $|n|$ distinct points, by the polar formula for complex roots. At each such point the complex derivative nz^{n-1} is nonzero. Multiplication by a complex number $a + \mathrm{i}b$ has real derivative matrix

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \qquad \det = a^2 + b^2 > 0.$$

Each inverse-image point therefore has local degree +1. The signed-count formula gives $\deg f_n = |n|$. In particular, $z \mapsto 1/z$ has degree +1 on $\mathbb{C}\mathrm{P}^1$. On the equatorial circle the same formula has degree n instead: a negative exponent reverses its one-dimensional orientation. Conjugation $z \mapsto \bar{z}$ has real matrix $\mathrm{diag}(1, -1)$, so it reverses the complex orientation and has sphere degree -1.

Self-check. Identify the degree of a smooth map $f\colon S^1 \to S^1$ with Lecture 3's winding number. In particular, compute it for $f(e^{\mathrm{i}\phi}) = e^{\mathrm{i}n\phi}$.

Answer. Use the normalized angular form $\alpha_{\mathrm{ang}}/(2\pi)$ on the target. A smooth angular lift ψ of the loop $\phi \mapsto f(e^{\mathrm{i}\phi})$ has $\psi(2\pi) - \psi(0) = 2\pi\,\mathrm{wind}(f)$ by Lecture 3. Its pullback is $\psi'(\phi)\mathrm{d}\phi/(2\pi)$, so integration gives $\deg f = \mathrm{wind}(f)$. For the power map the lift is $\psi(\phi) = n\phi$, and the degree is n for every $n \in \mathbb{Z}$.

Self-check. State in one sentence what degree measures and why the integral-one top form was introduced. Include the hypotheses on the source and target.

Answer. For a smooth map between nonempty closed connected oriented manifolds of the same positive dimension, degree is the homotopy-invariant signed count of inverse images at a regular value expressed by integrating the pullback of a top form normalized to integral one. The decisive independence step is that two such forms differ by an exact form, whose pullback integrates to zero by Stokes.

Unit 14B De Rham cohomology and Part I synthesis

14.7 Periods and singular homology

Recall from Lecture 6 that a singular cycle is a chain with zero boundary, and that $H_k(M;\mathbb{R})$ denotes homology with real coefficients. For a smooth singular simplex $\sigma\colon \Delta^k \to M$, integration means $\int_{\Delta^k} \sigma^*\alpha$, extended linearly to chains. A smooth simplex here extends smoothly to a neighbourhood of its parameter simplex. Arbitrary continuous singular simplices cannot be integrated this way.

Theorem 14.14 (De Rham · *cited*)**.** *For every smooth manifold M and $k \geq 0$, integration induces an isomorphism*

$$H^k(M) \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(H_k(M;\mathbb{Z}),\mathbb{R}), \qquad [\alpha] \longmapsto \left([c] \longmapsto \int_c \alpha\right).$$

Here $\mathrm{Hom}_{\mathbb{Z}}$ *means homomorphisms of abelian groups. This includes replacing continuous*

cycles by smooth representatives. The cited construction uses Stokes on simplices, including their corners, and shows independence of all representatives. See Lee, Theorem 18.14, p. 484, with the comparison construction in Chapter 18.

A closed form with zero periods over all cycles is exact. Conversely, every homomorphism from integral homology to $\mathbb{R}$ is the period functional of a closed form. Thus the theorem gives both uniqueness of a class from its periods and existence for every such assignment.

Example 14.15 (Periods on the torus)**.** Write $T^2 = \mathbb{R}^2/(2\pi\mathbb{Z})^2$, oriented by the ordered angular coordinates (θ^1, θ^2). The forms $\mathrm{d}\theta^1$ and $\mathrm{d}\theta^2$ descend from $\mathbb{R}^2$ because translations preserve them. Let a and b be the positively oriented coordinate circles. For $\eta_1 = \mathrm{d}\theta^1/(2\pi)$ and $\eta_2 = \mathrm{d}\theta^2/(2\pi)$, the period matrix is

$$\begin{pmatrix} \int_a \eta_1 & \int_b \eta_1 \\ \int_a \eta_2 & \int_b \eta_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

For example, on a the angle θ^1 increases from 0 to 2π and θ^2 is constant. This gives the first column. Lecture 6 gives $H_1(T^2;\mathbb{Z}) \cong \mathbb{Z}^2$, generated by a and b. De Rham's theorem therefore makes $[\eta_1]$ and $[\eta_2]$ a basis of $H^1(T^2)$. The prescribed periods $7/3$ and $-5/2$ are realized by $(7/3)\eta_1 - (5/2)\eta_2$. Neither angle is a global real-valued function on the torus. A nonzero period proves that neither angle form is exact.

The wedge $\eta_1 \wedge \eta_2$ has integral one on the oriented torus, since integration over the angular square gives $(2\pi)^2/(2\pi)^2 = 1$. Together with $H_2(T^2;\mathbb{Z}) \cong \mathbb{Z}$, this gives $H^2(T^2) \cong \mathbb{R}$.

Definition 14.16 (Singular cochains)**.** A real singular k-cochain is a real-linear functional $\varphi\colon C_k(M;\mathbb{R}) \to \mathbb{R}$. Its *coboundary* is the $(k+1)$-cochain given by

$$\varphi \circ \partial_{k+1}\colon C_{k+1}(M;\mathbb{R}) \longrightarrow \mathbb{R}, \qquad c \longmapsto \varphi(\partial_{k+1}c).$$

A cochain is *closed* when its coboundary is zero. A k-cochain is a *coboundary* when it has the form $\psi \circ \partial_k$ for a $(k-1)$-cochain ψ. In degree zero the only coboundary is zero.

The identity $\partial_k\partial_{k+1} = 0$ makes every coboundary closed. For a smooth singular $(k+1)$-chain c and a k-form α, Stokes gives

$$\int_c \mathrm{d}\alpha = \int_{\partial c} \alpha.$$

Thus integration respects the passage from forms to cochains in the cited comparison theorem.

We identify the cohomology of real cochains with the real dual of homology. Write $Z_k = \ker \partial_k$ for cycles and $B_k = \operatorname{im} \partial_{k+1}$ for boundaries. A closed cochain vanishes on B_k, so restriction to Z_k gives a well-defined functional on $Z_k/B_k = H_k(M;\mathbb{R})$. Conversely, composing a functional on this quotient with the quotient map gives a functional on Z_k that vanishes on B_k. Extending a basis of Z_k to a basis of $C_k(M;\mathbb{R})$ extends this functional to a closed cochain. This proves surjectivity of restriction.

For its kernel, let a closed cochain φ vanish on all cycles. For $k \geq 1$, define a functional first on $\operatorname{im} \partial_k$ by

$$\psi(\partial_k c) = \varphi(c), \qquad c \in C_k(M;\mathbb{R}).$$

Two chains with the same boundary differ by a cycle, so this is well defined. Extend ψ to $C_{k-1}(M;\mathbb{R})$ by a basis extension. The defining equality gives $\varphi = \psi \circ \partial_k$, a coboundary. Conversely, a coboundary vanishes on cycles because their boundary is zero. For $k = 0$, every

chain is a cycle because $\partial_0 = 0$, so the kernel consists only of the zero cochain. Restriction therefore identifies closed cochains modulo coboundaries with the real homology dual.

For integer chains, real-valued cochains are additive homomorphisms $C_k(M;\mathbb{Z}) \to \mathbb{R}$. Each extends uniquely to a real-linear functional on $C_k(M;\mathbb{R})$ by its values on singular simplices. The coboundary operations agree. The same restriction and factorization argument for integer chains uses a cited extension property in place of basis extension: every homomorphism from a subgroup of an abelian group to $\mathbb{R}$ extends to the whole group. This property is called injectivity of $\mathbb{R}$ and is stated in Lee, p. 473. It supplies both extensions in the argument above. Thus both homology duals compute the cohomology of the same real-valued cochain complex. This proves their identification given that cited property.

Definition 14.17 (Betti numbers and Euler characteristic)**.** For a closed manifold, its *Betti numbers* and *Euler characteristic* are

$$b_k(M) = \dim_{\mathbb{R}} H_k(M;\mathbb{R}), \qquad \chi(M) = \sum_{k=0}^{\dim M} (-1)^k b_k(M).$$

These dimensions are finite by the cited finite-dimensionality of de Rham groups (Lee, Problem 18-8, p. 489, an exercise with hints) and the comparison just proved. For a finite Δ-complex model, Theorems 6.13 and 6.14 identify this with the count of Lecture 6. In finite dimension the comparison gives $\dim H^k(M) = b_k(M)$. The real coefficients matter: an integer homology class annihilated by $m \geq 1$ has period P with $mP = 0$, hence $P = 0$. Differential forms therefore do not recover torsion classes. For example, Lecture 6's integral homology calculation gives $H_1(\mathbb{R}\mathrm{P}^2;\mathbb{Z}) \cong \mathbb{Z}_2$. Every homomorphism from this group to $\mathbb{R}$ is zero, since its value P on the generator satisfies $2P = 0$. De Rham's theorem gives $H^1(\mathbb{R}\mathrm{P}^2) = 0$, although its fundamental group is nontrivial.

14.8 Loops, homology and periods in Part I

The fundamental group records based loops and their composition. Passing to first homology makes that composition abelian, by the cited comparison in Lecture 6. Higher homology uses cycles of higher dimension. De Rham cohomology describes their real-valued period functionals. The diagram separates the loop comparison from the period comparison. In its last line, M is closed and X is smooth with isolated zeros.

$$\pi_1(M) \xrightarrow{\text{abelianization}} H_1(M;\mathbb{Z})$$

$$H^k(M) \xrightarrow{\text{integration: isomorphism}} \mathrm{Hom}_{\mathbb{Z}}(H_k(M;\mathbb{Z}),\mathbb{R})$$

$$\chi(M) = \textstyle\sum_k (-1)^k b_k(M) = \sum_{p:X_p=0} \mathrm{ind}_p X$$

Figure 14.10: For a path-connected space, first integral homology is the abelianization of the fundamental group. For a smooth manifold, de Rham classes correspond to real period functionals in every degree.

The sphere and torus calculations distinguish several roles of these invariants. The groups in each triple below are in degrees 0, 1 and 2; cohomology is de Rham cohomology.

Space	π_1	(H_0, H_1, H_2) over $\mathbb{Z}$	(H^0, H^1, H^2)
S^1	$\mathbb{Z}$	$(\mathbb{Z}, \mathbb{Z}, 0)$	$(\mathbb{R}, \mathbb{R}, 0)$
S^2	$\{1\}$	$(\mathbb{Z}, 0, \mathbb{Z})$	$(\mathbb{R}, 0, \mathbb{R})$
T^2	$\mathbb{Z}^2$	$(\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z})$	$(\mathbb{R}, \mathbb{R}^2, \mathbb{R})$
$\mathbb{R}\mathrm{P}^2$	$\mathbb{Z}_2$	$(\mathbb{Z}, \mathbb{Z}_2, 0)$	$(\mathbb{R}, 0, 0)$
$\mathbb{R}^2 \setminus \{0\}$	$\mathbb{Z}$	$(\mathbb{Z}, \mathbb{Z}, 0)$	$(\mathbb{R}, \mathbb{R}, 0)$

On the circle, a normalized angular period is both winding and degree. On the sphere, the normalized area period computes degree, while the sum of vector-field indices is the Euler characteristic, equal to two. On the torus, two independent angular periods give two classes in degree one. Its Euler characteristic is zero, consistent with a nonzero constant field descending from $\mathbb{R}^2$. The projective plane retains an order-two loop in integral homology, although every closed real 1-form has zero period around it.

Charts and tangent spaces made calculus possible on these spaces. Exterior differentiation and Stokes connected that calculus to cycles. The index of a zero and the degree of a map are integers obtained from this connection. Lecture 15 introduces a metric to measure lengths and define the Hodge star. These measurements add geometric information to the topological invariants above.

Self-check. Consider the real chain complex with $C_2 = \mathbb{R}$, $C_1 = \mathbb{R}^3$ and $C_0 = \mathbb{R}$, whose boundary matrices are

$$\partial_2 = \begin{pmatrix} 1 \\ 5 \\ 1 \end{pmatrix}, \qquad \partial_1 = \begin{pmatrix} 2 & -1 & 3 \end{pmatrix}.$$

For the 1-cochains $\varphi = (7, 1, -12)$ and $\eta = (-8, 4, -12)$, decide which are closed and which are coboundaries. Use the cycle $z = (1, 2, 0)^{\mathsf{T}}$ to distinguish their homology functionals.

Answer. The product $\partial_1 \partial_2 = 2 - 5 + 3 = 0$ verifies the chain condition. Both cochains are closed because $\varphi \partial_2 = 7 + 5 - 12 = 0$ and $\eta \partial_2 = -8 + 20 - 12 = 0$. The vector z is a cycle since $\partial_1 z = 2 - 2 = 0$. The value $\varphi(z) = 7 + 2 = 9$ prevents φ from being a coboundary: every cochain of the form $\psi \circ \partial_1$ vanishes on z. The equality $\eta = -4\partial_1$ exhibits η as a coboundary. Its homology functional is zero.

Summary. On a nonempty closed connected oriented manifold, top cohomology is measured by integration. Between such manifolds of equal positive dimension, degree is a homotopy invariant integer, computed by signed inverse images at a regular value. It gives the even-sphere hairy-ball theorem for smooth fields and the smooth Brouwer theorem. Poincaré–Hopf relates zero indices to the Euler characteristic. De Rham's theorem identifies cohomology with real period functionals on homology. Part II adds a metric, connections and physical fields to these topological constructions.

Reading. Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 17.30–17.31, pp. 454–455; Theorem 17.35, pp. 457–458; Theorems 17.38–17.39, pp. 459–460; Theorem 18.14, p. 484. Milnor, *Topology from the Differentiable Viewpoint*, §6, pp. 32–35 (online scan).

Part II

Geometry and Topology in Physics

Part II adds a metric (Lecture 15) and a way to compare vectors at different points (Lecture 16). Comparison around a loop gives curvature (Lectures 17 and 18), and Einstein's equation relates curvature to matter (Lecture 19). Bundles and connections (Lectures 20 and 21) carry the same construction over to phases, and integrals of their curvature give integers (Lectures 22 and 23). These integers then govern Berry's phase, band insulators, defects, solitons and zero modes (Lectures 24 to 28).

Part II assumes Part I, special relativity at the level of Lorentz transformations, the Euler–Lagrange equations and their field version, and the quantum mechanics of finite systems: states, Hermitian Hamiltonians and spin one-half. Lecture 28 also assumes the Dirac equation. Each of these is recalled where it is used. From Part I it uses chiefly forms, Stokes' theorem, cohomology and the degree.

Lecture 15

Metrics, the Hodge star and Maxwell's equations

Objectives. *A metric measures tangent vectors and selects a volume form once an orientation is chosen. The Hodge star then turns a form into a form of complementary degree. On spacetime this operation packages Maxwell's four equations into two equations for forms. The monopole flux shows why closed field forms need not have global potentials. On a closed oriented Riemannian manifold, the metric selects a harmonic representative of each de Rham class.*

By the end of this lecture you will be able to:

- Use a metric to lower indices and compute the Hodge star with orientation and signature specified.
- Derive all four Maxwell equations and charge conservation from the form equations.
- Use a nonzero sphere period to rule out a global monopole potential.
- Compute the codifferential and the harmonic representative of a form on the circle.

Unit 15A Metrics and the Hodge star

15.1 Metrics and the identification of vectors with covectors

Let M be a smooth n-manifold.

Definition 15.1. A *Riemannian metric* g is a smoothly varying positive-definite symmetric bilinear form g_p on each tangent space T_pM.

Definition 15.2. A *pseudo-Riemannian metric* requires nondegeneracy instead of positive definiteness. Nondegeneracy means that $g_p(v, w) = 0$ for all $w \in T_pM$ implies $v = 0$.

We use g for the metric throughout Part II.

By Sylvester's law of inertia, a nondegenerate symmetric form admits a basis with diagonal entries $+1$ and -1.

Definition 15.3. Such a basis is called *pseudo-orthonormal*. The *signature* is $(n - s, s)$, where s counts the negative entries. A *Lorentzian metric* in dimension $n \geq 2$ has signature $(1, n - 1)$.

The two counts are independent of the basis. This is the signature result imported from *Linear Algebra for Physics*, Lecture 11, the example *Signature of a form*. For a smooth nondegenerate metric, the signature is constant on each connected component: eigenvalues vary continuously and cannot cross zero.

In coordinates, with $i, j \in \{1, \dots, n\}$, $g = g_{ij}\mathrm{d}x^i \otimes \mathrm{d}x^j$. Its inverse matrix is written (g^{ij}).

Definition 15.4. For $X \in T_pM$ and $\alpha \in T_p^*M$, the mutually inverse *musical maps* are the maps

$$\flat \colon T_pM \to T_p^*M, \qquad \sharp \colon T_p^*M \to T_pM,$$

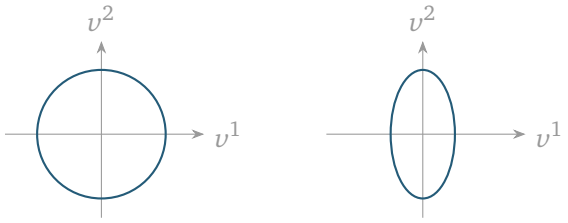
given by

$$X^\flat = g(X, \cdot), \qquad (X^\flat)_i = g_{ij}X^j, \qquad (\alpha^\sharp)^i = g^{ij}\alpha_j.$$

These maps lower and raise an index using a metric. The coordinate relabelling $x_i \mapsto x^i$ in Lecture 7 did not apply a metric. For example, in one tangent space take the following metric matrix and vector components. The resulting lower components are

$$(g_{ij}) = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \qquad (X^i) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad ((X^\flat)_i) = \begin{pmatrix} 4 \\ 7 \end{pmatrix}.$$

The vector has squared length $4+14 = 18$. The figure shows how a positive metric determines its unit vectors.



$g = \mathrm{diag}(1, 1)$ $\qquad$ $g = \mathrm{diag}(4, 1)$

Figure 15.1: At a point, Euclidean unit vectors form a circle. For $g = 4\mathrm{d}x \otimes \mathrm{d}x + \mathrm{d}y \otimes \mathrm{d}y$ they form an ellipse with horizontal radius $1/2$.

Proposition 15.5 (Existence and induced metrics · *proved given partitions of unity*)**.** *Every smooth manifold admits a Riemannian metric. An immersion into a Riemannian manifold induces a Riemannian metric by pullback.*

Proof. On each coordinate domain use $g_a = \sum_i \mathrm{d}x_a^i \otimes \mathrm{d}x_a^i$. Choose a subordinate partition ρ_a from Theorem 12.8 and put $g = \sum_a \rho_a g_a$, extending each summand by zero. Supports make the extensions smooth. Local finiteness makes the sum smooth. For nonzero v, every summand $\rho_a g_a(v, v)$ is nonnegative, and at least one is positive because $\sum_a \rho_a = 1$. Thus the metric is positive definite. For an immersion f, the differential is injective, so $(f^*g)(v, v) = g(\mathrm{d}f(v), \mathrm{d}f(v)) > 0$ for $v \neq 0$. Smoothness and symmetry are preserved by pullback. ∎

Definition 15.6. An *isometry* is a diffeomorphism $f \colon (M, g_M) \to (N, g_N)$ with $f^*g_N = g_M$.

A pseudo-Riemannian metric can restrict degenerately: for the matrix $\mathrm{diag}(1, -1, -1, -1)$, the vector $(1, 1, 0, 0)$ spans a line on which the restricted form is zero. So the induced-metric assertion used positivity.

We compute the induced metric on the unit sphere using the parametrization

$$p(\theta, \phi) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta), \qquad 0 < \theta < \pi,$$

where ϕ lies in an angular chart. The coordinate tangents have dot products

$$\partial_\theta p \cdot \partial_\theta p = 1, \qquad \partial_\theta p \cdot \partial_\phi p = 0, \qquad \partial_\phi p \cdot \partial_\phi p = \sin^2\theta.$$

Thus the induced metric is

$$g = d\theta \otimes d\theta + \sin^2\theta \, d\phi \otimes d\phi.$$

The figure compares the lengths of the coordinate tangents at two points.

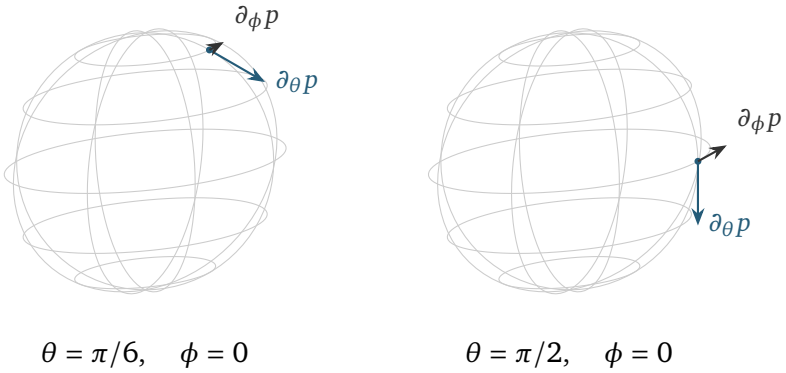


Figure 15.2: On the unit sphere at $\phi = 0$, the coordinate tangents are $\partial_\theta p = (\cos\theta, 0, -\sin\theta)$ and $\partial_\phi p = (0, \sin\theta, 0)$. Foreshortening in the receding y-direction preserves the ratio between the two $\partial_\phi p$ arrows, whose lengths on the sphere are $1/2$ at $\theta = \pi/6$ and 1 at $\theta = \pi/2$.

Definition 15.7. The *length* of a piecewise smooth curve $\gamma : [a, b] \to M$ in a Riemannian manifold is

$$\ell(\gamma) = \int_a^b \sqrt{g(\dot\gamma, \dot\gamma)}\, dt.$$

A meridian from the north pole to the south pole has length $\int_0^\pi 1 \, d\theta = \pi$. A latitude at $\theta = \theta_0 \in (0, \pi)$ has length $\int_0^{2\pi} \sin\theta_0 \, d\phi = 2\pi \sin\theta_0$. The meridian length includes the endpoints by continuity of the induced metric.

15.2 Volume forms and the Hodge star

Definition 15.8. On an oriented pseudo-Riemannian n-manifold, the *metric volume form* is, in positive coordinates,

$$\mu = \sqrt{|\det(g_{ij})|}\, dx^1 \wedge \cdots \wedge dx^n.$$

The defining normalization is $\mu = \theta^1 \wedge \cdots \wedge \theta^n$ when $\theta^1, \dots, \theta^n$ is dual to a positive pseudo-orthonormal basis.

To justify the coefficient, choose such a basis with metric matrix $S = \operatorname{diag}(\varepsilon_1, \dots, \varepsilon_n)$, where $\varepsilon_i = \pm 1$. If the coordinate basis has columns A in that basis, its metric matrix is $A^T S A$. Hence $|\det g| = (\det A)^2$. The dual bases satisfy $dx^1 \wedge \cdots \wedge dx^n = (\det A)^{-1} \theta^1 \wedge \cdots \wedge \theta^n$. Since $\det A > 0$, the coefficient gives the required normalization. Under a positive coordinate change the square root of the absolute metric determinant contributes the absolute Jacobian and the top wedge contributes its inverse. The expressions therefore agree. They are smooth because the determinant never vanishes. This is the chosen volume form μ of Lecture 11, now fixed by metric and orientation. Without orientation the same coefficient defines a density.

For the sphere metric just computed, $\sqrt{\det(g_{ij})} = \sin\theta$, so $\mu = \sin\theta\, d\theta \wedge d\phi$. This is the outward area form from Lecture 10.

In Lecture 10, a spatial vector field gave both a work 1-form and a flux 2-form. The metric identifies the vector with the work form. The metric and orientation then determine the flux form. The Hodge star will give this second passage in every dimension and degree.

For $0 \le k \le n$, the inverse metric gives a bilinear pairing of covectors. Its induced pairing on decomposable k-forms is

$$\langle \alpha_1 \wedge \cdots \wedge \alpha_k, \beta_1 \wedge \cdots \wedge \beta_k \rangle_g = \det\left[g^{-1}(\alpha_i, \beta_j)\right]_{i,j=1}^{k},$$

extended bilinearly. The determinant is multilinear and alternating in each list of covectors, so it depends only on the two wedges. On functions it is ordinary multiplication. For example, in Euclidean three-space,

$$\langle dx \wedge dy, dx \wedge dy \rangle_g = \det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1.$$

This uses Lecture 10's determinant wedge, with no additional normalization from the fermionic wedge of *Quantum Mechanics I*. The pairing is nondegenerate: in a pseudo-orthonormal dual basis the basis wedges are orthogonal and each has norm squared +1 or −1.

Definition 15.9. On an oriented pseudo-Riemannian n-manifold, the *Hodge star* is the pointwise real-linear map

$$\star \colon \Omega^k(M) \longrightarrow \Omega^{n-k}(M)$$

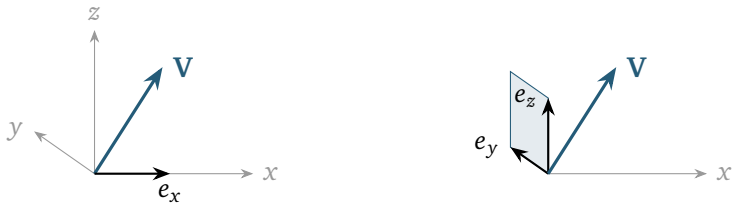
characterized by

$$\alpha \wedge \star\beta = \langle \alpha, \beta \rangle_g \mu \qquad (\alpha, \beta \in \Omega^k(M)).$$

In oriented Euclidean $\mathbb{R}^2$, the basis covectors are orthonormal. For example, $dy \wedge (-dx) = dx \wedge dy$ fixes the sign of $\star dy$. The defining equation gives

$$\star 1 = dx \wedge dy, \qquad \star dx = dy, \qquad \star dy = -dx, \qquad \star(dx \wedge dy) = 1.$$

In Euclidean three-space, the star sends the work 1-form to the flux 2-form. The figure compares their evaluations for one vector.



$\mathbf{V}^\flat(e_x) = 2$ $\qquad$ $(\star\mathbf{V}^\flat)(e_y, e_z) = 2$

Figure 15.3: For $\mathbf{V} = (2, 1, 2)$, evaluation of $\mathbf{V}^\flat$ on e_x and evaluation of $\star\mathbf{V}^\flat$ on (e_y, e_z) both give 2. The segment measures a component along a path; the oriented square measures flux.

Theorem 15.10 (Hodge star). *Let (M, g) be an oriented pseudo-Riemannian n-manifold and $0 \le k \le n$. There is a unique pointwise real-linear operator, smooth in position, $\star\colon \Omega^k(M) \to \Omega^{n-k}(M)$ such that*

$$\alpha \wedge \star\beta = \langle \alpha, \beta \rangle_g \mu \qquad (\alpha, \beta \in \Omega^k(M)).$$

If the metric has s negative directions, then on k-forms

$$\star\star\alpha = (-1)^{k(n-k)+s}\alpha.$$

Proof. We prove existence and uniqueness by the wedge pairing at a point. We then compute the square on basis wedges. The wedge pairing between degrees k and $n-k$ is nondegenerate: pairing with a basis wedge selects the coefficient of its complementary wedge. It therefore represents the functional $\alpha \mapsto \langle \alpha, \beta \rangle_g \mu$ uniquely. The coordinate linear system uses smooth inverse-metric coefficients and nonzero volume coefficient, so its solution varies smoothly.

For the sign, use a positively oriented pseudo-orthonormal dual basis $\theta^1, \dots, \theta^n$ at one point. For an increasing index set I, write θ^I for its wedge, I^c for its ordered complement, $\varepsilon_I = \prod_{i \in I} \varepsilon_i$, and σ_I for the sign of the permutation (I, I^c). The defining equation gives

$$\star\theta^I = \varepsilon_I \sigma_I \theta^{I^c}.$$

The two metric signs multiply to $\prod_i \varepsilon_i = (-1)^s$. Interchanging the blocks I and I^c uses $k(n-k)$ transpositions, so $\sigma_I \sigma_{I^c} = (-1)^{k(n-k)}$. Applying the formula twice proves the square identity on every basis wedge, hence on every form. ∎

For example, in oriented Euclidean $\mathbb{R}^3$, $\star dx = dy \wedge dz$ and $\star(dy \wedge dz) = dx$. The complementary plane is shown in the figure.

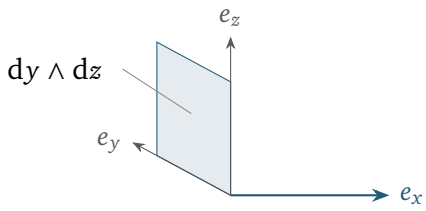


Figure 15.4: In Euclidean three-space, the metric identifies the covector dx with e_x. Its star is the oriented area form $dy \wedge dz$ on the complementary plane.

Reversing orientation negates μ and therefore negates the star. For a compactly supported real form α, the integral $\int_M \alpha \wedge \star\alpha$ is unchanged: the integrand and the integration orientation both change sign. The pairing need not be positive in indefinite signature. For complex forms extend the star complex-linearly and use $\bar{\alpha} \wedge \star\beta$ for the Hermitian pairing, antilinear in its first argument.

15.2.1 The spacetime star

Use coordinates $(x^0, x^1, x^2, x^3) = (t, x, y, z)$ on an open subset of spacetime $\mathbb{R}^4$. Greek indices run from 0 to 3 and spatial indices from 1 to 3. Take the Minkowski metric and orientation

$$\eta = \operatorname{diag}(1, -1, -1, -1), \qquad \mu = dt \wedge dx \wedge dy \wedge dz.$$

This is the metric recalled from *Symmetries in Physics*, Lecture 17, *Minkowski space from physics*. Units have $c = 1$ and rationalized sources. The metric has $s = 3$, so $\star^2 = -1$ on 2-forms. A real 2-form ω satisfying $\star\omega = \omega$ or $\star\omega = -\omega$ must therefore be zero. This distinguishes the Lorentzian fields here from the Euclidean self-dual fields of Lecture 23. Euclidean four-space instead has $\star^2 = +1$ on 2-forms.

The six basis 2-forms give a direct way to use the signature convention. For example, $dt \wedge dx$ has norm squared -1 and $(dt \wedge dx) \wedge (dy \wedge dz) = \mu$. Its star is therefore $-dy \wedge dz$. The pairings used here are

$$\langle dt \wedge dx, dt \wedge dx \rangle_\eta = \det \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = -1, \qquad \langle dy \wedge dz, dy \wedge dz \rangle_\eta = 1.$$

The same rule gives the table:

α	$\star\alpha$	α	$\star\alpha$
$dt \wedge dx$	$-dy \wedge dz$	$dy \wedge dz$	$dt \wedge dx$
$dt \wedge dy$	$-dz \wedge dx$	$dz \wedge dx$	$dt \wedge dy$
$dt \wedge dz$	$-dx \wedge dy$	$dx \wedge dy$	$dt \wedge dz$

The degree-one stars are:

$$\star dt = dx \wedge dy \wedge dz, \quad \star dx = dt \wedge dy \wedge dz,$$
$$\star dy = dt \wedge dz \wedge dx, \quad \star dz = dt \wedge dx \wedge dy.$$

Self-check. In oriented Euclidean $\mathbb{R}^3 \setminus \{0\}$, put $r = \sqrt{x^2 + y^2 + z^2}$ and take $\mathbf{E}^\flat = (x\,dx + y\,dy + z\,dz)/r^3$. Compute $\star\mathbf{E}^\flat$. Use its outward sphere period to decide whether it has a global 1-form primitive.

Answer. The Euclidean rules give

$$\star\mathbf{E}^\flat = \frac{x\,dy \wedge dz + y\,dz \wedge dx + z\,dx \wedge dy}{r^3}.$$

This is the Coulomb flux form of Lecture 12, written here as β_C. Its outward unit-sphere period is 4π, so Stokes rules out a global primitive. Here the spatial Euclidean star is used, not the four-dimensional spacetime star.

Unit 15B Electromagnetic fields and global potentials

15.3 Maxwell's equations as equations for forms

Physical input. We use classical electromagnetism in vacuum, with smooth electric field $\mathbf{E}$, magnetic field $\mathbf{B}$, charge density ρ and spatial current $\mathbf{J}$. Electric work per unit charge along a spatial path is $\int E_i dx^i$. Magnetic flux through an oriented spatial surface is $\int \mathbf{B} \cdot \mathbf{n}\, dS$. Charge in a spatial region is the integral of ρ over its volume. The outward current flux measures charge leaving per unit time. In the units specified above, we take the vacuum Maxwell laws with sources as physical input:

$$\operatorname{div}\mathbf{B} = 0, \quad \partial_t\mathbf{B} + \operatorname{curl}\mathbf{E} = 0, \qquad \operatorname{div}\mathbf{E} = \rho, \quad \operatorname{curl}\mathbf{B} - \partial_t\mathbf{E} = \mathbf{J}.$$

This is the electromagnetic model of Nakahara, §7.9, Example 7.16, expressed with our signature and source convention.

The Cartesian spatial components use the Euclidean spatial metric. They are not obtained by lowering spacetime indices with η. For a spatial vector V, we write its flux 2-form as

$$\mathcal{B}(V) = V_x \mathrm{d}y \wedge \mathrm{d}z + V_y \mathrm{d}z \wedge \mathrm{d}x + V_z \mathrm{d}x \wedge \mathrm{d}y.$$

Magnetic flux integrates a 2-form over a spatial surface. Electric work integrates a 1-form over a spatial path. Faraday's law relates the change of magnetic flux to electric circulation. On a spacetime wall swept out by a path, the electric term therefore has both a spatial differential and a time differential.

Definition 15.11. The real *field-strength 2-form* is

$$F_{\mathrm{em}} = E_i \mathrm{d}x^i \wedge \mathrm{d}t + B_x \mathrm{d}y \wedge \mathrm{d}z + B_y \mathrm{d}z \wedge \mathrm{d}x + B_z \mathrm{d}x \wedge \mathrm{d}y.$$

Definition 15.12. The *current 1-form* and *current 3-form* are

$$j = \rho \mathrm{d}t - J_i \mathrm{d}x^i, \qquad \mathcal{J} = \star j.$$

The subscripts em distinguish these real electromagnetic forms from later connection forms. The current 1-form is the metric dual of the charge-current vector:

$$j = (\rho \partial_t + J_i \partial_i)^\flat.$$

On a fixed-time surface, F_{em} restricts to $\mathcal{B}(\mathbf{B})$. It also satisfies $F_{\mathrm{em}}(\partial_i, \partial_t) = E_i$. For a compact fixed oriented surface Σ, put $\Phi(t) = \int_\Sigma \mathcal{B}(\mathbf{B}(t))$. For $t_1 < t_2$, orient $[t_1, t_2] \times \Sigma$ by $\mathrm{d}t$ followed by the orientation of Σ. We assume that this cylinder lies in the field domain. Let s be positive arclength along $\partial\Sigma$. The outward-first boundary rule gives

$$\partial([t_1, t_2] \times \Sigma) = \Sigma_{t_2} - \Sigma_{t_1} - [t_1, t_2] \times \partial\Sigma.$$

On the wall with product orientation, the electric term is $-\mathrm{d}t \wedge E_i \mathrm{d}x^i$. The induced wall orientation contributes a second minus sign. The figure records the resulting signed integrals.

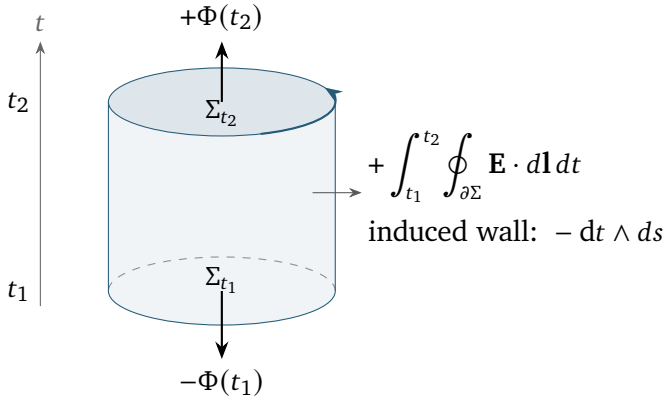


Figure 15.5: The spacetime cylinder over a positively oriented disc has top flux $+\Phi(t_2)$ and bottom flux $-\Phi(t_1)$. The induced wall contribution is $+\int_{t_1}^{t_2} \oint_{\partial\Sigma} \mathbf{E} \cdot d\mathbf{l}\, dt$, with $\partial\Sigma$ counter-clockwise viewed from above.

Stokes therefore turns the homogeneous form equation into integrated Faraday's law:

$$0 = \int_{\partial([t_1,t_2]\times\Sigma)} F_{\mathrm{em}} = \Phi(t_2) - \Phi(t_1) + \int_{t_1}^{t_2} \oint_{\partial\Sigma} \mathbf{E}\cdot d\mathbf{l}\, dt.$$

Theorem 15.13 (Maxwell's equations in forms)**.** *On the specified spacetime, the Maxwell laws stated as physical input are equivalent to*

$$\mathrm{d}F_{\mathrm{em}} = 0, \qquad \mathrm{d}{\star}F_{\mathrm{em}} = \mathcal{J}.$$

These equations imply charge conservation, $\partial_t\rho + \operatorname{div}\mathbf{J} = 0$. When these equations hold, every point has a neighbourhood with $F_{\mathrm{em}} = \mathrm{d}A_{\mathrm{em}}$ for a real 1-form A_{em}.

Proof. The spacetime star table gives

$$\begin{aligned}
{\star}F_{\mathrm{em}} &= B_i\mathrm{d}t\wedge\mathrm{d}x^i + E_x\mathrm{d}y\wedge\mathrm{d}z + E_y\mathrm{d}z\wedge\mathrm{d}x + E_z\mathrm{d}x\wedge\mathrm{d}y,\\
\mathcal{J} &= \rho\mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z - J_x\mathrm{d}t\wedge\mathrm{d}y\wedge\mathrm{d}z + J_y\mathrm{d}t\wedge\mathrm{d}x\wedge\mathrm{d}z - J_z\mathrm{d}t\wedge\mathrm{d}x\wedge\mathrm{d}y.
\end{aligned}$$

In the flux notation, the source form is

$$\mathcal{J} = \rho\mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z - \mathrm{d}t\wedge\mathcal{B}(\mathbf{J}).$$

Differentiation gives

$$\begin{aligned}
\mathrm{d}F_{\mathrm{em}} &= (\operatorname{div}\mathbf{B})\mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z + \mathrm{d}t\wedge\mathcal{B}(\partial_t\mathbf{B} + \operatorname{curl}\mathbf{E}),\\
\mathrm{d}{\star}F_{\mathrm{em}} &= (\operatorname{div}\mathbf{E})\mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z + \mathrm{d}t\wedge\mathcal{B}(\partial_t\mathbf{E} - \operatorname{curl}\mathbf{B}).
\end{aligned}$$

For instance, the coefficient of $\mathrm{d}t\wedge\mathrm{d}y\wedge\mathrm{d}z$ in the first equation is $\partial_t B_x + \partial_y E_z - \partial_z E_y$. The other cyclic coefficients give the other components of the curl. The magnetic term in the star contributes

$$\mathrm{d}(B_i\mathrm{d}t\wedge\mathrm{d}x^i) = \partial_j B_i\mathrm{d}x^j\wedge\mathrm{d}t\wedge\mathrm{d}x^i = -\mathrm{d}t\wedge\mathcal{B}(\operatorname{curl}\mathbf{B}).$$

The minus sign comes from moving $\mathrm{d}x^j$ past $\mathrm{d}t$. Independence of the four basis 3-forms gives the homogeneous equations. For the source equation the same comparison gives

$$\operatorname{div}\mathbf{E} = \rho, \qquad \partial_t\mathbf{E} - \operatorname{curl}\mathbf{B} = -\mathbf{J}.$$

The second equality is the stated Ampère–Maxwell law. Applying d to the source equation gives $\mathrm{d}\mathcal{J} = 0$. Differentiating the displayed current form gives $\mathrm{d}\mathcal{J} = (\partial_t\rho + \operatorname{div}\mathbf{J})\mu$, which proves conservation. Finally choose a star-shaped coordinate neighbourhood and apply Theorem 13.11 to the closed 2-form. Its primitive is the required local potential. ∎

15.3.1 Gauss's law, charge conservation and covariance

On a fixed-time slice, $\mathrm{d}t$ pulls back to zero. The forms ${\star}F_{\mathrm{em}}$ and $\mathcal{J}$ restrict to $\mathcal{B}(\mathbf{E})$ and $\rho\mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z$. For a compact smooth spatial region V in this slice, Stokes gives Gauss's law:

$$\int_{\partial V} \mathbf{E}\cdot\mathbf{n}\, dS = \int_V \rho\, dx\, dy\, dz.$$

Thus electric flux through the outward boundary equals the enclosed charge. The integral of $\mathcal{J}$ over an oriented spacetime hypersurface measures signed net charge crossing that

hypersurface. Write $Q(t) = \int_V \rho(t)\,dx\,dy\,dz$ for a fixed region V. Assume that $[t_1, t_2] \times V$ lies in the field domain. Applying Stokes to $\mathrm{d}\mathcal{J} = 0$ on this cylinder gives

$$Q(t_2) - Q(t_1) + \int_{t_1}^{t_2} \int_{\partial V} \mathbf{J} \cdot \mathbf{n}\, dS\, dt = 0.$$

Positive outward current therefore decreases the charge in V. The charge cylinder shows the same boundary signs for a current 3-form.

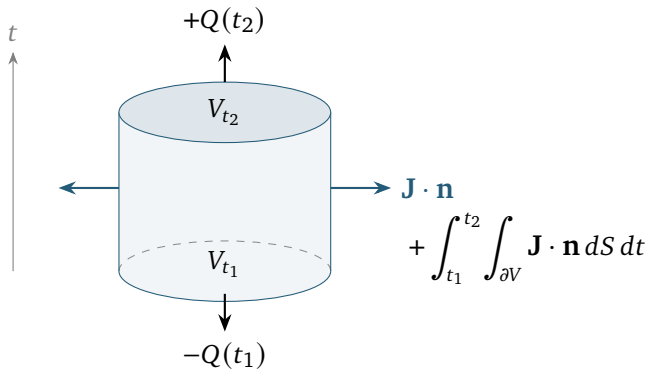


Figure 15.6: The top and bottom of a swept spatial region contribute $+Q(t_2)$ and $-Q(t_1)$, while the wall contributes positive outward current. This is a schematic with the third spatial direction suppressed.

The homogeneous equation $\mathrm{d}F_{\mathrm{em}} = 0$ needs neither a metric nor an orientation. The source equation uses the metric and orientation through the star. An orientation-preserving isometry f preserves the form pairing and volume. Pulling back the defining star identity and using uniqueness gives $f^*(\star\alpha) = \star(f^*\alpha)$. Pullback also commutes with d by Lecture 10. Hence f^*F_{em} and $f^*\mathcal{J}$ satisfy the same form equations. In particular, proper Lorentz transformations preserve Maxwell's equations. Their pullbacks mix the time-space and space-space components of F_{em}, which are its electric and magnetic parts.

15.3.2 Potentials and gauge transformations

Write the local potential in terms of its scalar and spatial parts as

$$A_{\mathrm{em}} = -\phi \mathrm{d}t + A_i \mathrm{d}x^i.$$

Taking its derivative gives $E_i = -\partial_i\phi - \partial_t A_i$ and $\mathbf{B} = \operatorname{curl}\mathbf{A}$, where $\mathbf{A} = (A_x, A_y, A_z)$.

Definition 15.14. For a smooth real function χ, a *gauge transformation* is $A_{\mathrm{em}} \mapsto A_{\mathrm{em}} + \mathrm{d}\chi$.

This transformation leaves F_{em} unchanged because $\mathrm{d}^2\chi = 0$. Its component form is $A_i \mapsto A_i + \partial_i\chi$ and $\phi \mapsto \phi - \partial_t\chi$. This recalls *Theoretical Mechanics*, Lecture 3, *The charged particle*: the same transformation changes the particle Lagrangian by a total derivative.

The Landau–Lifshitz lower potential has the opposite sign at matched units.[1]

[1] Thus $A^{\mathrm{LL}} = -A_{\mathrm{em}}$ and $F^{\mathrm{LL}} = -F_{\mathrm{em}}$. The subsection *Conventions in electromagnetic sources* at the end of §15.4 translates the source equations of Nakahara and Landau–Lifshitz.

15.4 The monopole obstruction

Let $r = \sqrt{x^2 + y^2 + z^2}$ and let $\Phi_m \in \mathbb{R} \setminus \{0\}$ be the magnetic flux. We consider a static *magnetic monopole* field on $\mathbb{R} \times (\mathbb{R}^3 \setminus \{0\})$. Its magnetic field is $\mathbf{B} = \Phi_m(x, y, z)/(4\pi r^3)$. Its electric field is zero. Its field 2-form is

$$F_{em} = \frac{\Phi_m}{4\pi} \frac{x\,dy \wedge dz + y\,dz \wedge dx + z\,dx \wedge dy}{r^3}.$$

Proposition 15.15 (No global monopole potential · *proved given partitions of unity*)**.** *This form is closed and has period Φ_m on the outward unit sphere in each fixed-time slice. It has local potentials but no global 1-form potential on $\mathbb{R} \times (\mathbb{R}^3 \setminus \{0\})$, spacetime with the monopole worldline removed.*

Proof. The form is time-independent, with no time differential. Its spatial part is the Coulomb flux form of §12.7 times $\Phi_m/(4\pi)$. Lecture 12 proved closedness and sphere period 4π for that form, giving the stated period. Local potentials follow from the Poincaré lemma. If $F_{em} = dA_{em}$ globally, restriction to a fixed-time sphere and Stokes would give zero period, contradicting $\Phi_m \neq 0$. ∎

This is a conditional field model; no existence of a physical monopole is assumed. The nonzero sphere period says that $[F_{em}] \neq 0$ in $H^2_{dR}(\mathbb{R} \times (\mathbb{R}^3 \setminus \{0\}))$. It is the Coulomb cohomology class computed in Lecture 13, with the stated scaling. The figure shows the removed worldline and a sphere that detects the class.

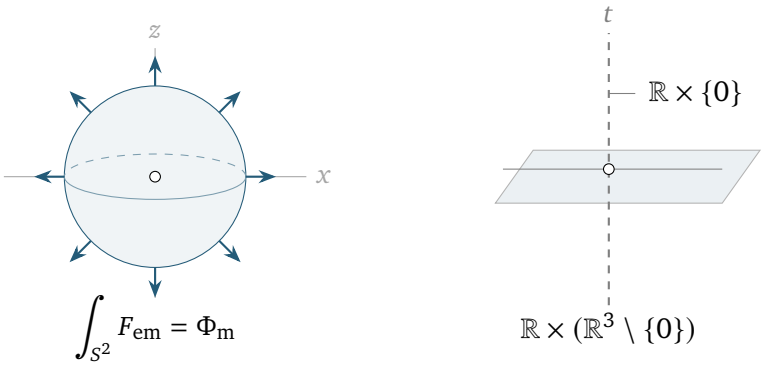


Figure 15.7: The omitted set is the whole time axis through the spatial origin. Each outward fixed-time sphere has flux Φ_m, with the arrows shown for positive flux and reversed when $\Phi_m < 0$.

The obstruction prevents describing the field by one smooth potential on its whole domain. Local potentials must instead be related on their overlaps. No integer condition has been imposed on Φ_m. Lecture 22 studies the additional condition from a bundle and a charged particle.

15.4.1 Vacuum duality

Example 15.16 (An electric field and its magnetic dual)**.** Take a constant electric field $\mathbf{E} = E_0 e_x$, with $E_0 \neq 0$, and zero magnetic field on Minkowski space. Its potential and field are

$$A_{em} = E_0 x\,dt, \qquad F_{em} = E_0 dx \wedge dt.$$

All field derivatives vanish, so Maxwell's equations hold with zero sources. The table gives $\star F_{\mathrm{em}} = E_0 \mathrm{d}y \wedge \mathrm{d}z$. This is the field of a constant magnetic field $\mathbf{B}' = E_0 e_x$ and zero electric field. A potential for it is $E_0 y \mathrm{d}z$. Thus the transformation $F_{\mathrm{em}} \mapsto \star F_{\mathrm{em}}$ changes this electric solution into a magnetic solution.

Definition 15.17. For a source-free field, the transformation $F_{\mathrm{em}} \mapsto \star F_{\mathrm{em}}$ is *vacuum duality*.

Indeed, $\mathrm{d}F_{\mathrm{em}} = 0$ and $\mathrm{d}\star F_{\mathrm{em}} = 0$ imply $\mathrm{d}(\star F_{\mathrm{em}}) = 0$ and $\mathrm{d}\star(\star F_{\mathrm{em}}) = -\mathrm{d}F_{\mathrm{em}} = 0$. The transformed components are $\mathbf{E}' = -\mathbf{B}$ and $\mathbf{B}' = \mathbf{E}$. Applying duality twice gives $-F_{\mathrm{em}}$, rather than F_{em}.

Example 15.18 (The Coulomb field and the monopole). On $\mathbb{R} \times (\mathbb{R}^3 \setminus \{0\})$, let $Q \in \mathbb{R} \setminus \{0\}$ and take

$$\mathbf{E} = \frac{Q(x,y,z)}{4\pi r^3}, \qquad \mathbf{B} = 0, \qquad \phi = \frac{Q}{4\pi r}.$$

This static electric field has global potential $A_{\mathrm{em}} = -\phi \mathrm{d}t$. Indeed, $-\partial_i \phi = Q x^i/(4\pi r^3)$, so $F_{\mathrm{em}} = \mathrm{d}A_{\mathrm{em}} = E_i \mathrm{d}x^i \wedge \mathrm{d}t$. The field is source-free on this domain, since the spatial divergence and curl of $\mathbf{E}$ vanish away from the origin. Its spacetime star is

$$\star F_{\mathrm{em}} = \frac{Q}{4\pi} \frac{x\mathrm{d}y \wedge \mathrm{d}z + y\mathrm{d}z \wedge \mathrm{d}x + z\mathrm{d}x \wedge \mathrm{d}y}{r^3}.$$

This is the monopole form with $\Phi_{\mathrm{m}} = Q$. On a fixed-time outward sphere, the period of F_{em} is zero because $\mathrm{d}t$ pulls back to zero. The period of $\star F_{\mathrm{em}}$ is Q. Thus vacuum duality can take an exact field to a closed field with no global potential.

Conventions in electromagnetic sources

Source signs must be translated together. Nakahara's Example 7.16 uses signature $-+++$. For fixed orientation, metric reversal multiplies the k-form pairing and its star by $(-1)^k$. Thus the star on 2-forms is unchanged, while Nakahara's current is $j_{\mathrm{src}} = -j$. It follows that $\star_{\mathrm{src}} j_{\mathrm{src}} = \mathcal{J}$. His source equation is $\mathrm{d}^\dagger F_{\mathrm{em}} = j_{\mathrm{src}}$. Definition 7.6 gives $\mathrm{d}^\dagger = \star_{\mathrm{src}} \mathrm{d} \star_{\mathrm{src}}$ on Lorentzian 2-forms. Apply the source star and use its square $+1$ on 3-forms to obtain $\mathrm{d}\star F_{\mathrm{em}} = \mathcal{J}$.

The lower potential $\phi \mathrm{d}t - A_i \mathrm{d}x^i$ of Landau–Lifshitz has the opposite sign to A_{em} at matched units. In these units, $A^{\mathrm{LL}} = -A_{\mathrm{em}}$ and $F^{\mathrm{LL}} = -F_{\mathrm{em}}$. To convert instead to Gaussian fields with $c = 1$, set $F_{\mathrm{G}} = \sqrt{4\pi} F_{\mathrm{em}}$ and $\mathcal{J}_{\mathrm{G}} = \mathcal{J}/\sqrt{4\pi}$. Then $\mathrm{d}\star F_{\mathrm{G}} = 4\pi \mathcal{J}_{\mathrm{G}}$. The Gaussian lower-potential convention $F_{\mathrm{G}}^{\mathrm{LL}} = -F_{\mathrm{G}}$ therefore gives $\mathrm{d}\star F_{\mathrm{G}}^{\mathrm{LL}} = -4\pi \mathcal{J}_{\mathrm{G}}$. This converts the field sign and source normalization together.

Self-check. On Minkowski space, take

$$F_{\mathrm{em}} = (2t + 3x)\mathrm{d}y \wedge \mathrm{d}z + (5y - t)\mathrm{d}z \wedge \mathrm{d}x + 7z\mathrm{d}x \wedge \mathrm{d}y - 2y\mathrm{d}x \wedge \mathrm{d}t.$$

Compute $\star F_{\mathrm{em}}$ and the coefficient of $\mathrm{d}t \wedge \mathrm{d}y \wedge \mathrm{d}z$ in $\mathrm{d}F_{\mathrm{em}}$. Decide whether this field can satisfy Maxwell's equations for any electric sources.

Answer. The table gives

$$\star F_{\mathrm{em}} = (2t + 3x)\mathrm{d}t \wedge \mathrm{d}x + (5y - t)\mathrm{d}t \wedge \mathrm{d}y + 7z\mathrm{d}t \wedge \mathrm{d}z - 2y\mathrm{d}y \wedge \mathrm{d}z.$$

The required coefficient is $\partial_t B_x + \partial_y E_z - \partial_z E_y = 2$. In fact, $\operatorname{div} \mathbf{B} = 3 + 5 + 7 = 15$ and $\partial_t \mathbf{B} + \operatorname{curl} \mathbf{E} = (2, -1, 2)$. Thus $\mathrm{d}F_{\mathrm{em}} \neq 0$, independently of the electric source form.

Unit 15C Harmonic representatives

15.5 Harmonic representatives

A de Rham class consists of closed forms differing by exact forms. A manifold is *closed* if it is compact and has no boundary. On a closed oriented Riemannian manifold, we can compare the lengths of these forms using an integrated inner product. Harmonic representatives minimize this norm.

Definition 15.19. For real forms of equal degree on a closed oriented Riemannian manifold, the *integrated pairing* and its norm are

$$(\alpha,\beta)_{L^2}=\int_M \alpha\wedge\star\beta, \qquad \|\alpha\|_{L^2}^2=(\alpha,\alpha)_{L^2}.$$

Definition 15.20. The *codifferential* $\delta\colon \Omega^k(M)\to\Omega^{k-1}(M)$ is the formal adjoint of exterior differentiation, characterized by $(\mathrm{d}\alpha,\beta)_{L^2}=(\alpha,\delta\beta)_{L^2}$. We set it to zero on functions.

This operator differs from the connecting map δ_{MV} of Lecture 13. We first compute this adjoint on the unit circle $\mathbb{R}/(2\pi\mathbb{Z})$, with metric $\mathrm{d}\theta\otimes\mathrm{d}\theta$ and positive form $\mathrm{d}\theta$. The angle θ is local, but $\mathrm{d}\theta$ is a global 1-form. The star gives $\star 1=\mathrm{d}\theta$ and $\star\mathrm{d}\theta=1$. For smooth periodic functions f and a, integration by parts gives

$$(\mathrm{d}f,a\mathrm{d}\theta)_{L^2}=\int_0^{2\pi} f'a\,d\theta=-\int_0^{2\pi} fa'\,d\theta.$$

Hence $\delta(a\mathrm{d}\theta)=-a'$.

On a closed oriented Riemannian n-manifold, Stokes gives the formula on k-forms β, for $1\le k\le n$:

$$\delta\beta=(-1)^{n(k+1)+1}\star\mathrm{d}\star\beta.$$

To derive it, take $\alpha\in\Omega^{k-1}(M)$. The graded product rule and the absence of boundary give

$$0=\int_M \mathrm{d}(\alpha\wedge\star\beta)=(\mathrm{d}\alpha,\beta)_{L^2}+(-1)^{k-1}\int_M \alpha\wedge\mathrm{d}\star\beta.$$

Thus $\star\delta\beta=(-1)^k\mathrm{d}\star\beta$. Applying the star and using its square on degree $k-1$ gives the printed formula. In signature with s negative directions, the corresponding formal-adjoint formula has sign $(-1)^{n(k+1)+s+1}$. We use positivity only in Riemannian signature.

Definition 15.21. The *Hodge Laplacian* on $\Omega^k(M)$ is

$$\Delta=\mathrm{d}\delta+\delta\mathrm{d}.$$

Definition 15.22. A form is *harmonic* if $\Delta\alpha=0$.

Definition 15.23. A form is *coclosed* if $\delta\alpha=0$.

The operators d, δ and Δ act on forms. They are distinct from coordinate partial derivatives. On the circle the formulas above give

$$\Delta f=-f'', \qquad \Delta(a\mathrm{d}\theta)=-a''\mathrm{d}\theta.$$

A periodic function with zero second derivative is constant. Thus the harmonic functions are constants, and the harmonic 1-forms are constant multiples of $d\theta$. A 1-form with period P has harmonic representative $(P/(2\pi))d\theta$. Indeed, subtracting this constant form leaves a periodic coefficient of mean zero, whose integral from 0 to θ is a periodic primitive.

The Riemannian pointwise pairing is positive definite. The integrated norm therefore vanishes only for the zero form. Adjointness gives

$$(\Delta\alpha, \alpha)_{L^2} = \|d\alpha\|_{L^2}^2 + \|\delta\alpha\|_{L^2}^2.$$

Hence a form is harmonic exactly when it is both closed and coclosed. In particular, a harmonic form represents a de Rham class. For a harmonic k-form α with $k \geq 1$ and any $\gamma \in \Omega^{k-1}(M)$, adjointness gives $(\alpha, d\gamma)_{L^2} = (\delta\alpha, \gamma)_{L^2} = 0$. Consequently,

$$\|\alpha + d\gamma\|_{L^2}^2 = \|\alpha\|_{L^2}^2 + \|d\gamma\|_{L^2}^2.$$

Equality with $\|\alpha\|_{L^2}^2$ holds only when $d\gamma = 0$. Thus α is the unique form of minimum norm in its class. For degree zero there are no nonzero exact forms, so uniqueness in a class is immediate. This argument proves uniqueness when a harmonic representative exists.

Theorem 15.24 (Hodge representatives · *cited*)**.** *Every de Rham class on a closed oriented Riemannian manifold has a unique harmonic representative.*

The analytic existence theorem is cited from Nakahara, Theorem 7.8, §7.9.6. The closedness hypothesis is inherited there from Theorem 7.7. The statement does not apply to the Lorentzian spacetime used above.

On the flat torus $T^2 = \mathbb{R}^2/(2\pi\mathbb{Z})^2$ with metric $dx \otimes dx + dy \otimes dy$ and orientation $dx \wedge dy$, the forms dx and dy descend from the plane. The coordinates x and y are not global real functions on the torus. The Euclidean star rules show that both forms are closed and coclosed. Their periods on the positive coordinate circles are $(2\pi, 0)$ and $(0, 2\pi)$. These periods distinguish their cohomology classes. On the round sphere, the metric volume form is also harmonic: its exterior derivative is zero by degree, and its star is the constant function 1.

Self-check. State in one sentence what a harmonic representative is and why we introduced it. On the unit circle, identify the harmonic representative of $(3 + 2\cos(2\theta) - 5\sin(3\theta))d\theta$ and justify that it is in the same class.

Answer. A harmonic representative is a closed and coclosed form selected by the metric as the unique form of minimum integrated norm in a de Rham class. The given form has period 6π, so its harmonic representative is $3d\theta$. The decisive exactness calculation is

$$(2\cos(2\theta) - 5\sin(3\theta))d\theta = d\Big(\sin(2\theta) + \frac{5}{3}\cos(3\theta)\Big).$$

The primitive is periodic, and $\delta(3d\theta) = 0$.

Summary. A Riemannian metric exists by partitions of unity and identifies tangent vectors with covectors. A metric and orientation define volume and the Hodge star. Its square depends on degree, dimension and the number of negative metric directions. With signature $+---$, Maxwell's equations are $\mathrm{d}F_{\mathrm{em}} = 0$ and $\mathrm{d}{\star}F_{\mathrm{em}} = \mathcal{J}$. They imply charge conservation and local potentials. Monopole flux prevents a global potential. On closed oriented Riemannian manifolds, the cited Hodge theorem selects one harmonic form in each cohomology class. Lecture 16 uses a metric to differentiate tangent fields and define geodesics.

Reading. Nakahara, *Geometry, Topology and Physics*, second edition, §7.1 and §7.9, especially (7.172)–(7.177), Example 7.16, and Theorems 7.7–7.8. Tu, *Differential Geometry*, Theorem 1.12, pp. 6–7, and §16.4–§16.5, pp. 132–135.

Lecture 16

Connections and geodesics

Objectives. *Differentiating a tangent field involves both its components and its coordinate basis. A connection accounts for this change. A metric selects one connection by compatibility and zero torsion. Its parallel transport preserves inner products, and its geodesics have parallel velocity.*

By the end of this lecture you will be able to:

- Construct the Levi-Civita connection from a nondegenerate metric.
- Compute parallel transport and geodesic equations in coordinates.
- Find a conserved geodesic quantity from a Killing field.
- Explain why the exponential is onto for a compact connected Lie group.

Unit 16A Connections, transport and geodesics

16.1 Affine connections

We first differentiate a constant vector in a moving basis. In the punctured Euclidean plane, the polar unit vectors are

$$e_r = (\cos\phi, \sin\phi), \qquad e_\phi = (-\sin\phi, \cos\phi), \qquad \partial_r = e_r, \quad \partial_\phi = r e_\phi.$$

Along the unit circle $\gamma(t) = (\cos t, \sin t)$, the constant Cartesian vector $V = (1, 0)$ has changing components:

$$V = \cos t\, e_r - \sin t\, e_\phi, \qquad \frac{de_r}{dt} = e_\phi, \quad \frac{de_\phi}{dt} = -e_r.$$

The ordinary product rule includes the derivatives of the basis vectors:

$$\frac{dV}{dt} = -\sin t\, e_r + \cos t\, e_\phi - \cos t\, e_\phi + \sin t\, e_r = 0.$$

Differentiating only the components would miss this cancellation. The figure shows the constant vector in the two polar bases.

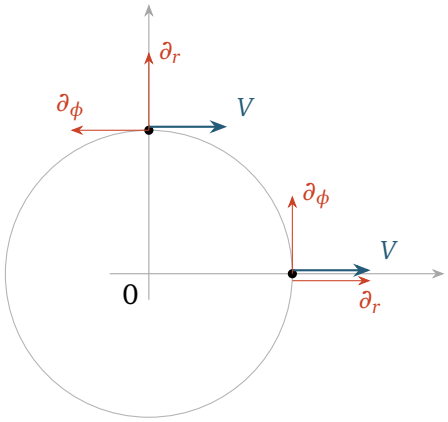


Figure 16.1: The same Cartesian vector $(1, 0)$ at two points of the unit circle. Its polar components change from $(1, 0)$ at $(1, 0)$ to $(0, -1)$ at $(0, 1)$.

On a surface in Euclidean space, a tangent field's ambient derivative can have a normal component. Tangential projection keeps its derivative within the tangent space. On the unit sphere, take the equator $\gamma(t) = (\cos t, \sin t, 0)$ and the field $V(t) = (1 + t)\dot{\gamma}(t)$. At $p = \gamma(0)$ its ambient derivative is $(-1, 1, 0)$, with tangent part $(0, 1, 0)$ and normal part $(-1, 0, 0)$. The figure also shows the field at $q = \gamma(\pi/3)$.

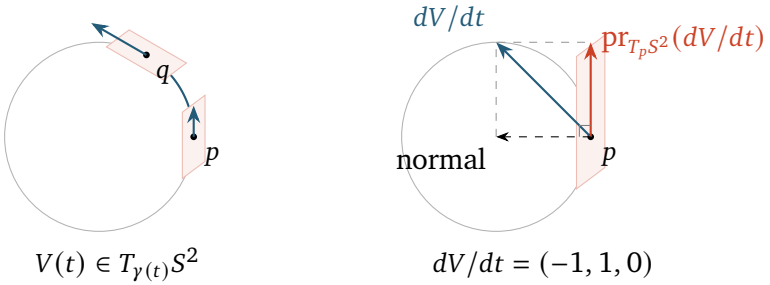


Figure 16.2: Field values at p and q belong to their own tangent planes. At p, the ambient derivative splits into its tangent and normal components.

For the velocity field $V = \dot{\gamma}$ alone, the ambient derivative is $\ddot{\gamma} = -\gamma$. Its tangent projection vanishes. A connection gives a derivative on a manifold without requiring an ambient space.

Let M be a smooth n-manifold without boundary and let $\mathfrak{X}(M)$ denote its smooth vector fields, as in Lecture 9.

Definition 16.1. An *affine connection* is an $\mathbb{R}$-bilinear map

$$\nabla \colon \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$$

that satisfies, for $f \in C^\infty(M)$,

$$\nabla_{fX} Y = f \nabla_X Y, \qquad \nabla_X (fY) = X(f) Y + f \nabla_X Y.$$

The field $\nabla_X Y$ is the *covariant derivative* of Y in direction X.

The first rule makes the derivative depend only on the direction at the point. The second rule differentiates scalar multiples of the field by the product rule. For example, on Cartesian $\mathbb{R}^n$, the usual directional derivative $\nabla_X Y = X^i \partial_i Y^j \partial_j$ satisfies both rules.

The rules make the derivative local. Suppose Y vanishes on a neighbourhood of p. Take a cutoff function f at p with support in that neighbourhood (Definition 12.12). Then $fY = 0$, and the Leibniz rule at p gives $\nabla_X Y|_p = 0$. The first rule proves the same locality in X. Thus local fields can be extended with cutoffs before taking their derivative.

In coordinates $x^1, \dots, x^n$, write $\partial_i = \partial/\partial x^i$.

Definition 16.2. The *Christoffel symbols* Γ_{ij}^k are defined by

$$\nabla_{\partial_i}\partial_j = \Gamma_{ij}^k \partial_k, \qquad (\nabla_X Y)^k = X^i(\partial_i Y^k + \Gamma_{ij}^k Y^j).$$

Repeated indices range from 1 to n. The second formula follows from the two connection rules and shows that the symbols determine the connection. In a chart, this expression uses only X_p, Y_p and the first derivatives of Y at p.

Definition 16.3. The *torsion* is

$$T(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y].$$

The connection is *torsion-free* when $T = 0$.

It is a tensor: the derivative term in $\nabla_Y(fX) = Y(f)X + f\nabla_Y X$ cancels the derivative term in $[fX,Y] = f[X,Y] - Y(f)X$. Thus $T(fX,Y) = fT(X,Y)$. Antisymmetry gives linearity over functions in the other slot. To see why function-linearity gives a pointwise tensor, restrict locally using cutoffs. Expanding $Z = Z^i\partial_i$ in any function-linear slot gives $A(Z)(p) = Z^i(p)A(\partial_i)(p)$, which depends only on Z_p.

Since coordinate fields commute, zero torsion is equivalent to $\Gamma_{ij}^k = \Gamma_{ji}^k$.

Definition 16.4. For a pseudo-Riemannian metric g, a *metric connection* satisfies

$$Xg(Y,Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z) \qquad (X,Y,Z \in \mathfrak{X}(M)).$$

This identity differentiates the inner product using the connection.

For a parametrized surface $p(x^1,x^2)$, the equality of mixed partials gives $\partial_i\partial_j p = \partial_j\partial_i p$. Tangential projection preserves this equality. Zero torsion expresses this property intrinsically. Metric compatibility alone does not force it. On Euclidean $\mathbb{R}^3$, the connection $\nabla_{\partial_i}\partial_j = \varepsilon_{ijk}\partial_k$ is metric because $\varepsilon_{ijk} + \varepsilon_{ikj} = 0$. Its torsion is $T(\partial_i,\partial_j) = 2\varepsilon_{ijk}\partial_k$. The alternating symbol has $\varepsilon_{123} = 1$.

16.2 The Levi-Civita connection

We derive the connection from metric compatibility and zero torsion. The cyclic sum in Koszul's formula determines it uniquely.

Definition 16.5. A torsion-free metric connection is called the *Levi-Civita connection*.

Theorem 16.6 (Levi-Civita). *Every pseudo-Riemannian manifold has a unique torsion-free metric connection. It satisfies the Koszul formula*

$$\begin{aligned} 2g(\nabla_X Y, Z) = {} & Xg(Y,Z) + Yg(X,Z) - Zg(X,Y) \\ & - g(X,[Y,Z]) + g(Y,[Z,X]) + g(Z,[X,Y]). \end{aligned}$$

Its coordinate symbols are

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell}(\partial_i g_{\ell j} + \partial_j g_{\ell i} - \partial_\ell g_{ij}).$$

Proof. For uniqueness, metric compatibility gives

$$\begin{aligned} Xg(Y,Z) &= g(\nabla_X Y, Z) + g(Y, \nabla_X Z), \\ Yg(X,Z) &= g(\nabla_Y X, Z) + g(X, \nabla_Y Z), \\ Zg(X,Y) &= g(\nabla_Z X, Y) + g(X, \nabla_Z Y). \end{aligned}$$

Add the first two identities and subtract the third. Replace $\nabla_Y X$ by $\nabla_X Y - [X,Y]$, and do the same for the other reversed pairs. The result is

$$\begin{aligned} Xg(Y,Z) + Yg(X,Z) - Zg(X,Y) = {} & 2g(\nabla_X Y, Z) - g([X,Y],Z) \\ & + g(Y,[X,Z]) + g(X,[Y,Z]). \end{aligned}$$

Rearranging gives Koszul's formula. Nondegeneracy determines $\nabla_X Y$ uniquely.

For existence, denote the right side by $\mathcal{K}(X,Y,Z)$. Expand derivatives by the product rule and brackets by $[X, fY] = X(f)Y + f[X,Y]$. The derivative terms cancel to give

$$\begin{aligned} \mathcal{K}(X,Y,fZ) &= f\mathcal{K}(X,Y,Z), \\ \mathcal{K}(fX,Y,Z) &= f\mathcal{K}(X,Y,Z), \\ \mathcal{K}(X,fY,Z) &= f\mathcal{K}(X,Y,Z) + 2X(f)g(Y,Z). \end{aligned}$$

For example, in the first identity the extra terms are $X(f)g(Y,Z) + Y(f)g(X,Z) - Y(f)g(X,Z) - X(f)g(Y,Z) = 0$. The cutoff and coordinate-expansion argument of §16.1 therefore makes $\mathcal{K}(X,Y,\cdot)/2$ a smooth covector field. Apply the inverse metric to define $\nabla_X Y$. The remaining identities give linearity in X and the Leibniz rule in Y. Additivity and real bilinearity follow directly from the formula.

Finally, exchanging the indicated slots in $\mathcal{K}$ gives

$$\begin{aligned} \mathcal{K}(X,Y,Z) - \mathcal{K}(Y,X,Z) &= 2g([X,Y],Z), \\ \mathcal{K}(X,Y,Z) + \mathcal{K}(X,Z,Y) &= 2Xg(Y,Z). \end{aligned}$$

The first identity proves zero torsion by nondegeneracy. The second proves metric compatibility. Substituting coordinate fields in Koszul, whose brackets vanish, gives $2g_{k\ell}\Gamma_{ij}^k = \partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}$. Multiplication by the inverse metric yields the stated symbols. ∎

For a constant metric matrix the symbols vanish in those coordinates. In particular, Minkowski geodesics will be straight lines. For an embedded surface in Euclidean three-space, the connection is $\nabla_X Y = \mathrm{pr}_{TM}(D_X Y)$, the tangential projection of the ambient derivative. Indeed, projection preserves the derivative rules. The ambient product rule gives metric compatibility because normal components pair to zero with tangent fields. The ambient identity $D_X Y - D_Y X = [X,Y]$ gives zero torsion. Uniqueness identifies this connection.

Self-check. Compute the plane's Levi-Civita symbols in polar coordinates $r > 0$. Compare with Cartesian coordinates to test whether Christoffel symbols are a tensor.

Answer. The metric is $g = \mathrm{d}r \otimes \mathrm{d}r + r^2 \mathrm{d}\phi \otimes \mathrm{d}\phi$. Its only nonzero derivative is $\partial_r g_{\phi\phi} = 2r$. The formula gives

$$\Gamma^r_{\phi\phi} = -r, \qquad \Gamma^\phi_{r\phi} = \Gamma^\phi_{\phi r} = 1/r.$$

All other symbols vanish. The Cartesian symbols are zero, whereas these are nonzero at the same points. A tensor with zero components in one basis has zero components in every basis. Thus Christoffel symbols are not tensor components.

16.3 Parallel transport and geodesics

A curve can meet the same point at different parameter values with different field values. Such a field need not extend to a field on a neighbourhood. We define its derivative along the curve and check that the definition agrees in different bases.

Definition 16.7. A *field along a curve* $\gamma\colon [a,b] \to M$ assigns $V(t) \in T_{\gamma(t)}M$ smoothly. Its *covariant derivative along the curve* is given in a coordinate segment by

$$\frac{DV}{dt} = (\dot V^k + \Gamma^k_{ij}(\gamma(t))\dot x^i V^j)\partial_k.$$

Equivalently, write this expression as $\dot v^j \partial_j + v^j \nabla_{\dot\gamma}\partial_j$ for $V = v^j\partial_j$ along the curve. In its first argument, the connection uses only the value of the direction field at the point.

Definition 16.8. A *frame* is a local smooth basis, written as the row $e = (e_1, \dots, e_n)$.

Multiplication of this row by a coefficient column means $ev = \sum_j v^j e_j$. For a new frame $e' = eA$ and coefficient columns $v = Av'$, define C by $\nabla_{\dot\gamma} e = eC$. The Leibniz rule and the coefficient change give

$$\nabla_{\dot\gamma}(eA) = e(\dot A + CA),$$
$$e(A\dot v' + \dot A v' + CAv') = e(\dot v + Cv).$$

The first line differentiates the new basis. The second compares the two expressions for DV/dt. Here $\dot A$ is the derivative along γ. The Leibniz rule supplies this term because $v = Av'$ gives $\dot v = \dot A v' + A\dot v'$. Thus the derivative is independent of the frame, including at repeated points of the curve. For $V = Y \circ \gamma$ it equals $\nabla_{\dot\gamma} Y$.

Definition 16.9. A field is *parallel* when $DV/dt = 0$.

Definition 16.10. The *parallel transport* along $\gamma\colon [a,b] \to M$ sends an initial vector to the endpoint of its parallel extension, written

$$P_\gamma\colon T_{\gamma(a)}M \to T_{\gamma(b)}M, \qquad P_\gamma(v) = V(b), \quad V(a) = v, \quad DV/dt = 0.$$

Proposition 16.11 (Parallel transport · *proved given linear ODE existence and uniqueness*)**.** *For a smooth curve $\gamma\colon [a,b] \to M$ and $v \in T_{\gamma(a)}M$, there is a unique parallel field with $V(a) = v$. The map P_γ is a linear isomorphism. For a metric connection it preserves the metric.*

Proof. The coordinate equation is a linear ODE $\dot{V} = -C(t)V$, where $C^k{}_j = \Gamma^k_{ij}\dot{x}^i$. On a compact coordinate segment its smooth coefficients are bounded, so linear ODE existence and uniqueness give a solution throughout that segment. A finite subdivision places each closed segment in a chart. Propagate the initial value through the subdivision. Coordinate independence and uniqueness glue the solutions. The same uniqueness proves linearity. Reversing the curve gives the inverse map. In coordinates, metric compatibility reads $\partial_i g_{k\ell} = \Gamma^m_{ik} g_{m\ell} + \Gamma^m_{i\ell} g_{km}$. Differentiate $g_{k\ell}V^kW^\ell$ along the curve and substitute this identity. It gives the along-curve product rule. For parallel fields,

$$\frac{d}{dt}g(V,W) = g(DV/dt, W) + g(V, DW/dt) = 0.$$

Thus endpoint inner products agree. ∎

This is the ODE input already used in Theorem 9.8.

Example 16.12 (Transport around a plane circle)**.** We transport the initial vector with polar components $(1,0)$ along $r = 1$ and $\phi = t$. The polar symbols computed above give

$$\dot{V}^r - V^\phi = 0, \qquad \dot{V}^\phi + V^r = 0.$$

The solution with these initial data is

$$V^r(t) = \cos t, \qquad V^\phi(t) = -\sin t.$$

Substitution gives $-\sin t - (-\sin t) = 0$ and $-\cos t + \cos t = 0$. Reconstruction gives $V = \cos t\, e_r - \sin t\, e_\phi = (1,0)$. Thus the changing polar components describe a constant Cartesian vector.

Definition 16.13. A *geodesic* is a parametrized curve with $D\dot{\gamma}/dt = 0$.

A geodesic has zero covariant acceleration. On an embedded surface, its ambient acceleration is normal to the surface. Its coordinate equation is

$$\ddot{x}^k + \Gamma^k_{ij}\dot{x}^i\dot{x}^j = 0.$$

The local second-order ODE theorem gives a unique maximal geodesic γ_v with $\gamma_v(0) = p$ and $\dot{\gamma}_v(0) = v \in T_pM$. Here *maximal* means that its interval of definition admits no extension. The geodesic equation is a first-order field on TM, with bundle components $(\dot{x}^k, \dot{v}^k) = (v^k, -\Gamma^k_{ij}v^iv^j)$.

Definition 16.14. The *connection exponential* is the map

$$\operatorname{Exp}_p \colon \mathcal{D}_p \to M, \qquad \operatorname{Exp}_p(v) = \gamma_v(1),$$

where $\mathcal{D}_p \subset T_pM$ consists of initial velocities for which the geodesic exists through time 1.

Definition 16.15. The connection is *geodesically complete* if every maximal geodesic is defined on all of $\mathbb{R}$.

The open smooth flow domain in Theorem 9.8 makes this map smooth. The domain is open as the time-1 slice of that flow domain, restricted to T_pM. It contains zero because the constant curve is a geodesic. For a geodesically complete connection, $\mathcal{D}_p = T_pM$. For the Levi-Civita connection, $g(\dot{\gamma}, \dot{\gamma})$ is constant by metric compatibility. In the Riemannian case, its square root is the speed.

Definition 16.16. The *energy* of a curve on $[a, b]$ is

$$E(\gamma) = \frac{1}{2}\int_a^b g(\dot\gamma, \dot\gamma)\, dt.$$

A curve is *stationary* when $\frac{d}{ds}\big|_0 E(\gamma_s) = 0$ for every smooth family γ_s with $\gamma_0 = \gamma$ and fixed endpoints.

Physical input. For a free unit-mass particle constrained to a Riemannian manifold, with no potential, we take the kinetic Lagrangian to be $L = \frac{1}{2}g_{ij}\dot x^i \dot x^j$. The physical paths satisfy the Euler–Lagrange equations recalled from *Theoretical Mechanics*, Lecture 2, *Lagrange's equations*.

Proposition 16.17 (Energy equations · *sketched; coordinate equivalence proved*)**.** *The Levi-Civita geodesics are precisely the stationary curves of the energy with fixed endpoints.*

The Euler–Lagrange equation is $d(\partial L/\partial \dot x^k)/dt - \partial L/\partial x^k = 0$. For $L = \frac{1}{2}g_{ij}\dot x^i \dot x^j$, its left side is

$$g_{k\ell}\ddot x^\ell + \partial_i g_{kj}\dot x^i \dot x^j - \frac{1}{2}\partial_k g_{ij}\dot x^i \dot x^j = g_{k\ell}(\ddot x^\ell + \Gamma^\ell_{ij}\dot x^i \dot x^j).$$

The contraction uses the symmetry of $\dot x^i \dot x^j$:

$$\partial_i g_{kj}\dot x^i \dot x^j = \frac{1}{2}(\partial_i g_{kj} + \partial_j g_{ki})\dot x^i \dot x^j.$$

Nondegeneracy gives the geodesic equation. For a fixed-endpoint variation, write $W^k = \partial_s x^k_s\big|_{s=0}$. Differentiating the energy and integrating by parts gives

$$\begin{aligned}\left.\frac{dE(\gamma_s)}{ds}\right|_0 &= \int_a^b \left(\frac{1}{2}\partial_k g_{ij} W^k \dot x^i \dot x^j + g_{kj}\dot W^k \dot x^j\right) dt \\ &= [g_{kj}W^k \dot x^j]_a^b - \int_a^b g_{k\ell}(\ddot x^\ell + \Gamma^\ell_{ij}\dot x^i \dot x^j) W^k\, dt.\end{aligned}$$

The boundary term vanishes since $W(a) = W(b) = 0$. Arbitrary compactly supported coordinate variations make the integral vanish exactly when its coefficient vanishes. Subdivision into coordinate segments gives the same conclusion for the whole curve. Hence the free particle of the physical input moves along Levi-Civita geodesics, with constant speed.

Definition 16.18. On a connected Riemannian manifold, use Lecture 15's length $\ell(\gamma)$ of a piecewise smooth curve. The distance between two points is the infimum of lengths of piecewise smooth curves joining those points. A *minimizing geodesic* attains that infimum.

For $b > a$, apply Cauchy–Schwarz to the constant function 1 and the speed. This gives

$$\ell(\gamma)^2 \le 2(b-a)E(\gamma),$$

with equality exactly at constant speed. Energy therefore also selects a parametrization. Stationary energy does not imply a global length minimum. No minimizing assertion follows in indefinite signature.

On the unit sphere, tangential projection gives $D\dot\gamma/dt = \ddot\gamma + |\dot\gamma|^2\gamma$. Indeed, differentiating $\gamma \cdot \gamma = 1$ twice gives $\gamma \cdot \ddot\gamma = -|\dot\gamma|^2$. For $v \in T_pS^2$ with $c = |v| > 0$, the unique solution is

$$\gamma_v(t) = \cos(ct)p + \frac{\sin(ct)}{c}v.$$

It lies on the sphere, has the prescribed initial data, and satisfies $\ddot{\gamma} = -c^2\gamma$. Its image is a *great circle*, the intersection with a plane through the origin. For $v = 0$ the geodesic is constant. Evaluating at time 1 gives the sphere exponential

$$\operatorname{Exp}_p(v) = \cos|v|\, p + \frac{\sin|v|}{|v|} v \quad (v \neq 0), \qquad \operatorname{Exp}_p(0) = p.$$

The figure follows radial lines of initial velocities into meridians.

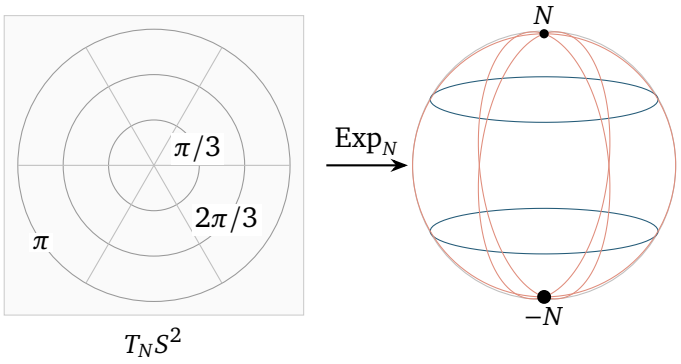


Figure 16.3: At the north pole, tangent circles of radii $\pi/3$ and $2\pi/3$ map to latitudes. The whole circle of radius π maps to the south pole.

In Minkowski coordinates, the zero Christoffel symbols instead give $\gamma_v(t) = p + tv$ for every initial vector, including zero and *null vectors*, which satisfy $\eta(v, v) = 0$.

16.4 Killing fields and conserved quantities

A metric symmetry gives a conserved pairing with geodesic velocity. For a field K with local flow φ_t, define the metric Lie derivative by $L_K g = \frac{d}{dt}\big|_0 \varphi_t^* g$.

Definition 16.19. A *Killing field* satisfies $L_K g = 0$.

A useful coordinate test is independence of a metric coordinate. If $\partial_a g_{ij} = 0$ for all i and j, translation in x^a preserves the metric, so $K = \partial_a$ is Killing. For the kinetic Lagrangian its pairing is the corresponding momentum:

$$g(\partial_a, \dot{\gamma}) = g_{aj}\dot{x}^j = \frac{\partial L}{\partial \dot{x}^a}.$$

A coordinate absent from L is called *cyclic*. The flow law gives $\frac{d}{dt}\varphi_t^* g = \varphi_t^*(L_K g)$, as for forms in Lecture 11. Thus the local flow of a Killing field consists of isometries, and conversely. For local fields Y and Z, put $Y_t = (\varphi_{-t})_* Y$ and $Z_t = (\varphi_{-t})_* Z$. The definition of tensor pullback gives $g(Y, Z) \circ \varphi_t = (\varphi_t^* g)(Y_t, Z_t)$. Differentiate at zero. The product rule and Theorem 9.12 give $Kg(Y, Z) = (L_K g)(Y, Z) + g([K, Y], Z) + g(Y, [K, Z])$. Rearranging proves $(L_K g)(Y, Z) = Kg(Y, Z) - g([K, Y], Z) - g(Y, [K, Z])$. Metric compatibility and zero torsion turn this into

$$(L_K g)(Y, Z) = g(\nabla_Y K, Z) + g(Y, \nabla_Z K).$$

Hence K is Killing exactly when the bilinear form $(Y, Z) \mapsto g(\nabla_Y K, Z)$ is antisymmetric.

Proposition 16.20 (Killing conservation). *Along a Levi-Civita geodesic γ, every Killing field K gives the constant quantity $g(K, \dot\gamma)$.*

Proof. Metric compatibility and the geodesic equation give

$$\frac{d}{dt} g(K, \dot\gamma) = g(\nabla_{\dot\gamma} K, \dot\gamma) = \frac{1}{2}(L_K g)(\dot\gamma, \dot\gamma) = 0. \qquad \blacksquare$$

For example, a constant translation field in Minkowski space is Killing because the metric is constant. Its pairing with $\dot\gamma = v$ is constant, as the straight-line solution also shows.

16.5 The round sphere in coordinates

For the unit sphere, the spherical parametrization is

$$p(\theta, \phi) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta), \qquad 0 < \theta < \pi.$$

Here ϕ lies in an angular chart. Lecture 15 computed its induced metric:

$$(g_{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2\theta \end{pmatrix}.$$

The figure compares a great circle with a latitude.

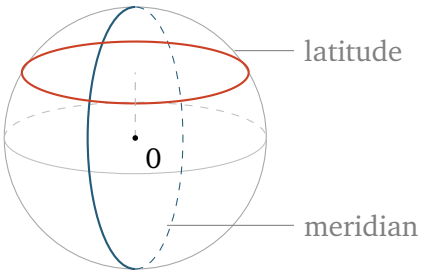


Figure 16.4: A meridian lies in a plane through the sphere's centre. A northern latitude lies in a plane parallel to the equator, away from the centre.

Example 16.21 (Sphere connection and angular momentum). Only $\partial_\theta g_{\phi\phi} = 2\sin\theta\cos\theta$ is nonzero among the metric derivatives. The Christoffel formula gives

$$\Gamma^\theta_{\phi\phi} = -\frac{1}{2}\partial_\theta g_{\phi\phi} = -\sin\theta\cos\theta,$$
$$\Gamma^\phi_{\theta\phi} = \Gamma^\phi_{\phi\theta} = \frac{1}{2\sin^2\theta}\partial_\theta g_{\phi\phi} = \cot\theta.$$

All other symbols vanish. Substitution gives

$$\ddot\theta - \sin\theta\cos\theta\,\dot\phi^2 = 0, \qquad \ddot\phi + 2\cot\theta\,\dot\theta\dot\phi = 0.$$

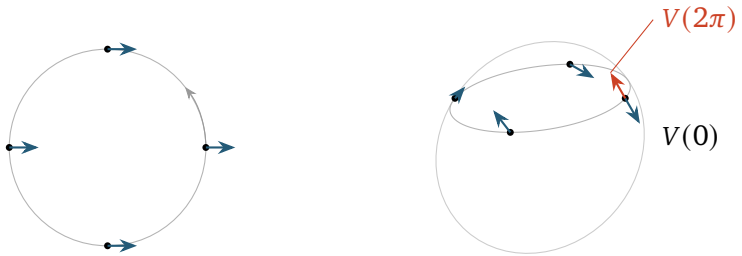
For the latitude circle $\theta = \theta_0$ traversed at constant nonzero angular speed, the first equation requires $\sin\theta_0\cos\theta_0 = 0$. Within this chart only $\theta_0 = \pi/2$ works: the equator is a geodesic. The coordinate poles are points, not nonconstant latitude curves. A meridian with constant ϕ and constant $\dot\theta$ solves both equations: $\ddot\theta = 0$ and $\dot\phi = \ddot\phi = 0$.

The field $K = \partial_\phi$ generates rotations about the vertical axis, which preserve the metric. Its conserved quantity and its derivative are

$$g(K, \dot\gamma) = \sin^2\theta\,\dot\phi, \qquad \frac{d}{dt}(\sin^2\theta\,\dot\phi) = \sin^2\theta(\ddot\phi + 2\cot\theta\,\dot\theta\dot\phi) = 0.$$

This is the axial angular momentum for the unit-mass energy Lagrangian. The coordinate singularity at a pole does not make the smooth connection singular.

For a latitude we use the positively oriented orthonormal frame $e_\theta = \partial_\theta$ and $e_\phi = (\sin\theta)^{-1}\partial_\phi$. The following figure compares transport around a plane circle and a spherical latitude.



plane: $V = (1, 0)$ sphere: $a = \pi/3$

Figure 16.5: The plane arrows are constant Cartesian vectors. On the latitude $a = \pi/3$, the solution's frame angle is $-t/2$, and the returning vector is reversed.

Example 16.22 (Transport around a latitude)**.** We transport $V(0) = e_\theta$ along $\theta = a$ and $\phi = t$, with $0 < a < \pi$. The spherical Christoffel symbols give

$$\dot V^\theta - \sin a\cos a\,V^\phi = 0, \qquad \dot V^\phi + \cot a\,V^\theta = 0.$$

Put $W = \sin a\,V^\phi$, the component along e_ϕ. The equations and their solution become

$$\dot V^\theta = \cos a\,W, \qquad \dot W = -\cos a\,V^\theta, \qquad (V^\theta, W) = (\cos(t\cos a), -\sin(t\cos a)).$$

Both differential equations follow by differentiating this pair. The frame returns to its initial value after one circuit. Thus the vector returns rotated by $-2\pi\cos a$, modulo 2π, relative to the positive frame (e_θ, e_ϕ). At the equator the frame components stay fixed. At $a = \pi/3$ the angle is $-\pi$.

The sphere's rotational symmetries also give the conserved vector $\gamma \times \dot\gamma$. We recall the rotation fields from Lecture 9. For the field $K_b(p) = b \times p$, with fixed $b \in \mathbb{R}^3$, its pairing is $g(K_b, \dot\gamma) = b \cdot (\gamma \times \dot\gamma)$. The figure shows the resulting bound on the latitude of a tilted great circle.

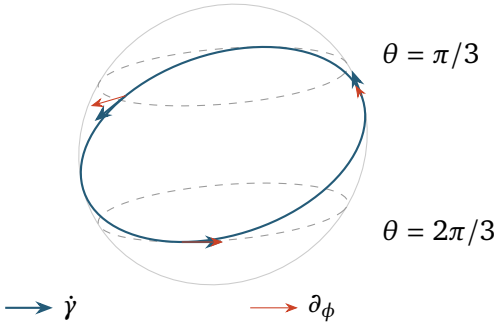


Figure 16.6: The unit-speed great circle $\gamma(t) = (\cos t, \frac{\sqrt{3}}{2}\sin t, \frac{1}{2}\sin t)$ has $\sin^2\theta\,\dot\phi = \sqrt{3}/2$ and $\pi/3 \le \theta \le 2\pi/3$.

For this curve $z = \frac{1}{2}\sin t$, so the polar angle has the stated range. The axial pairing is $x\dot y - y\dot x = \sqrt{3}/2$.

A geodesic need not minimize length on its entire interval. The following great-circle arcs have the same endpoints.

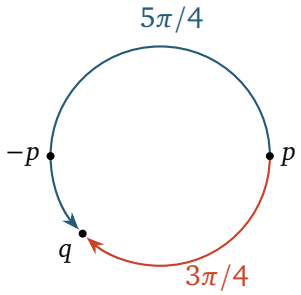


Figure 16.7: From p to q, the increasing-angle unit-circle arc has length $5\pi/4$. The decreasing-angle arc has length $3\pi/4$. The longer geodesic has passed the antipode $-p$.

The antipode is reached after length π along a unit-speed great circle. For an arc of length between π and 2π, the complementary arc has length $2\pi - \ell < \ell$. The longer arc in the figure therefore cannot minimize length. No local geodesic equation implies a global minimum.

Self-check. On the unit sphere, transport $V(0) = 2e_\theta + e_\phi$ along $\theta = \pi/3$ from $\phi = 0$ to $\phi = 2\pi/3$. Give its final coordinate components and check its norm.

Answer. The orthonormal components rotate through $-(2\pi/3)\cos(\pi/3) = -\pi/3$. Thus

$$(V^\theta, W) = \left(1 + \frac{\sqrt{3}}{2}, \frac{1}{2} - \sqrt{3}\right), \qquad V^\phi = \frac{W}{\sin(\pi/3)} = \frac{1}{\sqrt{3}} - 2.$$

The squared norm is $(V^\theta)^2 + W^2 = 5$, equal to $2^2 + 1^2$ initially.

Unit 16B Geodesics and exponentials on Lie groups

16.6 Bi-invariant metrics and the group exponential

For a compact connected Lie group, every element is the exponential of an initial tangent vector at the identity. We prove this by identifying group exponentials with geodesics for a bi-invariant metric. Averaging constructs such a metric, and the cited completeness theorem joins the identity to every point.

On a Lie group G, write $\ell_h(g) = hg$ and $r_h(g) = gh$.

Definition 16.23. A metric is *bi-invariant* if every left and right translation is an isometry.

Recall from Lecture 9 that each $X \in \mathfrak{g} = T_eG$ defines a complete left-invariant field X^L, whose identity curve $c_X(t)$ is a one-parameter subgroup.

Definition 16.24. The *group exponential* is the map

$$\exp\colon \mathfrak{g} \to G, \qquad \exp X = c_X(1).$$

Scaling the ODE gives $c_X(t) = \exp(tX)$, and the flow of X^L is $r_{\exp(tX)}$. This map differs in definition from Exp_p. For a matrix Lie group it agrees with the matrix exponential of Lecture 9.

Theorem 16.25 (Bi-invariant geodesics · *proved given smooth ODE uniqueness*)**.** *For a bi-invariant pseudo-Riemannian metric, $\nabla_{X^L} Y^L = \frac{1}{2}[X,Y]^L$. The geodesics through the identity are exactly $t \mapsto \exp(tX)$, and $\operatorname{Exp}_e = \exp$ on all of $\mathfrak{g}$.*

Proof. The inner products of left-invariant fields are constant. Right translations are isometries, so X^L is Killing. The metric Lie derivative identity above therefore gives $g([X^L,Y^L],Z^L) + g(Y^L,[X^L,Z^L]) = 0$. Using this identity in the three bracket terms of Koszul gives

$$2g(\nabla_{X^L}Y^L, Z^L) = g([X^L,Y^L],Z^L).$$

The fields Z^L span each tangent space, so nondegeneracy proves the formula. Along c_X, its velocity is X^L, hence its acceleration is $\frac{1}{2}[X,X]^L = 0$. Uniqueness of the geodesic ODE gives all identity geodesics. Completeness of X^L gives their full real domains and the exponential equality. ∎

For compact G, a bi-invariant Riemannian metric exists. We construct it using Lecture 12's integration. A left-translated basis of $\mathfrak{g}$ gives a global frame and a positive, left-invariant volume form μ, normalized to have integral 1. Let b be any positive inner product on $\mathfrak{g}$. Define Ad_h to be the differential at the identity of $g \mapsto hgh^{-1}$. Average by

$$B(X,Y) = \int_G b(\operatorname{Ad}_{g^{-1}} X, \operatorname{Ad}_{g^{-1}} Y)\, \mu(g).$$

This is positive definite: a nonzero vector has a strictly positive integrand. Conjugation composes, so substituting $g = hk$ and using left-invariance of volume gives $B(\operatorname{Ad}_h X, \operatorname{Ad}_h Y) = B(X,Y)$. Extend B by left translation to a smooth metric. The identity $r_h \circ \ell_g = \ell_{gh} \circ (g' \mapsto h^{-1}g'h)$ shows that right translation acts on left-translated components by $\operatorname{Ad}_{h^{-1}}$. The Ad-invariance of B proves that the metric is bi-invariant.

For a connected Riemannian manifold, the distance defined in Unit 16A is the infimum of lengths of joining curves. The cited Hopf–Rinow theorem states that completeness of this distance is equivalent to geodesic completeness, and that on a complete connected Riemannian manifold every two points are joined by a minimizing geodesic (Tu, Theorem 16.6, p. 131). The distance induces the manifold topology (Lee, Lemma 6.2, pp. 94–95), so compactness makes it complete. Reparametrize the joining geodesic to $[0,1]$ by a linear change of time. Thus a compact connected G has every element $g = \operatorname{Exp}_e v = \exp v$ for some $v \in \mathfrak{g}$. This surjectivity conclusion is proved given integration and Hopf–Rinow; it does not assert injectivity of the exponential.

Example 16.26 (The rotation group $SO(2)$)**.** The angular metric on $SO(2)$ is invariant under left and right translations, which add a constant angle. For the generator

$$X = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \exp(tX) = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}.$$

Every plane rotation occurs in this formula. The parameters t and $t + 2\pi$ give the same element, so the exponential is onto and is not injective. The identity geodesics are exactly these constant-angular-speed rotations.

Self-check. State in one sentence what a connection does and why we introduced it. Explain how the two exponential maps become equal at a Lie group's identity.

Answer. A connection differentiates tangent fields by including the change of basis, so that differentiation is independent of the chosen coordinates. For a bi-invariant metric, $\nabla_{X^L} X^L = \frac{1}{2}[X,X]^L = 0$. The group curve $\exp(tX)$ is therefore the geodesic with initial velocity X. Uniqueness gives $\operatorname{Exp}_e X = \exp X$ at time 1.

Summary. A nondegenerate metric determines the torsion-free metric connection by Koszul's formula. Parallel transport solves a linear ODE and preserves the metric. Geodesics have parallel velocity and constant metric norm. Killing fields give conserved pairings with that velocity. For a bi-invariant metric the identity geodesics are one-parameter subgroups. Compact averaging and the cited Hopf–Rinow theorem make the group exponential onto for compact connected groups. Lecture 17 computes curvature from this connection.

Reading. Tu, *Differential Geometry*, §6, §13–§15, especially Problems 15.4–15.5, and Theorem 16.6, p. 131. Lee, *Riemannian Manifolds: An Introduction to Curvature*, Theorem 5.4, pp. 68–70, and Lemma 6.2, pp. 94–95. Nakahara, §7.2, §7.4 and §7.7.

Lecture 17

Curvature

Objectives. *Curvature measures the change in a vector transported around a small loop. We compute it from Christoffel symbols and from differential forms in an orthonormal frame. The sphere fixes the sign used for surface geometry and gravitation.*

By the end of this lecture you will be able to:

- Compute curvature with the output and direction indices specified.
- Prove tensoriality, the metric antisymmetries and the first Bianchi identity.
- Use Cartan's equations to find the curvature of a surface.
- Distinguish zero curvature from zero Christoffel symbols in one chart.

Unit 17A The curvature tensor and its contractions

17.1 The curvature tensor

Parallel transport compares tangent vectors along a path, as in Lecture 16.

Definition 17.1. Let M be a smooth manifold with an affine connection ∇. The *curvature operator* is defined on smooth fields by

$$R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z.$$

On the unit sphere, take the loop from the north pole $N = e_z$ to e_x, then to e_y, then back to N, along the shorter great-circle arcs. We orient the sphere by its outward normal. This loop is counter-clockwise when viewed from outside.

The vector initially equal to e_x reaches the vertex e_x as $-e_z$. It stays equal to $-e_z$ along the equator and returns to N as e_y. Along a great circle its tangent is parallel. A constant vector perpendicular to that circle's plane is also parallel, because its ambient derivative is zero. These observations verify the stated transports by tangent projection. The return vector has rotated by $+\pi/2$ in $T_N S^2$. The figure compares this with transport around a planar triangle.

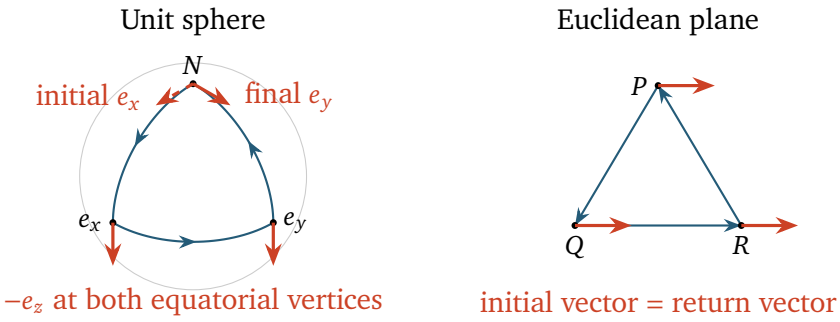


Figure 17.1: Parallel transport around the oriented octant sends e_x to e_y at N. The planar comparison returns its initial vector unchanged.

The octant has area $\int_0^{\pi/2}\int_0^{\pi/2}\sin\theta\,d\theta\,d\phi = \pi/2$. Lecture 18 relates the rotation to the integral of surface curvature.

For the Cartesian connection on $\mathbb{R}^n$, the components of $R(X,Y)Z$ are $XY(Z^k) - YX(Z^k) - [X,Y](Z^k) = 0$. The bracket correction removes the failure of the direction fields to commute.

Curvature measures the leading change under transport around a small loop. Choose local coordinates $(x, y, \ldots)$ about p. Traverse the coordinate square in the order $+x, +y, -x, -y$, with side $\varepsilon > 0$ and the other coordinates fixed. Write F_{xy} for the matrix of $R(\partial_x, \partial_y)$ at p in a chosen basis of T_pM. The parallel-transport matrix around this square satisfies

$$\mathrm{Hol}(\square_\varepsilon) = \mathbb{1} - \varepsilon^2 F_{xy} + O(\varepsilon^3).$$

The coordinate directions commute, so the square closes. Lecture 21 proves this small-square formula. The tensoriality proved next says that the curvature coefficient depends only on the vectors at p, independently of their extensions.

Theorem 17.2 (Tensoriality of curvature)**.** *Curvature is linear over $C^\infty(M)$ in all three arguments. It therefore gives a smooth tensor of type $(1,3)$.*

Proof. Real multilinearity is immediate from the connection rules. For $f \in C^\infty(M)$, the Leibniz rule and $[fX,Y] = f[X,Y] - Y(f)X$ give

$$R(fX,Y)Z = fR(X,Y)Z - Y(f)\nabla_X Z + Y(f)\nabla_X Z = fR(X,Y)Z.$$

Antisymmetry in X and Y gives linearity in Y. Expansion in the last slot gives

$$R(X,Y)(fZ) = fR(X,Y)Z + \big(XY(f) - YX(f) - [X,Y](f)\big)Z = fR(X,Y)Z.$$

The cross terms $Y(f)\nabla_X Z$ and $X(f)\nabla_Y Z$ cancel between the two second derivatives. By the cutoff and coordinate-expansion argument of §16.1, the value therefore depends only on the three vectors at the point. Smoothness follows from the local coefficients of the connection. ∎

Definition 17.3. The tensor of Theorem 17.2 is called the *Riemann curvature tensor*.

Write $\partial_\mu = \partial/\partial x^\mu$ in a chart. Greek indices here range from 1 to n. The index order is fixed by the action

$$R(\partial_\mu, \partial_\nu)\partial_\sigma = R^\rho{}_{\sigma\mu\nu}\partial_\rho.$$

Applying the connection product rule twice gives

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu\Gamma^\rho_{\nu\sigma} - \partial_\nu\Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}.$$

For example, applying the product rule gives

$$\nabla_{\partial_\mu}\nabla_{\partial_\nu}\partial_\sigma = (\partial_\mu\Gamma^\rho_{\nu\sigma} + \Gamma^\lambda_{\nu\sigma}\Gamma^\rho_{\mu\lambda})\partial_\rho.$$

Subtract the reversed expression. The bracket term vanishes because coordinate fields commute. Curvature is a tensor even though the Christoffel symbols are not. For the Euclidean plane, Lecture 16 gives the polar symbols $\Gamma^r_{\phi\phi} = -r$ and $\Gamma^\phi_{r\phi} = \Gamma^\phi_{\phi r} = 1/r$. For example, the curvature component is

$$R^r{}_{\phi r\phi} = \partial_r(-r) - (-r)(1/r) = -1 + 1 = 0.$$

All curvature components vanish because the tensor is zero in Cartesian coordinates.

Definition 17.4. A connection is *flat* when its curvature is zero.

Example 17.5 (The sphere in coordinates)**.** We compute a curvature component on the sphere of radius $a > 0$. Its spherical metric is

$$g = a^2 \mathrm{d}\theta \otimes \mathrm{d}\theta + a^2 \sin^2\theta \mathrm{d}\phi \otimes \mathrm{d}\phi, \qquad 0 < \theta < \pi.$$

The constant scale does not change Lecture 16's Christoffel symbols. Substituting them into the coordinate formula gives

$$\begin{aligned} R^\theta{}_{\phi\theta\phi} &= \partial_\theta(-\sin\theta\cos\theta) - (-\sin\theta\cos\theta)\cot\theta \\ &= (\sin^2\theta - \cos^2\theta) + \cos^2\theta = \sin^2\theta. \end{aligned}$$

Lowering the output with the metric gives $R_{\theta\phi\theta\phi} = a^2\sin^2\theta$. This positive component fixes the operator sign on the unit sphere.

17.2 Metric symmetries and the first Bianchi identity

Now let ∇ be the Levi-Civita connection of a pseudo-Riemannian metric. Lower the output index first: $R_{\rho\sigma\mu\nu} = g_{\rho\lambda}R^\lambda{}_{\sigma\mu\nu}$. For the proofs, abbreviate the four-field expression by $C(X,Y,Z,W) = g(R(X,Y)Z,W)$.

Proposition 17.6 (Curvature symmetries · *sketched; antisymmetries and first Bianchi proved*)**.** *The curvature satisfies*

$$\begin{aligned} C(X,Y,Z,W) &= -C(Y,X,Z,W) = -C(X,Y,W,Z), \\ R(X,Y)Z + R(Y,Z)X + R(Z,X)Y &= 0, \\ C(X,Y,Z,W) &= C(Z,W,X,Y). \end{aligned}$$

The middle equation is the first Bianchi identity.

Proof and sketch. The first antisymmetry follows from the operator definition. For the other one, apply $XY - YX - [X,Y]$ to $g(Z,W)$. It vanishes on this scalar function. Metric compatibility expands the result as

$$0 = g(R(X,Y)Z,W) + g(Z,R(X,Y)W),$$

with the two cross products of first derivatives cancelling. This proves antisymmetry in the last pair.

For Bianchi, sum cyclically over (X, Y, Z). The six second derivatives combine by zero torsion as follows:

$$\begin{aligned}\nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z + \nabla_Y\nabla_Z X - \nabla_Z\nabla_Y X + \nabla_Z\nabla_X Y - \nabla_X\nabla_Z Y \\ = \nabla_X(\nabla_Y Z - \nabla_Z Y) + \nabla_Y(\nabla_Z X - \nabla_X Z) + \nabla_Z(\nabla_X Y - \nabla_Y X) \\ = \sum_{\text{cyc}} \nabla_X [Y, Z].\end{aligned}$$

Relabel the bracket-direction terms to obtain

$$\sum_{\text{cyc}} R(X, Y)Z = \sum_{\text{cyc}} \big(\nabla_X [Y, Z] - \nabla_{[Y,Z]} X\big) = \sum_{\text{cyc}} [X, [Y, Z]] = 0.$$

The last equality is Jacobi's identity.

For pair exchange, first pair Bianchi with W:

$$C(X, Y, Z, W) + C(Y, Z, X, W) + C(Z, X, Y, W) = 0.$$

Write this scalar identity for four cyclic permutations of the four fields. Add the identities and use the two antisymmetries to cancel terms. The remaining equality gives pair exchange after renaming the fields. This sketches the algebraic step; see Lee, Proposition 7.4, pp. 121–123. ∎

In the lowered-output convention, these identities give $R_{abcd} = -R_{bacd} = -R_{abdc}$ and $R_{abcd} = R_{cdab}$. Our lowered indices correspond to the action $g(R(\partial_c, \partial_d)\partial_b, \partial_a)$.[1] Changing the operator sign negates curvature values. Changing slot order alone can also introduce a sign.

17.3 Ricci, scalar and sectional curvature

Definition 17.7. For a Riemannian metric and a tangent 2-plane $P \subset T_pM$, its *sectional curvature* is

$$K(P) = \frac{g(R(u, v)v, u)}{g(u, u)g(v, v) - g(u, v)^2}, \qquad P = \operatorname{span}\{u, v\},$$

where u and v are linearly independent.

The denominator is positive. On changing the basis of P, antisymmetry in both pairs multiplies the numerator by the square of the basis-change determinant. The Gram determinant in the denominator has the same factor. Thus the quotient is independent of the basis. For orthonormal u and v, it is $g(R(u, v)v, u)$. In indefinite signature the same quotient defines sectional curvature only for nondegenerate planes, whose Gram determinant is nonzero. For the sphere in Example 17.5, the Gram determinant is $a^4 \sin^2\theta$, so its tangent plane has $K = 1/a^2$.

Definition 17.8. The *Ricci tensor* is the trace

$$\operatorname{Ric}(X, Y) = \operatorname{tr}(Z \mapsto R(Z, Y)X).$$

Definition 17.9. The *scalar curvature* R_{sc} is the metric trace of Ric.

[1] Lee uses $\operatorname{Rm}_{\text{Lee}}(X, Y, Z, W) = C(X, Y, Z, W)$. Thus $R_{abcd} = \operatorname{Rm}_{\text{Lee}}(\partial_c, \partial_d, \partial_b, \partial_a)$, with the same operator sign.

In coordinates these traces are

$$\mathrm{Ric}_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}, \qquad R_{\mathrm{sc}} = g^{\mu\nu}\,\mathrm{Ric}_{\mu\nu}\,.$$

Pair exchange makes Ricci symmetric: in $\mathrm{Ric}_{ij} = g^{ab}R_{aibj}$, exchange the curvature pairs and then rename a and b. The symbol R_{sc} denotes scalar curvature, while a denotes the sphere radius.

For a Riemannian metric, let $(e_1, \dots, e_n)$ be an orthonormal basis at p. The trace definitions give

$$\mathrm{Ric}(e_1, e_1) = \sum_{i=2}^{n} g(R(e_i, e_1)e_1, e_i) = \sum_{i=2}^{n} K(\mathrm{span}\{e_1, e_i\}),$$
$$R_{\mathrm{sc}} = 2\sum_{i<j} K(\mathrm{span}\{e_i, e_j\}).$$

The omitted term in the first trace is zero by direction antisymmetry. Each plane occurs twice in the second trace, once in each order. Thus Ricci sums the plane curvatures through a unit direction. Scalar curvature sums over all orthogonal basis planes, with this factor of two. On a surface there is only one tangent 2-plane, so $\mathrm{Ric} = Kg$ and $R_{\mathrm{sc}} = 2K$. The first identity follows from the directional trace and polarisation. These unsigned sums require a positive metric. The metric traces remain the definitions in indefinite signature. Lecture 19 uses Ricci and scalar curvature in the gravitational field equation.

Curvature of a bi-invariant metric

The connection of Lecture 16 also gives curvature directly from Lie brackets.

Proposition 17.10 (Bi-invariant curvature). *For a bi-invariant pseudo-Riemannian metric on a Lie group and left-invariant fields X, Y and Z,*

$$R(X,Y)Z = -\frac{1}{4}[[X,Y],Z].$$

For a Riemannian metric and an orthonormal left-invariant pair X and Y,

$$K(\mathrm{span}\{X,Y\}) = \frac{1}{4}\|[X,Y]\|^2.$$

Proof. Lecture 16 gives $\nabla_X Y = \frac{1}{2}[X,Y]$ for left-invariant fields. All brackets below are again left-invariant, so

$$\begin{aligned} R(X,Y)Z &= \frac{1}{4}\big([X,[Y,Z]] - [Y,[X,Z]]\big) - \frac{1}{2}[[X,Y],Z] \\ &= \frac{1}{4}[[X,Y],Z] - \frac{1}{2}[[X,Y],Z] = -\frac{1}{4}[[X,Y],Z]. \end{aligned}$$

The second equality is Jacobi's identity. The metric invariance proved in Lecture 16 gives $g([Y,B],X) + g(B,[Y,X]) = 0$ for $B = [X,Y]$. Consequently,

$$g(R(X,Y)Y,X) = -\frac{1}{4}g([B,Y],X) = \frac{1}{4}g(B,B).$$

This is the sectional-curvature formula for the orthonormal pair. ∎

Example 17.11 (A bi-invariant metric on $SU(2)$)**.** We use the real inner product $g(A,B) = -2\operatorname{tr}(AB)$ on the traceless anti-Hermitian matrices $\mathfrak{su}(2)$. It is positive because $-\operatorname{tr}(A^2) = \operatorname{tr}(A^\dagger A)$. Trace invariance under conjugation makes its left-translated metric bi-invariant. An orthonormal basis is

$$E_1 = \frac{1}{2}\begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}, \quad E_2 = \frac{1}{2}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad E_3 = \frac{1}{2}\begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}.$$

Matrix multiplication gives $[E_1, E_2] = E_3$ and its cyclic counterparts. Thus $K(\operatorname{span}\{E_1, E_2\}) = \frac{1}{4}$. The positive value agrees with our sphere sign convention.

Self-check. State in one sentence what the curvature tensor is and why it was introduced. Explain why a nonzero Christoffel symbol does not by itself establish nonzero curvature.

Answer. The curvature tensor is the function-linear operator measuring the leading failure of parallel transport to return a vector unchanged around small coordinate loops. The polar plane has nonzero Christoffel symbols but $R^r{}_{\phi r\phi} = -1 + 1 = 0$. The derivative and product terms cancel, so a connection coefficient alone does not measure curvature.

Unit 17B Moving frames and surface curvature

17.4 Moving frames and Cartan's equations

The spherical coordinate vectors have varying lengths. An orthonormal frame separates these lengths from the rotation of the frame. Its connection forms record that rotation, while its coframe records the metric.

Definition 17.12. A *local frame* $e = (e_1, \dots, e_n)$ is a smooth basis of tangent vectors at every point of an open set U. It is a row of fields.

Definition 17.13. Its *dual coframe* is the column of 1-forms $\theta = (\theta^1, \dots, \theta^n)^T$ with $\theta^i(e_j) = \delta^i_j$.

Definition 17.14. For a Riemannian metric, a frame is *orthonormal* when $g(e_i, e_j) = \delta_{ij}$.

Gram–Schmidt applied smoothly to a coordinate frame supplies local orthonormal frames. The construction divides only by positive smooth lengths. The dual coframe then gives $g = \sum_i \theta^i \otimes \theta^i$. The figure compares coordinate vectors with unit frame vectors on a sphere.

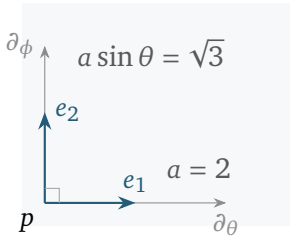


Figure 17.2: For a radius-a sphere, the spherical coordinate vectors have lengths a and $a\sin\theta$ (drawn at $a = 2$ and $\theta = \pi/3$). Dividing by those lengths gives the orthonormal frame e_1 and e_2 on a chart away from the poles.

Definition 17.15. The *connection forms* $\omega^i{}_j$ are specified by

$$\nabla_X e_j = e_i \omega^i{}_j(X).$$

Definition 17.16. The *curvature forms* $\Omega^i{}_j$ are specified by

$$R(X,Y)e_j = e_i \Omega^i{}_j(X,Y).$$

Linearity of the connection in its direction makes the first coefficients 1-forms. Tensoriality and direction antisymmetry make the second 2-forms. In an orthonormal frame, $\omega^i{}_j(X)$ is the component along e_i of the rate of change of e_j in direction X. The first structure equation below expresses zero torsion. The second computes curvature from those rates. Write ∇V for the vector-valued 1-form $X \mapsto \nabla_X V$. The matrices act on coefficient columns: for $V = ev$, $\nabla V = e(\mathrm{d}v + \omega v)$. For matrices of forms, the wedge includes matrix multiplication: $(A \wedge B)^i{}_j = \sum_k A^i{}_k \wedge B^k{}_j$. The exterior derivative acts on each entry.

Lemma 17.17 (Exterior derivative on fields)**.** *For a smooth 1-form α and smooth fields X and Y,*

$$(\mathrm{d}\alpha)(X,Y) = X(\alpha(Y)) - Y(\alpha(X)) - \alpha([X,Y]).$$

Proof. The right side is function-linear in X and Y by the product and bracket rules. On coordinate fields it agrees with Lecture 10's definition of $\mathrm{d}\alpha$. Tensoriality therefore gives equality for all fields. For example, when $\alpha = f\mathrm{d}g$, the right side is $X(f)Y(g) - Y(f)X(g)$, the evaluation of $\mathrm{d}f \wedge \mathrm{d}g$. ∎

Theorem 17.18 (Cartan's structure equations)**.** *On a Riemannian manifold, the Levi-Civita connection in a local orthonormal frame satisfies*

$$\mathrm{d}\theta = -\omega \wedge \theta, \qquad \Omega = \mathrm{d}\omega + \omega \wedge \omega.$$

The connection matrix ω and curvature matrix Ω are skew-symmetric. Conversely, a skew matrix ω satisfying the first equation is the unique Levi-Civita connection matrix for this coframe.

Proof. Differentiate $g(e_i, e_j) = \delta_{ij}$. Metric compatibility gives $\omega^i{}_j + \omega^j{}_i = 0$. For any connection matrix in this frame, the lemma gives

$$\theta^i(T(X,Y)) = (\mathrm{d}\theta^i + \omega^i{}_j \wedge \theta^j)(X,Y).$$

Indeed, expand $Y = \theta^j(Y)e_j$ in $\nabla_X Y$ and subtract the reversed expression and $[X,Y]$. Zero torsion therefore gives the first structure equation.

Conversely, a smooth matrix ω defines a connection by $\nabla_X(ev) = e(Xv + \omega(X)v)$. The product rule verifies both connection axioms. If ω is skew, the extra terms in differentiating $u^T v$ cancel because $u^T(\omega(X)^T + \omega(X))v = 0$. Thus this connection is metric. The displayed torsion identity makes it torsion-free when the first equation holds. Uniqueness in Theorem 16.6 identifies it with the Levi-Civita connection, proving the converse and uniqueness of the skew solution.

For the second, expand $\nabla_X \nabla_Y e_j$ using the Leibniz rule and subtract its reversed expression and $\nabla_{[X,Y]} e_j$. The coefficient of e_i is

$$X\omega^i{}_j(Y) - Y\omega^i{}_j(X) - \omega^i{}_j([X,Y]) + \omega^i{}_k(X)\omega^k{}_j(Y) - \omega^i{}_k(Y)\omega^k{}_j(X).$$

The first terms are $\mathrm{d}\omega^i{}_j(X,Y)$ and the last terms are $(\omega \wedge \omega)^i{}_j(X,Y)$. This proves the curvature equation. The last-pair metric antisymmetry already proved gives $\Omega^i{}_j = -\Omega^j{}_i$. ∎

17.5 Gaussian curvature of surfaces

Definition 17.19. On a Riemannian surface, the *Gaussian curvature* K is the sectional curvature of its tangent plane.

Choose a positively oriented orthonormal frame on a chart. Write its connection entry as $\omega^1{}_2$. Skew-symmetry makes the matrix

$$\omega = \begin{pmatrix} 0 & \omega^1{}_2 \\ -\omega^1{}_2 & 0 \end{pmatrix}.$$

Cartan's equations become

$$\mathrm{d}\theta^1 = -\omega^1{}_2 \wedge \theta^2, \qquad \mathrm{d}\theta^2 = \omega^1{}_2 \wedge \theta^1, \qquad \mathrm{d}\omega^1{}_2 = K\theta^1 \wedge \theta^2.$$

For the last equality, the diagonal-free 2×2 matrix has $(\omega \wedge \omega)^1{}_2 = 0$. The coefficient of $\Omega^1{}_2$ on (e_1, e_2) is $g(R(e_1, e_2)e_2, e_1) = K$. Thus the curvature coefficient has the displayed positive sign. In terms of the frame, the same convention is $\nabla e_1 = -\omega^1{}_2 e_2$ and $\nabla e_2 = \omega^1{}_2 e_1$.

Proposition 17.20 (Surface curvature)**.** *On a Riemannian surface, the lowered-output curvature satisfies*

$$R_{abcd} = K(g_{ac}g_{bd} - g_{ad}g_{bc}), \qquad \mathrm{Ric} = Kg, \qquad R_{\mathrm{sc}} = 2K.$$

The round sphere of radius $a > 0$ has $K = 1/a^2$. The Poincaré half-plane with metric $(\mathrm{d}x \otimes \mathrm{d}x + \mathrm{d}y \otimes \mathrm{d}y)/y^2$ has $K = -1$. The torus $\mathbb{R}^2/(2\pi\mathbb{Z})^2$ with its quotient Euclidean metric has $K = 0$.

In two dimensions, antisymmetry in both pairs leaves a single independent curvature component in an orthonormal basis. Its value is K. Tensoriality therefore gives the displayed formula in any coordinates. Contracting its first and third indices with the inverse metric gives $\mathrm{Ric} = Kg$ and $R_{\mathrm{sc}} = 2K$. This is intrinsic; no embedding is required.

Definition 17.21. For a surface embedded in $\mathbb{R}^3$ with a local unit normal n, its *principal curvatures* κ_1 and κ_2 at p are the eigenvalues of the shape operator $-\mathrm{d}n_p : T_pM \to T_pM$.

The normal sections in the principal directions have these signed curvatures. Tu's Theorem 12.3, p. 90, identifies intrinsic Gaussian curvature with $K = \kappa_1\kappa_2$.[2]

With outward normals, a unit sphere has both principal curvatures -1. A unit cylinder has principal curvatures -1 and 0. For the upward normal on $z = (x^2 - y^2)/2$, their values at the origin are 1 and -1. Their products give the signs shown in the figure.

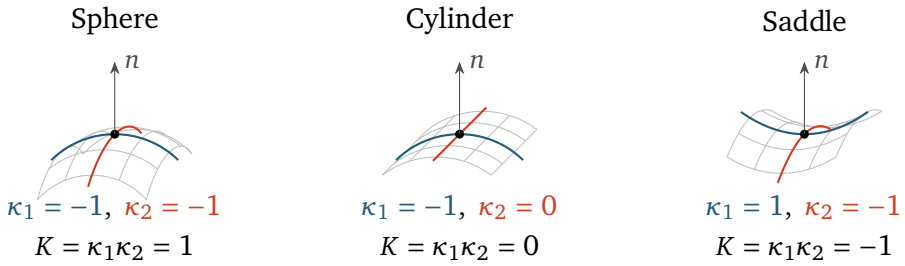


Figure 17.3: Normal sections in principal directions on a sphere, cylinder and saddle. With the indicated normals, their principal-curvature products are 1, 0 and −1.

[2] With our frame action, $\nabla e_1 = -\omega^1{}_2 e_2$ and $\nabla e_2 = \omega^1{}_2 e_1$. His (12.9) then reads $\mathrm{d}\omega^1{}_2 = +K\theta^1 \wedge \theta^2$.

The forms calculations for the sphere and half-plane follow in Examples 17.22 and 17.23. For the torus $\mathbb{R}^2/(2\pi\mathbb{Z})^2$, use the metric induced by the Euclidean plane under the quotient charts. Their transition maps are translations, so the metric agrees on overlaps. Its coordinate coefficients are constant, hence its Christoffel symbols and curvature vanish. The figure shows why the Cartesian frame also descends.

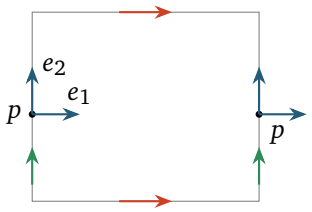


Figure 17.4: Opposite sides are identified by translations. The same Cartesian orthonormal frame matches across both pairs, giving the flat quotient metric.

This metric differs from the metric induced on a doughnut-shaped surface in $\mathbb{R}^3$.

17.6 The sphere and half-plane by forms

Example 17.22 (The sphere in an orthonormal frame)**.** We compute the curvature of the radius-a sphere of Example 17.5 using Cartan's equations. Here θ is the polar angle, while θ^1 and θ^2 are coframe entries, not powers of that angle. The coframe is

$$\theta^1 = a\mathrm{d}\theta, \qquad \theta^2 = a\sin\theta\mathrm{d}\phi.$$

Their derivatives are $\mathrm{d}\theta^1 = 0$ and $\mathrm{d}\theta^2 = a\cos\theta\mathrm{d}\theta\wedge\mathrm{d}\phi$. We solve for the connection coefficients by expanding in coordinate 1-forms. Write $\omega^1{}_2 = p\mathrm{d}\theta + q\mathrm{d}\phi$ for smooth functions p and q. The first structure equation gives

$$0 = -pa\sin\theta\mathrm{d}\theta\wedge\mathrm{d}\phi.$$

Since $a > 0$ and $0 < \theta < \pi$, this forces $p = 0$. The second equation then gives

$$a\cos\theta\mathrm{d}\theta\wedge\mathrm{d}\phi = q\mathrm{d}\phi\wedge a\mathrm{d}\theta = -qa\mathrm{d}\theta\wedge\mathrm{d}\phi.$$

Thus $q = -\cos\theta$, so the unique skew solution is

$$\omega^1{}_2 = -\cos\theta\mathrm{d}\phi.$$

It follows that

$$\mathrm{d}\omega^1{}_2 = \sin\theta\mathrm{d}\theta\wedge\mathrm{d}\phi = \frac{1}{a^2}\theta^1\wedge\theta^2.$$

Uniqueness of the Levi-Civita connection makes this the same curvature. Both routes give $\mathrm{Ric} = g/a^2$ and $R_{\mathrm{sc}} = 2/a^2$. These are smooth tensor identities on a dense coordinate domain, so continuity extends them, and $K = 1/a^2$, across the poles and angular seams.

A unit tangent vector at height y in the half-plane has Euclidean length y. The figure shows the corresponding tangent unit circles at several points.

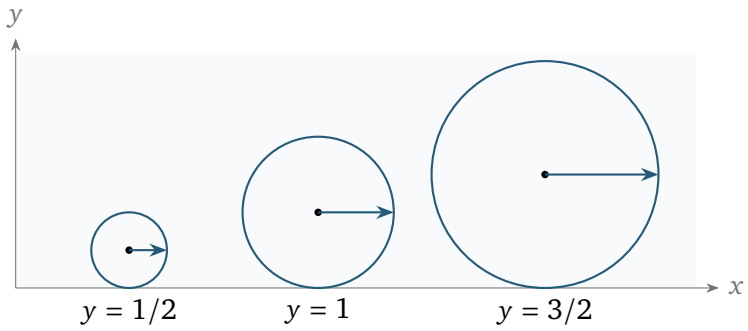


Figure 17.5: Each circle represents unit tangent vectors at its marked base point for $g = (\mathrm{d}x^2 + \mathrm{d}y^2)/y^2$; it is not a metric circle of points.

Example 17.23 (The Poincaré half-plane)**.** On the half-plane $y > 0$, choose $\theta^1 = \mathrm{d}x/y$ and $\theta^2 = \mathrm{d}y/y$. The derivatives and solution of the first equation are

$$\mathrm{d}\theta^1 = \frac{\mathrm{d}x \wedge \mathrm{d}y}{y^2}, \qquad \mathrm{d}\theta^2 = 0, \qquad \omega^1{}_2 = -\frac{\mathrm{d}x}{y}.$$

Taking its derivative gives $\mathrm{d}\omega^1{}_2 = -\mathrm{d}x \wedge \mathrm{d}y/y^2 = -\theta^1 \wedge \theta^2$, so $K = -1$.

The flat quotient torus above has a different metric from an embedded torus. For an embedded torus with major radius b and tube radius r, we use $b > r > 0$ and the parametrization

$$p(u, v) = ((b + r\cos v)\cos u, (b + r\cos v)\sin u, r\sin v).$$

The figure locates the regions of positive and negative curvature.

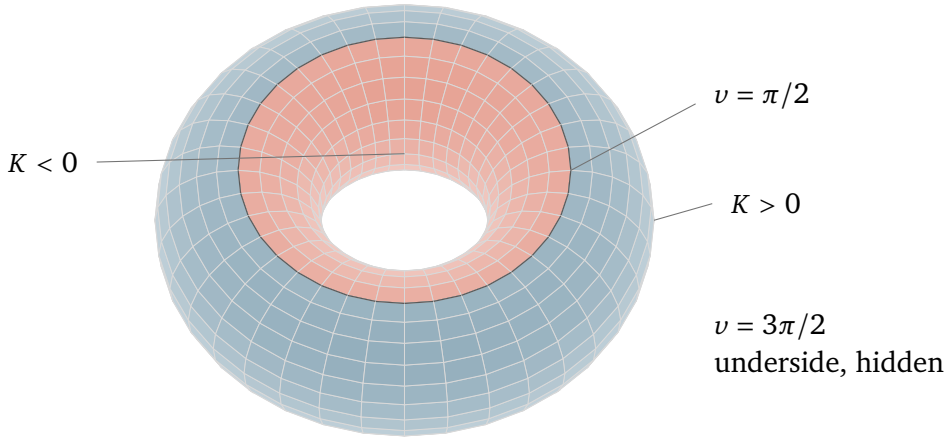


Figure 17.6: For $b = 2$ and $r = 1$, the outer region $\cos v > 0$ has positive curvature, and the inner region $\cos v < 0$ has negative curvature; the circles $v = \pi/2$ and $v = 3\pi/2$ have zero curvature.

Example 17.24 (An embedded torus)**.** Differentiating the parametrization gives the metric and coframe

$$g = r^2\mathrm{d}v \otimes \mathrm{d}v + (b + r\cos v)^2\mathrm{d}u \otimes \mathrm{d}u, \qquad \theta^1 = r\mathrm{d}v, \quad \theta^2 = (b + r\cos v)\mathrm{d}u.$$

We choose the chart orientation (v, u), so $\theta^1 \wedge \theta^2$ is positive. The derivatives are $\mathrm{d}\theta^1 = 0$ and $\mathrm{d}\theta^2 = -r\sin v\mathrm{d}v \wedge \mathrm{d}u$. Writing $\omega^1{}_2 = p\mathrm{d}v + q\mathrm{d}u$ in the first equation gives $p = 0$. The

second gives $-r\sin v = -rq$, hence $q = \sin v$. Therefore

$$\omega^1{}_2 = \sin v\,\mathrm{d}u, \qquad \mathrm{d}\omega^1{}_2 = \cos v\,\mathrm{d}v \wedge \mathrm{d}u, \qquad K = \frac{\cos v}{r(b + r\cos v)}.$$

The denominator is positive because $b > r > 0$. Thus the curvature is positive outside, negative inside and zero on the two marked circles.

The second Bianchi identity

Curvature forms satisfy a differential identity because $\mathrm{d}^2 = 0$. Lecture 19 contracts this identity to obtain a divergence-free tensor in the gravitational field equation. The curvature-form calculation used no metric hypothesis. Thus $\Omega = \mathrm{d}\omega + \omega \wedge \omega$ holds in any local frame for any affine connection. Differentiate a tensor by the connection product rule: $\nabla_X T$ differentiates the output and subtracts a term for the covariant derivative of each input. For curvature, this means

$$\begin{aligned}(\nabla_X R)(Y,Z)W = {} & \nabla_X(R(Y,Z)W) - R(\nabla_X Y, Z)W \\ & - R(Y, \nabla_X Z)W - R(Y,Z)\nabla_X W.\end{aligned}$$

Proposition 17.25 (Second Bianchi identity · *sketched; matrix-form identity proved*)**.** *For the Levi-Civita connection of a pseudo-Riemannian metric,*

$$\mathrm{d}\Omega + \omega \wedge \Omega - \Omega \wedge \omega = 0, \qquad \sum_{\mathrm{cyc}(X,Y,Z)} (\nabla_X R)(Y,Z) = 0.$$

Proof and sketch. The identity $\mathrm{d}^2\omega = 0$ gives $\mathrm{d}\Omega = \mathrm{d}\omega\wedge\omega - \omega\wedge\mathrm{d}\omega$. Substitute $\mathrm{d}\omega = \Omega - \omega\wedge\omega$; the cubic terms cancel. This proves the matrix equation.[3] Evaluating it on three directions and using zero torsion cancels their bracket terms, giving the stated cyclic covariant identity. This sketches the conversion from forms to the covariant expression. ∎

The cylinder's angular chart also gives coordinates on a Euclidean strip. The figure displays the local length-preserving map used in the self-check.

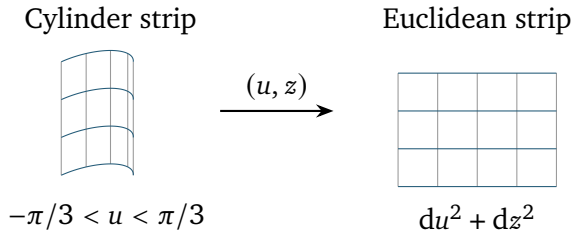


Figure 17.7: An open angular strip on the unit cylinder maps to a Euclidean strip by $(u, z) \mapsto (u, z)$; for radius a, the horizontal coordinate is au.

Self-check. Compute the intrinsic Gaussian curvature of the cylinder of radius $a > 0$.

Answer. The parametrization $p(u, z) = (a\cos u, a\sin u, z)$ induces $g = a^2\mathrm{d}u \otimes \mathrm{d}u + \mathrm{d}z \otimes \mathrm{d}z$. Use $\theta^1 = a\mathrm{d}u$ and $\theta^2 = \mathrm{d}z$ on an angular chart. Both forms are closed, so the skew connection matrix $\omega = 0$ satisfies Cartan's first equation and is the matrix of the unique metric torsion-free connection. Its curvature is zero, hence $K = 0$. The coordinates (au, z) make the metric

[3] Nakahara's (7.147b) states the matrix identity with the same action convention.

Euclidean on each angular chart. This is a local isometry, not a global identification of the cylinder with a plane. The cylinder bends in its embedding but has zero intrinsic curvature.

Self-check. On $\mathbb{R}^2$ with coordinates (x, y) and metric $g = \mathrm{d}x \otimes \mathrm{d}x + (1+x^2)^2 \mathrm{d}y \otimes \mathrm{d}y$, solve for $\omega^1{}_2$ in the coframe $\theta^1 = \mathrm{d}x$ and $\theta^2 = (1+x^2)\mathrm{d}y$. Find K. Repeat for the polar plane with $\theta^1 = \mathrm{d}r$ and $\theta^2 = r\mathrm{d}\phi$, $r > 0$.

Answer. For a coframe $\mathrm{d}x$, $f(x)\mathrm{d}y$ with $f > 0$, write $\omega^1{}_2 = p\mathrm{d}x + q\mathrm{d}y$. The first equation gives $pf = 0$, so $p = 0$. The second gives $f' = -q$. Thus $\omega^1{}_2 = -f'\mathrm{d}y$ and $\mathrm{d}\omega^1{}_2 = -f''\mathrm{d}x \wedge \mathrm{d}y$, so $K = -f''/f$. For $f(x) = 1 + x^2$, these formulas give

$$\omega^1{}_2 = -2x\mathrm{d}y, \qquad K = -\frac{2}{1+x^2}.$$

For the polar plane, $f(r) = r$ gives $\omega^1{}_2 = -\mathrm{d}\phi$ and $K = 0$. A nonzero connection form can therefore have zero curvature.

Summary. Curvature is the tensor $R(X,Y) = [\nabla_X, \nabla_Y] - \nabla_{[X,Y]}$. Metric compatibility and zero torsion give its antisymmetries and first Bianchi identity. For a Riemannian metric, Ricci sums sectional curvatures through a unit direction, and scalar curvature is twice the sum over basis planes. A bi-invariant metric gives $R(X,Y)Z = -\frac{1}{4}[[X,Y],Z]$ on left-invariant fields. In a local orthonormal frame, Cartan's equations replace Christoffel computations by $\mathrm{d}\theta = -\omega \wedge \theta$ and $\Omega = \mathrm{d}\omega + \omega \wedge \omega$. On a surface, $\mathrm{d}\omega^1{}_2 = K\theta^1 \wedge \theta^2$. The sphere gives positive curvature, the half-plane negative curvature, and the flat quotient torus and cylinder zero curvature. Lecture 18 integrates the surface equation to compute parallel transport around loops.

Reading. Tu, *Differential Geometry*, §10–§12, especially Theorems 11.1 and 11.7 and Theorem 12.3, p. 90. Lee, *Riemannian Manifolds: An Introduction to Curvature*, Chapter 7, Propositions 7.1, 7.4 and 7.5. Nakahara, §7.3 and §7.8.

Lecture 18

Holonomy and Gauss–Bonnet

Objectives. *Parallel transport around a loop can rotate a tangent vector. On a surface, the rotation is the enclosed curvature, modulo a full turn. Keeping track of the boundary tangent and its corner turns gives the Gauss–Bonnet formula. Summing that formula over a closed surface relates total curvature to its Euler characteristic. The ideal Foucault pendulum relates latitude holonomy to a slowly rotating swing line.*

By the end of this lecture you will be able to:

- Compute surface holonomy using a frame defined on the enclosed disk.
- Derive the angle excess and the boundary terms in local Gauss–Bonnet.
- Prove global Gauss–Bonnet given the cited topological inputs.
- Compute the latitude dependence of ideal Foucault precession.

18.1 Holonomy on an oriented surface

Let (M, g) be an oriented Riemannian surface with its Levi-Civita connection. Write $\mathrm{d}A$ for its metric area form. The notation $\mathrm{d}A$ names the area form, rather than the exterior derivative of a globally defined one-form A. In a positive orthonormal frame $e = (e_1, e_2)$, recall Lecture 17's convention

$$\nabla e_1 = -\omega^1{}_2 e_2, \qquad \nabla e_2 = \omega^1{}_2 e_1, \qquad \mathrm{d}\omega^1{}_2 = K\mathrm{d}A.$$

Definition 18.1. The positive quarter-turn J is defined by $Je_1 = e_2$ and $Je_2 = -e_1$.

It is independent of the positive orthonormal frame because plane rotations commute with this quarter-turn.

Definition 18.2. For a piecewise smooth loop γ based at p, its *holonomy* is the parallel-transport map of Lecture 16,

$$P_\gamma : T_pM \longrightarrow T_pM,$$

composed along its smooth segments.

Parallel transport preserves the metric and orientation, so holonomy is a rotation. To see the angle and its sign, write a transported unit vector along a framed segment as

$$V = \cos\alpha\, e_1 + \sin\alpha\, e_2, \qquad \frac{DV}{dt} = (\dot\alpha - \omega^1{}_2(\dot\gamma))JV.$$

The derivative follows by the product rule and the two frame actions above. The angle α has a real lift along the interval by Lecture 3's circle lifting theorem. Parallel transport means $\dot\alpha = \omega^1{}_2(\dot\gamma)$. Thus its matrix between the positive frames at the segment endpoints is the

rotation through $\alpha(b) - \alpha(a)$, measured counter-clockwise. The same construction works across finitely many curve segments. The figure illustrates the angle measured relative to a local frame.

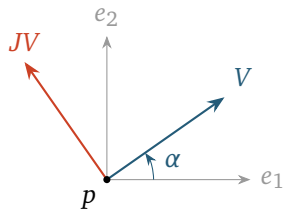


Figure 18.1: A unit vector has angle α from the first frame vector. Parallel transport changes this angle by the connection integral along the curve.

Theorem 18.3 (Surface holonomy · *proved given partitions of unity*)**.** *Let $D \subset M$ be a compact disk with smooth boundary, contained in an open set with one positive orthonormal frame. Parallel transport around ∂D, with its positive boundary orientation, rotates vectors by*

$$\Theta_D \equiv \int_D K \mathrm{d}A \pmod{2\pi}.$$

Proof. The frame agrees with itself when the loop returns to its starting point. Integrating the parallel-angle equation and using Stokes gives

$$\alpha(b) - \alpha(a) = \oint_{\partial D} \omega^1{}_2 = \int_D \mathrm{d}\omega^1{}_2 = \int_D K \mathrm{d}A.$$

A tangent-plane rotation records this real angle only modulo 2π. Linearity of transport extends the unit-vector conclusion to every vector. ∎

Reversing the loop negates the angle. A different positive frame along the loop changes the net lifted angle $\alpha(b) - \alpha(a)$ by an integer multiple of 2π. Indeed, its change-of-frame map is circle-valued, and its lifted angle has endpoint difference $2\pi m$ for some $m \in \mathbb{Z}$ by Theorem 3.11. The rotation is independent of that choice. Stokes requires a frame on the whole disk, not merely near its boundary.

For disks shrinking to p, choose the real holonomy angle tending to zero. The theorem identifies this lift with the curvature integral. Continuity gives

$$\left| \frac{\Theta_D}{\mathrm{Area}(D)} - K(p) \right| \leq \sup_{q \in D} |K(q) - K(p)| \longrightarrow 0.$$

Thus Gaussian curvature is the limiting rotation per unit area. Lecture 17's surface curvature operator is $R(e_1, e_2) = -KJ$. Its negative generates the infinitesimal holonomy rotation. For a positively traversed coordinate square of side ϵ, in coordinates orthonormal at p, the rotation expansion gives

$$P_{\square_\epsilon} = \mathbb{1} + \epsilon^2 K(p) J + O(\epsilon^3) = \mathbb{1} - \epsilon^2 \Omega_{\mathrm{curv}}(\partial_x, \partial_y)_p + O(\epsilon^3).$$

Here Ω_{curv} is Lecture 17's matrix of curvature two-forms, written Ω there. The subscript separates it from the solid angle Ω_D of §18.6. Lecture 21 proves the corresponding small-square formula for a general connection.

18.2 Geodesic triangles and angle excess

Definition 18.4. A *geodesic triangle* bounds an embedded disk by three regular geodesic segments, with nondegenerate corners. Its interior angles α_1, α_2 and α_3 lie in $(0, 2\pi)$.

These interior angles are distinct from the transported-vector angle α.

Definition 18.5. The signed *exterior turn* at a vertex is $\varepsilon_i = \pi - \alpha_i \in (-\pi, \pi)$. It is the change from the incoming to the outgoing tangent, positive counter-clockwise in a positive tangent frame.

Definition 18.6. The *angle excess* is $\alpha_1 + \alpha_2 + \alpha_3 - \pi$.

We use the cited *turning-angle theorem*: for the positive boundary of a disk in one positive frame extending over the disk, the continuous changes of tangent angle along the sides plus the exterior turns add to 2π. See Tu, Theorem 17.4, pp. 142–143. For the unit circle in a Cartesian frame, the positive unit tangent is $(-\sin t, \cos t)$, whose angle increases by 2π. This verifies the circle case. The general theorem, including nonconvex boundaries, remains a cited input. A frame defined only on an annulus does not satisfy its extension hypothesis. The latitude calculation in Example 18.16 compares zero tangent turning in the spherical frame with one full turn in a frame defined over the cap. For a proof in a chart, see Lee, Theorem 9.1 and Lemma 9.2, pp. 158–163.

Theorem 18.7 (Triangle excess · *proved given turning and partitions of unity*)**.** *If a geodesic triangle D lies in one framed neighbourhood as above, then*

$$\alpha_1 + \alpha_2 + \alpha_3 - \pi = \int_D K \mathrm{d}A.$$

Proof. Along each side, the unit tangent is parallel, so its frame angle changes by the integral of $\omega^1{}_2$ on that side. The turning theorem therefore gives

$$\oint_{\partial D} \omega^1{}_2 + \sum_i (\pi - \alpha_i) = 2\pi.$$

Stokes gives $\oint_{\partial D} \omega^1{}_2 = \int_D K \mathrm{d}A$. Rearranging proves the formula. For this piecewise smooth boundary, Stokes follows by rounding each corner in a coordinate ball of radius ϵ_{r}. Smoothness bounds the changed connection integral by $O(\epsilon_{\mathrm{r}})$ and the changed curvature integral by $O(\epsilon_{\mathrm{r}}^2)$. The replacement arcs have length $O(\epsilon_{\mathrm{r}})$, including at concave corners. Apply smooth-boundary Stokes to the rounded disk and take $\epsilon_{\mathrm{r}} \to 0$. ∎

Example 18.8 (Angle excess of the octant)**.** Lecture 17 transported $e_x \in T_N S^2$ around the octant boundary $N \to e_x \to e_y \to N$ in Fig. 17.1. It returned as e_y, a rotation by $+\pi/2$ in the outward orientation. The unit-sphere octant has curvature integral

$$\int_0^{\pi/2} \int_0^{\pi/2} \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi = \frac{\pi}{2}.$$

Its three right angles give the same excess $3\pi/2 - \pi = \pi/2$. This triangle is contained in a framed chart obtained by removing a point outside the octant from the sphere.

18.3 The local Gauss–Bonnet formula

Let $T = \dot\gamma$ be the unit tangent of a smooth boundary segment parametrized by arc length s.

Definition 18.9. The signed *geodesic curvature* of this segment is

$$k_g = g(\nabla_T T, JT).$$

For a positive boundary, JT points into the disk. Differentiating $g(T,T) = 1$ makes its covariant acceleration perpendicular to T. Writing $T = \cos\zeta\, e_1 + \sin\zeta\, e_2$ in the angle derivative already computed gives

$$\nabla_T T = (\zeta' - \omega^1{}_2(T))JT, \qquad k_g = \zeta' - \omega^1{}_2(T).$$

Thus geodesic sides have zero geodesic curvature. The corner diagram distinguishes a jump in the tangent from curvature on a smooth side.

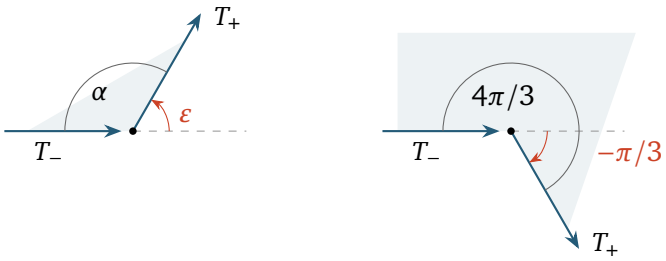


Figure 18.2: The exterior turn ε takes the incoming tangent to the outgoing tangent. The interior angle is $\alpha = \pi - \varepsilon$. The concave panel has $\alpha = 4\pi/3$ and exterior turn $-\pi/3$.

Theorem 18.10 (Local Gauss–Bonnet · *proved given turning and partitions of unity*)**.** *Let D be a compact embedded disk whose positive boundary has finitely many regular smooth sides and nondegenerate corners. Suppose one positive orthonormal frame is defined on a neighbourhood of all of D. Then*

$$\int_D K\,\mathrm{d}A + \oint_{\partial D} k_g\,\mathrm{d}s + \sum_i \varepsilon_i = 2\pi.$$

The side integrals exclude the corners. Their turns occur in the finite sum.

Proof. Integrate $k_g = \zeta' - \omega^1{}_2(T)$ on each side. The turning theorem gives $\sum \Delta\zeta + \sum_i \varepsilon_i = 2\pi$. Summing the side integrals therefore gives

$$\oint_{\partial D} k_g\,\mathrm{d}s = 2\pi - \sum_i \varepsilon_i - \oint_{\partial D} \omega^1{}_2.$$

Use Stokes, with the same corner-rounding limit as above, to replace the last integral by $\int_D K\,\mathrm{d}A$. ∎

Consider a northern unit-sphere cap $0 \le \theta \le a$, where $0 < a < \pi$. Its positive boundary runs in increasing ϕ. Use the unit spherical frame $e_\theta = \partial_\theta$ and $e_\phi = (\sin\theta)^{-1}\partial_\phi$. The unit tangent is $T = e_\phi$ and $JT = -e_\theta$. Lecture 16's symbols give $\nabla_T T = -\cot a\, e_\theta$, so $k_g = \cot a$. The two integrals are

$$\int_D K\,\mathrm{d}A = 2\pi(1 - \cos a), \qquad \oint_{\partial D} k_g\,\mathrm{d}s = \cot a\,(2\pi \sin a) = 2\pi\cos a.$$

Their sum is 2π, with no corner contribution. A smooth frame on this cap is exhibited in the latitude example below.

18.4 Gauss–Bonnet on a closed surface

Theorem 18.11 (Gauss–Bonnet · *proved given cited topology and Stokes inputs*)**.** *Every closed oriented Riemannian surface satisfies*

$$\int_M K \mathrm{d}A = 2\pi\chi(M).$$

Proof. We apply local Gauss–Bonnet to each triangular face and add. Shared-edge integrals cancel, and the angles at each vertex fill one full turn. The remaining count is $V - E + F$. We use a finite chart-contained smooth triangulation, with a compatible Δ-complex structure as in Lecture 6. The construction after this proof justifies these inputs and the hypotheses of the local formula.

Let V, E and F count vertices, edges and faces. Orient each face by the surface orientation and sum the local formula. Every shared edge has opposite boundary directions in its two faces. Reversing T leaves $\nabla_T T$ unchanged and negates JT, so it negates k_g. The two edge integrals cancel. At each vertex, the incident interior angles sum to 2π. Thus the sum of exterior turns over all face corners is $3\pi F - 2\pi V$. The figure displays these cancellations and the complete angle around a vertex.

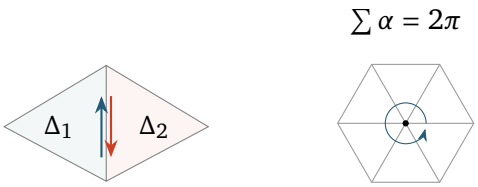


Figure 18.3: Adjacent faces traverse a shared edge in opposite directions. The incident face angles at an interior vertex fill one full turn.

The summed formula gives

$$\int_M K \mathrm{d}A = 2\pi F - (3\pi F - 2\pi V) = 2\pi V - \pi F = 2\pi(V - E + F).$$

The last equality uses $3F = 2E$, since each triangular side belongs to two faces on a surface without boundary. By Theorems 6.13 and 6.14, $V - E + F = \chi(M)$. For disconnected M, apply the argument to each component and add. ∎

Triangulations and the local hypotheses. Use the cited existence of a finite smooth triangulation of a compact surface (Lee, discussion before Theorem 9.7, pp. 166–167). In Lee's definition, each curved triangle and its enclosed region lie in one coordinate chart, with regular sides and no cusps (p. 162). We also use the cited planar *Schoenflies theorem*: a simple closed plane curve and its enclosed region form a closed topological disk (Stillwell, §0.3.9, pp. 35–36). Gram–Schmidt on the connected chart neighbourhood of the face gives a positive orthonormal frame, changing one vector's sign if needed. Thus each closed face is a disk to which the local formula applies. The edges need not be geodesics.

To obtain the required Δ-complex, order the vertices and choose one parametrization for each edge, from its lower to its higher vertex. These give a homeomorphism from the boundary of each standard triangle to its face boundary. Identify the standard triangle with a closed unit disk by radially rescaling rays from its centroid. Schoenflies supplies a disk homeomorphism. It can be chosen to match that boundary map. Indeed, any difference

on the boundary circle is a homeomorphism $\psi\colon S^1 \to S^1$, which extends to the disk by $r\zeta \mapsto r\psi(\zeta)$, with the centre fixed. Composing with this extension matches the chosen boundary map. The face maps now agree on shared edges and give a finite Δ-complex structure on M.

The integral does not change when the metric changes, although the pointwise Gaussian curvature and the area form both change. A sphere therefore has total curvature 4π for every Riemannian metric. A torus has total curvature zero for every Riemannian metric, even when its curvature is not identically zero. For the closed oriented genus-g surface Σ_g, Lecture 6 gives $\chi(\Sigma_g) = 2 - 2g$. Hence

$$\int_{\Sigma_g} K\,\mathrm{d}A = 4\pi(1-g).$$

The figure compares an embedded torus with a surface homeomorphic to the sphere. The colours distinguish regions of positive and negative Gaussian curvature.

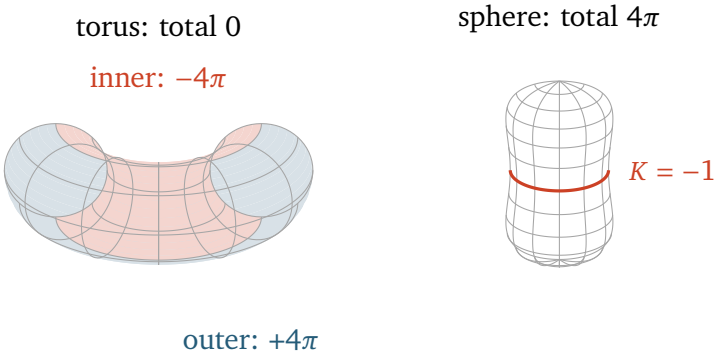


Figure 18.4: For the cutaway torus with $b = 2$ and $r = 1$, the outer and inner halves contribute $+4\pi$ and -4π. The surface with a waist has total curvature 4π and curvature -1 at its equator.

Example 18.12 (Curvature integrals on the embedded torus)**.** We use the torus of Example 17.24, with $b > r > 0$ and its chart orientation (v, u). Its positive area form is $\mathrm{d}A = r(b + r\cos v)\mathrm{d}v \wedge \mathrm{d}u$. That example found the Gaussian curvature. Multiplying by the area form gives

$$K = \frac{\cos v}{r(b + r\cos v)}, \qquad K\mathrm{d}A = \cos v\,\mathrm{d}v \wedge \mathrm{d}u.$$

Integrating over $-\pi/2 < v < \pi/2$ and one full u-circle gives $+4\pi$. Integrating over $\pi/2 < v < 3\pi/2$ and one full u-circle gives -4π. The two contributions cancel, although neither vanishes separately.

For the sphere panel, we revolve the meridian $(f(t), z(t))$, given by

$$f(t) = \sin t\big(1 - \frac{1}{2}\sin^2 t\big), \qquad z(t) = \cos t, \qquad 0 \le t \le \pi.$$

The surface is smooth at the two poles and has no self-intersections. To compute its curvature, put $h = \sqrt{f'^2 + z'^2}$. On $0 < t < \pi$, the coframe $(h\mathrm{d}t, f\mathrm{d}u)$ gives $\omega^1{}_2 = -(f'/h)\mathrm{d}u$. Thus

$$K = -\frac{1}{hf}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{f'}{h}\right).$$

At the waist $t = \pi/2$, the values are $f = 1/2$, $f' = 0$, $f'' = 1/2$ and $h = 1$, so $K = -1$. Gauss–Bonnet still gives total curvature 4π because the surface is homeomorphic to S^2.

18.5 The Euler number of the tangent sphere

Recall from §14.5 that the index of an isolated tangent-field zero is the winding of its normalized components on a small positive circle. The cited Poincaré–Hopf theorem, Theorem 14.9, states that the sum of these indices on a closed oriented surface is its Euler characteristic. In particular this sum is independent of the chosen field with isolated zeros.

Definition 18.13. For the tangent bundle of S^2, call this common sum its *Euler number*.

Corollary 18.14 (The tangent-sphere Euler number · *sketched given Poincaré–Hopf*)**.** *The Euler number of TS^2 is $\chi(S^2) = 2$.*

The field $X(p) = e_3 - p_3 p$ from Lecture 14 has its two zeros at the poles, with index +1 at each. It supplies a field to which the definition applies. Poincaré–Hopf makes every such index sum equal to 2. A nowhere-zero tangent field would have empty index sum zero, contradicting this value. Hence there can be no global tangent frame on the sphere. A disk in one coordinate chart has a framed neighbourhood by Gram–Schmidt. The local curvature formula requires that the chosen frame extend over that disk, rather than only around its boundary. The whole sphere admits no such frame. We can also compare the index sum directly with the curvature integral. Let X be a smooth tangent field on a closed oriented surface, with all zeros isolated. Its zero set is closed and discrete, hence finite by compactness. Choose disjoint small disks D_i around its zeros p_i, with positive orthonormal frames f_i defined over the disks. On $M' = M \setminus \bigcup_i \operatorname{int}(D_i)$ the vectors $e_1 = X/|X|$ and $e_2 = Je_1$ give a positive orthonormal frame. Write ω_X for its connection entry. Each boundary component of M' has the opposite orientation to ∂D_i. Stokes therefore gives

$$\int_{M'} K \mathrm{d}A = -\sum_i \oint_{\partial D_i} \omega_X .$$

Near ∂D_i, let ψ_i be a local lift of the angle from $f_{i,1}$ to e_1. Differentiating this frame rotation gives $\omega_X = \omega_{f_i} - \mathrm{d}\psi_i$. The angle differential agrees on overlapping lifts. Its integral is $2\pi \operatorname{ind}_{p_i} X$ by the definition of index. Stokes in the disk frame now gives

$$\oint_{\partial D_i} \omega_X = \int_{D_i} K \mathrm{d}A - 2\pi \operatorname{ind}_{p_i} X .$$

Adding the disk integrals to the punctured-surface integral yields

$$\frac{1}{2\pi} \int_M K \mathrm{d}A = \sum_i \operatorname{ind}_{p_i} X .$$

Thus the Euler number of TS^2 is also its total curvature divided by 2π. Together with Gauss–Bonnet, this argument proves the surface case of Lecture 14's cited Poincaré–Hopf theorem. Lecture 22 expresses the same integer using changes between local frames.

18.6 Latitude holonomy and the Foucault pendulum

Definition 18.15. The *solid angle* of a region on the unit sphere is its area.

Let the latitude be $\lambda \in (-\pi/2, \pi/2)$ and put $a = \pi/2 - \lambda$. Follow its circle once in increasing ϕ.

Example 18.16 (Latitude rotation in a cap frame)**.** Recall the transport around this latitude from Example 16.22. Its angle change in the spherical frame was $-2\pi\cos a = -2\pi\sin\lambda$. The northern cap's curvature integral is its solid angle,

$$\Omega_D = 2\pi(1-\cos a) = 2\pi(1-\sin\lambda).$$

This differs from Lecture 16's angle by 2π and represents the same holonomy. The spherical frame itself is undefined at the north pole, so Stokes cannot be applied to that frame across the cap.

To use a frame on the full cap, define the rotated vectors by

$$e_{N,1} = \cos\phi\, e_\theta - \sin\phi\, e_\phi, \qquad e_{N,2} = \sin\phi\, e_\theta + \cos\phi\, e_\phi.$$

We substitute $e_\theta = (\cos\theta\cos\phi, \cos\theta\sin\phi, -\sin\theta)$ and $e_\phi = (-\sin\phi, \cos\phi, 0)$. The identity

$$\frac{1-\cos\theta}{\sin^2\theta} = \frac{1}{1+\cos\theta}$$

removes the apparent singularity at the north pole. In Cartesian coordinates $p = (x, y, z)$ on the unit sphere, the columns are

$$E_N = \begin{pmatrix} 1 - x^2/(1+z) & -xy/(1+z) \\ -xy/(1+z) & 1 - y^2/(1+z) \\ -x & -y \end{pmatrix}.$$

The rotation makes it tangent, positive and orthonormal away from the poles. The displayed expression is smooth for $z > -1$ and at the north pole gives e_x and e_y. Continuity therefore preserves those properties there. Differentiating the rotated frame and using Lecture 17's spherical-frame form $\omega^1{}_2 = -\cos\theta\mathrm{d}\phi$ gives

$$\omega_N = \omega^1{}_2 + \mathrm{d}\phi = (1-\cos\theta)\mathrm{d}\phi = \frac{x\mathrm{d}y - y\mathrm{d}x}{1+z}.$$

This is smooth at the north pole. Its integral around the latitude is $2\pi(1-\cos a)$, so Stokes in this frame gives the cap's curvature integral. At the equator the holonomy is zero modulo 2π, while the cap integral is 2π. Holonomy alone does not retain that full turn.

On a positive latitude boundary, the unit tangent is $T = e_\phi$. Its components in the spherical frame are constant, so its tangent angle has total change zero. This frame does not extend to the north pole. In the smooth north frame, the same tangent is

$$T = -\sin\phi\, e_{N,1} + \cos\phi\, e_{N,2}.$$

Its lifted angle is $\phi + \pi/2$, with total change $+2\pi$. This verifies the extension hypothesis in the turning-angle theorem for this cap. The dials in the figure compare the tangent components in these two frames.

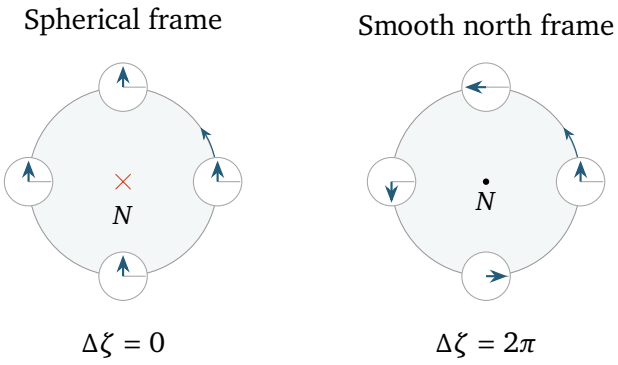


Figure 18.5: The same positively traversed latitude has constant tangent components in the singular spherical frame and one positive turn in the smooth north frame.

Physical input. We use the ideal small-swing pendulum model on a slowly rotating sphere. For the sphere's angular velocity vector $\boldsymbol{\nu}_\oplus$, the rotating-frame Coriolis acceleration is $-2\boldsymbol{\nu}_\oplus \times v$. For a small horizontal displacement ξ, the restoring acceleration is $-\nu_0^2\xi$, where $\nu_0 > 0$ is the natural angular frequency. The model retains terms linear in the horizontal displacement and the rotation rate. It neglects damping, anisotropy, centrifugal corrections and vertical motion at this order.[1]

Let $\nu_\oplus > 0$ be the sphere's angular rotation rate, with $\nu_\oplus \ll \nu_0$. In a positive east–north frame, write the small displacement as $\xi = \xi_E + i\xi_N$. Let n be the outward unit normal. The angular velocity decomposes as

$$\boldsymbol{\nu}_\oplus = (\boldsymbol{\nu}_\oplus \cdot n)n + \boldsymbol{\nu}_{\oplus,\tan}, \qquad b = \boldsymbol{\nu}_\oplus \cdot n = \nu_\oplus \sin\lambda.$$

For horizontal v, the cross product of $\boldsymbol{\nu}_{\oplus,\tan}$ with v is normal. Also $n \times v = Jv$ in the positive east–north frame. Hence the horizontal projection is

$$(-2\boldsymbol{\nu}_\oplus \times v)_{\tan} = -2bJv.$$

The figure shows the normal component and the resulting acceleration.

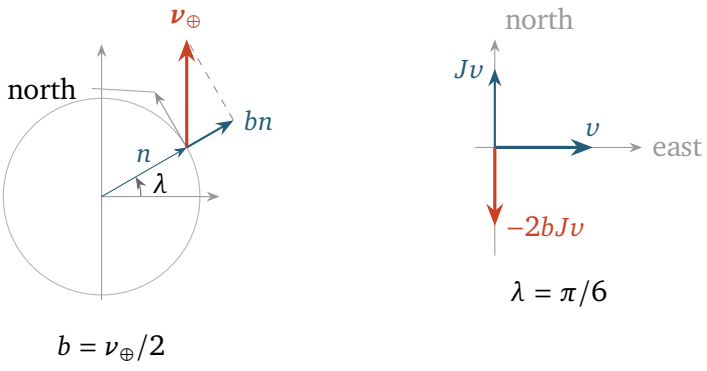


Figure 18.6: At latitude $\lambda = \pi/6$, the axis has normal component $\nu_\oplus/2$. In the positive east–north plane, an eastward velocity has southward Coriolis acceleration.

[1] See Fitzpatrick, *Newtonian Dynamics*, Foucault Pendulum, for the linear Coriolis model, and Dray, *The Foucault Pendulum* (1998), for its slowly changing direction and tangential-projection interpretation.

Multiplication by i represents J on $\xi_E + i\xi_N$. Adding restoring and Coriolis accelerations gives

$$\ddot{\xi} + 2ib\dot{\xi} + \nu_0^2 \xi = 0.$$

Substituting $\xi = e^{-ibt} w$ cancels the first-derivative terms and gives

$$\ddot{w} + (\nu_0^2 + b^2) w = 0.$$

The b^2 term is an exact consequence of solving this linear model. It does not account for all physical corrections of second order in rotation. An oscillation along a fixed line in the w plane therefore has slow orientation angle $-bt$. Its rate agrees with parallel transport:

$$\dot{\alpha} = -\cos a \, \dot{\phi} = -\nu_\oplus \sin \lambda = -b.$$

Over one rotation $2\pi/\nu_\oplus$, the change is $-2\pi \sin \lambda$. It is clockwise in the northern hemisphere and counter-clockwise in the southern. At latitude $\lambda = \pi/6$, the model predicts $-\pi$ in one rotation of the sphere. The oriented direction reverses while the swing line returns.

A geometric heuristic interprets the slowly varying swing direction after averaging over its fast oscillations. In this idealization, no force turns the swing line within the moving horizontal plane. Its ambient change is only the normal change needed to stay tangent to the sphere. By Lecture 16's tangential-projection description, this is parallel transport. The Coriolis calculation above supplies the rate within the stated model. An oriented swing direction records angles modulo 2π. The physical unoriented swing line records them modulo π. For a spin-1/2 ground state aligned with a field direction traversing this latitude, Lecture 24 derives the Berry phase $-\Omega_D/2$ modulo 2π.

Self-check. Check global Gauss–Bonnet for the round sphere and the flat quotient torus. Decide whether zero total curvature forces pointwise zero curvature.

Answer. On a sphere of radius R, curvature $1/R^2$ times area $4\pi R^2$ gives $4\pi = 2\pi\chi(S^2)$. On the flat torus $K = 0$, so the integral is $0 = 2\pi\chi(T^2)$, using $\chi(T^2) = 0$ from Lecture 6. Zero total curvature does not force pointwise zero curvature. The embedded torus above has positive outer curvature and negative inner curvature, whose integrals are $+4\pi$ and -4π.

Self-check. On the unit sphere, take the sector $0 \le \theta \le \arccos(2/5)$ and $0 \le \phi \le 3\pi/5$. Compute its curvature integral, its smooth-side geodesic-curvature integral and its total exterior turn. Verify local Gauss–Bonnet and give its holonomy angle.

Answer. Put $a = \arccos(2/5)$. The area integral is $(3\pi/5)(1 - 2/5) = 9\pi/25$. The meridians have zero geodesic curvature. On the latitude arc, the integral is

$$\int_0^{3\pi/5} \cot a \sin a \, d\phi = \frac{3\pi}{5}\frac{2}{5} = \frac{6\pi}{25}.$$

The interior angles are $3\pi/5$ at N and $\pi/2$ at each latitude endpoint. Thus the exterior turns sum to $2\pi/5 + \pi/2 + \pi/2 = 7\pi/5$. Their sum with the two integrals is 2π. The holonomy is $9\pi/25$ modulo 2π by the curvature integral.

Self-check. State in one sentence what holonomy is and why we introduced it. Explain why a holonomy rotation alone cannot recover the full curvature integral.

Answer. Holonomy is parallel transport around a based loop as a map $P_\gamma : T_pM \to T_pM$, introduced to compare a vector with its return and measure curvature through the resulting

rotation. A rotation retains only its angle modulo 2π. For example, an equator has identity holonomy although its northern cap has curvature integral 2π.

Summary. Surface transport has angle $\int_D K\mathrm{d}A$ modulo 2π when one positive orthonormal frame covers the disk. The turning theorem adds the boundary tangent and corner turns, giving local Gauss–Bonnet. Using the cited topology inputs and summing over the triangulation yields $\int_M K\mathrm{d}A = 2\pi\chi(M)$ on every closed oriented Riemannian surface. The tangent sphere has Euler number 2 by the cited Poincaré–Hopf theorem. Latitude transport gives the ideal Foucault angle $-2\pi\sin\lambda$ modulo 2π. Lecture 19 uses curvature and geodesics in a Lorentzian spacetime.

Reading. Tu, *Differential Geometry*, §17, especially Theorems 17.4–17.5 and 17.9, pp. 142–146. Lee, *Riemannian Manifolds: An Introduction to Curvature*, Ch. 9, pp. 158–168, including the smooth triangulation discussion. Stillwell, *Classical Topology and Combinatorial Group Theory*, §0.3.9, pp. 35–36 (Schoenflies theorem).

Lecture 19

Lorentzian geometry and general relativity

Objectives. *A Lorentzian metric specifies elapsed time and the paths of freely falling particles. Einstein's equation relates its curvature to matter. We recover Newton's acceleration in a weak field and verify the Schwarzschild vacuum metric. Its conserved quantities then reduce a massive particle's motion to an orbit equation.*

By the end of this lecture you will be able to:

- Distinguish timelike, null and spacelike tangent vectors in signature $+---$.
- Recover Newton's equation from a weak static metric and slow motion.
- Explain the action and conservation statements in Einstein's equation.
- Verify Schwarzschild vacuum curvature and derive the massive orbit equation.

Unit 19A Motion in a given metric

19.1 Lorentzian metrics and proper time

Definition 19.1 (Lorentzian spacetime)**.** A *Lorentzian spacetime* is a smooth four-dimensional manifold with a metric of signature $(+,-,-,-)$.

At each point an orthonormal frame identifies the metric with $\eta = \operatorname{diag}(1,-1,-1,-1)$. This recalls the flat interval in *Symmetries in Physics*, Lecture 17, *Minkowski space from physics*. Here the metric may vary from point to point. Assume the spacetime is oriented, so its metric volume form is defined.

Definition 19.2 (Causal type and time orientation)**.** A nonzero tangent vector v is *timelike* if $g(v,v) > 0$, *null* if $g(v,v) = 0$, and *spacelike* if $g(v,v) < 0$. The null vectors form the *light cone*. A *time orientation* is a continuous choice of one of the two timelike cones as future.

We assume such a choice. In an orthonormal frame adapted to this choice, future vectors have positive time component. The figure shows one space direction and the time direction.

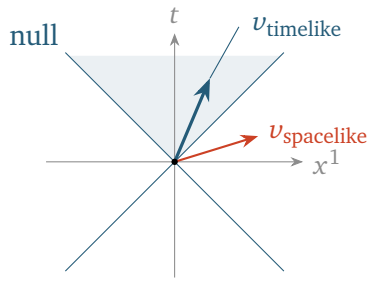


Figure 19.1: The null lines bound the future timelike cone in a tangent plane. The classification uses the metric at the point, rather than the drawing's Euclidean angles.

We use $c = 1$ and coordinates $x^0 = t$.

Definition 19.3 (Proper time and four-velocity)**.** The *proper time* along a future timelike curve $x(\sigma)$ is defined by

$$\tau(\sigma_1) - \tau(\sigma_0) = \int_{\sigma_0}^{\sigma_1} \sqrt{g_{\mu\nu} \frac{\mathrm{d}x^\mu}{\mathrm{d}\sigma} \frac{\mathrm{d}x^\nu}{\mathrm{d}\sigma}}\, \mathrm{d}\sigma.$$

Its *four-velocity* is $U = \mathrm{d}x/\mathrm{d}\tau$.

The definition gives $g(U, U) = 1$.

Physical input. The physical free-fall postulate says that an unforced test particle follows a timelike geodesic of the Levi-Civita connection. A light ray follows a null geodesic, using an affine parameter in place of proper time. These laws describe unforced structureless test bodies and light in the geometric-optics approximation. Universal free fall motivates assigning test-body trajectories to spacetime geometry. It does not derive the metric geodesic law. An ideal clock measures proper time along its worldline.

The existence and equation of these geodesics are the mathematical results of Lecture 16.

19.2 Newton's equation from a weak metric

Let $\Phi = \epsilon \varphi_{\text{pot}}(x^1, x^2, x^3)$ be a smooth static potential, where ϵ is a small parameter. Derivatives of Φ count as first order in ϵ. We write the metric as $\mathrm{d}s^2 = g_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu$. Products denote symmetric tensor products, so $\mathrm{d}t^2 = \mathrm{d}t \otimes \mathrm{d}t$ and $2\mathrm{d}v\,\mathrm{d}r = \mathrm{d}v \otimes \mathrm{d}r + \mathrm{d}r \otimes \mathrm{d}v$. In spatial Cartesian coordinates consider

$$\mathrm{d}s^2 = (1 + 2\Phi)\mathrm{d}t^2 - (1 - 2\Phi) \sum_{i=1}^{3} (\mathrm{d}x^i)^2.$$

Assume $|\Phi| < 1/2$, so the signature is Lorentzian. A clock at fixed spatial coordinates has rate

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = \sqrt{1 + 2\Phi} = 1 + \Phi + O(\epsilon^2).$$

Thus a clock at a lower potential accumulates less proper time over the same coordinate-time interval. The time coefficient determines the leading slow-motion acceleration. The spatial factor also enters curvature and the field equation in §19.3. We define the Euclidean spatial gradient by

$$\nabla_E \Phi = (\partial_1 \Phi, \partial_2 \Phi, \partial_3 \Phi).$$

The figure shows a trajectory in its leading Newtonian approximation.

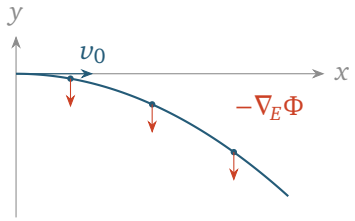


Figure 19.2: For a locally uniform downward acceleration, the leading Newtonian trajectory is $x = v_0 t$, $y = -at^2/2$, with $v_0 > 0$ and $a = |\nabla_E \Phi| > 0$. This curve represents the weak-field, slow-motion limit.

Proposition 19.4 (Weak-field free fall · *proved*)**.** *For a future timelike geodesic with small coordinate speed, the coordinate-time acceleration to first order in the potential and lowest order in speed is*

$$\frac{\mathrm{d}^2 x^i}{\mathrm{d}t^2} = -\partial_i \Phi, \qquad i = 1, 2, 3.$$

Proof. We use an affine parameter λ and write dots for its derivatives. Put $v^\mu = \mathrm{d}x^\mu/\mathrm{d}t$, so $v^0 = 1$. A timelike tangent has nonzero time component in this metric, since a nonzero tangent with zero time component has negative metric square. The chain rule and the affine time equation give

$$\ddot{x}^i = \frac{\mathrm{d}^2 x^i}{\mathrm{d}t^2}\dot{t}^2 + v^i \ddot{t}, \qquad \ddot{t} = -\Gamma^0_{\mu\nu} v^\mu v^\nu \dot{t}^2.$$

We substitute these identities in the spatial geodesic equation and divide by $\dot{t}^2$. This gives

$$\frac{\mathrm{d}^2 x^i}{\mathrm{d}t^2} = -\Gamma^i_{\mu\nu} v^\mu v^\nu + v^i \Gamma^0_{\mu\nu} v^\mu v^\nu.$$

The Christoffel formula gives

$$\Gamma^i_{00} = \frac{\partial_i \Phi}{1 - 2\Phi} = \partial_i \Phi + O(\epsilon^2), \qquad \Gamma^0_{0i} = \frac{\partial_i \Phi}{1 + 2\Phi}.$$

The static diagonal metric has $\Gamma^0_{00} = \Gamma^0_{ij} = \Gamma^i_{0j} = 0$. Write $\mathbf{v} = (v^1, v^2, v^3)$ for the coordinate spatial velocity. All remaining terms in the spatial equation contain at least two spatial velocity factors. The acceleration is $-\partial_i \Phi + O(\epsilon^2) + O(\epsilon |\mathbf{v}|^2)$, which proves the stated approximation. ∎

Physical input. For an attractive Newtonian point source of mass $m > 0$, $\Phi = -G_N m/r$, where $G_N > 0$ is Newton's constant. For static mass density ρ, Newton's field law is $\nabla_E^2 \Phi = 4\pi G_N \rho$.

Its negative gradient points toward the source.

Two nearby freely falling particles reveal the second derivatives of the potential. Let ξ^i be their infinitesimal separation in the weak static field. Subtracting their leading Newtonian equations and linearizing in separation gives

$$\ddot{\xi}^i = -\partial_i \partial_j \Phi \, \xi^j.$$

Here dots denote coordinate-time derivatives. Lecture 17's curvature formula gives, to first order in ϵ,

$$R^i{}_{0j0} = \partial_j \Gamma^i_{00} = \partial_i \partial_j \Phi, \qquad \ddot{\xi}^i = -R^i{}_{0j0} \xi^j.$$

These are tidal relative accelerations in the static weak-field, slow-motion approximation. Taking the trace gives $\mathrm{Ric}_{00} = \nabla_E^2\Phi$ at this order. For an initially comoving small ball of particles with volume V, the separation map at short elapsed time Δt is $\mathbf{1} - \frac{1}{2}(\Delta t)^2\,\mathrm{Hess}_E\,\Phi$ to second order. Differentiating its determinant at $\Delta t = 0$ gives

$$\left.\frac{\dot V}{V}\right|_{\Delta t=0} = 0, \qquad \left.\frac{\ddot V}{V}\right|_{\Delta t=0} = -\nabla_E^2\Phi.$$

The ball can deform even when this trace vanishes.

For the exterior potential $-G_N m/r$, the Hessian eigenvalues are $-2G_N m/r^3$ radially and $G_N m/r^3$ in each transverse direction. The relative acceleration therefore stretches the ball radially and squeezes it transversely. For comparison, a uniform-density Newtonian model has local potential $\Phi = 2\pi G_N \rho r^2/3 + \text{constant}$. Its Hessian is $(4\pi G_N\rho/3)\mathbf{1}$. For the drawing, we evaluate the vacuum map at $\Delta t\sqrt{G_N m/r^3} = 1/2$ and the matter map at $\Delta t\sqrt{4\pi G_N\rho/3} = 1/2$. The respective axis factors are $(5/4, 7/8, 7/8)$ and $(7/8, 7/8, 7/8)$. The figure compares these second-order separation maps.

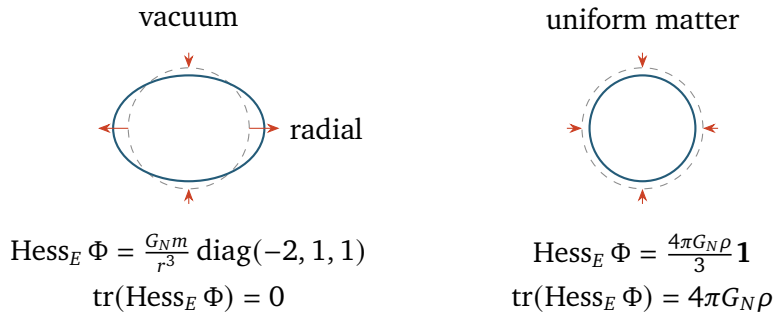


Figure 19.3: Dashed circles show initially comoving balls, solid curves their second-order separation maps, and arrows the relative acceleration in the weak static slow-motion approximation. The Hessian trace is zero in vacuum and $4\pi G_N\rho$ in uniform matter.

Unit 19B Einstein's equation and matter

19.3 Einstein's equation and its conservation identity

The given weak metric predicts motion. A field law must also constrain the metric from the energy and momentum of matter.

Definition 19.5 (Stress–energy tensor)**.** The symmetric *stress–energy tensor* T describes energy density, momentum density and their fluxes. In an orthonormal observer frame with $c = 1$, its upper components have the meanings

T^{00}	energy density
$T^{0i} = T^{i0}$	energy flux in direction i and momentum density
T^{ij}	flux of i-momentum across a j-normal face.

Stress includes pressure, the transport of normal momentum across a face. Lowering with signature $+---$ gives $T_{00} = T^{00}$, $T_{0i} = -T^{0i}$ and $T_{ij} = T^{ij}$. The figure fixes the face normal used to read a flux component.

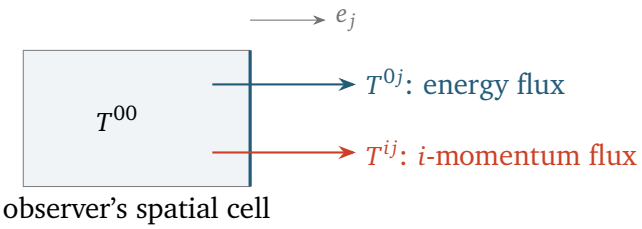


Figure 19.4: An observer's cell stores energy density T^{00}. Across its face of outward normal e_j, energy flux is T^{0j} and the flux of i-momentum is T^{ij}, in an orthonormal frame with $c = 1$.

Physical input. For pressureless matter (dust), the model is

$$T_{\mu\nu} = \rho U_\mu U_\nu, \qquad g(U,U) = 1.$$

Here ρ is rest energy density and U is the matter four-velocity. In its rest frame the stress tensor is $\operatorname{diag}(\rho, 0, 0, 0)$.

Recall Lecture 17's definitions $\mathrm{Ric}_{\mu\nu} = R^\rho{}_{\mu\rho\nu}$ and $R_{\mathrm{sc}} = g^{\mu\nu}\,\mathrm{Ric}_{\mu\nu}$.

Definition 19.6 (Einstein tensor)**.** The *Einstein tensor* is

$$G_{\mu\nu} = \mathrm{Ric}_{\mu\nu} - \frac{1}{2} R_{\mathrm{sc}} g_{\mu\nu}.$$

Physical input. We use Einstein's field law in four dimensions with zero cosmological constant:

$$G_{\mu\nu} = 8\pi G_N T_{\mu\nu}.$$

Conservation and the Newtonian limit motivate and check this choice. They do not prove that it is the unique possible gravitational field law.

The conservation identity follows from the covariant second Bianchi identity of Proposition 17.25, whose conversion from forms was sketched. We write $\nabla_\lambda T_{\dots} = (\nabla_{\partial_\lambda} T)_{\dots}$ and $\nabla^\mu = g^{\mu\lambda}\nabla_\lambda$. We contract the identity as follows. Use the lowered tensor $R_{ab\mu\nu}$ of Lecture 17. Metric compatibility and the two pair antisymmetries give

$$\begin{aligned} 0 &= g^{a\mu} g^{b\lambda} \left(\nabla_\lambda R_{ab\mu\nu} + \nabla_\mu R_{ab\nu\lambda} + \nabla_\nu R_{ab\lambda\mu} \right) \\ &= 2\nabla^\mu \,\mathrm{Ric}_{\mu\nu} - \partial_\nu R_{\mathrm{sc}}. \end{aligned}$$

The three contractions are respectively $\nabla^\mu \,\mathrm{Ric}_{\mu\nu}$, $\nabla^\mu \,\mathrm{Ric}_{\mu\nu}$ and $-\partial_\nu R_{\mathrm{sc}}$. Thus $\nabla^\mu G_{\mu\nu} = 0$. Einstein's equation therefore implies $\nabla^\mu T_{\mu\nu} = 0$. This is local covariant conservation. It alone does not assert a globally conserved energy on an arbitrary spacetime. For a Killing field K, the current $J_\nu = T_{\mu\nu} K^\mu$ obeys

$$\nabla^\nu J_\nu = T_{\mu\nu} \nabla^\nu K^\mu = 0.$$

The last equality contracts a symmetric tensor with the antisymmetric Killing derivative from Lecture 16. This supplies a conserved flux through hypersurfaces when the boundary flux vanishes and the integrals exist.

For dust, the product rule expands conservation as

$$0 = \nabla^\mu (\rho U_\mu U_\nu) = U_\nu \nabla_\mu (\rho U^\mu) + \rho U^\mu \nabla_\mu U_\nu.$$

Contracting with U^ν removes the acceleration term because metric compatibility gives $U^\nu U^\mu \nabla_\mu U_\nu = \frac{1}{2} U^\mu \nabla_\mu g(U,U) = 0$. Thus $\nabla_\mu(\rho U^\mu) = 0$. Where $\rho > 0$, substitution back gives $\nabla_U U = 0$. The dust conservation equation therefore yields both rest-density continuity and geodesic matter motion.

For the weak metric above, the additional first-order symbols are

$$\Gamma^i_{jk} = -\delta^i_k \partial_j \Phi - \delta^i_j \partial_k \Phi + \delta_{jk} \partial_i \Phi, \qquad \Gamma^\mu_{\mu i} = -2\partial_i \Phi.$$

Here and in the next calculation terms of order ϵ^2 are omitted. The quadratic Christoffel terms do not contribute at first order. Staticity removes time derivatives and gives $\mathrm{Ric}_{0i} = 0$. Thus $\mathrm{Ric}_{ij} = \partial_k \Gamma^k_{ji} - \partial_j \Gamma^\mu_{\mu i}$: the mixed second derivatives cancel, leaving $\delta_{ij} \sum_k \partial_k^2 \Phi$. The full contraction gives

$$\mathrm{Ric}_{00} = \nabla_E^2 \Phi, \qquad \mathrm{Ric}_{ij} = \delta_{ij} \nabla_E^2 \Phi, \qquad R_{\mathrm{sc}} = -2\nabla_E^2 \Phi, \qquad G_{00} = 2\nabla_E^2 \Phi.$$

Here $\nabla_E^2 = \sum_i \partial_i^2$ is the Euclidean spatial Laplacian. For example, at this order $\mathrm{Ric}_{00} = \partial_i \Gamma^i_{00}$. The 00 field equation for slowly moving matter gives $\nabla_E^2 \Phi = 4\pi G_N \rho$, the positive Newtonian Poisson equation. This checks both the sign and the coefficient of the field law.

The spatial perturbation in the weak metric contributes to this check. If we retained $g_{00} = 1 + 2\Phi$ but set $g_{ij} = -\delta_{ij}$, the linearized Einstein component G_{00} would instead vanish. The leading slow-motion acceleration alone does not determine the spatial metric.

19.3.1 The variational form

We take action integrals on a relatively compact coordinate domain. All variations vanish near its boundary. Let S_{m} be a matter action and let $\mu = \sqrt{|\det g|}\, \mathrm{d}t \wedge \mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3$ in positive coordinates.

Physical input. We normalize the matter action so that its response to compactly supported inverse-metric variations, with matter fixed, is

$$\delta S_{\mathrm{m}} = \frac{1}{2} \int T_{\alpha\beta}\, \delta g^{\alpha\beta} \mu.$$

This identifies the variational tensor with the physical stress–energy tensor.

Write $F_{\mu\nu}$ for the components of F_{em} in this paragraph.

Physical input. For the field of Lecture 15, we use the normalized Maxwell action

$$S_{\mathrm{em}} = -\frac{1}{2} \int F_{\mathrm{em}} \wedge \star F_{\mathrm{em}} = -\frac{1}{4} \int F_{\rho\sigma} F^{\rho\sigma} \mu.$$

This is Nakahara's (7.210b) in the present units.

Vary the inverse metric while fixing the potential, hence the covariant field components. The inverse-metric factors and the determinant variation give

$$\delta(F_{\rho\sigma} F^{\rho\sigma}) = 2 F_{\alpha\lambda} F_\beta{}^\lambda \delta g^{\alpha\beta}, \qquad \delta\mu = -\frac{1}{2} g_{\alpha\beta} \delta g^{\alpha\beta} \mu.$$

Substitution in the action and comparison with the stress definition give

$$T_{\mu\nu} = -F_{\mu\lambda} F_\nu{}^\lambda + \frac{1}{4} g_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta}.$$

In an orthonormal frame, $F_{0\lambda}F_0{}^{\lambda} = -|\mathbf{E}|^2$ and $F_{\alpha\beta}F^{\alpha\beta} = 2(|\mathbf{B}|^2 - |\mathbf{E}|^2)$. Hence $T_{00} = (|\mathbf{E}|^2 + |\mathbf{B}|^2)/2$ is nonnegative.

Physical input. We use the Einstein–Hilbert gravitational action

$$S_{\text{grav}} = -\frac{1}{16\pi G_N}\int R_{\text{sc}}\,\mu.$$

Theorem 19.7 (Variational form of Einstein's equation · *cited variation; algebraic consequence proved*)**.** *For compactly supported metric variations, the cited variation formula is*

$$\delta\int R_{\text{sc}}\,\mu = \int G_{\alpha\beta}\,\delta g^{\alpha\beta}\,\mu.$$

Stationarity of $S_{\text{grav}} + S_{\text{m}}$ *therefore gives*

$$\text{Ric}_{\mu\nu} - \frac{1}{2}R_{\text{sc}}g_{\mu\nu} = 8\pi G_N T_{\mu\nu}.$$

The variation formula is Nakahara, §7.10.2, equations (7.203)–(7.214), translated to our signature and matter convention.[1]

It discards a divergence because the variation has compact support. The displayed coefficient follows by setting $-G_{\mu\nu}/(16\pi G_N) + T_{\mu\nu}/2 = 0$. The action gives the stated physical law its variational form.

Self-check. State in one sentence what the Einstein tensor is and why we introduced it.

Answer. The Einstein tensor is the trace-adjusted Ricci tensor $G_{\mu\nu} = \text{Ric}_{\mu\nu} - \frac{1}{2}R_{\text{sc}}g_{\mu\nu}$, introduced as a curvature source tensor whose divergence vanishes. The decisive identity is $2\nabla^{\mu}\,\text{Ric}_{\mu\nu} = \partial_\nu R_{\text{sc}}$, which cancels the trace term's divergence by metric compatibility.

Unit 19C Schwarzschild geometry and orbits

19.4 The Schwarzschild vacuum metric

In vacuum, $T_{\mu\nu} = 0$ makes the Einstein tensor zero. Taking the metric trace in four dimensions gives $0 = R_{\text{sc}} - \frac{1}{2}(4)R_{\text{sc}} = -R_{\text{sc}}$. Thus the vacuum field equation is $\text{Ric}_{\mu\nu} = 0$.

We now use geometric units $G_N = c = 1$. Let $M > 0$ denote the mass parameter, equal to $G_N m/c^2$ before this unit choice. On the exterior $r > 2M$, set $f(r) = 1 - 2M/r$.

Definition 19.8 (Schwarzschild metric)**.** The *Schwarzschild metric* is

$$\mathrm{d}s^2 = f\,\mathrm{d}t^2 - f^{-1}\mathrm{d}r^2 - r^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2).$$

[1] Replacing a metric by its negative leaves its Christoffels, curvature operator and Ricci tensor unchanged. The scalar curvature reverses sign, so the Einstein tensor is unchanged. Nakahara's $-+++$ Einstein–Hilbert sign consequently becomes the minus sign above. For stress, his (7.211) varies the lower metric. Converting to his inverse metric contributes a minus sign by (7.203). Our inverse metric is its negative, giving a second minus sign. The two cancel, yielding our positive inverse-metric variation with the same lowered stress tensor. We use (7.211) for this translation. The inverse-metric prefactor in (7.212) needs a minus sign to agree with it. Carroll's free notes use the same curvature index order as ours. Carroll, *Lecture Notes on General Relativity* (1997), (3.67), (3.90)–(3.96), pp. 76, 81–82. His metric has the opposite sign. His Schwarzschild geodesic formulas (7.33)–(7.48), pp. 172–174, translate by this metric reversal.

The angular expression is the round-sphere metric, with usual overlapping charts at its coordinate poles.

Proposition 19.9 (Schwarzschild vacuum). *The exterior Schwarzschild metric has* $\mathrm{Ric} = 0$. *The fields* ∂_t *and* ∂_ϕ *are Killing fields. Along an affinely parametrized geodesic, the quantities*

$$E = f\dot{t}, \qquad L = r^2 \sin^2\theta\, \dot{\phi}$$

are constant.

Proof. First compute for any smooth positive function $f(r)$. Up to symmetry in the two lower indices, the nonzero Christoffels are

$$\begin{aligned} &\Gamma^t_{tr} = f'/(2f), && \Gamma^r_{tt} = ff'/2, && \Gamma^r_{rr} = -f'/(2f),\\ &\Gamma^r_{\theta\theta} = -rf, && \Gamma^r_{\phi\phi} = -rf\sin^2\theta, && \Gamma^\theta_{r\theta} = 1/r,\\ &\Gamma^\phi_{r\phi} = 1/r, && \Gamma^\theta_{\phi\phi} = -\sin\theta\cos\theta, && \Gamma^\phi_{\theta\phi} = \cot\theta. \end{aligned}$$

Substitute them into Lecture 17's Ricci contraction,

$$\mathrm{Ric}_{\mu\nu} = \partial_\rho\Gamma^\rho_{\nu\mu} - \partial_\nu\Gamma^\rho_{\rho\mu} + \Gamma^\rho_{\rho\lambda}\Gamma^\lambda_{\nu\mu} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\rho\mu}.$$

For example, $\Gamma^\rho_{\rho r} = 2/r$ and the tt entry is

$$\mathrm{Ric}_{tt} = \frac{(f')^2 + ff''}{2} + \frac{ff'}{r} - \frac{(f')^2}{2} = f\left(\frac{f''}{2} + \frac{f'}{r}\right).$$

For the other independent diagonal entries, substitution gives

$$\begin{aligned} \mathrm{Ric}_{rr} &= \partial_r\left(-\frac{f'}{2f}\right) + \frac{2}{r^2} - \frac{f'}{rf} - \left(\frac{(f')^2}{2f^2} + \frac{2}{r^2}\right),\\ \mathrm{Ric}_{\theta\theta} &= \partial_r(-rf) + \csc^2\theta - 2f - (-2f + \cot^2\theta). \end{aligned}$$

The off-diagonal entries vanish. Simplifying the diagonal entries gives

$$\mathrm{Ric}_{rr} = -f^{-1}\left(\frac{f''}{2} + \frac{f'}{r}\right), \qquad \mathrm{Ric}_{\theta\theta} = 1 - f - rf', \qquad \mathrm{Ric}_{\phi\phi} = \sin^2\theta\ \mathrm{Ric}_{\theta\theta}\,.$$

The equation $\mathrm{Ric}_{\theta\theta} = 0$ is $(rf)' = 1$. Integration gives $f = 1 - C/r$ for a constant C. This also satisfies $f''/2 + f'/r = 0$. Within this one-function ansatz, the vacuum equations therefore determine f. Comparison with $g_{00} = 1 + 2\Phi$ and $\Phi = -G_N m/r$ in units $c = 1$ identifies $C = 2G_N m = 2M$. This explains the mass parameter. It is not a proof that every static spherically symmetric metric has the assumed form. For $f = 1 - 2M/r$, the combinations are $f''/2 + f'/r = -2M/r^3 + 2M/r^3 = 0$ and $1 - f - rf' = 2M/r - 2M/r = 0$. Thus every Ricci component is zero. Smoothness extends this tensor equality across the angular chart poles.

The metric coefficients are independent of t and ϕ. The Lie derivative identity in Lecture 16, §16.4, is

$$(\mathcal{L}_K g)(Y,Z) = Kg(Y,Z) - g([K,Y],Z) - g(Y,[K,Z]).$$

For $K = \partial_t$ or $K = \partial_\phi$ and coordinate fields Y and Z, the brackets vanish. Independence of the metric coefficients therefore gives $\mathcal{L}_{\partial_t} g = \mathcal{L}_{\partial_\phi} g = 0$. Lecture 16's Killing conservation theorem makes $g(\partial_t, \dot{\gamma}) = f\dot{t}$ and $-g(\partial_\phi, \dot{\gamma}) = r^2\sin^2\theta\dot{\phi}$ constant. ∎

The vacuum exterior is not flat. Lecture 17's identity Ric = Kg applied only to surfaces. Here the Christoffel table gives

$$\begin{aligned} R^{\theta}{}_{\phi\theta\phi} &= (\sin^2\theta - \cos^2\theta) - f\sin^2\theta + \cos^2\theta \\ &= (1-f)\sin^2\theta = \frac{2M}{r}\sin^2\theta. \end{aligned}$$

This component is nonzero for $M > 0$ and $0 < \theta < \pi$. The Ricci contraction vanishes while the full curvature remains nonzero.

The divergence of f^{-1} at $r = 2M$ is a coordinate singularity. Indeed, put $v = t + r_*(r)$, where $r_*' = f^{-1}$. Substitution of $\mathrm{d}t = \mathrm{d}v - f^{-1}\mathrm{d}r$ gives

$$\mathrm{d}s^2 = f\,\mathrm{d}v^2 - 2\mathrm{d}v\,\mathrm{d}r - r^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2).$$

Its determinant is $-r^4\sin^2\theta$, nonzero at $r = 2M$ away from the angular coordinate poles. These coordinates extend the metric there. Radial null directions satisfy

$$0 = \mathrm{d}v(f\,\mathrm{d}v - 2\mathrm{d}r).$$

The ingoing family has $\mathrm{d}v = 0$, with future direction $\mathrm{d}r < 0$. The outgoing family has $\mathrm{d}r/\mathrm{d}v = f/2$, with $\mathrm{d}v > 0$. The future outgoing slope is positive outside $r = 2M$, zero on it and negative inside. Both future null directions decrease r inside. Their future radial cone contains no direction of increasing r. Angular motion contributes a nonpositive term to the metric, so a future causal direction with $\mathrm{d}v > 0$ satisfies $\mathrm{d}r/\mathrm{d}v \le f/2$. It also decreases r inside. The figure shows these local null directions in the extended coordinates.

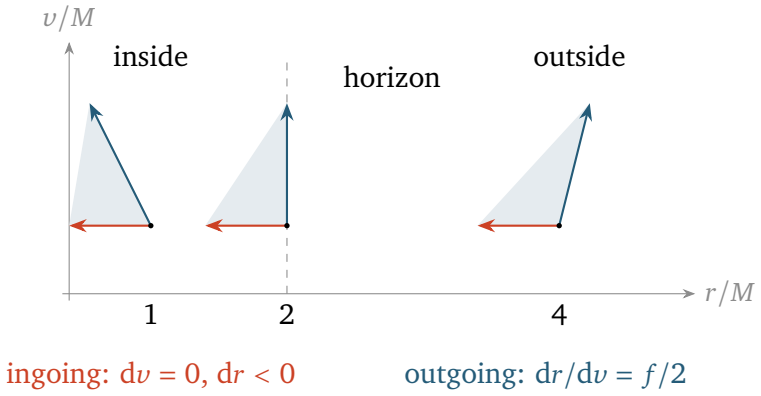


Figure 19.5: With v vertical and r horizontal, the future ingoing ray points left and the future outgoing ray has slope $\mathrm{d}r/\mathrm{d}v = f/2$. The shaded local future cones tilt inward for $r < 2M$.

Definition 19.10 (Event horizon)**.** The future *event horizon* is the boundary of the region from which future-directed light signals can reach arbitrarily distant observers.

In the Schwarzschild black-hole extension, the surface $r = 2M$ is this horizon. This global identification is cited from Carroll, §7, pp. 183–185. The local cone calculation establishes its one-way property. The curvature singularity at $r = 0$ can be detected by a scalar. In the exterior $r > 2M$, use the orthonormal frame

$$f^{-1/2}\partial_t, \qquad f^{1/2}\partial_r, \qquad r^{-1}\partial_\theta, \qquad (r\sin\theta)^{-1}\partial_\phi.$$

Label these vectors by 0, 1, 2 and 3 in the displayed order. Substituting the Christoffel table into Lecture 17's curvature formula and lowering the first index gives the independent nonzero components

$$(R_{0101}, R_{0202}, R_{0303}) = \frac{M}{r^3}(2, -1, -1),$$
$$(R_{1212}, R_{1313}, R_{2323}) = \frac{M}{r^3}(1, 1, -2).$$

All components not obtained from these by the curvature symmetries vanish. Raising all four indices leaves each listed component unchanged, because each index occurs twice. The two pair antisymmetries give four equal squares for each listed component. Thus

$$R_{abcd}R^{abcd} = 4(4 + 1 + 1 + 1 + 1 + 4)\frac{M^2}{r^6} = \frac{48M^2}{r^6}.$$

The ingoing metric and this scalar are analytic for $r > 0$, so the identity extends across $r = 2M$. The scalar diverges as $r \to 0$. This curvature singularity cannot be removed by a change of coordinates.

19.5 The massive orbit equation

Consider a future timelike Schwarzschild geodesic parametrized by proper time. Rotations of the angular coordinates are isometries because they preserve the round-sphere factor. Rotate them so the initial position and tangential velocity lie in the equatorial plane. The θ equation has the solution $\theta = \pi/2$, $\dot\theta = 0$ for these initial data. Uniqueness of the geodesic equation keeps the orbit in that plane. The constants E and L are its energy and angular momentum per unit rest mass.

Proposition 19.11 (Schwarzschild orbits · *proved*)**.** *A proper-time-parametrized equatorial timelike geodesic obeys*

$$\dot r^2 + f\left(1 + \frac{L^2}{r^2}\right) = E^2.$$

If $L \neq 0$, use ϕ as parameter and put $u = 1/r$. Then, with primes now denoting derivatives with respect to ϕ,

$$u'' + u = \frac{M}{L^2} + 3Mu^2.$$

Proof. The normalization $g(\dot\gamma, \dot\gamma) = 1$ gives

$$1 = f\dot t^2 - f^{-1}\dot r^2 - r^2\dot\phi^2 = \frac{E^2 - \dot r^2}{f} - \frac{L^2}{r^2}.$$

Rearranging gives the radial energy equation. To retain turning points, use the radial geodesic equation itself:

$$\ddot r = -\frac{ff'}{2}\dot t^2 + \frac{f'}{2f}\dot r^2 + rf\dot\phi^2 = -\frac{M}{r^2} + \frac{L^2}{r^3} - \frac{3ML^2}{r^4}.$$

The last equality substitutes the constants and the normalization just proved. It never divides by $\dot r$. Since $\dot\phi = Lu^2 \neq 0$, the chain rule gives $\dot r = -Lu'$ and $\ddot r = -L^2u^2u''$. Substitute in the radial equation and divide by $-L^2u^2$ to obtain the result. ∎

The Newtonian orbit equation lacks the term $3Mu^2$. We can see its effect perturbatively. Put $p = L^2/M$, $\epsilon_{\mathrm{orb}} = M/p$ and $w = pu$. The orbit equation becomes

$$w'' + w = 1 + 3\epsilon_{\mathrm{orb}} w^2.$$

For $0 < e < 1$ and $\epsilon_{\mathrm{orb}} \ll 1$, the Newtonian ellipse is $w_0 = 1 + e\cos\phi$. Writing $w = w_0 + \epsilon_{\mathrm{orb}} w_1 + O(\epsilon_{\mathrm{orb}}^2)$ gives

$$w_1'' + w_1 = 3w_0^2 = 3 + \frac{3}{2}e^2 + 6e\cos\phi + \frac{3}{2}e^2\cos(2\phi).$$

One particular solution is

$$w_1 = 3 + \frac{3}{2}e^2 - \frac{1}{2}e^2\cos(2\phi) + 3e\phi\sin\phi.$$

The term growing with ϕ is the first-order expansion of a shifted phase:

$$\cos((1 - 3\epsilon_{\mathrm{orb}})\phi) = \cos\phi + 3\epsilon_{\mathrm{orb}}\phi\sin\phi + O(\epsilon_{\mathrm{orb}}^2).$$

The remaining displayed terms in w_1 are bounded. Thus successive closest approaches, called perihelia, are separated by $2\pi/(1 - 3\epsilon_{\mathrm{orb}})$ to first order. The advance beyond a full turn is

$$\Delta\phi = 6\pi\epsilon_{\mathrm{orb}} + O(\epsilon_{\mathrm{orb}}^2) = \frac{6\pi M^2}{L^2} + O(\epsilon_{\mathrm{orb}}^2).$$

This is a weak-field prediction for bound test-particle orbits. The scale $M = G_N m/c^2$ is a length, so the approximation requires $p \gg M$. The figure plots the phase approximation with its bounded first-order corrections.

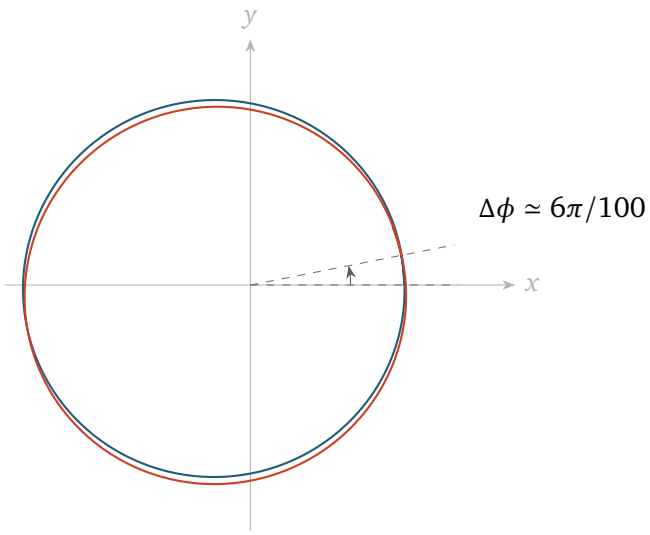


Figure 19.6: For $M = 1$, $p = 100$ and $e = 1/5$, the plotted first-order model is $w = 1 + e\cos((1 - 3\epsilon_{\mathrm{orb}})\phi) + \epsilon_{\mathrm{orb}}(3 + 3e^2/2 - e^2\cos(2\phi)/2)$, with $r = p/w$. The dashed rays mark successive perihelion directions to first order.

For radial motion $L = 0$, the radial energy equation still applies, but ϕ cannot serve as parameter.

19.6 A circular orbit and its effective potential

Definition 19.12 (Effective potential)**.** The *effective potential* for fixed L is

$$U_{\text{eff}}(r) = f(r)\left(1 + \frac{L^2}{r^2}\right).$$

The radial equation is $\dot{r}^2 + U_{\text{eff}} = E^2$ and $\ddot{r} = -U'_{\text{eff}}/2$. Radial motion is allowed exactly where $U_{\text{eff}} \le E^2$. For $M = 1$, $L = 4$ and $E^2 = 47/50$, the turning radii are the three roots $r_1 < r_2 < r_3$ of

$$3r^3 - 100r^2 + 800r - 1600 = 0.$$

They satisfy $2 < r_1 < 4 < r_2 < 12 < r_3 < 30$. The allowed exterior intervals are $2 < r \le r_1$ and $r_2 \le r \le r_3$. The outer interval describes a bound radial oscillation between closest and farthest approach. It does not imply a closed spatial orbit. For comparison, $E^2 = 51/50$ lies above the potential everywhere outside the horizon, since $1 - U_{\text{eff}} = 2(r-4)^2/r^3 \ge 0$ for these parameters. An inward orbit at this energy has no radial turning point before the horizon. The figure marks the energy levels and allowed intervals.

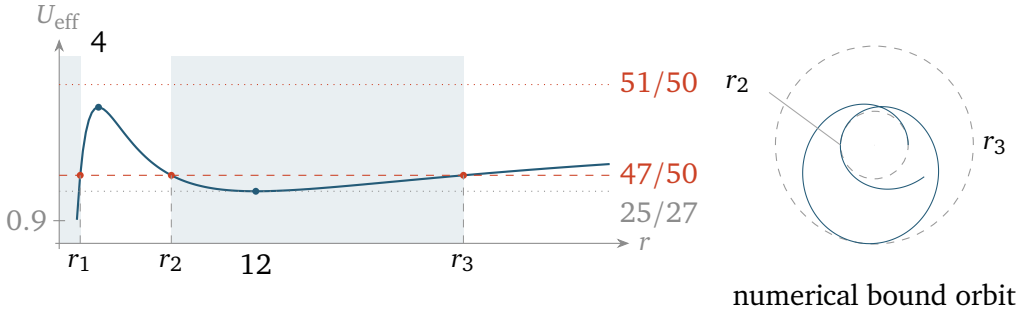


Figure 19.7: For $M = 1$ and $L = 4$, the potential has a maximum at $r = 4$ and a minimum at $r = 12$, with stable circular level $25/27$. The level $E^2 = 47/50$ gives three turning points and the shaded allowed radial intervals on the truncated potential plot, beside a numerical outer orbit between its turning circles.

Example 19.13 (A stable circular orbit)**.** Find a massive circular orbit at radius $r_0 = 12M$. For a general circular radius $r_0 > 3M$, solving $U'_{\text{eff}}(r_0) = 0$ and $E^2 = U_{\text{eff}}(r_0)$ gives

$$L^2 = \frac{Mr_0^2}{r_0 - 3M}, \qquad E^2 = \frac{(r_0 - 2M)^2}{r_0(r_0 - 3M)}.$$

With the angular momentum held fixed, the second derivative there is

$$U''_{\text{eff}}(r_0) = \frac{2M(r_0 - 6M)}{r_0^3(r_0 - 3M)}.$$

It is positive for $r_0 > 6M$, so the radial linearization has a restoring force. At $r_0 = 12M$ the constants are $L = 4M$ and $E^2 = 25/27$. This gives the potential minimum marked in the figure after setting $M = 1$.

Self-check. Recover the inward Newtonian acceleration from Γ^r_{tt} far from the Schwarzschild source.

Answer. The table gives $\Gamma^r_{tt} = ff'/2 = M(r-2M)/r^3 = M/r^2 + O(M^2/r^3)$. For slow motion, the geodesic equation contributes $-\Gamma^r_{tt}$ to the coordinate-time radial acceleration. Its leading value is $-M/r^2$, the attractive Newtonian result. It also matches $-\partial_r\Phi$ for $\Phi = -M/r$.

Self-check. A proper-time-parametrized equatorial timelike geodesic has $L = 4M$ and a radial turning point at $r = 9M$. Find E^2, $\ddot{r}$ and u'' there. Determine whether the turning point is a closest or farthest approach.

Answer. At the turning point $\dot{r} = 0$, so normalization gives $E^2 = (1-2/9)(1+16/81) = 679/729$. We use the radial geodesic equation, since differentiating the energy equation would give $0 = 0$ at this point:

$$\ddot{r} = -\frac{1}{81M} + \frac{16}{729M} - \frac{48}{6561M} = \frac{5}{2187M} > 0.$$

The chain rule $\ddot{r} = -L^2u^2u''$ then gives $u'' = -5/(432M)$. The positive radial acceleration makes this a closest approach.

Summary. A weak static Lorentzian metric gives Newton's equation for slow free fall. The cited Einstein–Hilbert variation gives Einstein's equation with positive $8\pi G_N T_{\mu\nu}$ in our conventions. Contracting the second Bianchi identity gives covariant conservation. The Schwarzschild exterior has zero Ricci tensor and nonzero full curvature. Its Killing constants, unit timelike normalization and radial geodesic equation give the massive orbit equation $u'' + u = M/L^2 + 3Mu^2$ in geometric units, for $L \neq 0$. Lecture 20 describes fields that require several local frames.

Reading. Nakahara, §7.10.1–2 and Example 7.15. Carroll, *Lecture Notes on General Relativity* (1997), §4, especially pp. 104–116, and §7, pp. 172–185.

Lecture 20

Fibre bundles

Objectives. *A vector field assigns a vector in a different tangent space at each point. A bundle describes these fibres and the changes between their local coordinates. We reconstruct bundles from transition functions and prove the global-section criterion for principal bundles. The Hopf bundle and the tangent sphere show how local products can fail to give a global product.*

By the end of this lecture you will be able to:

- Construct a smooth vector or principal bundle from a smooth cocycle.
- Relate local frames, section components and transition functions.
- Prove the principal-bundle section criterion and identify the Hopf bundle.
- Compute the tangent sphere's transition winding and compare it with Hopf's.

Unit 20A Bundles and gluing

20.1 Local products and fibres

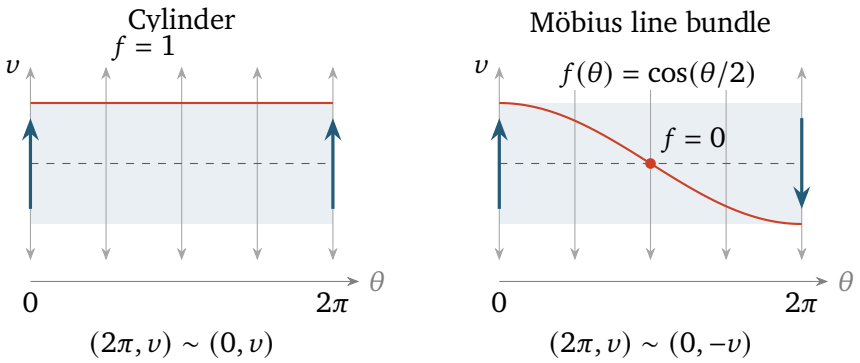
Example 20.1 (Cylinder and Möbius gluing)**.** We attach a real line to each point of a circle. The product $S^1 \times \mathbb{R}$ gives a cylinder. A different gluing gives the Möbius line bundle:

$$E_{\mathrm{M}} = ([0, 2\pi] \times \mathbb{R})/\!\sim, \qquad (2\pi, v) \sim (0, -v).$$

Over a small arc of the circle, both spaces are products with $\mathbb{R}$. A full circuit reverses the chosen fibre coordinate in E_{M}. The familiar finite strip contains only part of each full line fibre.

The cylinder has a continuous choice of a nonzero vector with constant coefficient 1. A continuous choice of a vector in each fibre of E_{M} has a coefficient f on $[0, 2\pi]$ with $f(2\pi) = -f(0)$. The intermediate value theorem forces a zero. Thus these local products have different global properties.

The cut-open products show the gluing and the full line fibres.



Shading: $|v| \le 1$; arrows continue along the full fibre $\mathbb{R}$.

Figure 20.1: The cylinder identifies equal fibre coordinates at its ends. The Möbius line bundle identifies opposite coordinates; shading marks only the strip $|v| \le 1$ in each full line fibre.

Definition 20.2 (Fibre bundle)**.** A smooth *fibre bundle* consists of a smooth surjection $\pi_E : E \to M$ and a manifold F, its *typical fibre*. Every point of the *base space* M has an open neighbourhood U_i and a diffeomorphism

$$\varphi_i : U_i \times F \longrightarrow \pi_E^{-1}(U_i), \qquad \pi_E(\varphi_i(x, v)) = x.$$

This is a *local trivialization*. The manifold E is the *total space*. The *fibre at* x is $E_x = \pi_E^{-1}(x)$.

All manifolds here are Hausdorff and second countable.

Definition 20.3 (Bundle isomorphism and triviality)**.** A *bundle isomorphism* over M is a diffeomorphism of total spaces commuting with the projections and preserving the specified fibre structure. A bundle is *trivial* if it is isomorphic to $M \times F$.

A local trivialization assigns a fibre coordinate independently at each base point of U_i.

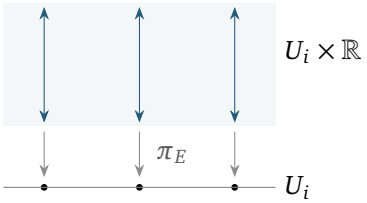


Figure 20.2: A local trivialization gives fibre coordinates over U_i. The projection sends each vertical fibre to its base point.

Definition 20.4 (Vector bundle)**.** A real or complex *vector bundle of rank* r has vector-space fibres and local trivializations linear on each fibre, with typical fibre $V_{\text{rep}} = \mathbb{R}^r$ or $\mathbb{C}^r$.

Definition 20.5 (Line bundle)**.** A rank-one vector bundle is a *line bundle*.

For example, TM of Lecture 8 is a rank-dim M real vector bundle.

Definition 20.6 (Section)**.** A *section over an open set* $U \subset M$ is a smooth map $s : U \to E$ such that $\pi_E \circ s = \mathrm{id}_U$. It is a *global section* when $U = M$.

Definition 20.7 (Local frame). A *local frame* of a rank-r vector bundle is a row of r smooth sections that form a basis in each fibre of its domain.

The frames of a single vector space have no preferred member. Given two row frames p and p', exactly one invertible matrix a satisfies $p' = pa$. Collecting the frames of all fibres gives a space over M. Its points are bases. Right multiplication changes those bases. A principal bundle records this space of choices and the group relating them.

Definition 20.8 (Principal bundle). Let G be a Lie group. A *principal G-bundle* is a bundle $\pi_E : P \to M$ with a smooth right action of G that is free and transitive on each fibre. Its local trivializations are required to satisfy $\varphi_i(x, ab) = \varphi_i(x, a)b$.

Freeness and transitivity say that any two points of a fibre differ by exactly one group element. Choosing $\varphi_i(x, e)$ chooses a reference in that fibre. The group identity belongs to this coordinate description. It is not a preferred point of the fibre. For the frame bundle, the reference is a basis. For the Hopf bundle, it will be a unit representative of a complex line.

Lecture 4's smooth covering examples are bundles with discrete fibre. The sheets over an evenly covered set give the local product. The antipodal cover $S^n \to \mathbb{R}\mathrm{P}^n$ is a principal $\mathbb{Z}_2$-bundle under the action $x \mapsto \pm x$. The rotation cover $S^3 \to \mathrm{SO}(3)$ has the same property. A general covering need not have a deck group transitive on its fibres.

20.2 Transition functions and reconstruction

Definition 20.9 (Transition functions and structure group). For a vector bundle, comparing trivializations defines smooth *transition functions* $g_{ij} : U_i \cap U_j \to \mathrm{GL}(r, \mathbb{K})$, where $\mathbb{K}$ is $\mathbb{R}$ or $\mathbb{C}$, by

$$\varphi_j(x, v) = \varphi_i(x, g_{ij}(x)v).$$

Thus the coordinates of the same vector satisfy $v_i = g_{ij}v_j$. The *structure group* is the group of allowed transition matrices, usually $\mathrm{GL}(r, \mathbb{R})$ or $\mathrm{GL}(r, \mathbb{C})$.

The symbols g_{ij} denote transitions, and h_i will denote frame changes. Restricting to oriented orthonormal frames makes the structure group $\mathrm{SO}(r)$. For a principal bundle, equivariance instead gives

$$\varphi_j(x, a) = \varphi_i(x, g_{ij}(x)a), \qquad g_{ij} : U_i \cap U_j \to G.$$

Indeed, compare the trivializations at the group identity and then act on the right by a. This also shows that the principal transitions are smooth. For a vector bundle, the standard bases give local row frames e_i with $e_j = e_i g_{ij}$. On a triple overlap, successive coordinate changes give

$$v_i = g_{ij}v_j = g_{ij}g_{jk}v_k.$$

Direct comparison gives $v_i = g_{ik}v_k$ for every $v_k \in V_{\mathrm{rep}}$. Thus composition forces the product law below.

Definition 20.10 (Cocycle). The *cocycle condition* is

$$g_{ij}g_{jk} = g_{ik}, \qquad g_{ii} = I, \qquad g_{ji} = g_{ij}^{-1}.$$

For a principal bundle, replace the identity matrix I by the group identity e.

The coordinate arrows go from k through j to i, in the order of matrix action on a column.

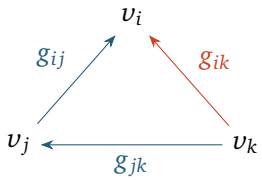


Figure 20.3: A fibre-coordinate column passes from region k to j and then to i by $g_{ij}g_{jk}$. Direct passage uses g_{ik}. All maps refer to the same base point in a triple overlap.

We describe the Möbius gluing by two base arcs. Let U_1 and U_2 be the images of the angle intervals $(-\pi/2, 3\pi/2)$ and $(\pi/2, 5\pi/2)$. Their overlap has a component A containing angle π and a component B containing angle 0. On A the angle representatives agree. On B they differ by 2π, so the Möbius seam reverses the coefficient. The transition is therefore

$$g_{12}|_A = +1, \qquad g_{12}|_B = -1.$$

For the cylinder both signs are $+1$. The picture records these overlaps and the effect of changing a frame on one whole arc.

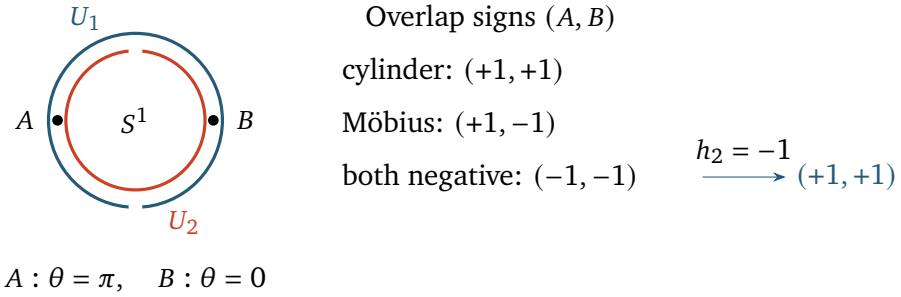


Figure 20.4: The arcs overlap near π in A and near 0 in B. Changing one arc's frame by -1 reverses both overlap signs.

Proposition 20.11 (Cocycle reconstruction · *proved*)**.** *Smooth transition matrices satisfying the cocycle condition on an open cover of M determine a smooth vector bundle, up to bundle isomorphism. The same construction with fibre a Lie group G and transition functions acting by left multiplication constructs a principal bundle.*

Proof. We glue product charts and verify that their quotient is a smooth manifold. Take the disjoint union of $U_i \times V_{\mathrm{rep}}$ and identify

$$(x, v)_j \sim (x, g_{ij}(x)v)_i.$$

The identity law gives reflexivity. The inverse law gives symmetry. The product law gives transitivity, so this is an equivalence relation. Project its quotient to M by the common base point. Over U_i, each class has a unique representative in the ith copy. The corresponding local coordinate map sends a representative in copy j to $(x, g_{ij}(x)v)$. It is a homeomorphism because the preimage of an open set in one copy is open in every other copy, since the transition map is a diffeomorphism. The quotient topology therefore agrees with these product charts. Their changes are smooth and linear on fibres.

Points with different base points have disjoint neighbourhoods lifted from M. Points with the same base point can be separated in one product chart. Thus the total space is Hausdorff. Second countability of M gives a countable subcover. Choose a basis element inside a cover member at each point and choose one member per basis element. The corresponding product charts, each second countable, give a countable basis for the total space. Hence the quotient is a smooth manifold.

The same argument with V_{rep} replaced by G gives smooth charts $U_i \times G$. The action $(x, a) \cdot b = (x, ab)$ is independent of the chart because left transition multiplication commutes with right multiplication. It is free and transitive on each fibre, as required in the principal definition.

If another bundle has trivializations φ_i with these transitions, send the quotient class $[(x, v)_i]$ to $\varphi_i(x, v)$. The defining transition relation makes this independent of its representative. In product charts it is the identity, so it is a smooth bundle isomorphism, equivariant in the principal case. This proves uniqueness. ∎

For the two-arc data $(+1, -1)$, the quotient is E_{M}. The product charts use exactly its angle representatives and seam relation.

Proposition 20.12 (Change of frame · *proved*). *Let $h_i \colon U_i \to \mathrm{GL}(r, \mathbb{K})$ be smooth frame changes. Under changes $e'_i = e_i h_i$, the transitions become*

$$g'_{ij} = h_i^{-1} g_{ij} h_j.$$

For a principal bundle, smooth maps $h_i \colon U_i \to G$ change the local sections by $s'_i = s_i h_i$ and give the same law. Conversely, transitions related in this way give isomorphic bundles.

Proof. The new frames satisfy $e'_j = e'_i h_i^{-1} g_{ij} h_j$, proving the displayed change law. For an isomorphism between the two constructions, use $(x, v)_i \mapsto (x, h_i^{-1} v)'_i$. The change law makes this map agree on overlaps. Its inverse uses h_i, so it is a smooth bundle isomorphism. For principal sections the computation is $s'_j = s_i g_{ij} h_j = s'_i h_i^{-1} g_{ij} h_j$. The same chartwise map, with v a group element, is equivariant. ∎

Proposition 20.13 (Common refinement · *proved*). *Two covers can be compared by restricting their trivializations to the pairwise intersections of their members.*

Proof. The pairwise intersections form an open cover of M refining both covers. Restrictions preserve the transition relations. The chartwise isomorphism in Proposition 20.11 compares each restricted reconstruction with its original bundle. ∎

20.3 Sections and frames of a vector bundle

In a local frame, write $s = e_i \sigma_i$. Its coefficient columns satisfy

$$\sigma_i = g_{ij} \sigma_j.$$

Conversely, smooth columns with this compatibility glue to a section. The zero columns always give a global zero section.

A global frame exists exactly when the vector bundle is trivial. Indeed, a frame defines the isomorphism $(x, v) \mapsto e(x)v$. Its inverse is smooth in local coordinates because the frame matrix is invertible. A trivialization supplies the converse frame by its standard basis. For

a line bundle, one nowhere-zero section is a frame. For higher rank, one section need not give an entire basis.

In the Möbius line bundle, the intermediate value argument excludes a nowhere-zero section, so no global frame exists. The zero section gives no basis of a fibre.

Self-check. Describe the Möbius line bundle by its seam and its two overlap signs. Show that it is nontrivial. Determine what happens if both overlap signs are -1.

Answer. The seam is $(2\pi, v) \sim (0, -v)$, with transition signs $(+1, -1)$. A nowhere-zero section would have a continuous coefficient satisfying $f(2\pi) = -f(0)$, which the intermediate value theorem excludes. A trivial line bundle has a nowhere-zero section, so the bundle is nontrivial. Its zero section still exists. For signs $(-1, -1)$, change the frame on U_2 by -1. The change law gives $1^{-1}(-1)(-1) = +1$ on both components. These are the cylinder transitions, so this bundle is trivial.

Self-check. State in one sentence what a fibre bundle is and why we introduced it when studying vector fields. Explain what local product charts do not determine by themselves.

Answer. A fibre bundle is a smooth space over a base that is locally a product with a fixed fibre, introduced to describe the varying spaces in which fields take values. The overlap identifications determine how the products fit together. The cylinder and the Möbius line bundle have the same typical fibre $\mathbb{R}$ but different overlap signs, so local product shape alone does not determine triviality.

Unit 20B Principal bundles, Hopf and the tangent sphere

20.4 Principal and associated bundles

Definition 20.14 (Oriented orthonormal frame bundle)**.** For an oriented Riemannian n-manifold, its *oriented orthonormal frame bundle* consists of all positive orthonormal frames in its tangent spaces.

A point of this total space is a base point together with an ordered positive orthonormal basis of its tangent space. Right multiplication of a row frame by $\mathrm{SO}(n)$ is free and transitive on each fibre. Local Gram–Schmidt frames give equivariant product charts. This makes it a principal $\mathrm{SO}(n)$-bundle, even when no global tangent frame exists.

The local sections $s_i(x) = \varphi_i(x, e)$ obey $s_j = s_i g_{ij}$. The group acts on the right, while g_{ij} changes fibre coordinates by multiplication on the left.

Proposition 20.15 (The section criterion · *proved*)**.** *A principal bundle is trivial if and only if it has a global section.*

Proof. A global section s gives a map $\Psi\colon M \times G \to P$ by $\Psi(x, a) = s(x)a$. Freeness and transitivity make it bijective and equivariant. In a local trivialization write $s(x) = s_i(x)b_i(x)$. Then $\Psi(x, a)$ has local coordinate $b_i(x)a$. Its inverse has coordinate $b_i(x)^{-1}a$, so both maps are smooth. Conversely, a trivialization gives the section $x \mapsto (x, e)$. ∎

Vector-bundle triviality requires a full frame, as above.

Definition 20.16 (Associated bundle)**.** Let $\rho\colon G \to \mathrm{GL}(V_{\mathrm{rep}})$ be a smooth representation. The *associated bundle* is

$$P \times_G V_{\mathrm{rep}} = (P \times V_{\mathrm{rep}})/\!\sim, \qquad (pa, v) \sim (p, \rho(a)v), \qquad [p, v] \longmapsto \pi_E(p).$$

For the oriented orthonormal frame bundle, take the standard representation of $\mathrm{SO}(n)$ on $\mathbb{R}^n$. A row frame p and a component column v specify the tangent vector pv. The quotient recovers the tangent bundle by the map

$$P \times_{\mathrm{SO}(n)} \mathbb{R}^n \longrightarrow TM, \qquad [p, v] \longmapsto pv.$$

The equality $(pa)v = p(av)$ explains the identification in the quotient. Replacing p by pa changes the same vector's components from v to $a^{-1}v$. Thus $[pa, a^{-1}v] = [p, v]$. In a local frame the displayed map and its inverse are product-coordinate identity maps. Hence it is a smooth vector-bundle isomorphism.

For a general representation, the local trivializations are $(x, v) \mapsto [s_i(x), v]$. Their transitions are $\rho(g_{ij})$, so reconstruction gives a smooth vector bundle.

Definition 20.17 (Equivariant map)**.** A smooth map $f\colon P \to V_{\mathrm{rep}}$ is *equivariant* for this right action if

$$f(pa) = \rho(a)^{-1} f(p).$$

Sections correspond to these equivariant maps. The section is $x \mapsto [p, f(p)]$ for any p over x: $[pa, f(pa)] = [p, \rho(a) f(pa)] = [p, f(p)]$. Conversely, a section uniquely determines these fibre coordinates $f(p)$. In a product chart they are $\rho(a)^{-1} f(s_i(x))$, which proves smoothness and the converse.

20.5 Homogeneous spaces as principal bundles

Proposition 20.18 (Homogeneous principal bundles · *proved given the cited homogeneous-space and inverse function theorems*)**.** *For a closed subgroup H of a Lie group G, the quotient $G \to G/H$ is a principal H-bundle under right multiplication.*

Proof. We construct local sections of the quotient submersion and use the group action to trivialize its fibres. For a closed subgroup H of a Lie group G, Theorem 8.25 gives the smooth quotient G/H and a submersion $\pi_E\colon G \to G/H$. The inverse function theorem gives local submersion coordinates. Choose base coordinates y and supplement the independent differentials of $y \circ \pi_E$ by complementary coordinates ζ. The map $(y \circ \pi_E, \zeta)$ has invertible differential, hence is a local coordinate system. In these coordinates $\pi_E(y, \zeta) = y$. Fixing ζ gives a smooth local section s near each base point. The maps

$$U \times H \longrightarrow \pi_E^{-1}(U), \qquad (x, h) \longmapsto s(x)h$$

are diffeomorphisms. Indeed, their inverse sends a to $(aH, s(aH)^{-1}a)$. The second entry lies in the embedded subgroup H and is smooth there. In submanifold charts its normal coordinates vanish. Thus $H \to G \to G/H$ is a principal H-bundle. ∎

For example, the quotient $\mathrm{SO}(3) \to S^2$, given by the last column, has fibre $\mathrm{SO}(2)$. The first two columns give a positive orthonormal tangent frame at that point. This identifies $\mathrm{SO}(3)$ with the oriented orthonormal frame bundle of the round sphere.

20.6 The Hopf bundle

Physical input. A pure qubit state is a complex line in $\mathbb{C}^2$. Its normalized vector representatives differ by a common phase in $\mathrm{U}(1)$.

A phase convention chooses one unit representative of each line in its domain. The real tautological line gives a simpler example of choosing vectors on lines. Over $\mathbb{R}\mathrm{P}^1$, its fibre at a line is that line itself. Write its unit vector as $q(\alpha) = (\cos\alpha, \sin\alpha)^T$ for $0 \le \alpha \le \pi$. The lines at 0 and π agree. Their unit representatives satisfy $q(\pi) = -q(0)$. Thus the coefficient in $vq(\alpha)$ changes sign at the seam. This is the Möbius line bundle over $\mathbb{R}\mathrm{P}^1 \cong S^1$. The figure shows the lines and their pairs of unit vectors.

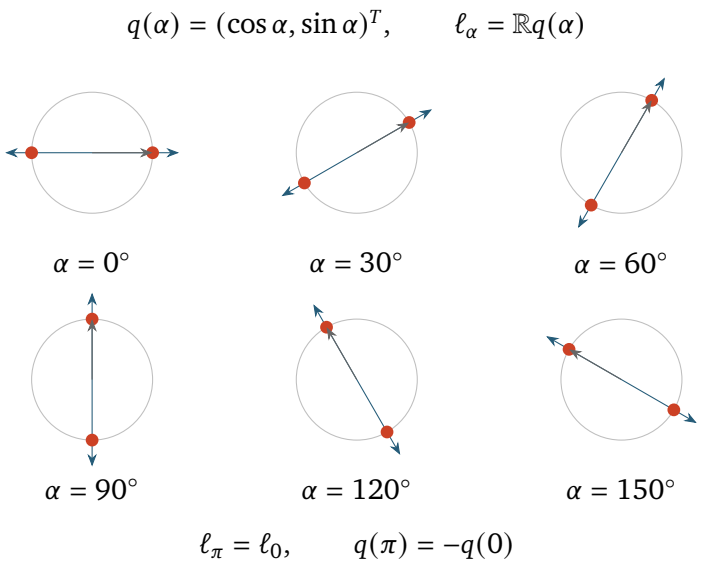


Figure 20.5: Each base point is a real line, and its fibre is that same line. The unit representatives change from $q(0)$ to $-q(0)$ as the line makes one circuit of $\mathbb{R}\mathrm{P}^1$.

Let $N = [1 : 0]$ and $S = [0 : 1]$ be the north and south poles of the Bloch sphere. The projective coordinate $z = b/a$ is defined on $U_N = \mathbb{C}\mathrm{P}^1 \setminus \{S\}$. The coordinate $w = a/b = 1/z$ is defined on $U_S = \mathbb{C}\mathrm{P}^1 \setminus \{N\}$. Unit representatives of their lines are

$$p_N(z) = \frac{(1, z)^T}{\sqrt{1 + |z|^2}}, \qquad p_S(w) = \frac{(w, 1)^T}{\sqrt{1 + |w|^2}}.$$

The northern representative tends to $(0, e^{i\phi})^T$ along $z = Re^{i\phi}$ as $R \to \infty$. Its limit at the south pole depends on the approach direction, so this phase convention does not extend there.

Definition 20.19 (Tautological line bundle)**.** The *tautological line bundle* is the set

$$\mathcal{T} = \{(\ell, v) \in \mathbb{C}\mathrm{P}^1 \times \mathbb{C}^2 : v \in \ell\}, \qquad (\ell, v) \longmapsto \ell.$$

The fibre at ℓ is the line ℓ itself, which explains the name. Its local coordinates are $v = cp_N(z)$ or $v = cp_S(w)$, with $c \in \mathbb{C}$. The inverse coefficient is $c = p_N^\dagger v$ or $c = p_S^\dagger v$, respectively. These maps give smooth product charts with linear overlap maps. On the overlap, $p_S = p_N|z|/z$. Orient the equator $|z| = 1$ by increasing ϕ, as the boundary of the northern cap. On this positive equator $z = e^{i\phi}$, the frame transition is $g_{NS} = e^{-i\phi}$, with winding -1. We keep the chart order (N, S). Reversing that order replaces the transition by its inverse and reverses its winding. For one fibre point, the phase coordinates obey

$$p_N e^{i\chi_N} = p_S e^{i\chi_S}, \qquad \chi_S = \chi_N + \phi \pmod{2\pi}.$$

The cap products glue by this change from northern to southern phase coordinates.

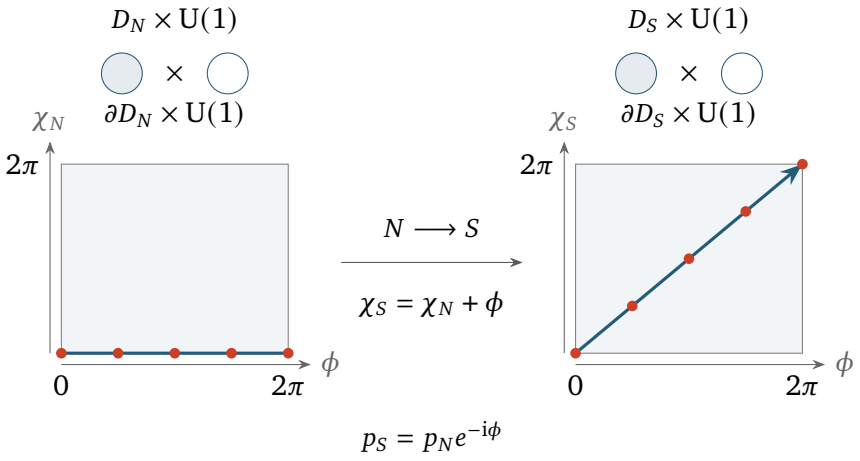


Figure 20.6: Each cap product has a boundary torus, shown cut open into a square. The same fibre point changes coordinates by $\chi_S = \chi_N + \phi$ from north to south, although the frame transition is $e^{-\mathrm{i}\phi}$.

The nontriviality proof compares the fundamental groups of a product and S^3. We complete Lecture 4's argument for Proposition 4.9, including the omitted-point step.

Lemma 20.20 (Sphere loops · *proved*)**.** *The spheres S^2 and S^3 are simply connected.*

Proof. To prove that S^n is simply connected for $n \in \{2, 3\}$, subdivide a loop into hemispherical pieces using Lemma 3.5. On each piece replace it by the normalized straight segment between its endpoints. Both curves have positive scalar product with the hemisphere centre, so their normalized straight interpolation is a homotopy fixing those endpoints. The resulting polygon lies in finitely many two-dimensional linear subspaces. Their union misses a point of the sphere. For each subspace choose a nonzero linear functional vanishing on it and evaluate these on $(1, t, \dots, t^n)$. Each gives a nonzero polynomial, so a parameter outside their finite root sets gives a vector outside the union. Normalize it to obtain the omitted sphere point. Stereographic projection from that point puts the polygon in $\mathbb{R}^n$, where straight contraction to its base point fixes that base point. Pasting the piece homotopies proves the assertion for $n = 2$ and $n = 3$. Both spheres are path-connected by short arcs, joining antipodes via a third point. ∎

Theorem 20.21 (The Hopf bundle · *proved given the cited homogeneous-space and inverse function theorems*)**.** *The map $\mathrm{SU}(2) \to \mathrm{SU}(2)/\mathrm{U}(1) \cong \mathbb{C}\mathrm{P}^1 \cong S^2$ is a nontrivial principal $\mathrm{U}(1)$-bundle. Under $\mathrm{SU}(2) \cong S^3$, it sends a unit column $(a, b)^T$ to its line $[a : b]$. Its associated complex line bundle for the standard $\mathrm{U}(1)$ representation is the tautological line bundle.*

Proof. Recall Lecture 8's explicit identification

$$(a, b) \longmapsto \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

Right multiplication by $\mathrm{diag}(u, u^{-1})$, for $u \in \mathrm{U}(1)$, multiplies the first column by u. Its orbits are exactly the unit representatives of each complex line. The homogeneous-bundle proposition proves the principal structure.

By Lemma 20.20, both S^2 and S^3 are simply connected. Loops and based homotopies in a product are precisely pairs of loops and homotopies. Thus $\pi_1(S^2 \times S^1) \cong \mathbb{Z}$, whereas $\pi_1(S^3) = 0$. A trivial Hopf bundle would give a diffeomorphism between these total spaces, which is impossible.

The map $[p, v] \mapsto ([p], vp)$ identifies the associated line with $\mathcal{T}$. It is well-defined because $(pu, v) \sim (p, uv)$. Its inverse exists on each fibre, and in the local unit representatives above both maps are product-coordinate identity maps. Hence it is a smooth bundle isomorphism. ∎

By the section criterion, Hopf nontriviality excludes a global smooth phase convention. A continuous global section would also give a homeomorphism with $S^2 \times S^1$: in local coordinates the inverse uses multiplication by the inverse of the section's continuous phase. The same fundamental-group contradiction therefore excludes a continuous global choice. The local choices p_N and p_S represent the same state through their phase transition. Lecture 24 studies transport of these phases. The phase circle below lies over a single fixed state line.

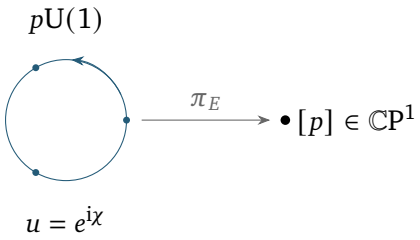


Figure 20.7: All unit vectors pu, with $u \in \mathrm{U}(1)$, lie over the same line $[p]$. The circle records their relative phase.

We recall from Lecture 3 that winding classifies loops in $\mathrm{U}(1)$ up to homotopy. For the bundle classification, we cite that bundles over either disk are trivial. Trivializing over the two caps leaves one transition on their overlap, which retracts onto the equator. Changing the section on a cap multiplies this transition by a phase that extends across a disk. Such a boundary phase has winding zero, since its loop contracts across that disk. Hence changing cap sections preserves the transition winding. The cited clutching theorem also says that homotopic equatorial transitions give isomorphic topological bundles. Together these facts explain why a single integer classifies them.

Proposition 20.22 (Clutching classification · *cited*)**.** *The underlying topological principal* $\mathrm{U}(1)$*-bundles over* S^2*, up to continuous equivariant bundle isomorphism, are classified by the winding of the transition* g_{NS} *on the positive equator. Every integer occurs.*

This is Naber, Theorem 4.4.3, pp. 231–232. That theorem concerns continuous bundles. No smooth classification proof is claimed here. Its integer invariant already distinguishes the smooth examples. On the overlap $z \neq 0$, the transitions $(z/|z|)^n$ give smooth representatives for every $n \in \mathbb{Z}$. The Hopf transition has winding -1, which gives a second proof of its nontriviality.

20.7 The tangent sphere and its transition winding

Use the oriented complex charts $z = x + \mathrm{i}y$ and $w = u + \mathrm{i}v = 1/z$ of Lecture 7 on $S^2 \cong \mathbb{C}\mathrm{P}^1$. Here u and v denote real chart coordinates. The symbol u in the Hopf action denoted a

phase, and v in the bundle charts denoted a fibre coordinate. Multiplication by i on tangent vectors is defined by $\mathrm{i}\partial_x = \partial_y$. The change $w = 1/z$ has complex-linear differential and preserves this rule. The real tangent planes therefore form a complex line bundle.

Example 20.23 (The tangent-sphere transition)**.** Find its frame transition and its component transition around the equator. Use complex frames $e_z = \partial_x$ and $e_w = \partial_u$, where a and b now denote real component coefficients. A complex coefficient $a + \mathrm{i}b$ means the real vector $a\partial_x + b\partial_y$ in the first frame. The coordinate derivative and its inverse are

$$\frac{\mathrm{d}w}{\mathrm{d}z} = -z^{-2}, \qquad \frac{\mathrm{d}z}{\mathrm{d}w} = -z^2.$$

The chain rule gives the frame relation and the component relation

$$e_w = e_z(-z^2), \qquad \sigma_w = (-z^{-2})\sigma_z.$$

The real plane rotation through β is

$$\mathrm{Rot}(\beta) = \begin{pmatrix} \cos\beta & -\sin\beta \\ \sin\beta & \cos\beta \end{pmatrix}.$$

Identify a complex section coefficient with the real column $(\mathrm{Re}\,\sigma, \mathrm{Im}\,\sigma)^T$. On $z = e^{\mathrm{i}\phi}$ the corresponding real matrices are

$$g_{zw} = -\mathrm{Rot}(2\phi), \qquad \sigma_w = -\mathrm{Rot}(-2\phi)\sigma_z.$$

The frame loop has lifted angle $\pi + 2\phi$ and winding +2. The component loop has angle $\pi - 2\phi$ and winding -2. The general component law is $\sigma_z = g_{zw}\sigma_w$. Solving for σ_w uses $g_{wz} = g_{zw}^{-1}$. There is one transition integer once the chart order is fixed. Here $(z, w) = (N, S)$, so the tangent transition has winding +2 and the Hopf transition has winding -1. The Bloch map is $(2x, 2y, 1 - |z|^2)/(1 + |z|^2)$. For the round metric, differentiating it gives

$$g = \frac{4(\mathrm{d}x^2 + \mathrm{d}y^2)}{(1 + |z|^2)^2}.$$

Normalizing the coordinate frames therefore replaces $-z^2$ by its phase $-z^2/|z|^2$ on the overlap. Its equatorial winding is still +2. A new unit frame on either cap changes this winding by zero, because its boundary phase extends across the disk and is null-homotopic by Lecture 3. Consequently the oriented orthonormal frame bundle is nontrivial by the cited classification. A nowhere-zero tangent field would give a unit frame and trivialize it, contradicting its winding. This is the hairy ball theorem of Lecture 14, now obtained from transition winding. A unit field and its positive quarter-turn would form a global oriented orthonormal frame. The integer 2 agrees with $\chi(S^2)$ in Lecture 18. Lecture 22 relates this transition winding to the curvature integral.

The angle lifts record the transition in chart order (z, w) and its inverse.

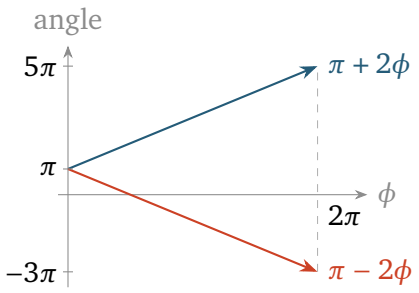


Figure 20.8: The transition g_{zw} has angle lift $\pi + 2\phi$. The inverse map g_{wz} has angle lift $\pi - 2\phi$ and reads components in the reverse direction.

On the equator, use the positive spherical frame (e_θ, e_ϕ), where e_θ points south and e_ϕ points in the direction of increasing ϕ. The normalized coordinate frames have the expressions

$$\widehat{e}_z = \cos\phi\, e_\theta - \sin\phi\, e_\phi, \qquad \widehat{e}_w = -\cos\phi\, e_\theta - \sin\phi\, e_\phi.$$

Their angles in this common frame are $-\phi$ and $\pi + \phi$. The relative angle is therefore $\pi + 2\phi$. The unrolled equator shows both varying arrows in the same tangent-plane coordinates.

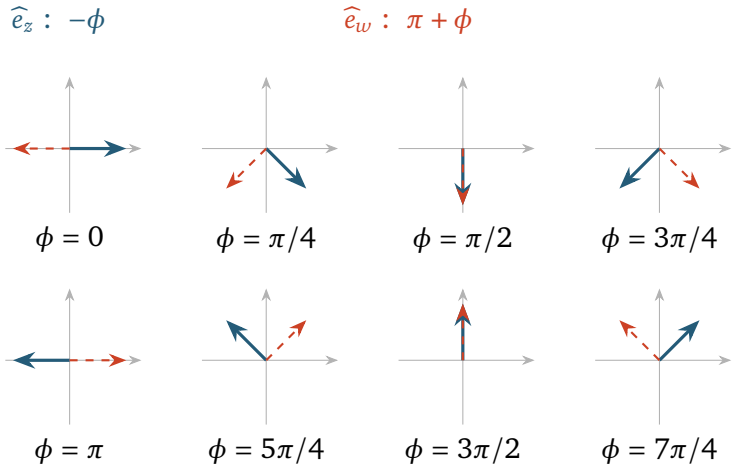


Figure 20.9: At each equatorial point, the horizontal axis is e_θ and the vertical axis is e_ϕ. The normalized z-frame turns through -2π and the w-frame through $+2\pi$, giving relative winding +2.

Self-check. At the equatorial point $z = e^{i\pi/3}$, a tangent vector has complex component $\sigma_z = 2 - i$ in the coordinate frame e_z. Find σ_w and verify that the two frames give the same vector.

Answer. The inverse transition gives

$$\sigma_w = -e^{-2i\pi/3}(2 - i) = 1 + \frac{\sqrt{3}}{2} + i\left(\sqrt{3} - \frac{1}{2}\right).$$

Multiplication by the frame transition gives $(-e^{2i\pi/3})\sigma_w = 2 - i$. Thus $e_w\sigma_w = e_z\sigma_z$. Using the frame transition itself on the component would give the wrong vector.

Summary. Transition functions reconstruct smooth vector and principal bundles. Frames transform by $e_j = e_i g_{ij}$. The same transition gives $\sigma_i = g_{ij}\sigma_j$. A principal bundle is trivial exactly when it has a global section. The Hopf bundle $S^3 \to S^2$ is nontrivial and its associated standard line is tautological. With chart order $(N, S) = (z, w)$, the tangent sphere has transition winding +2. The inverse component map has winding −2. Lecture 21 describes connections and transport in these local frames.

Reading. Nakahara, §9.2–9.4, especially Theorem 9.2. Naber, §§4.2–4.4, especially Theorem 4.3.4, pp. 223–224 (continuous principal reconstruction), and Theorem 4.4.3, pp. 231–232 (topological clutching). Smooth reconstruction follows Nakahara, §9.2.2, with the row-frame convention used here. The homogeneous-bundle proof uses Lee, *Introduction to Smooth Manifolds*, second edition, Theorem 4.5, for submersion coordinates. The associated-bundle section correspondence also appears in *Lie Groups*, Lecture 26, *From equivariant functions to sections*.

Lecture 21

Gauge fields: connections, curvature and holonomy

Objectives. *A bundle has local frames, but differentiating their coefficients depends on the frame. A connection corrects this derivative and specifies parallel transport. We derive its change law and show how curvature measures transport around a small loop. A flat connection can still have nontrivial transport around a hole. The Aharonov–Bohm phase is this effect for a charged particle.*

By the end of this lecture you will be able to:

- Derive the connection and curvature laws under a change of frame.
- Compute parallel transport with the correct multiplication order.
- Relate a small counter-clockwise loop to its curvature.
- Compute the flat Aharonov–Bohm holonomy and its gauge-invariant phase.

Unit 21A Connection and curvature

21.1 Covariant derivatives

Let $E \to M$ be a smooth rank-r complex vector bundle. Write $\Gamma(E)$ for its smooth sections and $\Omega^k(M;E)$ for its E-valued k-forms. In a frame, the latter are columns of ordinary k-forms. Recall Lecture 20's row frame e and coefficient column σ, with $s = e\sigma$. A change of frame $e' = eh$ gives $\sigma' = h^{-1}\sigma$. The ordinary derivative therefore obeys

$$\mathrm{d}\sigma' = h^{-1}\mathrm{d}\sigma + \mathrm{d}(h^{-1})\sigma.$$

The extra term means that the derivative of a coefficient column does not transform as an E-valued one-form. Even a constant column can become nonconstant after a frame change. A connection supplies a derivative independent of this choice. For a charged wave amplitude, it supplies a rule for comparing phases in different fibres.

Definition 21.1. A *connection* is a complex-linear map

$$\nabla\colon \Gamma(E) \longrightarrow \Omega^1(M;E), \qquad \nabla(fs) = \mathrm{d}f \otimes s + f\nabla s \quad \text{for } f \in C^\infty(M,\mathbb{C}).$$

Here $(\mathrm{d}f \otimes s)_p(X) = \mathrm{d}f_p(X)s(p)$ for $p \in M$ and $X \in T_pM$. Thus ∇s assigns a fibre vector to each tangent direction. We write $(\nabla s)(X) = \nabla_X s$, as for tangent fields in Lecture 16.

Locality. We construct a *bump function*, a smooth function equal to one near a chosen point $p \in M$, with support inside a chosen neighbourhood U. Take a coordinate ball centred at p with $\overline{B_R} \subset U$ and $0 < r < R$. Define $\rho(t) = e^{-1/t}$ for $t > 0$ and $\rho(t) = 0$ for $t \le 0$. Every positive-side derivative is $e^{-1/t}$ times a polynomial in $1/t$, by induction. All tend to zero at zero, so ρ is smooth. In the ball's coordinates the desired function is

$$f(x) = \frac{\rho(R^2 - |x|^2)}{\rho(R^2 - |x|^2) + \rho(|x|^2 - r^2)}.$$

The denominator is positive. The function is one on B_r and zero outside B_R. Its support is compact inside the chart, so extension by zero is smooth on M.

The connection is local. If s vanishes near p, choose such an f supported where $s = 0$. The Leibniz rule applied to $fs = 0$ gives $(\nabla s)_p = 0$. A section defined on U can instead be multiplied by a bump supported in U, equal to one near p, and extended by zero. Locality makes its derivative at p independent of the extension. This defines the local operator with the same Leibniz rule and commutes with restriction. In a local frame, define the matrix of one-forms A by $\nabla e = eA$. The Leibniz rule gives

$$\nabla s = eD\sigma, \qquad D\sigma = \mathrm{d}\sigma + A\sigma.$$

Definition 21.2. The matrix A in $\nabla e = eA$ is the local *gauge potential*.

It is a matrix of one-forms, rather than a single global matrix on a nontrivial bundle. Lecture 16's tangent connection is the real version of this definition, with real fibres, real linearity and real-valued functions. The figure shows one fixed vector in three rotating frames in this real rank-two case.

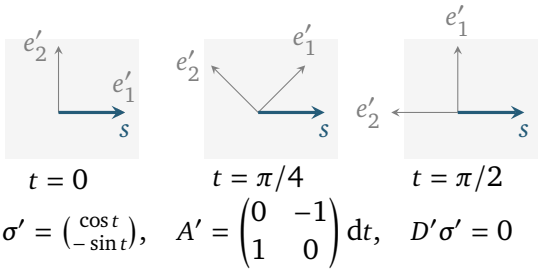


Figure 21.1: A fixed vector has changing coefficients in rotating frames. Its covariant derivative remains zero because the connection term cancels the ordinary derivative of its coefficients.

21.2 Frame changes and gauge transformations

Let $h\colon U \to \mathrm{GL}(r, \mathbb{C})$ be a smooth change of frame, $e' = eh$. The symbol h denotes a frame change. Planck's constant is h_{P}. The same section has coefficients $\sigma' = h^{-1}\sigma$. Differentiating the new frame gives

$$\nabla(eh) = e(Ah + \mathrm{d}h) = ehA', \qquad A' = h^{-1}Ah + h^{-1}\mathrm{d}h. \tag{21.1}$$

The extra derivative compensates for the changing frame. The following diagram expresses the resulting covariance.

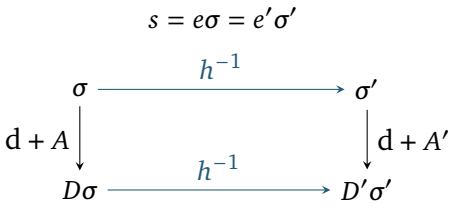


Figure 21.2: Both a section's coefficient column and its covariant derivative change by h^{-1}. The connection law makes the square commute.

Indeed, $\mathrm{d}(h^{-1}) = -h^{-1}(\mathrm{d}h)h^{-1}$ implies $D'\sigma' = h^{-1}D\sigma$ after the two derivative terms cancel. Conversely, local matrices obeying this change law on overlaps define a connection, because these local derivatives agree as bundle-valued one-forms. The compatibility law is Nakahara's (10.9).[1]

Definition 21.3. For an atlas with transitions in a matrix group G, its *G-frames* are the right translates $e_i a$, with $a \in G$. Require A to take values in its Lie algebra in these frames.

We use these frames for group-valued transport in §21.4.

The G-frames of E form a principal bundle by Lecture 20's cocycle construction.

Definition 21.4. A *gauge transformation* is an equivariant smooth automorphism of this principal bundle over the identity of M.

In local sections s_i, write it as $s_i(x)a \mapsto s_i(x)u_i(x)a$. The right-hand group coordinate is a. Since $s_j = s_i g_{ij}$, compatibility is precisely

$$u_j = g_{ij}^{-1} u_i g_{ij}.$$

A chosen global trivialization identifies these transformations with maps $M \to G$. For an abelian group the compatibility law reduces to $u_i = u_j$, even on a nontrivial bundle. For a nonabelian group, arbitrary global matrix maps need not satisfy it. A passive frame change describes the same connection. On E, the corresponding automorphism acts on columns by $\sigma_i \mapsto u_i\sigma_i$. Transporting the connection so that $D_i'(u_i\sigma_i) = u_i D_i \sigma_i$ gives

$$A_i' = u_i A_i u_i^{-1} + u_i \mathrm{d}(u_i^{-1}).$$

This follows by differentiating $u_i\sigma_i$ and cancelling $\mathrm{d}u_i$. It is the frame law with $h = u_i^{-1}$, the active description.

The passive and active descriptions keep different data fixed:

	Fixed data	Transformed data
Passive	section and connection	frame and coefficients
Active	frame	section and connection

A passive change can be chosen on one chart. An active transformation must satisfy the overlap compatibility law on the whole bundle. The equality of their local formulas does not remove this global requirement.

[1] We use row frames, so coefficient columns transform by h^{-1}.

Unitary connections.

Definition 21.5. A *Hermitian metric* is a smoothly varying positive Hermitian inner product on each fibre, antilinear in its first argument.

Definition 21.6. A connection is *unitary* if it differentiates this inner product by the product rule.

Definition 21.7. A *unitary frame* is an orthonormal basis in every fibre of its domain.

In such a frame that rule is equivalent to $A^\dagger = -A$. The covariant product rule differs from the ordinary derivative of $\sigma^\dagger\tau$ by the term $\sigma^\dagger(A^\dagger + A)\tau$. Here σ and τ are coefficient columns of two local sections. For a line bundle, A is therefore imaginary.

A charged particle. For charge $q \in \mathbb{R}$, the physical potential of Lecture 15 is the real one-form $A_{\mathrm{em}} = -\phi\mathrm{d}t + \mathbf{A}\cdot\mathrm{d}\mathbf{x}$. The connection convention and its component derivatives are

$$A_{\mathrm{conn}} = -\frac{\mathrm{i}q}{\hbar}A_{\mathrm{em}}, \qquad D_t = \partial_t + \frac{\mathrm{i}q}{\hbar}\phi, \qquad D_i = \partial_i - \frac{\mathrm{i}q}{\hbar}A_i.$$

Here $\hbar > 0$ remains explicit. Thus $-\mathrm{i}\hbar D_i = -\mathrm{i}\hbar\partial_i - qA_i$.

Definition 21.8. Replacing ordinary derivatives by these derivatives is *minimal coupling*.

The real electromagnetic potential and the imaginary connection potential are distinct.

Physical input. For a spinless nonrelativistic particle of mass $m > 0$ on a flat Cartesian patch, the local wave amplitude $\psi(t, \mathbf{x})$ is a coefficient of the charged section. Its squared modulus is the probability density. The charged Schrödinger equation is

$$\mathrm{i}\hbar D_t\psi = \frac{1}{2m}\sum_i(-\mathrm{i}\hbar D_i)^2\psi.$$

Each differential operator acts on everything to its right. In the ideal coherent-arm approximation, the electromagnetic action $q\int_\gamma A_{\mathrm{em}}$ multiplies an arm amplitude by $\exp(\mathrm{i}q\int_\gamma A_{\mathrm{em}}/\hbar)$. The remaining kinetic and apparatus phases are included in that arm's amplitude. We hold them fixed when comparing static solenoid fluxes.

Since $\mathrm{i}\hbar D_t = \mathrm{i}\hbar\partial_t - q\phi$, the same equation in ordinary derivatives is

$$\mathrm{i}\hbar\partial_t\psi = \frac{1}{2m}\sum_i(-\mathrm{i}\hbar\partial_i - qA_i)^2\psi + q\phi\psi.$$

The connection laws imply covariance of this equation.

For electromagnetism take a real smooth function χ and $h = e^{-\mathrm{i}q\chi/\hbar}$. The frame law then reads

$$A'_{\mathrm{em}} = A_{\mathrm{em}} + \mathrm{d}\chi, \qquad \psi' = e^{\mathrm{i}q\chi/\hbar}\psi, \qquad D'\psi' = e^{\mathrm{i}q\chi/\hbar}D\psi.$$

Here ψ is the charged section's coefficient, which transforms inversely to its frame. These equations include the charge sign and agree with the minimal-coupling derivatives. Applying the spatial covariance twice shows that each squared operator in the Schrödinger equation transforms by the same phase as ψ. With the original frame held fixed, the same local formulas describe the active map $u = e^{\mathrm{i}q\chi/\hbar}$. A global real function χ defines this automorphism even when no global frame exists.

21.3 Curvature and the Bianchi identity

For matrix-valued forms, wedge multiplication includes matrix multiplication: $(A \wedge B)_{ab} = \sum_c A_{ac} \wedge B_{cb}$. Extend D to coefficient-valued forms by $D\eta = \mathrm{d}\eta + A \wedge \eta$.

Definition 21.9. The *curvature two-form* is

$$F = \mathrm{d}A + A \wedge A, \qquad F_{ij} = \partial_i A_j - \partial_j A_i + [A_i, A_j].$$

Its action on a coefficient column is $D^2\sigma = F\sigma$. The components mean $F = \frac{1}{2} F_{ij} \mathrm{d}x^i \wedge \mathrm{d}x^j$.

Definition 21.10. For an endomorphism-valued k-form B, the *covariant exterior derivative* is

$$DB = \mathrm{d}B + A \wedge B - (-1)^k B \wedge A.$$

This derivative acts on both the output and input of an endomorphism.

Theorem 21.11 (Local connection laws · *proved*)**.** *A connection has local form $D = \mathrm{d} + A$ and the frame law* (21.1)*. Its curvature transforms by $F' = h^{-1} F h$ and satisfies the Bianchi identity*

$$DF = \mathrm{d}F + A \wedge F - F \wedge A = 0.$$

Proof. The first assertions follow from the Leibniz and frame calculations above. Expanding $D^2\sigma$ cancels $-A \wedge \mathrm{d}\sigma$ against $A \wedge \mathrm{d}\sigma$, leaving $F\sigma$. To check curvature covariance explicitly, put $B = h^{-1} A h$ and introduce the *Maurer–Cartan form* $\vartheta_{\mathrm{MC}} = h^{-1}\mathrm{d}h$. The product rule and the inverse derivative give

$$\mathrm{d}B = h^{-1}(\mathrm{d}A)h - \vartheta_{\mathrm{MC}} \wedge B - B \wedge \vartheta_{\mathrm{MC}}, \qquad \mathrm{d}\vartheta_{\mathrm{MC}} = -\vartheta_{\mathrm{MC}} \wedge \vartheta_{\mathrm{MC}}.$$

Since $A' = B + \vartheta_{\mathrm{MC}}$, substituting these identities into $\mathrm{d}A' + A' \wedge A'$ leaves only $h^{-1}(\mathrm{d}A + A \wedge A)h$. Finally,

$$\mathrm{d}F = \mathrm{d}A \wedge A - A \wedge \mathrm{d}A, \qquad A \wedge F - F \wedge A = A \wedge \mathrm{d}A - \mathrm{d}A \wedge A.$$

Adding gives the Bianchi identity. These are Nakahara's (10.38)–(10.47) in the present matrix notation. ∎

The value of a potential at a point. Within its matrix Lie algebra, the potential's value at a point can be changed arbitrarily. For a prescribed Lie-algebra-valued covector B_p, choose $h(p) = I$ and $\mathrm{d}h_p = B_p - A_p$. The frame law gives $A'_p = B_p$ and $F'_p = F_p$. Such a first derivative is realized locally by exponentiating a Lie-algebra-valued function that vanishes at p and has the prescribed derivative there. Thus A_p alone is not frame invariant. Curvature changes only by conjugation, so its traces are frame invariant. This is the bundle version of the distinction between Christoffel symbols and curvature in Lecture 16 and Lecture 17.

Dictionary of potentials and curvature. For U(1) the scalar one-form obeys $A \wedge A = 0$. The abelian Bianchi identity is $\mathrm{d}F = 0$. The physical and connection quantities are related by

$$F_{\mathrm{conn}} = -\frac{\mathrm{i}q}{\hbar} F_{\mathrm{em}}, \qquad \mathcal{J}_{\mathrm{conn}} = -\frac{\mathrm{i}q}{\hbar} \mathcal{J}.$$

The real source three-form $\mathcal{J}$ is Lecture 15's current. The connection-valued source $\mathcal{J}_{\mathrm{conn}}$ expresses the same source in the units of F_{conn} for the field equation in Lecture 23. For nonzero charge, the abelian Bianchi identity is precisely Lecture 15's homogeneous Maxwell

equation $\mathrm{d}F_{\mathrm{em}} = 0$. For zero charge the connection is zero and does not encode that electromagnetic equation. For a Hermitian matrix potential A_{phys} with $D = \mathrm{d} - \mathrm{i}g_{\mathrm{YM}}A_{\mathrm{phys}}$, the corresponding formulas are

$$A_{\mathrm{conn}} = -\mathrm{i}g_{\mathrm{YM}}A_{\mathrm{phys}}, \qquad F_{\mathrm{conn}} = -\mathrm{i}g_{\mathrm{YM}}(\mathrm{d}A_{\mathrm{phys}} - \mathrm{i}g_{\mathrm{YM}}A_{\mathrm{phys}} \wedge A_{\mathrm{phys}}).$$

The coupling g_{YM} is a scalar, never a gauge map. The commutator term remains in the nonabelian theory.

For the tangent bundle, the matrix A is Lecture 17's matrix of connection forms ω. The matrix F is its curvature matrix Ω. In particular,

$$A = (\omega^i{}_j), \qquad F = (\Omega^i{}_j), \qquad R(X,Y)(ev) = eF(X,Y)v.$$

Along a curve, the matrix $a(t) = A(\dot\gamma(t))$ is Lecture 16's $C(t)$. Lecture 16's frame-change matrix, there called A, is our h. The identity $DF = 0$ is Lecture 17's second Bianchi identity in this notation. On an oriented surface, write μ_g for the metric area form. This is the form written $\mathrm{d}A$ in Lecture 18. It is distinct from the exterior derivative $\mathrm{d}A$ of the connection potential here. The surface formulas of Lectures 17 and 18 become

$$A = -\omega^1{}_2 J, \qquad F = -K\mu_g J, \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Here $\omega^1{}_2$ is the scalar connection form and J is the positive quarter-turn. Lecture 18's rotation angle around a framed disk is $\Theta_D \equiv \int_D K\mu_g \pmod{2\pi}$.

Self-check. State in one sentence what a connection is and why an ordinary derivative of coefficient columns does not replace it.

Answer. A connection is a complex-linear Leibniz derivative of sections into bundle-valued one-forms, introduced to compensate for the term $\mathrm{d}(h^{-1})\sigma$ in the ordinary derivative after a frame change. That term prevents $\mathrm{d}\sigma$ from transforming as a bundle-valued one-form.

Unit 21B Transport, holonomy and the Aharonov–Bohm phase

21.4 Parallel transport and a small square

Definition 21.12. A section along a piecewise smooth path $\gamma\colon [0,1] \to M$ is *parallel* if its covariant derivative along the path vanishes.

In one local frame put $a(t) = A_{\gamma(t)}(\dot\gamma(t))$. Its coefficient equation is $\dot\sigma = -a(t)\sigma$.

Definition 21.13. The *transport matrix* $U(t)$ sends the initial column to this column.

Its initial value is I. Its equation is $\dot U = -a(t)U$. The coefficient equation gives the matrix equation column by column. Linear ODE existence and uniqueness give a solution throughout the compact parameter interval. Unlike Lecture 9's incomplete field $x^2\partial_x$, these linear transport equations have solutions on the whole path interval. The inverse solves $\dot V = Va(t)$, so $\frac{\mathrm{d}}{\mathrm{d}t}(VU) = 0$ and transport is invertible. On successive bundle charts, change coefficients at crossings and compose these solutions. The frame covariance proved above and uniqueness make this independent of subdivisions.

Iterating $U(t) = I - \int_0^t a(s)U(s)\mathrm{d}s$ gives

$$U(t) = I + \sum_{n\geq 1}(-1)^n \int_{0<t_1<\cdots<t_n<t} a(t_n)\cdots a(t_1)\mathrm{d}t_1\cdots\mathrm{d}t_n.$$

The nth term has norm at most $K^n t^n/n!$ when $\|a\| \leq K$.

Definition 21.14. The convergent series above is the *path-ordered exponential*.

Its norm is at most e^{Kt}, so no finite-time blow-up occurs. Later times stand on the left. In a unitary frame for a unitary connection, $\frac{\mathrm{d}}{\mathrm{d}t}(U^\dagger U) = 0$, so transport is unitary. For a matrix Lie group G, the equation $\dot{U}U^{-1} = -a(t) \in \mathfrak{g}$ is a right-invariant ODE on G whose G-valued solution extends over the path interval by uniform short-time existence at the identity and restarting by right translations on finitely many subintervals. Uniqueness of the linear ODE in $\mathrm{GL}(r,\mathbb{C})$ identifies this solution with the transport matrix, so transports and based-loop holonomies belong to G. For scalar A this reduces to $U(\gamma) = \exp(-\int_\gamma A)$ in a frame covering the path. This agrees with the U(1) transport of *Lie Groups*, Lecture 27.[2] The figure shows a parallel phase at successive parameter values.

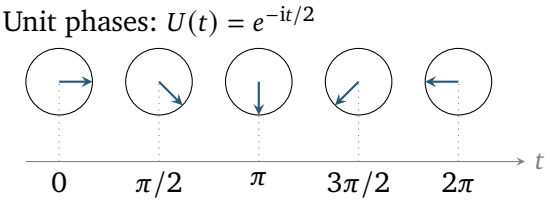


Figure 21.3: The circles show unit phases with $A = \mathrm{i}\mathrm{d}t/2$ and transport $U(t) = e^{-\mathrm{i}t/2}$. The phase turns clockwise and reaches -1 at $t = 2\pi$.

Definition 21.15. The *holonomy* $\mathrm{Hol}(\gamma)$ is the transport around a based loop.

Coefficient covariance $D'\sigma' = h^{-1}D\sigma$ and uniqueness show that changing the base frame conjugates it.

Definition 21.16. The trace of a holonomy is its *Wilson loop*.

Conjugation invariance of the trace makes the Wilson loop frame invariant. A loop crossing several frames requires their transition matrices as well.

Theorem 21.17 (Parallel transport · *proved*)**.** *Parallel transport is the unique solution*

$$\dot{U}(t) = -a(t)U(t), \qquad U(0) = I.$$

Under a change of frame it becomes

$$U'(\gamma) = h(\gamma(1))^{-1}U(\gamma)h(\gamma(0)).$$

For the counter-clockwise coordinate square of side ϵ based at 0,

$$\mathrm{Hol}(\square_\epsilon) = I - \epsilon^2 F_{xy}(0) + O(\epsilon^3).$$

[2] Both conventions use $D = \mathrm{d} + A$ and transport $\exp(-\int A)$.

Proof. The preceding construction proves existence, uniqueness and invertibility. The endpoint formula follows by differentiating $h(\gamma(t))^{-1}U(t)h(\gamma(0))$.

For a scalar connection, Stokes gives

$$\mathrm{Hol}(\square_\epsilon) = \exp\left(-\int_{[0,\epsilon]^2} F\right) = 1 - \epsilon^2 F_{xy}(0) + O(\epsilon^3).$$

For matrices, first-order terms from opposite edges still cancel. Transverse changes give the derivative terms in F_{xy}. Products of first-order terms give its commutator. We retain their order in the four edge expansions.

For the square, write $C = A_x(0)$ and $D = A_y(0)$. Write $C_x = \partial_x A_x(0)$ and $D_y = \partial_y A_y(0)$. Here D denotes a constant matrix only within this calculation. Starting along positive x, integrate each edge equation through second order. The series bound makes all terms with at least three integrals $O(\epsilon^3)$:

$$\begin{aligned}
U_1 &= I - \epsilon C + \frac{\epsilon^2}{2}(C^2 - C_x) + O(\epsilon^3),\\
U_2 &= I - \epsilon D + \epsilon^2(D^2/2 - \partial_x A_y(0) - D_y/2) + O(\epsilon^3),\\
U_3 &= I + \epsilon C + \epsilon^2(C^2/2 + C_x/2 + \partial_y A_x(0)) + O(\epsilon^3),\\
U_4 &= I + \epsilon D + \frac{\epsilon^2}{2}(D^2 + D_y) + O(\epsilon^3).
\end{aligned}$$

For example, on the second edge $a(t) = D + \epsilon\partial_x A_y(0) + tD_y + O(\epsilon^2)$, for $0 \le t \le \epsilon$. The first integral contributes $-\epsilon D - \epsilon^2\partial_x A_y(0) - \epsilon^2 D_y/2$. The iterated integral contributes $\epsilon^2 D^2/2$. The other edges follow from their signed tangents in the same way. The successive transport acts on the left, giving

$$\begin{aligned}
U_4U_3U_2U_1 &= I + \epsilon^2(\partial_y A_x(0) - \partial_x A_y(0) - [C, D]) + O(\epsilon^3)\\
&= I - \epsilon^2 F_{xy}(0) + O(\epsilon^3).
\end{aligned}$$

∎

The figure fixes the traversal order used in this calculation.

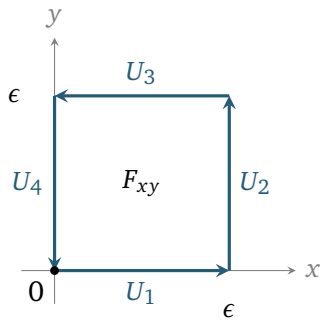


Figure 21.4: The path starts at 0 and traverses positive x, positive y, negative x, then negative y. Its transport is $U_4U_3U_2U_1$.

For the tangent connection, the square theorem measures the leading change $-\epsilon^2 R(\partial_x, \partial_y)$ of a transported vector. On a surface, choose positive coordinates orthonormal at the base point. The dictionary then gives $I + \epsilon^2 K(0)J + O(\epsilon^3)$, a rotation through the signed angle $K(0)\epsilon^2$ to leading order. This agrees with Lecture 18's curvature integral.

Recall that $\gamma \cdot \delta$ traverses γ first. Uniqueness therefore gives

$$U(\gamma \cdot \delta) = U(\delta)U(\gamma), \qquad U(\gamma^{-1}) = U(\gamma)^{-1}.$$

Transport reverses the order of the path product.

21.5 Flat connections and the Aharonov–Bohm phase

Definition 21.18. A connection is *flat* if $F = 0$.

This condition removes local loop curvature but need not remove holonomy around a hole.

Theorem 21.19 (Flat holonomy · *sketched; proved for U(1)*)**.** *For a flat connection, based-loop holonomy depends only on the based homotopy class. After choosing a base G-frame, the map*

$$[\gamma] \longmapsto \mathrm{Hol}(\gamma)^{-1}$$

is a homomorphism $\pi_1(M, x_0) \to G$, where G is the matrix structure group. For each based loop its conjugacy class is independent of the chosen base frame.

Proof for U(1). Write the based homotopy as $H(s,t)$, with t the path parameter. We subdivide its parameter square into rectangles on which transport depends only on endpoints. Each rectangle has unit boundary transport, so internal edges cancel. The extension to continuous paths makes this argument apply to continuous based homotopies. In a small ball with a unitary frame, $\mathrm{d}A = 0$. By Theorem 13.11, write $A = \mathrm{d}f$, with imaginary f. For endpoints p_0 and p_1 in this ball, transport is $\exp(f(p_0) - f(p_1))$. Equivalently, Stokes gives unit transport around every loop bounding a disk in that ball. The endpoint formula also defines local transport on continuous paths. Subdividing into these balls extends it to continuous paths, agreeing with ODE transport on piecewise smooth paths. Refinements do not change it, since endpoint factors cancel. The local frames $e_i e^{-f_i}$ have zero connection form. On a connected overlap they therefore differ by a constant phase, by the frame law. Either local formula gives the same fibre map.

The figure shows the parameter square, rather than a region embedded in M.

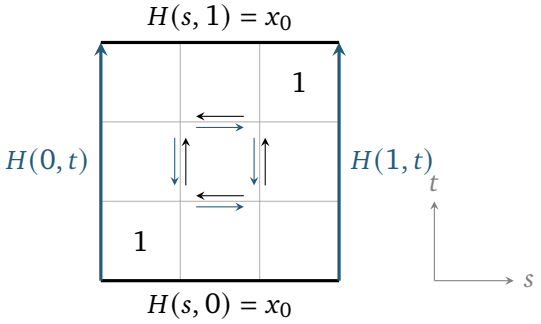


Figure 21.5: Each cell maps into one framed ball where $A = \mathrm{d}f$. After internal edges cancel in opposite pairs, only the two loop transports remain.

Given a continuous based homotopy, the Lebesgue lemma subdivides its parameter square into rectangles whose images lie in these balls. Choose a unit basis in the fibre at every grid vertex. Each rectangle has boundary product 1 by the local endpoint formula. Multiply the scalar boundary products. Every internal edge cancels with its reverse. The edges $t = 0$ and $t = 1$ are constant at the base point and have transport 1. The transports of the loops on $s = 0$ and $s = 1$ are therefore equal. Finally, taking inverses in the first-path-first composition rule gives the claimed homomorphism. This argument uses no global trivialization of the bundle. ∎

Lecture 5 assumed that a scalar statistical factor is homotopy invariant and multiplicative. By Theorem 21.19, inverse holonomy of a flat unitary line-bundle connection has both properties and is a unitary character of the fundamental group. Flatness remains a hypothesis on that statistical sector.

For a general matrix group, the variation of transport gives the corresponding sketch. Let $H(s,t)$ be a smooth based homotopy and $U(s,t)$ its transport from $t=0$ to t. In a pullback frame fixed along both endpoint edges, the based boundary conditions give

$$U(s,1)^{-1}\partial_s U(s,1) = -\int_0^1 U(s,t)^{-1}F(H_s,H_t)U(s,t)\mathrm{d}t.$$

To obtain it, differentiate $\partial_t U = -A(H_t)U$ in s and combine $\partial_s U + A(H_s)U$ before integrating in t. Patching frames and the passage to continuous homotopies are omitted here. If $F=0$, the formula makes the endpoint transport constant in s.

The angular one-form on the punctured plane is global although its angle is not:

$$\mathrm{d}\theta = \frac{-y\mathrm{d}x + x\mathrm{d}y}{x^2+y^2}, \qquad M = \mathbb{R}^2 \setminus \{0\}.$$

We model a long shielded solenoid by the removed point in this transverse plane. Its flux Φ is positive out of the page. The accessible exterior has zero electromagnetic field. A coherent source sends amplitudes through two slits, on opposite sides of the solenoid, to the same point on a screen. In this static model $\phi = 0$. The figure shows the two arms and a phase-resolved interference profile. The profiles assume equal unit arm magnitudes. Their apparatus phase difference is χ_{app}, and their flux phase is $\varphi_{\mathrm{AB}} = q\Phi/\hbar$.

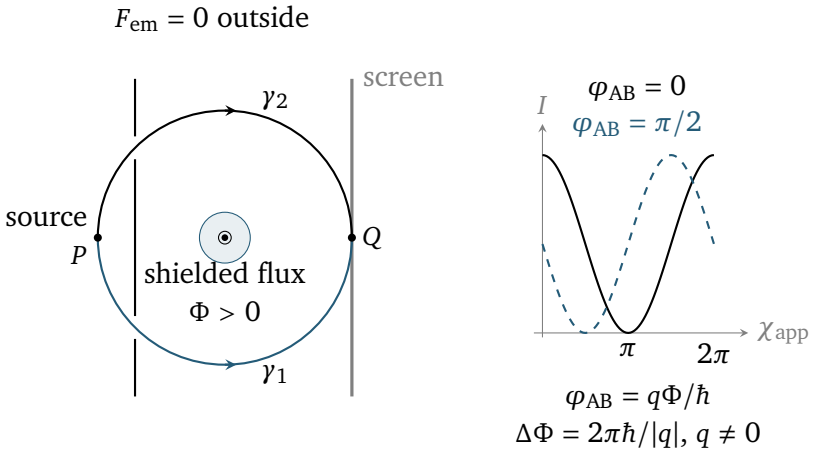


Figure 21.6: The field-free arms from source P to screen point Q form the counter-clockwise loop $\gamma_1 \cdot \gamma_2^{-1}$. The profiles plot $2+2\cos(\chi_{\mathrm{app}} + \varphi_{\mathrm{AB}})$ for $\varphi_{\mathrm{AB}} = 0$ and $\varphi_{\mathrm{AB}} = \pi/2$.

Example 21.20 (The Aharonov–Bohm connection)**.** Compute the holonomy on the trivial complex line over M with

$$A_{\mathrm{em}} = \frac{\Phi}{2\pi}\mathrm{d}\theta, \qquad A = -\frac{\mathrm{i}q\Phi}{2\pi\hbar}\mathrm{d}\theta.$$

Differentiation gives $\mathrm{d}(\mathrm{d}\theta) = 0$ on M, so $F = 0$ there. Let γ be the counter-clockwise unit circle, with $\theta(t) = 2\pi t$ and circulation 2π. Hence

$$\mathrm{Hol}(\gamma) = \exp\left(-\oint_\gamma A\right) = e^{\mathrm{i}q\Phi/\hbar}.$$

The loop does not bound a disk in M, so Stokes does not set this circulation to zero. The bundle is trivial, but the connection need not be gauge equivalent to the zero connection. Put $\alpha = q\Phi/(2\pi\hbar)$. On the angular cover $h = e^{i\alpha\theta}$ would cancel A. It descends to a single-valued map on M exactly when $\alpha \in \mathbb{Z}$. For an integer n, the genuine gauge map $h = e^{in\theta}$ changes α to $\alpha - n$ and leaves the holonomy unchanged. Thus these flat connections are distinguished by α modulo $\mathbb{Z}$.

The loop $\gamma_1 \cdot \gamma_2^{-1}$ winds once counter-clockwise, so its angular-form integral is also 2π by Lecture 3. Scalar transport gives $U(\gamma_1 \cdot \gamma_2^{-1}) = U(\gamma_1)U(\gamma_2)^{-1}$. By the coherent-arm physical input, the ratio of the two electromagnetic phases is $U(\gamma_1)U(\gamma_2)^{-1} = e^{iq\Phi/\hbar}$. For amplitudes c_1 and c_2 with the electromagnetic factors removed, the screen intensity is

$$\begin{aligned} I &= |c_1 U(\gamma_1) + c_2 U(\gamma_2)|^2 \\ &= |c_1|^2 + |c_2|^2 + 2\,\mathrm{Re}(c_1 \bar{c}_2 e^{iq\Phi/\hbar}). \end{aligned}$$

The phase is defined modulo 2π. The signed particle charge is q. For $q \neq 0$, the positive flux period is

$$\Delta\Phi = \frac{2\pi\hbar}{|q|}.$$

Changing flux changes the relative phase and hence shifts the interference fringes, although the field vanishes everywhere along both arms. The spatial displacement of the fringes also depends on the apparatus phases. The exterior curvature alone therefore does not determine the interference pattern. The holonomy retains the flux dependence. This is also the phase in *Quantum Mechanics II*, Lecture 3, for counter-clockwise forward hops on a ring.[3]

The phase lift in the figure displays the same holonomy on a cut angular interval.

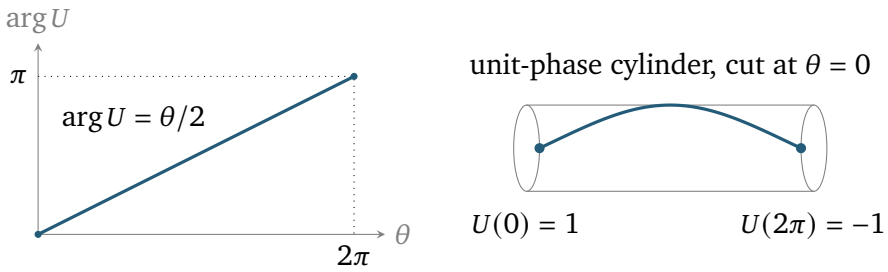


Figure 21.7: For $\alpha = 1/2$, transport along angle θ has phase $e^{i\theta/2}$. The two endpoints lie over the same base point but differ by the holonomy -1.

21.6 A constant nonabelian connection

The coefficient column σ is distinct from the Pauli matrices σ_x, σ_y and σ_z, recalled explicitly as

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

For a unit column $v \in \mathbb{C}^2$, recall the Bloch vector with components $r_a = v^\dagger \sigma_a v$, for $a = x, y, z$. The action of SU(2) on these vectors is the rotation map of Lecture 4.

[3] Nakahara, §10.5.3, (10.95)–(10.100), uses $\hbar = 1$. Our potential convention is $A_{\mathrm{conn}} = -iqA_{\mathrm{em}}/\hbar$.

Example 21.21 (Curvature from noncommuting transport)**.** Find the curvature and small-square holonomy of the connection on $\mathbb{R}^2 \times \mathbb{C}^2$ given, for $a, b \in \mathbb{R}$, by

$$A = C\mathrm{d}x + D\mathrm{d}y, \qquad C = -\frac{\mathrm{i}a}{2}\sigma_z, \qquad D = -\frac{\mathrm{i}b}{2}\sigma_x.$$

Both coefficient matrices are traceless and anti-Hermitian, so their transport lies in SU(2). Indeed, unitarity follows as above and $\frac{\mathrm{d}}{\mathrm{d}t}\det U = -\operatorname{tr}(A(\dot\gamma))\det U = 0$. The exterior derivative is zero, but $[\sigma_z, \sigma_x] = 2\mathrm{i}\sigma_y$ gives

$$F = [C, D]\mathrm{d}x \wedge \mathrm{d}y = -\frac{\mathrm{i}ab}{2}\sigma_y \mathrm{d}x \wedge \mathrm{d}y.$$

It is nonzero when $ab \neq 0$. The constant edge equations have exact solutions, so

$$\mathrm{Hol}(\square_\epsilon) = e^{\epsilon D}e^{\epsilon C}e^{-\epsilon D}e^{-\epsilon C} = I + \frac{\mathrm{i}ab\epsilon^2}{2}\sigma_y + O(\epsilon^3).$$

The coefficient is $-F_{xy}$, as the transport theorem requires. Thus vanishing ordinary derivatives of a nonabelian potential do not imply flatness. Under Lecture 4's SU(2) $\to$ SO(3) map, the positive x edge rotates the Bloch vector by $-a\epsilon$ about z. The positive y edge rotates it by $-b\epsilon$ about x. The two return edges have the opposite angles in the same traversal order. Their product is a rotation about y with leading angle $-ab\epsilon^2$. Rotations about different axes need not commute.

The figure shows exact Bloch directions for one finite square of the connection computed in the example.

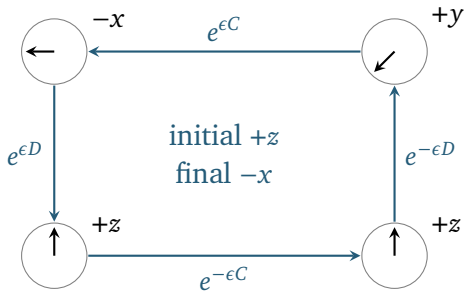


Figure 21.8: For $a = b = 1$ and $\epsilon = \pi/2$, the four transports send an initial Bloch direction $+z$ successively to $+z$, $+y$, $-x$ and $-x$. The final direction differs from the initial one despite the constant coefficients.

Self-check. Show that a pure gauge $A = h^{-1}\mathrm{d}h$ is flat and compute its transport.

Answer. The inverse derivative gives $\mathrm{d}(h^{-1}\mathrm{d}h) = -h^{-1}\mathrm{d}h \wedge h^{-1}\mathrm{d}h$, which cancels $A \wedge A$. This is the Maurer–Cartan identity for $\vartheta_{\mathrm{MC}} = h^{-1}\mathrm{d}h$. Differentiating $U(t) = h(\gamma(t))^{-1}h(\gamma(0))$ verifies the transport equation and its initial condition. If h is global, every closed loop has holonomy I. The Aharonov–Bohm cancellation uses only a local h unless $\alpha \in \mathbb{Z}$.

Self-check. On two successive segments, each parameterized by a unit interval, let the coefficient matrices be

$$a_1 = -\begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \qquad a_2 = -\begin{pmatrix} 0 & 0 \\ 3 & 0 \end{pmatrix}.$$

Compute the total transport. Then compute it in a new frame with endpoint values

$$h(0) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad h(1) = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}.$$

Here 0 and 1 label the endpoints of the complete path.

Answer. Each matrix squares to zero, so the edge transports are $I - a_1$ and $I - a_2$. Later transport multiplies on the left, giving

$$U = (I - a_2)(I - a_1) = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}, \qquad U' = h(1)^{-1} U h(0) = \begin{pmatrix} 1/2 & 3/2 \\ 3 & 10 \end{pmatrix}.$$

Reversing the factors instead gives $\begin{pmatrix} 7 & 2 \\ 3 & 1 \end{pmatrix}$, so the traversal order changes the answer.

Summary. A connection differentiates sections independently of their chosen frame. Its matrices change by $A' = h^{-1}Ah + h^{-1}\mathrm{d}h$. Curvature is $F = \mathrm{d}A + A \wedge A$, with $F' = h^{-1}Fh$ and $DF = 0$. Transport solves $\dot{U} = -A(\dot{\gamma})U$ and multiplies later segments on the left. A counter-clockwise small square has $\mathrm{Hol} = I - \epsilon^2 F_{xy} + O(\epsilon^3)$. A flat connection can retain nontrivial holonomy around a hole, as the Aharonov–Bohm phase $e^{\mathrm{i}q\Phi/\hbar}$ shows. Lecture 22 relates curvature flux to the transition winding of a line bundle.

Reading. Nakahara, §10.1.3–4, §10.3.4–5, §10.4.2 and §10.5.1, 10.5.3.

Lecture 22

Chern classes and the Dirac monopole

Objectives. *A connection can change while its bundle stays fixed. We extract a cohomology class from its curvature that survives this change. On a sphere its integral equals the transition winding and is therefore an integer. For a charged particle in a monopole field, this integer gives Dirac's quantization condition. The tangent sphere supplies a second example through Gauss–Bonnet.*

By the end of this lecture you will be able to:

- Prove that a line bundle's curvature class is independent of its connection.
- Recover its sphere Chern number from the oriented equatorial transition.
- Derive Dirac quantization from the two smooth monopole potentials.
- Relate the tangent sphere and the degree-n line to their Chern numbers.

Unit 22A Curvature classes and monopole flux

22.1 The curvature class of a line bundle

Let $E \to M$ be a complex line bundle with a Hermitian fibre metric and a unitary connection. In a unitary frame e_i, the connection's imaginary potential A_i has curvature $F_i = \mathrm{d}A_i$. Lecture 21's frame law on overlaps is

$$e_j = e_i g_{ij}, \qquad A_j = A_i + g_{ij}^{-1}\mathrm{d}g_{ij}, \qquad g_{ij} \colon U_i \cap U_j \to \mathrm{U}(1).$$

Locally write $g_{ij} = e^{\mathrm{i}\chi}$. Then $\mathrm{d}(g_{ij}^{-1}\mathrm{d}g_{ij}) = \mathrm{d}(\mathrm{i}\mathrm{d}\chi) = 0$. Thus the F_i agree and define a global imaginary two-form F. It is closed because $\mathrm{d}F_i = \mathrm{d}^2 A_i = 0$.

Definition 22.1. For a smooth map $f \colon N \to M$, the *pullback bundle* $f^*E \to N$ has fibre $E_{f(y)}$ at $y \in N$.

Pull back the product charts to $f^{-1}(U_i)$. Their transitions $g_{ij} \circ f$ satisfy the same cocycle, so Lecture 20 makes this a smooth bundle.

Definition 22.2. A *characteristic class* assigns a cohomology class to a bundle, invariant under bundle isomorphism and compatible with pullback.

Definition 22.3. Here the *real first Chern class* is

$$c_1(E) = \left[\frac{\mathrm{i}}{2\pi}F\right] \in H^2_{\mathrm{dR}}(M;\mathbb{R}).$$

The form inside the brackets is real since F is imaginary. We use de Rham classes throughout this lecture. Torsion classes, those killed by a nonzero integer, are not detected by these real forms. The integer sphere computation below needs no torsion comparison.

Theorem 22.4 (Chern–Weil · *cited; line-bundle first Chern class proved given partitions of unity*)**.** *Every complex line bundle has a Hermitian metric and a unitary connection. The class $c_1(E)$ is independent of both choices and is a characteristic class. More generally, on a unitary rank-r bundle, the forms $\operatorname{tr}(F^k)$ are closed and their classes in $H^{2k}_{\mathrm{dR}}(M;\mathbb{C})$ are independent of the connection, for every integer $k \geq 1$. Here powers use matrix wedge multiplication. This general assertion is cited.*

Proof for the line-bundle class. First fix a Hermitian metric and compare two unitary connections. In the same local unit frames, both potentials change by the same transition term. Their differences therefore glue to a global imaginary one-form B. Their curvatures satisfy

$$F^{(1)} - F^{(0)} = \mathrm{d}B, \qquad \frac{\mathrm{i}}{2\pi}(F^{(1)} - F^{(0)}) = \mathrm{d}\left(\frac{\mathrm{i}}{2\pi}B\right).$$

Thus changing the connection leaves the real cohomology class unchanged.

Next we construct the metric and a unitary connection. Lecture 12's cited partitions of unity supply the auxiliary choices. Take a locally finite partition ψ_i subordinate to product charts U_i. For $v, w \in E_x$, the weighted sum of their standard chartwise Hermitian products is

$$\langle v, w\rangle = \sum_i \psi_i(x)\langle v, w\rangle_i.$$

It is positive definite since the nonnegative weights sum to one. Their supports and local finiteness make the sum smooth. Normalize the local frames for this metric. In each unit frame let $\nabla^{(i)}$ be the connection with potential zero. The sum $\nabla s = \sum_i \psi_i \nabla^{(i)} s$ defines a global connection. Each summand extends smoothly by zero. The sum is locally finite. The Leibniz and metric product rules are affine in the connection, so their weighted sums give the same rules because $\sum_i \psi_i = 1$. Thus this connection is unitary.

Finally we compare connections unitary for possibly different metrics. Use the same complex frames for both, without requiring them to be unitary. The line frame law also holds for $h \in \mathbb{C}^\times$. Locally write $h = e^{b+\mathrm{i}\chi}$ with b and χ real. Then $h^{-1}\mathrm{d}h = \mathrm{d}b + \mathrm{i}\mathrm{d}\chi$ is closed, so curvature is unchanged in any such frame. Write the two potentials as $A_i^{(0)}$ and $A_i^{(1)}$ in these common frames. Their difference $B_i = A_i^{(1)} - A_i^{(0)}$ has no transition derivative term, so $B_j = B_i$. These complex one-forms glue to a global B. Both curvatures are imaginary. Taking real parts in their exact difference gives

$$F^{(1)} - F^{(0)} = \mathrm{d}B, \qquad \frac{\mathrm{i}}{2\pi}F^{(1)} - \frac{\mathrm{i}}{2\pi}F^{(0)} = \mathrm{d}\operatorname{Re}\left(\frac{\mathrm{i}}{2\pi}B\right).$$

Their real de Rham classes coincide. For a smooth map $f\colon N \to M$, the pullback bundle has pulled-back local potentials. Its curvature is f^*F because pullback commutes with d. Thus $c_1(f^*E) = f^*c_1(E)$. An arbitrary complex bundle isomorphism transports the metric and connection together. Independence of both choices then proves invariance even if the isomorphism does not preserve a previously chosen target metric. ∎

The general statement is Nakahara, Theorem 11.1 and (11.20)–(11.21). No Riemannian metric or orientation on the base entered this definition.

Definition 22.5. For a closed oriented surface Σ, the bundle's *first Chern number* is

$$C_1(E;\Sigma) = \frac{\mathrm{i}}{2\pi}\int_\Sigma F.$$

Changing the surface orientation reverses this number. Stokes makes it independent of the representative curvature in the same class.

22.2 Integrality on the sphere

Give S^2 its outward orientation, equal to the complex orientation of $\mathbb{C}\mathrm{P}^1$. Let D_N and D_S be the closed northern and southern hemispheres. Their common equator C has increasing azimuth ϕ as ∂D_N. The southern boundary is $-C$.

Lemma 22.6 (Cap frames · *proved*)**.** *A Hermitian line bundle with a unitary connection has a smooth unit frame on each pole-deleted chart of the sphere.*

Proof. Identify the sphere minus the opposite pole with $\mathbb{R}^2$. Choose one unit vector in the fibre over its coordinate origin and transport it along every radial segment to its endpoint. Lecture 21 gives unique unitary transport. Smooth dependence of linear ODE solutions on the endpoint gives a smooth frame. Locally in the endpoint, a finite chain of trivializations covers these compact segments, so this smooth-dependence argument also applies when no initial global frame is given. Each resulting chart contains its entire closed hemisphere. ∎

Theorem 22.7 (The sphere Chern number · *proved*)**.** *Let $E \to S^2$ be a Hermitian line bundle with a unitary connection. Choose smooth unitary frames over neighbourhoods of D_N and D_S. In the frame order $e_S = e_N g_{NS}$,*

$$C_1(E;S^2) = \frac{\mathrm{i}}{2\pi}\int_{S^2} F = \frac{1}{2\pi\mathrm{i}}\oint_C g_{NS}^{-1}\mathrm{d}g_{NS} = \operatorname{wind}(g_{NS}) \in \mathbb{Z}.$$

The figure represents the hemispheres as oriented disks.

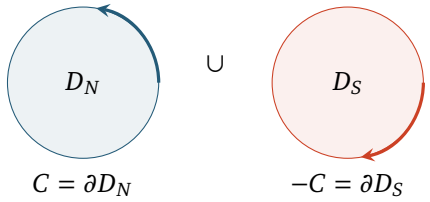


Figure 22.1: The two hemispheres induce opposite orientations on their common equator. Adding their Stokes formulas leaves $\oint_C (A_N - A_S)$, with both disks viewed from above the north pole.

Proof. In these frames, Stokes and the boundary orientations give

$$\frac{\mathrm{i}}{2\pi}\int_{S^2} F = \frac{\mathrm{i}}{2\pi}\left(\int_{D_N} \mathrm{d}A_N + \int_{D_S} \mathrm{d}A_S\right) = \frac{\mathrm{i}}{2\pi}\oint_C (A_N - A_S) = \frac{1}{2\pi\mathrm{i}}\oint_C g_{NS}^{-1}\mathrm{d}g_{NS}.$$

The last equality uses $A_S = A_N + g_{NS}^{-1}\mathrm{d}g_{NS}$ and $-\mathrm{i} = 1/\mathrm{i}$. Lecture 3's circle lifting theorem identifies the last integral with the integer winding. This proves the result for every such bundle, rather than only the monopole potentials. ∎

A different cap frame changes the transition by phases extending over the disks. Each such boundary phase has winding zero, so the integer is unchanged. Lecture 20's cited topological clutching classification applies to the principal $\mathrm{U}(1)$ bundle of unitary frames. It adds that equal integers give continuously isomorphic underlying bundles and that every integer occurs. The tautological line has transition $e^{-\mathrm{i}\phi}$, hence Chern number -1. The phase of the southern frame relative to the northern frame makes one or two positive turns in the following examples.

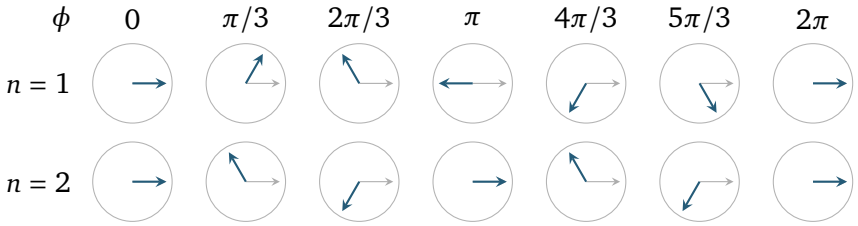


Figure 22.2: At each equatorial azimuth, the grey arrow represents e_N and the coloured arrow represents e_S relative to e_N. The transitions $e^{\mathrm{i}\phi}$ and $e^{2\mathrm{i}\phi}$ have windings +1 and +2.

22.3 The Dirac monopole

Definition 22.8. A *magnetic monopole* is a point source with nonzero outward magnetic flux through an enclosing sphere.

Let $q \neq 0$ be the signed particle charge and Φ_{m} the magnetic flux.

Physical input. We use the radial field model of Lecture 15 on the complement of the source. We describe a charged particle by local wave-function coefficients with the minimally coupled derivative $D = \mathrm{d} - (\mathrm{i}q/\hbar)A_{\mathrm{em}}$ of Lecture 21. Coefficients in overlapping frames describe one state and obey $\psi_S = g_{NS}^{-1}\psi_N$ for a single-valued phase transition. In the coherent-path model of Lecture 21, the electromagnetic phase along a path is the connection's transport phase. On a sphere enclosing the monopole, the real electromagnetic curvature is

$$F_{\mathrm{em}} = \frac{\Phi_{\mathrm{m}}}{4\pi}\sin\theta\,\mathrm{d}\theta\wedge\mathrm{d}\phi, \qquad \int_{S^2} F_{\mathrm{em}} = \Phi_{\mathrm{m}}.$$

For nonzero flux, Lecture 15's Stokes argument excludes a global potential on the sphere. The local coefficients therefore form a section of a line bundle rather than a function in a global frame. The *Wu–Yang potentials* on the two pole-deleted charts are

$$A_{\mathrm{em}}^{(N)} = \frac{\Phi_{\mathrm{m}}}{4\pi}(1-\cos\theta)\mathrm{d}\phi, \quad A_{\mathrm{em}}^{(S)} = -\frac{\Phi_{\mathrm{m}}}{4\pi}(1+\cos\theta)\mathrm{d}\phi,$$
$$A_N = -\frac{\mathrm{i}n}{2}(1-\cos\theta)\mathrm{d}\phi, \qquad A_S = +\frac{\mathrm{i}n}{2}(1+\cos\theta)\mathrm{d}\phi,$$

where $n = q\Phi_{\mathrm{m}}/(2\pi\hbar)$ is initially a real parameter. The conversion is $A = -(\mathrm{i}q/\hbar)A_{\mathrm{em}}$ and $F = -(\mathrm{i}q/\hbar)F_{\mathrm{em}}$. The imaginary connection potential includes $q/\hbar$ and is distinct from the real potential.

The polar coordinate ϕ fails at the poles, but these potentials are smooth on their claimed charts. For ambient unit-sphere coordinates (x, y, z),

$$(1 - \cos\theta)\mathrm{d}\phi = \frac{x\mathrm{d}y - y\mathrm{d}x}{1+z}, \qquad (1 + \cos\theta)\mathrm{d}\phi = \frac{x\mathrm{d}y - y\mathrm{d}x}{1-z}.$$

The first denominator is nonzero off the south pole, the second off the north pole. Both give the same smooth curvature $F = -(\mathrm{i}n/2)\sin\theta\mathrm{d}\theta \wedge \mathrm{d}\phi$.

The forms $(1 - \cos\theta)\mathrm{d}\phi$ and $(1 + \cos\theta)\mathrm{d}\phi$ equal $-4\pi a_U$ and $4\pi a_V$, respectively, for Lecture 13's cap primitives. Here U is the north chart and V is the south chart. Their rescalings satisfy

$$A_N = 2\pi\mathrm{i}n\, a_U, \qquad A_S = 2\pi\mathrm{i}n\, a_V, \qquad a_V - a_U = \frac{\mathrm{d}\phi}{2\pi}.$$

The de Rham calculation allowed every real multiple of the sphere class. For a charged bundle, the rescaled overlap difference must come from a single-valued phase transition.

Corollary 22.9 (Dirac quantization · *proved*)**.** *The Wu–Yang potentials define a unitary line-bundle connection for charge $q \neq 0$ exactly when*

$$q\Phi_{\mathrm{m}} = 2\pi\hbar n, \qquad n \in \mathbb{Z}.$$

Its sphere Chern number is n.

Proof. On the overlap, $A_S - A_N = \mathrm{i}n\mathrm{d}\phi$. The frame transition must therefore solve $g_{NS}^{-1}\mathrm{d}g_{NS} = \mathrm{i}n\mathrm{d}\phi$. It has the form $ce^{\mathrm{i}n\phi}$ for a constant $c \in \mathrm{U}(1)$. It is single-valued around the equator exactly when $e^{2\pi\mathrm{i}n} = 1$, which holds exactly for $n \in \mathbb{Z}$. For every such integer choose $g_{NS} = e^{\mathrm{i}n\phi}$ on the full overlap. This phase is smooth away from the poles. Lecture 20 reconstructs its smooth bundle. The connection law glues the two potentials. The sphere theorem gives the bundle's Chern number n. ∎

This is a conditional charge-quantization argument, not an assertion that monopoles exist. The potentials follow Nakahara, §10.5.2, with its imaginary-potential and north/south transition choices translated by the displayed conversion and frame order. For a fixed nonzero monopole flux, every allowed electric charge must therefore belong to $(2\pi\hbar/\Phi_{\mathrm{m}})\mathbb{Z}$.

Definition 22.10. The *Dirac string* is the singular negative z-axis of the radially extended northern potential in the one-potential description.

On a latitude traversed with increasing ϕ, its circulation is

$$\oint_{\theta=\text{constant}} A_{\mathrm{em}}^{(N)} = \frac{\Phi_{\mathrm{m}}}{2}(1 - \cos\theta) \longrightarrow \Phi_{\mathrm{m}} \quad \text{as } \theta \to \pi.$$

Lecture 21's holonomy formula therefore gives the limiting phase $e^{\mathrm{i}q\Phi_{\mathrm{m}}/\hbar}$ around the string. The string is invisible to this charged particle exactly when this phase is 1. This is the same quantization condition. The string is a gauge singularity, not an additional physical solenoid. For positive flux, the string flux in this singular description points toward the origin along the negative z-axis. The figure distinguishes this direction from the signed cap circulations.

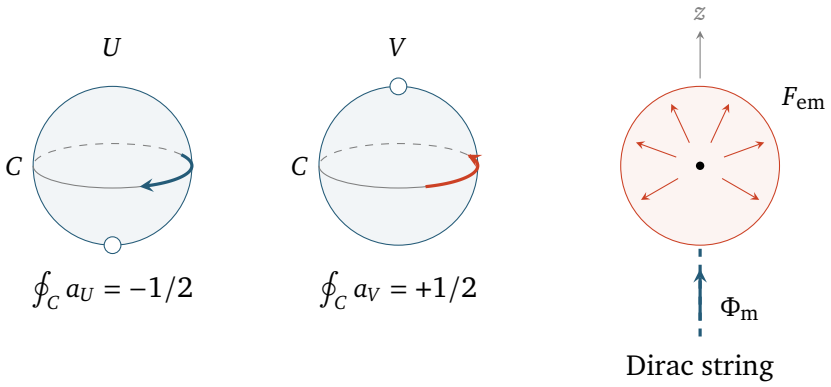


Figure 22.3: The cap primitives have opposite circulation signs along increasing azimuth. For positive Φ_{m}, the northern-potential description has a string on the negative z-axis with flux directed toward the origin.

A trial half-integer parameter gives smooth local potentials but fails the single-valued transition condition, as the phase paths show.

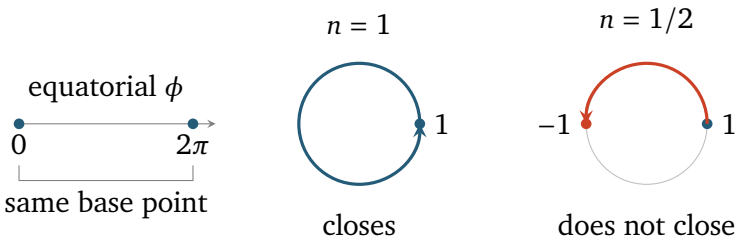


Figure 22.4: As ϕ runs from 0 to 2π, the phase $e^{\mathrm{i}n\phi}$ closes for $n = 1$. For the trial value $n = 1/2$, its final value is -1 instead of 1, so it cannot define a transition on the equator.

For $n = -1$, the monopole line is isomorphic to Lecture 20's tautological line. Both have transition $e^{-\mathrm{i}\phi}$ in the stated frame order. The principal circle bundle of its unit frames is the Hopf bundle $S^3 \to \mathbb{C}\mathrm{P}^1$. A unit frame is a unit vector in the tautological fibre, hence a point of S^3.

The cap integral need not be an integer. For a northern cap of polar radius $\alpha \in [0, \pi]$, the same curvature gives

$$\frac{\mathrm{i}}{2\pi} \int_{D_\alpha} F = \frac{n}{2}(1 - \cos\alpha).$$

The figure plots $n = 1$. Its hemisphere value is $1/2$, although the sphere number is 1.

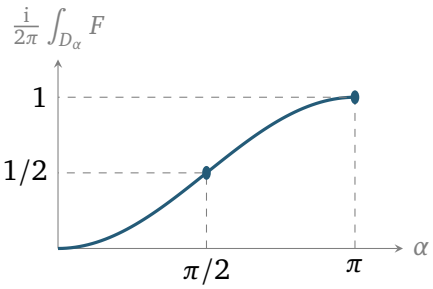


Figure 22.5: For the charge-one curvature, cap flux varies continuously with cap size. Only the full closed-sphere integral is constrained to the integer 1.

22.4 The tangent sphere as a complex line

On an oriented Riemannian surface Σ, let J rotate a positive orthonormal frame (e_1, e_2) by $Je_1 = e_2$ and $Je_2 = -e_1$. Positive orthonormal frame changes commute with this rotation, so it defines multiplication by i on every tangent plane. Its Hermitian fibre product is $\langle u, v\rangle = g(u, v) + \mathrm{i}g(Ju, v)$ for $p \in \Sigma$ and $u, v \in T_p\Sigma$. Lecture 17's Levi-Civita connection has $\nabla e_1 = -\omega^1{}_2 e_2$ and $\mathrm{d}\omega^1{}_2 = K\mathrm{d}\mathcal{A}$, where $\mathrm{d}\mathcal{A}$ denotes the oriented area form. This area symbol is distinct from the exterior derivative $\mathrm{d}A$ of a connection potential. The Levi-Civita connection's skew matrix commutes with J, hence the connection is complex-linear and unitary on this complex line.

Proposition 22.11 (The tangent Chern number · *proved*). *For a closed oriented Riemannian surface Σ, its tangent complex line has*

$$A = -\mathrm{i}\omega^1{}_2, \qquad F = -\mathrm{i}K\mathrm{d}\mathcal{A}, \qquad C_1(T\Sigma;\Sigma) = \frac{1}{2\pi}\int_\Sigma K\mathrm{d}\mathcal{A} = \chi(\Sigma).$$

In particular, $C_1(TS^2; S^2) = 2$.

Proof. The complex frame is e_1, with $e_2 = \mathrm{i}e_1$. Thus $\nabla e_1 = -\mathrm{i}\omega^1{}_2 e_1$, proving the connection formula. Differentiate and use Lecture 17's surface structure equation to obtain the curvature. Multiplication by $\mathrm{i}/(2\pi)$ gives $K\mathrm{d}\mathcal{A}/(2\pi)$. Lecture 18's Gauss–Bonnet theorem evaluates its integral as $\chi(\Sigma)$. For the sphere the Euler characteristic is 2, as required. ∎

This agrees with Lecture 20's tangent-frame winding +2. The inverse component transition winds −2. It is not the transition used in the Chern-number theorem.

22.5 Higher Chern classes

For higher-rank bundles, curvature matrices change by conjugation. Invariant polynomials give global scalar forms from these local matrices. The degree-four form gives the second Chern number used for nonabelian fields in Lecture 23. For a unitary rank-r bundle, use the defining $r \times r$ matrix curvature F. The frame law $F' = h^{-1}Fh$ and conjugation invariance of the determinant make the following form global. Even-degree forms commute as scalar entries.

Definition 22.12. The *total Chern form* is defined by

$$c(F) = \det\left(I + \frac{\mathrm{i}F}{2\pi}\right) = 1 + c_1(F) + c_2(F) + \cdots + c_r(F).$$

The component $c_j(F)$ has degree $2j$ and vanishes when $2j > \dim M$. The corresponding cohomology classes are the *Chern classes*.

The invariant-polynomial form of Chern–Weil, Nakahara Theorem 11.1, gives their closedness and connection independence. It applies to each determinant coefficient, not only the traces of powers. For two connections, their difference $B = A^{(1)} - A^{(0)}$ is an endomorphism-valued one-form. With $A_t = A^{(0)} + tB$ and curvature F_t, the cited relative transgression expresses the difference as an exterior derivative:

$$\operatorname{tr}\big((F^{(1)})^k\big) - \operatorname{tr}\big((F^{(0)})^k\big) = \mathrm{d}\left(k\int_0^1 \operatorname{tr}(B \wedge F_t^{k-1})\mathrm{d}t\right).$$

The trace expression inside is global, although the individual potentials are local. Nakahara's (11.20)–(11.21) supplies this assertion. Expanding the determinant through degree four gives

$$c_1(F) = \frac{\mathrm{i}}{2\pi}\operatorname{tr} F, \qquad c_2(F) = \frac{\operatorname{tr}(F\wedge F) - \operatorname{tr}F \wedge \operatorname{tr}F}{8\pi^2}.$$

This uses the elementary determinant coefficient $((\operatorname{tr}X)^2 - \operatorname{tr}(X^2))/2$ with $X = \mathrm{i}F/(2\pi)$. For an $\mathrm{SU}(r)$ connection, $\operatorname{tr}F = 0$, so $c_2(F) = \operatorname{tr}(F\wedge F)/(8\pi^2)$. Lecture 23 uses this normalization for its second Chern number.

Example 22.13 (A rank-two Chern form)**.** On the trivial rank-two bundle over $\mathbb{R}^4$, take the diagonal unitary potential

$$A = \operatorname{diag}(-2\pi\mathrm{i}x^1\mathrm{d}x^2, -2\pi\mathrm{i}x^3\mathrm{d}x^4).$$

Each diagonal one-form wedges with itself to zero, so $A \wedge A = 0$. Its curvature and total Chern form are

$$F = \operatorname{diag}(-2\pi\mathrm{i}\mathrm{d}x^1\wedge\mathrm{d}x^2, -2\pi\mathrm{i}\mathrm{d}x^3\wedge\mathrm{d}x^4),$$
$$c(F) = (1 + \mathrm{d}x^1\wedge\mathrm{d}x^2)(1 + \mathrm{d}x^3\wedge\mathrm{d}x^4).$$

The degree-four term is $c_2(F) = \mathrm{d}x^1\wedge\mathrm{d}x^2\wedge\mathrm{d}x^3\wedge\mathrm{d}x^4$. It equals $\mathrm{d}(x^1\mathrm{d}x^2\wedge\mathrm{d}x^3\wedge\mathrm{d}x^4)$, so its cohomology class is zero, as required for this trivial bundle.

Self-check. Show that a trivial Hermitian line bundle over a closed oriented surface has Chern number zero. Deduce what a nonzero Chern number says about the existence of a global unit frame.

Answer. Choose a global unit frame, so the connection has a global one-form A and $F = \mathrm{d}A$. Stokes gives $\int_\Sigma F = \int_{\partial\Sigma} A = 0$ because $\partial\Sigma$ is empty. A nonzero Chern number therefore excludes a global unit frame.

Self-check. A unitary line-bundle connection has cap potentials whose restrictions to the positive equator $C = \partial D_N$ are

$$A_N|_C = -\mathrm{i}(2+\cos\phi)\mathrm{d}\phi, \qquad A_S|_C = \mathrm{i}(1+\cos\phi)\mathrm{d}\phi.$$

Find one compatible equatorial frame transition and the sphere Chern number. Compute both hemisphere contributions. Explain why swapping the cap order leaves the sphere number unchanged. Decide whether $h = e^{\mathrm{i}\phi}$ is a smooth frame change on the whole northern chart.

Answer. The difference is $A_S - A_N = \mathrm{i}(3+2\cos\phi)\mathrm{d}\phi$. Thus $g_{NS} = e^{\mathrm{i}(3\phi+2\sin\phi)}$ satisfies the frame law. Its lifted phase increases by 6π, so the sphere number is 3. Stokes gives the two cap contributions

$$\frac{\mathrm{i}}{2\pi}\int_{D_N} F = 2, \qquad \frac{\mathrm{i}}{2\pi}\int_{D_S} F = 1.$$

Swapping caps inverts the transition and reverses the equator orientation. These two sign changes leave the number 3 unchanged. The proposed frame change has limit 1 along $\phi = 0$ and -1 along $\phi = \pi$ at the north pole, so it does not extend there.

Self-check. State in one sentence what the first Chern class is and why it was introduced. Explain why a curvature integral over a cap does not have the same integrality guarantee.

Answer. The first Chern class is the real cohomology class $[\mathrm{i}F/(2\pi)]$ of a line bundle, introduced to retain information about the bundle when its connection changes. For a fixed metric, two connections give curvatures differing by $\mathrm{d}B$ for a global imaginary one-form B, so the classes coincide. The sphere integrality proof uses a closed equator to measure transition winding. A cap has a boundary, and its integral alone measures no such winding. For $n = 1$, the hemisphere number is $1/2$.

Unit 22B Holomorphic monopole lines and their sections

22.6 The degree-n holomorphic line

We express the monopole line in frames where a covariant differential equation becomes ordinary holomorphicity of the coefficients. Global holomorphic sections must also satisfy the transition law at the opposite pole. We construct two such sections of $\mathcal{O}(1)$ below. Lecture 28 uses the same frames to turn a zero-mode equation into a count of sections.

Definition 22.14. In a complex chart $z = x + \mathrm{i}y$, a smooth function is holomorphic if $\partial_{\bar z} f = 0$, where $\partial_{\bar z} = \frac{1}{2}(\partial_x + \mathrm{i}\partial_y)$.

Definition 22.15. A *holomorphic line bundle* has local frames with holomorphic transition functions.

Definition 22.16. The $(0,1)$ *part* of a complex one-form is its $\mathrm{d}\bar z$ term. Write $\bar\partial f = (\partial_{\bar z} f)\mathrm{d}\bar z$.

The complex differentials satisfy

$$\mathrm{d}z = \mathrm{d}x + \mathrm{i}\mathrm{d}y, \qquad \mathrm{d}\bar z = \mathrm{d}x - \mathrm{i}\mathrm{d}y, \qquad \mathrm{d}f = (\partial_z f)\mathrm{d}z + (\partial_{\bar z} f)\mathrm{d}\bar z,$$

where $\partial_z = \frac{1}{2}(\partial_x - \mathrm{i}\partial_y)$. Use the projective coordinates z and $w = 1/z$ of Lecture 20. Here z denotes the complex coordinate rather than the ambient height used in §22.3.

Definition 22.17. Define the *degree* of a line over $\mathbb{C}\mathrm{P}^1$ to be its Chern number in the complex orientation. Define the *degree-n line* $O(n)$ for $n \in \mathbb{Z}$ by holomorphic frames with

$$e_S^{\mathrm{hol}} = e_N^{\mathrm{hol}} z^n.$$

Definition 22.18. A local section is *holomorphic* exactly when its holomorphic-frame coefficient satisfies $\partial_{\bar{z}} f = 0$. A global holomorphic section has such coefficients on every chart, obeying the section transition law.

Example 22.19 (Two global sections of $O(1)$)**.** Since $e_S^{\mathrm{hol}} = e_N^{\mathrm{hol}} z$, a section has coefficients related by $f_S(w) = w f_N(1/w)$. The northern coefficient $f_N(z) = 1$ therefore gives $f_S(w) = w$. It defines a global holomorphic section s_0, with its only zero at the south pole. The northern coefficient $f_N(z) = z$ gives $f_S(w) = 1$. This section s_1 has its only zero at the north pole. Their linear combinations have coefficients

$$f_N(z) = a + bz, \qquad f_S(w) = aw + b, \qquad a, b \in \mathbb{C}.$$

Both coefficients are holomorphic on their whole charts and satisfy the transition law. Evaluation at $z = 0$ and $z = 1$ shows that s_0 and s_1 are linearly independent.

The diagram shows the relation between holomorphic and unitary frames.

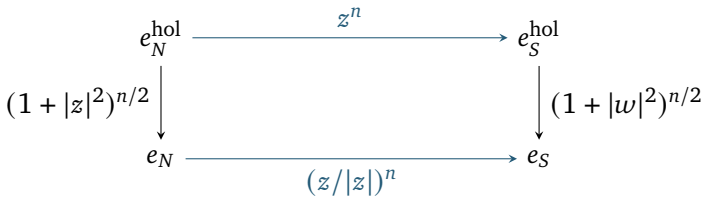


Figure 22.6: Positive real rescalings turn holomorphic frames into unitary frames. Their transition is the phase $(z/|z|)^n$, with winding n.

Example 22.20 (The monopole line as $O(n)$)**.** Identify the Wu–Yang monopole line with $O(n)$, for $n \in \mathbb{Z}$. Equip these frames with squared norms $\|e_N^{\mathrm{hol}}\|^2 = (1+|z|^2)^{-n}$ and $\|e_S^{\mathrm{hol}}\|^2 = (1+|w|^2)^{-n}$. They agree on the overlap since $|w| = 1/|z|$ and $e_S^{\mathrm{hol}} = e_N^{\mathrm{hol}} z^n$. The unitary frames are

$$e_N = (1+|z|^2)^{n/2} e_N^{\mathrm{hol}}, \qquad e_S = (1+|w|^2)^{n/2} e_S^{\mathrm{hol}}.$$

Their transition is $(z/|z|)^n$. In the northern unit frame the Wu–Yang connection is

$$A_N = -\frac{n}{2} \frac{\bar{z}\mathrm{d}z - z\mathrm{d}\bar{z}}{1+|z|^2}.$$

Changing to $e_N^{\mathrm{hol}} = (1+|z|^2)^{-n/2} e_N$ gives

$$A_N^{\mathrm{hol}} = A_N - \frac{n}{2}\mathrm{d}\log(1+|z|^2) = -\frac{n\bar{z}}{1+|z|^2}\mathrm{d}z.$$

In the southern holomorphic frame the same calculation gives

$$A_S^{\mathrm{hol}} = -\frac{n\bar{w}}{1+|w|^2}\mathrm{d}w = A_N^{\mathrm{hol}} + n\frac{\mathrm{d}z}{z} \quad \text{on the overlap.}$$

Both expressions have no antiholomorphic differential term and are smooth at their own chart origin. Thus the $(0,1)$ part of the covariant derivative is ordinary $\bar\partial$ in these frames. This equips the monopole line with the displayed holomorphic structure. Its Chern number is n by the equatorial winding theorem. For the coefficient ψ in the unit frame the same equation is $(\partial_{\bar z} + nz/[2(1+|z|^2)])\psi = 0$. This local equation will reappear in Lecture 28 with its stated spinor twist.

With the displayed metric for $O(1)$, the two sections have norms $\|s_0\| = \cos(\theta/2)$ and $\|s_1\| = \sin(\theta/2)$. The figure shows these lengths in separate fibres above points of the meridian $\phi = 0$.

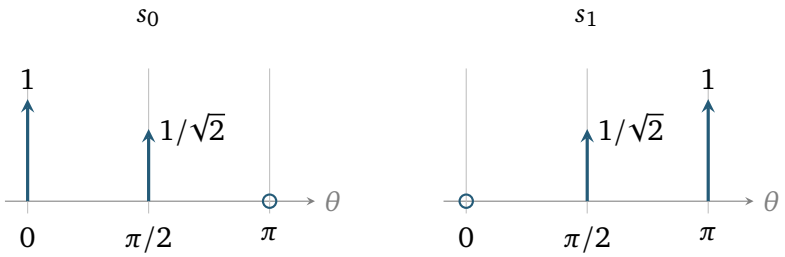


Figure 22.7: The two sections of $O(1)$ vanish at opposite poles. Vertical arrows lie in the line fibres, with lengths given by the displayed norms.

The cited holomorphic classification on $\mathbb{C}\mathrm{P}^1$ says that every holomorphic line is isomorphic to exactly one $O(n)$, with n its degree. See Gunning, *Some Topics in the Function Theory of Compact Riemann Surfaces*, preliminary version (2018), Theorem 2.2, (2.3) and Corollary 2.3, pp. 34–35. This is a holomorphic result in addition to Lecture 20's continuous clutching classification. The tautological frames $(1, z)^T$ and $(w, 1)^T$ have holomorphic transition z^{-1}. The tangent coordinate frames have holomorphic transition $-z^2$ by Lecture 20. Their degrees and the cited classification identify the tautological line with $O(-1)$ and the tangent line with $O(2)$.

Summary. A line bundle's real first Chern class is $[\mathrm{i}F/(2\pi)]$ and is independent of the connection. On the outward sphere its integral is the winding of the frame transition $e_S = e_N g_{NS}$ along the northern boundary. The monopole condition is $q\Phi_\mathrm{m} = 2\pi\hbar n$ with $n \in \mathbb{Z}$. The tangent sphere has Chern number 2, while $O(n)$ has number n. A given complex bundle's class needs no base metric or orientation. The tangent example uses an oriented metric to define its complex structure. Signed surface integrals need the orientation. Lecture 23 uses the second Chern form to study nonabelian fields.

Reading. Nakahara, §10.5.2 and §11.1–11.2. Tu, *Differential Geometry*, §25.8, p. 235. Gunning, Theorem 2.2, (2.3) and Corollary 2.3, pp. 34–35, for the holomorphic classification.

Lecture 23

Yang–Mills theory and instantons

Objectives. *Lecture 21 supplied the geometry of a gauge field. Lecture 22, in its section on higher Chern classes, gave $c_2(F) = \operatorname{tr}(F \wedge F)/(8\pi^2)$ for an $\mathrm{SU}(r)$ connection. We now choose an action and derive its field equation. For an $\mathrm{SU}(2)$ connection extending over the four-sphere, the curvature also gives an integer. This integer bounds the action and identifies solutions of a first-order equation. The BPST field fixes the orientation and normalization in a concrete example.*

By the end of this lecture you will be able to:

- Derive the vacuum Yang–Mills equation with the fundamental trace normalization.
- Prove the local Chern–Simons identity and distinguish it from global exactness.
- Complete the Euclidean action square and identify its equality condition.
- Compute the BPST action and charge and determine their behavior under orientation reversal.

Unit 23A The Yang–Mills field equation

23.1 The Yang–Mills action and equation

The passive frame change $e' = eh$ has $A^h = h^{-1}Ah + h^{-1}\mathrm{d}h$. We use this frame convention throughout. The derivative term distinguishes a potential from a form that transforms by conjugation alone.

Definition 23.1 (Adjoint-valued forms). Lie-algebra-valued forms that transform by conjugation, with no derivative term, are called *adjoint-valued*.

We use a connection for $\mathrm{SU}(r)$, with $r \geq 2$, in its fundamental representation. We choose anti-Hermitian generators T^a normalized by $\operatorname{tr}(T^aT^b) = -\delta^{ab}/2$.[1] For $\mathrm{SU}(2)$, these are $T^a = -\mathrm{i}\sigma_a/2$ with the Pauli matrices of Lecture 21.[2] For spatial labels $a = 1$, $a = 2$ and $a = 3$, the indexed Pauli matrices are respectively σ_x, σ_y and σ_z. In Minkowski coordinates $x^0 = t$, the metric has signature $(+---)$ and the volume orientation is $\mathrm{d}t \wedge \mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3$. The real component coefficients are defined by

$$A = T^a A^a_\mu \mathrm{d}x^\mu, \qquad F = \mathrm{d}A + A \wedge A = \frac{1}{2} T^a F^a_{\mu\nu} \mathrm{d}x^\mu \wedge \mathrm{d}x^\nu.$$

[1] The Hermitian-potential dictionary is $A = -\mathrm{i}A_{\mathrm{Tong}}$ and $F = -\mathrm{i}F_{\mathrm{Tong}}$ for Tong's unrescaled matrices. His rescaling $A_{\mathrm{phys}} = A_{\mathrm{Tong}}/g_{\mathrm{YM}}$ instead gives $A = -\mathrm{i}g_{\mathrm{YM}}A_{\mathrm{phys}}$. Tong, §2.1, uses the positive Hermitian trace in his component action.

[2] The real basis from *Lie Groups and Their Representation Theory* is related to these generators by $(\mathrm{i}\sigma_z, \mathrm{i}\sigma_y, \mathrm{i}\sigma_x) = (-2T^3, -2T^2, -2T^1)$. That basis has cyclic bracket factor 2. The generators T^a used here have cyclic bracket factor one.

Here a runs over the $r^2 - 1$ generators. Spacetime indices run from 0 to 3. The coupling $g_{\mathrm{YM}} > 0$ is a scalar, distinct from the metric g. We write vol for the metric volume form denoted by μ in Lecture 15. Assume the displayed action integrals converge, or work on a bounded region with variations supported in its interior. The curvature transforms by conjugation under a frame change. Taking a trace of its Hodge pairing gives a gauge-invariant four-form. This quadratic expression generalizes the Maxwell action of Lecture 19. The term $A \wedge A$ in the curvature makes the nonabelian theory nonlinear.

Physical input. In classical Yang–Mills theory, the gauge field is a connection. Its vacuum dynamics are the stationary points of the action defined below, under compactly supported variations. For gauge group SU(3), this is the gauge-field part of the theory of the strong interaction. These physical inputs are described in Tong, §2.1–2.1.1.

Definition 23.2 (Yang–Mills action)**.** The *Yang–Mills action* is the following metric-dependent functional in Minkowski signature:

$$S_M[A] = \frac{1}{g_{\mathrm{YM}}^2} \int_M \operatorname{tr}(F \wedge \star F) = -\frac{1}{4g_{\mathrm{YM}}^2} \int_M F^a_{\mu\nu} F^{a\mu\nu} \operatorname{vol}.$$

The covariant derivative on an endomorphism-valued k-form B is $DB = \mathrm{d}B + A \wedge B - (-1)^k B \wedge A$. Here k denotes the form degree.

Theorem 23.3 (The field equation · *proved*)**.** *The Minkowski action S_M is gauge invariant. Its stationary points under arbitrary smooth compactly supported connection variations satisfy $D\star F = 0$.*

Proof. The Hodge pairing of two two-forms contracts their components with a factor $1/2$. Multiplying this by $\operatorname{tr}(T^a T^b) = -\delta^{ab}/2$ gives the coefficient $-1/4$. On overlaps and under a frame change, $F^h = h^{-1} F h$. The star acts only on differential forms, so conjugation and cyclicity of the trace make the integrand global and gauge invariant.

The difference of two connections transforms by conjugation, with no derivative term. Let $a = \delta A$ be a compactly supported adjoint-valued one-form. This a denotes a variation, whereas the superscript in T^a labels a generator. Differentiate $\mathrm{d}(A + \varepsilon a) + (A + \varepsilon a)^2$ at $\varepsilon = 0$. The result is $\delta F = \mathrm{d}a + A \wedge a + a \wedge A = Da$. Symmetry of the two-form pairing gives

$$\delta S_M = \frac{2}{g_{\mathrm{YM}}^2} \int_M \operatorname{tr}(Da \wedge \star F),$$
$$\operatorname{tr}(Da \wedge \star F) = \mathrm{d} \operatorname{tr}(a \wedge \star F) + \operatorname{tr}(a \wedge D\star F).$$

The second line follows from the graded product rule. The connection commutators cancel inside the trace. Stokes removes the compactly supported exact term. The remaining pairing with every a vanishes exactly when $D\star F = 0$. The trace pairing is nondegenerate on $\mathfrak{su}(r)$, so it detects every component. ∎

With matter, define the connection-valued current three-form $\mathcal{J}_{\mathrm{conn}}$ by $\delta S_{\mathrm{matter}} = -(2/g_{\mathrm{YM}}^2) \int_M \operatorname{tr}(a \wedge \mathcal{J}_{\mathrm{conn}})$. The same variation then gives $D\star F = \mathcal{J}_{\mathrm{conn}}$.

For U(1) the generator is $-\mathrm{i}$, whose trace square is -1. We use the different prefactor $1/(2g^2)$, where $g = q/\hbar \neq 0$ is signed. This abelian coefficient differs from the positive nonabelian coupling g_{YM}. Lecture 21's physical-potential convention gives

$$A = -\frac{\mathrm{i}q}{\hbar} A_{\mathrm{em}}, \qquad F = -\frac{\mathrm{i}q}{\hbar} F_{\mathrm{em}}, \qquad S_M = \frac{1}{2g^2} \int_M F \wedge \star F = -\frac{1}{2} \int_M F_{\mathrm{em}} \wedge \star F_{\mathrm{em}}.$$

Let $\mathcal{J} = \star j$ be Lecture 15's real current and set $\mathcal{J}_{\mathrm{conn}} = -(iq/\hbar)\mathcal{J}$. For this prefactor, with $a = \delta A$, define the matter current by

$$\delta S_{\mathrm{matter}} = -\frac{1}{g^2}\int_M a \wedge \mathcal{J}_{\mathrm{conn}} = \int_M \delta A_{\mathrm{em}} \wedge \mathcal{J}.$$

The equation $\mathrm{d}\star F = \mathcal{J}_{\mathrm{conn}}$ becomes Lecture 15's $\mathrm{d}\star F_{\mathrm{em}} = \mathcal{J}$. The real current and the imaginary connection current are distinct.

Self-check. Show directly that $\mathrm{tr}(F \wedge \star F)$ is invariant under a frame change. Explain why this does not require the two local potentials to be equal.

Answer. The transformed product is $h^{-1}(F \wedge \star F)h$, since the star commutes with matrix conjugation. Its trace equals the original trace by cyclicity. The potentials obey the affine frame law, which also contains $h^{-1}\mathrm{d}h$. Gauge invariance concerns the traced curvature expression, rather than equality of potentials.

Unit 23B Euclidean duality, Chern–Simons and the bound

23.2 Euclidean action and duality

We now study a Riemannian variational problem on an oriented four-manifold. Its positive action compares fields with the same second Chern number. We replace the Minkowski metric by a positive Riemannian metric.

Definition 23.4 (Euclidean action). The *Euclidean action* is

$$S_E[A] = -\frac{1}{g_{\mathrm{YM}}^2}\int_M \mathrm{tr}(F \wedge \star F).$$

The minus sign makes the density nonnegative. In an orthonormal form basis, the matrix pairing is $-\mathrm{tr}(X^2) \geq 0$ for anti-Hermitian X. We name the fourth coordinate $\tau = x^4$ and use the spatial-first orientation $\mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3 \wedge \mathrm{d}x^4$ on $\mathbb{R}^4$. It is the negative of the time-first orientation. The coordinate τ belongs to this Riemannian problem.

Physical input. In the Euclidean path-integral formulation of quantum gauge theory, stationary Euclidean fields contribute to quantum amplitudes with a leading semiclassical weight $\sim e^{-S_E/\hbar}$. This is cited physics from Tong, §2.3.1–2.3.3, with $\hbar$ restored. Fluctuations supply additional factors, so this weight is not a transition probability. Interpreting τ as Euclidean time uses this additional formulation of the quantum theory.

Lecture 15 gives $\star^2 = 1$ on Euclidean two-forms.

Definition 23.5 (Duality). A curvature is *self-dual* if $F = \star F$ and *anti-self-dual* if $F = -\star F$.

Its orthogonal parts $F_+ = (F + \star F)/2$ and $F_- = (F - \star F)/2$ obey $\star F_+ = F_+$ and $\star F_- = -F_-$. Either duality equation implies the vacuum field equation, since $D\star F = \pm DF = 0$ by Lecture 21's Bianchi identity. Real Minkowski two-forms have $\star^2 = -1$. Imposing either real duality equation there forces $F = 0$.

Reversing orientation changes the star's sign and also changes integration's sign. Their product leaves S_E unchanged. The action depends on the metric through the star. An integral of a fixed four-form instead changes sign with orientation.

23.3 Chern–Simons and the second Chern number

In Lecture 22, the local identity $F = \mathrm{d}A$ turned a curvature integral over two caps into an integral of their potential difference along the equator. For the second Chern form, we seek a local three-form whose derivative is $\operatorname{tr}(F \wedge F)$. Its boundary integral will relate the four-dimensional charge to an equatorial transition. For matrix-valued forms of degrees p and q, cyclicity becomes $\operatorname{tr}(B \wedge C) = (-1)^{pq} \operatorname{tr}(C \wedge B)$. This follows by interchanging the scalar differential forms in each matrix entry.

Definition 23.6 (Chern–Simons form)**.** In a local frame define the *Chern–Simons three-form* by

$$\mathrm{CS}(A) = \operatorname{tr}\left(A \wedge \mathrm{d}A + \frac{2}{3} A \wedge A \wedge A\right).$$

The coefficient $2/3$ compensates for the derivative of the cubic term. We denote the Maurer–Cartan form by ϑ_{MC}, distinct from a coframe θ^a.

Theorem 23.7 (The local transgression · *sketched; local identities proved*)**.** *The local forms satisfy*

$$\mathrm{d}\,\mathrm{CS}(A) = \operatorname{tr}(F \wedge F), \qquad \mathrm{CS}(\vartheta_{\mathrm{MC}}) = -\frac{1}{3} \operatorname{tr}(\vartheta_{\mathrm{MC}}^3), \qquad \vartheta_{\mathrm{MC}} = h^{-1}\mathrm{d}h.$$

For a general frame change, the local gauge law is

$$\mathrm{CS}(A^h) = \mathrm{CS}(A) - \mathrm{d} \operatorname{tr}(\mathrm{d}h h^{-1} \wedge A) - \frac{1}{3} \operatorname{tr}(\vartheta_{\mathrm{MC}}^3).$$

The resulting global characteristic-class interpretation is cited.

Proof of the local identities. Write matrix wedge products as juxtaposition. The product rule gives $\mathrm{d} \operatorname{tr}(A\mathrm{d}A) = \operatorname{tr}((\mathrm{d}A)^2)$. For the cubic term it gives

$$\mathrm{d} \operatorname{tr}(A^3) = \operatorname{tr}(\mathrm{d}A\, A^2 - A\, \mathrm{d}A\, A + A^2 \mathrm{d}A) = 3 \operatorname{tr}(\mathrm{d}A\, A^2).$$

Moving the first A past the remaining total degree three changes sign. This proves the middle term's sign and also gives $\operatorname{tr}(A^4) = -\operatorname{tr}(A^4) = 0$. Consequently

$$\operatorname{tr}(F^2) = \operatorname{tr}((\mathrm{d}A)^2 + 2\mathrm{d}A\, A^2) = \mathrm{d}\,\mathrm{CS}(A).$$

The Maurer–Cartan equation is $\mathrm{d}\vartheta_{\mathrm{MC}} = -\vartheta_{\mathrm{MC}}^2$. Substitution into the definition gives $(-1 + 2/3) \operatorname{tr}(\vartheta_{\mathrm{MC}}^3)$, which is the claimed pure-gauge value. More generally, a coefficient c in place of $2/3$ would give $3c$ in front of $\operatorname{tr}(\mathrm{d}A\, A^2)$. Matching the curvature square requires $3c = 2$, hence $c = 2/3$. ∎

For the general law set $B = h^{-1}Ah$. Expand $\mathrm{CS}(B + \vartheta_{\mathrm{MC}})$ using $\mathrm{d}B = h^{-1}\mathrm{d}Ah - \vartheta_{\mathrm{MC}}B - B\vartheta_{\mathrm{MC}}$. Graded trace cyclicity groups the mixed terms into $-\mathrm{d} \operatorname{tr}(\vartheta_{\mathrm{MC}} \wedge B) = -\mathrm{d} \operatorname{tr}(\mathrm{d}h h^{-1} \wedge A)$. The unmixed terms give the two Chern–Simons forms above. This outlines the expansion. Nakahara, §11.5, supplies the general transgression framework. The local potential and its Chern–Simons form need not be global. On a closed connected oriented four-manifold, a nonzero integral of $\operatorname{tr}(F \wedge F)$ obstructs a global three-form primitive by Stokes. A local identity does not assert that such a primitive exists.

Definition 23.8 (Second Chern number)**.** For an SU(2) bundle over an oriented closed four-manifold, its *second Chern number* is

$$k = \int_M c_2(F) = \frac{1}{8\pi^2}\int_M \operatorname{tr}(F \wedge F).$$

Lecture 22's higher Chern forms show that this number is independent of the connection and metric. Its sign uses the orientation.

Definition 23.9 (Chern character)**.** The *Chern character* is the trace exponential, expanded in wedge powers:

$$\operatorname{ch}(F) = \operatorname{tr}\exp(\mathrm{i}F/(2\pi)).$$

The quadratic exponential term gives its degree-four component on an SU(2) bundle:

$$\operatorname{ch}_2(F) = -\frac{1}{8\pi^2}\operatorname{tr}(F \wedge F) = -c_2(F).$$

Thus our k is minus the integrated second Chern character. [3]

On S^4 the integer value of our Chern number can be read from an equatorial transition. Represent the two hemispheres by an inside and an outside four-ball as in the figure.

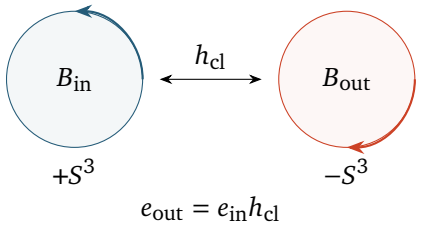


Figure 23.1: Two local Stokes integrals have opposite equatorial orientations. The circles represent the equator with signs relative to ∂B_{in} for the frame order $e_{\mathrm{out}} = e_{\mathrm{in}} h_{\mathrm{cl}}$.

Theorem 23.10 (The clutching integer · *sketched; global construction cited*)**.** *Let* $h_{\mathrm{cl}} \colon S^3 \to \mathrm{SU}(2)$ *glue the frames in the figure's order. Orient* SU(2) *as the outward unit sphere under* $(a_1, a_2, a_3, a_4) \mapsto a_4 I + \mathrm{i} a_a \sigma_a$*. Then* $k = -\deg h_{\mathrm{cl}} \in \mathbb{Z}$*.*

Sketch. Use local frames over the two balls, and Stokes with their opposite boundaries. The gauge law and the vanishing integral of an equatorial exact form give

$$k = \frac{1}{8\pi^2}\oint_{S^3}\left(\mathrm{CS}(A_{\mathrm{in}}) - \mathrm{CS}(A_{\mathrm{out}})\right) = \frac{1}{24\pi^2}\oint_{S^3}\operatorname{tr}\left((h_{\mathrm{cl}}^{-1}\mathrm{d}h_{\mathrm{cl}})^3\right).$$

For the identity unit-quaternion map, at $a_4 = 1$ the Maurer–Cartan form is $\mathrm{i}\sigma_a \mathrm{d}a_a$. The Pauli products give cubic trace $12\mathrm{d}a_1 \wedge \mathrm{d}a_2 \wedge \mathrm{d}a_3$. The outward sphere volume there is its negative, since the outward normal is last in the ordered ambient coordinates.[4] Both forms are left invariant, so everywhere $\operatorname{tr}(\vartheta^3_{\mathrm{MC}}) = -12\,\mathrm{vol}_{S^3}$. The sphere volume is $4\pi\int_0^\pi \sin^2\chi\,\mathrm{d}\chi = 2\pi^2$. Thus the normalized three-form $-\operatorname{tr}(\vartheta^3_{\mathrm{MC}})/(24\pi^2)$ has integral one. Lecture 14's degree formula identifies the displayed integral as minus the degree. The global two-ball construction is the cited input from Nakahara, §10.5.5. ∎

[3] Nakahara's (11.54)–(11.55) and (12.90) use the integrated second Chern character as their instanton number.

[4] In Nakahara (10.124)–(10.125), the equatorial north-pole volume is $+\mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3$. Our outward volume form at that pole is its negative.

On a region with boundary, the integral of the Chern form need not be an integer, just as the cap flux in Lecture 22 need not be integral.

The comparison with Lecture 22 records the local primitive and the boundary map in each dimension.

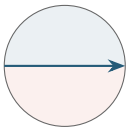
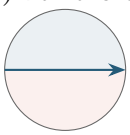
Line bundle on S^2	**SU(2) bundle on S^4**
	
$F = \mathrm{d}A$	$\operatorname{tr}(F \wedge F) = \mathrm{d}\,\mathrm{CS}(A)$
$e_S = e_N g_{NS}$	$e_{\mathrm{out}} = e_{\mathrm{in}} h_{\mathrm{cl}}$
$g_{NS} \colon S^1 \to \mathrm{U}(1)$	$h_{\mathrm{cl}} \colon S^3 \to \mathrm{SU}(2)$
$\frac{1}{2\pi\mathrm{i}} \oint g_{NS}^{-1} \mathrm{d} g_{NS}$	$\frac{1}{24\pi^2} \oint \operatorname{tr}((h_{\mathrm{cl}}^{-1} \mathrm{d} h_{\mathrm{cl}})^3)$
$C_1 = \operatorname{wind} g_{NS}$	$k = -\deg h_{\mathrm{cl}}$

Figure 23.2: The two-cap method uses a local primitive in each frame and the stated equatorial frame transition. The line-bundle equator is the northern boundary, and the four-dimensional equator is ∂B_{in}.

For a field on $\mathbb{R}^4$, the same integer applies when a change of frame near infinity extends the connection smoothly over the compactification S^4. If its asymptotic potential is $h_\infty^{-1}\mathrm{d}h_\infty$, the outer frame uses the inverse transition, so $k = \deg h_\infty$. This requires the stated asymptotic and extension conditions. Finite action alone does not supply them. The BPST potential in Unit 23C exhibits this asymptotic form explicitly.

23.4 The action bound

For an adjoint-valued two-form B, write $\|B\|^2 = -\int_M \operatorname{tr}(B \wedge \star B)$. This is nonnegative and vanishes only for $B = 0$.

Theorem 23.11 (The instanton bound · *proved*)**.** *For an* SU(2) *connection over Euclidean* S^4*,*

$$S_E \geq \frac{8\pi^2}{g_{\mathrm{YM}}^2} |k|.$$

Equality holds exactly for a self-dual or anti-self-dual curvature. In our negative-trace convention, a nonzero self-dual field has $k < 0$*, and a nonzero anti-self-dual field has* $k > 0$*.*

Proof. The Euclidean star is an isometry with square one on two-forms. Its two eigenspaces are orthogonal. Using the parts F_+ and F_- defined above gives

$$g_{\mathrm{YM}}^2 S_E = \|F_+\|^2 + \|F_-\|^2, \qquad 8\pi^2 k = \|F_-\|^2 - \|F_+\|^2.$$

The sum of two nonnegative numbers is at least the absolute value of their difference. Equality holds exactly when one number vanishes. For $k > 0$ this requires $F_+ = 0$, while for $k < 0$ it requires $F_- = 0$. For $k = 0$ equality forces $F = 0$, which obeys both duality equations.

The same decomposition gives the square completion directly. Expand the square for either $s = +1$ or $s = -1$ to obtain

$$-\operatorname{tr}(F \wedge \star F) = -\frac{1}{2}\operatorname{tr}\big((F - s\star F) \wedge \star(F - s\star F)\big) - s\operatorname{tr}(F \wedge F).$$

After integration this gives $g_{\mathrm{YM}}^2 S_E = \frac{1}{2}\|F - s\star F\|^2 - s8\pi^2 k$. Choose s so that $-sk = |k|$. Nonnegativity proves the bound. Equality requires $F = s\star F$. Conversely, that equation removes the square and gives equality. For a nonzero dual field, $8\pi^2 k = -s\|F\|^2$, proving the stated signs. ∎

Definition 23.12 (Instanton). An *instanton* here is a smooth finite-action Euclidean connection with self-dual or anti-self-dual curvature.

On S^4, or under the stated extension from $\mathbb{R}^4$, the bound makes it an action minimum within its fixed integer sector.

The norm plane shows how a fixed charge constrains the two curvature components.

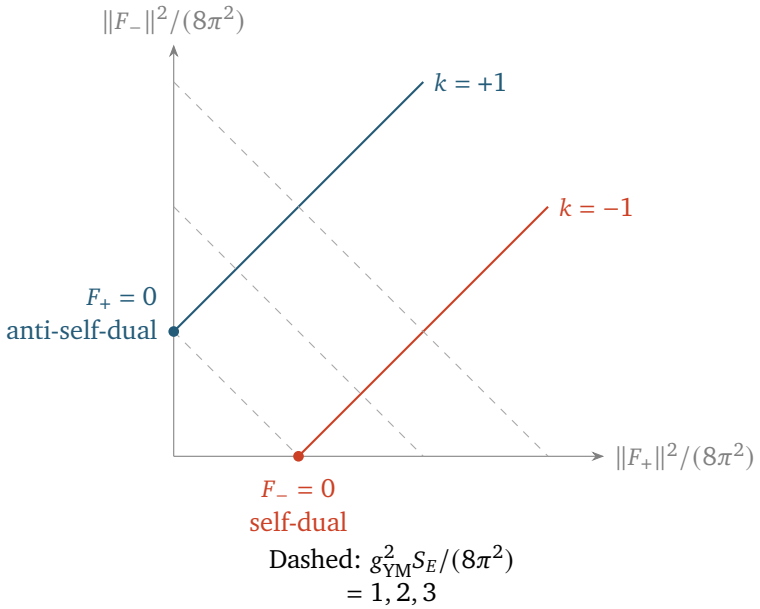


Figure 23.3: The coordinates are squared norms divided by $8\pi^2$, rather than spatial coordinates. Each fixed-charge line first meets an action contour on an axis, where one duality component vanishes.

The bound also compares the minimum actions in different integer sectors.

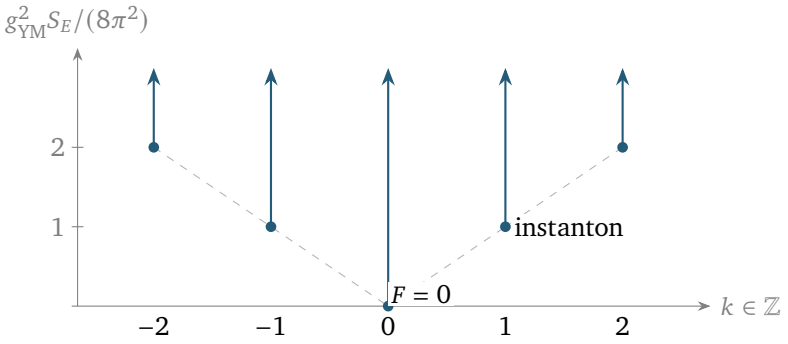


Figure 23.4: The bound is plotted at integer charges, with height $g_{\mathrm{YM}}^2 S_E/(8\pi^2)$. The vertical rays indicate the allowed inequality in each sector, and equality requires a dual curvature.

Self-check. State in one sentence what an instanton is and why it was introduced.

Answer. An instanton is a smooth finite-action Euclidean connection with self-dual or anti-self-dual curvature, introduced because the duality equation attains the least action in its fixed Chern-number sector on S^4, or under the stated smooth extension from $\mathbb{R}^4$. The decisive step is that one of $\|F_+\|^2$ and $\|F_-\|^2$ vanishes.

Unit 23C The BPST field, the topological term and principal connections

23.5 A charge-one field and a topological term

The *BPST instanton* is the following charge-one SU(2) example with size $\lambda > 0$. Recall $r^2 = \sum_{\mu=1}^{4} (x^\mu)^2$, with spatial indices $a, b, c \in \{1, 2, 3\}$. The potential is smooth at the origin and is given by

$$A = \frac{\mathrm{i}\sigma_a \left(x^4 \mathrm{d}x^a - x^a \mathrm{d}x^4 + \epsilon_{abc} x^b \mathrm{d}x^c\right)}{r^2 + \lambda^2}.$$

For $r > 0$, define the asymptotic unit-quaternion map

$$h_\infty : S^3 \to \mathrm{SU}(2), \qquad h_\infty(x/r) = \frac{x^4 I + \mathrm{i} x^a \sigma_a}{r}.$$

Its radial extension gives the same potential in the form

$$A = f(r)\vartheta_{\mathrm{MC}}, \qquad f(r) = \frac{r^2}{r^2 + \lambda^2}, \qquad \vartheta_{\mathrm{MC}} = h_\infty^{-1} \mathrm{d}h_\infty.$$

The scalar factor tends to one as $r \to \infty$. Thus the asymptotic pure gauge is explicit for this field.

The figure compares an exact exterior pure gauge in the plane with the BPST asymptotic gauge.

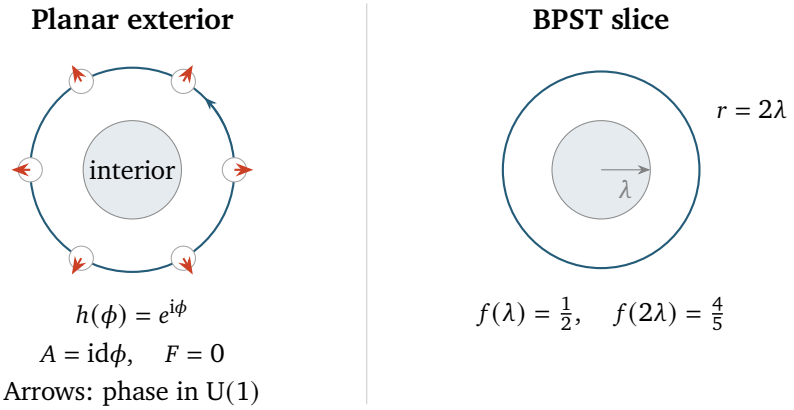


Figure 23.5: The planar exterior has $A = i\,d\phi$ and a single-valued phase of winding one. The BPST panel is a two-dimensional slice: its factor $f(r)$ approaches one, while its curvature remains nonzero at every finite radius.

Example 23.13 (The BPST curvature and action)**.** We compute the curvature, charge and action of this regular potential. We compute the component F_{14} from its derivative and commutator terms. We abbreviate the positive denominator by $d = r^2 + \lambda^2$ and write $t = x^4$. Here d is a scalar, distinct from the exterior derivative d. Let $\mathbf{x} = (x^1, x^2, x^3)$ and $\mathbf{b} = (t, x^3, -x^2)$. The potential has components

$$A_1 = \frac{i\boldsymbol{\sigma} \cdot \mathbf{b}}{d}, \qquad A_4 = -\frac{i\boldsymbol{\sigma} \cdot \mathbf{x}}{d}.$$

The Pauli commutator is $[i\boldsymbol{\sigma} \cdot \mathbf{u}, i\boldsymbol{\sigma} \cdot \mathbf{v}] = -2i\boldsymbol{\sigma} \cdot (\mathbf{u} \times \mathbf{v})$ for $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$. Differentiating the components and applying this commutator gives

$$\partial_1 A_4 - \partial_4 A_1 = -\frac{2i\sigma_x}{d} + \frac{2i}{d^2}\boldsymbol{\sigma} \cdot (x^1\mathbf{x} + t\mathbf{b}),$$
$$[A_1, A_4] = \frac{2i}{d^2}\boldsymbol{\sigma} \cdot (\mathbf{b} \times \mathbf{x}).$$

The vector terms cancel because

$$x^1\mathbf{x} + t\mathbf{b} + \mathbf{b} \times \mathbf{x} = r^2(1, 0, 0).$$

Hence the coefficient of $dx^1 \wedge dx^4$ is

$$F_{14} = -\frac{2i\sigma_x}{d} + \frac{2ir^2\sigma_x}{d^2} = -\frac{2i\lambda^2\sigma_x}{d^2}.$$

The spatial pair gives $F_{23} = 2i\lambda^2\sigma_x/d^2$. Cyclic permutation of the spatial labels gives the other pairs. The full curvature is

$$F = \frac{2i\lambda^2\sigma_a}{(r^2 + \lambda^2)^2}\left(\frac{1}{2}\epsilon_{abc}\,dx^b \wedge dx^c - dx^a \wedge dx^4\right).$$

For example, the coefficient pair multiplying σ_x is $dx^2 \wedge dx^3 - dx^1 \wedge dx^4$. The spatial-first star interchanges these two basis forms, so their difference is anti-self-dual. The other cyclic pairs have the same property. Taking the fundamental trace gives

$$\operatorname{tr}(F \wedge F) = -\operatorname{tr}(F \wedge \star F) = \frac{48\lambda^4}{(r^2 + \lambda^2)^4}\,\mathrm{vol}_{\mathbb{R}^4}.$$

Radial integration, followed by $w = r^2/\lambda^2$, yields

$$\int_{\mathbb{R}^4} \operatorname{tr}(F \wedge F) = 96\pi^2\lambda^4 \int_0^\infty \frac{r^3 \mathrm{d}r}{(r^2+\lambda^2)^4} = 48\pi^2 \int_0^\infty \frac{w\mathrm{d}w}{(1+w)^4} = 8\pi^2.$$

The last integral is $1/6$, by writing $w = (1+w) - 1$ and integrating the two powers. Consequently $k = 1$ and $S_E = 8\pi^2/g_{\mathrm{YM}}^2$, independent of the size.

The asymptotic map h_∞ has degree +1 in the stated orientation. At infinity the pure-gauge Chern–Simons integral therefore gives $k = -\int_{S^3} \operatorname{tr}((h_\infty^{-1}\mathrm{d}h_\infty)^3)/(24\pi^2) = +1$. The clutching map has the inverse order. The change of frame $h_{\mathrm{cl}} = h_\infty^{-1}$ makes the outer asymptotic potential zero. Its degree is -1, so the clutching formula gives the same charge. Reversing to time-first orientation makes this same field self-dual and changes k to -1.[5]

The normalized radial contribution for this example is plotted in the figure. Its maximum occurs at $u = \sqrt{3/5}$, so the size λ fixes the radial scale of the density.

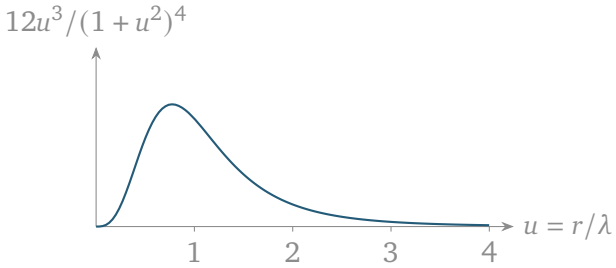


Figure 23.6: For $u = r/\lambda$, the BPST Chern-number contribution per du is $12u^3/(1+u^2)^4$. Its integral over $[0,\infty)$ is 1, and the plot shows the initial interval $0 \le u \le 4$.

Proposition 23.14 (BPST normalization · *sketched*). *For size $\lambda > 0$ on spatial-first $\mathbb{R}^4$, the regular BPST field is anti-self-dual, extends over the conformal compactification S^4, and has $k = +1$ and $S_E = 8\pi^2/g_{\mathrm{YM}}^2$. Reflection of one coordinate gives its self-dual partner with $k = -1$ and the same action.*

Extension and orientation

The extension over the compactification is a cited part of the BPST construction in Nakahara, §10.5.5, and Tong, §2.3.1. We assume this asymptotic regularity when treating an $\mathbb{R}^4$ field as a connection on S^4. Finite action alone is not used as a proof of that extension. The Euclidean star on two-forms is unchanged by a conformal metric rescaling, because its scale factor in dimension four is the zeroth power. Thus stereographic compactification preserves the duality equation and action. Give $S^4 \subset \mathbb{R}^5$ its outward orientation. Let $e_1, \dots, e_5$ be the standard basis of $\mathbb{R}^5$ and put $r^2 = \sum_{\mu=1}^4 (x^\mu)^2$. Use the inverse south-pole stereographic chart $X\colon \mathbb{R}^4 \to S^4 \setminus \{-e_5\}$, given by

$$X(x) = \frac{(2x^1, 2x^2, 2x^3, 2x^4, 1-r^2)}{1+r^2}.$$

At $x = 0$, the outward normal is e_5 and the four coordinate tangent vectors are $2e_i$. The ordered determinant is +16. The inverse $x^i = X^i/(1+X^5)$, for $i \in \{1,2,3,4\}$, shows that this

[5] Lecture 4 wrote $U = aI - \mathrm{i}v^a\sigma_a$. Our oriented quaternion coordinates there are $(a_4, \mathbf{a}) = (a, -\mathbf{v})$. Tong calls his time-first BPST field self-dual with $\nu = +1$. Our spatial-first orientation and translated anti-Hermitian field give anti-self-duality and $k = +1$.

is a smooth chart. Since $\mathbb{R}^4$ is connected, it preserves the spatial-first orientation everywhere. Consequently the computed charge on $\mathbb{R}^4$ is also the charge on this outward S^4.

Definition 23.15 (Theta term). For a fixed real coefficient θ_{YM}, distinct from the Maurer–Cartan form ϑ_{MC}, the *theta term* is

$$S_\theta = \frac{\theta_{\mathrm{YM}}}{8\pi^2}\int_M \operatorname{tr}(F\wedge F) = \theta_{\mathrm{YM}}k.$$

Locally its density is $\theta_{\mathrm{YM}}\mathrm{d}\,\mathrm{CS}(A)/(8\pi^2)$. Its variation is $\delta\operatorname{tr}(F\wedge F) = 2\mathrm{d}\operatorname{tr}(a\wedge F)$, using $DF = 0$. It therefore leaves the field equation unchanged for compactly supported variations, or on a closed manifold. On a manifold with boundary, boundary conditions still matter.

Physical input. In the quantum path integral, an added real-time action S_θ multiplies a field's weight by $e^{\mathrm{i}S_\theta/\hbar}$, as in Tong, §2.2, with $\hbar$ restored.

For integer k, this factor is $e^{\mathrm{i}(\theta_{\mathrm{YM}}/\hbar)k}$. It is unchanged when θ_{YM} increases by $2\pi\hbar$. Thus the dimensionless angle is $\theta_{\mathrm{YM}}/\hbar$. The topological term can change the relative quantum weights of sectors even though its bulk variation vanishes.

Self-check. For the BPST potential, take $\lambda = 2$ and $(x^1, x^2, x^3, x^4) = (1, 2, -1, 3)$. Compute $F_{23} = \partial_2 A_3 - \partial_3 A_2 + [A_2, A_3]$ by differentiating the potential before evaluating it at this point.

Answer. The denominator is $d = 19$. Differentiating first retains the derivatives of d and gives

$$\partial_2 A_3 - \partial_3 A_2 = \frac{\mathrm{i}}{361}(28\sigma_x - 2\sigma_y - 14\sigma_z),$$
$$[A_2, A_3] = \frac{\mathrm{i}}{361}(-20\sigma_x + 2\sigma_y + 14\sigma_z).$$

The last two Pauli components cancel, leaving $F_{23} = 8\mathrm{i}\sigma_x/361$. This agrees with the coefficient in the full curvature.

23.6 A connection on the principal bundle

The local potentials also have an intrinsic description on a principal bundle. Lecture 20's principal bundle $P \to M$ carries a right action of a Lie group G. Its Lie algebra is $\mathfrak{g}$.

Definition 23.16 (Fundamental vertical vector). For $\xi \in \mathfrak{g}$, the *fundamental vertical vector* at $p \in P$ is $\xi_p^{\#} = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_0 p\exp(t\xi)$.

It is tangent to the fibre.

Definition 23.17 (Principal connection). A *principal connection* is a smooth $\mathfrak{g}$-valued one-form ω_P on P with

$$\omega_P(\xi^{\#}) = \xi, \qquad R_a^*\omega_P = \mathrm{Ad}_{a^{-1}}\,\omega_P \quad (a \in G),$$

where $R_a(p) = pa$ and $\mathrm{Ad}_a\,\xi = a\xi a^{-1}$ for a matrix group.

Definition 23.18 (Gauge transformation of a principal bundle). Gauge transformations are bundle automorphisms over the identity that commute with the right action.

Lecture 21 represents them by compatible maps $u_j = g_{ij}^{-1}u_i g_{ij}$. A global map $M \to G$ suffices in a chosen trivialization.

Remark 23.19 (Principal connections)**.** The kernel of ω_P gives the horizontal tangent spaces, complementary to the vertical tangent spaces. In a local section $s_i : U_i \to P$, the potential is $A_i = s_i^* \omega_P$. For $s_j = s_i g_{ij}$ these potentials obey Lecture 21's frame law. A representation $\rho : G \to \mathrm{GL}(V)$ induces on the associated bundle the local derivative $\mathrm{d} + \rho_*(A_i)$, where ρ_* is the representation's differential at the identity. Thus one principal connection supplies the potentials in all representations. These intrinsic-to-local facts are cited from Tu, §28.1, §29.4 and §31.3–31.5.

Summary. The fundamental anti-Hermitian trace fixes the Minkowski prefactor and gives $D\star F = 0$. For U(1) the different trace normalization recovers Maxwell's action. Locally $\mathrm{d}\,\mathrm{CS}(A) = \mathrm{tr}(F \wedge F)$. The Chern–Simons form need not be global. On S^4, $k = -\deg h_{\mathrm{cl}}$ in the stated two-frame order. The Euclidean bound is $S_E \geq 8\pi^2|k|/g_{\mathrm{YM}}^2$, saturated exactly by dual fields. Regular spatial-first BPST is anti-self-dual with $k = +1$. The action needs the metric and is orientation independent. The Chern number has the opposite dependence. Lecture 24 returns to line-bundle connections generated by quantum eigenstates.

Reading. Tu, *Differential Geometry*, §28.1, §29.4 and §31.3–31.5. Nakahara, §10.5.4–10.5.5, §11.3.1 (11.54)–(11.55), §11.5 and §12.6 (12.90). Tong, *Gauge Theory*, §2.1–2.3, for the action, topological phase and semiclassical Euclidean input.

Lecture 24

Berry's phase

Objectives. *A slowly cycled Hamiltonian can return an eigenstate with a measurable relative phase. Part of this phase depends on the elapsed time. Another part depends on the path of the parameters. We identify the second part with line-bundle holonomy. For the aligned spin-1/2 ground state, it is minus half the signed solid angle enclosed by the field-direction loop. The eigenvalue gap states where this description applies.*

By the end of this lecture you will be able to:

- State the gap and slow-evolution hypotheses of the cited adiabatic theorem.
- Construct the connection and curvature of a smooth simple eigenline.
- Compute a closed-cycle geometric phase with the correct gauge convention.
- Derive the spin-1/2 phase and its ground-state Chern number -1.

Unit 24A Eigenline transport and phase

24.1 Slow evolution and the gap

Recall the finite-dimensional quantum notation from *Quantum Mechanics I*. A pure state is a complex line in $\mathbb{C}^N$. A unit representative is written as a ket $|\psi\rangle$. Its conjugate transpose is the bra $\langle\psi|$, a linear functional. The inner product $\langle\phi|\psi\rangle = \phi^\dagger\psi$ is antilinear in its first argument. Multiplying a representative by a phase leaves its line unchanged.

Physical input. The Hamiltonian is a Hermitian matrix. A state vector evolves by the Schrödinger equation

$$\mathrm{i}\hbar\,\partial_t\,|\psi\rangle = H(t)\,|\psi\rangle\,.$$

Let $s \in [0,1]$ describe a fixed path of external parameters. Executing it in a duration $T > 0$ means $s = t/T$ for $0 \le t \le T$. Write the evolving vector as $\psi_T(s)$.

Definition 24.1 (Adiabatic limit)**.** The *adiabatic limit* is $T \to \infty$ with this path fixed.

Definition 24.2 (Simple eigenvalue and gap)**.** An eigenvalue is *simple* if its eigenspace has dimension one. Its *gap* is its distance to the other eigenvalues.

We first derive the transport equation by projecting a leading adiabatic ansatz. Choose a smooth unit eigenvector $|n(s)\rangle$ with eigenvalue $E_n(s)$ and write

$$\psi_{\mathrm{app}}(s) = D(s)c(s)\,|n(s)\rangle\,, \qquad D(s) = \exp\left(-\frac{\mathrm{i}T}{\hbar}\int_0^s E_n(u)\mathrm{d}u\right).$$

The terms containing TE_n cancel in the Schrödinger residual, leaving

$$i\hbar\psi'_{\text{app}} - TH\psi_{\text{app}} = i\hbar D\big(c'\,|n\rangle + c\,|n'\rangle\big).$$

Requiring the component of this residual along $|n\rangle$ to vanish gives the projected equation

$$c' + \langle n|n'\rangle\, c = 0.$$

This is Lecture 21's scalar parallel-transport equation. The duration T no longer occurs in it. The orthogonal part of the residual generally remains nonzero. This calculation motivates the connection but does not prove convergence of the full evolution.

Theorem 24.3 (Adiabatic evolution · *cited*)**.** *Let $H(s)$ be a smooth family of Hermitian $N \times N$ matrices on $[0,1]$. Let $E_n(s)$ be a smooth simple eigenvalue separated from all other eigenvalues by a fixed $\delta > 0$. Start in a unit vector of its eigenspace and solve $i\hbar\,\psi_T'(s) = TH(s)\psi_T(s)$. There is a unique smooth unit vector $v(s)$ in that eigenspace with $v(0) = \psi_T(0)$ and $\langle v(s)|v'(s)\rangle = 0$. For each fixed $s \in [0,1]$, the cited theorem gives*

$$\left\| \psi_T(s) - \exp\left(-\frac{iT}{\hbar}\int_0^s E_n(u)\,du\right) v(s) \right\| \longrightarrow 0 \quad \text{as } T \to \infty.$$

This is the finite-dimensional form of Kato's adiabatic theorem. See Barry Simon, p. 2167, for the gap assumption and phase separation, and p. 2168 for the transport condition on the limit.

Definition 24.4 (Dynamical phase)**.** The exponential in Theorem 24.3 is the *dynamical phase factor*.

Definition 24.5 (Geometric phase)**.** If $H(1) = H(0)$ and the chosen eigenline returns to itself, then $v(1) = e^{i\gamma(C)}v(0)$ for a phase defined modulo 2π. This scalar is the *geometric phase factor*, or *Berry phase factor*.

The symbol C denotes the parameter cycle, whereas $\gamma(C)$ denotes its phase.

The theorem is a limit statement. A changing eigenvector need not solve the full Schrödinger equation exactly at finite T.

The figure separates a gap condition from the choice of eigenvectors.

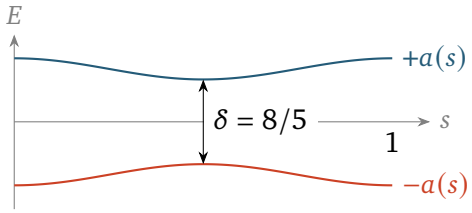


Figure 24.1: The diagonal family $H(s) = \operatorname{diag}(-a(s), a(s))$, with $a(s) = 1 + \cos(2\pi s)/5$, has a gap at least $8/5$. Its eigenspaces are constant, so its geometric phase is zero.

A varying eigenline occurs in the avoided-crossing family

$$H(s) = \begin{pmatrix} \Lambda(s-\frac{1}{2}) & \varepsilon \\ \varepsilon & -\Lambda(s-\frac{1}{2}) \end{pmatrix}, \qquad \Lambda > 0, \quad \varepsilon > 0,$$

$$E_\pm(s) = \pm\sqrt{\Lambda^2(s-\tfrac{1}{2})^2 + \varepsilon^2}.$$

Squaring the Hamiltonian gives the displayed energies. The minimum gap is $\delta = 2\varepsilon$ at $s = 1/2$. The lower eigenline varies along the path. The figure supplements the preceding constant-eigenspace family.

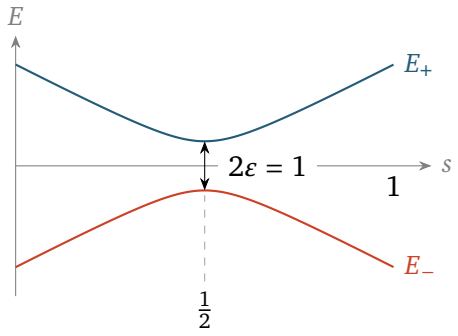


Figure 24.2: The exact levels for $\Lambda = 4$ and $\varepsilon = 1/2$ have minimum gap 1. The eigenlines vary even though the two levels never meet.

A heuristic scale for slow evolution is

$$\frac{\hbar \max_{s\in[0,1]} \|H'(s)\|}{T\delta^2} \ll 1.$$

To see the two gap factors, differentiate $H|n\rangle = E_n|n\rangle$ and project onto an orthogonal unit eigenvector $|r\rangle$ with eigenvalue E_r. The result is

$$\langle r|n'\rangle = \frac{\langle r|H'|n\rangle}{E_n - E_r}.$$

The relative dynamical oscillation has frequency $T(E_r - E_n)/\hbar$ in s. Together these suggest the displayed scale. For the avoided crossing, $\|H'\| = \Lambda$, so it reads $T \gg \hbar\Lambda/(4\varepsilon^2)$. This is a scale estimate, not a quantitative error theorem. Such a theorem also needs bounds on further derivatives and the endpoint behavior of the family. A zero gap invalidates the cited theorem's hypothesis. It does not rule out every exact eigenvector evolution through a crossing.

24.2 Eigenlines and their connection

Let $\mathcal{U}$ be an open region in a smooth parameter manifold, with a smooth family $H\colon \mathcal{U} \to \mathrm{Herm}(N)$ and a chosen smooth simple eigenvalue $E_n(R)$. Here $\mathrm{Herm}(N)$ is the real vector space of Hermitian matrices.

Definition 24.6 (Eigenline bundle). The *eigenline bundle* is the subset of $\mathcal{U} \times \mathbb{C}^N$ given by

$$L_n = \{(R, w) : R \in \mathcal{U},\ H(R)w = E_n(R)w\}.$$

Its fibre over R is the eigenspace, including its zero vector.

The index n labels the chosen energy branch. The integer N is the dimension of the ambient state space. Smoothness of the selected branch is an explicit hypothesis here. It holds for the separated branches used below. Along a smooth parameter path $R(s)$, the adiabatic theorem applies to $H(R(s))$ when its eigenvalue has the stated uniform gap.

Definition 24.7 (Berry connection)**.** The *Berry connection* of the eigenline bundle is the projected derivative $\nabla\xi = P\mathrm{d}\xi$, where $P(R)$ is the orthogonal projection onto the eigenline.

Lecture 16 constructed a surface connection by differentiating in $\mathbb{R}^3$ and projecting onto the tangent plane. Here we differentiate in $\mathbb{C}^N$ and project onto the eigenline. The figure compares these projections on exact real curves.

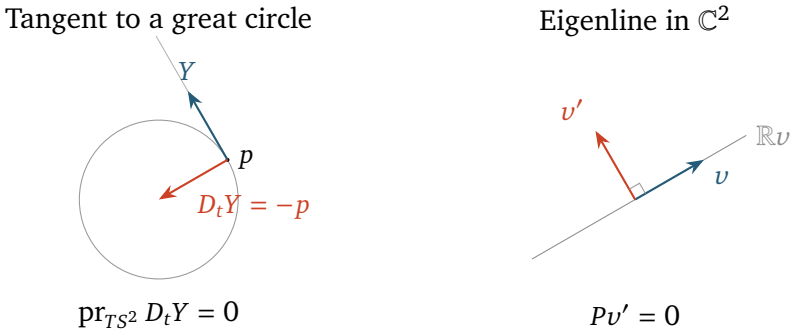


Figure 24.3: Along the unit sphere's great circle $p = (\cos t, \sin t, 0)$, the field $Y = p'$ has ambient derivative $Y' = -p$ and zero tangent projection. For the real unit vector $v = (\cos t, \sin t)$ in $\mathbb{C}^2$, the derivative is orthogonal to its complex line, so $Pv' = 0$.

Theorem 24.8 (The Berry connection · *proved geometrically; adiabatic interpretation given the cited theorem*)**.** *The eigenlines form a smooth Hermitian line bundle. For a smooth unit eigenvector $|n_i\rangle$ on $U_i \subset \mathcal{U}$, the* Berry connection *has imaginary local one-form*

$$\mathcal{A}_i = \langle n_i | \mathrm{d}n_i \rangle .$$

These forms define the projected derivative $\nabla\xi = P\mathrm{d}\xi$, where $P(R)$ is the orthogonal projection onto the eigenline. Its parallel transport is the vector $v(s)$ in the adiabatic theorem. Thus its holonomy around a cycle C is $e^{\mathrm{i}\gamma(C)}$.

Proof. The matrix $H(R) - E_n(R)I$ has rank $N-1$. Its adjugate, the transposed cofactor matrix, has a nonzero column at any fixed point. Its columns lie in the kernel of the original matrix, by the identity $B\,\mathrm{adj}(B) = \det(B)I$ for every square matrix B. Choose such a column. It stays nonzero nearby and is smooth in R. Divide it by its positive norm to obtain a smooth unit eigenvector $|n_i\rangle$. Its scalar multiples give a local product chart for L_n. On overlaps write $|n_j\rangle = |n_i\rangle\, g_{ij}$, where $g_{ij} = \langle n_i | n_j \rangle \in \mathrm{U}(1)$. These smooth transitions satisfy the cocycle law, so Lecture 20's charts give the bundle.

The projection is $P = |n_i\rangle\langle n_i|$ and is unchanged by this unit phase. It is therefore global. For a section $\xi = |n_i\rangle\, c$, direct differentiation gives

$$P\mathrm{d}\xi = |n_i\rangle \left(\mathrm{d}c + \langle n_i | \mathrm{d}n_i \rangle\, c\right).$$

The ordinary product rule gives the connection Leibniz rule because $P\xi = \xi$. Self-adjointness of P similarly gives the Hermitian metric product rule. Differentiating $\langle n_i | n_i \rangle = 1$ shows $\mathcal{A}_i + \overline{\mathcal{A}_i} = 0$, so the potential is imaginary. Finally, for a vector $v(s)$ in the eigenline, its projection of $v'(s)$ is $v(s)\,\langle v(s) | v'(s) \rangle$ when v is unit. Thus the adiabatic theorem's condition is exactly $\nabla_{\partial_s} v = 0$. Uniqueness of Lecture 21's transport identifies the two vectors and their endpoint phase. ∎

The bundle and connection depend only on the eigenspaces. The dynamical phase also uses the eigenvalues and traversal time.

24.3 Curvature, gauge changes and holonomy

A different unit vector on one chart has the form $|n'\rangle = |n\rangle\, h$, with $h\colon U \to \mathrm{U}(1)$. Differentiating the product and using unit norm gives

$$\mathcal{A}' = \mathcal{A} + h^{-1}\mathrm{d}h, \qquad \mathcal{A}' = \mathcal{A} + \mathrm{i}\mathrm{d}\chi \quad \text{if } h = e^{\mathrm{i}\chi}.$$

This is Lecture 21's passive line-frame law. The coefficient of a fixed section changes by $c' = h^{-1}c$.

Definition 24.9 (Real Berry potential)**.** The *real Berry potential* is $A_{\mathrm{B}} = \mathrm{i}\mathcal{A}$.

Definition 24.10 (Berry curvature)**.** The *Berry curvature* is the global two-form $\mathrm{d}\mathcal{A}$.

The real potential differs from the imaginary connection form used in the transport equation. Lecture 18 writes $\mathrm{d}A$ and Lecture 22 writes $\mathrm{d}\mathcal{A}$ for a surface area form. Neither denotes the Berry connection $\mathcal{A}$ or its curvature $\mathrm{d}\mathcal{A}$ used here. Differentiating the local connection yields the full local formula

$$\mathrm{d}\mathcal{A} = \langle \mathrm{d}n| \wedge |\mathrm{d}n\rangle = \sum_{a=1}^{N} \mathrm{d}\overline{n^a} \wedge \mathrm{d}n^a.$$

Here the bra–ket wedge denotes the componentwise sum on the right. The term with $\mathrm{d}^2 n$ vanishes. The gauge law shows that the curvature is unchanged on overlaps. Thus Lecture 22 supplies the real class $c_1(L_n) = [\mathrm{i}\mathrm{d}\mathcal{A}/(2\pi)]$.

In a periodic unit frame along a closed cycle, the scalar transport equation gives

$$e^{\mathrm{i}\gamma(C)} = \exp\left(-\oint_C \mathcal{A}\right), \qquad \gamma(C) = \mathrm{i}\oint_C \mathcal{A} = \oint_C A_{\mathrm{B}} \pmod{2\pi}.$$

Such a frame exists along the cycle even when none exists on all of $\mathcal{U}$. Indeed, transport a unit vector along the parameter interval to get $v(s)$. If $v(1) = e^{\mathrm{i}\alpha}v(0)$, the vector $e^{-\mathrm{i}\alpha s}v(s)$ is periodic. Alternatively, use local frames and the transition factors when changing charts, as in Lecture 21. Under any single-valued gauge map on the cycle, $\oint_C h^{-1}\mathrm{d}h = 2\pi\mathrm{i}\,\mathrm{wind}(h)$. The geometric phase changes by an integer multiple of 2π, leaving its factor invariant. No global real logarithm of h is required. A parallel unit vector can therefore have zero connection coefficient along the interval without being periodic at its endpoints. A periodic change of frame cannot remove a nontrivial closed-cycle holonomy.

If one frame covers an oriented compact spanning surface Σ with $\partial\Sigma = C$, Stokes gives $\gamma(C) = \mathrm{i}\int_\Sigma \mathrm{d}\mathcal{A}$ modulo 2π. The surface and frame hypotheses matter, as they did for the Aharonov–Bohm calculation. The real potential also gives the local phase convention used in Lecture 21: A_{B} corresponds to $(q/\hbar)A_{\mathrm{em}}$. This compares connections on different parameter spaces, rather than identifying the physical fields.[1]

Physical input. In an ideal coherent two-arm interferometer, the complex amplitudes of the arms add at recombination. The detection intensity is proportional to the squared modulus

[1] Berry's equation (6) uses our $\mathrm{i}\oint \mathcal{A}$. Tong uses $A_{\mathrm{Tong}} = -\mathrm{i}\mathcal{A} = -A_{\mathrm{B}}$. His factor $\exp(-\mathrm{i}\oint A_{\mathrm{Tong}})$ is the same holonomy.

of their sum. We assume that both arms end in the same internal eigenline with the same normalized representative.

We cycle the parameters in one arm and hold them fixed in the reference arm. If their energy histories have equal integrals and their durations agree, their dynamical phases cancel in the relative phase. For arm amplitudes a_{cyc} and a_{ref}, the detection intensity is proportional to

$$|a_{\mathrm{cyc}}e^{\mathrm{i}\gamma(C)} + a_{\mathrm{ref}}|^2 = |a_{\mathrm{cyc}}|^2 + |a_{\mathrm{ref}}|^2 + 2\,\mathrm{Re}\big(\overline{a_{\mathrm{ref}}}a_{\mathrm{cyc}}e^{\mathrm{i}\gamma(C)}\big).$$

Thus a phase of one arm changes interference with the other arm. For a constant energy $-b$, the dynamical phase angle is $bT/\hbar$. Doubling T doubles this angle while leaving the limiting geometric phase fixed. Reversing the loop with the same duration and the same energy history in reverse order leaves the dynamical integral unchanged. It reverses the geometric phase. Comparing these two traversals gives $2\gamma(C)$ modulo 2π. This comparison alone determines $\gamma(C)$ only modulo π. The figure shows the reference-arm scheme for the spin rotation computed in §24.4.

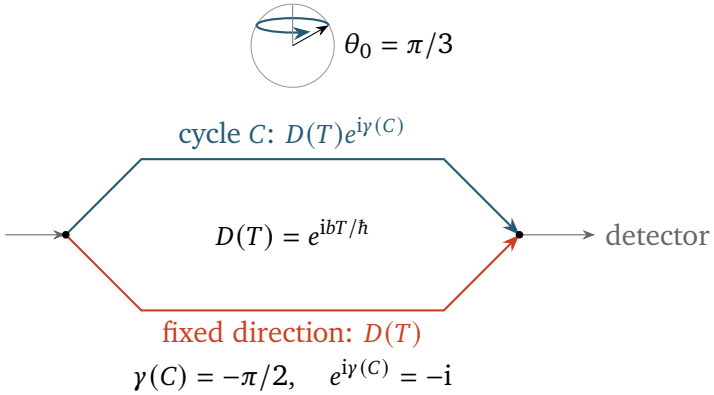


Figure 24.4: Both arms have energy $-b$ and duration T, so their common dynamical factor cancels. For the illustrated field-direction latitude $\theta_0 = \pi/3$, the cycled arm acquires the geometric factor $-\mathrm{i}$ in the adiabatic limit.

Self-check. State in one sentence what the Berry connection is and why it was introduced.

Answer. The Berry connection is the projected derivative $P\mathrm{d}$ on the eigenline bundle, introduced because the cited adiabatic theorem identifies its parallel transport with evolution after removal of the dynamical phase. Indeed, $P\mathrm{d}(|n\rangle\, c) = |n\rangle\,(\mathrm{d}c + \langle n|\mathrm{d}n\rangle\, c)$ gives the projected Schrödinger equation.

Self-check. For a smooth open path $C\colon [0,1] \to \mathcal{U}$, explain how $\mathrm{i}\int_C \mathcal{A}$ changes under $|n\rangle \mapsto e^{\mathrm{i}\chi}|n\rangle$, for a single-valued real χ on the path.

Answer. The gauge law gives

$$\mathrm{i}\int_C (\mathcal{A} + \mathrm{i}\mathrm{d}\chi) = \mathrm{i}\int_C \mathcal{A} - [\chi(C(1)) - \chi(C(0))].$$

Thus this integral alone is not an intrinsic open-path phase. An endpoint reference must also transform if an open-path interference measurement is specified. For a closed cycle the holonomy is intrinsic and its phase is defined modulo 2π.

Unit 24B Spin eigenlines and solid angle

24.4 A spin-$1/2$ in a rotating field

Recall the Pauli matrices and the spin operators S_a, for $a \in \{x, y, z\}$:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad S_a = \frac{\hbar}{2}\sigma_a.$$

The symbol σ_a denotes these matrices, distinct from Lecture 21's section-coefficient column. Their products obey $\sigma_a\sigma_b = \delta_{ab}I + \mathrm{i}\epsilon_{abc}\sigma_c$. Use an energy-scaled field $\mathbf{B} = b\mathbf{n}$, with $b > 0$ and $|\mathbf{n}| = 1$.

Physical input. We model a spin-1/2 in the energy-scaled field $\mathbf{B}$ by the Hamiltonian $H = -\mathbf{B} \cdot \boldsymbol{\sigma}$.

Its eigenvalues are $-b$ and $+b$ because $(\mathbf{n}\cdot\boldsymbol{\sigma})^2 = I$. Its gap is $2b$ everywhere on the direction sphere. The earlier avoided-crossing matrix is also of this Pauli form: $H(s) = \Lambda(s - \frac{1}{2})\sigma_z + \varepsilon\sigma_x$.

For a smooth loop $C\colon [0,1] \to S^2$ based at $p = C(0) = C(1)$, choose a continuous filling $f\colon [0,1]^2 \to S^2$, smooth on each rectangle of a finite subdivision, with

$$f(s,0) = C(s), \qquad f(s,1) = f(0,u) = f(1,u) = p.$$

Lecture 20's hemisphere, polygon and stereographic construction supplies such a filling, smooth on each rectangle of a finite subdivision. Orient the square by $\mathrm{d}s \wedge \mathrm{d}u$. Recall the outward unit-sphere area form β from Lecture 12.

Definition 24.11 (Signed solid angle). The *signed solid angle* of this filling is

$$\Omega_f(C) = \int_{[0,1]^2} f^*\beta.$$

The proof of Proposition 24.12 shows that its class modulo 4π is independent of the filling. We denote this class by $\Omega(C)$. For a simple loop bounding an outward-oriented region D on its left, one filling has signed area $\mathrm{area}(D)$. The complementary region must have its orientation reversed to have the same boundary. Its signed area is $\mathrm{area}(D) - 4\pi$. The resulting phases differ by 2π and give the same factor. The figure shows this ambiguity for a latitude.



$\gamma = -\pi/2$ $\qquad$ $\gamma = 3\pi/2$

Figure 24.5: At $\theta_0 = \pi/3$, the northern cap has signed area π and the reversed complementary cap has signed area -3π. Both have the same oriented boundary and give geometric factor $-\mathrm{i}$.

Proposition 24.12 (The spin eigenline · *proved*)**.** *For* $\mathbf{n} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$*, the aligned ground-state eigenline is Lecture 20's tautological line on* $\mathbb{C}\mathrm{P}^1$*. On the northern chart it has*

$$|n_N\rangle = \begin{pmatrix} \cos(\theta/2) \\ e^{\mathrm{i}\phi}\sin(\theta/2) \end{pmatrix}, \quad \mathcal{A}_N = \frac{\mathrm{i}}{2}(1-\cos\theta)\mathrm{d}\phi, \quad \mathrm{d}\mathcal{A} = \frac{\mathrm{i}}{2}\sin\theta\mathrm{d}\theta\wedge\mathrm{d}\phi.$$

In the outward orientation its Chern number is -1*. For an oriented loop, the geometric phase is* $\gamma(C) = -\Omega(C)/2$ *modulo* 2π*, where the solid angle is defined modulo* 4π*.*

Proof. The direction matrix is

$$\mathbf{n}\cdot\boldsymbol{\sigma} = \begin{pmatrix} \cos\theta & e^{-\mathrm{i}\phi}\sin\theta \\ e^{\mathrm{i}\phi}\sin\theta & -\cos\theta \end{pmatrix}.$$

Multiplying by the displayed vector gives $|n_N\rangle$, so $H|n_N\rangle = -b|n_N\rangle$. Its norm is one. In $z = \tan(\theta/2)e^{\mathrm{i}\phi}$ it is $(1,z)^T/\sqrt{1+|z|^2}$, the tautological unit frame of Lecture 20. Its Pauli expectations are

$$\langle n_N|\sigma_x|n_N\rangle = \sin\theta\cos\phi, \qquad \langle n_N|\sigma_y|n_N\rangle = \sin\theta\sin\phi, \qquad \langle n_N|\sigma_z|n_N\rangle = \cos\theta.$$

Thus $\mathbf{n}$ is the state's Bloch vector. The unit eigenvectors form Lecture 20's Hopf circle bundle $S^3 \to S^2$. The inner product with its θ derivative is $-\cos(\theta/2)\sin(\theta/2)/2 + \sin(\theta/2)\cos(\theta/2)/2 = 0$. The ϕ derivative gives $\mathrm{i}\sin^2(\theta/2) = \mathrm{i}(1-\cos\theta)/2$. Differentiating gives the curvature stated above.

The southern unit vector is $|n_S\rangle = e^{-\mathrm{i}\phi}|n_N\rangle$. It is smooth at the south pole and has $\mathcal{A}_S = \mathcal{A}_N - \mathrm{i}\mathrm{d}\phi$. Thus its frame transition has winding -1. These are exactly Lecture 22's Wu–Yang potentials with monopole integer $n = -1$. The eigenline is its degree--1 line $O(-1)$. Here the subscripts N and S name the northern and southern charts. Equivalently, direct curvature integration gives

$$\frac{\mathrm{i}}{2\pi}\int_{S^2}\mathrm{d}\mathcal{A} = -\frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi}\sin\theta\mathrm{d}\theta\mathrm{d}\phi = -1.$$

The nonzero Chern number prevents a smooth unit eigenvector on the whole sphere. To compute the phase for a general loop, we instead pull the eigenline and connection back to the filling square. We construct a unit frame $w(s,u)$ by vertical transport from a fixed vector on the top edge. The figure shows this construction and the boundary orientation used in Stokes.

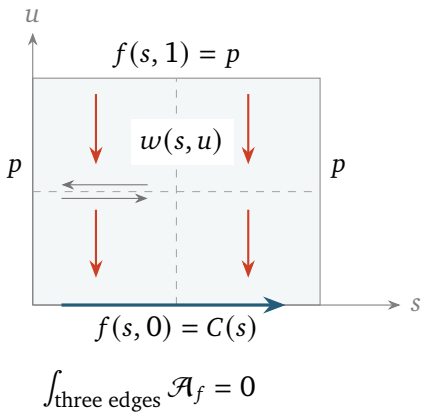


Figure 24.6: The bottom edge runs in increasing s for the orientation $\mathrm{d}s \wedge \mathrm{d}u$. Vertical transport from the fixed unit vector on the top edge makes $w(s,u)$ constant on the three edges mapped to p, so only the bottom integral survives after shared edges cancel.

Fix a unit vector over p. From the top edge, transport it down each vertical segment. Smooth dependence of transport on parameters gives a unit frame $w(s,u)$, continuous on the square and smooth on each rectangle of the finite subdivision. The top and side edges map constantly to p, so the frame is the same fixed vector on all three. In particular, its bottom restriction is a periodic frame along C.

Let $\mathcal{A}_f = \langle w|\mathrm{d}w\rangle$ be its connection form on the square. On each rectangle, its curvature is $\mathrm{d}\mathcal{A}_f = f^*\mathrm{d}\mathcal{A}$. The tangential restrictions on shared edges agree, so internal boundary integrals cancel. The top and side integrals vanish because their frame is constant. The positively oriented bottom edge runs in increasing s. Stokes on the rectangles therefore gives

$$\gamma(C) = \mathrm{i}\int_0^1 \mathcal{A}_f(\partial_s)(s,0)\mathrm{d}s = \mathrm{i}\int_{[0,1]^2} f^*\mathrm{d}\mathcal{A} = -\frac{1}{2}\Omega_f(C) \pmod{2\pi}.$$

Two fillings f and g compute the same intrinsic holonomy. Hence $\exp[-\mathrm{i}(\Omega_f(C)-\Omega_g(C))/2] = 1$, so their signed areas differ by an integer multiple of 4π. This proves both the solid-angle convention and the claimed phase formula. ∎

The figure fixes the orientation of a latitude and its northern cap.

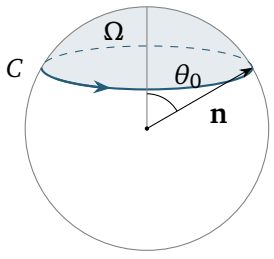


Figure 24.7: Increasing azimuth orients the latitude as the boundary of the northern cap of solid angle $\Omega = 2\pi(1-\cos\theta_0)$. Its spin ground-state geometric phase is $-\Omega/2$.

Example 24.13 (A closed field rotation)**.** Let $b > 0$ be fixed, choose $0 < \theta_0 < \pi$, and rotate the field with $\phi(t) = 2\pi t/T$ for $0 \le t \le T$. Start in $|n_N(\theta_0, 0)\rangle$ and take the adiabatic limit.

We use $H = -b\mathbf{n}\cdot\boldsymbol{\sigma}$. On the northern cap, the ground line is spanned by

$$|n_N(\theta,\phi)\rangle = \begin{pmatrix} \cos(\theta/2) \\ e^{\mathrm{i}\phi}\sin(\theta/2) \end{pmatrix}.$$

Differentiating this unit vector gives its connection

$$\mathcal{A}_N = \langle n_N|\mathrm{d}n_N\rangle = \mathrm{i}\sin^2(\theta/2)\mathrm{d}\phi = \frac{\mathrm{i}}{2}(1-\cos\theta)\mathrm{d}\phi.$$

The northern cap $D = \{0 \le \theta \le \theta_0\}$ carries this frame and has boundary C in increasing ϕ. Stokes on this cap gives

$$\mathrm{i}\int_D \mathrm{d}\mathcal{A}_N = -\frac{1}{2}\int_0^{2\pi}\int_0^{\theta_0}\sin\theta\mathrm{d}\theta\mathrm{d}\phi = -\pi(1-\cos\theta_0) = -\frac{1}{2}\Omega(C).$$

The northern unit frame is periodic on this path, so direct transport gives the same value:

$$\gamma(C) = \mathrm{i}\int_0^{2\pi}\frac{\mathrm{i}}{2}(1-\cos\theta_0)\mathrm{d}\phi = -\pi(1-\cos\theta_0).$$

The energy is $-b$, so the dynamical factor is $e^{+\mathrm{i}bT/\hbar}$. The final state is asymptotic to

$$e^{+\mathrm{i}bT/\hbar}e^{-\mathrm{i}\pi(1-\cos\theta_0)}\,|n_N(\theta_0,0)\rangle\,.$$

For $\theta_0 = \pi/3$ the geometric factor is $-\mathrm{i}$. The southern potential instead gives a phase larger by 2π, hence the same factor.

The orthogonal excited vector $(-e^{-\mathrm{i}\phi}\sin(\theta/2), \cos(\theta/2))^T$ has $\mathcal{A}_{\mathrm{exc}} = -\mathrm{i}(1-\cos\theta)\mathrm{d}\phi/2$. Its curvature and Chern number have the opposite sign, giving $+1$.

Using the same cap, Lecture 18's tangent-vector transport angle is Ω. The spin phase is its negative half before either angle is reduced modulo 2π. At the equator the vector holonomy is a full turn while the spin phase factor is -1.

In Lecture 18's smooth northern tangent frame, the connection entry is $\omega_N = (1-\cos\theta)\mathrm{d}\phi$. The complex tangent-line connection is $-\mathrm{i}\omega_N$, whereas $\mathcal{A}_N = \mathrm{i}\omega_N/2$. In these frames the aligned spin-line connection is minus half the tangent-line connection. For $\theta_0 = \pi/3$, parallel transport through azimuth ϕ has spin phase angle $-\phi/4$ and tangent-vector angle $\phi/2$ relative to these northern frames. The figure samples both angles along the same latitude.

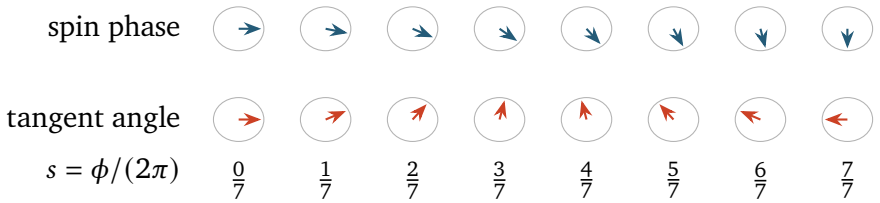


Figure 24.8: The eight samples have $\phi = 2\pi k/7$ for $k = 0,\dots,7$ and $\theta_0 = \pi/3$. The spin phase ends at $-\pi/2$, while the tangent vector ends at angle $+\pi$ in its northern frame.

The spin coefficient also describes a lift into the phase circles over the parameter loop. The figure opens the loop at its base point to display both endpoint representatives.

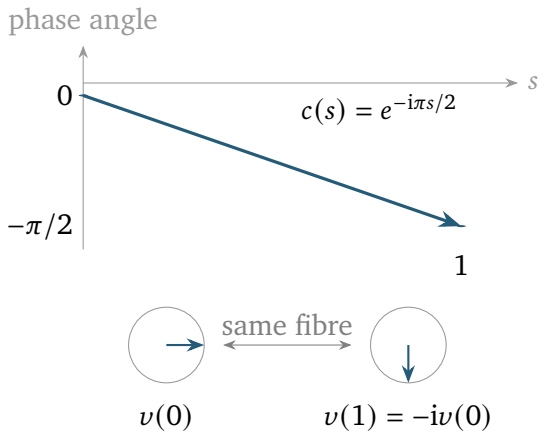


Figure 24.9: For the same latitude, $c(s) = e^{-i\pi s/2}$ gives the parallel vector $v(s) = c(s)\,|n_N(s)\rangle$. The endpoint circles represent the same fibre, but the parallel vector returns multiplied by $-i$.

24.5 The degeneracy and general spin

For $\mathbf{B} \neq 0$, the ground-state projector is $P(\mathbf{B}) = (I + \mathbf{B}\cdot\boldsymbol{\sigma}/|\mathbf{B}|)/2$. It depends only on the field direction. Pulling the sphere curvature back to $\mathbb{R}^3 \setminus \{0\}$ gives

$$\mathrm{d}\mathcal{A} = \frac{\mathrm{i}}{2|\mathbf{B}|^3}\left(B^1\mathrm{d}B^2 \wedge \mathrm{d}B^3 + B^2\mathrm{d}B^3 \wedge \mathrm{d}B^1 + B^3\mathrm{d}B^1 \wedge \mathrm{d}B^2\right).$$

The real curvature $\mathrm{d}A_{\mathrm{B}} = \mathrm{i}\mathrm{d}\mathcal{A}$ points inward. Its flux through an outward-oriented sphere is -2π. This is Lecture 22's $n = -1$ line in field-parameter space. The figure shows its inward real Berry curvature.

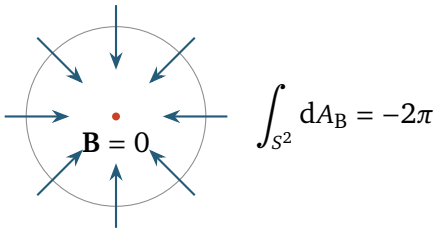


Figure 24.10: The real Berry curvature points inward in field-parameter space. Any outward sphere around the excluded degeneracy has $\int \mathrm{d}A_{\mathrm{B}} = -2\pi$ and ground-line Chern number -1: this is Lecture 22's $n = -1$ line.

At $\mathbf{B} = 0$ the two eigenvalues coincide. Every line is then an eigenline, so the selected ground line and its projector are not determined. For the identically zero Hamiltonian the exact evolution is constant. It does not force a prescribed changing eigenline to be followed. The gap hypothesis prevents this ambiguity.

A nonzero Chern number on the boundary of a parameter ball forces a degeneracy inside if the Hamiltonian and the selected eigenvalue branch are smooth throughout the ball. Indeed, suppose the branch stayed simple there. Radial transport from the centre would give a smooth unit frame on the ball, as in Lecture 22. Its connection form $\mathcal{A}$ would then

be global, so Stokes would give

$$C_1(\partial B^3) = \frac{\mathrm{i}}{2\pi}\int_{\partial B^3} \mathrm{d}\mathcal{A} = \frac{\mathrm{i}}{2\pi}\int_{B^3} \mathrm{d}^2\mathcal{A} = 0.$$

This contradicts a nonzero boundary Chern number. The figure compares a sphere enclosing the degeneracy with one bounding a ball that avoids it.

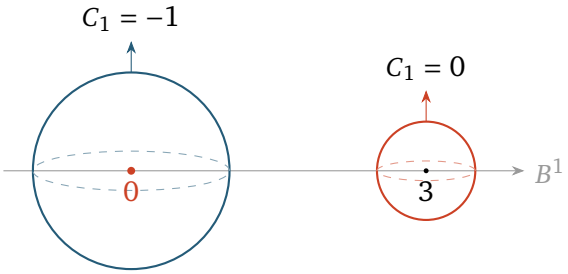


Figure 24.11: For $H = -\mathbf{B}\cdot\boldsymbol{\sigma}$, the unit sphere about the origin has ground-line Chern number -1. The sphere of radius $1/2$ about $(3,0,0)$ bounds a ball avoiding the origin and has Chern number zero.

Every Hermitian 2×2 matrix has the form

$$H = E_0 I + \mathbf{d}\cdot\boldsymbol{\sigma} = \begin{pmatrix} E_0 + d_z & d_x - \mathrm{i}d_y \\ d_x + \mathrm{i}d_y & E_0 - d_z \end{pmatrix}, \qquad E_0, d_x, d_y, d_z \in \mathbb{R}.$$

Its levels coincide exactly when $\mathbf{d} = 0$. For a smooth three-parameter family, assume that the derivative of $\mathbf{d}$ at such a zero is invertible. The inverse function theorem then makes $\mathbf{d}$ a local coordinate system. An enclosing small sphere has lower-line Chern number $+1$ or -1 according to the orientation of these coordinates. Indeed, its lower line pulls back the excited line of $-\mathbf{n}\cdot\boldsymbol{\sigma}$, whose Chern number is $+1$. Lecture 14's degree formula multiplies this by the sign of the coordinate Jacobian. The field model here has $\mathbf{d} = -\mathbf{B}$, which reverses orientation and gives -1. For a larger Hamiltonian this description applies to a separated smooth two-level block under the same invertibility hypothesis. Crossings constrained by symmetries or with singular derivative need not have this local form.

For general spin, we assume the following additional matrix data.

Proposition 24.14 (Spin j · *sketched*)**.** *Let $j > 0$ be an integer or half-integer. Assume Hermitian dimensionless spin matrices $J_a = S_a/\hbar$ on $\mathbb{C}^{2j+1}$, with $[J_a, J_b] = \mathrm{i}\epsilon_{abc}J_c$. Assume an orthonormal basis $|j,\ell\rangle$ with*

$$J_z\,|j,\ell\rangle = \ell\,|j,\ell\rangle\,, \qquad \ell \in \{-j, -j+1, \ldots, j\}.$$

On the outward direction sphere, for the eigenline of $\mathbf{n}\cdot\mathbf{J}$ of weight $m \in \{-j, -j+1, \ldots, j\}$,

$$\mathcal{A}_N = \mathrm{i}m(1-\cos\theta)\mathrm{d}\phi, \qquad \gamma(C) = -m\Omega(C) \pmod{2\pi}, \qquad C_1 = -2m.$$

Sketch. Choose $J_z\,|j,m\rangle = m\,|j,m\rangle$ and rotate the weight vector by

$$|n_N\rangle = e^{\mathrm{i}m\phi}e^{-\mathrm{i}\phi J_z}e^{-\mathrm{i}\theta J_y}\,|j,m\rangle\,.$$

The combined full-turn operator acts on every basis vector by

$$e^{2\pi\mathrm{i}m}e^{-2\pi\mathrm{i}J_z}\,|j,\ell\rangle = e^{2\pi\mathrm{i}(m-\ell)}\,|j,\ell\rangle = |j,\ell\rangle\,,$$

because $m - \ell \in \mathbb{Z}$. Thus the rotated vector is single-valued in ϕ. Differentiating the conjugated operators and using the commutators gives $e^{i\theta J_y} J_z e^{-i\theta J_y} = J_z \cos\theta - J_x \sin\theta$. In a J_z weight vector the expectation of any commutator with J_z is zero. The relations $[J_z, J_y] = -iJ_x$ and $[J_z, J_x] = iJ_y$ therefore give $\langle j, m| J_x |j, m\rangle = \langle j, m| J_y |j, m\rangle = 0$. The remaining expectation is $\langle j, m| J_z |j, m\rangle = m$. Thus the θ coefficient is zero and the ϕ coefficient is $im - im\cos\theta$. Differentiate, integrate over the sphere, and apply the transport formula to get the stated values. The stated matrix data suffice for these rotation and weight identities. Berry, §4, pp. 50–51, obtains the same spin phase using ladder operators. ∎

The quantity $2m$ is an integer, so changing the solid angle by 4π does not change the phase factor. For the ground line of $H = -\kappa \mathbf{B} \cdot \mathbf{J}$ with $\kappa > 0$, the weight is $m = j$ and the degeneracy is again at $\mathbf{B} = 0$.

Self-check. Take $\theta_0 = \arccos(1/3)$ and $\phi(s) = -4\pi s$ for $s \in [0, 1]$. Starting with $c(0) = 1$, compute the ground-state transport factor in both northern and southern frames. Find the signed solid angle for two clockwise traversals of the northern cap and the excited-state phase.

Answer. The azimuth changes by -4π. The northern potential is $id\phi/3$, so the transport equation gives $c(s) = e^{i4\pi s/3}$. The phase and factor are

$$\gamma_N = \frac{4\pi}{3}, \qquad e^{i\gamma_N} = -\frac{1}{2} - \frac{\sqrt{3}}{2} i.$$

The southern potential is $-2id\phi/3$ and gives $\gamma_S = -8\pi/3$. The phases differ by -4π, so their factors agree. The signed solid angle is $-2 \cdot 2\pi(1 - 1/3) = -8\pi/3$, verifying $\gamma = -\Omega/2$. The excited potential has the opposite sign and gives phase $-4\pi/3$ modulo 2π.

Summary. A smooth simple eigenline has the unitary connection $\mathcal{A} = \langle n|d n\rangle$ and curvature $\langle d n| \wedge |d n\rangle$. The cited gapped adiabatic theorem identifies its holonomy with the geometric factor after removing the dynamical phase. The imaginary connection and real potential satisfy $A_B = i\mathcal{A}$. For $H = -\mathbf{B} \cdot \boldsymbol{\sigma}$ with $\mathbf{B} \neq 0$, the ground line has Chern number -1 on an outward direction sphere and $\gamma = -\Omega/2$ modulo 2π. The excluded degeneracy is the source of this parameter-space curvature. Lecture 25 integrates the same curvature over the momentum torus of a gapped band.

Reading. Berry, *Quantal phase factors accompanying adiabatic changes*, §2 and §4, pp. 46–48 and 50–51. Simon, *Holonomy, the quantum adiabatic theorem and Berry's phase*, pp. 2167–2168. Nakahara, §10.6. Tong, *The Quantum Hall Effect*, current §1.5.1–1.5.2, pp. 28–34, with $A_{\text{Tong}} = -A_B$.

Lecture 25

Topological band theory

Objectives. *Lecture 24 attached a line bundle to an isolated simple eigenvalue. For a periodic crystal, the parameters are momenta on a torus. The normalized curvature integral on that torus is an integer. For a two-band model we compute it as a degree, then use the cited Hall-response law to relate this integer to a measured conductance.*

By the end of this lecture you will be able to:

- Construct a Bloch eigenline bundle for a smooth isolated simple band.
- Prove integrality of its normalized curvature flux on the oriented momentum torus.
- Compute the lower-band Chern number and gap closings of the QWZ model.
- State the filled-band Hall formula with its physical hypotheses and component convention.

Unit 25A Bloch bundles and the QWZ degree

25.1 Bloch Hamiltonians and the momentum torus

Consider a square lattice with cell label $r \in \mathbb{Z}^2$ and N orbitals in each cell. Use lattice spacing one and a fixed orthonormal orbital basis. A one-particle wavefunction has a column $\psi_r \in \mathbb{C}^N$ at each cell. A translation-invariant finite-range Hamiltonian is given by

$$(H\psi)_r = \sum_{R\in\mathbb{Z}^2} T_R \psi_{r+R}, \qquad T_{-R} = T_R^\dagger,$$

where only finitely many hopping matrices T_R are nonzero. The adjoint condition makes the operator Hermitian. Recall the positive-phase Bloch convention from *Quantum Mechanics I*, Lecture 8. Substitution of $\psi_r = e^{\mathrm{i}\mathbf{k}\cdot r} v$ gives a matrix acting on the orbital column.

Definition 25.1 (Bloch Hamiltonian)**.** The *Bloch Hamiltonian* in this convention is

$$H(\mathbf{k}) = \sum_R T_R e^{\mathrm{i}\mathbf{k}\cdot R}.$$

It is a smooth Hermitian $N \times N$ matrix. Adding 2π to either momentum component leaves this matrix unchanged.

Definition 25.2 (Brillouin torus)**.** The *Brillouin torus* is the momentum parameter space

$$\mathbb{T}^2 = \mathbb{R}^2/(2\pi\mathbb{Z})^2,$$

with orientation $\mathrm{d}k_x \wedge \mathrm{d}k_y$.

On an $L \times L$ periodic lattice, the allowed momenta have components $2\pi a/L$, with $a \in \{0, \dots, L-1\}$. The finite geometric sum makes Bloch waves with distinct momenta orthogonal. There are L^2 such momenta, each paired with the N orbital columns. The resulting Bloch waves form a basis of the NL^2-dimensional one-particle space. The substitution above therefore gives its full block decomposition. A finite sample therefore samples L^2 points of one smooth family $H(\mathbf{k})$. The family does not depend on L. Only the sampling changes. The eigenline bundle and its curvature belong to this continuous family. The infinite-lattice Bloch ansatz is not a normalized square-summable state.

Definition 25.3 (Band)**.** A *band* is an energy branch $E_n(\mathbf{k})$ of this family.

Definition 25.4 (Isolated band)**.** We call a chosen smooth simple band *isolated* when, at every momentum, its distance from every other eigenvalue is at least a fixed $\delta > 0$.

This is a separation at the same momentum. It does not yet assert an energy interval separating occupied and empty states at all momenta.

Physical input. We model a slowly driven electron by a wavepacket, a superposition of Bloch states concentrated near a central momentum $\mathbf{k}$ in an isolated band. For electron charge $-e$, with $e > 0$, a weak uniform electric field drives that momentum by

$$\hbar \dot{\mathbf{k}} = -e\mathbf{E}.$$

We assume interband transitions are negligible in this semiclassical approximation.

The field therefore changes the parameter of $H(\mathbf{k})$. Lecture 24's eigenline curvature can affect this slow evolution.

Proposition 25.5 (The Bloch bundle · *proved*)**.** *The eigenspaces of a smooth isolated simple band form a Hermitian line bundle $L_n \to \mathbb{T}^2$. For local unit eigenvectors $v_i(\mathbf{k})$, its connection is $\mathcal{A}_i = v_i^\dagger \mathrm{d}v_i$.*

Proof. The periodic matrix family is defined on the torus itself. Lecture 24's adjugate construction supplies smooth local eigenvectors there. Their unit-phase transitions give the line bundle and the projected Hermitian connection. Its curvature is global and imaginary. ∎

Definition 25.6 (Band Chern number)**.** The *band Chern number* of an isolated simple band is

$$C_n = \frac{\mathrm{i}}{2\pi} \int_{\mathbb{T}^2} \mathrm{d}\mathcal{A}.$$

The integral is real because the curvature is imaginary.

25.2 Integral flux and the two-band degree

Definition 25.7 (Smooth family)**.** A *smooth family* of Hermitian line bundles with connection here means one Hermitian line bundle with a smooth unitary connection on $\mathbb{T}^2 \times [0, 1]$, restricted to the momentum slices.

Proposition 25.8 (Integrality on the torus · *proved*)**.** *For a Hermitian line bundle with a unitary connection on the oriented torus, the number $\mathrm{i} \int_{\mathbb{T}^2} F/(2\pi)$ is an integer. It is unchanged in a smooth family as defined above.*

Proof. We cut the torus into a square and compare a unit frame across its paired edges. Stokes converts the curvature flux to edge-phase changes, as in Lecture 22. Compatibility at the corner forces their difference to be an integer multiple of 2π.

Pull the bundle and connection back along $\mathbb{R}^2 \to \mathbb{T}^2$. Transport a fixed unit vector at the origin along radial segments. Lecture 21's smooth parameter dependence gives a smooth unit frame on $\mathbb{R}^2$. Restrict it to the square $[0, 2\pi]^2$ and call its connection form A. Write $x = k_x$ and $y = k_y$ in this argument. After identifying equal torus fibres, the edge frames satisfy

$$e(2\pi, y) = e(0, y)h_x(y), \qquad e(x, 2\pi) = e(x, 0)h_y(x),$$

with smooth maps $h_x, h_y \colon [0, 2\pi] \to \mathrm{U}(1)$. Lecture 3's phase lifting on an interval gives real smooth functions χ_x and χ_y with $h_x = e^{\mathrm{i}\chi_x}$ and $h_y = e^{\mathrm{i}\chi_y}$. Smoothness follows locally by a smooth inverse of the exponential covering. Lecture 21 gives the frame-change law $A' = h^{-1}Ah + h^{-1}\mathrm{d}h$. For $h = e^{\mathrm{i}\chi}$ the added term is $\mathrm{i}\mathrm{d}\chi$. Thus $A_R - A_L = \mathrm{i}\mathrm{d}\chi_x$ along the vertical edges and $A_T - A_B = \mathrm{i}\mathrm{d}\chi_y$ along the horizontal edges. We write e_L, e_R, e_B and e_T for the restrictions of the square frame to its four sides. The figure shows these edge comparisons after equal torus fibres have been identified.

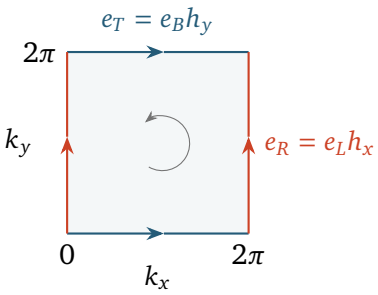


Figure 25.1: Opposite sides represent the same momenta. The phase functions compare the square frame across paired edges, and the circular arrow gives the positive boundary orientation.

Let $A_{R,y}$ and $A_{L,y}$ be the tangential coefficients on the vertical edges. Let $A_{T,x}$ and $A_{B,x}$ be those on the horizontal edges. The top and left sides have reversed boundary directions, so Stokes gives

$$\begin{aligned}\int_{\mathbb{T}^2} F = \oint_{\partial\square} A &= \int_0^{2\pi} (A_{R,y} - A_{L,y})\mathrm{d}y - \int_0^{2\pi} (A_{T,x} - A_{B,x})\mathrm{d}x \\ &= \mathrm{i}\Delta\chi_x - \mathrm{i}\Delta\chi_y,\end{aligned}$$

where $\Delta\chi = \chi(2\pi) - \chi(0)$. Consequently,

$$\int_{\mathbb{T}^2} F = \mathrm{i}(\Delta\chi_x - \Delta\chi_y), \qquad \frac{\mathrm{i}}{2\pi}\int_{\mathbb{T}^2} F = \frac{\Delta\chi_y - \Delta\chi_x}{2\pi}.$$

The right and top comparisons give the same frame at the last corner, so

$$h_y(0)h_x(2\pi) = h_x(0)h_y(2\pi).$$

Substitution of their lifted phases makes $\Delta\chi_x - \Delta\chi_y$ an integer multiple of 2π. This proves integrality. Neither edge phase is required to be a closed loop. In such a family, its global curvature restricts smoothly to the momentum slices, so the curvature integral varies continuously. An integer-valued continuous function on an interval is constant, proving the last claim. ∎

For a smooth Hamiltonian family whose chosen band stays isolated, the adjugate construction is smooth jointly in momentum and deformation parameter. The eigenline and projected connection of the chosen band give such a family.

For an example of integrality, the square potential $A = -\mathrm{i}nx\,\mathrm{d}y/(2\pi)$, with $n \in \mathbb{Z}$, has $h_x(y) = e^{-\mathrm{i}ny}$ and $h_y(x) = 1$. Its flux is $-2\pi\mathrm{i}n$, so its Chern number is n. The edge phase at $y = 2\pi$ agrees with its value at zero precisely because n is an integer. When $h_y = 1$, the corner identity makes h_x a closed loop. In that case $C = -\operatorname{wind}(h_x)$, as in Lecture 22's transition calculation.

A nonzero Chern number obstructs a smooth periodic unit eigenvector on the whole torus. Such an eigenvector would give a global form $\mathcal{A} = v^\dagger \mathrm{d}v$. Stokes on the closed torus would then give $\int_{\mathbb{T}^2} \mathrm{d}\mathcal{A} = 0$. Thus a nonzero number forces a phase mismatch between local eigenvectors.

Now take two orbitals per cell and a Hamiltonian $H(\mathbf{k}) = \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}$, where $\mathbf{d}\colon \mathbb{T}^2 \to \mathbb{R}^3 \setminus \{0\}$ is smooth. The Pauli product identity gives energies $-|\mathbf{d}|$ and $+|\mathbf{d}|$.

Definition 25.9 (Direction map)**.** The *direction map* is given by

$$\widehat{\mathbf{d}}\colon \mathbb{T}^2 \longrightarrow S^2, \qquad \widehat{\mathbf{d}}(\mathbf{k}) = \frac{\mathbf{d}(\mathbf{k})}{|\mathbf{d}(\mathbf{k})|}.$$

Theorem 25.10 (The lower band as a degree · *proved*)**.** *For the lower band and the outward orientation on S^2,*

$$\mathrm{d}\mathcal{A} = -\frac{\mathrm{i}}{2}\widehat{\mathbf{d}} \cdot (\partial_{k_x}\widehat{\mathbf{d}} \times \partial_{k_y}\widehat{\mathbf{d}})\,\mathrm{d}k_x \wedge \mathrm{d}k_y, \qquad C = \deg(\widehat{\mathbf{d}}).$$

Proof. Dividing H by $|\mathbf{d}|$ changes its energies but leaves its eigenvectors unchanged. The lower eigenspace of $\widehat{\mathbf{d}} \cdot \boldsymbol{\sigma}$ is the excited eigenspace of Lecture 24's $-\mathbf{n} \cdot \boldsymbol{\sigma}$. That sphere line has curvature $-\mathrm{i}\beta/2$, where β is the outward area form. Indeed, its northern potential in Lecture 24 is $-\mathrm{i}(1 - \cos\theta)\mathrm{d}\phi/2$. Differentiating gives $-\mathrm{i}\sin\theta\mathrm{d}\theta \wedge \mathrm{d}\phi/2$. Lecture 12's area form satisfies $\beta_p(v, w) = p \cdot (v \times w)$ for $p \in S^2$ and $v, w \in T_pS^2$. Its pullback therefore has coefficient $\widehat{\mathbf{d}} \cdot (\partial_{k_x}\widehat{\mathbf{d}} \times \partial_{k_y}\widehat{\mathbf{d}})$. The band line is its pullback by $\widehat{\mathbf{d}}$. Pullback of the projected connection pulls back its curvature, giving the displayed formula. Lecture 14's degree identity gives

$$C = \frac{1}{4\pi}\int_{\mathbb{T}^2} \widehat{\mathbf{d}}^*\beta = \deg(\widehat{\mathbf{d}}),$$

because $\int_{S^2} \beta = 4\pi$. ∎

The degree counts preimages of a regular sphere direction with their orientation signs. It need not count a one-to-one covering of the sphere. The upper band has the opposite curvature and Chern number. For more than two orbitals, the curvature definition still applies, but this particular map to a direction sphere is unavailable.

25.3 The Qi–Wu–Zhang model

We seek a finite-range two-band lattice model with a direction map of nonzero degree. Finite-range hopping gives trigonometric polynomials. We choose $d_x = \sin k_x$ and $d_y = \sin k_y$. Their simultaneous zeros occur when both momentum coordinates belong to $\{0, \pi\}$. The third component can change sign at these four points as a parameter varies.

Definition 25.11 (Qi–Wu–Zhang model)**.** The *Qi–Wu–Zhang model*, abbreviated QWZ, has direction vector

$$\mathbf{d} = (\sin k_x, \sin k_y, u + \cos k_x + \cos k_y), \qquad u \in \mathbb{R}.$$

Its Hamiltonian is the matrix

$$H(\mathbf{k}) = \begin{pmatrix} u + \cos k_x + \cos k_y & \sin k_x - \mathrm{i} \sin k_y \\ \sin k_x + \mathrm{i} \sin k_y & -u - \cos k_x - \cos k_y \end{pmatrix}.$$

The parameter u is an energy parameter, distinct from Lecture 19's inverse radius. In the lattice convention of the first section its only possibly nonzero hopping matrices are $T_0 = u\sigma_z$, $T_{(1,0)} = (\sigma_z - \mathrm{i}\sigma_x)/2$ and $T_{(0,1)} = (\sigma_z - \mathrm{i}\sigma_y)/2$, together with their adjoints. Thus it is a finite-range Hermitian lattice system.

This is a model lattice Hamiltonian, rather than a model of a specified material. We call a momentum with $d_x = d_y = 0$ a *vertical point*. At such a point, the value of d_z is its *mass*. The mass multiplies σ_z and opens the local Dirac gap of §25.5. We denote the four points by Γ, X_1, X_2 and M, as in the table.

Example 25.12 (The QWZ phase diagram · *proved*)**.** Find the lower-band Chern number and every gap closing as u varies. The energies are $E_- = -|\mathbf{d}|$ and $E_+ = |\mathbf{d}|$, so a gap closes exactly when $\sin k_x = \sin k_y = 0$ and $u + \cos k_x + \cos k_y = 0$. The relevant masses and transverse Jacobian signs are

point	(k_x, k_y)	d_z	$\cos k_x \cos k_y$
Γ	$(0, 0)$	$u + 2$	$+1$
X_1	$(\pi, 0)$	u	-1
X_2	$(0, \pi)$	u	-1
M	(π, π)	$u - 2$	$+1$

Hence the critical parameters are -2, 0 and 2. There are two distinct gap-closing points at zero. At every other parameter, compactness of the torus gives a positive minimum gap.

To compute the degree, count preimages of the north pole. A row contributes precisely when its mass d_z is positive. Near such a point the first two components of $\widehat{\mathbf{d}}$ are local sphere coordinates. Their Jacobian determinant is $\cos k_x \cos k_y / d_z^2$ at the point. These coordinates have the outward sphere orientation at the north pole. All selected preimages are therefore regular, with the signs in the last column. Lecture 14's signed-count theorem gives

$$C = \begin{cases} 0, & u < -2, \\ +1, & -2 < u < 0, \\ -1, & 0 < u < 2, \\ 0, & u > 2. \end{cases}$$

For $u < -2$ there are no north preimages. In the next interval only Γ contributes. Crossing zero adds both negatively oriented points. Crossing two adds the positive contribution at M. Thus the simultaneous closing at zero changes the integer by -2.

The square and the embedded torus in the figure describe the same periodic coordinates.

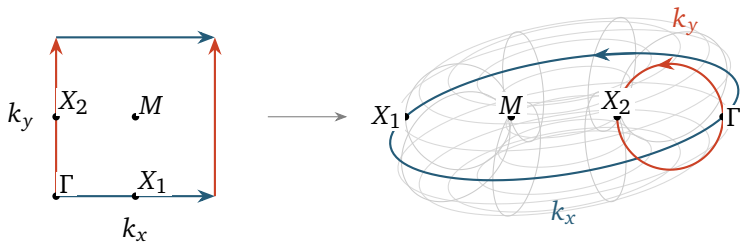


Figure 25.2: The parametrization $((2+\cos k_y)\cos k_x, (2+\cos k_y)\sin k_x, \sin k_y)$ embeds the momentum torus in $\mathbb{R}^3$. The marked points and the two coordinate circles agree with those on the square.

The vertical points and their orientation signs are now shown on the momentum square.

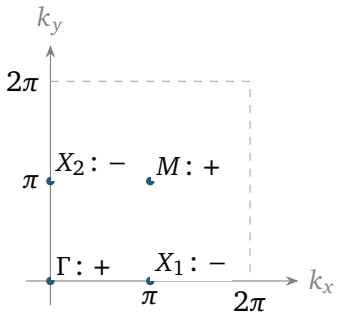


Figure 25.3: The four distinct vertical points in one torus cell. Each sign contributes to the north-pole count only when its mass is positive.

The direction maps show how these signed preimages give different degrees.

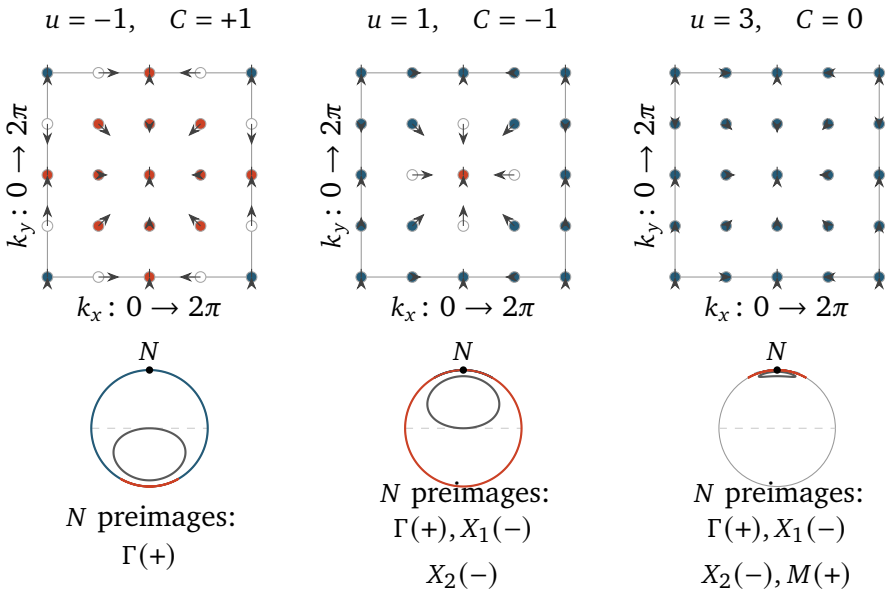


Figure 25.4: Arrows give $(\widehat{d}_x, \widehat{d}_y)$ on a grid with blue, red and white dots for positive, negative and zero $\widehat{d}_z$. The sphere curves show the images of $k_y = 0, \pi/2, \pi$ viewed along the y axis, with signed north-pole counts below each panel.

The phase diagram records the computed lower-band integer and its missing critical parameters.

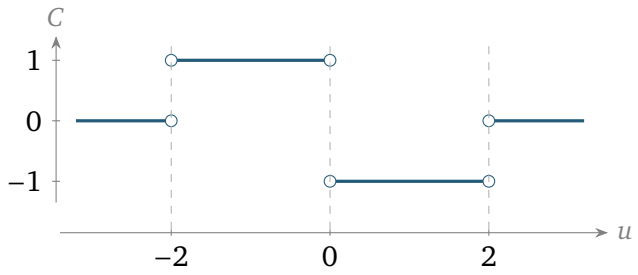


Figure 25.5: The QWZ lower-band Chern number as u increases. The number is undefined at $u = -2$, $u = 0$ and $u = 2$.

The band plots distinguish the two gapped examples from the critical parameters.

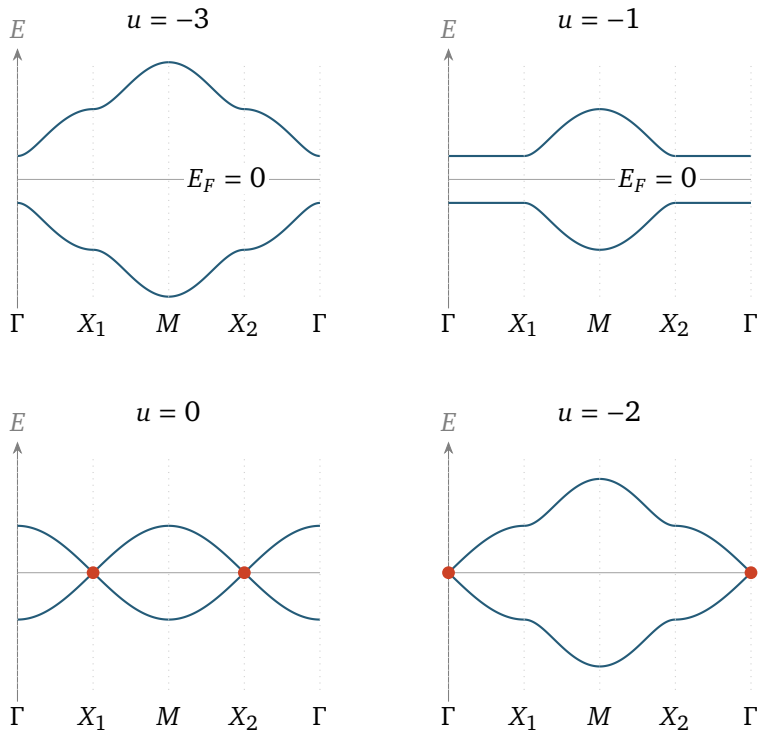


Figure 25.6: The energies E_- and E_+ along $\Gamma \to X_1 \to M \to X_2 \to \Gamma$ at four values of u. The closing at the repeated endpoint Γ is one torus point; the two X closings are distinct.

These values use $C = (\mathrm{i}/2\pi) \int \mathrm{d}\mathcal{A}$.[1]

The exact degree calculation requires no numerical mesh. An independent numerical check uses an $L \times L$ periodic momentum grid. Choose a unit lower-band vector w_j at each vertex, with arbitrary phase. Label a plaquette's four vertices counterclockwise. Multiply the forward overlaps $\langle w_1|w_2\rangle \langle w_2|w_3\rangle \langle w_3|w_4\rangle \langle w_4|w_1\rangle$ and divide each overlap by its magnitude, assuming none vanishes. Every vertex phase cancels from this product. With a

[1] Asbóth–Oroszlány–Pályi use the same real potential $A_\mathrm{B} = \mathrm{i}\mathcal{A}$. Their continuum definition is $Q = -(1/2\pi) \int \mathrm{d}A_\mathrm{B}$, so $Q = -C$ in their equations (2.38), (2.41) and (6.5).

counterclockwise plaquette, the product of normalized overlaps has argument approaching $\operatorname{Im} \int \mathrm{d}\mathcal{A}$. Thus the *negative* sum of these arguments, divided by 2π, approximates our C. At $u = -3, -1, 1, 3$, grids with 20×20 and 32×32 points give values close to $0, +1, -1, 0$, respectively, in agreement with a separate curvature integral. These numerical checks do not replace the degree proof.

Self-check. Compute the lower-band Chern number of the following Hamiltonian by counting south-pole preimages:

$$H_*(\mathbf{k}) = \mathbf{d}_*(\mathbf{k}) \cdot \boldsymbol{\sigma}, \qquad \mathbf{d}_* = (\sin(2k_x), \sin k_y, \frac{1}{2} + \cos(2k_x) + \cos k_y).$$

Determine the local signs using the outward orientation at the south pole.

Answer. The transverse components vanish for $k_x \in \{0, \pi/2, \pi, 3\pi/2\}$ and $k_y \in \{0, \pi\}$. The masses at these points are nonzero, so the band is isolated. Only $(\pi/2, \pi)$ and $(3\pi/2, \pi)$ have negative mass, equal to $-3/2$. At each of them the transverse Jacobian is $2\cos(2k_x)\cos k_y = 2$. The normalized chart Jacobian is therefore $2/(-3/2)^2 = 8/9$. The coordinates $(\widehat{d}_{*,x}, \widehat{d}_{*,y})$ have the opposite of the outward orientation at the south pole. Both local signs are -1, giving $C = -2$.

Self-check. Explain why the QWZ lower-band number is zero at $u = 3$, without counting preimages.

Answer. The component $d_z = 3 + \cos k_x + \cos k_y$ is at least one. Thus the direction map lies in the northern hemisphere and contracts there to the north pole. Its degree is zero by Lecture 14's homotopy invariance.

Unit 25B Hall response and gap closings

25.4 Filled bands and the Hall response

We consider a thin flat sample in the xy plane, with contacts that allow a transverse current. A voltage applied along x produces a weak field E_x. We measure the induced current per unit transverse length J_y. The *bulk* is the sample interior, away from its boundary. We call the periodic model *clean* when it has no disorder. Linear response is the term proportional to the applied weak field.

> **Definition 25.13** (Fermi energy)**.** The Fermi energy E_F is the zero-temperature occupation threshold.

Physical input. For noninteracting electrons at zero temperature, each one-particle state below E_F is occupied and each state above it is empty. Pauli exclusion permits one electron per state, with any spin already included among the orbitals.

> **Definition 25.14** (Band insulator)**.** An insulator has an open energy interval around E_F containing no bulk states.

This full energy gap is stronger than separation of a band at each momentum.

> **Definition 25.15** (Chern insulator)**.** A *Chern insulator* is a band insulator whose occupied bands have nonzero total Chern number.

For example, $H = 2\cos k_x I + \sigma_z$ has energies $2\cos k_x - 1$ and $2\cos k_x + 1$. Each band is isolated by a direct gap of two. Their energy ranges are $[-3, 1]$ and $[-1, 3]$, so no Fermi energy separates them completely. The Hamiltonian $H = \sigma_z$ instead has energies -1 and $+1$, with a full gap between them. The figure compares these periodic spectra, independent of k_y.

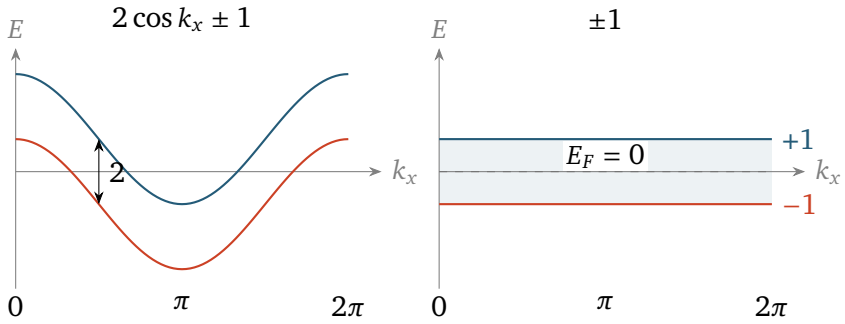


Figure 25.7: The spectra $2\cos k_x \pm 1$ overlap in energy despite their pointwise separation. The spectra ± 1 admit every Fermi energy in $(-1, 1)$.

The electron has signed charge $-e$, as fixed in §25.1. Write Planck's constant as $h_{\mathrm{P}} = 2\pi\hbar$, distinct from a frame-change map.

Definition 25.16 (Hall conductance). The scalar Hall conductance is defined by the transverse component convention

$$J_y = \sigma_{\mathrm{H}} E_x, \qquad J_x = -\sigma_{\mathrm{H}} E_y.$$

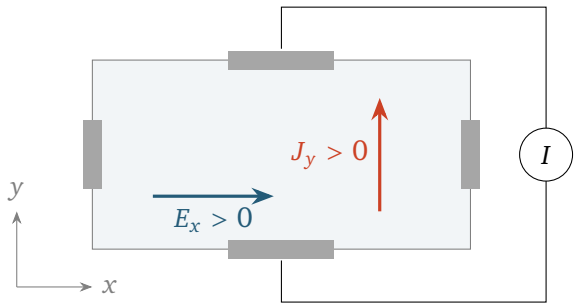
Physical input. **TKNN response 25.17** · *cited.* For a clean two-dimensional periodic noninteracting band insulator at zero temperature, with a bulk Fermi-energy gap and isolated occupied bands, the bulk linear Hall response is

$$\sigma_{\mathrm{H}} = \frac{e^2}{h_{\mathrm{P}}} \sum_{n \text{ occupied}} C_n.$$

For one filled band the sum has one term. In this ideal zero-temperature setting the longitudinal dc response vanishes.

We cite the filled-band response as TKNN. The formula also applies to periodic band insulators with zero magnetic flux through a unit cell, under the stated response hypotheses. The Kubo linear-response argument is cited rather than proved here (Tong, *The Quantum Hall Effect*, §2.3.1, equation (2.20), pp. 63–65). The response components follow the displayed definition. QWZ has no imposed uniform magnetic flux. The full bulk gap is compatible with the transverse filled-band response above. The integer stays fixed under deformations preserving the response hypotheses.

The sample sketch fixes the driving and measured directions for positive Hall conductance.



$$\sigma_H = Ce^2/h_{\mathrm{P}} > 0$$

Figure 25.8: A field along $+x$ produces transverse current along $+y$ when $C > 0$ for one occupied band. Transverse contacts measure the current indicated by the interior response arrows.

Physical input. For a boundary to a trivial insulator, bulk–boundary correspondence is additional cited physics (Asbóth–Oroszlány–Pályi, §6.2–6.3). For local noninteracting charge-conserving models, it fixes the net chirality of edge branches by the difference of occupied-band Chern numbers, with a fixed boundary orientation. Extra pairs of opposite-moving branches are possible. Edge-localized energies can cross E_F while the bulk gap remains open.

We derive the anomalous velocity as a non-rigorous first-order adiabatic approximation. We use a packet narrow in momentum, a smooth isolated nondegenerate band and slow driving. We omit higher-order terms and sustained transitions between bands.

Physical input. In the fixed orbital and cell-position convention of §25.1, the velocity operator is $\widehat{v}_a = \partial_a H/\hbar$, where $\partial_a = \partial/\partial k_a$ for $a \in \{x, y\}$. The filled-band current is the charge times the packet velocities summed with momentum density $\mathrm{d}^2k/(2\pi)^2$.

We choose local orthonormal eigenvectors $|n\rangle$ and $|m\rangle$. Along the path, we use a parallel gauge $\langle n|\dot{n}\rangle = 0$. After factoring the dynamical phase, the Berry phase is included by this gauge choice. We write the first-order state as

$$|\psi\rangle \simeq e^{-\mathrm{i}\int E_n \mathrm{d}t/\hbar}\left(|n\rangle + \sum_{m\neq n} c_m\,|m\rangle\right), \qquad c_m = \frac{\mathrm{i}\hbar\,\langle m|\dot{n}\rangle}{E_m - E_n}.$$

Indeed, projection of Schrödinger's equation onto $|m\rangle$ gives $(E_m - E_n)c_m = \mathrm{i}\hbar\,\langle m|\dot{n}\rangle$ to first order. Differentiating the eigenvalue equation and using orthogonality gives

$$\langle n|\,\partial_a H\,|m\rangle = (E_n - E_m)\,\langle \partial_a n|m\rangle\,.$$

Thus the energy denominators cancel in the velocity correction:

$$\begin{aligned}\hbar\langle\widehat{v}_a\rangle &\simeq \partial_a E_n + \sum_{m\neq n}\left(c_m\,\langle n|\,\partial_a H\,|m\rangle + \text{complex conjugate}\right)\\ &= \partial_a E_n - \mathrm{i}\hbar\left(\langle\partial_a n|\dot{n}\rangle - \langle\dot{n}|\partial_a n\rangle\right).\end{aligned}$$

Completeness supplies the last line. The omitted n term cancels in the antisymmetric difference because $\langle n|\partial_a n\rangle$ is imaginary. We define the real curvature components by

$$\Omega_{ab} = \mathrm{i}\left(\langle\partial_a n|\partial_b n\rangle - \langle\partial_b n|\partial_a n\rangle\right), \qquad a, b \in \{x, y\}.$$

The first-order velocity is therefore

$$\langle \widehat{v}_a \rangle \simeq \frac{\partial_a E_n}{\hbar} - \sum_{b \in \{x,y\}} \Omega_{ab} \dot{k}_b .$$

Let $\Omega_{\mathrm{B}} = \Omega_{xy}$, so $\mathrm{d}A_{\mathrm{B}} = \Omega_{\mathrm{B}} \mathrm{d}k_x \wedge \mathrm{d}k_y$. In two dimensions the second term is $-\dot{\mathbf{k}} \times (\Omega_{\mathrm{B}} \mathbf{e}_z)$, where $\mathbf{e}_z$ is the unit normal to the sample. For $E_x > 0$ and $\Omega_{\mathrm{B}} > 0$, the acceleration law gives negative $\dot{k}_x$. The anomalous electron velocity points along $-y$, so its negative charge produces current along $+y$. The figure separates momentum drift from real-space motion.

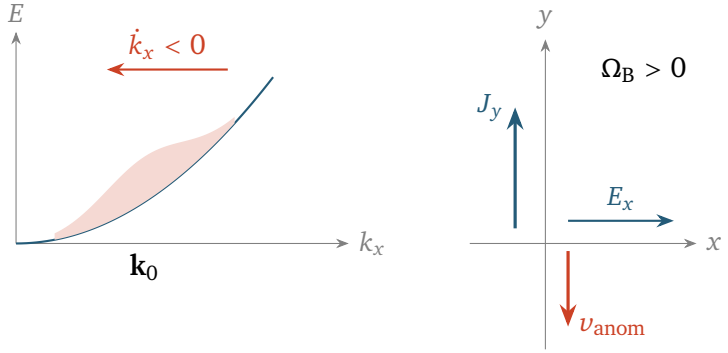


Figure 25.9: A packet near $\mathbf{k}_0$ drifts toward $-k_x$ under a positive E_x. For positive Ω_{B} its anomalous velocity is along $-y$, while its charge current is along $+y$.

Multiplication of this anomalous velocity by charge $-e$ gives a current contribution proportional to $(-E_y, E_x)$. For a filled band the group-velocity integral vanishes by periodicity:

$$\int_{\mathbb{T}^2} \partial_a E_n \mathrm{d}^2 k = 0, \qquad a \in \{x, y\}.$$

The momentum density of states is $\mathrm{d}^2k/(2\pi)^2$, so the anomalous contribution gives

$$\mathbf{J} = \frac{e^2}{\hbar} \int_{\mathbb{T}^2} \frac{\mathrm{d}^2 k}{(2\pi)^2} \Omega_{\mathrm{B}}(-E_y, E_x) = \frac{e^2}{h_{\mathrm{P}}} C(-E_y, E_x).$$

This checks the component sign and units of the cited formula. It is not a derivation of the Kubo theorem. For QWZ with only the lower band filled, the four insulating intervals have the conductances 0, $+e^2/h_{\mathrm{P}}$, $-e^2/h_{\mathrm{P}}$ and 0, in increasing u. Indeed, $\min|\mathbf{d}| > 0$ off the critical parameters, so $E_F = 0$ lies in the full gap between $-\min|\mathbf{d}|$ and $+\min|\mathbf{d}|$. At a critical parameter the gap hypothesis fails and the response law does not assign that quantized value.

25.5 Gap closings and Dirac cones

Definition 25.18 (Dirac cone). A *Dirac cone* is a linear two-band crossing with local Hamiltonian

$$H_{\mathrm{D}}(\mathbf{q}) = v_x q_x \sigma_x + v_y q_y \sigma_y + m\sigma_z, \qquad v_x v_y \neq 0.$$

Here $\mathbf{q}$ measures displacement from the crossing momentum. The energies of this local Hamiltonian at $m = 0$ are $\pm\sqrt{v_x^2 q_x^2 + v_y^2 q_y^2}$. The coefficients v_x and v_y are the local slopes, distinct from a unit eigenvector v. The mass m opens a gap. Integrating the lower-band

curvature density over a disk gives a finite-cutoff contribution. For $v_x = v_y = 1$ and $m \neq 0$, the contribution inside a circular disk is

$$C_{\text{cone}}(R) = \frac{1}{2}\operatorname{sgn}(m)\left(1 - \frac{|m|}{\sqrt{R^2 + m^2}}\right).$$

Here R is a momentum cutoff radius, rather than a lattice hopping vector. The figure shows this function against $R/|m|$.

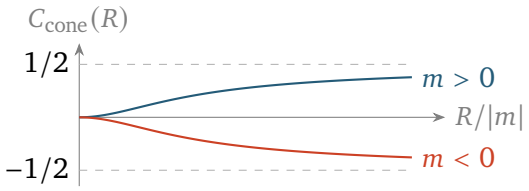


Figure 25.10: The continuum-cone contribution inside a disk of radius R, for unit positive slopes and $m \neq 0$. The limits are half-integers because the momentum plane is not a closed torus.

Proposition 25.19 (Changes at a closing · *sketched*)**.** *A smooth deformation of an isolated band cannot change its Chern number without losing its isolation. When isolated linear Dirac crossings are the only gap closings, their signed unit changes add. For a cone whose mass increases through zero, the change is* $\operatorname{sgn}(v_x v_y)$.

Sketch. Away from a closing, the eigenline curvature varies smoothly, so the integer remains constant by Proposition 25.8. Near a cone, the degree density gives the plane contribution

$$C_{\text{cone}} = \frac{1}{4\pi}\int_{\mathbb{R}^2} \frac{m v_x v_y \, \mathrm{d}q_x \mathrm{d}q_y}{(v_x^2 q_x^2 + v_y^2 q_y^2 + m^2)^{3/2}} = \frac{1}{2}\operatorname{sgn}(v_x v_y)\operatorname{sgn}(m).$$

Indeed, rescale the two momentum axes and use polar coordinates. The remaining radial integral is $\int_0^\infty |m| r (r^2 + m^2)^{-3/2} \mathrm{d}r = 1$. Its change across zero is the claimed signed unit. The curvature outside small crossing neighbourhoods has the same limiting integral from either side. In the local rescaling limit each neighbourhood has the displayed unit jump. Disjoint neighbourhoods add their contributions. This local limiting argument sketches the jump. It does not assign an integer to an isolated continuum cone. ∎

For QWZ the velocities at a vertical point are $v_x = \cos k_x$ and $v_y = \cos k_y$. Each local mass increases with u. The jumps are therefore +1 at −2, $-1 - 1 = -2$ at zero and +1 at two, in agreement with the full degree proof.

The direction map of $(v_x q_x, v_y q_y, m)$ maps the plane to an open hemisphere when $m \neq 0$. For positive $v_x v_y$, the north image at positive mass has signed area 2π. The south image at negative mass has signed area -2π. Division by 4π explains the two half-integers. The equator is approached only at infinite momentum. At $m = 0$, the origin is undefined and the punctured plane maps to the equator. The cone sections and their direction images are shown together.

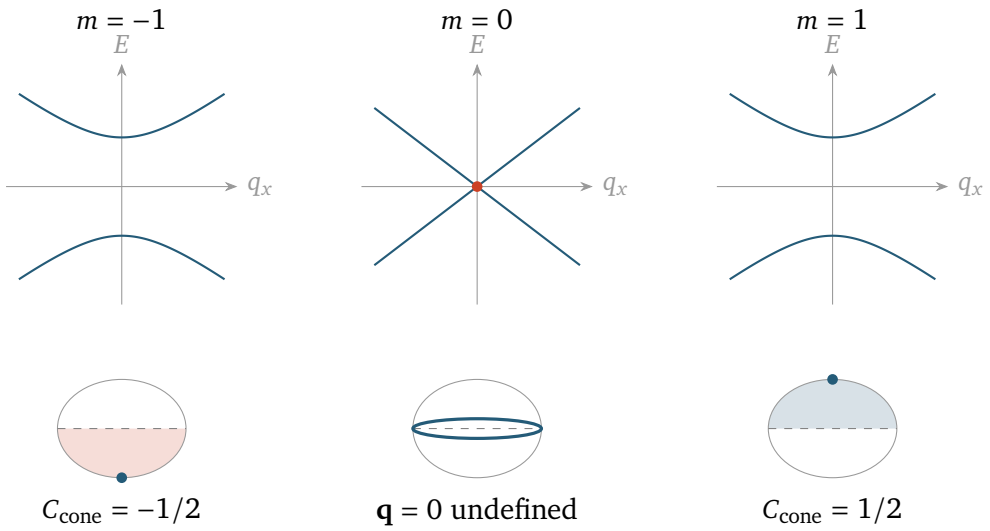


Figure 25.11: For unit positive slopes, the sections $q_y = 0$ have energies $\pm\sqrt{q_x^2 + m^2}$. The direction images are open hemispheres with signed flux $-1/2$ and $+1/2$ for negative and positive mass, and the equator with undefined origin at zero mass.

For QWZ, we can also view a closing in the parameter space $\mathbb{T}^2 \times [u_1, u_2]$. We give this space orientation $\mathrm{d}k_x \wedge \mathrm{d}k_y \wedge \mathrm{d}u$ and take gapped endpoint slices. The normalized pullback area form $\widehat{\mathbf{d}}^* \beta/(4\pi)$ is closed away from the zeros of $\mathbf{d}$. We remove small disjoint balls around its isolated zeros. Stokes gives

$$C(u_2) - C(u_1) = \sum_j \deg\left(\widehat{\mathbf{d}}\big|_{\partial B_j}\right).$$

The top slice has the positive torus orientation and the bottom has its negative. Each small sphere in this display has its outward orientation. Its orientation as an inner boundary of the remaining region is the opposite. This gives the sign in the display. Near a QWZ zero, the linear part is $(v_x q_x, v_y q_y, u - u_c)$. Its determinant is $v_x v_y$, so its small-sphere degree is $\operatorname{sgn}(v_x v_y)$. Higher-order terms do not change the degree on a sufficiently small sphere, by homotopy invariance. This is Lecture 24's parameter-space monopole applied to the lower eigenline of $H = \mathbf{d} \cdot \boldsymbol{\sigma}$. It is not a real-space magnetic monopole. The figure locates the four charges between the gapped slices.

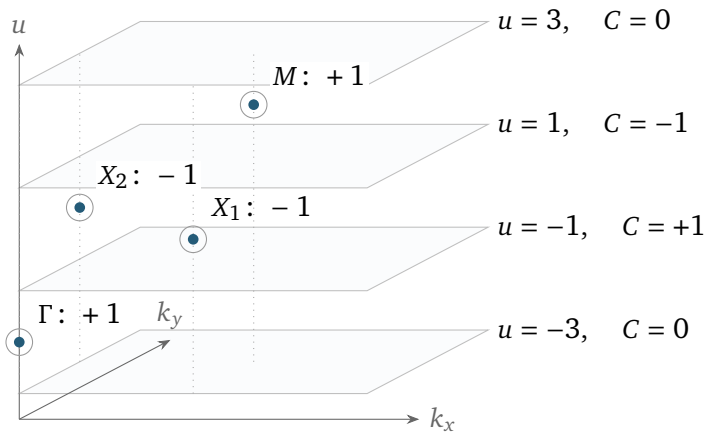


Figure 25.12: The axes have positive order (k_x, k_y, u), and each small sphere has its outward orientation. The four charges $+1, -1, -1, +1$ sum to the slice changes.

A deformation between different slice Chern numbers must therefore pass through a gap closing.

Self-check. State in one sentence what the band Chern number is and why it was introduced. Explain why a nonzero value prevents a smooth periodic unit eigenvector.

Answer. The band Chern number is the integer normalized curvature flux of an isolated eigenline over the momentum torus, introduced to classify gapped bands and determine their filled-band Hall response. A global periodic eigenvector would make $\mathcal{A}$ global, so Stokes on the closed torus would force that flux to vanish.

Summary. An isolated simple Bloch band defines a line bundle on the momentum torus. Its normalized curvature integral is an integer, fixed during any smooth deformation preserving band isolation. For $H = \mathbf{d} \cdot \boldsymbol{\sigma}$ with $\mathbf{d} \neq 0$, the lower-band number is $C = \deg(\widehat{\mathbf{d}})$. For QWZ its values are $0, +1, -1, 0$ in increasing u, with undefined numbers at $-2, 0, 2$. The cited filled-band response law gives the Hall conductance in units of e^2/h_{P}. The same principle of a homotopy class obstructing deformation will classify defects in Lecture 26.

Reading. Asbóth–Oroszlány–Pályi, *A Short Course on Topological Insulators*, §2.1.3–2.1.4 for the discrete sign; §2.2.1, §2.2.4 and §6.1, pp. 80–82 for QWZ, with continuum $Q = -C$. Thouless et al., *Quantized Hall Conductance in a Two-Dimensional Periodic Potential*, pp. 405–408, especially equation (5) on p. 406.

Lecture 26

Higher homotopy groups and topological defects

Objectives. *A loop detects whether a line defect can be removed while the surrounding medium stays ordered. A sphere provides the corresponding test for a point defect. We extend the fundamental group to maps from higher-dimensional spheres. We then distinguish based charges from the free homotopy classes that describe local defects. This distinction identifies opposite point charges for unoriented axes, but preserves their sign for oriented magnetic directions.*

By the end of this lecture you will be able to:

- Prove the group laws and commutativity of higher homotopy groups.
- Use the cited sphere groups and fibration sequence in the stated cases.
- Choose the linking circle for a line defect and sphere for a point defect.
- Compute vortex, nematic and sphere-valued charges with the required basepoint qualification.

Unit 26A Sphere maps and their groups

26.1 Ordered values and two defects

Definition 26.1 (Order parameter)**.** An *order parameter* records the local ordered state of a medium.

Definition 26.2 (Order-parameter space)**.** The allowed ordered values form the *order-parameter space* $\mathcal{M}$.

When a symmetry group G acts transitively on the uniform ordered states, Lecture 8 identifies this space with G/H, where H fixes a reference state.

Definition 26.3 (Director)**.** A director is an unoriented unit axis: $\mathbf{n}$ and $-\mathbf{n}$ describe the same state.

Definition 26.4 (Texture)**.** The word *texture* means a continuous spatial configuration of the order parameter.

Physical input. We model a superfluid by a complex field whose ordered value is its phase in S^1. We model a Heisenberg ferromagnet by a unit direction in S^2. We model a uniaxial

nematic by a director in $\mathbb{R}\mathrm{P}^2$. These descriptions restrict the field to ordered values away from defect cores.

Definition 26.5 (Defect core)**.** A *defect core* is a region where the chosen ordered description fails.

Outside the cores the order parameter is a continuous map into $\mathcal{M}$.

Definition 26.6 (Linking circle and sphere)**.** A small circle surrounding a line in three-dimensional space is a *linking circle*. A small sphere surrounding an isolated point is its *linking sphere*.

Definition 26.7 (Codimension and transverse disk)**.** The *codimension* of a smooth core is the spatial dimension minus the core dimension. A *transverse disk* meets the core at one point in directions complementary to its tangent directions.

The figure separates these two domains of the order-parameter map.

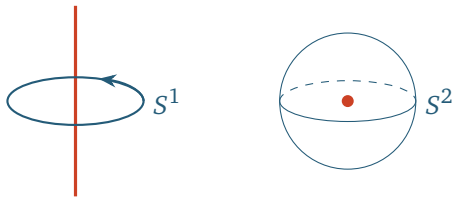


Figure 26.1: A circle links a line core, while a sphere encloses a point core. The order parameter is required to remain in $\mathcal{M}$ on the linking domain throughout a deformation.

For a straight superfluid core, a counter-clockwise linking circle can carry $\phi \mapsto e^{i\phi}$. Its winding is +1 by Lecture 3. It cannot extend to a continuous phase on the transverse disk because such an extension would contract that loop. For a magnetic point core, the field $\mathbf{n}(\mathbf{r}) = \mathbf{r}/|\mathbf{r}|$ on $\mathbb{R}^3 \setminus \{0\}$ restricts to the identity direction map on an outward linking sphere. Its degree is +1 by Lecture 14. We need sphere maps to describe this point obstruction.

A model energy explains the restriction to ordered values.

Physical input. For a complex field ψ on a spatial region U, take the model energy

$$E[\psi] = \int_U \big(\kappa|\nabla\psi|^2 + W(|\psi|)\big)\mathrm{d}^d x, \qquad W(R) = c(R^2 - \psi_0^2)^2, \qquad \kappa, c, \psi_0 > 0.$$

Here d is the spatial dimension and $R \geq 0$ is the field amplitude.

The potential vanishes precisely on the circle $|\psi| = \psi_0$. A fixed displacement of the amplitude from that circle raises the potential energy. At fixed nonzero amplitude, spatial variation of an initially uniform phase raises the gradient energy. A spatially uniform phase shift leaves this energy unchanged. A continuous nonzero complex field on a disk has a continuous normalized phase there, so its boundary winding is zero. Nonzero boundary winding therefore forces a zero of the complex field inside. Topology does not determine the size or lifetime of the region where the amplitude departs from ψ_0. The figure shows the potential and an explicit trial core profile.

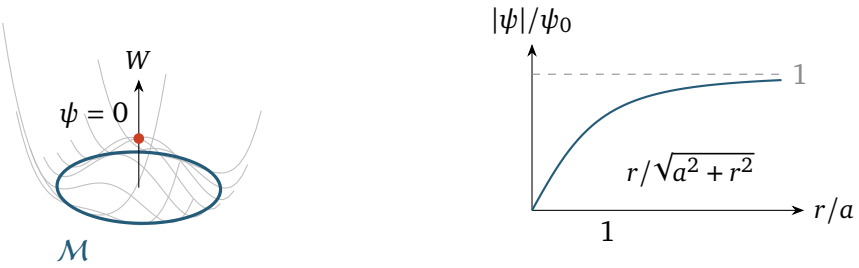


Figure 26.2: The model potential has a circle of minima. The trial field $\psi = \psi_0(x + iy)/\sqrt{a^2 + r^2}$, with $r^2 = x^2 + y^2$ and $a > 0$, has a zero at the centre and approaches the ordered amplitude.

26.2 Maps from cubes and higher homotopy groups

Let X be a space with basepoint $x_0 \in X$. For $n \geq 1$, put $I^n = [0,1]^n$.

Definition 26.8 (Higher homotopy group). The *higher homotopy group* $\pi_n(X, x_0)$ is initially the set of continuous maps $f\colon I^n \to X$ sending the entire boundary to x_0, modulo homotopies that keep that boundary at x_0.

Collapsing the boundary identifies $I^n/\partial I^n$ with S^n. Thus the same classes are based maps from a sphere. The quotient figure marks the entire boundary as one point.

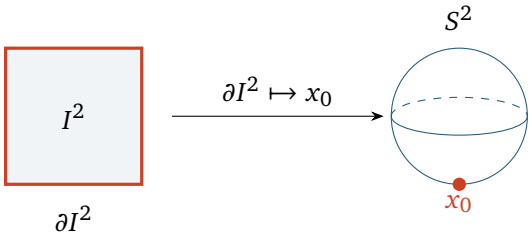


Figure 26.3: Collapsing the square boundary produces a sphere. All boundary points have one image, which is the sphere's basepoint.

For $n = 1$ this is Lecture 3's fundamental group.

Definition 26.9 (Path components). The symbol $\pi_0(X)$ instead denotes the set of path components. It is not generally a group.

Write $t' = (t^2, \dots, t^n)$. Concatenation in the first coordinate is given by

$$(f+g)(t^1, t') = \begin{cases} f(2t^1, t'), & 0 \leq t^1 \leq 1/2, \\ g(2t^1 - 1, t'), & 1/2 \leq t^1 \leq 1. \end{cases}$$

Both seam values are x_0, so pasting gives a continuous map relative to the boundary. For $n = 1$, this operation traverses f first, in the course's path convention. The plus sign denotes concatenation even when the group is nonabelian. We write γ^{-1} for the reversed path denoted $\bar{\gamma}$ in Lecture 3.

Proposition 26.10 (The group structure · *proved*)**.** *Concatenation makes $\pi_n(X, x_0)$ a group for every $n \geq 1$. Its identity is the constant map and its inverse is represented by $f^-(t^1, t') = f(1 - t^1, t')$. For $n \geq 2$ the group is abelian.*

Proof. Pasting homotopies on the two halves shows that concatenation is well-defined on classes. The two bracketings of three maps occupy first-coordinate slabs of lengths $(1/4, 1/4, 1/2)$ and $(1/2, 1/4, 1/4)$. For a homotopy parameter $u \in [0, 1]$, use slab lengths $((1+u)/4, 1/4, (2-u)/4)$ and rescale the first coordinate on each slab affinely to the unit interval. All lengths are positive. Adjacent maps agree at their boundary value x_0. This continuous reparametrization proves associativity.

A constant first slab is a pause before f. Shrink its length a continuously from $1/2$ to zero, using the value x_0 for $t^1 \leq a$ and $f((t^1 - a)/(1 - a), t')$ for $t^1 \geq a$. This is a homotopy relative to the cube boundary and proves the left unit law. Shrinking a final pause proves the right unit law. For $f + f^-$ the explicit retracing contraction is

$$K_u(t^1, t') = f\big((1 - u)\min\{2t^1, 2 - 2t^1\}, t'\big).$$

At $u = 0$ it is the concatenation and at $u = 1$ it is the constant boundary value. For $f^- + f$, replace the first argument by $1 - (1 - u)\min\{2t^1, 2 - 2t^1\}$, ending at the other boundary face. This proves both inverse laws.

For $n \geq 2$, let $\star$ concatenate in the second coordinate. The same proof gives it the same identity, denoted 0 at the class level. Pasting four maps into quadrants gives the exact interchange identity shown in the figure:

$$(a + b) \star (c + d) = (a \star c) + (b \star d).$$

Here $a, b, c, d \in \pi_n(X, x_0)$. Its two specializations give

$$a{+}b = (a{\star}0){+}(0{\star}b) = (a{+}0){\star}(0{+}b) = a{\star}b, \qquad a{+}b = (0{\star}a){+}(b{\star}0) = (0{+}b){\star}(a{+}0) = b{\star}a.$$

The operations therefore agree and are commutative. ∎

The figure uses only the first two coordinates. The other coordinates stay fixed.

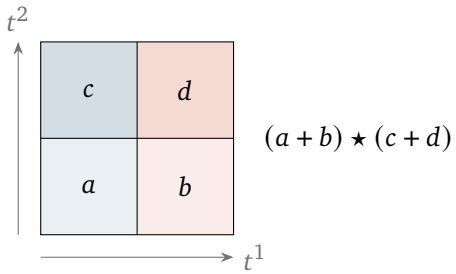


Figure 26.4: The four cube maps have the same placement whether one first joins horizontal pairs or first joins vertical pairs. All seam faces map to the basepoint.

For example, based maps $S^2 \to S^2$ add by joining two sphere maps at their basepoint. The next section identifies this group law with addition of degrees. The extra coordinate also gives a geometric explanation of commutativity. We can compress the two maps into disjoint small squares, with value x_0 outside them. We move the squares around each other in the first two coordinates. All other coordinates stay fixed. There is no second coordinate

for this interchange when $n = 1$. The figure shows disjoint square positions throughout the exchange.

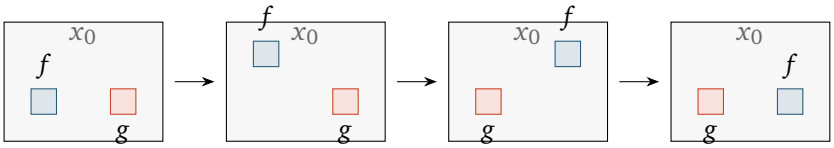


Figure 26.5: The square carrying f moves above the square carrying g before they exchange horizontal positions. The complement maps to x_0 throughout.

26.3 Sphere groups and the Hopf map

The degree from Lecture 14 is invariant under homotopy. The converse for sphere maps is an additional theorem: equal degrees imply based homotopy. We cite it here rather than infer it from the integral formula alone.

Theorem 26.11 (Homotopy groups of spheres · *cited*). *For every $n \geq 1$, the degree map gives $\pi_n(S^n) \cong \mathbb{Z}$. For $1 \leq k < n$, the group $\pi_k(S^n)$ vanishes. The group $\pi_3(S^2)$ is infinite cyclic and is generated by the Hopf map $S^3 \to S^2$ of Lecture 20.*

The precise sources are Hatcher, Corollary 4.9, p. 349, Corollary 4.25, p. 361, and Example 4.51, p. 380. The degree is Lecture 14's integral degree.[1] Thus the identity sphere map has charge +1 and generates $\pi_n(S^n)$. For smooth representatives constant near the cube boundary, choose a common regular value different from x_0. Its preimages in $f + g$ lie in the two half-cubes. Both first-coordinate rescalings preserve orientation. The signed preimage counts therefore give

$$\deg(f + g) = \deg f + \deg g.$$

We first compress the cube map into its interior to make it constant near the boundary. We then use smooth approximation relative to that neighbourhood (Lee, Theorem 6.26, recalled in Lecture 13).

A smooth map $S^k \to S^n$ with $k < n$ misses a point: its entire image consists of critical values and has measure zero by Sard's theorem recalled in Lecture 14. The complement of a point in S^n is contractible by stereographic projection. The contraction can fix the image basepoint, so the map is based null-homotopic. Smoothing supplies the same conclusion for

[1] Hatcher's homological degree is the integer multiplier d on $H_n(S^n;\mathbb{Z}) \cong \mathbb{Z}$. This definition is given in §2.2, p. 134. This multiplier does not depend on a choice of generator. For smooth maps, Lecture 14's cited de Rham identification intertwines this action with pullback of forms: integrating a form over a composed simplex equals integrating its pullback over that simplex. Thus the homological degree agrees with the integral degree of Theorems 14.3 and 14.14. To verify this, let c be the period of a normalized volume form ν on a generator $[S]$. Its de Rham class is nonzero since $\int_{S^n} \nu = 1$, so injectivity of the pairing gives $c \neq 0$. The top-cohomology argument in Lecture 14 gives $[f^*\nu] = (\deg f)[\nu]$. Naturality and the vanishing of exact-form periods therefore give

$$dc = \int_{f_*[S]} \nu = \int_{[S]} f^*\nu = (\deg f)c.$$

Cancellation of c identifies the two degrees. For continuous maps, Hatcher's definition uses homology directly.

continuous maps. There is no missing-point argument for the Hopf map, which is onto S^2. Its nontrivial class in $\pi_3(S^2)$ is the cited higher-dimensional example.

Lecture 3's winding belongs to $\pi_1(S^1)$. Lecture 14's index of a zero in dimension $n \geq 2$ is the degree of a map in $\pi_{n-1}(S^{n-1})$. Lecture 22's transition winding belongs to $\pi_1(\mathrm{U}(1))$. Lecture 23's integer k is the degree of the asymptotic map $S^3 \to \mathrm{SU}(2)$, a class in $\pi_3(\mathrm{SU}(2)) \cong \mathbb{Z}$ under that lecture's orientation convention.

We will also use the covering-space result in Hatcher, Proposition 4.1, p. 342: a covering induces an isomorphism on π_n for $n \geq 2$. Sphere maps and their homotopies lift, which explains this result. The contractible cover $\mathbb{R} \to S^1$ gives $\pi_n(S^1) = 0$ for $n \geq 2$. The cover $S^2 \to \mathbb{R}\mathrm{P}^2$ gives $\pi_2(\mathbb{R}\mathrm{P}^2) \cong \mathbb{Z}$ after choosing a lifted basepoint. That choice will matter when the basepoint is allowed to move.

26.4 The fibration sequence and two consequences

Definition 26.12 (Fibration). A *fibration* is a continuous map $p\colon E \to B$ with the homotopy lifting property: for every space Y, a homotopy $Y \times [0,1] \to B$ with a chosen lift at its initial time has a continuous lift throughout that restricts to the chosen lift at time zero.

For the theorem below it suffices to require this property for disks of every dimension. Fibre bundles have this property for disks by Hatcher's Proposition 4.48, p. 379. Choose a basepoint $b_0 \in B$ and a point e_0 in its fibre $F = p^{-1}(b_0)$. All homotopy groups below use these compatible basepoints.

Theorem 26.13 (The fibration sequence · *cited*). *For a fibration over a path-connected base, the homotopy groups have a long exact sequence. Its relevant segment is*

$$\cdots \longrightarrow \pi_2(F) \longrightarrow \pi_2(E) \longrightarrow \pi_2(B) \xrightarrow{\partial_{\mathrm{fib}}} \pi_1(F) \longrightarrow \pi_1(E) \longrightarrow \pi_1(B) \longrightarrow \cdots .$$

Exactness means that the image of each arrow is the kernel of the next.

This is Hatcher's Theorem 4.41, p. 376. The symbol ∂_{fib} names the connecting homomorphism, distinct from a geometric boundary. Represent a based sphere by a square with its boundary collapsed. Lifting this square leaves a boundary loop in the fibre. Its construction and the general exactness proof are cited. The figure distinguishes the collapsed boundary in the base from its lifted loop in the fibre.

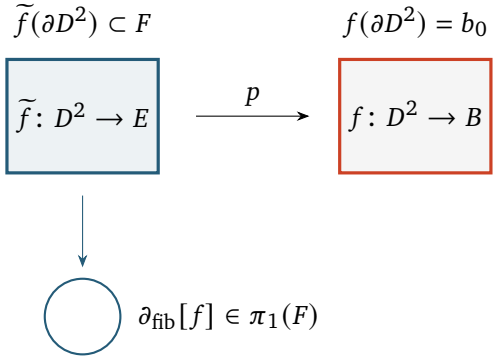


Figure 26.6: A lifted disk has boundary in the fibre because the base disk boundary maps to b_0. That fibre loop represents $\partial_{\mathrm{fib}}[f]$.

For the Hopf bundle, $F = S^1$, $E = S^3$ and $B = S^2$. The cited sphere groups give $\pi_2(S^3) = 0$ and $\pi_1(S^3) = 0$. The figure isolates the resulting segment.

$$0 = \pi_2(S^3) \longrightarrow \pi_2(S^2) \xrightarrow[\cong]{\partial_{\mathrm{fib}}} \pi_1(S^1) \longrightarrow \pi_1(S^3) = 0$$

Figure 26.7: Exactness makes the Hopf connecting map an isomorphism: its kernel is zero and its image is the entire fibre-loop group.

Hence $\pi_2(S^2) \cong \pi_1(S^1) \cong \mathbb{Z}$. The connecting map gives more information than the abstract integer group: it assigns a fibre loop to a sphere map. For the Hopf bundle, a degree-n sphere map gives a fibre loop of winding $\pm n$. The sign depends on the orientation used to define the connecting map. Lecture 22 instead fixes a sign by its explicit northern and southern frame order when it computes the winding of the equatorial transition.

More generally, let G be a simply connected Lie group with $\pi_2(G) = 0$ and let H be a closed Lie subgroup. Lecture 20 gives the principal bundle $H \to G \to G/H$. Exactness gives $\pi_2(G/H) \cong \pi_1(H)$ because both adjacent groups from G vanish. Here the basepoint of H is its identity, so its loops lie in its identity component. Both hypotheses $\pi_2(G) = 0$ and $\pi_1(G) = 0$ are needed for this exact-sequence argument. For $G = \mathrm{SU}(2) \cong S^3$ and $H = \mathrm{U}(1)$, both vanish by Theorem 26.11. The quotient is $\mathrm{SU}(2)/\mathrm{U}(1) \cong S^2$. Thus its point charge corresponds to a loop in the subgroup that fixes a direction. Lecture 27 uses this quotient for the Yang–Mills–Higgs monopole.

Self-check. State in one sentence what $\pi_n(X, x_0)$ is and why we introduced it for defects.

Answer. The group $\pi_n(X, x_0)$ consists of based homotopy classes of based maps $S^n \to X$, so it records whether a linking sphere map can contract while its image basepoint stays fixed. A nonzero class obstructs extension over the transverse disk.

Unit 26B Free defect classes and based charges

26.5 Nematic lifts and moving basepoints

A director map from a linking S^2 lifts to the internal unit-vector sphere because its domain is simply connected, by Lecture 4's lifting theorem. There are exactly two lifts, $\widetilde{f}$ and $-\widetilde{f}$. They are maps into the internal sphere, not tangent vector fields on the physical sphere. The antipodal map on the target has degree -1, so these lifts have degrees d and $-d$. The physical linking sphere keeps its outward orientation. The radial director field therefore has both an outward and an inward vector lift. The spatial comparison loop in the figure carries a half-turn of the director. The figure separates the source orientation from the target sign.

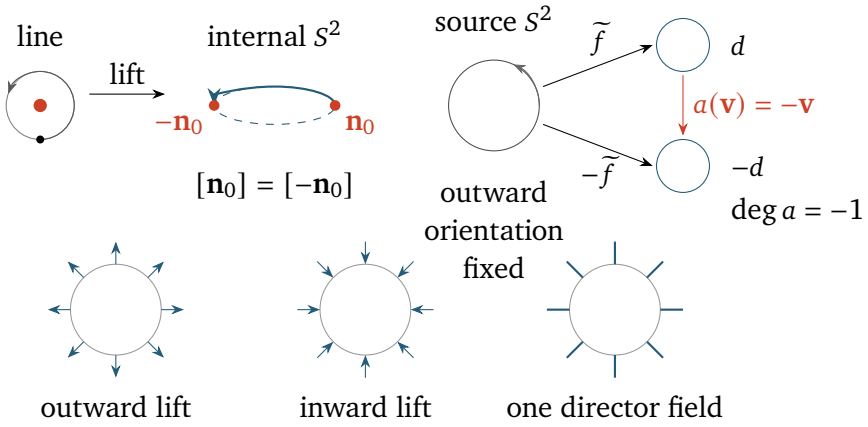


Figure 26.8: A nontrivial director loop lifts from $\mathbf{n}_0$ to $-\mathbf{n}_0$. Changing the target lift reverses its degree while the physical source orientation stays fixed.

For an ordinary based charge, a chosen point of the domain sphere must have a prescribed image x_0. A deformation of the physical texture can move that image. We therefore need the effect of moving the basepoint. Let $\gamma\colon [0,1] \to \mathcal{M}$ run from x_0 to x_1. Rescale the domain of a cube map based at x_1 onto a smaller concentric cube. On each radial segment of the surrounding shell, insert γ, running from x_0 at the outer boundary to x_1 at the inner boundary. The figure depicts this construction.

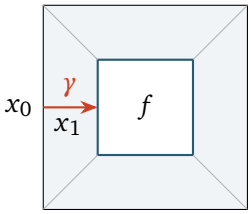


Figure 26.9: The shell changes the inner map's basepoint from x_1 to x_0 along γ. The outer boundary maps constantly to x_0.

Proposition 26.14 (The basepoint action · *sketched*)**.** *The shell construction gives an isomorphism $\beta_\gamma\colon \pi_n(\mathcal{M}, x_1) \to \pi_n(\mathcal{M}, x_0)$. For loops at x_0, it gives an action of $\pi_1(\mathcal{M}, x_0)$ on $\pi_n(\mathcal{M}, x_0)$. For $n = 1$ this is conjugation.*

Sketch. A homotopy of the path or inner map gives a homotopy of the shell map. Reversing the path gives the inverse isomorphism, since the retraced shell contracts. Two nested shells concatenate their paths. Consequently $\beta_{\gamma\eta} = \beta_\gamma \beta_\eta$ in the first-path-first convention. Joining inner cube maps shows that the induced maps preserve the group operation. For $n = 1$ the shell map is the path $\gamma f \gamma^{-1}$, as in Lecture 3. These constructions give the asserted action. They are described in Hatcher, pp. 341–342. ∎

Even though π_n is abelian for $n \geq 2$, this external action can be nontrivial. Commutativity of its own group law does not remove the basepoint issue.

26.6 Local defects and linking spheres

For a core of codimension c, the local transverse boundary is S^{c-1}. This explains which homotopy group occurs. The physical sphere's orientation is fixed. Reversing it can reverse an integer charge.

Definition 26.15 (Topological defect)**.** A *topological defect* has a nontrivial linking homotopy class that obstructs removal of its core within $\mathcal{M}$.

The table distinguishes a linking boundary from the compactification of an entire space.

Space dimension	Core	Codimension	Domain	Datum
3	wall	1	S^0	endpoint components
3	line	2	S^1	loop class
3	point	3	S^2	sphere class
2	line	1	S^0	endpoint components
2	point	2	S^1	loop class
d	none	—	$\mathbb{R}^d \cup \{\infty\} = S^d$	based $\pi_d(\mathcal{M})$

For a wall, the two points of S^0 record an ordered pair of components in $\pi_0(\mathcal{M})$. A nontrivial wall separates distinct components, so this case requires a disconnected target. Lecture 27's kink has this endpoint datum. For line and point defects, free classes require the qualification in the following theorem. The texture row fixes a uniform value at infinity.

Theorem 26.16 (Local defect classes · *sketched*)**.** *Let $\mathcal{M}$ be a path-connected order-parameter manifold. Classify restrictions to a linking domain by continuous homotopy through maps into $\mathcal{M}$. Without a fixed image basepoint, line defects in three dimensions have the conjugacy classes of $\pi_1(\mathcal{M})$, and point defects have the orbits of $\pi_2(\mathcal{M})$ under the $\pi_1(\mathcal{M})$ action. A two-dimensional texture with a fixed uniform value at spatial infinity has a based class in $\pi_2(\mathcal{M})$.*

Sketch. The first assertions concern free homotopy classes of maps from S^1 and S^2. To convert a free homotopy to a based one, follow the moving image of the chosen domain point. Its path is precisely the shell change of basepoint. Thus representatives of two based classes are freely homotopic exactly when this action identifies them. For loops the action is conjugation, recovering Lecture 3's statement: two loops based at x_0 are freely homotopic exactly when their classes are conjugate. For spheres this is its higher-dimensional version (Mermin, §VII.D, p. 631). For the texture, adding the point at infinity compactifies the oriented plane to S^2. The fixed limiting value supplies its target basepoint. This gives the stated based class. ∎

If the value at infinity is also allowed to move, the texture likewise needs action orbits. A nonzero linking class obstructs extension across the core within $\mathcal{M}$: an extension to the transverse disk would contract the boundary map radially. A zero class admits such an extension because a representative map is null-homotopic. This is a topological obstruction, not a claim about the energy or time needed to remove the core. The local boundary classes do not classify knotting or all configurations with several defects.

Moving a linking circle through a region without cores preserves its free class. Indeed, composing the spatial deformation with the order-parameter map gives a homotopy. Fix a common spatial point p and paths η_i from p to two small positive loops ℓ_i. These paths are called *whiskers*. In the arrangement shown, the positive enclosing loop deforms to the ordered loop

$$(\eta_1 \ell_1 \eta_1^{-1})(\eta_2 \ell_2 \eta_2^{-1}).$$

We traverse the right loop first and the left loop second. Composing with the field gives the corresponding ordered product in $\pi_1(\mathcal{M}, x_0)$, where x_0 is the field value at p. For S^1 the integers add. For a nonabelian group, the free conjugacy labels alone do not specify this product. The figure fixes the whiskers and the traversal order.

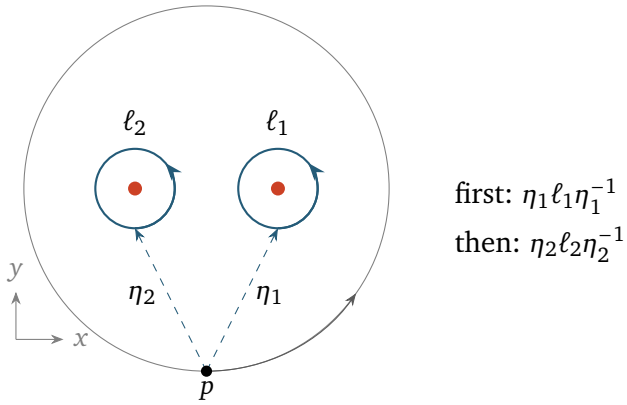


Figure 26.10: All circles run counter-clockwise in the oriented transverse plane. The whiskers give based loops, with loop 1 on the right traversed before loop 2 on the left.

For a concrete nonabelian comparison, take loops in Lecture 5's wedge of two circles. Its fundamental group is the free group F_2 on a and b. The elements b and aba^{-1} are distinct reduced words, but are conjugate by a. They therefore represent one free loop class. This example illustrates the conjugation mechanism independently of a particular medium.

26.7 Vortices, nematics and magnetic textures

The figure compares an oriented phase field with an unoriented director field on a linking circle.

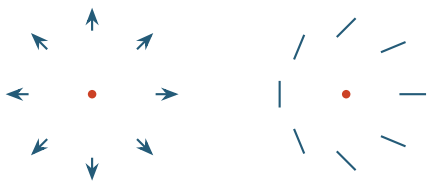


Figure 26.11: The superfluid phase makes a full turn, while the nematic axis closes after a half-turn. Arrows show the phase as a direction in the internal circle, not in space.

Example 26.17 (Superfluid vortices)**.** Determine the local line and point classes for a superfluid. A superfluid has a phase $e^{i\alpha} \in S^1$. On an oriented linking circle a vortex of winding $n \in \mathbb{Z}$ is represented by $\phi \mapsto e^{in\phi}$ for $0 \le \phi \le 2\pi$. Lecture 3 gives $\pi_1(S^1) = \mathbb{Z}$. This group is abelian, so its conjugacy classes are its individual integers. Based joining adds windings. Thus the enclosing winding of two vortices, joined as in the whiskered-loop figure, is the sum of their individual windings. Two vortices of winding +1 have total winding 2 and cannot cancel within S^1. Opposite windings have total zero and have no such obstruction. The group $\pi_2(S^1)$ vanishes, so there is no corresponding point charge.

For a neutral superfluid the phase also determines the circulation.

Physical input. Away from cores, we use the neutral-superfluid relation $\mathbf{v} = (\hbar/m)\nabla\alpha$, where $m > 0$ is the mass of the condensed boson and α is a locally chosen phase.

For $\alpha = n\phi$ on a transverse plane, the velocity is tangential:

$$\mathbf{v} = \frac{n\hbar}{mr}\mathbf{e}_\phi, \qquad \oint \mathbf{v} \cdot \mathrm{d}\boldsymbol{\ell} = \frac{2\pi n\hbar}{m}.$$

Here $r > 0$ is distance from the core and $\mathbf{e}_\phi$ is the counter-clockwise unit tangent. The circulation has the sign of the winding.

Definition 26.18 (Disclination)**.** The nontrivial line defect in a uniaxial nematic is called a *disclination*.

Example 26.19 (Nematic lines and point charges)**.** Determine the local line and point classes for a uniaxial nematic. A uniaxial nematic has $\mathcal{M} = \mathbb{R}\mathrm{P}^2$. Lecture 4 gives $\pi_1(\mathbb{R}\mathrm{P}^2) = \mathbb{Z}_2$. A representative director is $[\cos(\phi/2) : \sin(\phi/2) : 0]$. Its vector lift starts at a unit vector and ends at its negative. The director nevertheless closes, since antipodal vectors give the same axis. Joining two such loops gives a closed full-turn vector loop on S^2. That loop contracts because S^2 is simply connected. Thus the two nontrivial disclinations have total class zero and can cancel topologically.

The chosen sphere cover gives $\pi_2(\mathbb{R}\mathrm{P}^2) \cong \mathbb{Z}$. Changing the lifted basepoint around the nontrivial director loop applies the antipodal deck map. That map has degree -1 on S^2, so its action sends n to $-n$. Hence free point-defect classes identify opposite integers. They can be labelled by $|n|$, including zero. There is no single signed addition law on these orbits without the joining-path data. This is the nematic qualification in Mermin, p. 631. Two local free labels $|1|$ and $|1|$ allow comparison representatives $(+1, +1)$ or $(+1, -1)$ after paths bring them to a common basepoint. Their based sums are 2 and 0. Changing one comparison path by the nontrivial director loop reverses that representative. The local absolute values

therefore do not determine even the magnitude of a total charge. The full configuration and the chosen paths supply additional information. This calculation makes no prediction about the dynamics of bringing the defects together.

A full-turn vector loop can extend across a disk by tilting into the third internal direction. An explicit extension on $x^2 + y^2 \le 1$ is

$$\mathbf{n}_{\mathrm{esc}}(x, y) = \frac{(2x, 2y, 1 - x^2 - y^2)}{1 + x^2 + y^2}.$$

It has unit norm, points along $+\mathbf{e}_z$ at the centre, and equals $(\cos\phi, \sin\phi, 0)$ on the boundary circle. The formula also extends the projected full-turn nematic loop. It does not extend the half-turn disclination generator. The figure shows the internal directions along a radial cross-section.

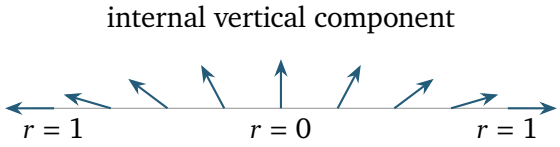


Figure 26.12: The full-turn vector loop has a smooth disk filling in S^2. The arrows tilt to the vertical at the centre, so there is no core.

Definition 26.20 (Hedgehog). The *hedgehog* is the field $\mathbf{n}(\mathbf{r}) = \mathbf{r}/|\mathbf{r}|$ outside the origin.

Definition 26.21 (Skyrmion texture). A *skyrmion texture* is a nonsingular plane texture with a nonzero sphere degree and a fixed uniform value at infinity.

The next figure compares the missing origin of a hedgehog with a smooth texture. For the plane texture, take a scale $a > 0$ and radial distance $r = \sqrt{x^2 + y^2}$. The texture arrows use the formula in the example below.

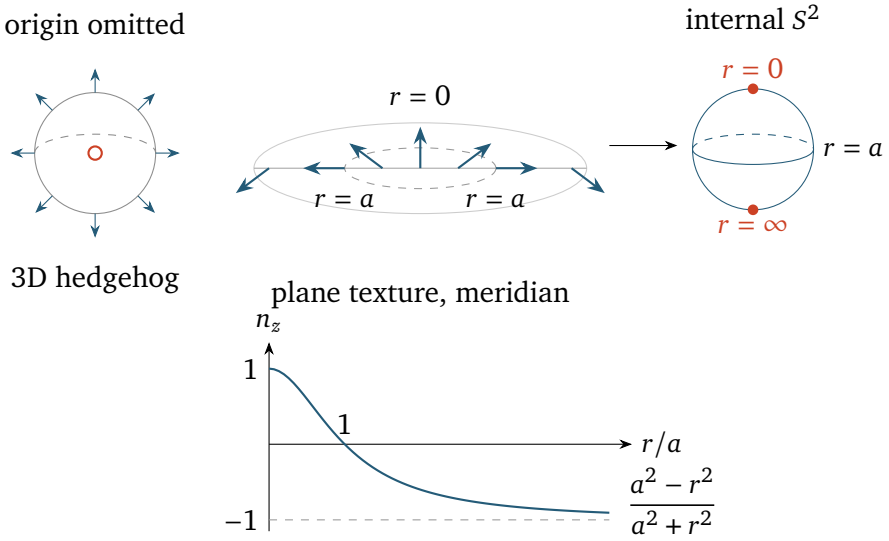


Figure 26.13: The hedgehog has a point core in physical three-space. The plane texture is smooth at its centre and covers the internal direction sphere with degree +1.

Example 26.22 (Magnetic point charge and plane texture)**.** Determine the magnetic point charge and the degree of a compactified plane texture. A Heisenberg ferromagnet has $\mathcal{M} = S^2$. The fundamental group of this target is zero, so there is no local line obstruction of this type. Its point classes are the signed integers $\pi_2(S^2) = \mathbb{Z}$, with no nontrivial basepoint action. The hedgehog restriction to an outward linking sphere is the identity direction map and has degree +1.

For $a > 0$, an explicit degree-one example is

$$\mathbf{n}(x, y) = \frac{(2ax, 2ay, a^2 - x^2 - y^2)}{a^2 + x^2 + y^2}, \qquad \mathbf{n} \cdot (\partial_x \mathbf{n} \times \partial_y \mathbf{n}) = \frac{4a^2}{(a^2 + x^2 + y^2)^2}.$$

Its norm is one. It tends to the south pole at infinity. With $u = x/a$ and $v = y/a$, this is Lecture 10's inverse stereographic map with its third component reversed. The reflection changes the negative area density in that lecture to the positive density here. The inversion coordinates below are the stereographic chart transition from Lecture 7. For the density calculation, write the field's numerator as $\mathbf{d}$ and denominator as $D = a^2+x^2+y^2$. Differentiating $\mathbf{n} = \mathbf{d}/D$ gives

$$\mathbf{n} \cdot (\partial_x \mathbf{n} \times \partial_y \mathbf{n}) = \frac{\mathbf{d} \cdot (\partial_x \mathbf{d} \times \partial_y \mathbf{d})}{D^3}, \qquad \mathbf{d} \cdot (\partial_x \mathbf{d} \times \partial_y \mathbf{d}) = 4a^2 D.$$

The terms proportional to $\mathbf{d}$ in either derivative drop out of the determinant. In inversion coordinates $X = x/(x^2 + y^2)$ and $Y = y/(x^2 + y^2)$, the same texture is

$$\mathbf{n} = \frac{(2aX, 2aY, a^2(X^2 + Y^2) - 1)}{1 + a^2(X^2 + Y^2)}.$$

It extends smoothly to $X = Y = 0$, the point at infinity. Its degree is

$$\frac{1}{4\pi} \int_{\mathbb{R}^2} \frac{4a^2 \mathrm{d}x\mathrm{d}y}{(a^2 + x^2 + y^2)^2} = \int_0^\infty \frac{2a^2 r \mathrm{d}r}{(a^2 + r^2)^2} = 1.$$

This is a texture, with no singular core in the plane. The degree labels its topology. No energy-minimizing field equation has been imposed.

Self-check. An ordered medium has $\mathcal{M} = \mathrm{SO}(3)$. Choose the linking domain in three-dimensional space. Find its local line classes and the result of joining two nontrivial lines.

Answer. The linking domain is a circle transverse to the line. Lecture 4 gives $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}_2$. It is abelian, so there are two free line classes. The sum of two nontrivial classes is zero, removing this local topological obstruction. This does not assert that a particular physical evolution makes the defects annihilate.

Self-check. A nematic configuration has two point defects. Chosen comparison paths give lifted degrees 3 and −2 at a common basepoint. Find the total based charge and its free label. Repeat when only the second path is changed by the nontrivial director loop. Explain why the local labels 3 and 2 do not determine a total free label.

Answer. The first total is $3 + (-2) = 1$, with free label 1. The changed path acts by the target antipodal map, so the second representative becomes +2. The new total is $3 + 2 = 5$, with free label 5. The same local absolute values have lost the comparison-path information.

Summary. The groups π_n classify based sphere maps and are abelian for $n \geq 2$. The cited sphere results and fibration sequence compute the groups used here. Local free defect classes still require conjugacy for loops and the fundamental-group action for spheres. Superfluid vortices retain signed integer winding. Nematic disclinations have $\mathbb{Z}_2$ charge. Free nematic point charges identify n with $-n$. Sphere-valued hedgehogs and based plane textures have signed degree. Lecture 27 adds boundary conditions and energy to this topological classification.

Reading. Hatcher, §4.1, pp. 340–343 and 349; §4.2, Corollary 4.25, p. 361, Theorem 4.41, p. 376, Proposition 4.48, p. 379, and Example 4.51, p. 380. Mermin, *The topological theory of defects in ordered media*, §VII.D–E, pp. 630–632. Nakahara, §4.8–4.9.

Lecture 27

Solitons and topological charge

Objectives. *Lecture 26 classified maps into an order-parameter space. A field theory also assigns an energy to each configuration. We separate the boundary conditions that define a topological sector from the field equations that select stationary configurations. The kink supplies a complete energy-bound calculation. The gauge vortex shows how a phase winding becomes a quantized magnetic flux.*

By the end of this lecture you will be able to:

- Identify a sector from its stated continuous vacuum boundary data.
- Derive the kink bound and its saturating profile with the signed charge.
- Prove Derrick's obstruction for the stated linear-target scalar energy, including $d = 2$.
- Derive vortex flux quantization under the stated asymptotic conditions.

Unit 27A Kinks, boundary sectors and scalar scaling

27.1 The scalar kink and the Bogomolny bound

Definition 27.1. A *static soliton* is a spatially localized, finite-energy stationary solution of the field equations.

Here static means time independent. Localization concerns the energy density: the field itself may approach different nonzero vacua at the two ends.

Take one real field on the line and constants $\mu, v > 0$.

Physical input. We choose the relativistic scalar model with Lagrangian density $\mathcal{L} = \frac{1}{2}\phi_t^2 - \frac{1}{2}\phi_x^2 - V(\phi)$, in units with light speed one. Its energy density is $\frac{1}{2}\phi_t^2 + \frac{1}{2}\phi_x^2 + V(\phi)$. The potential is the double well $V(\phi) = \frac{1}{2}\mu^2(v^2 - \phi^2)^2$.

Recall the field Euler–Lagrange equation: differentiating this Lagrangian gives

$$\phi_{tt} - \phi_{xx} + V'(\phi) = 0.$$

A static field therefore satisfies $\phi'' = V'(\phi)$. We choose a smooth function W whose derivative squares to twice this potential. This auxiliary function is called a *superpotential*. Define W and the static energy by

$$W(\phi) = \mu\left(v^2\phi - \frac{\phi^3}{3}\right), \qquad V(\phi) = \frac{1}{2}\big(W'(\phi)\big)^2, \qquad E = \int_{\mathbb{R}} \left(\frac{1}{2}(\phi')^2 + V(\phi)\right) \mathrm{d}x.$$

Here W' differentiates with respect to the field value, while ϕ' differentiates with respect to x.

Definition 27.2. The zero set $\mathcal{V} = V^{-1}(0)$ is the *vacuum manifold* when it is a manifold.

The vacua of this double well are $-v$ and $+v$. The potential and the kink's localized energy density are shown in the next figure.

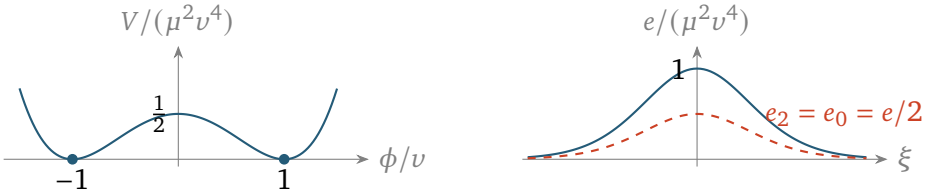


Figure 27.1: The double well vanishes at $\phi/v = \pm 1$. The kink has energy density $e/(\mu^2 v^4) = \operatorname{sech}^4 \xi$, where $\xi = \mu v(x - x_0)$, with equal gradient and potential halves.

Assume vacuum limits ϕ_- and ϕ_+ at the two ends:

$$\lim_{x \to -\infty} \phi(x) = \phi_-, \qquad \lim_{x \to +\infty} \phi(x) = \phi_+, \qquad \phi_-, \phi_+ \in \{-v, +v\}.$$

The figure shows the increasing kink profile centred at zero.

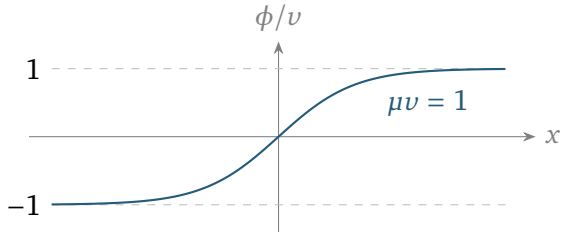


Figure 27.2: The exact kink $\phi/v = \tanh(\mu v x)$ joins $-v$ to $+v$. The increasing profile has charge +1.

Definition 27.3. A *soliton sector* fixes the boundary homotopy data, which here are the ordered vacuum endpoints.

Definition 27.4. A *topological charge* is an invariant used to label such a sector.

Definition 27.5. A *Bogomolny bound* is an energy bound fixed by boundary topological data.

Definition 27.6. The first-order equation $\phi' = sW'(\phi)$, for $s \in \{+1, -1\}$, is the *Bogomolny equation* for this model.

For this model, the difference of the endpoint values of W gives a Bogomolny bound.

Proposition 27.7 (The kink bound · *proved*). *Every smooth finite-energy configuration with these limits satisfies $E \geq |W(\phi_+) - W(\phi_-)|$. In the sector $\phi_- = -v$, $\phi_+ = v$, equality is attained by*

$$\phi_{\text{kink}}(x) = v \tanh\big(\mu v(x - x_0)\big), \qquad x_0 \in \mathbb{R}, \qquad E_{\text{kink}} = \frac{4\mu v^3}{3}.$$

Its charge $Q = (\phi_+ - \phi_-)/(2v)$ is +1. The decreasing antikink has charge -1 and the same energy.

Proof. For either sign $s \in \{+1, -1\}$, completing the square gives

$$E = \frac{1}{2}\int_{\mathbb{R}} (\phi' - sW'(\phi))^2 \mathrm{d}x + s\big(W(\phi_+) - W(\phi_-)\big).$$

The last term follows from the chain rule on a finite interval and then the stated limits. Choose s so its product with this difference is nonnegative. The square is nonnegative, proving the bound. Equality requires $\phi' = sW'(\phi)$, the Bogomolny equation.

For the increasing sector, this equation is $\phi' = \mu(v^2 - \phi^2)$. Separating variables on $-v < \phi < v$ gives $\operatorname{artanh}(\phi/v) = \mu v(x - x_0)$, hence the displayed profile. Its derivative is $\mu v^2 \operatorname{sech}^2(\mu v(x - x_0)) = W'(\phi)$. It has the required endpoint limits, so it saturates the bound. Direct endpoint evaluation gives $W(v) - W(-v) = 4\mu v^3/3$. Differentiating the first-order equation gives $\phi'' = W''W' = V'(\phi)$, the static field equation. Thus this saturating configuration is a solution, not only a trial field. Changing its sign gives the antikink and reverses the ordered vacuum pair and charge. ∎

Equal endpoint vacua give charge zero. The bound then permits the vacuum configuration. The square completion proves a global energy minimum in each kink sector. It does not imply that every configuration in that sector is stationary.

The static equation also gives a first integral, because

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{2}(\phi')^2 - V(\phi)\right) = \phi'(\phi'' - V'(\phi)) = 0.$$

Finite energy gives a sequence tending to infinity along which ϕ' tends to zero. The vacuum limit makes $V(\phi)$ tend to zero. Thus the constant in this first integral is zero. Along the kink we obtain $\frac{1}{2}(\phi')^2 = V(\phi)$. The gradient and potential contributions to its energy are equal.

For this double well, no nonconstant finite-energy static solution has equal endpoint vacua. Indeed, a nonconstant field with equal limits attains an interior maximum or minimum different from that common limit. At this turning point the first integral forces $V = 0$, so the field is at one of the vacua with derivative zero. ODE uniqueness for the smooth static equation then forces the constant vacuum solution. This conclusion uses the field equation and its first integral, rather than charge zero alone.

The kink also gives a moving localized solution. For a speed $\beta \in (-1, 1)$, define

$$\phi_\beta(t, x) = \phi_{\mathrm{kink}}\big(\gamma(x - \beta t)\big), \qquad \gamma = (1 - \beta^2)^{-1/2}.$$

The wave operator applied to this field gives $\gamma^2(\beta^2 - 1)\phi''_{\mathrm{kink}} = -\phi''_{\mathrm{kink}}$, so the static equation proves that the moving field solves the same evolution equation. Write E_2 and E_0 for the kink's rest-frame gradient and potential energies. Changing variables to $\gamma(x - \beta t)$ in the energy gives

$$E_\beta = \gamma(1 + \beta^2)E_2 + \frac{E_0}{\gamma} = \gamma(E_2 + E_0), \qquad E_2 = E_0.$$

The minimum energy in the ordered kink boundary class is positive while the continuous vacuum endpoints remain fixed. The translating energy density and the relativistic energy formula give the kink a particle interpretation, with rest mass equal to its rest energy in these units.

27.2 Vacuum boundary data and sectors

Recall that a classical field configuration assigns a value to each spatial point. For a scalar field $\phi\colon \mathbb{R}^d \to \mathbb{R}^N$, take a smooth potential $V \geq 0$. Its vacuum set is again $\mathcal{V} = V^{-1}(0)$. A constant field in $\mathcal{V}$ has zero potential and gradient energy.

In one dimension, assume limits ϕ_- and ϕ_+ in $\mathcal{V}$ at $-\infty$ and $+\infty$. Call their path components $\mathcal{V}_-$ and $\mathcal{V}_+$. In dimensions two and three, assume a continuous angular limit $\phi_\infty : S^{d-1} \to \mathcal{V}$, with convergence uniform in direction.

For an evolving field, require that its limiting boundary map be jointly continuous in angle and time. In one dimension require each endpoint limit to be continuous in time. The potential assigns positive energy density to values outside $\mathcal{V}$. This motivates vacuum limits, which we impose as boundary assumptions. Finite energy alone does not imply them. The vacuum manifold $\mathcal{V}$ is the order-parameter space $\mathcal{M}$ of Lecture 26 in these models. The sphere at infinity surrounds the entire localized region. A linking sphere from Lecture 26 carries the same class if it can be deformed to this sphere through a region on which the normalized field stays defined. In one dimension the sphere S^0 consists of the two distinguished endpoints. A homotopy of its maps moves each endpoint within its path component. The figure compares these boundary maps in the kink, vortex and hedgehog examples.

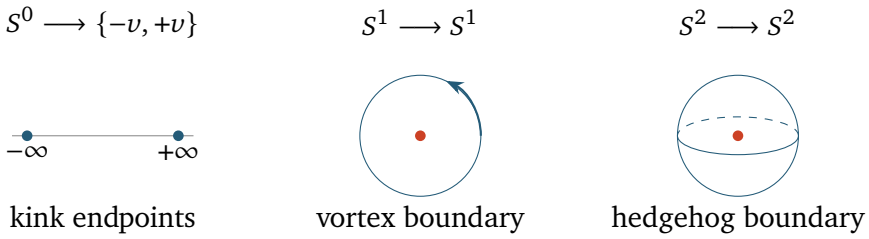


Figure 27.3: The boundary spheres have dimensions zero, one and two. Their maps take values in the vacuum set or target sphere of the corresponding model.

The figure preserves the order of the one-dimensional endpoints.

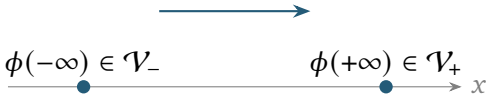


Figure 27.4: The left and right vacuum components form an ordered pair. Interchanging the two ends can reverse a kink's charge.

Proposition 27.8 (Boundary sectors · *proved*)**.** *Under these boundary conditions, the one-dimensional datum is the ordered pair of path components of $\mathcal{V}$ containing ϕ_- and ϕ_+. In dimensions two and three it is the free homotopy class of ϕ_∞. An evolution with jointly continuous vacuum boundary data preserves these data.*

Proof. At either end in one dimension, the time-dependent limiting value is a path in $\mathcal{V}$. It cannot leave its path component. The two ends remain distinguished, giving an ordered pair. For $d = 2$ or $d = 3$, the jointly continuous boundary map is exactly a homotopy of maps $S^{d-1} \to \mathcal{V}$. Its free homotopy class is therefore fixed. Lecture 26 expresses these classes using conjugacy or the fundamental-group action when needed. ∎

An integer charge exists in the kink and vortex examples, but the general sector datum need not be an integer. It constrains allowed deformations. Energy and the field equations determine whether a soliton exists.

A texture has different data: its value at infinity is fixed, and the degree belongs to the map on the compactified whole plane, as in Lecture 26. Its constant boundary map alone does not determine that degree.

27.3 Derrick's scaling obstruction

For a smooth linear-target field $\phi\colon \mathbb{R}^d \to \mathbb{R}^N$, use

$$E = E_2 + E_0, \qquad E_2 = \frac{1}{2}\int_{\mathbb{R}^d} |\nabla\phi|^2 \mathrm{d}^d x, \qquad E_0 = \int_{\mathbb{R}^d} V(\phi)\mathrm{d}^d x, \qquad V \geq 0.$$

The subscript counts derivatives in the energy term. A spatial dilation is $\phi_\lambda(x) = \phi(\lambda x)$ for $\lambda > 0$. The chain rule contributes λ^2 to the gradient term. The change of variables contributes λ^{-d} to both terms. Thus

$$E(\lambda) = \lambda^{2-d}E_2 + \lambda^{-d}E_0, \qquad E'(1) = (2-d)E_2 - dE_0.$$

The figure shows this expression with $E_2 = E_0 = 1$.

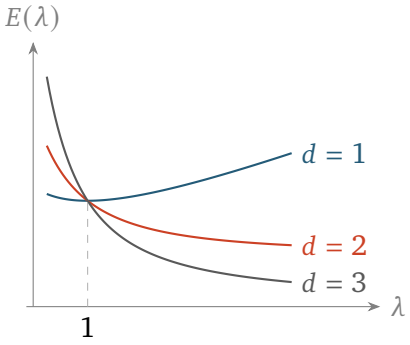


Figure 27.5: Dilation energy for the stated linear-target scalar model. Only the one-dimensional curve has a stationary point for positive E_2 and E_0.

Proposition 27.9 (Derrick's theorem · *proved*). *Let $V\colon \mathbb{R}^N \to [0,\infty)$ be smooth. Every smooth finite-energy static solution $\Delta\phi = \nabla V(\phi)$ of this model on $\mathbb{R}^d$, for $d \geq 2$, is a constant vacuum.*

Proof. **Step 1: the stress identity.** The field equation gives stationarity against compactly supported variations. A dilation changes the field arbitrarily far away, so this fact does not yet justify $E'(1) = 0$. Its infinitesimal variation is $x^j\partial_j\phi$, where x^j are the spatial coordinates. We cut off this variation at large radius. The stress tensor writes the resulting integration by parts as a divergence. Use the Euclidean metric to raise indices and let $e = |\nabla\phi|^2/2 + V(\phi)$. The stress tensor and its divergence are

$$T^i{}_j = \partial^i\phi\cdot\partial_j\phi - \delta^i_j e, \qquad \partial_i T^i{}_j = (\Delta\phi - \nabla V(\phi))\cdot\partial_j\phi = 0.$$

To see the cancellation, differentiate the first term by the product rule. Its second-derivative term $\partial^i\phi\cdot\partial_i\partial_j\phi$ is the derivative of $|\nabla\phi|^2/2$ and cancels that part of $\partial_j e$. The remaining potential derivative is $\nabla V\cdot\partial_j\phi$.

Step 2: the cutoff limit. Fix one smooth bump χ from Lecture 21, equal to one on the unit ball and zero outside the radius-two ball. Put $\chi_R(x) = \chi(x/R)$. The chain rule gives

$|\nabla\chi_R| \le C/R$ with $C = \sup|\nabla\chi|$. Integrate the divergence of $T^i{}_j x^j \chi_R$ and use compact support to obtain

$$\int_{\mathbb{R}^d} \chi_R T^i{}_i \mathrm{d}^d x = -\int_{\mathbb{R}^d} T^i{}_j x^j \partial_i \chi_R \mathrm{d}^d x.$$

The absolute value of the right side is bounded by a constant times $\int_{R<|x|<2R}(|\nabla\phi|^2 + V)\mathrm{d}^d x$, which tends to zero by finite energy. The trace is integrable, so dominated convergence on the left gives $(2-d)E_2 - dE_0 = 0$. For $d > 2$, nonnegativity forces $E_2 = E_0 = 0$, hence a constant vacuum.

Step 3: dimension two. For $d = 2$ the gradient term has zero scaling weight. The identity gives $E_0 = 0$ and leaves E_2 undetermined. We use the field equation to control this remaining term. Continuity and nonnegativity imply $V(\phi(x)) = 0$ everywhere. Each such field value minimizes V, so $\nabla V(\phi(x)) = 0$. The field equation becomes $\Delta\phi = 0$. Let u be any component of a derivative $\partial_i\phi$. It is harmonic and square-integrable. Its spherical mean about any point p is independent of radius: its derivative is the normalized outward flux of ∇u, zero by Stokes and $\Delta u = 0$. The mean tends to $u(p)$ at radius zero, so averaging the sphere means radially gives the ball mean. Cauchy–Schwarz therefore yields

$$|u(p)| = \left|\frac{1}{|B_R|}\int_{B_R(p)} u \mathrm{d}^d x\right| \le |B_R|^{-1/2}\|u\|_{L^2(\mathbb{R}^d)} \longrightarrow 0.$$

Thus all derivatives vanish and ϕ is a constant vacuum in this case too. ∎

A constrained sphere-valued sigma model is outside the linear-target hypothesis: its field equation is not $\Delta\phi = \nabla V$ for a potential on all of $\mathbb{R}^N$. Lecture 26's degree-one plane texture has this constrained target. Gauge fields and higher-derivative energies also have different scaling powers. They need not satisfy this obstruction. Our dilation is $\phi(\lambda x)$.[1] In one dimension the same stationary scaling identity reads $E_2 = E_0$, as for the kink.

For a smooth unit-vector field $\mathbf{n}\colon \mathbb{R}^2 \to S^2$ extending smoothly to the compactified plane, consider the finite Dirichlet energy

$$E_\sigma = \frac{1}{2}\int_{\mathbb{R}^2} (|\partial_x \mathbf{n}|^2 + |\partial_y \mathbf{n}|^2)\mathrm{d}x\mathrm{d}y.$$

This energy is invariant under spatial dilation in dimension two. Since $\mathbf{n}\cdot\partial_i\mathbf{n} = 0$, the cross product with $\mathbf{n}$ preserves the length of $\partial_i\mathbf{n}$. For $s \in \{+1, -1\}$ we therefore have

$$|\partial_x \mathbf{n} - s\mathbf{n}\times\partial_y\mathbf{n}|^2 = |\partial_x\mathbf{n}|^2 + |\partial_y\mathbf{n}|^2 + 2s\mathbf{n}\cdot(\partial_x\mathbf{n}\times\partial_y\mathbf{n}).$$

Lecture 26 identifies the integral of the last density with $4\pi Q$. Integration gives

$$E_\sigma = \frac{1}{2}\int_{\mathbb{R}^2} |\partial_x\mathbf{n} - s\mathbf{n}\times\partial_y\mathbf{n}|^2 \mathrm{d}x\mathrm{d}y - 4\pi s Q \ge 4\pi|Q|$$

when $s = -\operatorname{sgn}(Q)$ for nonzero charge. For zero charge, either sign gives the bound zero. For Lecture 26's degree-one texture of scale $a > 0$, differentiation gives

$$|\partial_x\mathbf{n}|^2 = |\partial_y\mathbf{n}|^2 = \frac{4a^2}{(a^2+x^2+y^2)^2}, \qquad \partial_x\mathbf{n} = -\mathbf{n}\times\partial_y\mathbf{n}.$$

Thus its energy is 4π for every scale a. This family attains the lower bound, so it is stationary under smooth compactly supported variations within the unit sphere. Its size is not fixed by

[1] Manton–Sutcliffe, equations (4.17)–(4.19), pp. 84–85, use this dilation. Inverting the scale parameter changes the derivative sign but not the obstruction.

this energy. For Lecture 26's three-dimensional hedgehog, the same Dirichlet model instead has $|\nabla \mathbf{n}|^2 = 2/r^2$. The energy in $r_0 < r < R$ is $4\pi(R - r_0)$, which diverges as R grows.

Self-check. For the scalar kink model, list the possible values of $Q = (\phi_+ - \phi_-)/(2v)$ for vacuum endpoints. Count the ordered endpoint sectors and decide whether the charge distinguishes all of them.

Answer. The four ordered pairs are $(-v, -v)$, $(-v, +v)$, $(+v, -v)$ and $(+v, +v)$. Their charges are respectively zero, +1, −1 and zero. The two equal-endpoint pairs are different sectors because a continuous vacuum-valued endpoint cannot move between the disconnected vacua. Thus the charge is an invariant, but is not a complete label of the sector.

Self-check. State in one sentence what a static soliton is and why we introduced its energy. Explain why a nonzero charge is insufficient to identify one.

Answer. A static soliton is a localized finite-energy stationary field, and its energy distinguishes stationary profiles and sector minima among fields with the same boundary topology. A charge labels allowed deformations, whereas the field equation must still be satisfied.

Unit 27B Gauge vortices and monopole charge

27.4 The abelian Higgs vortex and its flux

Definition 27.10. A *Higgs field* here is a charged complex scalar $\phi\colon \mathbb{R}^2 \to \mathbb{C}$.

The symbol ϕ denotes the field, and θ denotes the polar angle.

Physical input. We use the static abelian Higgs energy below as a field model. In its Ginzburg–Landau interpretation, the complex field describes an electron-pair condensate in a superconductor (Manton–Sutcliffe, Chapter 1 and §7.1). The pair charge is $q = -2e$, where $e > 0$ is the elementary charge. For fields constant along a straight vortex line, the plane is a transverse section and E is energy per unit length.

Let $q \neq 0$ be its signed charge and put $a = q/\hbar$. Use a smooth real potential A_{em} on the plane and Lecture 21's derivative $D = \mathrm{d} - \mathrm{i}aA_{\mathrm{em}}$. Write $A_{\mathrm{em}} = A_x \mathrm{d}x + A_y \mathrm{d}y$ and $B = \partial_x A_y - \partial_y A_x$. The symbol $v > 0$ again denotes the vacuum magnitude. Here $\lambda > 0$ is the scalar coupling, distinct from the dilation parameter in the preceding section. The static abelian Higgs energy is

$$E = \int_{\mathbb{R}^2} \left[\frac{1}{2} B^2 + \frac{1}{2}\left(|D_x\phi|^2 + |D_y\phi|^2\right) + \frac{\lambda}{8}\left(|\phi|^2 - v^2\right)^2 \right] \mathrm{d}x\mathrm{d}y.$$

Its vacuum field values have modulus v, a circle. Write $r = \sqrt{x^2 + y^2}$. Without a gauge field, the winding tail $\phi = ve^{\mathrm{i}n\theta}$ outside a core has gradient energy

$$E_{\mathrm{tail}} = \frac{1}{2} \int_{r_0}^{R} \int_0^{2\pi} \frac{v^2 n^2}{r^2} r \mathrm{d}\theta \mathrm{d}r = \pi v^2 n^2 \log(R/r_0), \qquad 0 < r_0 < R.$$

For nonzero winding this diverges as $R \to \infty$. The gauge potential permits cancellation of the angular phase derivative in $D\phi$. This cancellation makes finite planar energy possible.

The gauge model also changes the scaling argument. Write E_B, E_D and E_V for its magnetic, covariant-gradient and potential terms. For a dilation factor $\kappa > 0$, put $\phi_\kappa(x) = \phi(\kappa x)$ and

let $(A_{\mathrm{em}})_\kappa$ be the pullback of A_{em} under $x \mapsto \kappa x$. The potential components gain one factor of κ, and the magnetic field gains two. In dimension two the energy therefore scales as

$$E(\kappa) = \kappa^2 E_B + E_D + \kappa^{-2} E_V, \qquad E'(1) = 2(E_B - E_V).$$

The two weighted terms can balance, in contrast to the scalar obstruction. The figure shows the exact curve for $E_B = E_D = E_V = 1$.

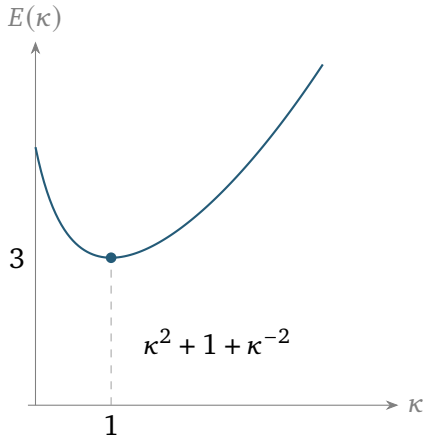


Figure 27.6: The magnetic and potential energies have opposite scaling weights. Their sum has a minimum at $\kappa = 1$ for these equal coefficients.

For $r = R$, write $\phi_R(\theta) = \phi(R\cos\theta, R\sin\theta)$. The symbol D_θ means the covariant derivative along the angular parameter, so $D_\theta \phi_R = \partial_\theta \phi_R - \mathrm{i}a(A_{\mathrm{em}})_\theta \phi_R$. It includes the radius factor relative to a unit tangent derivative.

The winding tail suggests a vanishing angular covariant derivative. We impose the precise integrated condition in the theorem as a boundary assumption.

Theorem 27.11 (Vortex flux · *sketched; flux proved*). *Assume B is integrable, $|\phi_R| \to v$ uniformly, and $\int_0^{2\pi} |D_\theta \phi_R| \mathrm{d}\theta \to 0$ as $R \to \infty$. Let n be the winding of $\phi_R/|\phi_R|$ for large R. The total flux is*

$$\Phi = \int_{\mathbb{R}^2} B \mathrm{d}x\mathrm{d}y = \frac{2\pi n}{a} = \frac{2\pi\hbar n}{q}.$$

At the critical coupling *$\lambda = a^2$, finite-energy fields in this sector obey $E \geq \pi v^2 |n|$.*

Flux proof. The amplitude is bounded below by $v/2$ outside a sufficiently large disk. Thus the nonzero fields on any two large circles are homotopic through the intervening annulus, and their winding n agrees. On each circle, the imaginary part of the logarithmic derivative gives

$$a \oint_{r=R} A_{\mathrm{em}} = 2\pi n - \int_0^{2\pi} \mathrm{Im}\left(\frac{D_\theta \phi_R}{\phi_R}\right) \mathrm{d}\theta.$$

The last integral tends to zero by the amplitude lower bound and the assumed angular derivative integral. Stokes on the disk gives $\oint A_{\mathrm{em}} = \int_{r \leq R} B \mathrm{d}x\mathrm{d}y$. Integrability of B gives the total flux in the limit, proving the formula. ∎

These asymptotic conditions are part of the sector. Finite energy alone was not used to infer pointwise limits. The magnetic flux now has the same integer origin as Lecture 3's winding and Lecture 22's quantization. Lecture 21 gives the limiting holonomy $e^{\mathrm{i}a\Phi} = e^{2\pi \mathrm{i} n} = 1$. Thus

the limiting Aharonov–Bohm phase of the condensate is trivial. For $q = -2e$, the magnitude of a single flux quantum is

$$\frac{2\pi\hbar}{2e} = \frac{h_{\mathrm{P}}}{2e}, \qquad h_{\mathrm{P}} = 2\pi\hbar.$$

A nowhere-zero field on a disk gives a disk extension of its normalized boundary map. Lecture 3 then gives winding zero. Consequently nonzero winding forces at least one zero of the Higgs field in the plane. Gauge maps in this discussion are smooth on the whole plane. Their restriction to a circle contracts through the disk to the value at its centre. Its winding is therefore zero, so these gauge changes preserve n.

For a sign $s \in \{+1, -1\}$, we obtain the bound by writing squares plus a boundary term, as for the kink. The square $|(D_x + \mathrm{i}sD_y)\phi|^2$ produces a mixed derivative term. The commutator converts that term into a curl and a multiple of $B|\phi|^2$. At critical coupling the magnetic and potential terms form a second square whose mixed term cancels this multiple. Only a flux term remains after integration.

Let $j_i = \mathrm{Im}(\bar{\phi} D_i \phi)$, for $i \in \{x, y\}$, be the components of a real boundary-current one-form, distinct from the physical current of Lecture 15. Set $\mathrm{curl}\, j = \partial_x j_y - \partial_y j_x$. On a radius-$R$ circle its circulation satisfies

$$\left| \oint_{r=R} j \right| \leq \sup_{\theta} |\phi_R| \int_0^{2\pi} |D_\theta \phi_R| \mathrm{d}\theta \longrightarrow 0.$$

The sector assumptions therefore remove this boundary term. Using $[D_x, D_y]\phi = -\mathrm{i}aB\phi$ gives $\mathrm{curl}\, j = 2\,\mathrm{Im}(\overline{D_x\phi} D_y \phi) - aB|\phi|^2$. For either sign $s \in \{+1, -1\}$, the resulting square completion at $\lambda = a^2$ is

$$E = \frac{1}{2} \int_{\mathbb{R}^2} \left[|(D_x + \mathrm{i}sD_y)\phi|^2 + \left(B + \frac{sa}{2}(|\phi|^2 - v^2) \right)^2 \right] \mathrm{d}x\mathrm{d}y + \frac{sav^2}{2}\Phi.$$

Obtain this identity first on disks using Stokes for the curl term. The boundary-current circulation tends to zero by the estimate above. Finite energy and integrability of the magnetic field give the limit. Choosing $s = \mathrm{sgn}(n)$ for $n \neq 0$ gives the bound. The first-order equations that set both squares to zero are

$$(D_x + \mathrm{i}sD_y)\phi = 0, \qquad B = -\frac{sa}{2}(|\phi|^2 - v^2).$$

Their solutions saturate the bound. Their existence requires an analytic input. Away from the zeros of ϕ, these first-order equations reduce to one nonlinear elliptic equation for $\log|\phi|^2$. The prescribed zeros supply its point-source conditions. Solving that boundary problem constructs vortices. The analytic existence step is cited from Manton–Sutcliffe, §7.8, pp. 199–200. Their normalization is $a = v = 1$, with critical $\lambda = 1$ and bound $\pi|n|$.[2]

The figure shows the smooth winding-one trial field calculated in the example, for $a > 0$. Its radial arrows show the phase as a direction in the internal complex plane. The separately labelled tangent arrow gives the circulation of A_{em}.

[2] In general their connection potential is aA_{em}. Its curvature is aB, so potential rescaling alone does not equate their unit-coupling energy with the general-a energy printed here.

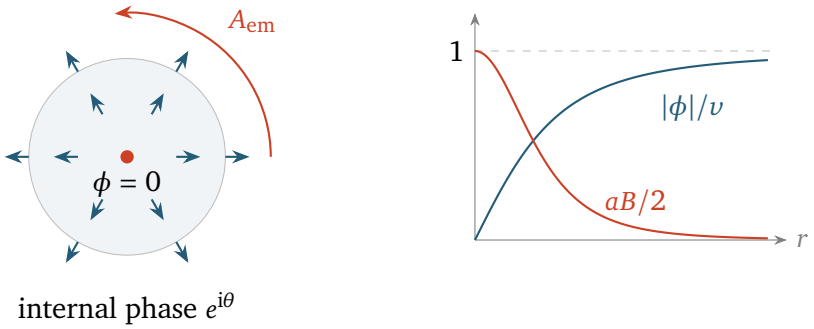


Figure 27.7: The trial field has internal phase $e^{i\theta}$ and positive potential circulation for $a > 0$. The exact radial profiles approach vacuum amplitude and zero magnetic field, respectively; this trial field is not a solution of the first-order equations.

Example 27.12 (From asymptotic phase to flux). Assume $\phi_R \to v e^{in\theta}$ with its first angular derivative, and $D_\theta \phi_R \to 0$ uniformly. The limiting derivative equation then gives $(A_{\mathrm{em}})_\theta \to n/a$ in this gauge. With B integrable, the flux proof needs only the integrated angular condition, and gives $\Phi = 2\pi n/a$. A negative signed charge reverses the magnetic flux for the same positive winding.

For a concrete smooth configuration of winding one, use

$$\phi(x,y) = \frac{v(x+iy)}{\sqrt{1+x^2+y^2}}, \qquad A_{\mathrm{em}} = \frac{-y\,dx + x\,dy}{a(1+x^2+y^2)}, \qquad B = \frac{2}{a(1+x^2+y^2)^2}.$$

It has a Higgs zero at the origin and no singular potential there. On a radius-R circle, $\phi_R = vRe^{i\theta}/\sqrt{1+R^2}$ and $(A_{\mathrm{em}})_\theta = R^2/[a(1+R^2)]$. Thus $D_\theta \phi_R = i\phi_R/(1+R^2)$ tends uniformly to zero. The flux is

$$\Phi = \frac{4\pi}{a}\int_0^\infty \frac{r\,dr}{(1+r^2)^2} = \frac{2\pi}{a}.$$

Direct differentiation of the displayed fields gives

$$D_x\phi = \frac{v}{(1+r^2)^{3/2}}, \qquad D_y\phi = \frac{iv}{(1+r^2)^{3/2}}.$$

Thus the sum of the squared norms of these two derivatives is $2v^2/(1+r^2)^3$. Substitution of $t = 1+r^2$ makes the radial integrals of $r/(1+r^2)^p$ equal to $1/[2(p-1)]$, for $p \in \{2,3,4\}$. Each energy density term is integrable. The total energy is

$$E = \pi\left[\frac{2}{3a^2} + \frac{v^2}{2} + \frac{\lambda v^4}{8}\right].$$

At critical coupling we can compare this trial with the bound in a fixed theory by varying its width $\ell > 0$. Replace its fields by

$$\phi_\ell = \frac{v(x+iy)}{\sqrt{\ell^2+r^2}}, \qquad (A_{\mathrm{em}})_\ell = \frac{-y\,dx + x\,dy}{a(\ell^2+r^2)}.$$

The same radial integrals give

$$E(\ell) = \pi\left[\frac{2}{3a^2\ell^2} + \frac{v^2}{2} + \frac{a^2v^4\ell^2}{8}\right].$$

The first and last terms have the form $A/\ell^2 + C\ell^2 \geq 2\sqrt{AC}$. They are equal at the minimizing width, so

$$\ell^2 = \frac{4}{\sqrt{3}\,a^2v^2}, \qquad E_{\text{trial,min}} = \pi v^2\left(\frac{1}{2} + \frac{1}{\sqrt{3}}\right) > \pi v^2.$$

The field demonstrates the finite-energy flux sector but does not attain its Bogomolny bound. It is a trial configuration, not an asserted solution of the vortex field equations.

27.5 The monopole charge *(optional)*

Lecture 28's zero-mode count does not depend on this section.

Remark 27.13 (The Yang–Mills–Higgs monopole)**.** In the SU(2) Yang–Mills–Higgs model, an adjoint Higgs field is a vector in the Lie algebra that transforms by conjugation. Its fixed-norm vacuum directions form $\mathrm{SU}(2)/\mathrm{U}(1) \cong S^2$. With a continuous nonzero Higgs limit at infinity, the normalized field defines a map from the outward spatial sphere to this vacuum sphere. Its degree is the topological monopole charge $Q \in \mathbb{Z}$, as in Lecture 26. Orient the Lie algebra by Lecture 23's basis T^1, T^2, T^3. The hedgehog with coefficients x^a/r in that basis restricts to the identity direction map between outward spheres, so it has $Q = +1$. The remaining U(1) connection acts on charged line bundles at infinity. Lecture 22 then requires $q\Phi_{\text{m}} = 2\pi\hbar n$ for each such charge convention. Relating that integer to the degree also requires specifying the chosen charged representation.

Physical input. The SU(2) Yang–Mills–Higgs equations admit smooth finite-energy monopoles (Manton–Sutcliffe, §8.2). Their Higgs amplitude vanishes at the core while the normalized field approaches the hedgehog at infinity. The residual abelian field approaches a Dirac monopole field at large distance. These are cited properties of those solutions.

Lecture 22's abelian description has a singular central point. The Yang–Mills–Higgs solution instead has smooth full fields at its core. The following figure compares only their direction maps on a surrounding sphere.

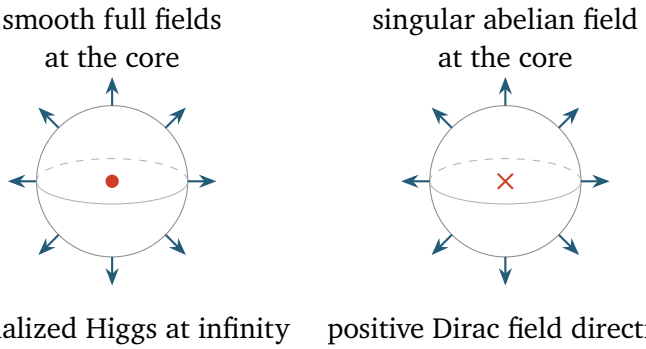


Figure 27.8: The normalized asymptotic Higgs map is the outward identity direction map of degree +1. A positive-flux Dirac field also has outward directions, but its abelian field is singular at the origin.

The degree describes the monopole boundary topology. The analytic monopole construction and its mass bound are treated in Manton–Sutcliffe, §8.2.

Self-check. A vortex has winding $n = -3$ and signed charge $q = -2e$ with $e > 0$. Determine its signed flux and critical-coupling energy bound. Then consider the printed width-one trial field in a different model, with $a = -2$ and $v = 3$ at critical coupling. Test the magnetic first-order equation at the origin and decide whether this trial attains its bound.

Answer. Substitution of both negative signs gives $\Phi = 3\pi\hbar/e = 3h_{\mathrm{P}}/(2e)$, and the bound is $E \geq 3\pi v^2$. The trial has winding +1, so its saturating sign is $s = +1$. At the origin its field vanishes and its magnetic field is $B(0) = 2/a = -1$. The first-order equation instead requires $B(0) = sav^2/2 = -9$. Thus this trial does not attain the bound, although it satisfies the sector assumptions.

Summary. Continuous vacuum boundary data define sectors that an evolution with jointly continuous vacuum boundary data cannot change. Finite energy alone does not supply those limits. The scalar kink saturates $E \geq |\Delta W|$ and has signed charge +1. Derrick's theorem excludes nonconstant smooth finite-energy static solutions of the stated nonnegative-potential linear-target scalar model for $d \geq 2$. The gauge vortex lies outside that model and has flux $2\pi\hbar n/q$ under the stated asymptotic conditions. Its critical-coupling bound follows by square completion. Existence uses the cited analytic input. Lecture 28 uses a quantized sphere flux to count chiral quantum zero modes.

Reading. Manton–Sutcliffe, Chapters 1 and 6; §4.2, pp. 84–85; §5.1–5.2, pp. 110–113; §7.3, pp. 164–165; §7.8, pp. 197–200; §8.2.

Lecture 28

The index on the sphere

Objectives. *Lecture 22 associated an integer with the flux through a sphere. We now use that integer to count quantum zero modes of a Dirac operator. The spin connection changes the line bundle in which one chiral component lives. On the round sphere this turns the count into a polynomial calculation. The cited index theorem explains why the difference of the two chiral counts is topological.*

By the end of this lecture you will be able to:

- Specify the two chiral components and the sign of their index.
- Count the holomorphic sections of $O(m)$ for every $m \in \mathbb{Z}$.
- Compute both monopole zero-mode counts using the stated cited identifications.
- Distinguish the Dirac count from the spinless Landau-level count.

Unit 28A Index on the sphere

28.1 Spinors and the canonical line on the sphere

We use the unit round sphere with its outward orientation. We identify it with $\mathbb{C}\mathrm{P}^1$ using Lecture 20's complex coordinates z and $w = 1/z$. The complex orientation agrees with this outward orientation.

Definition 28.1 (Spinor field). A *spinor field* is a section of a spinor bundle, the geometric bundle on which the Dirac equation acts.

Its construction is a cited input (Lawson–Michelsohn, Chapter II and Appendix D). On an oriented surface it has two complex line components S^+ and S^-.

Definition 28.2 (Chirality). The *chirality* of a spinor is its grading with eigenvalues +1 and −1 on S^+ and S^-.

Spinor chirality is distinct from even and odd differential-form degree.

Definition 28.3 (Canonical bundle). The *canonical bundle* K_{S^2} is the line of holomorphic cotangent vectors.

Its local frames are $\mathrm{d}z$ and $\mathrm{d}w$. Since $\mathrm{d}w = -z^{-2}\mathrm{d}z$, absorbing the constant minus sign into one frame gives $K_{S^2} \cong O(-2)$ in Lecture 22's frame convention. We choose the positive spinor line to be its square root:

$$S^+ = K_{S^2}^{1/2} \cong O(-1), \qquad (K_{S^2}^{1/2})^{\otimes 2} \cong K_{S^2}.$$

This square-root choice is part of the cited spinor identification.

We can model these two lines by the spin eigenlines from Lecture 24. At a point $\mathbf{n} \in S^2$, the matrix $\mathbf{n} \cdot \boldsymbol{\sigma}$ has eigenvalues +1 and −1. The aligned eigenline has Chern number −1 and models S^+. The opposite eigenline has Chern number +1 and models S^-.

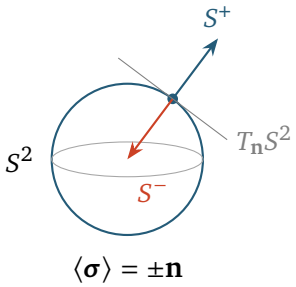
In the northern unit frame, Lecture 24 gives the aligned connection $\mathcal{A}_N = \mathrm{i}(1-\cos\theta)\mathrm{d}\varphi/2$. This is half the canonical-line connection, because the canonical line is the dual of the tangent line of Lecture 22.

The projected eigenline connections give the spin connections in this sphere model. The identification of their Dirac operator with the intrinsic operator remains the cited spinor input. A spin expectation direction alone does not establish that operator identity.

Recall that $L_n = \mathcal{O}(n)$ is the monopole line of Chern number $n \in \mathbb{Z}$. Twisting means taking the tensor product with this charged line. The positive twisted line is $S^+ \otimes L_n \cong \mathcal{O}(n-1)$. The full twisted spinor bundle has the cited description

$$S \otimes L_n \cong \left(\Lambda^{0,0} \oplus \Lambda^{0,1}\right) \otimes \mathcal{O}(n-1).$$

Here a $(0,0)$-form is a function and a $(0,1)$-form is locally a multiple of $\mathrm{d}\bar{z}$. The positive part is $\mathcal{O}(n-1)$. The negative part is $\Lambda^{0,1} \otimes \mathcal{O}(n-1)$. The figure shows the normal-spin model and the degree change under twisting.



Line	Chern number
TS^2	+2
K_{S^2}	−2
S^+	−1
S^-	+1
L_n	n
$\mathcal{O}(n-1)$	$n-1$

$$\boxed{S^+ = \mathcal{O}(-1)} \quad \otimes \quad \boxed{L_n = \mathcal{O}(n)} \quad \cong \quad \boxed{\mathcal{O}(n-1)}$$

Figure 28.1: The arrows show spin aligned with the outward normal and spin opposite to it. The degree table and the tensor product give the positive twisted line $\mathcal{O}(n-1)$.

These identifications are stated in Lawson–Michelsohn, Appendix D, equation (D.16), p. 396. Walpuski, Propositions 70.3 and 70.6, pp. 76–78, give the form decomposition and square-root construction explicitly.

28.2 The Dirac operator and zero energy

Physical input. For a charged spin-$\frac{1}{2}$ particle on the plane, minimal coupling gives $D_j = \partial_j - \mathrm{i}a(A_{\mathrm{em}})_j$, with $a = q/\hbar$. The spatial Dirac operator is $D = -\mathrm{i}(\sigma_x D_x + \sigma_y D_y)$. The massless Dirac equation is

$$\mathrm{i}\hbar\partial_t \Psi = \hbar v D\Psi, \qquad v > 0,$$

where v is the propagation speed. For a nonrelativistic particle of mass $m_{\mathrm{phys}} > 0$ with gyromagnetic factor two, the Pauli Hamiltonian is $H_{\mathrm{P}} = \hbar^2 D^2/(2m_{\mathrm{phys}})$.

The magnetic field is $B = \partial_x (A_{\mathrm{em}})_y - \partial_y (A_{\mathrm{em}})_x$. The commutator and the Pauli matrix product give

$$[D_x, D_y] = -\mathrm{i}aB, \qquad \sigma_x \sigma_y = \mathrm{i}\sigma_z,$$
$$D^2 = -D_x^2 - D_y^2 - \mathrm{i}\sigma_z [D_x, D_y] = -D_x^2 - D_y^2 - aB\sigma_z.$$

For a uniform field with $aB > 0$, choose $(A_{\mathrm{em}})_x = 0$ and $(A_{\mathrm{em}})_y = Bx$. On a coefficient $e^{\mathrm{i}ky} g(x)$, the scalar part becomes the oscillator $-g'' + (k - aBx)^2 g$. The Gaussian $g(x) = \exp[-aB(x - k/(aB))^2/2]$ has eigenvalue aB. The operators

$$b = \frac{\partial_x + aBx - k}{\sqrt{2aB}}, \qquad b^* = \frac{-\partial_x + aBx - k}{\sqrt{2aB}}$$

satisfy $[b, b^*] = 1$ and factor the scalar operator as $2aBb^*b + aB$. The Gaussian obeys $bg = 0$. Applying b^* successively therefore gives eigenvalues $(2r + 1)aB$, for integers $r \geq 0$. The spin term subtracts aB for $\sigma_z = +1$ and adds aB for $\sigma_z = -1$. The lowest orbital and spin contributions therefore cancel in the positive component. The figure records the resulting squared-Dirac energies on the plane.

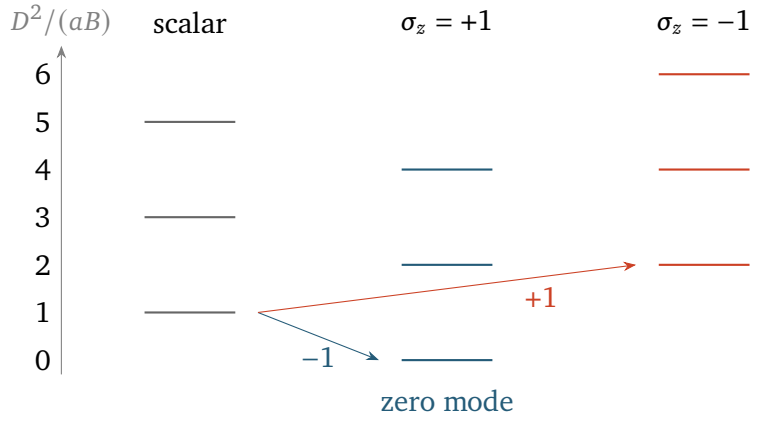


Figure 28.2: The scalar levels are shifted down or up by the spin term for $aB > 0$. The lowest positive-chirality level has zero energy; physical Pauli energies multiply these values by $\hbar^2/(2m_{\mathrm{phys}})$.

The sphere count concerns the curved Dirac operator rather than this plane oscillator spectrum. We express the curved operator using the two spinor lines.

We write $\Omega^{0,k}(\mathcal{O}(n-1))$ for the smooth $\mathcal{O}(n-1)$-valued $(0,k)$-forms.

Definition 28.4 (Dolbeault operator). The *Dolbeault operator* $\bar{\partial}_A$ is the $(0,1)$ part of the covariant derivative:

$$\bar{\partial}_A \colon \Omega^{0,0}(\mathcal{O}(n-1)) \longrightarrow \Omega^{0,1}(\mathcal{O}(n-1)).$$

In a holomorphic frame it sends a coefficient f to $(\partial_{\bar{z}} f)\mathrm{d}\bar{z}$, as in Lecture 22. Its adjoint $\bar{\partial}_A^*$ is defined by the Hermitian L^2 inner products from the bundle metric and sphere area. The first slot of these inner products is antilinear. There are no $(0,2)$-forms on a complex curve.

Proposition 28.5 (The sphere Dirac operator · *sketched*). *With the above chirality, the twisted Dirac operator is*

$$D = \sqrt{2}(\bar\partial_A + \bar\partial_A^*) = \begin{pmatrix} 0 & D^- \\ D^+ & 0 \end{pmatrix}, \qquad D^+ = \sqrt{2}\bar\partial_A, \qquad D^- = (D^+)^* = \sqrt{2}\bar\partial_A^*.$$

The block D^+ maps $\Omega^{0,0}(\mathcal{O}(n-1))$ to $\Omega^{0,1}(\mathcal{O}(n-1))$. The other block maps in the reverse direction.

Explanation. The cited spinor identification turns Clifford multiplication into exterior multiplication by an antiholomorphic covector and its adjoint contraction. Contracting the spin connection and monopole connection with it gives the displayed operator. This identification, including its $\sqrt{2}$ normalization, is Walpuski's Proposition 70.3. For a flat metric $\mathrm{d}x^2 + \mathrm{d}y^2$, the unit antiholomorphic coframe is $\mathrm{d}\bar z/\sqrt{2}$. Thus the positive block has coefficient $2\partial_{\bar z}$ and the negative block has coefficient $-2\partial_z$. Their product is $-4\partial_z\partial_{\bar z} = -\partial_x^2 - \partial_y^2$, the nonnegative scalar Laplacian of Lecture 15. To compare with the familiar flat Dirac equation, let σ_x and σ_y be the Pauli matrices. The corresponding self-adjoint spatial operator is $-\mathrm{i}(\sigma_x\partial_x + \sigma_y\partial_y)$. Multiplying its negative component by i gives the displayed chiral blocks. This verifies the local normalization of the cited operator identification. ∎

Definition 28.6 (Zero mode). A *zero mode* is a smooth spinor annihilated by D.

For a positive spinor in a holomorphic frame, the zero-mode equation is

$$D^+\psi = 0 \quad \Longleftrightarrow \quad \partial_{\bar z} f = 0.$$

This is the Cauchy–Riemann equation for the local coefficient f. Its solutions are holomorphic sections of the twisted line $\mathcal{O}(n-1)$. The transition between charts determines which local solutions extend over the sphere.

28.3 Polynomial sections and monopole zero modes

Definition 28.7 (Holomorphic sections and Dolbeault cohomology). For a holomorphic line E, the space of global holomorphic sections is $H^0(E) = \ker \bar\partial_E$. Its degree-one Dolbeault cohomology is

$$H^1(E) = \Omega^{0,1}(E)/\bar\partial_E\Omega^{0,0}(E).$$

Every $(0,1)$-form is Dolbeault-closed in complex dimension one, explaining this quotient. The cited Dolbeault Hodge theorem identifies each class with a unique harmonic representative. In this dimension that representative is annihilated by $\bar\partial_E^*$. Consequently the cited spinor and Hodge inputs identify $\ker D^+ \cong H^0(\mathcal{O}(n-1))$ and $\ker D^- \cong H^1(\mathcal{O}(n-1))$ (Walpuski, Proposition 70.3, p. 77).

We now count H^0 directly, including the global condition at infinity. Let $m \in \mathbb{Z}$ and recall the frames $e_S = e_N z^m$ for $\mathcal{O}(m)$. A section with coefficients f_N and f_S satisfies

$$f_S(w) = w^m f_N(1/w).$$

Both coefficients must be holomorphic at their own chart origin. The figure displays the monomial test for this condition.

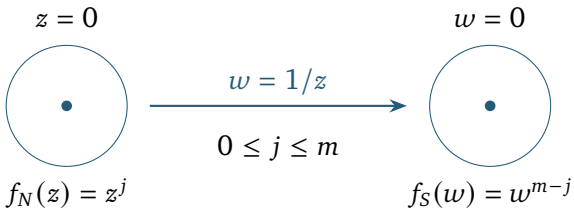


Figure 28.3: A northern monomial z^j becomes w^{m-j} in the southern frame. Holomorphicity at both poles requires $0 \le j \le m$.

We derive the needed expansion of a holomorphic function from Stokes' theorem of Lecture 12. A function is *holomorphic* here if it is smooth and satisfies $\partial_{\bar z} f = 0$. An *entire* function is holomorphic on all of $\mathbb{C}$.

Let f be holomorphic on a neighbourhood of the closed disc $|\zeta| \le R$. For a point z inside the disc, the one-form $f(\zeta)\mathrm{d}\zeta/(\zeta - z)$ is closed away from z. Indeed its exterior derivative is the $\partial_{\bar\zeta}$ derivative of its coefficient times $\mathrm{d}\bar\zeta \wedge \mathrm{d}\zeta$, which vanishes.

We apply Stokes on the disc with a small disc about z removed. The inner boundary is clockwise, so the outer counter-clockwise integral equals the counter-clockwise small-circle integral.

On the small circle $\zeta = z + \varepsilon e^{\mathrm{i}t}$, this last integral is $\mathrm{i}\int_0^{2\pi} f(z + \varepsilon e^{\mathrm{i}t})\mathrm{d}t$. Continuity and the limit $\varepsilon \to 0$ give Cauchy's formula:

$$f(z) = \frac{1}{2\pi\mathrm{i}} \oint_{|\zeta|=R} \frac{f(\zeta)}{\zeta - z}\mathrm{d}\zeta, \qquad |z| < R.$$

For $|z| < R$, the geometric series for this kernel converges uniformly on the integration circle. Termwise integration therefore gives the Taylor series and its coefficient bound:

$$\frac{1}{\zeta - z} = \sum_{j\ge 0} \frac{z^j}{\zeta^{j+1}}, \qquad f(z) = \sum_{j\ge 0} a_j z^j,$$
$$a_j = \frac{1}{2\pi\mathrm{i}} \oint_{|\zeta|=R} \frac{f(\zeta)}{\zeta^{j+1}}\mathrm{d}\zeta, \qquad |a_j| \le R^{-j} \max_{|\zeta|=R} |f(\zeta)|.$$

For an entire function, Stokes on an annulus makes each a_j independent of R. Hence this series represents the function on every disc.

Proposition 28.8 (Sections of a degree-m line · *proved*). *For $m \ge 0$, the holomorphic sections of $O(m)$ have northern coefficients equal to the polynomials of degree at most m. Their dimension is $m + 1$. For $m < 0$ there is no nonzero holomorphic section.*

Proof. The northern coefficient is entire, so the Stokes–Cauchy argument gives $f_N(z) = \sum_{j\ge 0} a_j z^j$ on all of $\mathbb{C}$. The southern coefficient is bounded by some $C > 0$ on a sufficiently small closed disc about zero. By the coefficient relation, the northern coefficient satisfies

$$f_N(z) = z^m f_S(1/z), \qquad |f_N(z)| \le C|z|^m \quad \text{for sufficiently large } |z|.$$

For all sufficiently large R, the Cauchy estimate gives $|a_j| \le CR^{m-j}$. Letting $R \to \infty$ shows that $a_j = 0$ whenever $j > m$. For $m < 0$ this kills every coefficient. For $m \ge 0$ it leaves exactly $1, z, \dots, z^m$. Conversely each of these monomials gives a nonnegative southern power and hence a global section. Their distinct Taylor coefficients prove linear independence, so the dimension is $m + 1$. ∎

We can also see the negative-chirality equation directly by integration by parts. In a holomorphic frame e_N of $E = \mathcal{O}(m)$, put $h = \|e_N\|^2$ and write a negative spinor as $u = f\mathrm{d}\bar z \otimes e_N$. For a conformal metric $\rho(\mathrm{d}x^2 + \mathrm{d}y^2)$, the area form is $\rho \mathrm{d}x\mathrm{d}y$ and $|\mathrm{d}\bar z|^2 = 2/\rho$. For a test section ge_N supported in this chart, these factors cancel:

$$\langle \bar\partial_E(ge_N), u\rangle = 2\int \overline{\partial_{\bar z} g}\, fh \mathrm{d}x\mathrm{d}y = -2\int \bar g\, \partial_z(fh)\mathrm{d}x\mathrm{d}y.$$

Thus $\bar\partial_E^* u = 0$ exactly when $v = h\bar f$ is holomorphic. The frame and covector transitions give

$$f_S = -\bar z^2 z^{-m} f_N, \qquad h_S = |z|^{2m} h_N, \qquad v_S = -w^{-m-2} v_N.$$

After absorbing the constant minus sign into a frame, this is the coefficient law of $\mathcal{O}(-m-2)$. The correspondence is antilinear and bijective: its inverse is $f = h^{-1}\bar v$ in each chart. It explains locally why the negative count equals $\dim H^0(\mathcal{O}(-m-2))$. For example, when $n = -2$, the positive twisted degree is $m = -3$ and the dual degree is one. The choices $v = 1$ and $v = z$ give the negative coefficients $(1+|z|^2)^{-3}\mathrm{d}\bar z$ and $\bar z(1+|z|^2)^{-3}\mathrm{d}\bar z$ in the holomorphic frame.

The corresponding global cohomology statement is *Serre duality*, a cited theorem for compact complex curves, whose sphere case is $H^1(E) \cong H^0(K_{S^2} \otimes E^*)^*$. Here E^* is the dual line and the outer star denotes the dual vector space. See Walpuski, Remark 70.5, p. 77, or Gunning, *Some Topics in the Function Theory of Compact Riemann Surfaces*, preliminary version (2018), Theorem 1.17, p. 28. For $E = \mathcal{O}(m)$ this gives $\dim H^1(\mathcal{O}(m)) = \dim H^0(\mathcal{O}(-m-2))$. The polynomial result therefore determines both counts.

Theorem 28.9 (Monopole zero modes · *sketched*). *On the round sphere with the monopole connection on $L_n = \mathcal{O}(n)$, for every $n \in \mathbb{Z}$,*

$$\dim\ker D^+ = \max(n,0), \qquad \dim\ker D^- = \max(-n,0),$$
$$\dim\ker D^+ - \dim\ker D^- = n.$$

The total number of zero modes is $|n|$. For $n > 0$ they all have positive chirality. For $n < 0$ they all have negative chirality. For $n = 0$ both kernels vanish for this sphere operator.

Proof. The positive component is $\mathcal{O}(n-1)$, so the polynomial count gives $\max(n,0)$. Serre duality identifies the negative count with $\dim H^0(\mathcal{O}(-n-1)) = \max(-n,0)$. Subtracting gives n for every integer n. Adding gives $|n|$. The global kernel identifications are the cited analytic steps in this derivation. ∎

Example 28.10 (Three zero modes in a monopole field). We take the sphere monopole with $n = 3$ and the nonnegative squared operator $H = D^2$. The physical Pauli Hamiltonian is a positive multiple of this operator, so their kernels agree. We determine its zero-energy states and their chirality. For a smooth spinor Ψ, the identity $\langle \Psi, D^2\Psi\rangle = \|D\Psi\|^2$ shows that $\ker H = \ker D$. The theorem gives three positive modes and no negative mode. Here the positive line has $m = n - 1 = 2$.

For a general $m \geq 0$, use Lecture 22's unit frame for $\mathcal{O}(m)$. The local positive zero-mode equation and its monomial solutions are

$$\left(\partial_{\bar z} + \frac{mz}{2(1+|z|^2)}\right)\psi = 0, \qquad \psi_j(z) = (1+|z|^2)^{-m/2} z^j, \quad 0 \leq j \leq m.$$

Differentiating the weight in ψ_j gives $-mz\psi_j/[2(1+|z|^2)]$, which cancels the connection term. These are unit-frame coefficients of sections of $\mathcal{O}(n-1)$. They are not coefficients

relative to $dz^{1/2}$ or to a coordinate density. The spin connection has already entered through the degree shift from n to m. For $m = 2$ the northern coefficients are

$$\frac{1}{1+|z|^2}, \qquad \frac{z}{1+|z|^2}, \qquad \frac{z^2}{1+|z|^2}.$$

Their southern holomorphic coefficients are w^2, w and 1, respectively, so each section is holomorphic at the south pole.

We normalize these states using the round area $dA = 4dx\,dy/(1+|z|^2)^2$, with $z = x + iy$. Here dA is an area form, whereas the subscript A in $\bar\partial_A$ names a connection. The angular integral makes distinct monomials orthogonal. With $r = |z|$, the three radial norms are

$$8\pi \int_0^\infty \frac{r^{2j+1}}{(1+r^2)^4} dr = \begin{cases} 4\pi/3, & j = 0, 2, \\ 2\pi/3, & j = 1. \end{cases}$$

Substitution $t = r^2$ reduces these to the integrals of $4\pi t^j/(1+t)^4$. Next put $u = (1+t)^{-1}$. The three integrals divided by 4π become $\int_0^1 u^2 du = 1/3$, $\int_0^1 u(1-u)du = 1/6$ and $\int_0^1 (1-u)^2 du = 1/3$, giving the displayed values. Dividing each coefficient by the square root of its stated squared norm gives an orthonormal basis. All three are zero-energy states of positive chirality.

We can locate the three normalized states on the sphere. With $z = \tan(\theta/2)e^{i\varphi}$, their unnormalized coefficients are $\cos^2(\theta/2)$, $\frac{1}{2}\sin\theta e^{i\varphi}$ and $\sin^2(\theta/2)e^{2i\varphi}$. Their normalized pointwise densities are

$$\varrho_0 = \frac{3(1+\cos\theta)^2}{16\pi}, \qquad \varrho_1 = \frac{3\sin^2\theta}{8\pi}, \qquad \varrho_2 = \frac{3(1-\cos\theta)^2}{16\pi}.$$

Each density integrates to one. The first is largest at the north pole, the second at the equator and the third at the south pole. The sum is $3/(4\pi)$, independent of position. The figure uses these densities with a common scale.

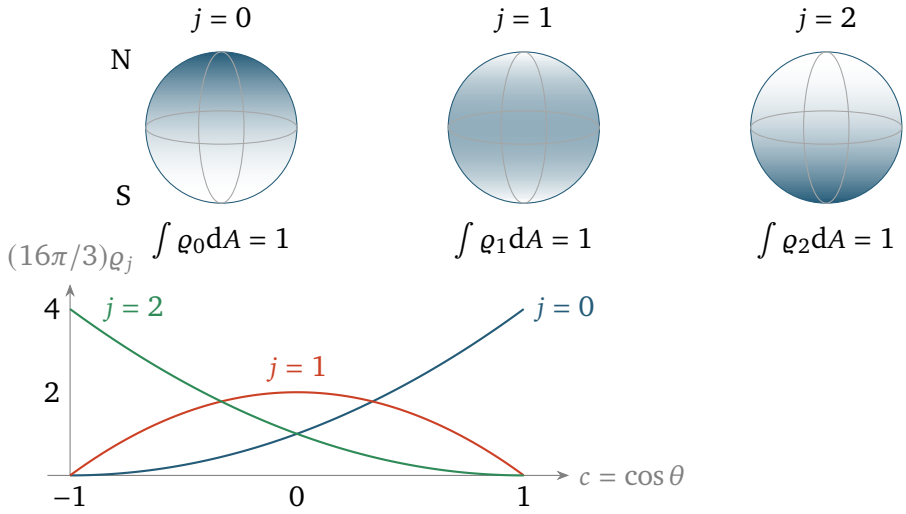


Figure 28.4: The shading and latitude curves show the three normalized $n = 3$ mode densities. The common plotted scale is $(16\pi/3)\varrho_j$, and each mode has integral one.

Rotations of the sphere lift to the monopole line through SU(2). To see their action on the positive zero modes for $n > 0$, write $m = n - 1$ and homogenize a polynomial:

$$F(Z_0, Z_1) = Z_0^m f_N(Z_1/Z_0) = \sum_{j=0}^{m} a_j Z_0^{m-j} Z_1^j.$$

Linear SU(2) substitutions in (Z_0, Z_1) preserve this degree-m space. On the monomial $Z_0^{m-j} Z_1^j$, the diagonal spin generator has weight $m/2 - j$. The operators $Z_0 \partial_{Z_1}$ and $Z_1 \partial_{Z_0}$ connect adjacent monomials with coefficients j and $m - j$. Thus these monomials form the symmetric-power spin-$m/2$ representation, with $m + 1$ states. For $n = 3$ the zero modes therefore form a spin-one triplet under these lifted rotations.

Definition 28.11 (Lowest Landau level)**.** The *lowest Landau level* is the lowest scalar magnetic-kinetic eigenspace.

Physical input. For a spinless particle on the round sphere with nonnegative monopole magnitude n, the lowest Landau level is represented by holomorphic sections of $\mathcal{O}(n)$. This sphere model is described by Greiter, §II, equations (20)–(21).

The polynomial count therefore gives $n + 1$ scalar states, compared with n spinor zero modes for $n > 0$. For $n = 3$, the scalar coefficients have degrees zero through three, giving four states. The spin connection reduces the degree by one in the Dirac problem. It cancels the +1 in the sphere's polynomial dimension formula. This comparison concerns degeneracy; the scalar lowest Landau energy is not the Dirac zero energy.[1]

28.4 The surface index theorem

The square D^2 preserves chirality. Its positive-energy states pair across the two components: if $D^- D^+ u = \epsilon u$ with $\epsilon > 0$, then $D^+ u/\sqrt{\epsilon}$ is a normalized state of $D^+ D^-$ with the same eigenvalue. Its norm is one because $\|D^+ u\|^2 = \langle u, D^- D^+ u \rangle = \epsilon$ for $\|u\| = 1$. The reverse map is $D^-/\sqrt{\epsilon}$. The figure shows why a difference can remain only at zero.

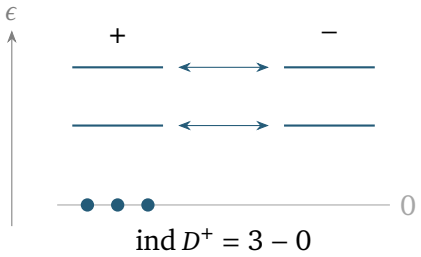


Figure 28.5: Positive squared-Dirac eigenvalues pair between chiralities. The schematic zero eigenspace has three positive modes and no negative mode, so its index is three.

The kernels are finite-dimensional by the cited elliptic theory of Lawson–Michelsohn, *Spin Geometry*, Chapter III, §5.

[1] Greiter uses negative electron charge and monopole magnitude $2s_0$. His lowest level has angular momentum s_0 and dimension $2s_0 + 1$. Here $n = 2s_0$ in the scalar comparison is a flux magnitude rather than a signed Chern convention.

Definition 28.12 (Chiral index)**.** The *index* is the signed difference

$$\operatorname{ind} D^+ = \dim\ker D^+ - \dim\ker D^-.$$

Both dimensions are over $\mathbb{C}$.

For the monopole connection, the computed kernel dimensions give $\operatorname{ind} D^+ = n$. An index of zero alone does not say that either kernel vanishes.

The polynomial count determines the left side of an index theorem for this sphere connection. The right side is the flux integer already computed in Lecture 22.

Theorem 28.13 (Atiyah–Singer on a surface · *cited*)**.** *Let M be a closed oriented Riemannian spin surface and L a Hermitian line bundle with unitary connection of curvature F_{conn}. Choose positive spinors in the square root of the canonical line for the chosen spin structure. For the corresponding chiral twisted Dirac operator,*

$$\operatorname{ind} D^+ = \int_M c_1(L) = \frac{\mathrm{i}}{2\pi}\int_M F_{\mathrm{conn}}.$$

This is the surface specialization of the twisted Dirac index theorem (Nakahara, equations (12.85)–(12.86)). The curvature here is anti-Hermitian, as in Lecture 22. For the sphere monopole, $F_{\mathrm{conn}} = -\mathrm{i}n\sin\theta\,\mathrm{d}\theta\wedge\mathrm{d}\varphi/2$. Its curvature integral gives $\mathrm{i}(-2\pi\mathrm{i}n)/(2\pi) = n$, agreeing with the kernel count. The theorem applies beyond this symmetric monopole connection. It fixes the difference of the kernels under smooth changes of the metric and connection on the same graded bundle. It does not generally fix either kernel dimension separately.

We use the self-adjoint realization of D on the closed sphere with the first-order Sobolev domain defined next. This is supplied by cited elliptic spectral theory (Lawson–Michelsohn, Theorem 5.8, pp. 196–197), with the domain identified by the elliptic estimate in Theorem 5.2.

Definition 28.14 (First-order Sobolev domain)**.** The first-order Sobolev domain consists of square-integrable sections whose first derivatives, understood by integration by parts, are also square-integrable in local charts.

Definition 28.15 (Fredholm operator)**.** An operator between the first-order Sobolev domain and L^2 is *Fredholm* if it has closed range and finite-dimensional kernel and cokernel.

The chiral block is Fredholm by the cited elliptic theory. The cokernel is the target modulo the range, and is identified with $\ker D^-$ by the adjoint. These analytic facts follow from cited elliptic theory (Lawson–Michelsohn, Chapter III, Theorem 5.2, pp. 193–194).

Along a norm-continuous path of Fredholm chiral blocks from the fixed Sobolev domain to L^2, the index is constant by the cited Fredholm stability theorem (Lawson–Michelsohn, Proposition 7.1 and Corollary 7.4, pp. 201–202). Each kernel dimension can change, with equal changes in the two counts.

The finite-dimensional model $T_t\colon \mathbb{C}^2 \to \mathbb{C}$, given by $T_t(a,b) = ta$, illustrates this: for $t \neq 0$ its kernel and adjoint kernel have dimensions 1 and 0. At $t = 0$ they have dimensions 2 and 1. Their difference stays one. This finite-dimensional family illustrates index stability. The separate counts in the symmetric sphere family were fixed by the explicit count. Perturbations that destroy the chiral block structure are outside this statement.

Self-check. For $n = 0$, determine the sphere index and both zero-mode counts. Explain what follows from the index alone.

Answer. The sphere theorem gives index zero and both kernel dimensions zero. The index alone says only that the two dimensions agree. Their vanishing uses the sphere-specific polynomial and Serre-duality count.

Self-check. For $n = -3$, determine both chiral counts and give northern negative-mode coefficients in a holomorphic frame. Also decide whether a northern coefficient z^5 is a global positive mode when $n = 5$.

Answer. The positive degree is -4, so there are no positive modes. The adjoint equation uses $O(2)$, whose coefficients are $1, z, z^2$. Since $h = (1+|z|^2)^4$, the three negative coefficients are $(1+|z|^2)^{-4}\bar{z}^j \mathrm{d}\bar{z}$, for $j = 0, 1, 2$. For $n = 5$ the positive degree is four. The proposed z^5 becomes w^{-1} in the southern frame and is not holomorphic there.

Self-check. State in one sentence what the chiral index is and why we introduced it.

Answer. The chiral index is the difference of the complex zero-mode dimensions, introduced because positive squared-Dirac eigenvalues pair across chiralities and only the zero modes can leave an imbalance. The surface index theorem identifies that difference with the curvature integer.

Topological counts across the course

A continuous integer-valued function on a connected parameter space is constant. We used this fact to prove stability once an integer had been defined. Lifts, Stokes' theorem and comparison on overlaps gave concrete ways to compute several of these integers. The circle row concerns a smooth map $f\colon S^1 \to S^1$. The Euler row concerns a closed oriented surface Σ. The first Chern row concerns a line bundle L over the sphere with the northern-boundary equator convention. The band row concerns Lecture 25's two-band direction map $\widehat{\mathbf{d}}$. The instanton row uses Lecture 23's SU(2) clutching map h_{cl}. The index row concerns the closed spin surface M and twisted Dirac operator of the surface theorem. The figure records the corresponding equalities.

$\mathrm{wind}(f)$	$=$	$\deg f$	Lecture 3, 14
$\chi(\Sigma)$	$=$	$\frac{1}{2\pi}\int_\Sigma K \mathrm{d}A$	Lecture 6, 18
$\int_{S^2} c_1(L)$	$=$	$\mathrm{wind}(g_{NS})$	Lecture 22
C	$=$	$\deg \widehat{\mathbf{d}}$	Lecture 25
k	$=$	$-\deg h_{\mathrm{cl}}$	Lecture 23
$\mathrm{ind}\, D^+$	$=$	$\int_M c_1(L)$	Lecture 28

Figure 28.6: Each equality connects two descriptions of the same count, under the hypotheses printed in the cited lectures. The instanton row uses the clutching map and its stated equator orientation.

Winding in Lecture 3 records the endpoint of a lifted circle loop. Degree in Lecture 14 counts signed preimages between closed connected oriented manifolds of the same dimension. For sphere maps, Lecture 26 also identifies degree with the corresponding sphere homotopy class.

The Euler characteristic joins homology, indices of zeros and total curvature in Lectures 6, 14 and 18. The line-bundle integer joins transition winding and curvature in Lecture 22. Lecture 25 applies the same construction to a band eigenline. In Lecture 23 the instanton integer is the degree of the asymptotic $S^3 \to \mathrm{SU}(2)$ map, with the opposite sign for the clutching map.

Defect data in Lecture 26 are homotopy classes, sometimes with a fundamental-group action; they need not be integers. The kink and vortex of Lecture 27 have integer charges under their stated boundary conditions.

The sphere Dirac calculation adds a spectral count to this list. Its index fixes the difference of zero-mode dimensions, while the separate dimensions come from the explicit polynomial and duality calculation. Berry holonomy remains a phase rather than an integer.

Unit 28B Chiral anomaly

28.5 The two-dimensional chiral anomaly *(optional)*

The sphere count, the surface index theorem and the closing synthesis do not depend on this section.

For this application we reinterpret two dimensions as Euclidean spacetime. The spatial-first coordinates are (x, τ) with orientation $\mathrm{d}x \wedge \mathrm{d}\tau$ and $z = x + \mathrm{i}\tau$. The Wick rotation is $t = -\mathrm{i}\tau$, so the local holomorphic argument becomes $z = x - t$. A fixed value of $x - t$ moves toward increasing x. Positive Euclidean chirality therefore continues to right movers. This local convention matches the Euclidean grading to the Lorentzian one.

Definition 28.16 (Chiral anomaly)**.** An *anomaly* is a failure after quantization of a classical conservation law. For the massless Dirac field with vector gauge coupling, the chiral anomaly concerns the separate right- and left-moving fermion-number currents.

Remark 28.17 (Axial charge and index · *cited*)**.** Let j_R and j_L be those currents, and let Q_R and Q_L be their integrated net numbers of particles minus antiparticles. The vector and axial charges are

$$Q = Q_R + Q_L, \qquad Q_5 = Q_R - Q_L, \qquad j_5 = j_R - j_L.$$

Physical input. Classically the massless model conserves the right- and left-moving number currents separately. Quantization preserving the vector current can violate axial-current conservation in a gauge background. For one Dirac fermion, with the chirality convention above and with no axial charge escaping through a spatial boundary, the cited integrated law is

$$\Delta Q_5 = 2 \operatorname{ind} D^+, \qquad \Delta Q = 0.$$

This is the chiral anomaly input from Bilal, §3.5–3.7, with the even-dimensional extension

in equations (3.44)–(3.47).[2]

If the Euclidean index is N, the sum and difference equations give $\Delta Q_R = N$ and $\Delta Q_L = -N$. An index +1 background therefore changes the imbalance by +2, while preserving the vector charge. The Euclidean spacetime background in this statement is distinct from the spatial sphere used to count monopole states.

We illustrate the sign with a ring of circumference $\ell > 0$.

Physical input. Take noninteracting massless right and left movers with speed $v > 0$ on the ring, coupled to a spatially uniform A_x in temporal gauge. The chiral Hamiltonians are $v(-i\hbar\partial_x - qA_x)$ and its negative. The reference vacuum fills the negative-energy states. During the flux change we transport occupations of the uncoupled momentum modes. Net particles and holes are counted relative to the final negative-energy vacuum.

We impose antiperiodic boundary conditions to avoid zero-energy endpoints. The allowed momenta and dimensionless flux are

$$k_j = \frac{2\pi}{\ell}\left(j + \frac{1}{2}\right), \quad j \in \mathbb{Z}, \qquad \alpha = \frac{a\ell A_x}{2\pi}, \quad a = q/\hbar.$$

In units of $2\pi\hbar v/\ell$, the two spectra are

$$\varepsilon_{R,j} = j + \frac{1}{2} - \alpha, \qquad \varepsilon_{L,j} = -\left(j + \frac{1}{2} - \alpha\right).$$

We increase $a\ell A_x$ by $+2\pi$, so α increases from zero to one. The endpoints are gauge equivalent through the single-valued phase $e^{2\pi i x/\ell}$. The figure follows the levels without changing their occupations.

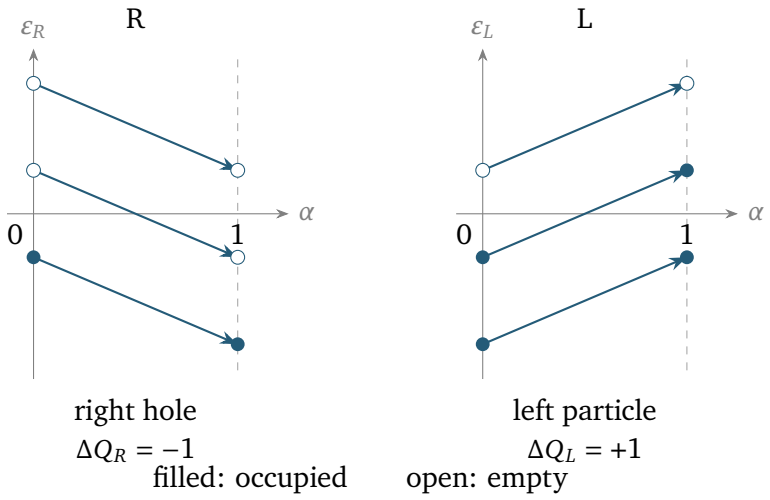


Figure 28.7: Increasing $a\ell A_x$ by 2π moves one empty right level down and one filled left level up through zero. Relative to the final negative-energy sea, they give one right hole and one left particle.

[2] Bilal's equation (3.30) writes the current divergence as minus his anomaly function. His equation (3.41) gives the integral of that function as minus twice the index in four dimensions. The dimension-independent regulated trace in equations (3.39)–(3.40) pairs nonzero modes and counts zero modes with a sign. The fermion and antifermion measures give the factor two. We use the two-dimensional consequence with positive Euclidean chirality continued to right movers.

Initially the occupied right levels have $j \le -1$, whereas the final right sea has $j \le 0$. The transported state is missing its $j = 0$ particle, so $\Delta Q_R = -1$. Initially the occupied left levels have $j \ge 0$, whereas the final left sea has $j \ge 1$. The transported state has one extra $j = 0$ particle, so $\Delta Q_L = +1$. All other occupation differences vanish. This subtraction gives finite changes without subtracting two infinite totals:

$$\Delta Q = 0, \qquad \Delta Q_5 = -2.$$

For the corresponding real Euclidean interpolation $A_x(\tau)$, the orientation is $\mathrm{d}x \wedge \mathrm{d}\tau$. Its curvature is $F_{\mathrm{em}} = -A_x'(\tau)\mathrm{d}x \wedge \mathrm{d}\tau$. Gluing the gauge-equivalent endpoints gives a spacetime torus. The surface index theorem gives

$$\operatorname{ind} D^+ = \frac{a}{2\pi}\int F_{\mathrm{em}} = -\frac{a\ell}{2\pi}\Delta A_x = -1.$$

Thus this flux direction gives $\Delta Q_5 = 2 \operatorname{ind} D^+ = -2$. Reversing the insertion gives index $+1$ and axial change $+2$.

Summary. The positive spinor line on the outward sphere is $O(-1)$. Twisting by the monopole line $O(n)$ gives $O(n-1)$. The Stokes–Cauchy polynomial proof and cited spinor, Hodge and Serre identifications give chiral counts $\max(n, 0)$ and $\max(-n, 0)$. Their difference is n, agreeing with the cited surface index theorem and the flux Chern number. For positive n, the spinor count is n and the spinless lowest Landau-level count is $n + 1$. With the stated spacetime, chirality and boundary conventions, the cited axial anomaly gives $\Delta Q_5 = 2 \operatorname{ind} D^+$. The ring insertion of $a\ell A_x$ by $+2\pi$ has index -1 and axial change -2.

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- *Quantum Mechanics I*. Course notes, cited also as *QM-I*. Lectures 8 and 14 are cited for the Bloch and fermionic conventions.
- *Quantum Mechanics II*. Course notes. Lecture 3 is cited for the phase on a ring.
- *Symmetries in Physics*. Course notes. Lecture 17 is cited for Minkowski space.
- *Theoretical Mechanics*. Course notes. Lectures 2 and 3 are cited for Lagrange's equations and the charged particle.

Index of notation

The symbols are listed alphabetically by meaning. The last column gives the lecture that first defines the symbol or declares its role. A dash means that the lectures do not give a separate definition. Local coordinate letters retain the meanings stated where they occur.

Symbol	Meaning	First definition
Ad_h	Adjoint action	Lecture 16
∇	Affine connection	Lecture 16
$\varepsilon_{ijk},\ \epsilon_{abc}$	Alternating symbol	Lecture 16
L	Angular momentum per rest mass	Lecture 19
F_{conn}	Anti-Hermitian connection curvature	Lecture 21
A_{conn}	Anti-Hermitian connection potential	Lecture 21
T^a	Anti-Hermitian Lie generators	Lecture 23
$\partial_{\bar{z}}$	Antiholomorphic derivative	Lecture 22
$P \times_G V_{\mathrm{rep}}$	Associated bundle	Lecture 20
Q_5	Axial charge	Lecture 28
C_n	Band Chern number	Lecture 25
$\gamma(C)$	Berry phase angle	Lecture 24
b_k	Betti number	Lecture 6
$B_k(X;R)$	Boundary group	Lecture 6
$D_k,\ r_k$	Boundary matrix and its rank	Lecture 6
T	Boundary unit tangent	Lecture 18
λ	BPST size	Lecture 23
B_n	Braid group	Lecture 5
$\mathbb{T}^2$	Brillouin torus	Lecture 25
F	Bundle curvature two-form	Lecture 21
$\bar{\partial}_A$	Bundle Dolbeault operator	Lecture 28
$\mathrm{Hol}(\gamma)$	Bundle holonomy	Lecture 21
π_E	Bundle projection	Lecture 20
g_{ij}	Bundle transition functions	Lecture 20
$\Omega^k(M;E)$	Bundle-valued k-forms	Lecture 21
K_{S^2}	Canonical line on the sphere	Lecture 28
θ	Canonical one-form	Lecture 11
$\omega = \mathrm{d}q^i \wedge \mathrm{d}p_i$	Canonical symplectic form	Lecture 11
∂_k	Chain boundary map	Lecture 6

Continued on the next page

Index of notation (continued)

Symbol	Meaning	First definition
$C_k(X;R)$	Chain group	Lecture 6
ρ	Charge density	Lecture 15
$U,\ U_i$	Chart domains	Lecture 7
$\mathrm{ch}(F)$	Chern character	Lecture 23
$\mathrm{CS}(A)$	Chern–Simons three-form	Lecture 23
$Q_R,\ Q_L,\ j_R,\ j_L$	Chiral charges and currents	Lecture 28
$\operatorname{ind} D^+$	Chiral Dirac index	Lecture 28
$D^+,\ D^-$	Chiral Dirac operators	Lecture 28
$S^+,\ S^-$	Chiral spinor lines	Lecture 28
Γ^k_{ij}	Christoffel symbols	Lecture 16
π	Circle constant	—
$p(t)$	Circle covering projection	Lecture 3
D^n	Closed unit disc	Lecture 2
$\overline{A}$	Closure of a subset	Lecture 1
δ	Codifferential	Lecture 15
R	Coefficient ring	Lecture 6
$\operatorname{coker} T$	Cokernel	Lecture 13
$\mathbb{C}\mathrm{P}^1$	Complex projective line	Lecture 7
Exp_p	Connection exponential	Lecture 16
$\omega^i{}_j$	Connection forms	Lecture 17
$\mathcal{J}_{\mathrm{conn}}$	Connection-valued current	Lecture 21
e_x	Constant path	Lecture 3
ι_X	Contraction with a field	Lecture 11
∂_i	Coordinate tangent derivative	Lecture 8
T^*M	Cotangent bundle	Lecture 10
T^*_pM	Cotangent space	Lecture 10
β_C	Coulomb flux form	Lecture 12
$\frac{DV}{dt}$	Covariant derivative along a curve	Lecture 16
$D,\ D_A$	Covariant exterior derivative	Lecture 21
j	Current one-form	Lecture 15
$\Omega^i{}_j$	Curvature forms	Lecture 17
$R(X,Y)Z$	Curvature operator	Lecture 17
E	Curve energy	Lecture 16
$Z_k(X;R)$	Cycle group	Lecture 6
$H^k(M)$	De Rham cohomology	Lecture 13
$[\alpha]$	De Rham cohomology class	Lecture 13
$\mathrm{Deck}(p)$	Deck transformation group	Lecture 4
$\deg f$	Degree of a map	Lecture 14

Continued on the next page

Index of notation (continued)

Symbol	Meaning	First definition
$O(n)$	Degree-n holomorphic line	Lecture 22
ω	Differential form	Lecture 10
$\mathrm{d}f_p$	Differential of a smooth map	Lecture 8
α	Dimensionless ring flux	Lecture 28
D	Dirac operator	Lecture 28
$\coprod$	Disjoint union	Lecture 4
$d(x, y)$	Distance in a metric space	Lecture 1
D_p	Distribution at a point	Lecture 9
$\mathrm{div}_\mu X$	Divergence relative to a volume form	Lecture 11
$H^0(E),\ H^1(E)$	Dolbeault cohomology	Lecture 28
$\bar{\partial}_E$	Dolbeault operator on a line bundle	Lecture 28
θ^a	Dual basis and coframe	Lecture 15
L_n	Eigenline bundle	Lecture 24
$G_{\mu\nu}$	Einstein tensor	Lecture 19
$\mathbf{E}$	Electric field	Lecture 12
F_{em}	Electromagnetic flux two-form	Lecture 13
χ	Electromagnetic gauge function	Lecture 15
ϕ	Electromagnetic scalar potential	Lecture 15
$[\gamma]$	Endpoint-fixed path class	Lecture 3
E_n	Energy eigenvalue	Lecture 24
E	Energy per unit rest mass	Lecture 19
S_E	Euclidean Yang–Mills action	Lecture 23
χ	Euler characteristic	Lecture 6
D^+_{dR}	Even-to-odd de Rham operator	—
$\psi_{t,s}$	Evolution of a time-dependent field	Lecture 11
d	Exterior derivative	Lecture 10
$\Lambda^k V^*$	Exterior power	Lecture 10
E_F	Fermi energy	Lecture 25
tr	Finite defining trace	Lecture 22
c_1	First Chern class	Lecture 22
$C_1(E;\Sigma)$	First Chern number	Lecture 22
φ_t	Flow of a vector field	Lecture 9
$\bar{\partial}^*_A$	Formal adjoint of the Dolbeault operator	Lecture 28
$U = \mathrm{d}x/\mathrm{d}\tau$	Four-velocity	Lecture 19
F_k	Free group	Lecture 5
$G * H$	Free product	Lecture 5
$\pi_1(X, x_0)$	Fundamental group	Lecture 3
$\xi^\#_p$	Fundamental vertical vector	Lecture 23

Continued on the next page

Index of notation (continued)

Symbol	Meaning	First definition
K	Gaussian curvature	Lecture 17
$\mathrm{GL}(n, \mathbb{K})$	General linear group	Lecture 8
Σ_g	Genus-g surface	Lecture 5
k_g	Geodesic curvature	Lecture 18
∂M	Geometric boundary	Lecture 12
$E_2,\ E_0$	Gradient and potential energies	Lecture 27
e	Group identity	Lecture 8
$\langle a_i \mid r_j \rangle$	Group presentation	Lecture 5
σ_{H}	Hall conductance	Lecture 25
H	Hamiltonian function	Lecture 11
X_H	Hamiltonian vector field	Lecture 11
$\mathrm{Herm}(N)$	Hermitian matrices	Lecture 24
A_{phys}	Hermitian physics potential	Lecture 21
ϕ	Higgs scalar	Lecture 27
λ	Higgs scalar coupling	Lecture 27
$\pi_n(X, x_0)$	Higher homotopy group	Lecture 26
$\mathcal{H}$	Hilbert space of edge qubits	Lecture 6
Δ	Hodge Laplacian	Lecture 15
$\star$	Hodge star	Lecture 15
∂_z	Holomorphic derivative	Lecture 22
$X \cong Y$	Homeomorphic spaces	Lecture 1
$[a : b]$	Homogeneous coordinates	Lecture 7
G/H	Homogeneous coset space	Lecture 8
$H_k(X; R)$	Homology group	Lecture 6
$f \simeq g$	Homotopic maps	Lecture 3
K_F	Homotopy operator	Lecture 13
$\mathbb{1}$	Identity matrix	—
$\mathcal{A}$	Imaginary Berry connection form	Lecture 24
$\mathrm{ind}_p X$	Index of an isolated zero	Lecture 14
f_*	Induced fundamental-group map	Lecture 3
k	Instanton number	Lecture 23
$\mathbb{Z}_m$	Integers modulo m	Lecture 5
$\mathbb{Z}_2$	Integers modulo two	Lecture 4
$\int_M \alpha$	Integral of a top form	Lecture 12
$(\alpha, \beta)_{L^2}$	Integrated pairing of forms	Lecture 15
$\mathrm{int}\, A$	Interior of a subset	Lecture 1
$u = 1/r$	Inverse radial coordinate	Lecture 19
$\mathcal{U}$	Isolated-eigenvalue parameter region	Lecture 24

Continued on the next page

Index of notation (continued)

Symbol	Meaning	First definition
$\ker f$	Kernel of a homomorphism	Lecture 5
K	Killing vector field	Lecture 16
Q	Kink topological charge	Lecture 27
ℓ_g	Left translation	Lecture 9
A^L	Left-invariant field from a tangent vector	Lecture 9
$\ell(\gamma)$	Length of a curve	Lecture 15
$[A, B]$	Lie algebra bracket	Lecture 8
$\mathfrak{g}$	Lie algebra of a Lie group	Lecture 8
L_X	Lie derivative of a field	Lecture 9
$L_X\alpha$	Lie derivative of a form	Lecture 11
G	Lie group	Lecture 8
exp	Lie group exponential	Lecture 9
$\tilde{f}$	Lift of a map	Lecture 3
$b_{\text{impact}}\ (b)$	Light-ray impact parameter	—
$\Omega^{0,k}(E)$	Line-valued forms of type (0,k)	Lecture 28
X_A	Linear vector field	Lecture 9
$\operatorname{sgn}\det \mathrm{d}f_x$	Local degree sign	Lecture 14
h_i	Local frame changes	Lecture 20
u_i	Local gauge maps	Lecture 21
$\mathbb{H}^n_-$	Lower half-space	Lecture 12
$\mathbf{B}$	Magnetic field	Lecture 15
$M,\ N$	Manifold and target manifold	Lecture 7
$\dim M$	Manifold dimension	Lecture 7
ϑ_{MC}	Maurer–Cartan form	Lecture 21
$\mathcal{D}$	Maximal flow domain	Lecture 9
I_p	Maximal time interval	Lecture 9
δ_{MV}	Mayer–Vietoris connecting map	Lecture 13
g	Metric tensor	Lecture 15
η	Minkowski metric	Lecture 15
S_M	Minkowski Yang–Mills action	Lecture 23
$n \in \mathbb{Z}$	Monopole Chern integer	Lecture 22
n	Monopole flux parameter	Lecture 13
$\flat,\ \sharp$	Musical maps	Lecture 15
G_N	Newton constant	Lecture 19
Φ	Newtonian gravitational potential	Lecture 19
$\langle\langle r_i \rangle\rangle$	Normal closure	Lecture 5
ω_N	Northern surface connection	Lecture 18
c_k	Number of k-simplices	Lecture 6

Continued on the next page

Index of notation (continued)

Symbol	Meaning	First definition
$B_r(x)$	Open metric ball	Lecture 1
Tr	Operator trace	—
X/G	Orbit space	Lecture 2
$\mathcal{M}$	Order-parameter space	Lecture 26
$\mathrm{Conf}_n^{\mathrm{ord}}(X)$	Ordered configuration space	Lecture 5
$\mathrm{O}(n)$	Orthogonal group	Lecture 7
p_γ	Outgoing permutation	Lecture 5
Φ_{m}	Outward monopole flux	Lecture 13
ω_{S^2}	Outward sphere area form	Lecture 10
P_γ	Parallel transport of tangent vectors	Lecture 16
ρ_i	Partition of unity functions	Lecture 12
γ	Path	Lecture 2
$\pi_0(X)$	Path component set	Lecture 26
$\gamma \cdot \delta$	Path concatenation	Lecture 3
$\mathcal{P}\exp$	Path-ordered exponential	Lecture 21
σ_a	Pauli matrices	Lecture 4
$X_e,\ Z_e$	Pauli operators on an edge	Lecture 6
S_n	Permutation group	Lecture 5
h_{P}	Planck constant	Lecture 21
$\{f, g\}$	Poisson bracket	Lecture 11
θ	Polar angle	Lecture 12
$e > 0$	Positive elementary charge	Lecture 25
$O(-1)$	Positive spinor line on the sphere	Lecture 28
J	Positive surface quarter-turn	Lecture 18
ω_P	Principal connection form	Lecture 23
$\kappa_1,\ \kappa_2$	Principal curvatures	Lecture 17
τ	Proper time	Lecture 19
$f^*\alpha$	Pullback of a form	Lecture 10
$\widehat{L_a}$	Quantum angular momentum	Lecture 9
$B([a : b])$	Qubit Bloch vector	Lecture 7
G/N	Quotient group	Lecture 5
$X/\sim$	Quotient set	Lecture 2
u	QWZ mass parameter	Lecture 25
A_{B}	Real Berry potential	Lecture 24
$\mathcal{J}$	Real current three-form	Lecture 15
A_{em}	Real electromagnetic potential	Lecture 13
$\mathbb{R}\mathrm{P}^n$	Real projective space	Lecture 2
Sym_n	Real symmetric matrices	Lecture 7

Continued on the next page

Index of notation (continued)

Symbol	Meaning	First definition
$\hbar$	Reduced Planck constant	Lecture 9
V_{rep}	Representation fibre	Lecture 20
$\bar{\gamma}$	Reversed path	Lecture 3
Ric	Ricci tensor	Lecture 17
$\mathrm{SO}(3)$	Rotation group	Lecture 2
$\mathfrak{so}(3)$	Rotation Lie algebra	Lecture 8
L_a	Rotation vector field	Lecture 9
R_{sc}	Scalar curvature	Lecture 17
$\bar{\partial}$	Scalar Dolbeault operator	Lecture 22
$V(\phi)$	Scalar potential energy density	Lecture 27
U_{eff}	Schwarzschild effective potential	Lecture 19
$f(r) = 1 - 2M/r$	Schwarzschild lapse function	Lecture 19
M_{BH} (M)	Schwarzschild mass parameter	Lecture 19
c_2	Second Chern class	Lecture 22
σ_i	Section coefficient column	Lecture 20
$K(P)$	Sectional curvature	Lecture 17
$F_+,\ F_-$	Self-dual and anti-self-dual parts	Lecture 23
q	Signed particle charge	Lecture 13
$\varphi \circ \partial_{k+1}$	Singular cochain coboundary	Lecture 14
$\Gamma(E)$	Smooth sections of a bundle	Lecture 21
$\mathfrak{X}(M)$	Smooth vector fields	Lecture 16
Φ	Solenoid flux	Lecture 21
Ω	Solid angle	Lecture 18
$\Omega^k(M)$	Space of k-forms	Lecture 10
$\mathbf{J}$	Spatial current vector	Lecture 15
λ	Spatial dilation parameter	Lecture 27
$\mathrm{SO}(n)$	Special orthogonal group	Lecture 8
$\mathrm{SU}(2)$	Special unitary group	Lecture 4
b	Spin field magnitude	Lecture 24
$j,\ m$	Spin label and weight	Lecture 24
S_a	Spin operators	Lecture 7
Δ^k	Standard simplex	Lecture 6
$T_{\mu\nu}$	Stress–energy tensor	Lecture 19
$W(\phi)$	Superpotential	Lecture 27
$\operatorname{supp} X$	Support of a vector field	Lecture 9
$\omega^1{}_2$	Surface connection entry	Lecture 17
g	Surface genus	Lecture 5
TM	Tangent bundle	Lecture 8

Continued on the next page

Index of notation (continued)

Symbol	Meaning	First definition
e	Tangent frame	Lecture 16
P_γ	Tangent holonomy	Lecture 18
T_pM	Tangent space	Lecture 8
$\mathcal{T}$	Tautological line bundle	Lecture 20
θ_{YM}	Theta-term coefficient	Lecture 23
$\mathcal{B}$	Topological basis	Lecture 1
T^2	Topological torus	Lecture 2
$\mathcal{T}$	Topology	Lecture 1
C	Toric code subspace	Lecture 6
$A_v,\ B_p$	Toric-code stabilizers	Lecture 6
$Z(c)$	Toric-code Z-string	Lecture 6
$T(X,Y)$	Torsion of a connection	Lecture 16
$c(F)$	Total Chern form	Lecture 22
$U(t)$	Transport matrix	Lecture 21
$\alpha_i,\ \varepsilon_i$	Triangle angles and exterior turns	Lecture 18
$S \otimes L_n$	Twisted spinor bundle	Lecture 28
$\mathbf{d}(\mathbf{k})$	Two-band Bloch vector	Lecture 25
$\widehat{\mathbf{d}}$	Two-band direction map	Lecture 25
$S^1,\ S^n$	Unit sphere	Lecture 1
$\chi : G \to \mathrm{U}(1)$	Unitary character	Lecture 5
$\mathrm{U}(n)$	Unitary group	Lecture 7
$\mathrm{Conf}_n(X)$	Unordered configuration space	Lecture 5
$\mathbb{H}^n$	Upper half-space	Lecture 12
v	Vacuum magnitude	Lecture 27
$\mathcal{V}$	Vacuum manifold	Lecture 27
∇s	Vector-bundle covariant derivative	Lecture 21
e_i	Vector-bundle frame	Lecture 20
X^i	Vector-field components	Lecture 9
$[X,Y]$	Vector-field Lie bracket	Lecture 9
μ	Volume form	Lecture 11
$\alpha \wedge \beta$	Wedge product	Lecture 10
$\mathrm{wind}\,\gamma$	Winding number	Lecture 3
g_{YM}	Yang–Mills coupling	Lecture 21

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